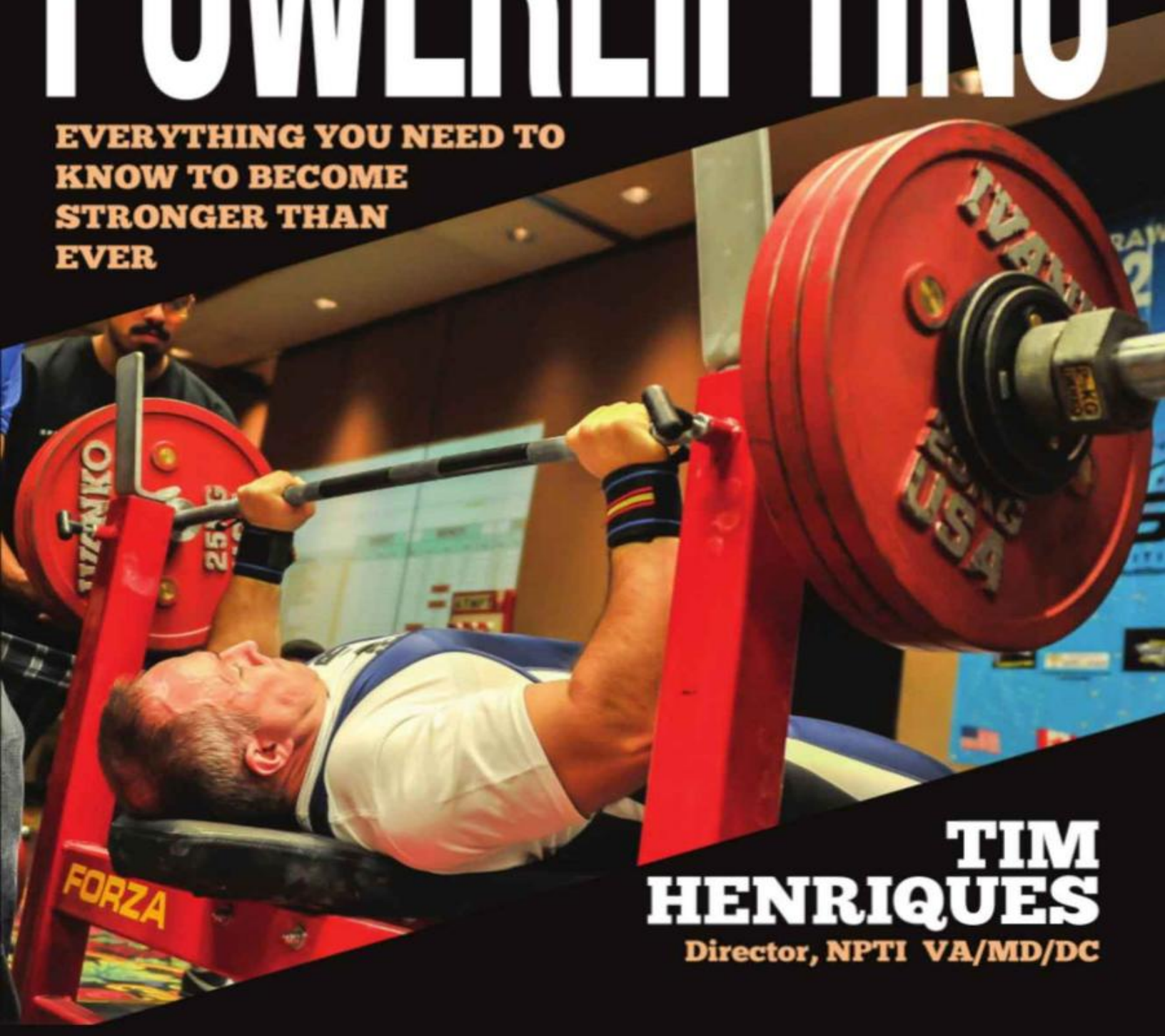
ALL ABOUT

POWERLIFTING

EVERYTHING YOU NEED TO KNOW TO BECOME STRONGER THAN EVER



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This book is dedicated to the sport of powerlifting and all of the athletes, officials, volunteers, and fans who help make it possible.

Disclaimer Statement

This book is for educational and informative purposes. This book is not meant to diagnose or treat any medical condition. Consult a medical professional and a qualified fitness instructor before beginning any exercise or nutrition program. All physical activity has inherent risks. Powerlifting, intense weight training, and other modes of training for sport carry exaggerated risks. If something hurts when you are training, cease that activity. Use all recommended safety equipment, safety apparatuses, and suggested precautions when exercising. References, web sites, and workout programs are provided as a guide, their inclusion does not constitute an endorsement from the author and any affiliates. Individual responses vary dramatically to exercise and nutrition guidelines. Users use these plans and knowingly participate at their own risk. The author, publishing company, any affiliates and associates of those parties are not responsible for any injury, damages, disability or death resulting from the information presented within this book.

Acknowledgments

It is hard to construct an appropriate thank you letter when literally the significant majority of your entire life has been spent working toward this goal. By “this goal” I mean being proficient enough in powerlifting and strength training to feel as though I can give something back to that community.

My first thank you is to the sport itself and all those involved in it. I was an athletic late bloomer in a household full of boys and a dad who loved competition. I was never a soccer or football or basketball star — indeed when I was 12 I remember my dad complaining to the youth league basketball coach that I didn’t play the minimum time necessary: 2 quarters. I didn’t blame the coach; I sucked, and I would rather win with me sitting on the bench than lose if that meant I got more playing time. I am pretty sure I scored a basketball on my team’s goal that season when I got so excited the ball was in my hands I just threw it at the hoop, forgetting, of course, I was on the wrong side of the court to make a shot.

Weight training and powerlifting entered the scene when I was 14, and I got a membership to Gold’s, joining as my childhood friend’s “cousin” to get a discount on the price. Fast forward 3 years and I was one of the strongest kids in my high school, I played varsity football, and I was awarded Health and PE Student of the Year my senior year.

I want to thank all of the regular workout partners I have had throughout my lifting career. I would not be where I am today without them. My best friend and collegiate workout partner Chris (how many people would fly on their own dime to 2 collegiate Nationals just to help their partner out?); my high school partners George, Matt, Brian, and Mike; my other college partners, Sean who showed me what true strength was, Mike, Dave (you are missed), Danny, and Ken; and my Gold's buddies Antonio, Mike, Carlos and Charles.

Right now I am blessed to coach a kick-ass team and have great workout partners. I want to thank all of Team Force, I have learned a ton from you and hopefully you have learned something from me. I particularly want to acknowledge my 2:30 crew of Sharif, Vadim, Jason and Rick. Looking back on things it is tough not to think of those workout times as some of the best times of my life; hopefully, we have more memories to create in the future.

I want to thank the meet directors, federation officials, volunteers, and judges who allow me to partake in the sport I love — without them there would be no sport and certainly no book. I have had great experiences with all of them, but I want to particularly acknowledge John Shifflett for always putting on great meets and Paul Bossi for the time and effort he puts into running his federation.

I want to thank my parents for their support during the years — often my big Christmas present was a plane ticket to Collegiate Nationals, and that was a big help. I want to particularly thank my Mom for teaching me to believe in myself and to my Dad for always pushing me to achieve more. And to my brothers — they are each unique and have blazed paths of success in their area of expertise. They have also set examples for me on how to be a man, a father, and a member of the community.

I want to thank Margaret Bishop for helping me edit this work and

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I want to thank those who contributed photos to this work, including hookgrip.com and all of the lifters I was able to interview. I want to particularly acknowledge Doug Jantz for his exceptional pictures; he is a photographer able to be hired for special events, including powerlifting competitions if you are interested in his services. I also want to thank my workout partner Vadim Snitkovsky for all of the high-quality images I was able to access from his files.

Finally I want to thank my family — my boys Nathan, Ryan, Collin, and my wife. Luckily I was able to work on this a good chunk before they were born, but still I have had to sacrifice time with them to work on this project to bring it to fruition. I want to thank them for allowing me to be selfish and focus on a goal that is important to me. I want to particularly acknowledge my wife, Christina. We live in a world based part in fantasy and part in reality. Part of us dreams about meeting one's soulmate and living happily ever after; yet practically we need a solid life partner who is strong and sturdy and who can pull more than their fair share. I have been lucky to find both of those things wrapped up in one amazing woman.

Introduction

The purpose behind this book is very simple. I love powerlifting, and I want to get you to like, and then ultimately love, powerlifting as well. I believe powerlifting is a very easy sell. What other activity improves your health; builds impressive amounts of muscle; makes you actually stronger than you look; improves athletic performance; builds confidence; relieves stress; fights depression; teaches life lessons; and improves one's quality of life? And this is an activity that you can perform throughout your life, from 15 to 85! I believe once you learn a little bit about powerlifting, you will like it. Hence the name of this book: *All About Powerlifting*. I believe the more you learn about powerlifting, the more you will like it. Take that first leap of faith and step up on the platform and you might just find yourself saying that you love powerlifting as well.

This book was written for those lifters who are interested in powerlifting or those who have actually competed. Lifters with fewer than 10 meets under their belt will likely get the most out of it, but I feel there is enough advanced information presented here for even elite athletes to get something from it — and at that stage what would you pay to take that bench from 450 to 500? You'll find that information and much more contained within these pages.

This book was written for those "pretty strong" lifters — the weekend warriors, the meatheads (I proudly consider myself one of that group) —

considering taking their training to the next level. If you are one of the strongest people in your gym, or you wish to become so, the information provided within will prove invaluable in that quest.

This book was written for athletes who include resistance training in their programs with the knowledge that the stronger the athlete, the better the athlete. You lift for a reason — why not maximize your gains and take advantage of your full athletic potential? If you are going to squat and bench and deadlift in your training, you might as well get the most out of the exercises while staying injury free. This is your guide, and you may be amazed at the difference it makes in how you perform. Unlock your ability by harnessing your full strength and power.

This book was written for those who have a powerlifter in their life, be it their husband or wife, son or daughter, father or mother, or just a person who they have a strong bond with. If you want to know what it is like to be a powerlifter, if you want to learn what is necessary to succeed in this sport and how you can support the one you love, you'll find that information inside.

This book was written for those who want to get a glimpse as to what it is like to be a truly elite powerlifter. I have included numerous, in-depth interviews with literally some of the best powerlifters to ever walk the planet. Honestly their information alone is worth the price of admission. Even if you don't get much from what I wrote, you can't help but to pick up valuable insights as these incredible lifters share what makes them tick, their ups and downs, and ultimately what they learned in their climb to the top.

Ask a strong person, someone who had to work hard for what they have, and ask them if it was worth it. Was that pursuit of strength worth the time and effort, the blood and the sweat, the sacrifices they had to make? Would they do it all over again? I've yet to meet someone who says no — it isn't worth it. That is as ringing of an endorsement as you can find. If people walk

the path you are interested in, if they get to the end of the path and then beckon you to follow, if they assure you the struggles and the challenges will be worth it, how can one not be inclined to follow?

Finally this book was written for me and for the sport itself. I feel that I have been so blessed to be able to compete in powerlifting for two decades; the sport has given so much to me that I wanted to give something back. This is my contribution to the sport. If I could step back in time and give this book to my 17-year-old self, the one who was nervous and excited about making that 4 hour drive alone to my first competition, I like to think that kid would have gotten something out of having this book as a guide for that long journey that was to come. I like to think that you will get something out of it as well.

A Note about the Pictures

Throughout this book you will see a variety of powerlifters performing various lifts. A few of the lifters — mainly the ones I had the pleasure of interviewing — are well known and are World Class lifters. Many of the other lifters, including myself, are not. I would have loved to include more pictures of elite powerlifters, both from the past and the present, but simple logistics make that difficult. The images taken from the cameras of a decade or two ago do not look great in print, and lifting pictures from the Internet and other media only works so well and that is often not a feasible option. Without a plethora of high-quality, easily available pictures of elite powerlifters doing their thing, I had to come up with another option.

That meant I had to use images I had ready access to, which were primarily images of myself and my teammates competing. Indeed I felt lucky to have access to the large number of high-resolution pictures that I did, in part thanks to my teammates with their excellent equipment and fine camera work during various competitions. I do hope to show with those images that powerlifting is an inclusive sport — there is room on the platform for the 10-year-old kid who wants to be like his dad, the teenager just getting started, the female showing that weights aren't just for the boys, the elite powerlifter lifting massive poundages, and the master lifter demonstrating that strength doesn't have to disappear as the candles accumulate on the birthday cake.

Ultimately powerlifting is just about you versus you — can you lift more than you have before? It is my hope that these images add entertainment and educational value to the text, and perhaps a bit of motivation as well. If you have high-resolution images that you think would add to the text and you would like them to be included in future editions, contact me and we'll see what we can do.

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About the Author

Chapter 1

What Is Powerlifting?

Powerlifting is the greatest sport there is. That is a pretty bold statement, I know, but I stand by it. How can that be true? It is certainly not the most popular sport there is, particularly in America. In Europe, weight lifters and powerlifters are held in higher esteem than they are in North America, but they are still not as popular as the best soccer or basketball stars.

Elite Lifter Mike Eaton locks out a heavy deadlift
Photo credit: Vadim Snitkovsky

Powerlifting competitions are not the most exciting competitions to watch. I would not call them boring, although I have heard them described that way. And they can be long; a short powerlifting competition lasts six hours, and a long one can run for more than twelve hours. Powerlifting is a relatively low-skill sport. When people watch a gymnast or a downhill skier or a football player they often marvel — how can they do that? The thought of doing that activity is beyond the realm of the possible for most individuals. But powerlifting is not that hard. We perform a squat and a deadlift every day, each time you stand up after you sit down and each time you pick something up. A bench press is easy to do; anybody can lie down and press something straight up. Because everybody can powerlift, everybody can, at some level, relate to the lifts. The question isn't if you *can* squat, it is *How much* can you squat? Powerlifting is a relatively low skill activity taken to the

extreme.

If powerlifting is not the most popular sport, and if it is not the most exciting sport to watch (although it does have its moments), how can I say it is the greatest sport? It is the greatest sport to *train* for, and let's face it, what do athletes spend the vast majority of their time doing? Training, of course. Even in a sport with frequent competitions, most athletes will still spend many more hours training than they will competing. If training is how you are going to spend most of your time, you want the training to be enjoyable. And in my opinion, there is no sport that is more fun to train for than powerlifting.

What makes training for powerlifting so great? There are many reasons. From a practical point of view, training for powerlifting is pretty cheap. You need access to a barbell and some weights. You can buy your own, but almost all powerlifters train in some sort of gym (homemade or commercial), and most gym memberships run from $20 to $100 a month, with about $30 being pretty average. For a dollar a day, you can have access to thousands of dollars of equipment. There are gyms all over the country and the world; weight training is not just for the elite or the rich. Anyone is able to powerlift. Men, women, kids, older adults, disabled people, special Olympians, all can train for and compete in powerlifting. Powerlifting is usually performed inside in a gym or fitness facility; thus it is not weather dependent. There is no season for powerlifting; it can and should be practiced year round. A better way of putting it is that in powerlifting, there is no off season.

> ***The question isn't if you* can *squat, it is* How much *can you squat?***

Weight training in general is a very safe activity, particularly once a basic understanding of proper form and technique is achieved. It is true that

powerlifting is weight training taken to the extreme, but compared to almost all other sports, powerlifting is quite safe. The rates of injuries in contact sports such as soccer and football are very high, but powerlifting injuries are relatively rare given the number of people who lift weights, many without any real instruction. Even when a person is injured, weight training is often prescribed as part of their physical therapy program, and with a proper understanding of anatomy, you can normally work around injuries until they heal.

Powerlifting at its heart, and upon the competitive platform, is an individual sport, but almost all successful powerlifters have people with whom they regularly train. When people go through a stressful event and come out okay on the other end, they develop a strong bond. Intense weight training sessions can serve as that stressful event, and one can develop a deep sense of camaraderie with a regular workout partner. Powerlifters often form teams or clubs to put like-minded people together. Working out in an atmosphere where other strong, intense, and enthusiastic people are training can be very invigorating; you can feel the energy in the air and people will feed off of each other. I believe that humans have a desire to be social, and having a workout partner can help fulfill that need. There are stories and movies about the bonds warriors develop as they go to war with their friends by their sides. Powerlifting can serve as a substitute to that war — a healthier and safer *war with the iron* that allows similar bonds to be built.

The qualities needed to succeed in powerlifting are admirable and transfer well to other areas of life. Powerlifting involves a tremendous amount of discipline, to go to the gym X number of times a week, every week, for months and years on end. Powerlifting

The results you get will mirror the effort you put into it.

involves a lot of hard work. Lifting heavy weights, particularly on big exercises, isn't easy, and you know the results you get will mirror the effort you put into it. This develops a good work ethic and work capacity. Powerlifting teaches patience, which is sometimes lost in this world of quick fixes. Gains come slowly and over time, and there is no fast way to achieve success in lifting. Powerlifting workouts can be brutal, and regularly experiencing difficult workouts will raise a person's constitution, pain threshold, and mental toughness — all desirable qualities.

Powerlifting, as with most forms of exercise, helps teach kinesthetic awareness, which is the ability to control your body in space. As our society becomes more and more technologically advanced, we have seen a decrease in motor control and coordination, simply because people move so much less. Powerlifting helps teach that control and understanding of how your body moves, which will transfer over to other skills and sports. Powerlifting alone will not normally make a person a great overall athlete (nor will any single skill), but powerlifting combined with practice in another sport will almost always yield a higher performance level than just practicing the sport by itself. The inclusion of serious weight training in any high-level sport preparation is one reason why so many athletic records have been broken in the last several decades.

Powerlifting is a very objective sport. If you perform the lift under a certain set of rules, which are pretty easy to follow, you will get credit for the lift. If you fail to do so, you will not get credit for the lift. Weights do not play any favorites, and it is not easy to deceive yourself about your ability. Three hundred pounds is three hundred pounds, no more and no less. Sometimes the objectivity can be painful, but ultimately it leads to an honest evaluation of one's abilities. This increased self-awareness can also transfer into other areas of life, and powerlifters can become more globally capable of

assessing what they are and are not able to do. This prevents someone from grossly overestimating their abilities, and it also helps individuals who tend to underestimate themselves.

Because powerlifting is objective, and therefore easily measurable, it lends itself to proper goal setting. Setting goals can be a powerful motivator for behavior, and achieving those goals can be extremely rewarding. Regularly achieving difficult and challenging goals will lead to increased self-confidence; the athlete begins to feel that they are capable of meeting life's demands. The individual nature of the sport adds to that increased confidence because the lifter, alone, was responsible for their success or failure. That can be scary, but ultimately it is very empowering as individuals realize that they can count on the most important person in their world, which is themselves.

Weights do not play any favorites.

Powerlifting is simple to understand but challenging to excel at. It is an activity that takes a day to learn and a lifetime to master. As a person increases their lifting ability, if they wish to make further progress they will have to increase their knowledge of fitness. Powerlifting often leads one to learn about human anatomy (which in my opinion is another fascinating subject), nutrition (the basics of which will be covered in this book, and that information is useful throughout your lifetime), and the general theories and practices of fitness as a field. While the stereotype of the weight lifter can be that of a dumb meathead, in reality, the athletes who excel in their respective sports usually have a higher than average intelligence.

All of the above benefits of powerlifting are significant, and we have not even touched upon the health benefits of regularly lifting weights. Working out has a beneficial impact on almost every system in the body. A few of the health benefits include: stronger muscles with a decreased chance for injury,

stronger bones which have a decreased risk of osteoporosis, stronger heart and lungs which can help prevent coronary artery disease, decreased total cholesterol and increased good cholesterol (HDL), decreased incidence of insomnia, improved regularity of bowel movements, improved sexual functioning, increased blood flow to the brain, improved feeling of well-being, decreased risk of diabetes, decreased risk of stroke, decreased bodyfat, and improved endurance.

A good powerlifter is a fit person. They will have a high work capacity, which is useful in the gym and is even more useful in life when you can work long hours and still have energy left to spare. Everyone has certain activities of daily living (ADLs) that they must regularly perform, and being strong makes those ADLs seem very easy. If you are strong, carrying your suitcases through the airport or putting them in the overhead bin is not a challenge. Climbing five flights of stairs is not hard. Picking up your dog or your kid or your laundry basket is easy. You don’t need help carrying in your groceries or loading your car. Moving furniture in your house or performing a push-up or a pull-up is no longer challenging. When you are fit, you are never worried that your physical capabilities will limit what you can do, and that is a good feeling. Overall, our quality of life is improved by being fit, particularly as we get older. I will venture to say that remaining independent and healthy for as long as possible is a goal that many people have as they age, and regular exercise can greatly improve your chances of reaching that goal.

It should be clear now that powerlifting is not just a sport but a lifestyle. It promotes healthy eating at regular intervals, clean living with abstinence from drug use and either little or no consumption of alcohol. It rewards dedication and hard work. It provides the physical activity that we, as humans, need to function properly and that we often don’t get in today’s society. The benefits of powerlifting transcend the sport itself. They are

greater than just the ability to lift weights. Indeed they often have the power to change a person's life, and that is no small thing.

> ***Powerlifting takes a day to learn and a lifetime to master.***

What Is Powerlifting?

If you wanted to identify the strongest person in the world, how would you go about doing that? First, you would need to define strength. Muscular strength has several definitions, but for our purposes, it can be defined as the ability of the muscle to exert maximal force against a resistance or the ability of the muscle to contract one time. Then, you would need to find a way to express or test that strength. What one single exercise will accurately assess how strong a person is? If you are thinking about that question and trying to come up with an answer, I applaud you. However, people have been trying to answer that question for a very long time, and as of today there is no single best exercise to measure a person's total body strength. Choosing only one exercise means some muscle, body part, or ability is getting overlooked. There are lots of great exercises that stress the majority of the body, but nothing really hits the entire body equally with just one movement. What do we do?

> ***The solution is powerlifting.***

In order to test total body strength, we need to rely on more than one exercise — it is just that simple. If you really want to measure total body strength as accurately as possible, you would want to perform at least one exercise for each major muscle group in the body. However, this becomes impractical if you think about testing a number of athletes in a given time period on eight or ten or twelve different exercises, and even that would not test the ability of the body to function as a whole. What is the solution? The solution is powerlifting.

Powerlifting is a sport made up of three different events; the squat, the bench press, and the deadlift. The exercises are always performed in that order. The athlete gets three attempts at each lift, with the highest successful

lift counting in each exercise. Then the three best lifts are added together to form the athlete's Total. The Total is your score in powerlifting. If your best squat was 400 lbs, your best bench was 300 lbs, and your best deadlift was 500 lbs you would have a total of 1200 lbs. If someone else had a squat of 300, a bench press of 500 lbs, and a deadlift of 300 lbs they would have a total of 1100 lbs. Twelve hundred pounds is higher, so your total would win. Even though the other person had a very impressive bench press, it is the overall score that matters the most.

In order to make the sport objective, every lift must be officially judged. Each lift always has 3 judges; if 2 or 3 of the judges indicate the lift was successful (by lighting up a white light on the scoreboard), the lift counts. If 2 or 3 judges indicate the lift was not successful (by lighting up a red light on the scoreboard), then the lift doesn't count.

Why those three exercises — the squat, bench press, and deadlift? Those three exercises provide a practical but still comprehensive measure of a person's total body strength. As previously mentioned, one exercise is simply not sufficient to test a person's total body strength. Even two exercises will leave some important area of the body relatively untested. But with three exercises, you can get a reasonably accurate assessment of total body strength.

To perform a squat, a person places a loaded barbell on their back. They squat down (hence the name) quite low, and then they stand back up. The squat primarily tests the strength of the legs. To perform a bench press, a person lies down on a bench facing up, takes a bar at arm's length, brings it down to their chest, pauses for a moment, and then presses the bar back up until the arms are straight again. The bench press primarily measures upper body strength. In the final exercise, the deadlift, a person approaches a loaded bar on the ground. The lifter picks the bar up off the ground and stands erect

with the arms hanging straight. The deadlift primarily tests the strength of the upper and lower back.

In many powerlifting competitions, it is possible for an athlete to compete in just one or two of the lifts instead of all three. If an athlete competes in one lift, it is called a single lift competition. In a single lift competition, you may choose to compete in just the squat, just the bench press, or just the deadlift. Bench press single lift competitions are the most popular. Sometimes a bicep curl is included as an option for a single lift competition. Single lift competitions don't measure total body strength but instead measure strength in one specific area.

If two lifts are contested, the two lifts are almost always the bench and the deadlift. This is called a push/pull competition (the bench is the push and the deadlift is the pull) which is sometimes referred to as an Ironman competition. These are quite popular, particularly with people who have an injury that prevents them from squatting with maximal weights.

Another type of powerlifting competition is a hybrid form called Powersports. In a Powersport competition, the lifters compete in three events: the bench press, the deadlift, and the standing bicep curl. In a similar but different variation, there is the CrossFit Total, which is composed of the total from the squat, standing press, and the deadlift.

Powerlifting is available to both men and women of any age. Athletes compete in various categories based on gender, bodyweight, and age.

The squat, bench press, and deadlift.

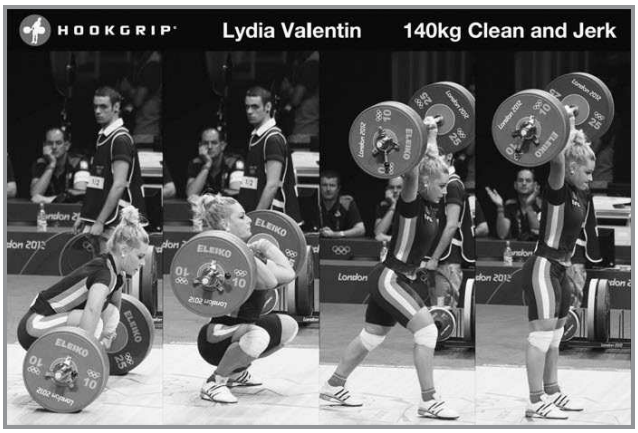
The three powerlifts (squat, bench, and deadlift) give a pretty accurate assessment of a person's total body maximal strength. However, even in those three lifts there are still some areas of the body that are not tested as fully. Those areas include the biceps, the calves, the hip flexors, and the rear

deltoids.

To summarize, powerlifting is a sport that best measures a person's total body maximal muscular strength. In powerlifting, a lifter will max out on three exercises: the squat, the bench press, and the deadlift. The highest lift in all three exercises are added together to form the lifter's Total Score, which is then compared and ranked to other lifters in the same weight/age class. The lifter with the highest Total Score wins.

What Powerlifting Is Not

Now that we know what powerlifting is, let's take a look at what powerlifting is not. Powerlifting (PL) falls under the general classification of strength sports. This category of strength sports contains several other sports that measure strength but have distinct differences from powerlifting.



An Olympic Weightlifter performing the Clean and Jerk

Weight Lifting — Often referred to as Olympic Weight Lifting or simply Olympic Lifting (OL), it is the sport that is most commonly confused with powerlifting. In Olympic Lifting the athletes compete in two exercises: the snatch and the clean and jerk. In both movements the bar ends up over the person's head with the arms straight. Olympic Lifting used to involve a press, a version of the military or overhead press, but it was dropped from competitions several decades ago. In the snatch you have to get the bar from the ground over your head in one motion, and you end up standing straight with your arms straight over your head. The second exercise is the clean and jerk. A clean is where you pick up a bar from the ground and lift it up to your shoulders. A jerk is where you start with the bar on your shoulders and then you drive it over your head, a little bit like an explosive overhead press. A clean and jerk is simply putting the two exercises together, so the bar starts on the ground, it is lifted (cleaned) up to the shoulders, and then the athlete drives it up over their head. The lifter ends standing straight with the arms straight overhead. A clean and jerk is easier to perform than a snatch because the bar can rest on the shoulders, thus generally more weight

is lifted in a clean and jerk than a snatch. Just like powerlifting, the lifter gets three tries at each lift, the highest score for each lift is recorded, and then the two scores in each lift are added together to form the lifter's Total Score. The highest Total Score wins the competition.

An Olympic Weightlifter performing the Snatch
Photo credit: hookgrip.com

There are several important differences between the two sports. In powerlifting more total weight is lifted than in Olympic Lifting. A good powerlifter will be able to squat and deadlift (and perhaps even bench press) more weight than an Olympic Lifter can snatch or clean and jerk. Actually the term "powerlifting" is a bit of a misnomer compared to Olympic Lifting, because the speed of the bar involved in powerlifting with maximal weights tends to be slow, which means not as much power is produced in powerlifting as is produced in Olympic Lifting.

Olympic Lifting is more technical and involves more skill than powerlifting. It takes longer to learn

An Olympic powerlifter does not exist.

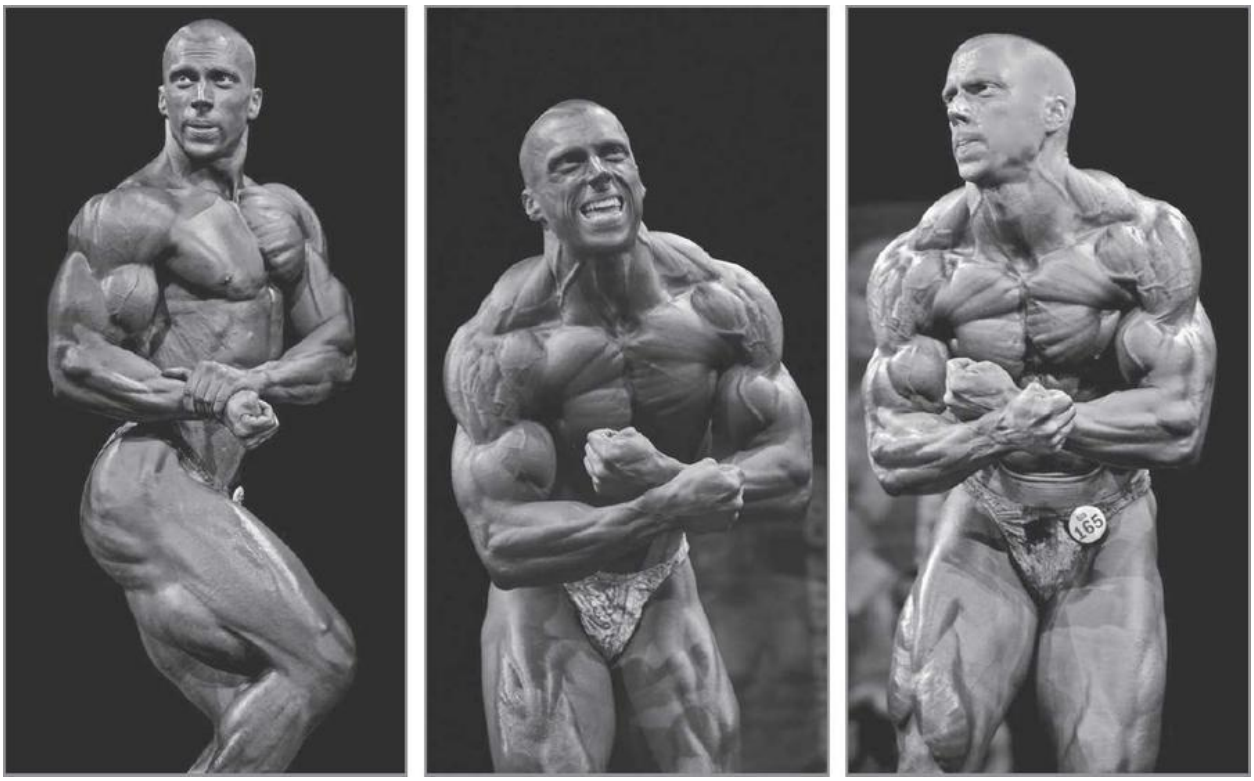
proper form on the exercises, and it requires a little more specialized equipment, namely in the form of rubber weights that many gyms (at least in America) do not have. Olympic lifting also involves other components of fitness such as speed, agility, and flexibility to a greater degree than powerlifting. Because the two lifts are somewhat similar, more major muscle groups are left out in Olympic Lifting than in powerlifting; indeed, being too developed in certain muscle groups can be a detriment in Olympic Lifting. For example, the pecs are not stressed too much in Olympic Lifting, nor are the biceps, and being large or tight in either muscle can negatively affect of the performance of the Olympic Lifts. Finally, as the name implies, Olympic Lifting is the type of weight lifting that is performed in the Olympics. Powerlifting thus far never has been and currently is not in the Olympics. An Olympic powerlifter does not exist.

Of course, you can't help but ask the age-old question, "Who is stronger, a powerlifter or an Olympic Lifter?" The answer to that question depends on how you define strength. Of course, a powerlifter is better at powerlifting and an Olympic Lifter is better at Olympic Lifting. A powerlifter lifts more weight than an Olympic Lifter, so in one sense they are stronger. However, an Olympic Lifter's training makes them relatively proficient at the powerlifts, with the exception of the bench press. Most Olympic Lifters are quite good at squats and deadlifts. A powerlifter's training may have very little to do with the Olympic Lifts. You can be an excellent powerlifter and you may not be good at a snatch or a clean and jerk. Indeed many powerlifters would be too tight to properly perform a heavy Olympic Lift. If there were a competition that included all five lifts — the snatch, clean and jerk, squat, bench press, and deadlift — most people believe a good Olympic Lifter would have the highest total for all five lifts. In that sense, an Olympic Lifter might be stronger than a powerlifter.

A strongman athlete flipping a tire

Strongman — Strongman competitions, as the name implies, involve various tests of strength. They usually involve odd lifts such as a log press, car deadlift or a stone lift. The goal was for this event to be a better test of brute strength, since Olympic lifting involves a tremendous amount of skill and powerlifting involves a fair amount of skill, especially at elite levels. However, many of the strongman events involve other components of fitness beside strength, such as cardiovascular endurance, muscular endurance, and speed. Many of their events involve performing multiple repetitions or performing something for a set time, for instance, one minute, and that slightly decreases the reliance on pure strength.

Powerlifters and Olympic Lifters are drawn to strongman competitions. Regular broadcasting on ESPN and other networks has greatly increased the popularity of these types of competitions. While the athletes who perform strongman competitions do indeed possess great strength, in and of itself, a strongman competition is the not the best test of pure maximal strength. Powerlifting and Olympic Lifting would be better choices.



Doug Miller, a top-level tested bodybuilder
Photo credit: Doug Miller

Bodybuilding — Bodybuilding is a sport where aesthetics plays the key role. In bodybuilding, an athlete is judged on how they look — the muscularity of the body — how much muscle do they have? The proportion of the body — is the body developed in the proper proportions? Do the legs match the arms? The symmetry of the body — does one side mirror the other side? And the overall definition a person has — how lean are they? How well can you see their muscles?

Actual physical performance in terms of muscular strength or any other component of fitness has nothing to do with the outcome of a bodybuilding show. One bodybuilder might be able to bench press 350 lbs, and another might be able to press 450 lbs, but if the one who can lift 350 lbs looks better, is in better proportion and is leaner, then he will win the competition. Of course, bodybuilders do get quite strong in the pursuit of building up larger

muscles, just as powerlifters usually get larger muscles in the pursuit of getting stronger. In general, due primarily to the type of training that is necessary to succeed at each sport, the strongest powerlifters are stronger than the strongest bodybuilders. On balance, the best bodybuilders look better in bodybuilding terms than the best powerlifters. It is certainly possible to be quite good at both sports.

Bodybuilding is open to both men and women of various ages. In addition, two hybrid forms of bodybuilding have recently become very popular. Fitness competitions involve females competing on both their looks and their fitness ability, as demonstrated by a routine consisting of mandatory moves, such as a one-arm push-up. Figure competitions are just like fitness competitions, but there is no specific fitness assessment; it is based purely on looks. It is a bit like a beauty pageant, but the competitors are judged on how much muscle they have and how lean they are, not just how "pretty" they are. In bodybuilding the competitors are to be judged only from the neck down; in fitness and figure competitions, overall looks, including the attractiveness of the competitor's face, are judged. Bodybuilding as a sport is significantly more subjective than either powerlifting or Olympic Lifting.

The thing that all of those sports have in common — powerlifting, bodybuilding, strongman competitions, and weight lifting — is that they are all practiced in a gym. The athletes in each sport must train with weights in order to succeed at their given endeavor, but there are significant differences between them. Just because someone regularly trains with a barbell does not mean that they automatically will be good at every exercise you can do with that barbell. It is similar to running. Jogging a marathon and running a 100 M

Powerlifters usually get larger muscles in the pursuit of getting stronger.

sprint are both forms of running, but each requires very specific training, and being good at one of those forms does not mean that a person will be good, or even adequate, at the other. It is perfectly reasonable that a powerlifter need never perform a snatch to become proficient at powerlifting. Each sport has its own requirements, and an athlete can be good at one sport without excelling at the other.

For the purposes of this introduction, it is important to understand the differences between the various “gym” sports but also recognize that these sports have more in common than not. The skills required to succeed, the energy and drive required to change your own body are very similar, and these connect the athletes of different disciplines. In a sense, all athletes who train hard with weights could be considered *Brothers of Iron,* and in my opinion, it is better to respect the similarities within the sports than disrespect the small differences between them.

All athletes who train hard with weights could be considered **Brothers of Iron.**

Is Powerlifting a Sport?

Throughout this chapter you may have noticed that I have referred to powerlifting as a sport and to powerlifters, weight lifters, etc., as athletes. For some reason there is dissension as to what is and is not a sport. People always want the activity that they personally do to be a sport and many other things to not be classified as a sport.

In order to solve this type of dilemma, we need to know the definitions of the words we are discussing. What is an athlete? Ask ten people, and you will probably get ten different answers. *Webster's* definition of an athlete is this:

> *One who is trained or skilled in exercises, sports, or games requiring physical strength, agility, or stamina.*

By that definition, powerlifters clearly can and should be classified as athletes.

Is powerlifting a sport? Turning to *Webster's*, using sport as a noun, the definition is as follows:

> *A source of diversion: recreation. Physical activity engaged in for pleasure.*

If we go with that definition, powerlifting is clearly physical activity, and it, or at least aspects of it, can be pleasurable for some people, so, according to *Webster's* powerlifting is a sport. However, I will be the first to say that in my opinion that definition seems lacking. Allow me to present another definition of a sport.

> *A sport is any physical activity that involves the coordinated*

activity of the large muscle groups of the body that is contested under specific rules, upon which the performance of the participant can be ranked in comparison to others, and the outcome is unknown at the start of the event. Organized sports should also have divisions in which the participant can progress, from smaller, local (amateur) competitions, to larger, regional, national or world competitions (professional).

It is my opinion that this definition of a sport serves as a better litmus test for identifying what is a sport and what is not. Simply pick an activity and see if it applies. Football, soccer, basketball, hockey, baseball, all the traditional sports clearly fall under this category of sport. Some more subjective sports such as golf, driving a race car, darts and billiards would still be considered sports because those activities all require coordination of the larger muscle groups in the body. However activities such as chess, checkers, and scrabble are not sports. That does not lessen the importance or the validity of those activities; it is simply not appropriate to classify them as sports because they are not physical activities. Instead those activities are games. In comparing sports and games, one activity is not inherently better or more worthwhile than another.

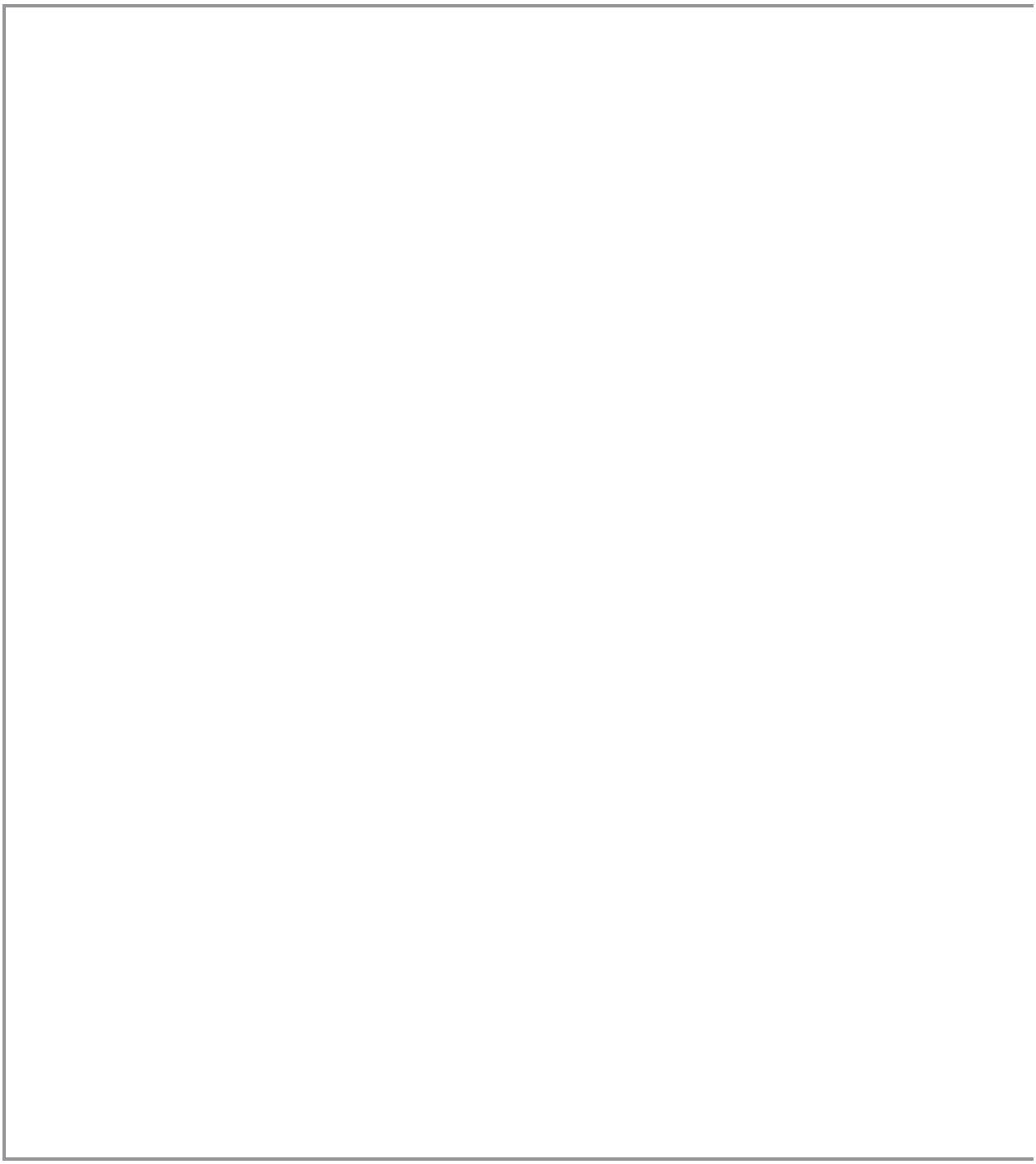
Using the above definition, powerlifting is once again clearly defined as a sport. Olympic Weight Lifting, strongman competitions, bodybuilding, and fitness competitions would also be considered sports. Figure competitions are more of a gray area, but they would not be classified as a sport because there is no required physical activity in those competitions other than walking. A Miss America competition would not be considered a sport.

Powerlifting is clearly a sport, and a competitive powerlifter is clearly an athlete.

What then is the definition of an athlete? We can use the definition provided earlier by *Webster's* or we can make it even simpler:

An athlete is anyone who competes in a sport.

To summarize, powerlifting is clearly a sport, and a competitive powerlifter is clearly an athlete. Any case to be made in opposition to those points would require an operational definition of the word "sport" and the word "athlete," and I think someone would be hard pressed to create a definition for those words that included the traditional sports but did not include powerlifting or weight lifting.



Roger Estep prepares for a big pull

Chapter 2

The History of Powerlifting

Full Disclosure to the Reader: This chapter is meant to provide a basic overview of some of the history of powerlifting. While I do consider myself an expert on the sport of powerlifting, I don't consider myself an expert on the history of powerlifting. Writing this book is not my full-time job, and it is not how I support myself — that is just an excuse, but it is what it is. It is my hope that perhaps someone reading this text with much more knowledge on the history of powerlifting than myself will contact me and we can give this topic the time and energy it deserves in future editions. Until that time this brief summary will have to suffice. In addition I include links at the end of the chapter to more detailed information if you — the reader — wish to learn more.

It might surprise you to learn that dumbbells were actually invented before barbells, although as one might expect both have gone through a transformation as they have evolved into the tools we are familiar with today.

The term "dumbbell" is actually a literal interpretation of a dumb, meaning silent, bell. Approximately 500 years ago, during the Tudor Era, very large church bells were rung by pulling ropes. It was noted that regular

performance of this activity led to increased strength and fitness. Ringing the bells, and other fitness types of activities, was associated with the lower classes. Those in the upper class wanted a way of staying physically fit but didn't want to be associated with using the same methods that the lower class people incorporated. To solve this issue, they set up smaller (but still large) church bells with ropes that they could pull on as a form of exercise. Of course, the user didn't want to announce to the world every time they were exercising, and the inside part of the bell (the clapper) was removed. Now the user was exercising by pulling on the ropes and moving a silent bell, or a dumbbell. Thus the idea of "working out with dumbbells" was born.

About 150 years ago George Barker Windship invented the first classic barbell. It was a bar that was set up, and graduating plates could be loaded onto it to change the load. Over time the bars and the weights became more standardized and over the last 50 years or so the weight, length and dimensions of barbells and Olympic plates have become relatively uniform.

Olympic Weight Lifting first entered into the Olympics in 1896, although it was not the standardized version that we have today and included many one-arm versions of the lifts. It was in and out of the Olympics for 2 decades. In 1928 OL was standardized into the 3 main lifts (clean and press, clean and jerk, and the snatch). 1972 was the last year that the clean and press were included as part of Olympic Weight Lifting; today the sport just consists of 2 lifts, the snatch and the clean and jerk. In my opinion, dropping the clean and press was a shame. As I see it, if you have to pick just one lift that best measures total body strength, the clean and press is it. To me the idea of someone cleaning and pressing 500 lbs over their head is simply astounding. True, that lift is harder to judge, and some serious issues had arisen because of that, but in my mind tightening up the rules would have made more sense than abandoning a great lift and exercise.

George Barker Windship invented the first barbell.

Olympic Weight Lifting started to fade in popularity and by the 1950s it was not as popular, particularly with Americans. In the ’50s and ’60s other exercises — referred to as “Odd Lifts” — started to become more popular. There were many of them — 42 different lifts had official standards. Initially the three contested lifts were the Squat, Bench Press, and the Curl (it may surprise some of the “curl haters” to realize that the curl was included in the most original version of powerlifting) — these lifts were called the Strength Set. In 1966 the curl was dropped and was replaced by the deadlift, leading to the classic competitive exercises that are still factored into one’s total today.

In the late 1960s powerlifting competitions were becoming increasingly popular, but there wasn’t one official governing body that established rules and put forth exact standards for all the lifts. In the early 1970s the first real World Championships occurred, and the IPF was formed as the primary governing body. There wasn’t any drug testing and with the exception of knee wraps the lifts were raw. Initially only men competed, in the late ’70s as powerlifting became more established female competitors were allowed to enter the competitions as well.

In the early 1980s it became clear that steroids were a big part of the sport at the top levels of competition. Not all lifters thought this was fair and drug free (or more accurately drug tested) federations appeared (using steroids didn’t become illegal in America until the 1980s). Also in the early ’80s the first real piece of powerlifting gear was introduced — the bench shirt invented by John Inzer. Initially technology advanced very slowly with powerlifting gear; a bench shirt might add 10 lbs to a lifter’s bench — more than likely it seemed to help them overcome the pause. As the technology advanced, that also spurred more dissent, and many different powerlifting

federations were created. Now some bench shirts add an additional 30–50% more weight than what the lifter is capable of lifting raw! The federations were generally divided along the following lines:

- Drug Tested or Untested
- Raw; Single Ply (one layer of material); or Multi-ply (multiple layers of material)
- Strict Judging or simply going for huge numbers

Stereotypically, lifters who prefer to compete in drug-tested organizations also tend to prefer raw competitions with reasonably strict judging (below parallel squats, long press commands on the bench, etc.) while those lifters who compete untested tend prefer to multi-ply material and less strict judging.

In the late ’90s and early 2000s the curl was added back in as a potential fourth contested lift in some federations although it is not factored into one’s total in any federation.

It isn’t feasible and it would be unfair to attempt to list all of the really impressive lifts from the start of powerlifting until now (unfair because I am sure I would leave out so many great lifts). I will, however, highlight just a few of these lifts to make it clear just how strong these early pioneers of powerlifting were and, in some ways, to show how little progress we have actually made in the last several decades.

In 1972 at one of the first real championship events hosted by the AAU, the following lifts were performed:

- Jim Williams benched 675 lbs raw and almost had 700

- Larry Pacifico benched 515 lbs at 198 lbs bodyweight (and he had a phenomenal total to boot)
- John Kuc totaled 2350 lbs, raw (likely with weak knee wraps)
- Jon Cole totaled 2370 lbs, raw (likely with weak knee wraps) (he did this at a previous meet earlier in the year)

  (Note both of those totals are higher than the All-Time official raw totals as of this day!)

Vince Anello deadlifting in the early '70s
Photo credit: Vince Anello

If any reader is looking for additional information on the history of powerlifting, the following links might be useful:

Dr. Ken Lestner's "History of Powerlifting" (this is a series of articles — more than 60 of them! — that go into the history of powerlifting, weight lifting, and strength training).
http://titansupport.com/blog/category/history-of-powerlifting-weightlifting-and-strength-training/

Wikipedia Page on Powerlifting
http://en.wikipedia.org/wiki/Powerlifting

Jennifer Thompson gets ready for a triple bodyweight squat
Photo credit: Jennifer Thompson

Chapter 3

The Squat

Some people feel that the squat is the king of all exercises. It is a grueling, demanding exercise that is difficult to perform correctly, but it provides great results. Not very many people squat regularly, and of those who do, not very many squat to competition standards. Knowing how to squat and doing it well puts you in a rather exclusive club.

The squat is the king of all exercises.

Types of Squats

There are many different types of squats that lifters can perform. The main form, and the one practiced in a powerlifting competition, is called a back squat, which simply means the bar is on the person's back. If you just see *squat* written down or referred to as an exercise, it is implied that it is a back squat.

There are two main types of back squats: high-bar and low-bar squats. This indicates the bar's position on the back. In a high-bar squat, the bar is right at the base of the neck. This position is usually more comfortable for beginners, and it helps you maintain an upright position. It is also the main kind of squat that Olympic Lifters use in their training; it is sometimes referred to as an Olympic squat. If a high-bar squat is taken as low as the lifter can go (again common in the Olympic lifts), it is also called an Ass to the Grass squat or ATG squat.

The main drawback to a high-bar squat is its level of difficulty; it is harder than a low-bar squat. The bar is farther away from your center of gravity. With light weights it is easy to stay upright and have good form, but heavy weights tend to make you fall forward and/or round your upper back. In a powerlifting competition we are trying to lift as much weight as possible; thus the vast majority of people use a low-bar squat.

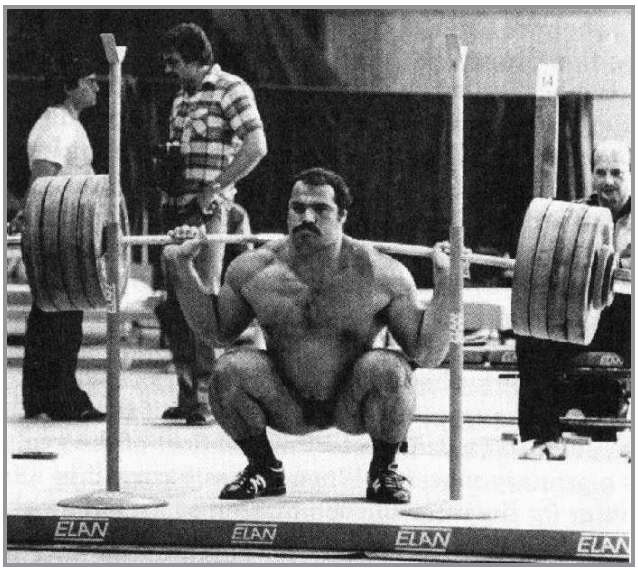
A low-bar squat is positioned lower on the back, about 3–4 inches below a high-bar squat. It will go across your middle traps and just above your rear delts. The high-bar squat is focused more on a specific point on your back; with a low-bar squat the bar is spread out over a larger area. A low-bar squat tends to cause the lifter to lean forward more, even with light weight, but it usually allows lifters to lift heavier weights, often 50–100+ pounds more.

If you switch from a high-bar to a low-bar position (or vice versa), expect the bar to feel uncomfortable for a month or so until you get used to it.

Initially your weight will go down and it will feel painful on your back, but quickly you will develop an internal callus on your back and soon the bar will feel very comfortable there. Ultimately you will be able to put much more weight than you can squat there and it will still feel comfortable.

Do not put a pad on the bar or wrap a towel around the bar if you are interested in competing; you simply have to get used to the feel of the bar on your back. Pads can allow the bar to roll on your back, and in a competition padding is not allowed. I would suggest you wear a regular T-shirt when you squat and not a tank top because you will be wearing a T-shirt in a competition and because the material can provide just a bit of cushioning. Some lifters like to wear thicker shirts like a polo shirt with a collar but those are not legal in most powerlifting federations. Note the bar should always be above the spine of the scapula; there is such a thing as it being too low on your back. When the bar is too low it is hard on the shoulders and illegal in competitions. If the bar is too low it will feel like it is constantly trying to roll down your back, and if it does serious injury can occur.

A powerlifter squats Ass to the Grass style
Photo credit: Vadim Snitkovsky



An Olympic Lifter squats High Bar, Ass to the Grass style
Photo credit: Vadim Snitkovsky

Equipment Set-up

When you squat in a competition you will squat out of squat racks or squat stands. You will lift the weight off the racks, walk back into position (this is called the walkout) and then perform the squat. Once you successfully complete the squat, the judge will tell you to "Rack" and you will take a step forward, at which point the spotter can help you back to the rack.

Because the competition uses squat stands, some lifters like to use them in practice to get used to them. I can certainly understand that point of view, and in general, any time you can make your practice more like your competition, that is a good thing. However, one serious drawback is that the regular squat stands do not have any safety system built in. In a competition you normally have 3–5 spotters; if something bad happens during the lift you should be fine. In a gym it can be hard to find one good spotter, let alone the 3 or more that you really need to be safe without any specific safety equipment set up. It is for that reason that I recommend that you squat in a power rack. You can adjust the safety pins so that they are just a little bit lower than the bar at the bottom of the squat, and that way if you do fail for some reason the safety pins are there.

Your walkout should be the same as if you were squatting in a competition when you use the rack. Another benefit of squatting in a power rack is that you are blocked off from most gym members. When you squat in the open, someone could move into you or something could roll into your area, both of which could be a serious hazard. Remember though, the results are the most important thing. If you love practicing squatting using the squat stands, go ahead and do so — just be mindful of the safety situation.

In a meet you will be asked what your rack height is; this way the squat stands can be adjusted to the height you

I recommend you squat in a power rack.

desire. This is how high the weight will be when you go to lift it off. You want the height that you practice with in the gym to be very close to or the same as the height that you will use in a meet. The standard guideline is that the bar should be below your shoulders, or about mid-chest level, when you are standing next to it. If you are squatting high-bar style it can be a little bit higher. When you lift the bar out you should have to do somewhere between a quarter and an eighth of a squat to get the bar out of the racks. If you have to go up on your tip toes to unrack the bar, or if you have to kind of throw it up to rerack the bar, the rack is too high.

Remember that unracking 135 lbs and unracking 495 lbs may not feel like the same thing. The heavier weight will sink into you more, it will bend you forward a little more, and the bar may bend — make sure the rack is at a height that is comfortable to you when you lift a heavy weight. It is better to be a little bit too low than too high. Generally if two people are working out together the rack height is set up to the height of the shorter person. Once you find a height that works for you in the gym, the simplest solution is to just measure the distance from the top of the bar to the ground and then set up the squat racks in a competition to that same height.

In a competition, you have the option of the rack being in or out. This means the squat stands can be closer together or out wide. Out wide is the usual position. If you choose the out position that means that your hands will be inside the stands. This position generally closely mimics the position of the power rack at a gym. If you choose the in position, the stands are slid inward, and it lets you put your hands out wide, generally where the stand normally is. Unless you feel like you really need to have a wide grip, go with the out position.

An example of the squat racks in the in position
Photo credit: Vadim Snitkovsky

Proper Technique

The squat is the highest skilled lift of any of the exercises performed in a powerlifting competition. It takes a while to learn and much longer to master. When first learning how to perform the squat most people feel and look awkward, but if you stick with the exercise it can yield impressive results.

There are many different goals and reasons to squat. Since this is a book about Powerlifting, my assumption is that you are trying to squat as much weight as possible for a 1 rep max (1RM). These guidelines are given to help you do just that.

Hand Position

Your grip on the bar is quite important, even though you are not lifting the bar with your arms. Your grip can affect how tight your upper body is, and how easy or hard it is for you to roll forward and stay upright. You want to walk up to the bar and grab the bar with your hands. Use the rings on the bar to line yourself up. A simple starting point is pinky on the rings, or wherever you grip the bar to perform the bench press. However, this is just a starting point; each person will want to adjust their grip to maximize their own power. If you are broad and/or you have inflexible shoulders, you will need to take a wider grip.

In a competition, you must have your hands on the bar, so don't get into the habit of holding the sleeves on the bar (thick part where the weight goes) or holding the weights themselves. Generally, if you need a wider grip, put your pointer finger on the ring and go from there. If you are still super tight, work your way out, or if that feels too loose, move your hands in a finger width at a time. Often people, particularly large males, are tight in this position the first time they try it (especially with a low-bar style); however, after practice, flexibility can be developed here as in any other position and with time it should become more comfortable. If you find yourself loosening up over time, feel free to bring your grip in slightly to adjust for that.

Females and narrower or more flexible men can often go with a closer grip and generally that is a good idea. A closer grip is usually pinkies on the ring or narrower. Having a narrow grip on the bar forces your traps and other back and upper body muscles to bunch up and get tight. This helps you hold the bar in position, and it can help keep the upper body in the right position. A negative aspect of an especially close grip is that it can bother your shoulders. In addition, the bar might roll slightly on your back and with a close grip this can aggravate your elbows.

I would suggest in training that you find a grip width that works for your normal sets. This grip should be pretty tight but not so much so that it is all you think about during the set. Over the course of time if you get more flexible you can always bring your grip in a bit. Then on very heavy sets or on meet days you can bring your grip in a finger or two to really make it tight. Again, watch for sore elbows when doing heavy squats, even a few days afterward.

The low-bar position can put a fair amount of stress on the shoulders and wrists. If it is really hurting your shoulders, you can widen your grip and that should help correct the issue. I would also recommend stretching and/or foam rolling that area (chest/delts/lats) and even using a bar or a broomstick and putting it back there to get used to that position. If your wrists are bending backwards and taking a lot of the weight then put your thumb on the same side of your fingers (open grip); this will help keep the thumb in line with the forearm. Unless you lift in raw divisions, which usually don't allow wrist wraps, you will be permitted to wear wrist wraps in a competition, and they can take a lot of pressure off of the wrists.

The squat is the highest skilled lift.

Foot Position

How wide you like your feet when squatting is going to be somewhat personal. In general to squat heavy you want a reasonably wide position, defined as wider than shoulder width apart. This helps decrease the overall range of motion you have to move the bar, and it helps you get down into the proper position. A relatively close stance, defined as shoulder width or narrower, will make use of the quads more, but it will make it hard to go low enough unless you are very flexible. Most people are not that flexible and a narrow stance can lead you to round forward or shoot the knees forward. A narrow stance increases the ROM necessary to complete the lift.

A very wide foot position, defined as heels being in line with the elbows or wider at the start of the squat (with your hands on the bar), has advantages and disadvantages as well. The main advantage is the large decrease in ROM, meaning less overall work is performed. It also places more emphasis on the glutes and adductors while placing less emphasis on the quads. It is easier to stay more upright with a wider squat; however, it places more stress on the hip joint. Some powerlifting organizations are stricter than others when it comes to judging the squat. A full below parallel squat is difficult to do with a very wide stance, especially combined with heavy weight. However, if you only need to go low enough so that the middle or bottom of the leg is parallel to the ground, the hips can often handle that depth with a wide foot position. If you are squatting to improve looks or general athletic performance then a shoulder width or slightly wider stance is best; the super-wide stance will probably not have much carry-over to your specific sport.

The angle of the feet themselves is worth mentioning. There are two basic positions — feet pointing either straight ahead or pointing out; no one

A narrow stance increases the ROM necessary to complete

recommends a pigeon-toed squat. In general, you will probably be most comfortable squatting with your feet in the same position they are in when you walk. For most people this will be slightly turned out. Some people like having their toes pointing straight ahead as they squat, which makes you tighter and gives you more of a rebound at the bottom. It also makes it harder to go all the way down, especially if you have to go deep, and your knees are inclined to come in toward each other, which is not desirable. The knees generally go where the toes are pointing or in slightly; it is rare (although possible) that the knees will flare out excessively when a person squats heavy. If you point your toes out, the question then becomes how much? The answer is “a little bit,” 10–20 degrees and no more than 45 degrees. Pointing the toes out helps most people go lower in the squat, and it can help most people keep their knees out as they squat. The majority of people seem to prefer pointing their toes out slightly when they squat for comfort and performance.

the lift.

Pointing the toes out helps most people go lower in the squat.

Head Position

Head position is a point of contention among experts when it comes to the squat. The standard recommendation is to look straight ahead or up slightly. Not everyone agrees with this. Most people look somewhere in a range of 90 degrees, with looking straight ahead representing the midpoint of that. Some experts suggest that you look down slightly, at a spot that is perhaps ten to fifteen feet in front of you. Others suggest looking up and still others suggest looking straight ahead. All recommendations have merit and are probably dependent on an individual's biomechanics.

Looking down is good because it can help you drive your neck into the bar and it can help you push up and backward, which is usual for squatting. However, looking down can also round your upper back which can cause you to round over and fall forward. Some people like to look down slightly on the way down and then straight ahead or up slightly on the way up.

An example of using mats to cover the mirror

Looking up is helpful because it drives the head up which tends to drive the chest up, however sometimes people actually lean forward more when they look up and thus increase the possibility of very significant forward lean. With very heavy weight this can make the rep difficult to complete.

Looking straight ahead is convenient, is easy to do, and normally feels pretty natural. If you are squatting in front of a mirror you can look straight ahead to watch your form. However, do not rely on the mirror for feedback over too long of a

period of time, or you may become dependent on the mirror to know how deep you are. There are no mirrors in a meet, and this can throw you off. I used to actually cover up the mirror at my school gym with a large black sheet just so I didn’t get reliant on it for depth (I worked there, so that helped in being able to do that). Later on I would just use stretching mats to block the mirrors in commercial gyms.

General Form for Performing the Squat

Squatting with a very heavy weight is an intense activity. The set-up and positioning of your body is important to maximize the benefits. Here are some tips to follow when squatting for maximal performance.

Take a grip on the bar and then walk your feet under the bar, preferably into the position you will use to lift the bar up and out of the racks. Then duck under the bar and put the bar on your back. Make sure your chest is up high and your head is back. To get set, once I grab the bar I get up under it and then try to stand up without moving the bar, as if I were trying to scratch the middle of my back with the bar. Because my hands are still on the bar, this causes me to arch my back and tighten everything up. Then I maintain that position while I lower myself back under the bar. You don't want to start a heavy squat all hunched over. Once in position drive the base of your neck back into the bar as if you were trying to push it off your back (think of trying to give yourself a double chin). This will help keep your center of gravity back and it will help keep the bar stable. Maintain that position throughout the lift.

Squatting with a very heavy weight is an intense activity.

Once your body is set, bring your feet underneath your body, if you have not done so already. They should be right under you so you will have a strong walkout. A walkout is the act of lifting the bar up and walking it out of the racks. If your feet are too far in front of you or behind you it will be tough to lift the bar up. Some people stagger their feet for the lift off, but once you begin to lift heavy weights (400+ lbs) this is not advisable. To unrack the bar you are performing a partial squat. We all know how easy partial squats are so you are doing a quick partial with the weight you are really going to lift. Position your body for this — your

feet are under you, even and symmetrical, and about shoulder width apart or a bit narrower.

Once your feet are under you, take a big breath in, get in proper position, tighten everything up, and then pop the bar off the racks. This should be forceful; you want your first thought to be, “I am going to destroy this” as opposed to, “Oh my god, this is heavy.” Almost any weight feels heavy when it is sitting on your relaxed back, but if everything is tight and you launch into it, the weight will not feel so bad. You want the walkout to build your confidence, not ruin it.

Once the bar is up and off the racks, you need to walk into position. Take controlled steps, and try to develop a routine where you take the same number of steps for each walkout. Practice this with every set, even with a light weight, so it becomes natural for you. Try to minimize how many steps you take. Some people can get set in two steps, which is the minimum it will take. I always take two steps and a slight shuffle with each foot to get in the right position. You should not take more than 4 steps. The more steps you take, the more energy you burn. This is not a big deal when you are lifting light weights, but if you are going heavy and hope to get really good at squats, you want to refine your walkout.

Try to minimize how many steps you take.

It should be noted that some powerlifting federations use a monolift. A monolift is a piece of equipment that holds the squat bar. You get up under it, placing yourself in the position you want to be when you actually begin the squat. Then when you are ready, you lift the weight up slightly and the arms that are holding the weights are moved out of the way. Once that happens, you can begin the squat without performing a walkout. While it doesn’t make a gigantic difference, avoiding the walkout can save precious energy,

especially with huge weights, and using the monolift is easier than performing a walkout once you get used to it. As a side note if you compete in a competition that has a monolift but you are not used to it, you can still walk the bar out of the monolift just like normal. It is probably better to do that than to try to do something new for the first time with maximal weight.

After the walkout is completed and you are standing straight, exhale but keep your body tight and then inhale again as much as you can. As you inhale attempt to lift your chest even more and look straight ahead or up slightly, but no more than 45 degrees. Make sure not to lose your good position as you walk the bar out. Once you inhale and are set, hold your breath. You are now ready to begin the descent; break with your hips by pushing them back and then bend your knees. Keep your chest up and keep looking straight ahead, go lower than you think you need to go, and then when you are ready, blast into the bar as if you were trying to launch it up in the air with your legs. Keep a tight grip on the bar, squeeze your core so that your trunk is like a strong column for support, and keep your chest up and neck back as you drive upwards. Exhale as you complete the lift or once you know that you will be able to do it. Even as Powerlifters, we do not hold our breath on the normal sets, but it is okay to hold the breath and produce the Valsalva Maneuver for a 1 rep max, especially if you are competing. At the completion of the lift stand up straight with your legs straight and then take a strong step forward and place the bar on the racks when the judge signals you to do so. Watch your fingers as you return the bar if you have a wide grip. In a competition the referee will tell you when to move forward.

Go lower than you think you need to go.

A significant majority

of good squatters will break first at the hip.

Squat Form Specifics

What is actually going on in a squat? As you start the descent, when you are standing tall with your chest up, you want to push your hips back as you arch your back and break at the hips. Breaking at the hips first, as opposed to the knees, helps maintain and cue the arch in the lower back. Not everyone breaks first at the hips; some prefer the knees, so again you must find what works for you. I would say that a significant majority of good squatters will break first at the hip. The taller you are, the more likely you will need to break at the hips first. Once you break at the hips by pushing them backwards, you then push your knees forward slightly. The hip and knee break are close to being simultaneous — as you push your hips back you tend to bend at the knee and that is fine. The amount the knees must go forward will be somewhat determined by limb length. If you have long legs and/or a short torso they will need to move forward somewhat. If you are built perfectly for squatting, your shin will stay basically upright and vertical throughout the lift. The knees do not need to move forward much as you start the descent; if necessary they can move more forward as you keep going down.

The hips push back, the knees bend, and now you begin to drop into the squat. You will normally want to hold your breath during the descent, unless you are completing a lot of reps, in which case you might inhale during the descent. If it is a 1RM then definitely hold your breath on the way down to help keep your torso rigid. As you drop down you want to think about keeping your torso relatively straight, but not perpendicular to the floor. No one can squat with their torso perfectly vertical and that is not the goal. A goal is to keep the alignment of the torso the same during the lift, so that you are simply inclining forward at the hips. If you think about the torso position that you have during heavy tricep pushdowns (leaning forward about 15

degrees or so, chest up, looking basically straight ahead), that is a good start. If there is writing across the chest of your shirt, you should be able to read that writing in the mirror during the entire lift. If you round forward too much, you will not be able to read the writing. Another way to think about it is that you should be able to see your nipples in the mirror during the squat; that helps keep the chest up and the shoulders retracted and down.

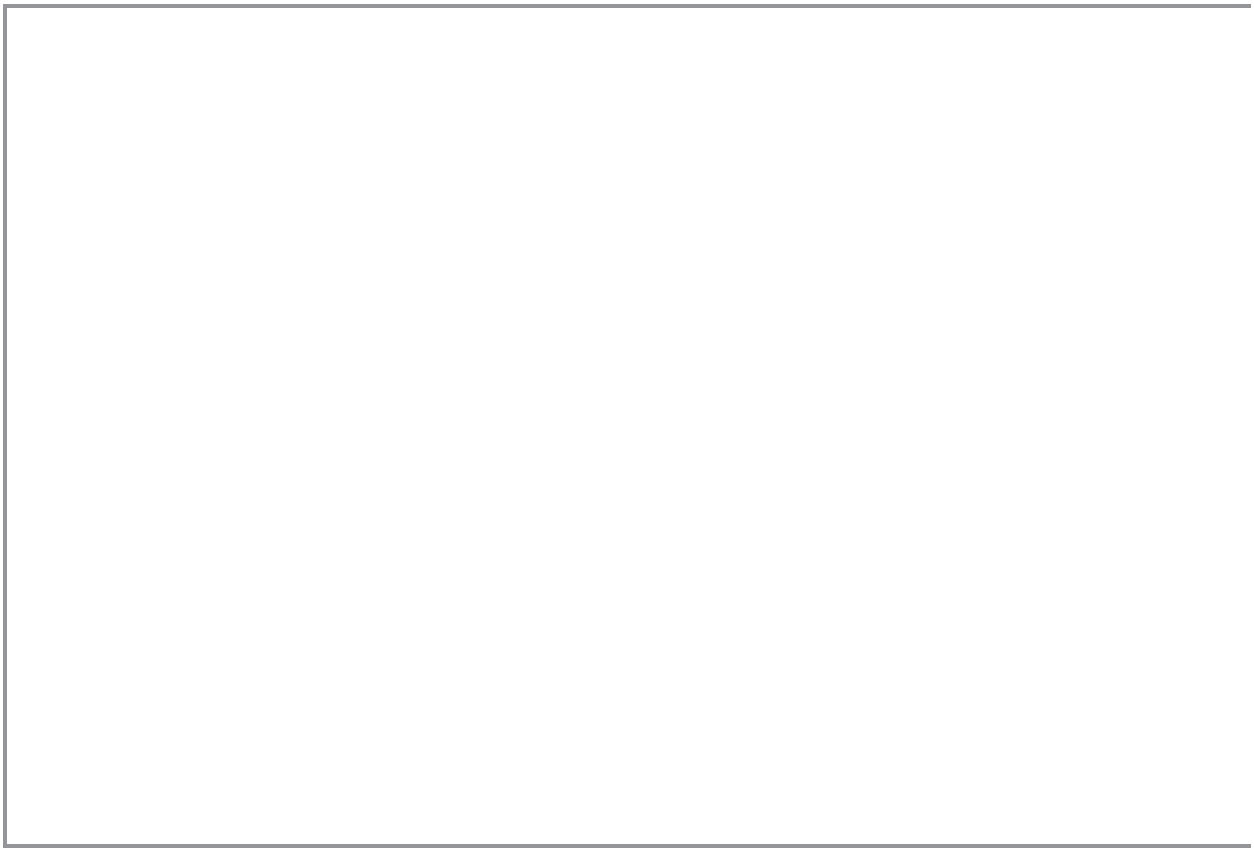
What to do with your arms during the squat can be another point of contention. There are two schools of thought regarding arm position. Some people suggest that you pull your elbows down and under the bar, much like you were trying to do a lat pulldown with the bar on your back. This can help activate your lats which can assist in keeping your trunk stable. This can also help keep your chest up and forward, which can help keep you from rolling forward.

Others suggest that you lift your elbows up toward the ceiling with the bar on your back. This can create a better shelf to hold the bar, and pushing the elbows up and back helps move your center of gravity slightly backward. However, this form can promote you to hunch forward if you are not flexible enough in the shoulders to do it, or if you are overly flexible and/or weak in the mid-back area. I personally don't agree with this method and I think the elbows should be pulled down. However, the bottom line is results. I would suggest you try both methods and see which one works better for you. It is doubtful that one is better to use in all situations for all lifters.

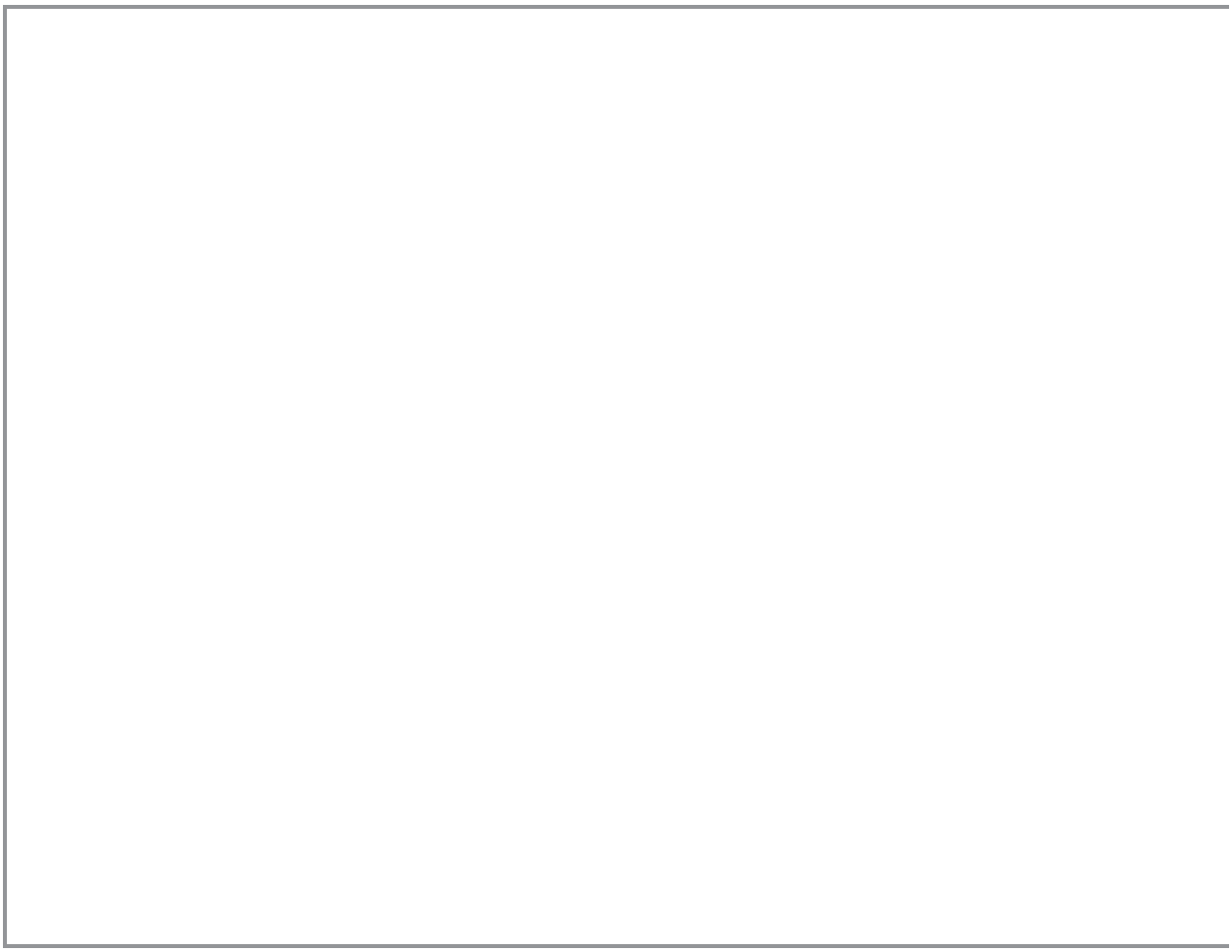
The bottom line is results.

Perhaps the hardest part of a squat is to drop down low enough for the squat to count in a competition. Usually, for a squat to count, you must have the crease of your hip (top of the thigh near your side when sitting) go below the top of your knee. This is referred to as a below parallel squat, and it is

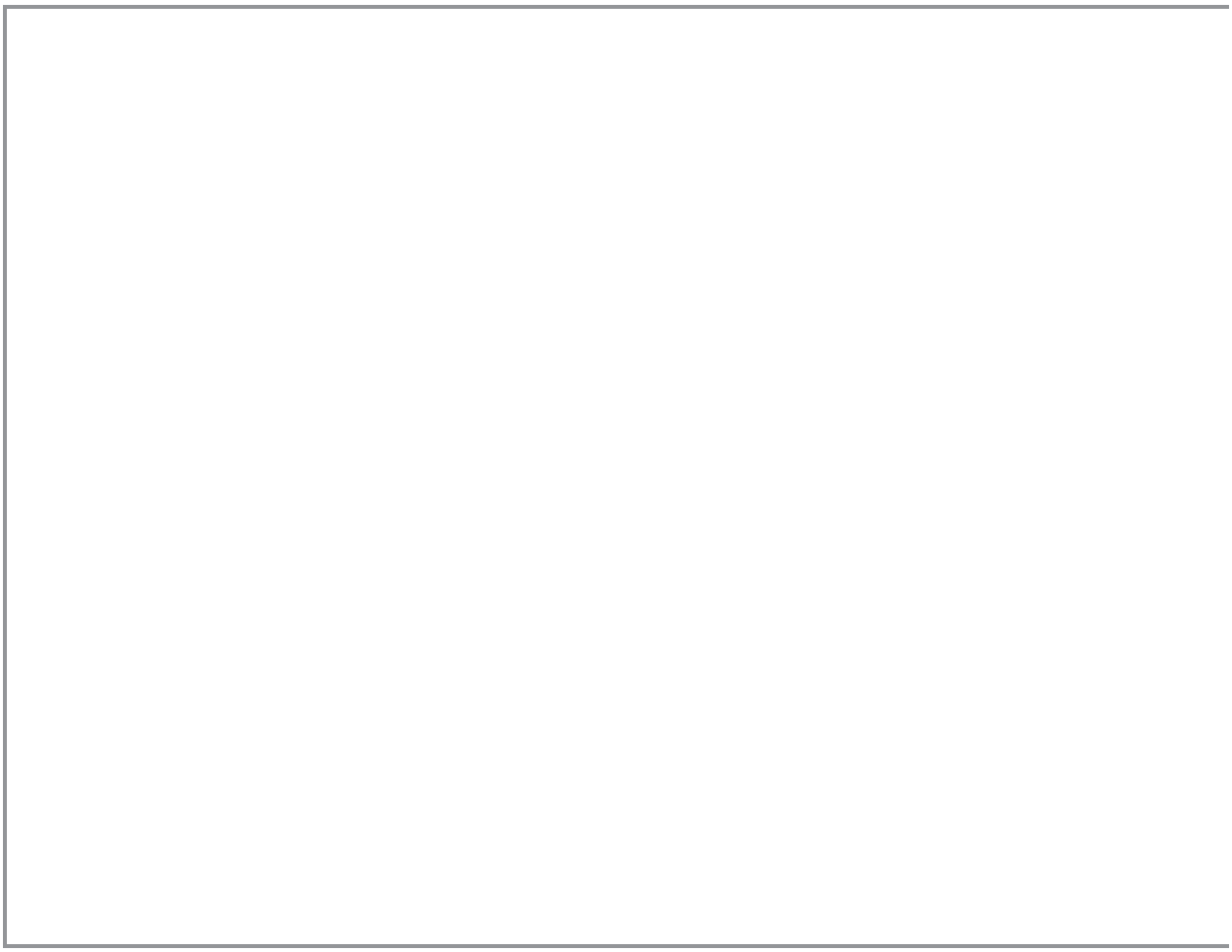
required to get white lights in a powerlifting competition. The vast majority of regular squats performed in gyms are not low enough to count as full squats; they are usually half or three-quarter squats. This is often the toughest thing for a new powerlifter to learn and get used to. The bottom few inches of a squat have a tremendous effect on strength, making the lift much more difficult. It can be hard for a person's ego to think they are squatting a certain weight, only to find out they can really only lift half or three quarters of that weight. But that is the way it is. Everyone else must squat that low, and so must you.



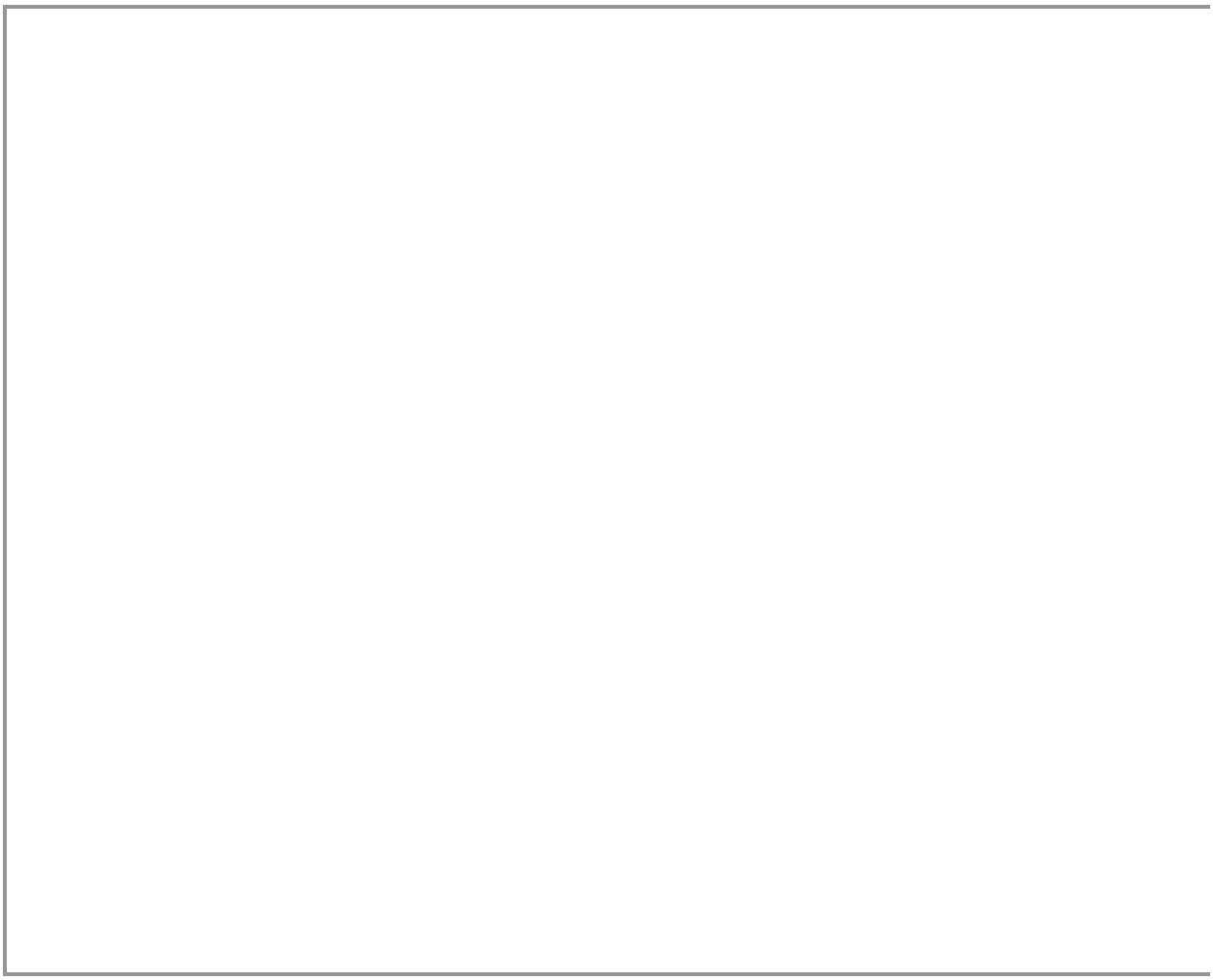
A powerlifter in the hole
Photo credit: Vadim Snitkovsky



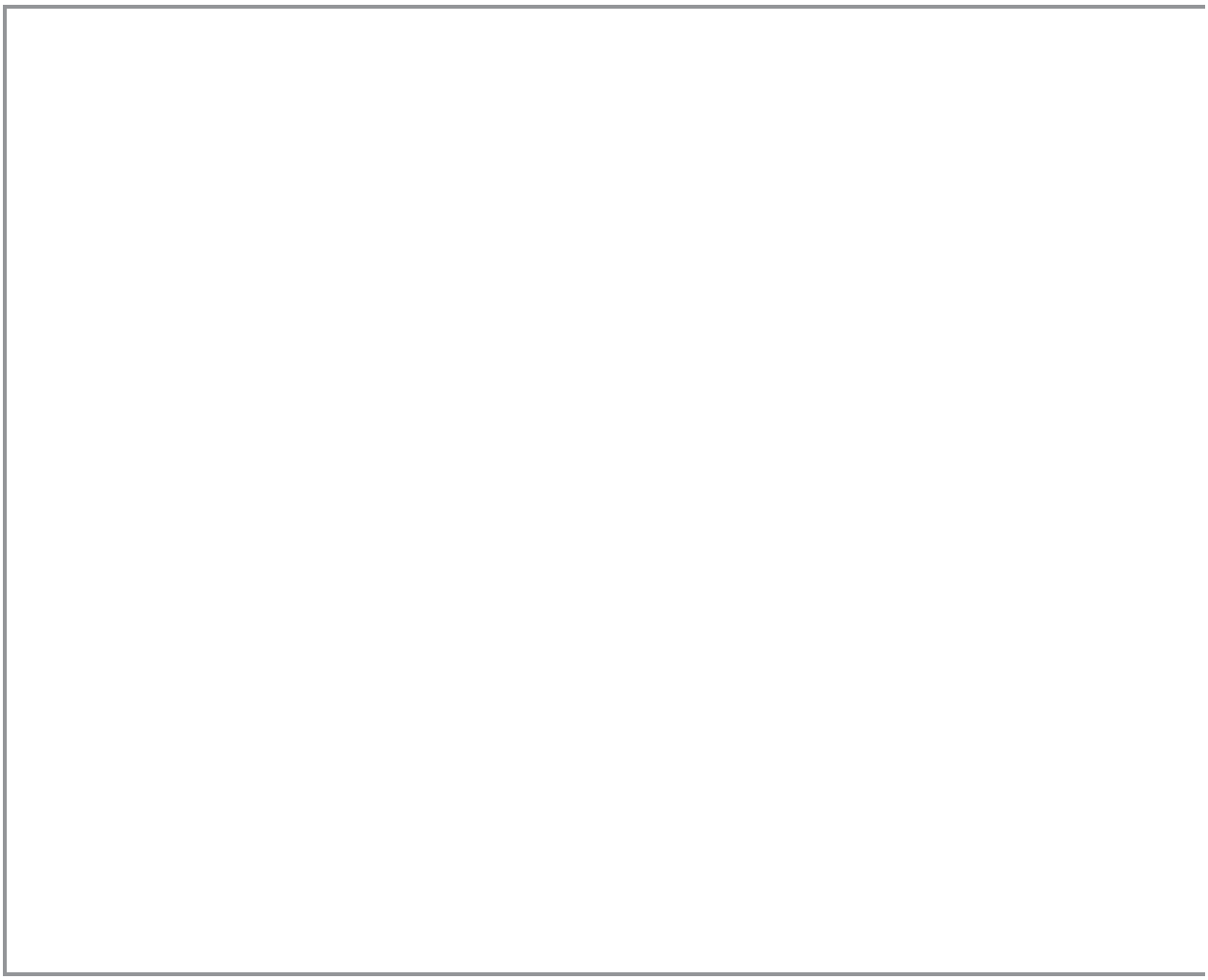
▶ **1: Take grip, get set under the bar, stay tight**



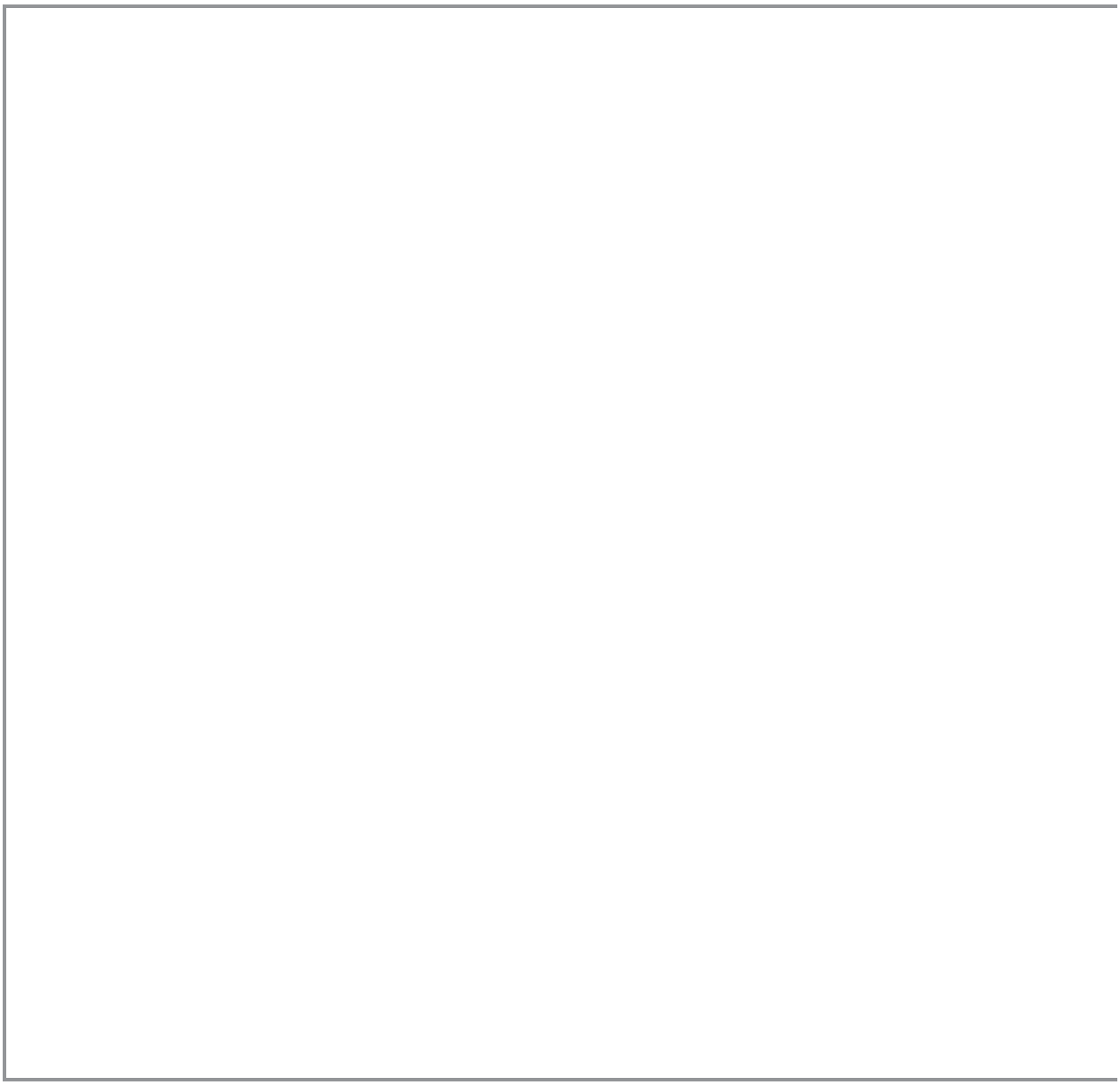
▶ **2: Lock bar into body in proper position**



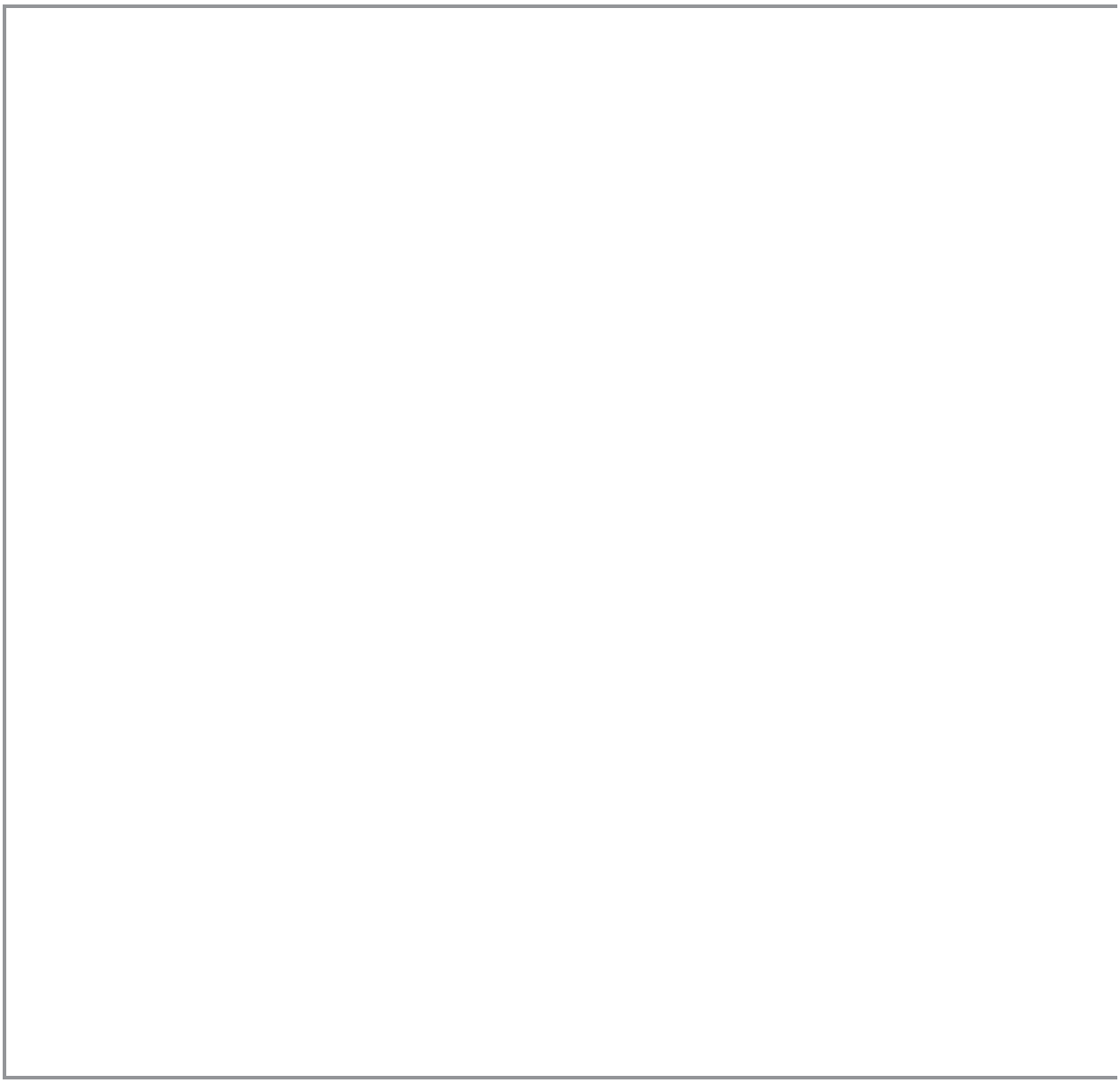
▶ **3: Lift bar off the racks, let it settle**



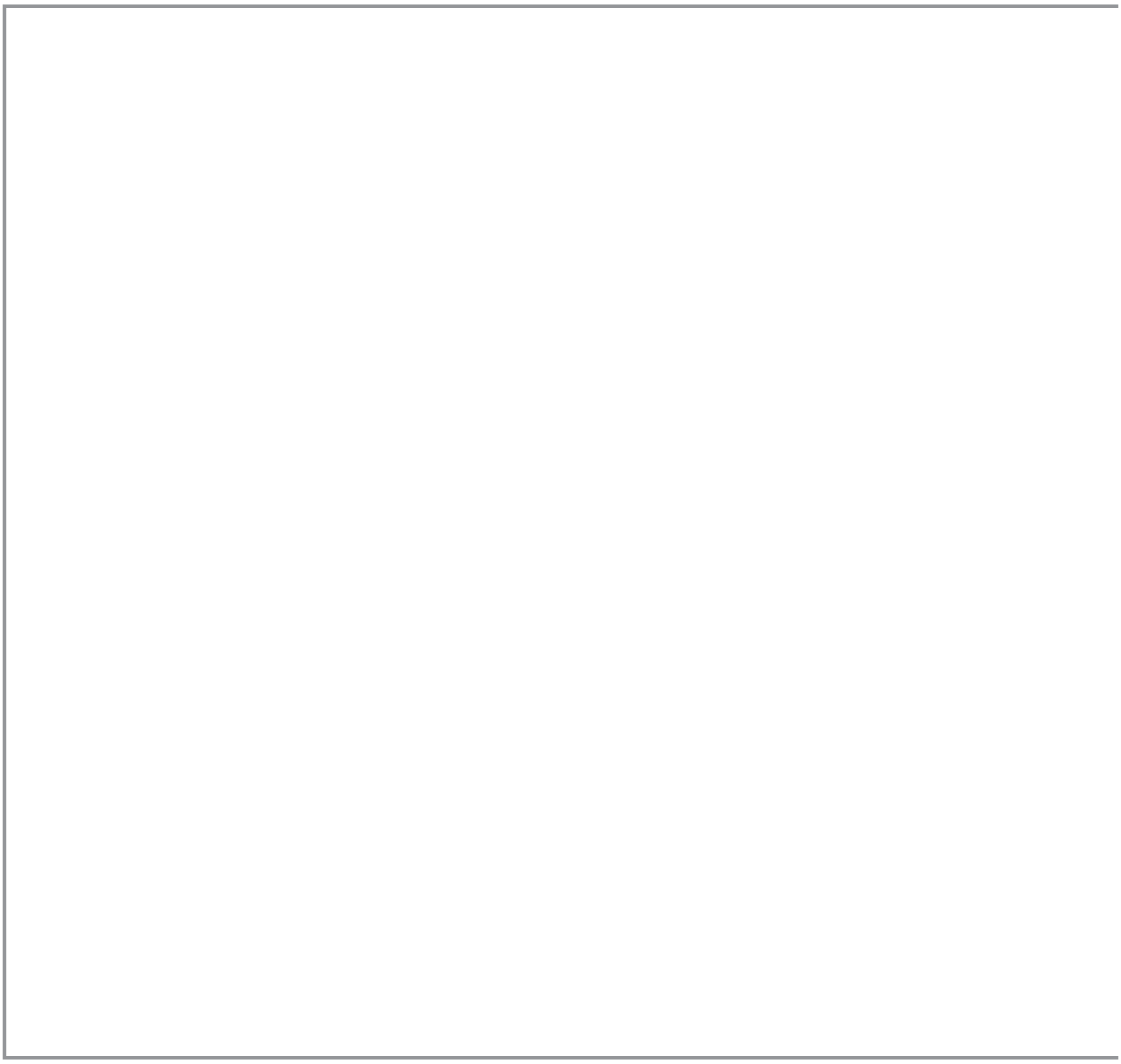
▶ **4: Walk back into position, wait for Squat command, big inhale**



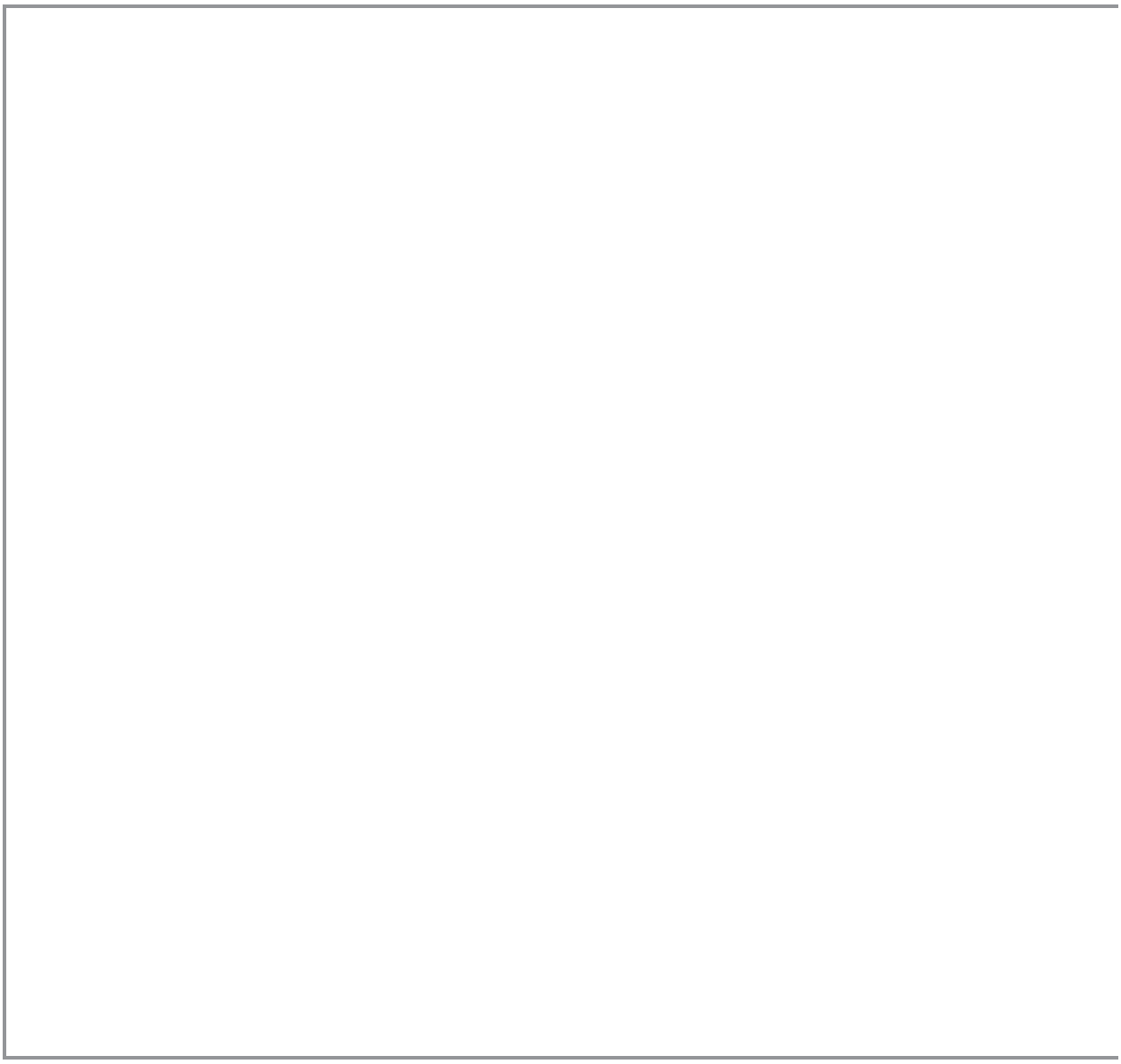
▶ **5: Begin descent, hips back**



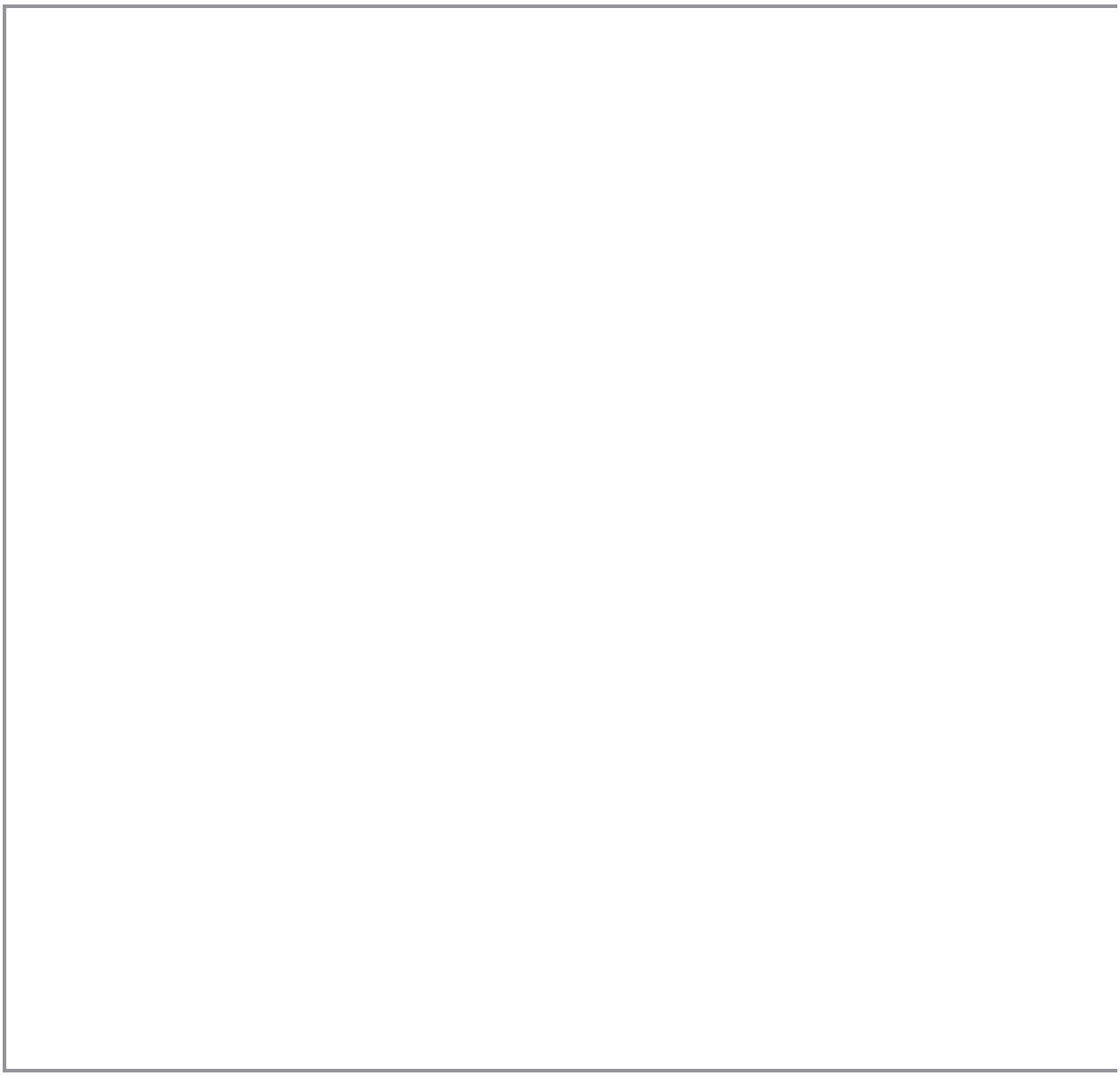
▶ **6: Controlled descent, weight on back of heels**



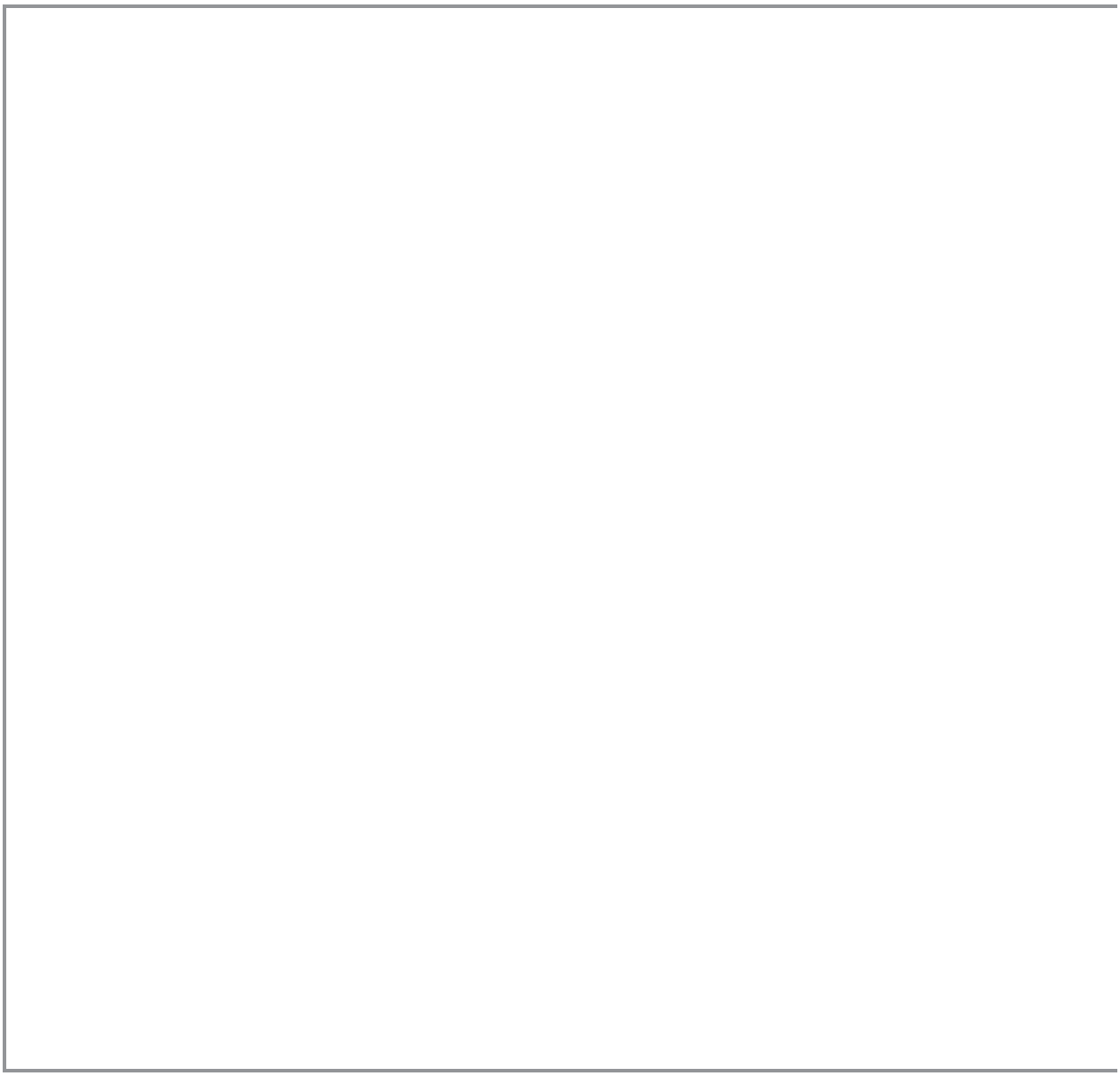
▶ **7: Stay tight, elbows under the bar and chest up**



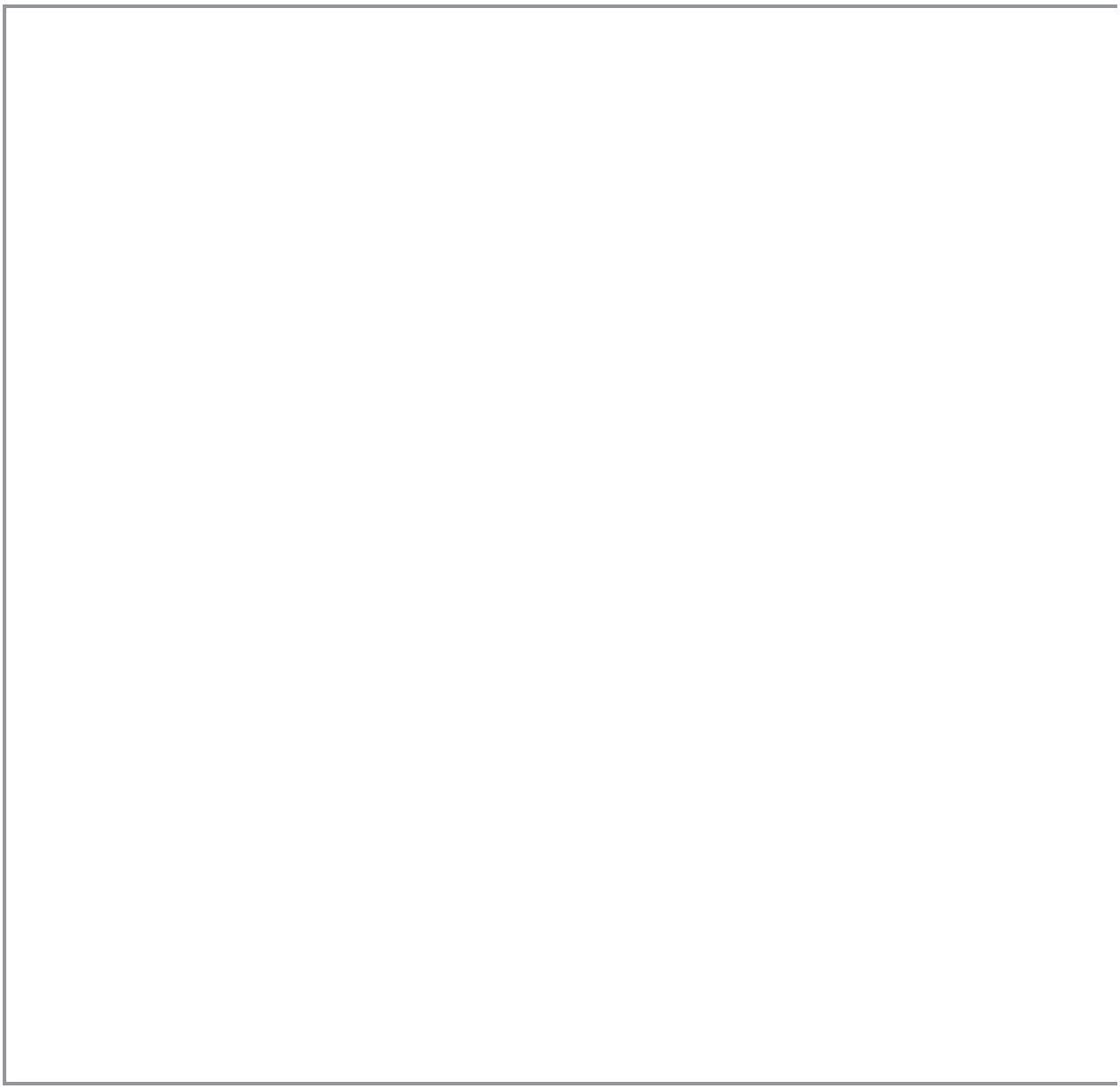
▶ **8: Hit depth — leave no doubt**



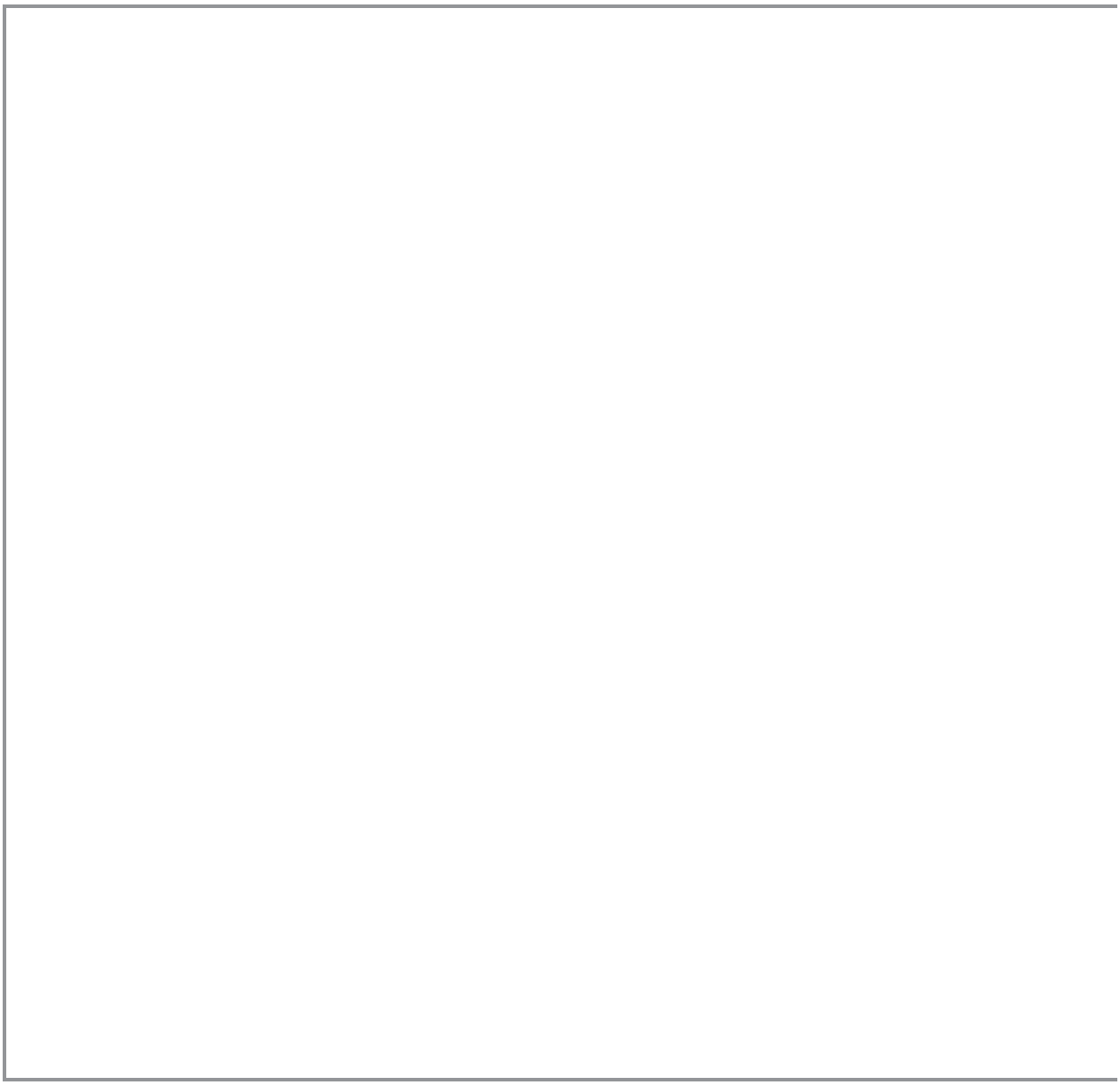
▶ **9: Shift weight back if necessary**



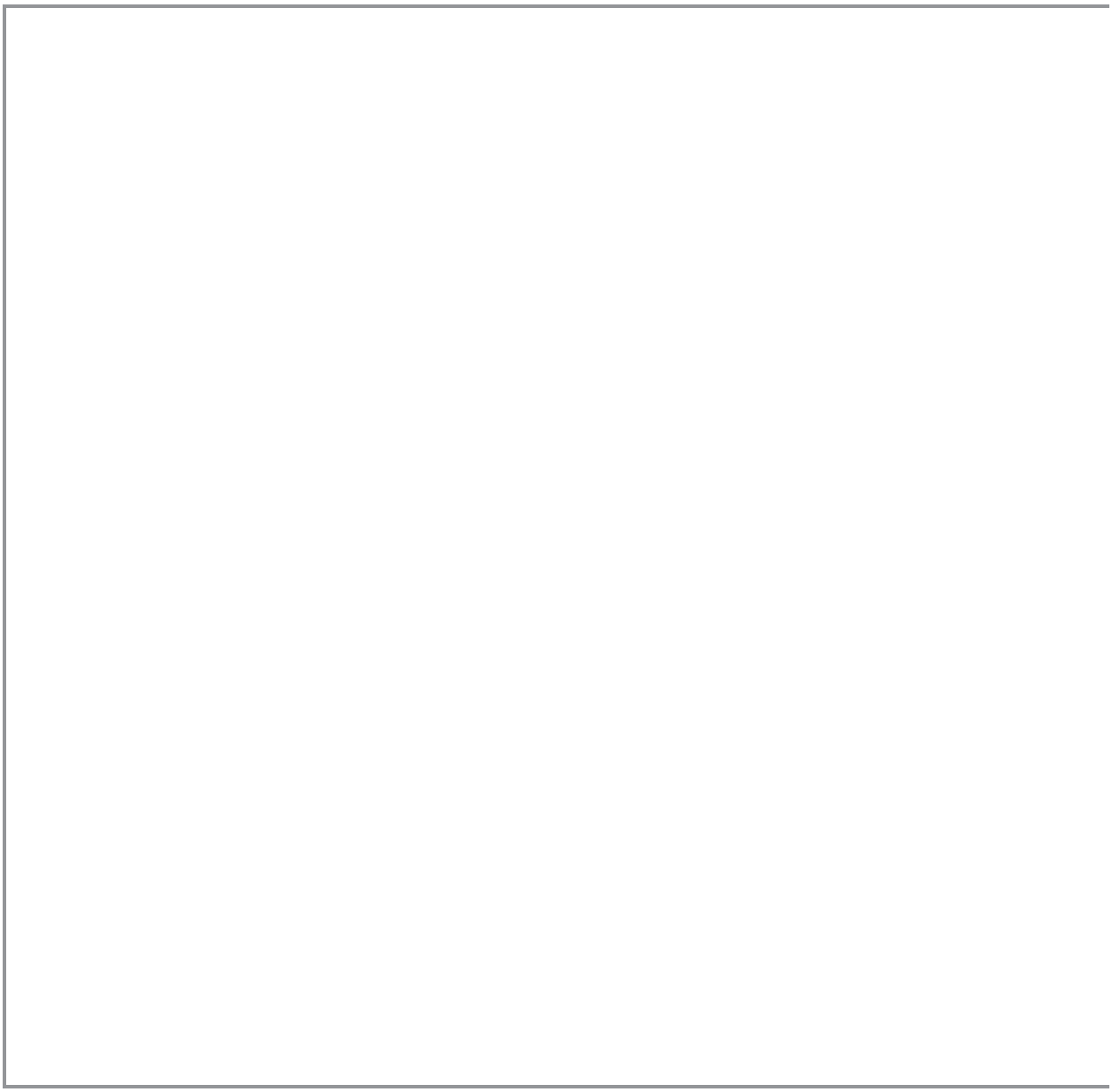
▶ **10: Drive up and back**



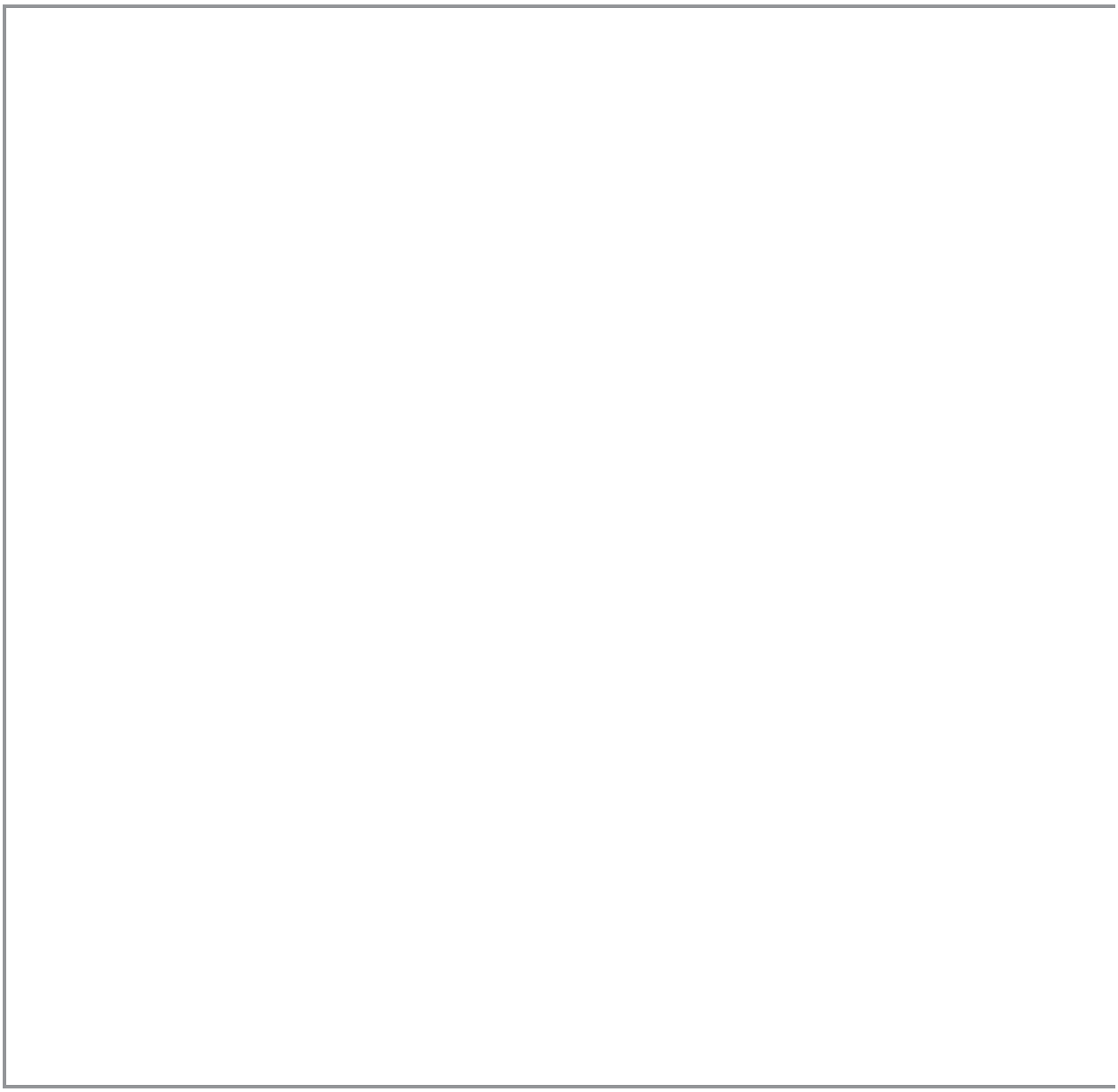
▶ **11: Knees out, tight core**



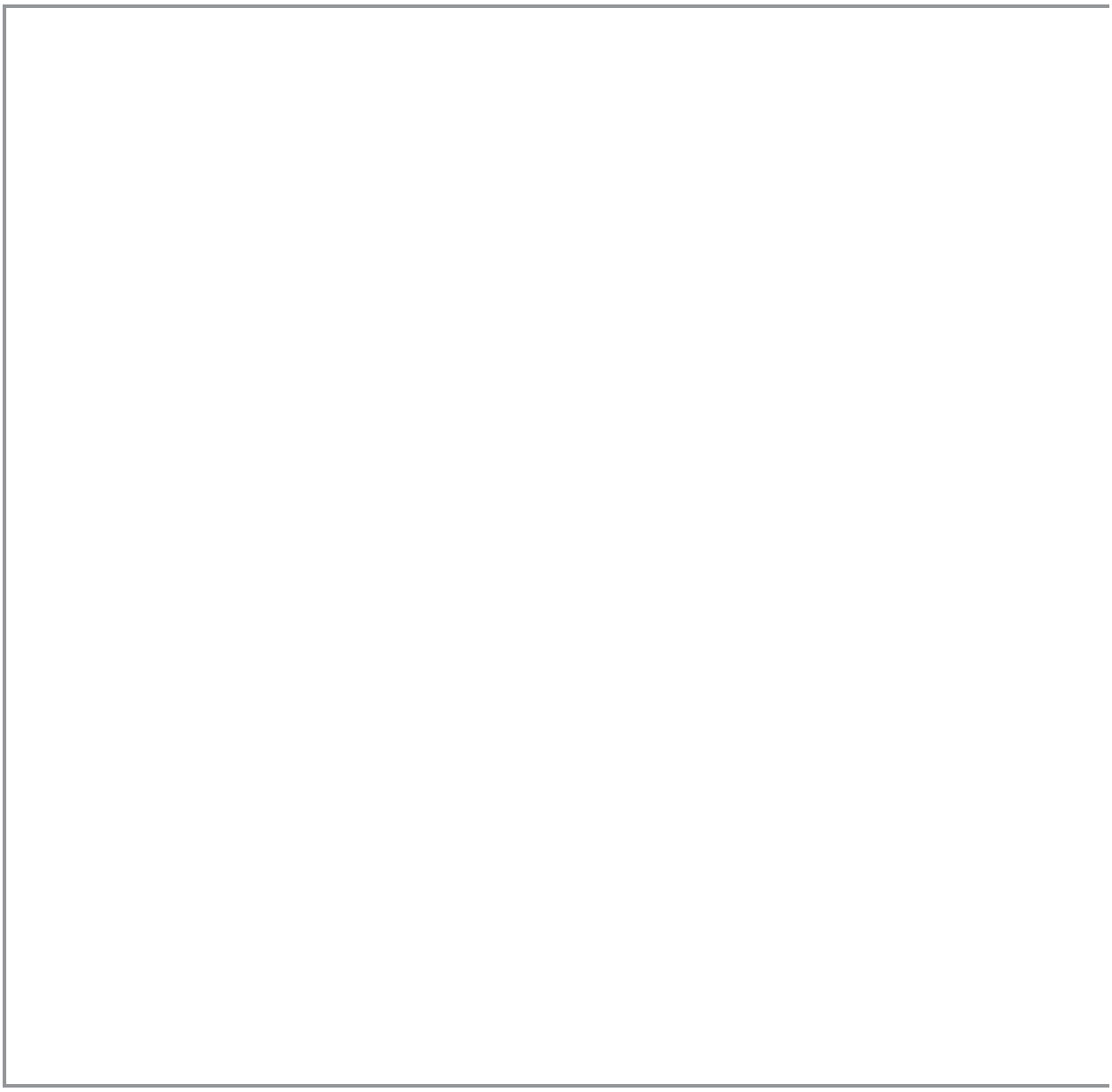
▶ **12: Fight toward lockout**



▶ **13: Keep extending**



▶ **14: Near the finish position**



▶ **15: Push hips forward, lock knees straight, wait for Rack command**
Photo credit: Vadim Snitkovsky

It should be noted that since the comparison is the top of the thigh to the top of the knee, those are the only two points that really count when it comes to depth. How low the bar goes is not that important. Don't imagine you have to get the bar down to a certain height when you squat, or your tendency will be to lean forward. Also don't just worry about the knees or think of a 90-

degree angle or anything like that. If your knees push too far forward you can have a very significant bend in the knee and still not be low enough. For help learning how to squat, continue reading for some tricks for getting low enough.

Blast "out of the hole."

Once you are low enough in the squat, which is a learned position, you then want to blast "out of the hole," as it is known, and stand upright. To do so you want to push your hips up and backward slightly. If going straight up was a 90-degree angle to the ground and going straight backward was a 0-degree angle this might be a 75-degree angle to the ground; you are pushing the hips back slightly toward your spotter as you come up. This movement helps put you in a powerful position to finish the squat, but it can cause you to round over, so you must keep the chest up as you do it. Once the hips have moved up and back just a bit, now you want to come up and forward with the hips while powerfully straightening your legs. This activates the most powerful muscles in your body—the glutes and quads—which you will need to squat the most weight. On the way up it is often useful to think about pushing your knees out, if for no other reason than to keep them from coming in. Pushing the knees out helps bring the hips under the body, especially when using a wider stance. As you near the top of the lift, continue to keep the chest up and drive forward with the glutes until your body is straight at the top position, with your hips under your body, not arched behind the body. In a competition lock the legs straight and stand upright and remain motionless until the head judge commands you to "Rack" the bar and then take one solid step forward. Once you make a good attempt to step forward the spotters can then assist you in returning the bar to the rack.

Things to do while Squatting

- Keep your chest up
- Keep a flat or slightly arched lower back
- Make sure you are even under the bar
- Place the bar on the top of your upper back
- Go down so that your femur is below parallel to the ground
- Break first at the hip, then at the knees
- Look straight ahead
- Keep your feet flat on the floor
- Keep weight in the back of your heels
- Keep your knees out and in line with your feet

Things NOT to do while Squatting

- Look up or down excessively
- Round your back
- Let the bar roll on your back
- Set up unevenly
- Only go down halfway
- Bend your knees first
- Let your knees go far in front of the toes
- Go up on your toes
- Let your knees move toward each other

> ***Lifters commonly have flexibility issues that negatively affect their squat.***

Flexibility/Mobility Problems

Lifters commonly have flexibility issues that negatively affect their squat. Any tight muscle can throw off the form, but three big problem areas are the calves, quads, and shoulders. If your calves are tight you won't be able to push your knees forward; this will make you sit back farther which can be okay but if you are not built for that then you will end up leaning too far forward with your upper body to compensate. It may make you go up on your toes during a squat or it might make your feet turn out as you descend into the squat. To improve flexibility in the calves try both static and dynamic stretches for them. Do not squat with anything under your heels or in shoes with a significant heel; this will just make the problem worse. Get a good stretch when you train your calves by going down low on the calf raises.

Tight quads can make it difficult to achieve the proper depth in a squat and they can contribute to knee problems. The entire quad might be tight or it might just be Rectus Femoris (RF) and the hip flexor area. To stretch the quads you can lie on your stomach and bring your heel back into your butt. To emphasize the RF you can place a pad under your knee to raise your thigh up and/or you can prop yourself up on your elbows while you do this stretch. Another good quad stretch is to kneel down and then sit back so your butt is on your heels, toes pointing away from you. To intensify this stretch lean backward. Dynamic stretches such as backward leg swings, with the knee both straight and bent, are good as well. Foam rolling before you squat can be very useful to loosen up this area along with the calves.

Tight shoulders don't immediately come to mind as a problem when squatting, but this can affect a lot of people. If you can't grip the bar comfortably then squatting will not feel right. If you are super tight, start off in a high-bar position with your hands as far apart as you can get them, probably next to the sleeve on the bar. Then gradually bring the hands in, a

finger width or two at a time. You can practice this anytime by placing an empty bar on your back or using a broomstick at home and just holding the bar in position on your back. Generally if the shoulders are tight during a squat it means the pecs, lats, teres major, front delts, subscapularis, and biceps are tight so you want to stretch those muscles. Good stretches are the wall glide, the floor glide, and the open book stretch. As a person becomes more flexible, they should be able to move to a low-bar position (which is harder on the shoulders), although they may have to move their hands out again and then bring them back in a little bit at a time.

Foam rolling before you squat can be useful.

Common Problems

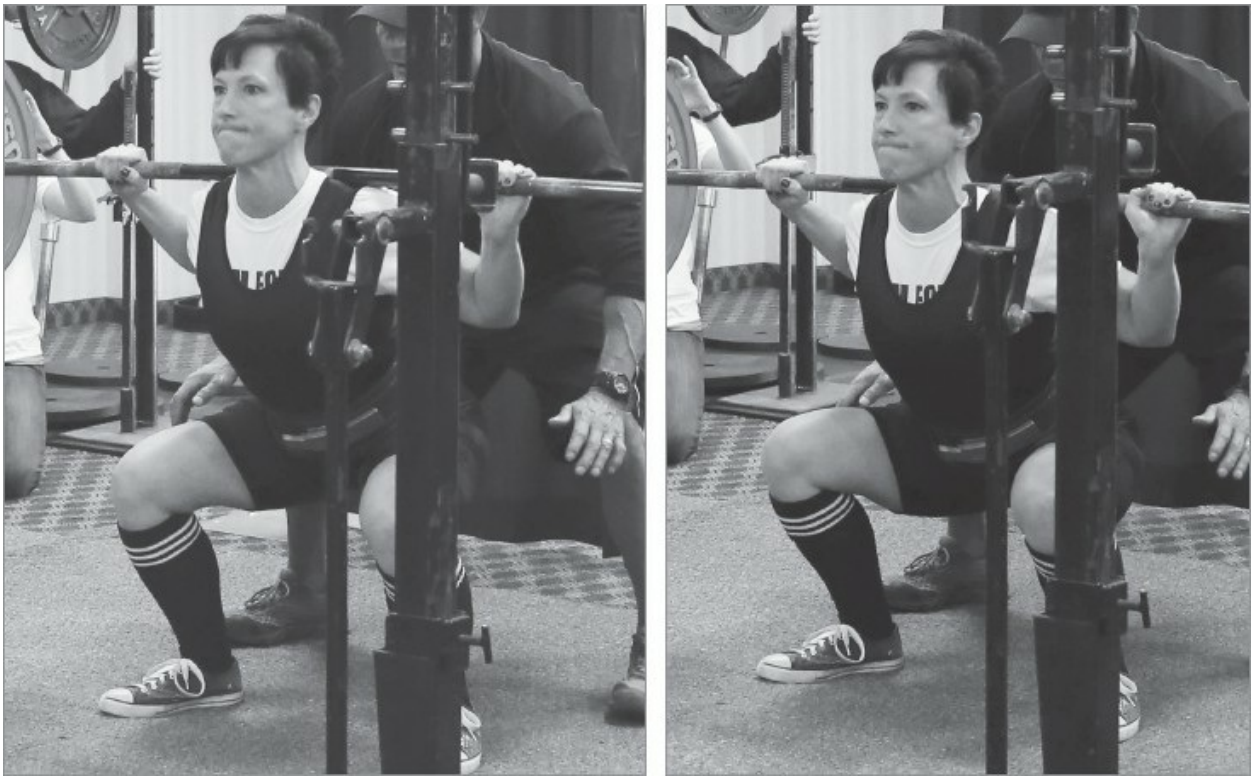
There are some common problems that affect a lot of people during the squat. One of them is falling forward as the weight gets heavy. There are several things you can do to prevent this from occurring. The first is to make sure that you have the bar as low as you can while keeping a good position. Keep your chest up and look straight ahead; you can try the different head and elbow positions discussed in this chapter. If you regularly fall forward try something different with your elbows, whatever you are not currently doing, and see if that helps or makes it worse. As you start the ascent make sure your first move is to push your hips up and back and not up and forward. Sometimes people shoot up and forward and as they follow that trajectory their center of gravity begins to move too far forward. It is better to feel like you are almost falling backward than forward when you squat. You can try to video your squat from the side to watch your form. Make a note of where the bar is in relation to your knees and ankles. If the bar is in front of your ankles, you are quite likely to fall forward.

Many people feel that weak abs and obliques contribute to falling forward. Some exercises can help prevent you from falling forward, either by strengthening certain muscles or promoting good form. Good mornings can help you catch yourself if you get in a bad position. Front squats and overhead squats will force you to remain more upright. Cable crunches and dumbbell side bends can help strengthen the abdominal area. Don't forget that sometimes it is poor flexibility that is forcing you into a bad position and making you fall forward; look for any weaknesses there.

The other huge problem in a squat is people simply don't go low enough for the squat to count in a competition. The squat is tough because there is no

People simply don't go low enough.

automatic measure that confirms it was performed correctly. In the bench press and the deadlift it is extremely easy for the lifter to know intuitively if they did the lift correctly. You know whether you touched your chest or not, and when you finish a deadlift there is a distinct feeling you get when you are locked out that is quite clear. With the squat there is not always a distinct feeling of knowing you went as deep as you needed to go. You don't want to go too deep in a competition (ATG squats) because that is making it unnecessarily hard. That would be like pausing the bench press for 30 seconds on your chest when you only needed to pause it for 1 second.

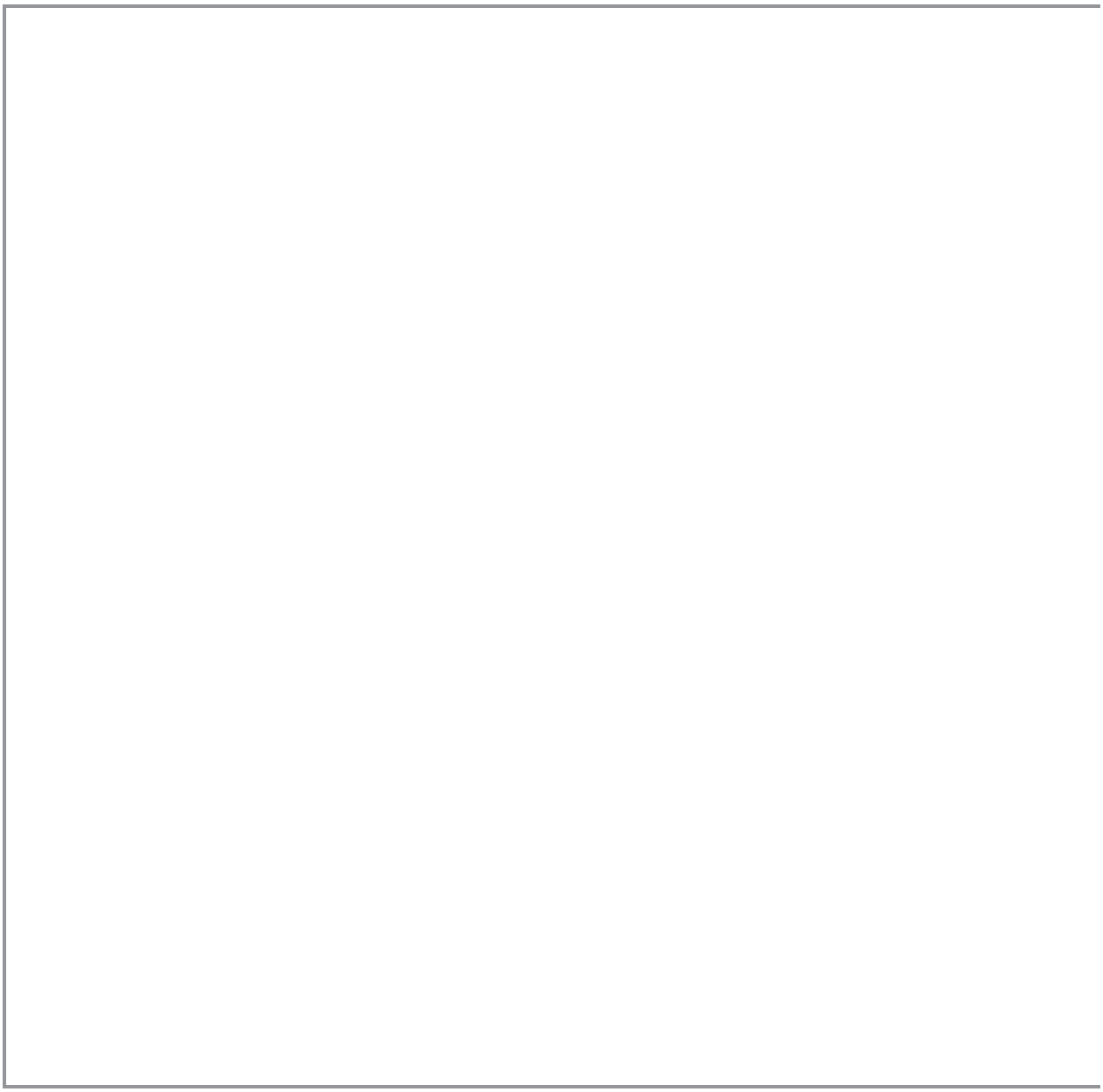


(Left) NOT Deep Enough

(Right) Deep Enough

The best way to learn this proper squat depth (legal depth but not too deep) is simply to practice it all the time. You want every rep to be good.

Verbal coaching from workout partners and other gym members can be very helpful. Those partners have to be able to be honest with you and tell you when you did not successfully complete the lift. Nobody wants to be the one to see their friend set a new personal record (PR) and then have to tell them that it was two inches too high. However, it is better that you do that in the gym where they can learn and adapt, than for the judges to tell them in a meet where they either perform poorly or bomb out (bombing out is when you don’t complete any lifts in an exercise in the competition).



A lifter loses their squat at the top as the bar slides down their back
Photo credit: Vadim Snitkovsky

Don’t expect regular gym members who have not been instructed in a proper powerlifting squat to give you good feedback. Generally your squat will be much deeper than what they are used to seeing, so it will look good to them, even if it is a bit high. You need an honest, brutally honest, training partner who knows what to look for and will give you the information

straight.

Finally lifters will sometimes lose their squat at the top. This typically happens with novice to intermediate-level lifters squatting light to moderate weights. The lifter will be very fired up (from the adrenaline of competing at a meet) and they will shoot the squat up faster than they expected (at least in the top part of the ROM). The bar will literally bounce off their shoulders at the top and then come down and sometimes it lands just posteriorly to its original position. Because of this the bar will sometimes roll down the back toward the butt. This is particularly common when wearing a singlet which is slicker than a traditional T-shirt. This can be corrected by cueing the lifters to maintain control at the top of the range of motion. This is cured by not driving the weight up as fast as possible, just fast enough to complete the lift, and cueing the lifter to pull the bar down on to their back to prevent the bar from "jumping." In addition, I encourage all of my lifters to squat in their singlets for the last 3–4 weeks prior to the competition to make sure they are used to how that singlet feels when the bar is on their back.

> ***You need an honest, brutally honest, training partner.***

Common Cues for the Squat

When squatting it is useful to have some key cues in your head or called out to you from your coaches and training partners, to help you keep your form under big weight. Outlined below are some common cues you will hear; they are generally presented in order of how the squat is performed. It would not work to focus on all of them, but pick 2–4 that seem to work well for you and zero in on them during your lift.

Get Tight
Tight Back
Strong Walkout
Good Setup
Screw your feet to the floor
Wait
Big Air
Hips Back
Back of the Heel
Outside of the Foot
Sink It
Drive
Brace the core
Knees Out
Finish
Hold

> ***If you have to go up on your toes, the rack is too high.***

Common Competition Mistakes in the Squat

There are some specific problems that lifters often have with the squat in a competition. The first is proper rack height. As addressed previously, the rack height should be low enough so that you can easily lift the bar off the racks and walk it out without the bar banging into racks. If you have to go up on your toes, the rack is too high. Remember, the bar will sit lower on the rack with several hundred pounds than it will when it is empty.

A lifter sinks a nice raw squat
Photo credit: Vadim Snitkovsky

The biggest problem is depth, and this can only be learned by regular practice. Not hitting proper depth is compounded by nerves. The squat is the first lift in any powerlifting meet and certainly the vast majority of powerlifters that I have spoken to find it the most nerve-wracking lift, especially the first attempt, the one that gets you in the meet. And let's not forget you are trying for a 1RM. If you go too low, you almost for sure won't get the lift, as an extra inch or two of depth can make the lift significantly harder. Of course if you cut it high it doesn't matter if you can lift it or not because it won't count in a meet. My philosophy is that it is better to go too low and fail because of lack of strength than to go too high and fail because

of improper depth. Practicing good form while lifting heavy weights is the only way to learn the proper depth.

Probably the next biggest problem is moving your feet once you get the "Squat" command. As discussed in the rules presented below, you will walk the bar out of the racks and position yourself the way you want. Then, when the judge believes you are ready to go, he will give you the "Squat" command, which is the signal to squat down. Once you get that signal, you cannot adjust your foot position at all. Sometimes lifters will shift their feet because it doesn't feel just right and that will disqualify them. The key is to practice your walkout and set-up so you are right all the time. You also don't want the judge to think you are ready before you are. Most lifters do this by looking down slightly as they walkout and get set, and then they look up or straight ahead to signify they are ready to go. Others give a little nod of the head to indicate they are ready; both methods work well. Find a pattern that works well for you and stick with it.

You will probably see these judges at your next meet.

A final problem which can occur on all the lifts is failure to wait for the commands. When the judge says "Bar is loaded" that means the bar is ready to go and you should walk up on the platform, get in position, and walk the bar out of the racks. Then the judge will say "Squat." If you squat down before he says so, you will be disqualified automatically. Once you squat, you go down to the proper depth and come back up. There is no up signal and you do not have to pause at the bottom, just drop down and come up. Once you are standing up after the squat you must lock your legs out and stay there for a second. Then the judge will say "Rack" and you must take a step forward toward the rack. Once you move forward the spotters can help you return the weight to the racks, and the lift is

complete. There are three judges on the squat; if two or three of them give you white lights, the lift was successful. If two or three of them give you red lights, the lift was not. If you did receive a red light and you do not know why, you can politely ask the judges why you received that light. Remember you may be emotional at this moment but still be polite. If you are rude it is just human nature that they will judge you harder next time. And even if it is your last lift, still always be polite because you will probably see these judges at your next meet.

Technical Rules of Performance

What follows are the official rules for the Squat. Rules can vary from federation to federation; these particular rules apply to the 100% RAW Powerlifting Federation. A simple explanation follows if necessary.

Squat:

1. **The lifter shall face the front of the platform. The bar shall be held horizontally across the shoulders, hands and fingers gripping the bar, and the top of the bar not more than the thickness of the bar below the outer edge of the shoulders. The hands may be positioned anywhere on the bar inside and/or in contact with the inner collars.**

The lifter faces the head judge, the bar goes on your back, and your hands must grip the bar itself.

2. **After removing the bar from the racks (the lifter may be aided in removal of the bar from the racks by the spotter/loaders), the lifter must move backwards to establish the starting position on his/her own. When the lifter is motionless, erect with knees locked, and the bar properly positioned, the Head Referee will give the signal to begin the lift. The signal shall consist of a downward movement of the arm and the audible command "Squat." Before receiving the signal to "squat" the lifter may make any position adjustments within the rules, without penalty. For reasons of safety the lifter will be requested to replace the bar, together with a backward movement of the arm, if after a period of five seconds he is not in the correct position to begin the lift. The Head Referee will then convey the reason why the signal to "Squat" was not given.**

The lifter will walk the bar out, get set, stand with legs locked and motionless and wait for the squat command to be given.

3. **Upon receiving the Head Referee's signal, the lifter must bend the knees and lower the body until the top surface of the legs at the hip**

joint is lower than the top of the knees. Only one descent attempt is allowed. The attempt is deemed to have commenced when the lifter's knees have unlocked. The bar may move from its starting position downwards on the lifters back the thickness diameter of the bar during the performance of the lift.

The lifter will squat once the command is given; once the descent starts the full lift must be completed successfully. The bar may roll very slightly on the back.

4. **The lifter must recover at will to an upright position with the knees locked. Double bouncing at the bottom of the squat attempt or any downward movement is not permitted. When the lifter is motionless (in the apparent final position) the Head Referee will give the signal to rack the bar.**

The lifter will rise from the bottom and stand with legs locked, standing straight. Once you start coming up you can't go back down and come up again. The lifter needs to wait for the rack command to move forward.

5. **The signal to rack the bar will consist of a backward motion of the arm and the audible command "Rack." The lifter must then move forward and return the bar to the racks. For reasons of safety the lifter may request the aid of the spotter/loaders in returning the bar to, and replacing it in the racks. The lifter must stay with the bar during this process.**

The lifter will get the rack command and move forward; you can't fall backward

Causes for disqualification of a Squat:

1. **Failure to observe the Chief Referee's signals at the commencement or completion of a lift.**
2. **Double bouncing at the bottom of the lift, or any downward movement during the ascent.**

3. Failure to assume an upright position with the knees locked at the commencement or completion of the lift.
4. Stepping backward or forward although lateral movement of the sole and rocking the feet between the ball and heel is permitted.
5. Failure to bend the knees and lower the body until the top surface of the legs at the hip joint is lower than the top of the knees, as in the diagram (in the rules booklet).
6. Any movement of the bar on the back more than the diameter/ thickness of the bar below its starting position.
7. Contact with the bar or the lifter by the spotter/loaders between the Head Referee's signals in order to make the lift easier.
8. Contact of the elbows or upper arms with the legs. Slight contact is permitted if there is no supporting that might aid the lifter.
9. Any dropping or dumping of the bar after completion of the lift.
10. Failure to comply with any of the items outlined under Rules of Performance for the squat.

A powerlifter completes a PR squat
Photo credit: Vadim Snitkovsky

Benefits of Squatting

If the squat is that hard, there had better be some good benefits from it, or else why do it, right? I am not a believer of suffering for no reason, but luckily the squat has its payoffs. From a powerlifting perspective, during the squat you will be using either the most weight or the second most weight of the three lifts, so it is a crucial lift. It sets the tone for the day since it is the first exercise, and after the squats are complete you can often have a very solid lead or be so far in the hole that you will not be able to make up the difference with the other two lifts. Training the squat hard will also help your deadlift, no matter what your style is, so it will have a beneficial effect on another key exercise. If there was just one exercise you could do to get good at powerlifting, it would be the squat, no doubt.

From a health point of view the squat has a great effect on the body. Of course it works the lower body and the core, particularly training the glutes, quads, hamstrings, adductors, erectors and even the abs. Because it uses large muscles and a large amount of weight, it stimulates the release of powerful, beneficial hormones such as testosterone and growth hormone. It will help strengthen the bones of the body, particularly the spine, hips, and legs. It can improve jumping ability, sprinting ability, the power of the hips and legs, and the size and shape of the hips and legs. It can even have a beneficial effect on the cardiovascular system when performed with a moderate to a high number of reps. It also helps improve basically every leg exercise there is, from deadlifts to leg press to leg extension to lunges. You name it and the squat will probably improve it. The squat will help build muscle mass in the entire body, particularly in the legs. Extra muscle will increase the metabolism and that will make it easier to lose

The squat can improve jumping and sprinting ability

or maintain weight; even 5 lbs of new muscle will make a little bit of a difference, which adds up over time. Muscular legs are leaner and denser than fat legs; a 50 lb leg that is made mainly of muscle will take up less space than a 50 lb leg made up mainly of fat.

Equipment and Apparel for the Squat

When squatting, it is important to be appropriately dressed and ready for a big lift. In addition, if you compete, you can wear powerlifting-specific apparel. The benefits of each of these things are discussed below.

Shoes – The proper type of shoes are important in a squat. You want a shoe with a firm sole and some ankle support. Running shoes are not ideal for a squat, and if you have the air-filled ones a heavy squat can pop that bubble. A flat-soled high top is good, such as the old school Converses. Deadlifting shoes (wrestling shoes) can work well for the squat but most people like more ankle stability and a very slight heel to squat in. In addition, you can buy actual squat shoes to squat in. They are kind of expensive ($100+), are not comfortable to walk around in, and they look a little weird, but they feel good for a heavy squat. Olympic Lifting shoes can also work well for the squat particularly if the lifter has tight ankles.

Running shoes are not ideal for a squat.

T-Shirt – It is a good idea to wear a T-shirt and not a tank top when squatting. A T-shirt gives your skin a little more protection so the bar isn't just digging into your skin. If you compete, it is against the rules to wear anything other than a T-shirt with your singlet. Oddly, sometimes the singlet can be a bit slippery and it can make the barbell feel slightly different on your back when you squat. I like to have my lifters wear their singlets for at least the last 3 heavy squats before a competition to make sure they are used to that feeling.

Shorts – When squatting most people like to wear shorts, preferably ones that don't go below the knee because otherwise they tend to get caught up on the knee as you go down. In addition, wearing shorts as opposed to pants

makes it easier to see how low you are going. In a competition you will be wearing a singlet and no shorts, although many people leave the shorts on while they are warming up.

Underwear – When choosing underwear for the squat you want to choose something that is comfortable, allows you to complete the range of motion, and keeps your hips warm. Briefs, boxer briefs, and Under Armour type underwear all seem fine. Boxers are generally too loose and not a good idea. In a competition, sometimes it is illegal to wear any briefs that have legs to them, even if they are non-supportive.

Belt – Most competitive lifters will choose to wear a belt when squatting heavy weights. The belt can increase intra-abdominal pressure and add stability to the trunk and core. It can also serve as a cue to remember to brace your core and it can serve as something to brace your core against.

Wrist Wraps – The squat can place a lot of pressure on the wrists particularly when the bar is in low-bar position. Wrist wraps can help ease that tension and are often used in the squat both during training and competitions.

The belt can add stability to the trunk and core.

Summary

Performing the squat in a competition can be a nerve-wracking experience. Perhaps this is because it is the most dangerous of the three lifts — the weight literally could crush you to the floor. Perhaps it is because it is the most subjective of the three lifts, and that knowledge that you hit proper depth does not come as easily as knowing you touched your chest in a bench or locked out the deadlift. Perhaps it is simply because it is the first lift performed in a competition and it would be nerve wracking no matter what it was. More likely it is a combination of all these things. The good news is that once you get that first squat down, the jitters, if you have any, go away and you can focus on the more important matters at hand—squatting double or triple your bodyweight and setting yourself up for a PR total.

Beginning Squat program for a normal adult male

Week 1	45x12	95x10	95x10	95x10
Week 2	45x12	95x10	95x10	105x10
Week 3	45x12	95x10	105x10	115x10
Week 4	45x12	95x10	115x10	125x10
Week 5	45x12	95x10	115x10	135x10

Beginning Squat program for a normal adult female

Week 1	25x12	45x10	45x10	45x10
Week 2	25x12	45x10	45x10	50x10
Week 3	25x12	45x10	50x10	55x10
Week 4	25x12	45x10	55x10	60x10
Week 5	25x12	45x10	55x10	65x10

Many people reading this will be well beyond the beginner stage, in which case they should refer to program design chapters in this book for a more detailed and advanced squat routine.

Interview with Kirk Karwoski

“Just don’t fuck it up for 20 seconds”

Kirk Karwoski has won multiple National and World Powerlifting titles. He is the first man in the IPF to squat more than 1000 lbs, and he is likely most famous for his video of him squatting 1000 for 2 reps in training for that competition.

This interview was conducted in person with Kirk. I asked him questions and wrote down his answers. This is what he had to say:

Provide us with a brief history of yourself.

I was born in 1966 in Penn State University, the result of too much alcohol and a missing condom. Currently I am a mechanic working on hydraulic machines, and I live in Maryland.

Kirk Karwoski doing what he does best

Photo credit: Kirk Karwoski

What are your best lifts?

In 1995 I hit a 1003 lb squat in competition; it was with the USPF, which at the time was part of the IPF. That was the first 1000 lb squat done officially with the IPF to their standards and tested. I was in the 275 lb weight class. That was with old school single ply squat suit and knee wraps. My best bench ever was 578 in a meet in an old single ply bench shirt. I deadlifted 777; both of those lifts were also at 275.

In the gym I hit a 1005 for 1, and I am probably most famous for my 1000x2 squat that I did in prep for the 1003 one in competition. I also hit 645x8 raw, no belt, smoked it, and I did 800x5 raw with just a belt. After I retired I came back and hit a competition squat with the AAU. I was 242, and I squatted 826, which was an All-Time best lift. I am very proud of that 826 raw squat because I felt it helped give raw lifting a push to prominence. I wish I had hit a max raw squat in my prime; there is no doubt in my mind I would have hit 903 to depth, raw, with just a belt.

I deadlifted 800 lbs a few times in the gym but never did it in a meet. I also did a curl competition; it was a standing strict curl (not against the wall) with an EZ bar, and I got 220 which I believe was a National Record at the time.

List some of the titles that you have won.

I won Teenage Nationals 3 times and Open Nationals 7 times. I won Junior Worlds in ’89, I believe, and then lost Open Worlds the next year by 5 lbs; then I won IPF Worlds. I was 6 time IPF World Champion. This was back when there was pretty much just one National and one World Championships for everybody, not one per federation like there is today. I retired in 1995.

When did you start training; when did you first compete; what were your first competition lifts?

At age 8 I saw the Incredible Hulk on TV and instantly realized that was the answer to all my problems. I fell in love with muscles and lifting, but I didn't get a barbell set until I was 12. I did a little bench press competition when I was very young; it was your bodyweight for reps. I think I got like 148 lbs for 10–11 reps or something. My first real competition I was 14–15 years old, I was a light 165er, and I hit a 420 Squat, 300 Bench, and 400 deadlift. I did a 450 deadlift but didn't wait for the down command; then I tried to pull 475 to go for the win and missed it.

How much weight did you lift the first few times you tried the squat?

Pretty much the first time I really tried to squat I hit 300 lbs. It was in high school and I had to squat that to get my name on the wall. I trained at home — it is tougher to practice the squat with minimal equipment — the coaches didn't think I could do it but I did. I weighed about 150 at that point.

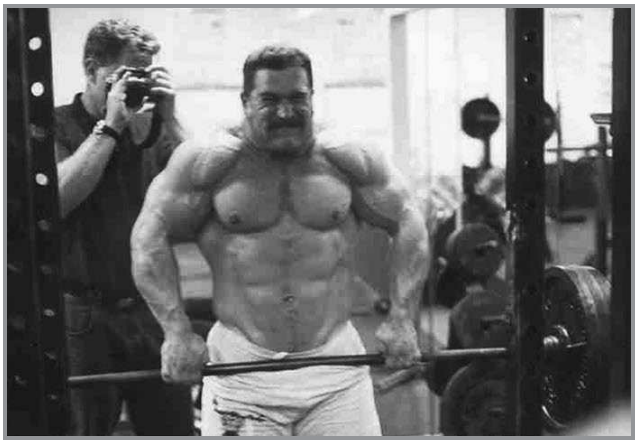
At my house I used to take my Dad's bad-weather tires, tie them together, lay a board on them, and that was my bench press. Of course, you didn't use collars; you just went until you failed, tilted the weight until one side unloaded, then it flipped over, you got up, rested a bit, and did it all over again.

I did a lot of track in high school and I really think that helped me as an athlete. I did the shot put and discus but I sucked at them; I was better at running. I could do a sub 11 second 100 M sprint, sub 23 second 200 M, and sub 55 second 400 M. I ran a 4.6 in the 40 yard dash after practice wearing everything but shoulder pads.

When I graduated high school I was about 200 lbs, I could squat and deadlift 600 lbs and bench 400 lbs. I could also do a 325 clean and jerk, a 245 snatch, and a 245 clean and press.

What was your first training program like?

I just did the workout the coaches gave us in high school; the main things to do were the powerlifts and the Olympic Lifts. I think I still have copies of those old handouts lying around somewhere.



Kirk performing his Karwoski Shrug
Photo credit: Kirk Karwoski

How much were you squatting when you first hit a plateau in the exercise? How long did you plateau there? What did you do to get past that plateau?

I hit a big plateau trying to get an 800 lb squat in a meet; that took me like 2 years. I finally hit like 804 — it was weird; there was a power failure, the lights were out, I still did my squat, it took me like 10 seconds to complete the lift, and then I just blacked out. This was at Nationals. I had bombed out before trying to get that squat; initially I wasn't a super consistent lifter.

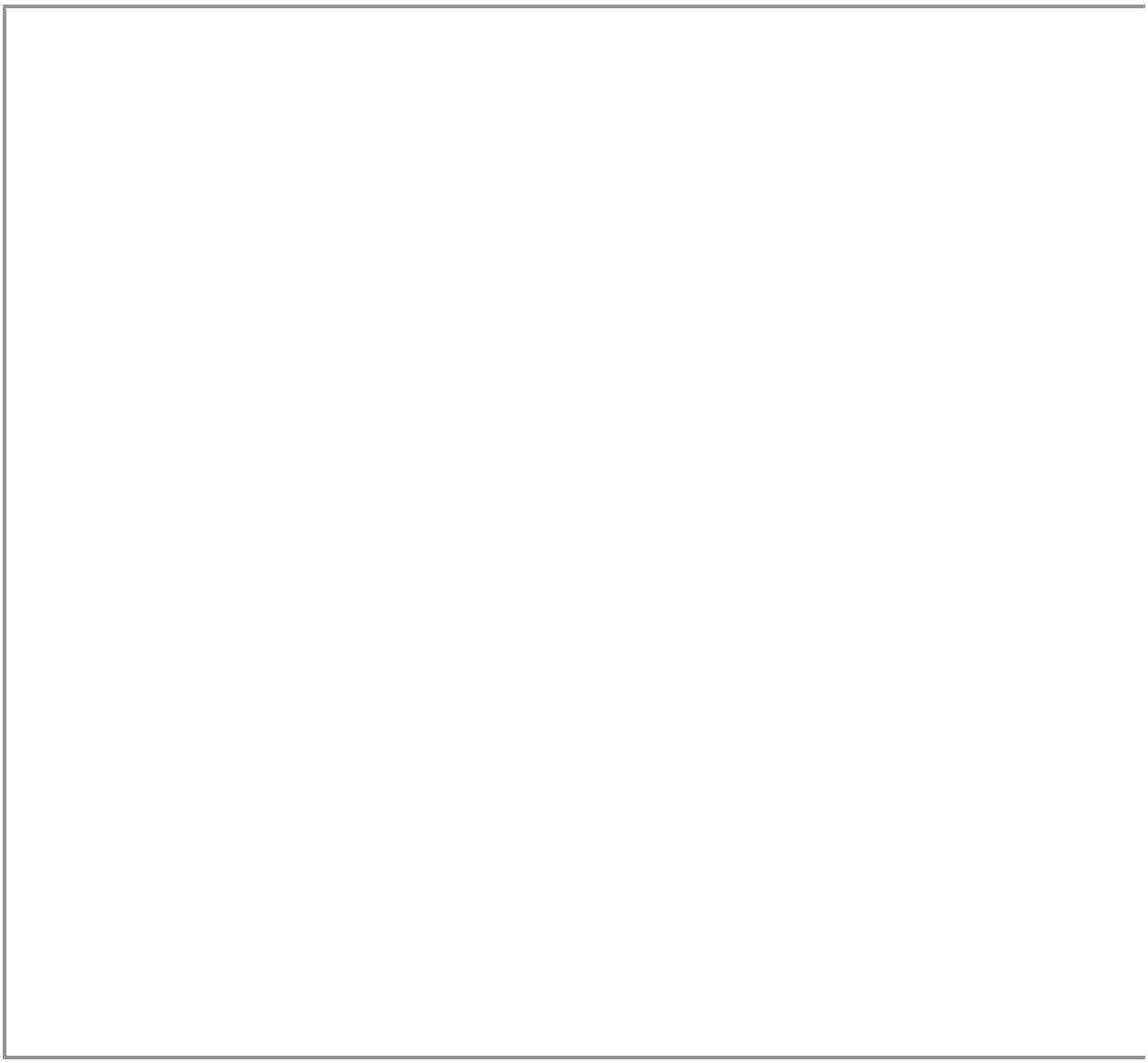
For me the plateau forced me to examine myself. I refined my technique and started to train smarter. I may be a big shaved ape but I am not stupid and this isn't rocket science. It forced me to become a better lifter. I also gained some weight and that weight gain helped me feel more stable in the bottom of the squat.

Give a history of the progress you have made in the squat.

Around 1985 when I was a teen I was squatting 620ish raw. In the late ’80s I was stuck with that 800 plateau. In 1990 I hit 900 and in 1995 I hit the 1003.

What do you feel is key to being successful in the squat?

In every rep, you must duplicate your form. It must become automatic and the form must be perfect. For me it is actually painful to squat 245 with perfect form because it is so light, but you have to make every rep the same. Don’t be sloppy with something and then think “I’ll tighten up my form once the weight gets heavy.” Every rep is a chance to practice perfect form.



In every rep, duplicate your form
Photo credit: Kirk Karwoski

What do you feel the best way to train for the squat is for a normal powerlifter (routine, days per week, exercises, sets, reps, rest, etc.)?

I think the periodization plan of working something hard once a week works really well. I think the newer ideas incorporating Prilepin's chart and stuff can be good too, but I had a lot of success with basic linear periodization and overload every week.

In the ideal world my training would look like this:

Monday: Squat

Tuesday: Closegrip bench and arms

Thursday: Deads and Back

Saturday: Chest and Shoulders

I would train hard as shit for about 90 minutes or so, I would be sore as hell. If there was a meet 15 weeks away, I would be sore all the time for those 15 weeks. I really think if you go and beat yourself up hard once a week in each area, I don't think you need more than that. Even the arm day above was a bonus, but when I did it I felt I was more muscular, I was stronger, I got so physically hard that I could handle that extra weight and the closegrip bench really helped out. I always believed that pain and strength were the same thing. The more hurt I was, the stronger I was.

For a while I had pretty good results just training twice a week when that was all I had time to do. It would go like this:

Day 1: Squat and Bench

Day 2: Squat light (mainly warm-up sets to get loose), Bench, and Deadlift.

My specific plan would go something like this. I would work up to a heavy set of 8 reps, raw no belt, over 4 weeks. Then I put on the belt, dropped to 5 reps and worked up to a heavy set of 5. Going up about 10–20 lbs a

week, again for 4 weeks. Then you go suit on, wraps on, straps down for triples, and then suit on all the way, each one of those for 3–4 weeks. When you add some gear (like the belt, suit, etc) you make bigger jumps. For me for example if I worked up to 645x8 then I would go 705x5 with a belt as an easy first week and then progress from there.

What do you think of training with a high frequency (3+ times per week) in the squat? Have you done this, what were your results, and do you believe in it or not?

I never tried this. I have a hard time imagining it working, I could barely climb a flight of stairs the next day — how I was going to squat 800 lbs again? I always had very physically demanding jobs. I don't think that would have worked well with those jobs and the extra labor I had to do.

What do you think of training with a medium frequency (2 times per week) in the squat? Have you done this, what were your results, and do you believe in it or not?

I did a second squat a week for a while with just warm-ups; it was okay but in the ideal world I preferred just training it hard once a week.

What do you think of training with a low frequency (1 time per week or less) in the squat? Have you done this, what were your results, and do you believe in it or not?

This is what always worked the best for me and this was the most popular method of training when I was competing; it was what most of the top guys were doing.

What are your favorite assistance exercises for the squat?

Honestly squats are the best thing. I liked front squats and would do those in high school; I would also do step ups then. I remember loading up 4 plates, putting it in front squat position and performing a step up on a bench or whatever. But as I got bigger front squats just choked me out.

Leg presses are okay but I don't think they are worth the effort. We had only so many plates in our gym; lugging them from the squat rack to the leg press, loading them up — it was more cardio than anything. Same thing for Hacks. I don't think boxes or bands and chains are necessary; that shit just overcomplicates it. It is a squat; move the bar in a straight line in the right way — that is what you need to learn how to do. It isn't that hard.

What are your thoughts on training until failure on the competition lifts?

Every week I had a plan. I would post up on the fridge what I was supposed to lift that week. Let's say it is 800x5. That is my mission, that is what I am doing. When I do that, I come home, circle it, and then the next date and rep goal goes up on the sheet. Once I had that goal, I would do everything I could to meet it — if I failed going for 800x5 so be it, although there was hell to pay for myself and everyone around me if I didn't hit that goal. I was not afraid of failure, but if I hit that goal rep I could stop; I wouldn't go for extra reps just because I was successful with the set of 5. But once that weight and reps were on the plan, you just do it, no question.

What injuries have you faced, and how did you overcome them?

When I was competing I had lots of minor injuries — strains, tears, etc. You just deal with them, work around them, etc. I never had any surgery. Since retiring I have gotten a bit banged up from stupid shit; my bicep pulled off the bone flipping a tire, I had surgery on my knee because it got infected, and now I am just trying to put it together so I can still play and throw around some weights when I want to.

How important do you feel that nutrition is to powerlifting performance?

After 1993 I learned how important nutrition was. Before that I would eat anything just to put on weight. I would drink 4 gallons of milk a day. I would spread peanut butter on cheese and wrap it up like a burrito, anything to get the calories in. Then I had a bodybuilder friend of mine ask if he could tighten me up so I did it. I felt a lot better, so much stronger because I had so much more muscle after a year or so. I thought about doing a bodybuilding show but the one I was interested in was too close to Nationals and I am not into beauty pageants anyway. I wanted to go set some World Records.

Initially I think he had me on 1.2 gr of protein per pound, 2 gr of carbs per pound and .25 gr of fat per pound, each day. Gradually we moved up to 2 gr of protein per pound. There was a period when I was pretty ripped for being 250–275 lbs.

What do you usually do with your body weight and nutrition to prepare for a powerlifting competition (drop 10 lbs, stay the same, consume high carbs, etc.)?

I would usually walk around 10–15 lbs heavier than I competed (290 for the 275 lb weight class). Then I would take out a few meals each day for a week and a half before the meet and that was all I had to do to make weight.

Once I was 256 the day before I was trying to make 242 and that was tougher, I had to flush all that out with tons and tons of water, but I made it just barely.

How important do you feel that supplementation is to powerlifting performance?

I am not a supplement guy really. I think the vitamin packs are good. I used some Met-Rx stuff. The supps are tough to trust; you don't know what you are getting — at least if you use the same thing all the time then you have a standard. Honestly I think if you just eat 2 cans of tuna a day you will be fine.

These legs were made for squattin'
Photo credit: Kirk Karwoski

What are your thoughts on powerlifting equipment (gear) in powerlifting? How do you incorporate gear into your training? Are you sponsored by any equipment manufacturers and if so, who are they?

I was sponsored by Titan back in the day and they treated me well, but I am not a fan of the gear. Get rid of it and go raw — keep the belt and wrist wraps, knee wraps maybe but I don't care about that. I could go either way with those.

How do you feel about the effectiveness of drug tests for catching those who use steroids?

I think drug testing is a good thing; it helps create a level playing field. If all of the athletes have to meet a certain standard and they know what that standard is, it helps keep everything the same, as long as those standards are being applied to everyone. I knew all the protocols when it came to drug testing. I could teach a class on that. It is important that the testers follow those standards. I also think it helps build credibility and it is necessary if we want to get powerlifting into the Olympics.

What do you think the key to unification of powerlifting is?

Unification won't happen unfortunately. There are too many petty, meathead, test-laden, ego-trippin' people running things to allow that happen.

Powerlifting really isn't a good spectator sport. The squat racks and the bench press are very intrusive; they get in the way of the viewer. You have all these spotters around; the audience is wondering what is happening. If we could fix all of that and go raw then I think it could possibly get into the Olympics, which it should be. It deserves that recognition. One of my biggest regrets is that I don't have an Olympic Gold Medal to show off — plus then I could do an underwear commercial saying "I squatted 1000 lbs in Hanes briefs" and I wouldn't have to work and I could hang out and drink beer all the time and that would be pretty cool, too.

What is your thought about the importance of having a workout partners) or teammates in helping you train for powerlifting?

Training partners are very important; being around the right group of people means everything. Back in the early ’90s at the MAC (Maryland Athletic Club), man we had an awesome environment. You get those guys together — I don’t want to get all sappy, but most will say that was the best time of their lives.

What is your thought about keeping a training journal while powerlifting?

I had a journal, I also had a sheet of paper on the fridge. I do think it is important to write stuff down and to make sure you know what you are lifting that week.

What books, websites, or coaches do you suggest or follow in your lifting, and what would you suggest other lifters do to learn more about lifting?

There weren't any books that I really followed; the Internet wasn't around back then. My coach was Marty Gallagher (author of the *Purposeful Primitive).* Everybody knows Marty; he helped turn me into that athlete who studied the lifts, he helped refine my technique, he was awesome.

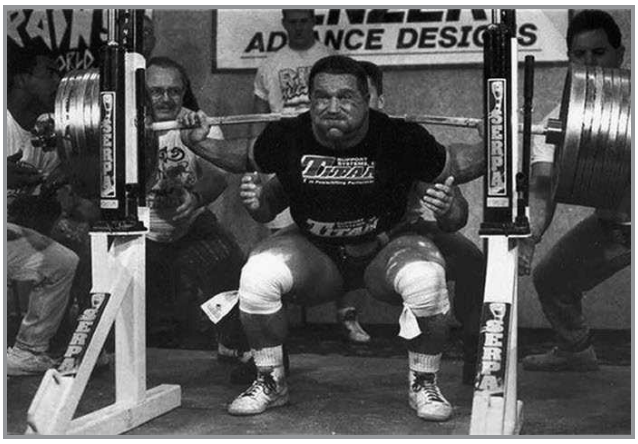
What do you attribute your personal success in powerlifting to?

Being too stupid to know when to quit, but seriously being determined. I just knew, I mean I knew in my core, powerlifting was my thing. I love it.

What do you feel is crucial to being successful in powerlifting, both in and out of the gym?

Consistency. Powerlifting is demanding both mentally and physically. After Worlds I would usually take a full month off just so I didn't have to think about it. It was exhausting getting ready for Nationals and then Worlds a few months later.

And sometimes having bigger balls than brains helps, too.



Coaches can be crucial to powerlifting success

Photo credit: Kirk Karwoski

What advice would you give to someone who was just beginning to take up powerlifting?

Get strong; you don’t need any gear or equipment at this stage. I think that really helped me in the long run. I think a 3 x bodyweight squat and deadlift and a 2 x bodyweight bench is a good standard before you use anything, I did all of that with a $10/belt and that was it.

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

Videotape and watch your lifts, work on consistency — that is the key. Stick with the basics; don’t overcomplicate it.

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

Basic tweaks are still very helpful at this stage. Find a coach; you may have to pay them for their knowledge, but if they up your squat by 70 lbs I would say that is worth something.

Are there any changes that you would make to powerlifting if you had the power to do so?

Gear is wrecking the sport but on the other hand it is also the only thing bringing monetary value to the sport, I am not sure what I would do. Really it is the sloppy judging and bullshit lifts that bother me the most. I see all of these ¾ squats posted, hundreds of people congratulating the lifter. I can barely get on Facebook anymore without getting super pissed. Last time I checked a squat was when your hip had to be below your knee. What is so hard about that? Walking the bar out is crucial and now that is gone with the monolift. I could teach a full-day seminar on just how to walk it out; with the big weights, that is super important. If you can set it up, you ride it.

I don't really know the answer. I do know that powerlifting should be in the Olympics and I would support whatever will make that happen.

Set up is crucial for a big squat
Photo credit: Kirk Karwoski

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

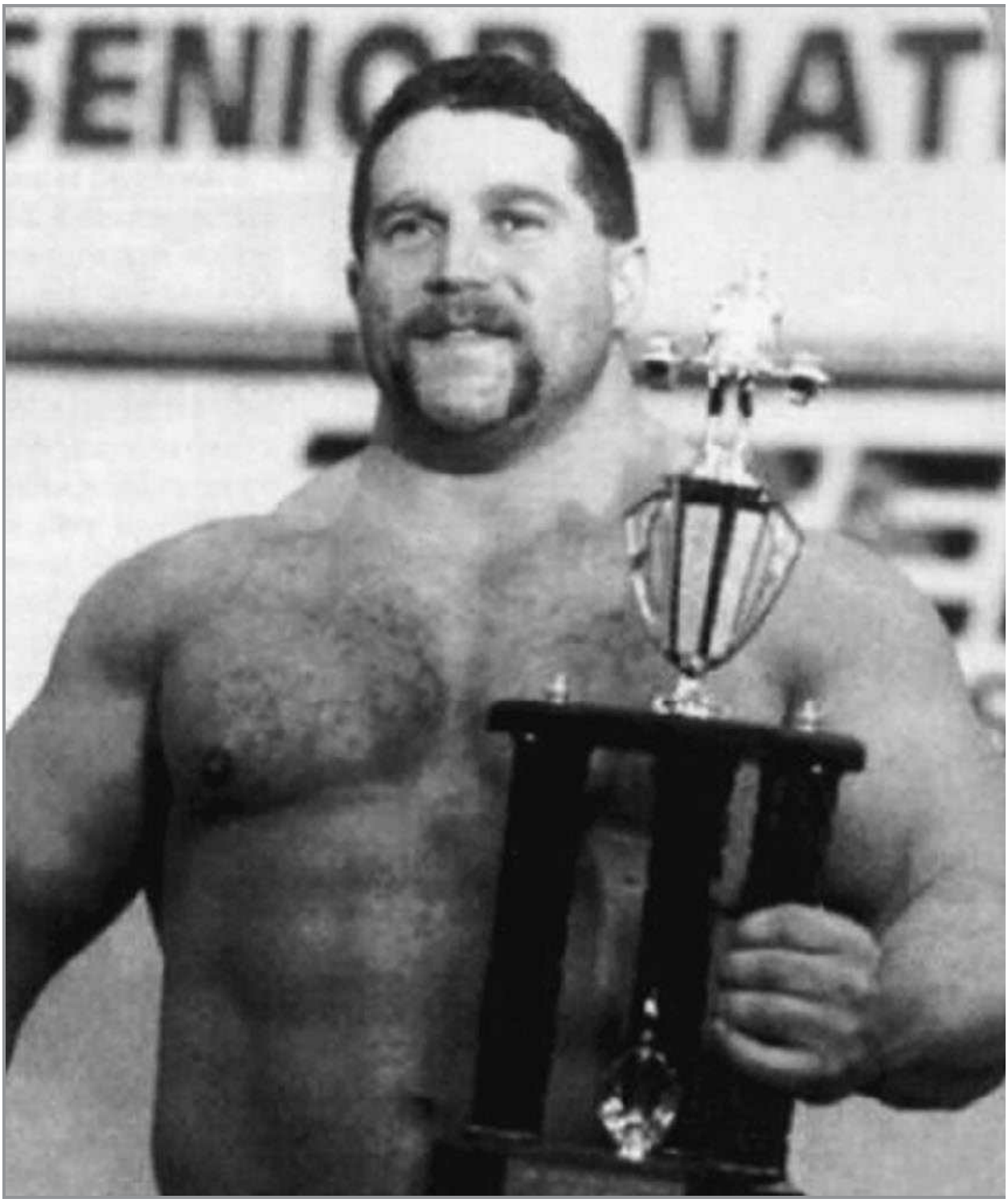
If you embrace it and love it, powerlifting will beat the shit out of you, but it will treat you well.

My favorite quote which I think helps put the sport in perspective is this:

> "Just don't fuck it up for 20 seconds."

Can you do that? If you can do that, you can be successful in powerlifting.

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.



Hard work will pay off
Photo credit: Kirk Karwoski

Chapter 4

Increasing the Squat

The name of the game in powerlifting is to lift the most weight possible.

Be prepared to spend hundreds if not thousands of hours pursuing that endeavor if you want to become truly strong; there is no way around it. The goal of this chapter is to help you improve your squat.

Jennifer Thompson sinks a 402 lb squat nice and deep
Photo credit: Jennifer Thompson

When looking at how to improve a lift, I find it most useful to examine why a lifter fails at a certain weight. All lifters have their maximums, be it a 185 lb squat for a newbie or a 675 lb squat for a total stud. In either scenario, when you add another 10 lbs, the lifter fails. Why? Finding out the answer to that question, I believe, then helps dictate the proper course of action to fix that problem. Lifters fail for a number of reasons, simply saying the person isn't strong enough is usually not enough. I am going to list some of the possible reasons for failures and then provide

> ***Be prepared to spend hundreds if not thousands of hours if you want to become truly strong***

ways to fix those problems. At the end of the chapter I am going to provide some sample routines that I have used that have helped improve the squat.

Technique is not static

Technique

By now you should know the proper form on a squat. If you don't, please refer back to Chapter 3 for detailed instructions. However, since technique is so important and all of the other methods don't really matter much if your technique is faulty, here is a brief overview of the key points.

- Proper height of the bar in the rack — just below shoulders
- Proper positioning on the bar on your back — generally low-bar position, centered
- Proper positioning of the hands on the bar — in as close as possible, open grip for most lifters
- Stand up strong and confident with the bar
- Proper walkout — two to four steps to get your position, do the same thing every time
- Stand tall, chest up, legs locked, elbows tight and down, neck rigid
- Screw feet into the ground
- Fix eyes on an unmoving point in front of you
- Push hips back and unlock knees
- Controlled drop into the descent, keeping knees out and chest up
- Hit proper depth
- Drive up with the hips
- Keep weight on the back of the heel
- Keep feet flat at all times
- Keep knees out on the upward phase
- Keep chest up and elbows down and core tight on the way up
- Drive through the ground with your legs to straighten your legs
- Stand tall at the end of a completed rep

If you follow those key points your technique should be solid in the

squat. Technique is not static; even once you have it down there are going to be periods of time when you are more or less confident. Filming yourself regularly on all the lifts is important, but it is probably the most important in the squat. Use different angles to get a clearer picture of your form. When performing multiple reps the form may not be absolutely perfect on every single rep, but it should be pretty close; remember, you will play like you practice — if you practice with shitty form expect that to come out on the platform. I always make it a goal of mine to make the last rep perfect. If you do that you will always be happy with your form.

Neuromuscular Coordination Techniques

To help ingrain the technique listed above, there are certain methods and procedures lifters use to practice the proper squat. First, all reps from the warm-ups to the heavy set should look the same and they should look good. Second, many lifters find that box squats are a useful tool to learn and/or improve their form on the squat.

The box squat offers several benefits. One main benefit is that when squatting to a box it helps lifters sit back more instead of sitting down. This activates the posterior chain, and that area is crucial in getting a barbending squat. The second benefit is that the box can be adjusted to the desired height and it can teach the lifter what it feels like to be in the correct depth (most lifters will find that a box of 10–15″ is the proper height for competition depth squats). A third benefit of a box is that lifters must develop explosive force from the bottom because you don't bounce off of the box; instead you sit on it and pause for a moment, and then you stand up. Finally the form right as a lifter is coming off the box is usually ideal form, and it can reinforce what that ideal form is for that lifter.

Make the last rep perfect.

Box squats do have their negatives. This includes the fact that there is no box in a competition and lifters can become reliant on the box for proper depth or because the box "catches" them at the bottom. Normal box squats involve relaxing a bit and pausing at the bottom, but this does not take place in a barbell squat. The barbell squat involves the stretch reflex, which can be trained, and the box squat removes that same reflex. Finally if a lifter sits too hard on the box, or possibly falls because of the box, then injury can occur. One should never try to bounce a heavy weight off of the box as that is a recipe for a disc injury in the lower back.

Many lifters find that having a speed, power, or technique day is useful for the squat. The squat is a high skill exercise, and it is nice to be able to practice that technique with non-maximal weights. Call this day whatever you want; it involves using lighter weights (40–80%), and the lifter performs the squat, but the training is not so intense that form breaks down. Think of this as a practice day; it is a day to practice improving your squatting skill. You can also experiment a bit with foot position and stance to see how things feel on this day, but keep in mind sometimes things feel good with light weight and not so good with heavy weight.

Training with bands and/or chains can help improve, or at least alter, the neuromuscular coordination involved in the lift. The theory is that as the bands are stretched or the chains are lifted up off the floor, the weight on the bar increases and so the lifter must continually accelerate the bar to complete the lift. In essence the strength curve of the lift has changed. The goal is to teach the lifter to accelerate through the sticking point once you take the bands or chains off the bar.

You get stronger only in the ROM you train in.

Principle of Specificity

The principle of specificity certainly applies to the squat, as with all exercises. If you want the lift you are doing to improve the squat, it has to train some specific part of that lift. Exercises that most closely match the squat will include a barbell traveling in a similar bar path. They may match the ROM on the squat or they may just include part of it. Remember, the problem with partials is that you get stronger only in the ROM you train in — if you spend all of your time performing half squats then you will get stronger in half squats, but it will not improve your full squat very much.

Exercises and their benefits

Full Squat – this is a competition-style powerlifting squat, and it should make up the bulk of your training.

ATG Squat – ass to grass squat, which is as low as you can go, emphasizes the glutes and builds power out of the hole, don't change form to get super low. Requires good flexibility.

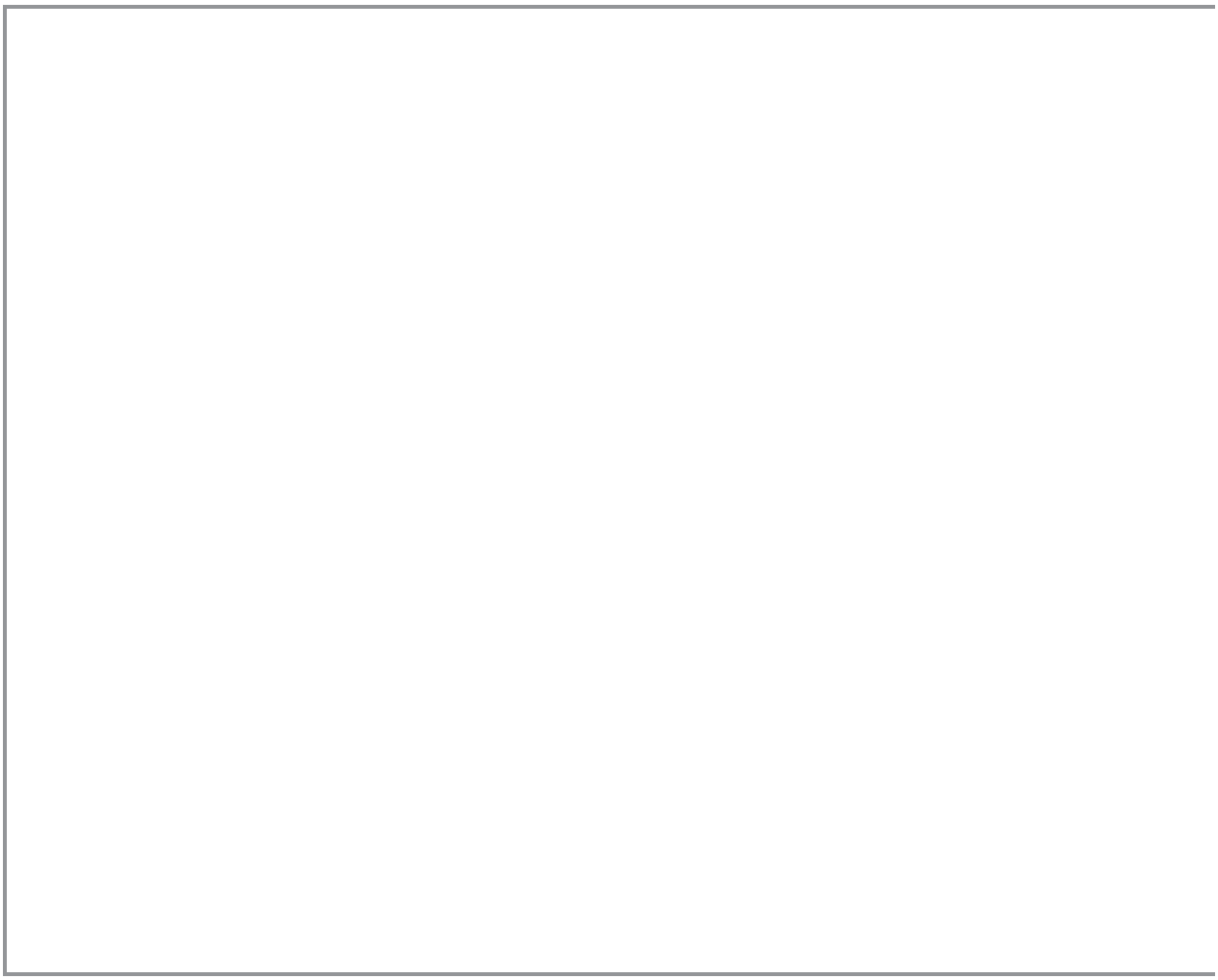
¾ Squat – a squat ¾ of the way down, which is about a bottom of the leg parallel to the ground squat. Helps the lifter get used to heavier weight but can teach the lifter to cut depth.

½ Squat – a squat ½ of the way down; this is what most people in the gym do and say they are squatting. Much easier, one can often lift 20–50% more with this style; it emphasizes mainly the quads, but there is not much transfer over to a full squat without also performing a lot of full squats.

¼ Squat – a squat ¼ of the way down. This is good to work with heavy weights to practice walkouts and build confidence, but there is little transfer over to full squats without also performing full squats.

Box Squat – as explained previously, sit on a sturdy box or bench and then stand back up. An aerobics step with risers works well if the gym doesn't have a box. Most boxes are adjustable; the lower the box, the harder it is, and the higher, the easier.

Front Squat – place the bar in front of your neck (a regular squat is sometimes called a back squat because the bar is on your back), on your shoulders near your clavicle. Hold the bar in rack position or with arms crossed. Perform a full squat. Significantly harder than a back squat; forces the lifter to stay more upright. Works more quads and less glutes; can help the lifter to stay upright during the squat. Maybe an uncomfortable position for those with inflexible wrists, elbows, and shoulders. Generally lift about 10–30% less than a back squat depending on how much you practice front squats.



A front squat held in the rack position

Safety Squat – Squats using a safety squat bar. This bar allows you to sit back more; you can either hold on to the bar or hold on to the rack; holding onto the rack makes it much easier. May be useful if the lifter has bad shoulders or knees as the pressure is placed differently on the body.

Cambered Bar Squat – Squats with the cambered (very bent) bar; changes the center of gravity; easier on the shoulders.

Squat w/bands – Regular squats (free or on a box) with bands attached to force the lifter to accelerate through a sticking point. Might be useful if the

lifter is slow or has a high sticking point.

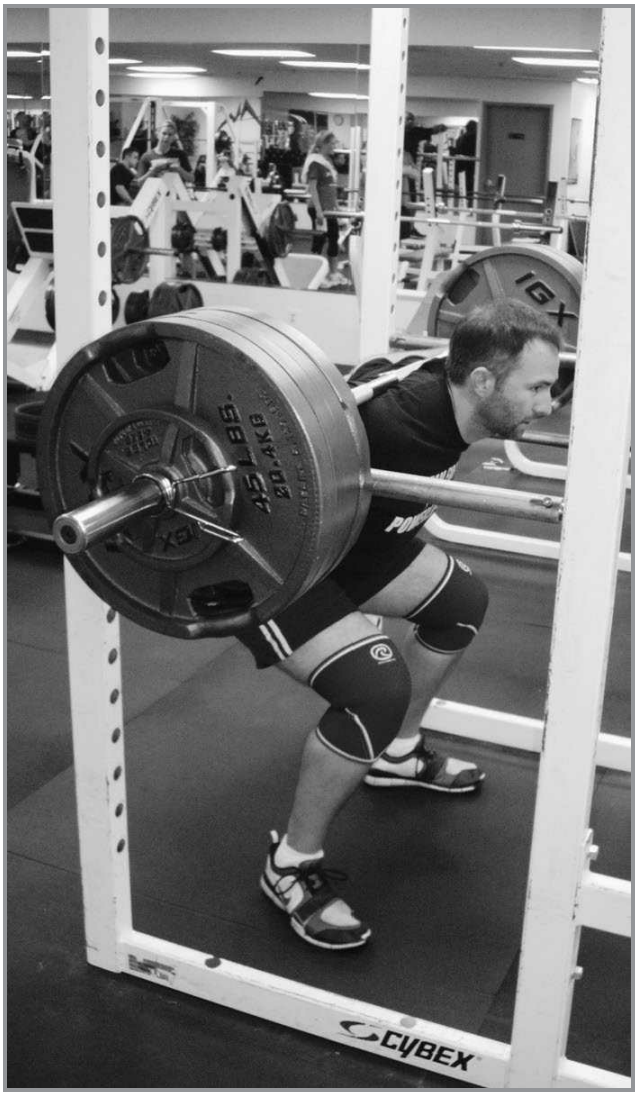
Reverse Squat w/bands – Squats with bands attached to the top of the rack, usually from the pull-up bar, that actually pull the lifter up and assist the lifter. This is easier than a normal squat. Most of the help is given at the bottom of the ROM with little help at the top. Useful for getting the lifter familiar with more weight; not useful if the lifter is weak at the bottom.

Squat w/chains – Regular squats (free or on a box) with chains attached, same principle as bands but generally the chains add weight more gradually.

Wide Stance – Squats with a wider than normal stance. This places more emphasis on the adductors, inner quads, and glutes if you go low. Use caution with a significantly wider than normal stance and heavy weight; build up to it. Wide squats place more stress on the hip joint and less on the back as the lifter can stay more vertical; the form becomes similar to a sumo deadlift. A very wide stance usually feels powerful at the top and weak at the bottom. Often it will be difficult for a lifter to achieve proper full squat depth with a very wide stance. There is little evidence to show that very wide stance squats have a positive transfer to other sporting activities.

Narrow Stance – A stance more narrow than normal, usually about shoulder width apart or slightly more narrow. Difficult to do without decent flexibility; don't let the heels come up when you do this. This places more emphasis on the quads, particularly the outer quad and the abductors. Foot position often mimics jumping position. Might benefit leg drive in a deadlift since the stance is similar.

Rack Squat – A squat performed from the bottom up starting in a power rack, sometimes called a concentric squat. The lifter places the pins at the



A rack squat, sometimes referred to as a concentric squat

desired height, sets the bar there, gets under the bar and then stands up with it. Builds starting strength. Make sure form mimics a real squat and not a good morning or something similar. Challenging to do with deep squats.

High Bar Squat – The bar is placed high on the back, usually near the base of the spine. Sometimes referred to as an Olympic squat. It is often more comfortable to learn this form as the lifter can stay more upright. Uses more quads and less glutes. Disadvantage is that the amount of weight lifted is almost always lower than a low-bar squat so it is not favored among powerlifters. When combined with ATG form it can help maintain flexibility.

Low Bar Squat – The bar is placed low on the back, usually 2–4″ below the base of the neck. Usually feels more comfortable once someone is used to it; there is more tissue to spread the weight of the bar over, and this reduces the moment arm of the resistance thus increasing leverage. This places more load on the lower back and glutes as it forces the lifter to lean forward more. This is the normal position used when powerlifting although it is not mandatory.

There are no mirrors in a competition.

Depth

One of the major problems that lifters have in competitions is not squatting to the proper depth. I think this is due to a few key reasons. The first reason is that in the squat, unlike the bench press and the deadlift, there is not a crystal clear bottom position. The bottom position is an ass to the grass squat, but if a lifter performs that in a competition they will limit the weight they can lift. Squatting a few inches lower than necessary will often cost 10–20% or more of the weight lifted, and that is just too big of a decrease for competitive athletes hoping to beat their competition. The bench press has a clear transition point — touch your chest — and on the deadlift you either touch the bar to the floor or not. But with the squat it is tricky to know how low to go.

Lifters often solve this problem in the gym by relying on what they look like in the mirror to know if they are going deep enough. The problem with this method is that there are no mirrors in a competition; indeed relying on a mirror and then lifting without it can actually make the lift more challenging. I actually suggest you cover the mirror up with pads, mats, a sheet, whatever so you get used to *feeling* a proper squat instead of *seeing* a proper squat.

To assist in the feeling of a squat at proper depth, you can place a band or tube strung across the power rack at proper squat depth. When the lifter hits the correct depth, his butt or upper hamstrings will touch the band, and he will know that his depth is correct. You can use this in training for a while to teach and/or reinforce what proper depth is; you either touched the band or you did not. Of course, the band must be set at the proper height or the whole exercise is pointless. The point is not to try to use the band to rebound at the bottom; it doesn’t need to be very tight, and you do have to make sure lifters are not tripping over the band when they perform their walkouts. You can’t use this system during a meet, but I have found it is helpful in getting lifters

to squat the right depth, and then once they have the technique down you can remove the band.

Hitting the tube signifies proper depth has been reached

Another way of solving the depth issue is to have a partner call out to you when your depth is correct. This method works but it does have some potential flaws. First, your partner had better know what correct squat depth is. Second, the timing of the call is important. With light weights the lifter is almost immediately able to reverse the direction of the weight, but with heavy weights it takes longer. If the call comes after you have already hit depth you will be sinking too deep and most likely wasting valuable energy, but if it comes too soon then the lift will not count. The caller can sometimes get overexcited (like during a meet) and call it too soon as well. This method can work but it can also cost you some lifts. A good squatter should be able to feel proper depth and should not need an external measure to know if it was acceptable or not.

Filming your squats is a great way to monitor your depth. Set up the camera at the proper depth when you are at the bottom (usually just putting the camera on a bench works fine), film mainly from the side and just film your sets. Most digital cameras these days have film features that allow at least a few minutes of film and that is usually all you need. Filming is great for many reasons. You can watch your form and look for flaws; I think seeing yourself

Simply train with proper depth.

lift heavy is a good confidence builder, and it is a way of recording hard work. Knowing you are on film tends to motivate the lifter to try harder, and it is a great way of truly charting your progress. And as you get older you will enjoy having the film record so you can show family, friends, or yourself some of your more truly impressive lifts.

My last suggestion for learning proper depth is simply to train with proper depth. It is human nature when the weight gets heavy and when one is tired to cheat a bit. That is why you need trained partners (not just cheerleaders) and cameras to keep you honest. Most regular people in the gym, even those who lift hard, usually can't tell the difference between a competition squat and one that was 2″ too high; it is still way lower than everybody else goes. And it is much easier to congratulate the person after a tough set than to say, "Sorry, only 3 out of those 8 reps counted because the rest were high" but if your partners don't do it, the judges will. Go to proper depth on the warm-ups, go to proper depth on the work sets, go to proper depth on the max outs. Remember, make the last rep perfect and you will be happy with the results.

(Left) This squat is NOT deep enough to count in a competition

(Middle) This squat is iffy; it might get reds, might get whites

(Right) This squat is deep enough to count in a competition
Photo credit: Vadim Snitkovsky

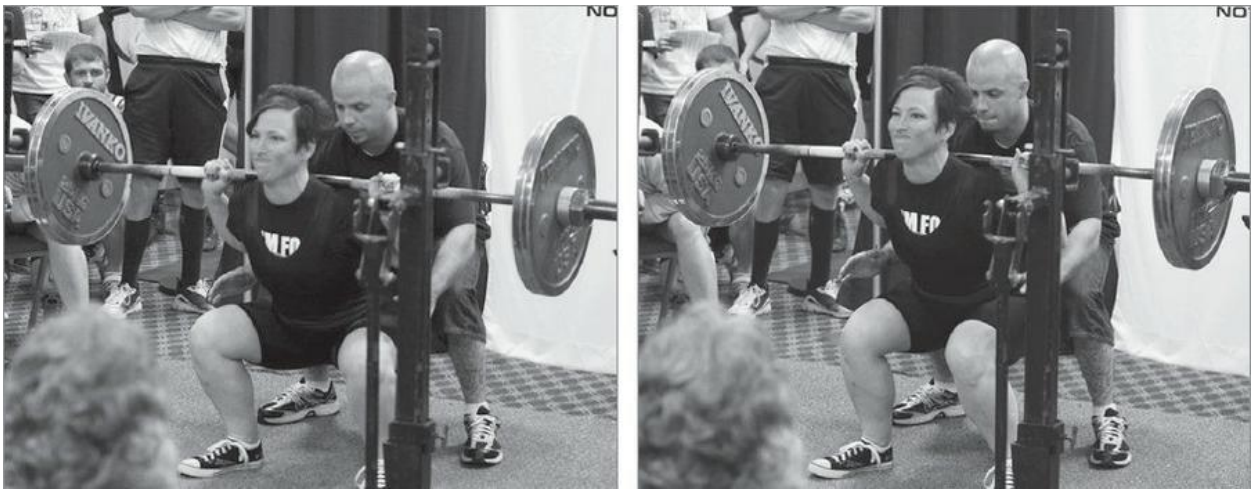
Joint Stability

If a joint is unstable, due either to being put in an unstable position or to an injury, sensors in and around that joint will regulate the amount of muscle force that crosses the joint, and they will limit the force to decrease the chance of injury. This is why squatting on a balance board or a BOSU ball is harder than squatting on the ground. The muscles haven't changed, and the ROM is the same, but because the joints don't feel stable they limit the amount of the force the muscles can generate, and squatting 135 on a balance board feels a lot harder than squatting 135 standing on the ground.

The squat is by nature a moderately stable exercise, but it does place significant load on a large number of joints in the body. If any one of those joints is injured (and it doesn't have to a big, obvious injury — it could be a smaller, internal injury) then that joint can limit the weight lifted on a squat. The 3 main joints that can limit a squat are the ankle, the knee, and the hip/lower back complex.

It is important that the foot remain completely flat on the ground when squatting, and the ankle should be relatively stable. The foot should not turn in or out or roll in or out at all during the range of motion. If it does adjusting the foot position and stance can help, lifting in more solid shoes can help, wearing orthotics can help, and simply making the lifter aware of this can help so they can try to actively correct the problem (again filming can be useful). Usually the foot doesn't hurt when squatting but if it is out of alignment it will force the other joints to compensate and that can cause pain at those joints. The ankle should be stable but also flexible enough to allow the shin to move forward as necessary while keeping the heel flat. Tight Achilles Tendons make this challenging; keeping the calves flexible is important, especially if you perform a lot of calf raises. Regular stretching and foam rolling can help this area. Massage of the bottom of the feet

(accomplished by rolling the foot on a tennis ball) can also help keep that area flexible. If the ankles roll in (evert) while squatting, this has a strong tendency to cause the knees to move toward each other. This will make squatting big weights both tougher to do and more dangerous and thus should be avoided. Lifters can think of driving the knees out and keeping the weight on the outside/back of their heels to help prevent this form fault.



(Left) The lifter descends with proper form and knees are out

(Right) The weight shifts to the inner foot and the knees collapse during the ascent

Photo credit: Vadim Snitkovsky

The knee is the joint that most often gives lifters problems during the squat. The knee is the largest and most complex in the body and I do not claim to know everything there is to know about knee problems. Knee injuries are usually either compression based (it hurts when additional weight compresses the knee joint, like standing with weight on your back) or shearing (it hurts when the femur pulls on the tibia, as in a leg extension or the bottom of a squat). Most lifting-related problems are shearing in nature. Knee problems are also either focused around the front of the knee (above or below the patella) and associated with the quads and knee extension, or they

can be focused around the back of the knee and associated with the hamstrings and knee flexion. For lifters the most common problems are with the front of the knee.

Knee issues can arise for a bunch of reasons: Poor technique, flat feet, immobile ankles, weak/tight hip flexors, tight quads, weak ab/adductors, poor quad/hamstring strength ratio, weak Rectus Femoris, an imbalance in the quads themselves, overuse, bow legs, and the genetic structure of the joint itself. Some of those issues are correctable with training, some with mechanical assistance (like orthodics for flat feet), and others are unchangeable. Listed below are some suggestions for helping with certain issues.

Immobile ankles – Stretch Achilles Tendons regularly, foam roll the area, perform ankle mobility drills.

Tight quads – Stretch the quads regularly, particularly the Rectus Femoris through both hip extension and knee flexion at the same time; foam roll the area regularly; heel should be able to touch your butt when you are stretching it.

Weak ab/adductors – Train the adductors through use of wide stance squats and leg press, sumo deadlifts, ball squeezes, and the adductor machine. Train the abductors through the use of narrow stance squats and leg presses, single leg squats and leg presses, step downs, fire hydrants, penguin walks, lying leg raises, and the abductor machine. Most males have weak abductors.

Weak/tight hip flexors –Stretch the hip flexor regularly, trying to activate the glutes at the same time. Strengthen the hip flexors through sit-ups, L-holds, banded mountain climbers, and whatever else seems to make you feel that area working.

Poor quad/hamstring ratio – Research is unclear as to the exact best ratio of the quads to hamstrings. Most suggest the hamstrings be at least ⅔ as strong as the quads, I would suggest for powerlifters the hamstrings should be .8–1.0 of the strength of the quads. It is important to measure this on similar machines to get an accurate reading. Your leg press should be about twice your squat; if it is not, most likely your quads are relatively weak. You should be able to leg curl your bodyweight for reps or perform RDLs with double bodyweight; if not, most likely your hamstrings are weak.

Weak Rectus Femoris – The Rectus Femoris muscle is one of the 4 quads but it is the only two joint muscle of the quads. It works to extend the knee and also flex the hip/trunk. You can place additional emphasis on this muscle with the following exercises: decline sit-ups, inverted sit-ups, lying leg extensions (often hard to find), L-holds, hanging leg raises. If you have a hard time holding your legs out straight in an L hold for 30 seconds then your RF is most likely weak (it may be tight as well; see above).

Imbalance in the Quads – The quads pull on the kneecap (patella), which in turn pulls on the tibia and makes the leg move. If the quads are not in reasonable balance then the patella will not slide evenly and this can cause problems over time. The two main quads to focus on are Vastus Medialis (the teardrop quad on the inside upper part of the knee) and the Vastus Lateralis (the sweep of the quad on the outer part of the thigh). To emphasize Medialis you want to take a wider stance, with your toes pointed out, and focus more on the top half of the ROM on leg exercises. To emphasize Lateralis you want to take a more narrow stance with toes straight ahead and focus more on the bottom half of the ROM on leg exercises. Stereotypically females have a weak Medialis. Males can have either; however, powerlifting puts a lot of focus on the Medialis with the tendency to train with wider stance squats, leg

presses, and sumo deadlifts. You can use a visual assessment to judge the size difference of the quads, but you will probably need a physical therapist to make an accurate assessment of any possible imbalance.

Keep the cardio low impact.

Overuse – It is easy to overtrain the knee, either with weights, with high impact power exercises like jumping and sprinting, with a lot of cardio, or a combo of any of the above. The bigger you are in terms of weight, the more likely this becomes. First, get your proper squatting frequency down (see following chart) and if you are not sure what that is, err on the low side if your knees are bothering you. Since we are focused on powerlifting, if your knees are bothering you eliminate all extra stuff and just complete the main leg exercises (squats, deadlifts, and maybe one other thing you know doesn't hurt your knee). Keep the cardio low impact, like walking and the elliptical, drop any high intensity stuff, and see how the knee feels. If it is feeling good then gradually introduce new exercises and up the workload. If you find something bothers it, then eliminate that exercise. If you can only do a few exercises pain free or if the minimum level bothers you, go see a physical therapist who is used to dealing with athletes or someone trained in Active Release Therapy and get some help; it is not going to go away on its own.

The hip and lower back complex are also involved in a squat. The hips are more reliant on bones and ligaments for stability than the shoulder; if you are having an issue in the hip it will most likely require some assistance to fix it. You could be having an issue with your adductors, abductors, hip flexors, erectors, QL, Multifidus, RF, glutes, even your core, or it could be structural. Remember that wide stance squatting really puts a beating on the hips, as do sumo deads; if you do a lot of those you might try bringing in your stance and see if that relieves the pressure on the hips.

Joint instability can also be signaled by some other signs and symptoms, one of which is poor eccentric strength. When you are squatting, if you feel like you could not reverse the squat at any time on the way down, then it is likely you have some joint instability somewhere. If your eccentric strength is not significantly higher than your concentric strength, that can signal a problem. If you need a tremendous amount of time to warm-up, or if the warm-up weights feel worse, harder, or slower than the regular weights, that could indicate an instability. My college workout partner and I used to keep track of what we called cold maxes, which indicated how much weight we thought we could lift if we just walked up and did it with no warm-up of any sort. We didn't actually test this by doing it, but we just thought about what we would be comfortable doing. I know personally my cold maxes were always pretty high on most lifts like bench and deadlifts but low on squats. Over time I developed the most issues with my knees and the most issues with squats, and I think there is a correlation there. I know for me it got to a point where the idea of squatting 135 without a 20 minute warm-up was intimidating, and that, of course, is not a good sign. It is time, actually past time, to get help from a medical/health professional if you are at that stage on any of your lifts.

You have to try to fix the problem soon.

Joint instability sucks for several reasons. First, it makes you weak. Second, you can't train the surrounding muscles the way you want to. Third, it will most likely show up as a significant injury if you do intense stuff outside the gym like play football or basketball or whatever. Fourth, you will most likely develop compensations to work around the injury. This works in the short term, but over time usually problems develop in the surrounding areas. If your knees hurt you might be able to do a good morning squat and that is

okay for a while, but then most likely something else will hurt. You have to try to fix the problem soon; don’t deal with it for years and then try to fix it when it is so bad you can barely lift at all because then it might be too late.

Remember that a joint will shut off the muscular power available to the muscles that cross that joint if that joint hurts. Not only is it possible to cause further damage to the area by training through joint pain, it simply isn’t very effective because the muscles are not working like they would during a regular workout anyway.

Bodyweight

One's bodyweight will have a powerful effect on the squat. Generally as a person gets larger their absolute strength increases. This is most likely due to several factors. First, as we gain weight some of that weight is usually muscle, and that additional muscle contributes to the lift. Sometimes we can use our bodyweight as part of the lift or exercise so the extra bodyweight can used for leverage. And as the previous section discussed, joint stability is a very significant determinant that affects strength. As one gets larger there is more tissue and other "stuff' surrounding the joint, which in essence acts like a wrap and the joint is more stable. The more stable the joint, the more muscular force that it allows to act on it. This is one reason why powerlifting gear is so effective. Not only does the gear provide an actual spring or rebound effect which moves some weight, the gear actually adds considerably to the joint stability so it allows the lifter's muscles to fire even stronger and thus more weight can be lifted.

The squat is heavily affected by bodyweight, second only to the bench press. Usually relative strength will decrease as bodyweight goes up, but most lifters can expect a 1–3 lb increase for every pound of bodyweight gained. If I weigh 200 lbs and I can squat 400 lbs (relative strength of 2.0), if I go up to 220 lbs most likely I will now be squatting somewhere between 420 and 440 lbs. Whether that move makes you more competitive in a competition is a different story, but the bottom line is you will almost always squat more as your weight goes up.

The squat places tremendous load on many muscles in your body.

Muscles Involved

The squat places tremendous load on many muscles in your body, particularly in your legs. Listed below are the muscles working a squat with their relative contribution on a scale from 1–5, with 5 being the most involved, and then some effective exercises in strengthening those muscles.

Muscle	Contribution	Exercises to Strengthen the Muscle
Glute Maximus	5 – Hip Extension	Deep Squats (any), Deep Leg Press, Hip Thrusts, High Step-ups, Sumo Deads, Jumps, Sprints
Glute Med/Min	3 – Hip Stabilization	Fire Hydrants, Penguin Walks, Lying Leg Raises, Single Leg Work, Step Downs, Martial Arts Training (kicks)
Quads	5 – Knee Extension	Leg Press, Squats, Partial Squats, Smith Machine Squats, Hack Squats, Front Squats, Leg Extension
Hamstrings	4 – Hip/Trunk Extension	RDLs, Stiff DL, Glute/Ham Raise, Deadlifts, Good Mornings, Leg Curl
Adductors (magnus)	4 – Hip Extension/ Stabilization	Sumo Deads, Wide/Deep Squats, Wide/Deep Leg Press, Adductor Machine
Erectors	3 – Trunk Extension	Deads (any version), Good Mornings, Hypers, Reverse Hypers
Core (abs, obliques)	2 – Trunk Stabilization	Inverted Sit-ups, L-Holds, Cable Crunch, Ab Wheel, Hanging Leg Raise, Landmines, Rotations

Gastroc	1 – Knee Stabilization	Standing Calf Raise, Leg Press Calf Raise, Donkey Calf Raise
Hip Flexors	2 – Hip/Trunk Stabilization	Decline Sit-ups, Hanging Knee Raises

Jennifer Thompson gets mentally focused for a big squat
Photo credit: Jennifer Thompson

Mental Attitude

The squat is a brutal exercise, no doubt about it. When you perform a tough squat workout or head into a competition, you must be mentally ready for it. Once you decide that you are going to complete that workout or that set, you pretty much need to have a do-or-die attitude. The squat is a relatively high-skill exercise. You need to have some control of your rage, otherwise your energy will not be channeled appropriately. At the same time it is not a super high-skill movement such as a dart throw or a golf swing so it is important to be pretty fired up to perform well. The squat is exhausting for the body; it will usually leave you huffing and puffing and maybe even seeing stars. Training the lower body is definitely more demanding than training the upper body. Sometimes, people even get sick and throw up after or during a squat workout; however, this is certainly not imperative or even desirable on a regular basis. Sometimes you do need to push the limits of your physical capacity so you know where they are and how hard you are really training, but at the same time you do not need to reach or exceed those limits on a regular basis.

> ***Most people undertrain.***

Optimal Frequency of Training for the Squat

With squats, as with all lifts, you need to put in the right amount of effort to get optimal results. Most people undertrain, meaning they don't train hard enough, especially with legs, which can take a tremendous beating. It is possible to overtrain, however; the most concrete way of knowing that is happening is by training hard and having your lifts plateau or, even worse, go down. Overtraining also can lead to an injury. Presented below is a chart that details the ranges of optimal frequency of training for squats. Squats are a high skill exercise so they need a reasonable amount of practice. In addition your ability on squats tends to go away relatively quickly if you stop squatting. If you take a month off you can bet your max squat is most likely going to drop.

Something that is very important to note on the chart above is that I use the term “intensity” a little differently than some authors. The intensity % is not just a measure of your 1 rep max; that is important, but if you only look at that variable you are missing a big piece of the puzzle, and that is how many reps were performed at that intensity. If I told you that 500 lbs was my 1RM on the squat and I was squatting 350 (70%), you might think it was a relatively low intensity day. But if I told you that I was squatting 350 for 20 reps straight, suddenly we have a super high intensity day that will be just as demanding and draining as lifting near my 1RM would be. To accurately figure out your total intensity % as indicated by the above chart, you need to know what % of your 1RM you are lifting, you need to know how many reps you are doing, and you need to know your own personal value of the lbs/rep at that weight (see Chapter 16 for more details). Here is the formula written

out:

Total Intensity % = (weight lifted + lbs/rep x extra reps completed)/1RM

Here is the above example fleshed out:

Total Intensity %= (350 lbs + 8 lbs/rep x 19 extra reps)/500

= (350 + 152)/500

= 502/500

Total Intensity % = 100.4% so this was a 100% effort day, meaning most likely the lifter did not have anything left in the tank (I could not have performed 25 reps for example). The 8 lbs/rep value was one I got as an average of that weight from Chapter 16; low reps are more valuable, higher reps are less valuable — for this all you need is a rough average.

Intensity trumps frequency.

The question asked is how often should I squat? A better question is how often should I squat at what intensity? You could squat 100 lbs every day if you wanted to, and it would not be beneficial to squat your max every day, so we must balance frequency with intensity. Generally intensity trumps frequency and it is the most important thing to focus on. The question is "How often can I squat intensely and still recover?" The second question then becomes, "Is that enough practice to keep your neuromuscular coordination at optimal levels?" As you can see from the previous chart the optimal frequency for the squat most likely changes with training age and skill level.

Initially beginners should squat 1–3 x week and they do not need to squat intensely at all; simply practicing the movement and gradually increasing the weight is fine. Then as they get better at squats and they can start to lift heavier weights, that frequency will usually go down, especially if it was on the high end of the range. If a person started squatting 3 times a week, then they should drop down to 2 times a week after a while (probably

about 3 months of training) and then they should most likely drop down to one time a week (probably after about a year of training). The intermediate stage (about 1–3 years of solid hard training) is generally the perfect set up for a lifter to squat one time a week, hard, every week. Most lifters will be able to make consistent, near linear progress in the squat like this. If you can squat more than your bodyweight but not double it yet, then you are most likely an intermediate level and generally those lifters benefit just from solid hard training done week in and week out. Once you get beyond that stage and your squat starts to become significant, then it usually becomes important to start modulating the intensity. Often squatting heavy every week becomes too draining, but at the same time squatting only once every week or less is not enough practice with the exercise. It is usually at this time that lifters start to increase their frequency again, with a majority of their squat days being "medium" intensity (below 90% total intensity) and then with a few heavy days and light days thrown in there for good measure. As the lifter becomes more and more proficient, the heavy days will decrease in frequency, but the overall frequency may climb (assuming the lifter is not suffering from any overuse injuries).

> ***The platform will tell you if you made the right decision or not.***

Specific Training Plans

Outlined and linked below are 3 specific training protocols tailored to the squat. One routine has the lifter squatting once a week, the second has the lifter squatting twice a week, and the third has the lifter squatting 5 times a week. You can decide which one of these best fits your needs or you can, of course, follow another routine, either of your own design or from someone else. In the end the platform will tell you if you made the right decision or not.

1 Time a Week

This is a very solid 6 week program that I and others have gotten quite good results from; it should have you performing your 1RM for a double by the end of the plan. This plan is performed just once a week and it can be used with almost any barbell lift; it seems to work the best with squats, bench, and curls. This type of plan can also be followed in the general training phase if you want to up the intensity a notch.

http://www.t-nation.com/free_online_article/most_recent/a_6week_squat_bench

2 Times a Week

Here is a program that involves squatting 2 times a week to prep for a max squat that is usually good for a 15–40 lb increase on your squat.

http://www.t-nation.com/free_online_article/most_recent/10_weeks_to_squat_p

5 Times a Week

Sometimes to really make progress you have to do something different. This program has you squatting (reasonably heavy) every day you workout, which is

Train legs consistently, train them hard; don't miss sessions and your legs will get strong.

5 days a week in this instance. One of my lifters hit the second biggest squat in the federation, of any weight class (she was 165 lbs), after following this routine.

If you want to try a high frequency plan for a big lift, the key is to start light and gradually increase everything. You do not want to get super sore or beat your body up. This program has you squatting every day (5 days a week), you can use the same principle on the bench press as well.

http://www.t-nation.com/free_online_article/most_recent/40_squat_workouts_in

Summary

The squat will reward good technique combined with consistent, hard training. One major thing that separates powerlifters from regular gym rats is the vast difference in leg strength. Strong legs come at a price, but the good news is they are generally available if one is willing to pay that price. The price is the blood, sweat, vomit, and tears of the lifter. Train legs consistently, at least once a week, and train them hard (pretty much harder than everybody else in the gym); don’t miss sessions and your legs will get strong. 50 hard squat sessions a year will add significant pounds on your total. 50 relatively hard squat sessions for 5 years straight will have you squatting at the end of your flight.

Interview with Wade Hooper

Wade Hooper is a multi-time National and World Champion. To say he was proficient in all the lifts would be an understatement, but he is best known for his amazing squat — *over 4.5x bodyweight in old single ply gear!* This interview was completed over the phone with me asking Wade questions and then writing down his responses.

Squat icon Wade Hooper comes out of the hole with a big squat
Photo credit: Wade Hooper

Provide us with a brief history of yourself.

I am 42 years old. I was born in 1971 in the Ancon Canal Zone near the Panama Canal. I have spent most of my life in Louisiana and Texas. I currently teach advanced mathematics at the high school level, and I have been doing that for 13 years; before that I was a strength coach.

What are your best lifts?

Listed below are my best competition lifts:

At 148 bodyweight – 672 SQ, 429 BP, 573 DL (single ply gear performed in the USPF)

At 165 bodyweight – 777 SQ, 534 BP, 611 DL (single ply gear performed in the USAPL)

At 181 bodyweight – 815 SQ, 567 BP, 600 DL (single ply gear performed in the USAPL)

I would say I am most proud of my 777 Squat at 165; that was a world record at the time and I was in a heated battle with Alexey Sivokon. We were trading records back and forth.

In the gym I did a raw 606 squat and a raw 606 deadlift. I also hit a raw 450 bench, all at one 165 *(author's note: these lifts are very close to the All-Time raw world records for all athletes; if combined they would yield an All-Time raw total, and they would significantly eclipse the All-Time raw drug-tested squat, bench, and total record at 165).*

I was never much of a rep guy so I didn't keep many rep records of my lifts, I puked too many times when I was younger to be fond of that type of work. We used to do the deck of cards workout where you have to lift a certain number of reps based on the card you draw. I think my best was a 315x15 bench press. I didn't usually do high rep squat stuff. I did a lot of volume but that was made up over many sets.

List some of the titles and awards that you have won.

- 2 time Collegiate National champion — USPF
- 2 time Junior National Champion — USPF
- 13 Open National Titles (USPF and USAPL)
- 148 Junior World Champion with the IPF
- 148 Open World Champion with the IPF, 1996
- 2 time 165 Open World Champion with the IPF, 2004 and 2006
- 181 Master's World Champion with the IPF, 2011

Something of note is that I am the only lifter to win a Junior, Open, and Master world title with the IPF.

I am currently retired from powerlifting. If I choose to compete again it would likely be in a bench only meet. I might try to do that as a Master. I am also reasonably banged up at this point so I see it as unlikely that I would squat or deadlift again competitively.

When did you start training; when did you first compete; what were your first competition lifts?

When I was 12, I got my first barbell set; it was one of those BP sets with the concrete weights that a lot of kids had in their basements. I would just lift for hours on end with that. It was mainly focused on upper body stuff; I think that gave me a good base for my lower body training which started in high school.

When I was a freshman in high school I had a 215 squat and a 205 bench press and I weighed 132 lbs. When I started squatting in high school I started off lifting right — no half squats for me — I just had a pretty good, natural groove right from the start. When I was a senior in high school I hit a 540 squat, 330 bench press, and a 450 deadlift at 165, all in a competition. At the time I was doing high-bar squats; that may have given me a good base because you are so upright, it does a good job of building erector and core strength in the hole. However, later on in my training I didn't find that high-bar squats did much to help my low-bar squat.

I didn't really have much support while lifting in high school; I pretty much had to figure out everything on my own. It wasn't until I got to college that I started learning from and being around pretty knowledgeable lifters. In college I became more of a technician. My own groove in the squat was good. I was able to squat with my shins perpendicular, hit pretty good depth. Squat form always came reasonably easy to me.

How much were you squatting when you first hit a plateau in the exercise? How long did you plateau there? What did you do to get past that plateau?

I had a pretty long plateau with 644 in the squat; I think I was stuck with that weight for about 5 meets, which is more than 2 years. I had to re-examine my training during that time, although I used basic periodization to blast through. I also think maturity, just plain maturity, helped me get through that plateau.

I had some issues going from multi-ply to single ply gear. I found that was hard to do. For a while I lifted in the WPO which was a multi-ply federation that paid the lifters. The money was nice while it lasted but it was tough; it was like you had to learn a totally new form to lift in that type of equipment.

Wade takes Gold at World's
Photo credit: Wade Hooper

What do you feel is key to being successful in the squat?

Squat a lot — you must practice the exercise. I found there to be only 3 really good exercises to help my squat: Squats, squats with chains, and pause squats. This was true once I was already a pretty good squatter. You must constantly work on your form and you have to be honest with yourself. Workout partners — good, honest partners — are key; you need someone that will give you the right feedback. It can be hard to get that; as you get stronger some lifters are simply in awe of you and your strength but come training time everyone is on the same level. You have to be consistent in your training and your actions.

What is your current training program like (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I made the most progress when I was following a Sheiko style routine, which I did from 2006 on. This has you squatting 3 or more times a week. I took the base template and adapted it for myself. I was actually able to talk with Boris Sheiko in person to learn more about his mindset and his goal in creating the program. I think a lot of people don't really follow his programs correctly. I had a bunch of different maxes (with gear or not, etc.) and I didn't always get the same result when I repeated the program. I could handle a really high volume in the squat. I found that once I was 38 years old or so I needed more recovery. When I would tweak the routines I would usually keep the main stuff the same but I would change the assistance work and generally make it easier. I would usually follow 6 week long cycles with this style of working out.

How did you warm-up for a heavy set of squats?

When it was time for me to squat I would usually spend 5–10 minutes riding the bike; then I would perform some basic stretching. When it was time to actually squat, my warm-up protocol was reasonably simple. Let's say I was working with 680 in the squat; that would be about 85% of my max give or take. I would warm-up like this:

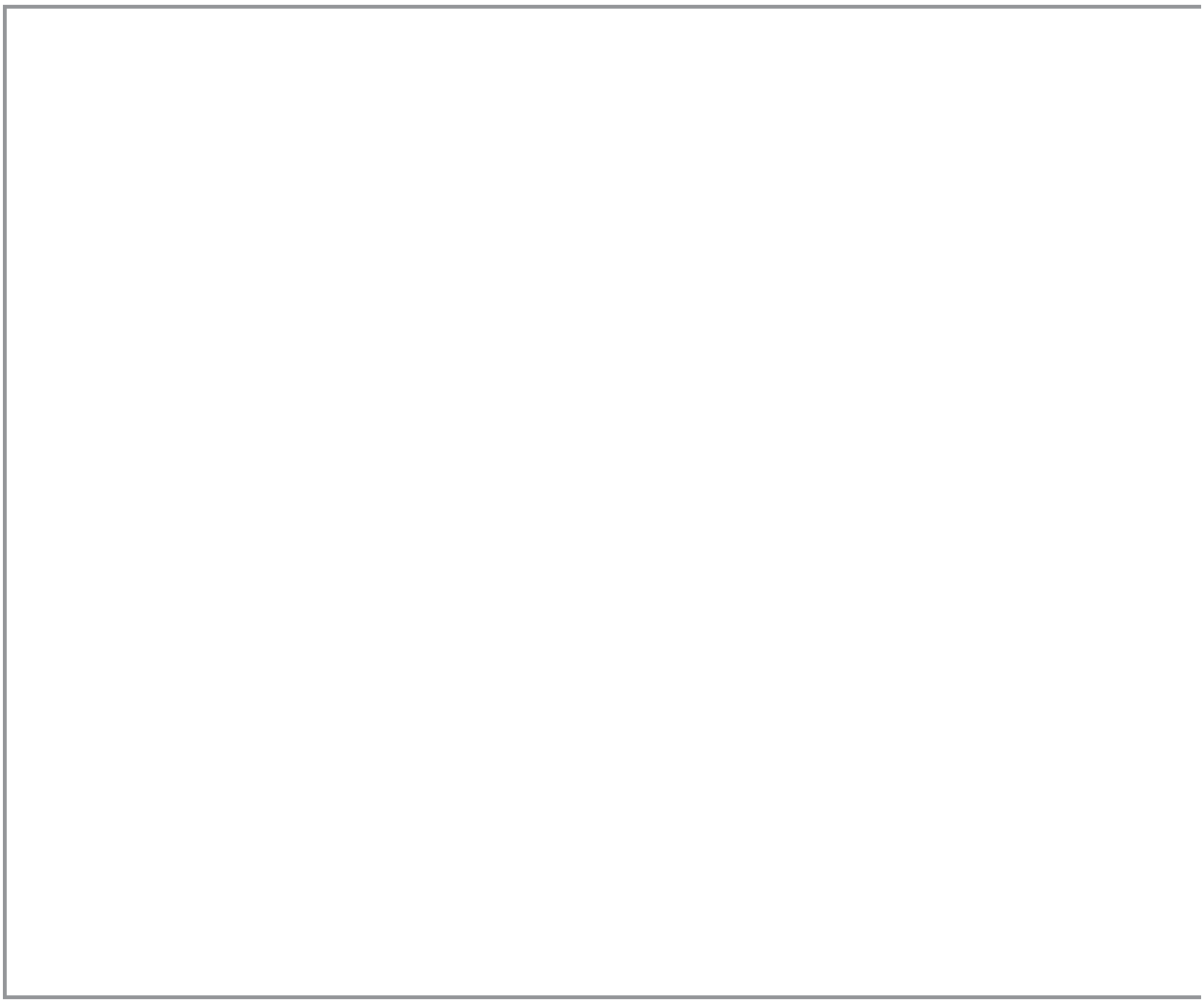
squat 135x10, 225x6, 315x2, 405x2, 500x1, 600x1, work sets

I took reasonable rest time during the squats and then I would move fast through the assistance stuff. No matter how you do it, the Sheiko style routines are long. I might spend 2 hours or more just squatting that day. When I was peaking for a meet I would spend 20–30 hours a week training. Sometimes my wife would get pissed; it would be 9:30 and I wouldn't be home — I would still be squatting. "I'm almost done, dear, just one more set." That is what I would say when she would call me up asking me where I was.

What do you think of training with a high frequency (3+ times per week) in the squat? Have you done this, what were your results, and do you believe in it or not?

I believe in this: you need to do it a lot to be good. It doesn't always have to be heavy. Powerlifting is a sport; the lifts involve some skill. Just like any other sport, to master it you must do the reps. I am also not a fan of doing too many partials because I think that can mess up the skill. Powerlifting is pretty simple; if you are not doing those 3 lifts you are not going to improve.

High frequency training can work for beginners but you really have to get the volume right. I think beginners should establish a solid base, and then reduce the frequency and volume when dealing with them. I think beginners should squat once a week for at least a year. It took me 3 years to fully transition to and understand the Sheiko workouts.



Wade just about to lock out a nice squat
Photo credit: Wade Hooper

What do you think of training with a low frequency (1 time per week or less) in the squat? Have you done this, what were your results, and do you believe in it or not?

I did once a week basic linear periodization for a long time and had good results with it, but after a point I had to change it up and that is when I started going higher frequency. Ed Coan is the only guy I know who can follow the same routine over and over again and still get good results from it. More power to him or anyone if they can do that; I had to change things up.

What are your favorite assistance exercises for the squat? What areas do you feel they work? How often do you do them? How heavy do you go? Do you feel there is a direct correlation to any of those exercises and your lift (e.g., your squat is always 100 lbs heavier than X exercise).

My 3 key exercises were squats, chain squats, and pause squats. For the chain squats I would either have a set amount of weight and then just keep adding chains until I got really heavy, or I would have the same amount of chains (say 200 lbs) and then add straight weight.

With the pause squats I would descend normally (which for me is a reasonably controlled descent anyway; if you squat with a dive-bomb style like Shane Hamman I would not recommend this method); I would count to myself “One and two and three” and then squat up. That count wasn’t super slow so it was probably 1.5 seconds real time, but it teaches you how to stay tight, keep your knees out, your hips open, and it forces you to stay upright in the hole.

I also really like front squats but as my career advanced I got too tight to perform them well; the bar would keep getting forward on me. In my earlier lifting days I did a lot of leg curls and leg extensions — a fair amount of leg presses, too. I would usually do sets of 10–12 on the machines. I wasn’t as concerned with weight; these were my isolation exercises and I treated them as such. I also noticed because I traveled a lot the weight on one machine might not match another one. I could leg press heavy but that varied as well from machine to machine; on mine I could do 1200 pounds for sets of 5 pretty regularly.

I also liked performing stiff legged deadlifts and stiff legged good mornings, not super heavy. I got a lot of conditioning from my assistance work.

What are your thoughts on training until failure on the competition lifts? How often would you do that? What about training until failure on the assistance lifts? How often would you do that?

I never trained for failure on squats, although I did sometimes fail, but that was rare. I could probably count on one hand how many reps I have missed in the gym. I think training to failure on the big lifts, and even on the little stuff, leads to poor form and overtraining. Ed Coan said it best so I always try to repeat and follow his advice: leave 1–2 reps left in the tank. I just didn't see training to failure as a good tool, I might use that in the off season every once in a while but never when prepping for a meet. That is training different systems.

How did you plan out your attempts for a meet?

For my attempts I generally followed a pretty set plan. I would think about a good third attempt based on my goals and what I did in the gym. Let's say I was going for an 800 lb squat. I usually opened up around 90% of that, maybe a bit higher. I liked to make a 20 kg jump from my 1st to my 2nd attempt. I might open with 739 and then go for 777; when I hit my 777 that is what I did. I had to repeat the weight that day because I cut my second, I think I was good for 800 then. Once you make your second then it is about a 10 kg jump for thirds.

I have found that littler guys need to make smaller jumps in weight. When I was 148 I couldn't handle making big jumps so I would make smaller jumps; this actually resulted in my opening heavier percentage wise.

Wade utilizes the sumo deadlift technique

Photo credit: Wade Hooper

What injuries have you faced and how did you overcome them?

I have had a few significant injuries. In 2000 shortly before Nationals I tore a ligament or tendon in the small of my back. Actually lifting with it wasn't so bad but it hurt to set up with the weight on squats and deads.

In 1998 the bar slipped down my back when squatting; this caused a 2nd degree separation of the AC joint (where your collarbone meets your shoulder). This really messed with my bench for a while. Initially I was going to do surgery but once they told me they were just going to fully open up that joint and shave it down that didn't sound appealing.

In 2009 I had a 3rd degree strain of my bicep, which means it was just holding on by a few threads. I hurt this when I was moving a TV — which weighed probably all of 60 lbs — with a friend after a workout. This was just 1 month before the World Games. Initially I wasn't going to lift but I had already paid for the tickets and the doctor said I couldn't hurt it anymore anyway, so I did. Surprisingly I tied my best total and hit a PR bench press. The injury didn't mess me up per se but dealing with the pain was the worst part. The first week I couldn't do anything but then it started getting better. I ended up having surgery when I was done to fix it.

How important do you feel that nutrition is to powerlifting performance?

I wish I could say I was just perfect with my nutrition but I wasn't. When I was 148 I had to monitor it more closely, I didn't watch every calorie but there was no bread, no soda, no pasta, and I ate higher protein. I could have focused on nutrition more throughout my career but honestly there is only so much time one can devote to this.

What do you usually do with your bodyweight and nutrition to prepare for a powerlifting competition?

In the off season my weight was 10–12 lbs heavier for 148s and about 5–7 lbs heavier for 165s. If I am doing 181 I don't need to watch it all.

My "in-season" for powerlifting was generally a 12–15 week prep for Nationals, and then Worlds was a little more than 2 months after that. I found it most useful to focus on 2, no more than 3 meets a year. The travel and prep gets expensive and it just wears you out.

How important do you feel that supplementation is to powerlifting performance? What kind of supplement program did you follow? Are you sponsored by any supplement companies and if so, what are they?

When I was competing I was sponsored by Quest Nutrition (their team won the USAPL Nationals team championships 8 times between 2000 and 2010). Sherman Ledford is the founder, and he always treated me well, I believed in their products. I would say creatine is the main thing I would take prepping for the meet. Specifically I would take their Hardcore — Recovery, JSF joint formula, the Quest Whey, and the Quest Plex during the in-season. I always liked to take some time after Worlds to let my body cleanse itself before I started the whole thing all over again. If I started lifting heavy again present day I would just contact Quest and start taking their stuff again.

It takes 5 people to spot this Quest Nutrition sponsored athlete

Photo credit: Wade Hooper

What are your thoughts on powerlifting equipment (gear) in powerlifting?

The raw movement started too late for me. I was essentially done competing by the time it really started gaining speed. I would have liked to do a raw meet in my prime. When I was competing the gear helped but it helped everybody pretty much equally; the strongest guy in the gear was the strongest guy raw. Now I feel like it is has gotten a little bit out of hand, especially on the bench. I wore the gear because everybody else did and there was no way at that level that you could give that much of a competitive advantage to somebody else by not wearing it. And I want to be clear, I did everything I could to maximize the gear because it was within the rules and I wanted to win. I do wish they hadn't opened Pandora's box though.

I do think gear is here to stay. The gear companies make too much money, and the federations make too much money from those companies to get rid of it altogether.

I wasn't a huge fan of the double ply stuff, at least for me. I didn't mind the gear so much when it just helped me lift, but the double ply stuff, it was like I had to relearn the lifts, I had to change my groove. I didn't want to change my groove to match the gear, I wanted the gear to match my groove. It just took me too long to figure that stuff out.

What are your thoughts on steroids in powerlifting? Do you compete in drug-tested competitions? How do you feel about the effectiveness of drug tests for catching those who use steroids?

I think that drug testing works; it seems to get better every year. I think there should be more standardization among countries when it comes to testing their lifters; right now it is up to the country to enforce the testing and I think some countries do a better job of that than others. I think the WADA is a good thing and it helps with the testing but again I think it should be consistent: either all the top people get tested or get rid of it all together. You always know it is possible that you may be competing against people who are drugged; you just do the best you can. I think the testing has improved and I think one can be fairly sure that they are competing on the same playing field.

When I was competing out of meet, testing was a pretty regular occurrence. It was unannounced for the most part; I would have people just show up at my gym and I had to do the test. I had to fill out an Athlete Locator Sheet that listed where I was during the day for months in advance so they could keep track of me. Some people seem to worry a lot about out of meet testing, but it never really bothered me; they could test me any time they wanted to. I had nothing to hide.

I do think that athletes should get to know the drug-testing protocol; they need to learn their rights as athletes. Don’t ignore that stuff or you can be taken advantage of.

How would you feel about powerlifting being united? What do you think the key to unification is?

I just don't think it will happen. If it were to happen, a few things need to take place. First, there needs to be acceptance of the double ply material. Second, there needs to be adherence to the regular rules of powerlifting, if they don't do that they should call it something else. Last time I checked, for a squat to count, the hips need to be below parallel, I don't see anything wrong with that. If the IPF were to get IOC approval, which I don't really see happening, then that might unify things.

Wade hits depth with a 4x bodyweight single ply squat
Photo credit: Wade Hooper

Some people think the raw movement will be the thing to do it, but I am not so sure; that might just be another division. I do think unification would be great for competition. When I would go to meets and you would really see the best 2 or 3 lifters in that weight class in country, or even the World, on the same platform battling it out, that was really something special to watch. Now everything is watered down.

There are way too many egos involved. My attitude was if you are the best you want to go up against the best; if you get beat it makes you a better lifter. There were many times I got my ass handed to me but you suck it up and modify things and you get better; you shouldn't just pull out of an organization and start a new one to avoid competition.

What is your thought about the importance of having a workout partner(s) or teammates in helping you train for powerlifting?

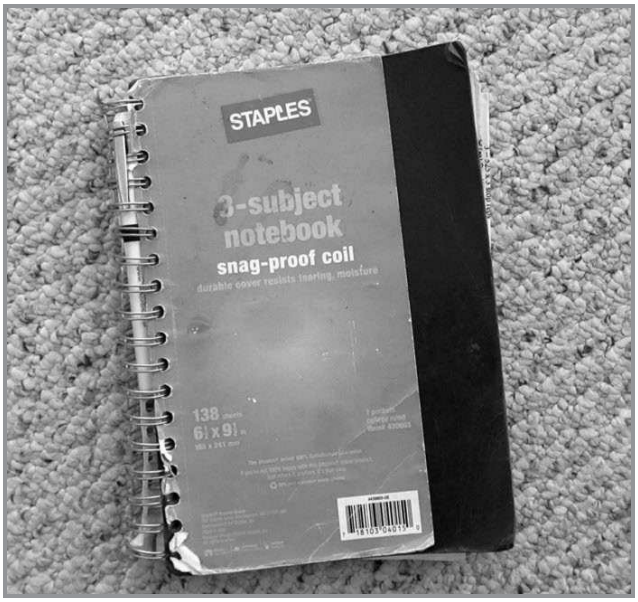
I believe it is very important to have good people around you. They are there to watch and help with technique, to give you feedback. I have been lucky to have some really great training partners and that has really helped me. I worked with LSU and Louisiana Tech; Jeff Douglas was my training partner. I think being around good people is key.

What is your thought about keeping a training journal while powerlifting?

I kept training journals for sure; starting with my freshman year of college I wrote down essentially everything. I still have them all. I think they are important.

What books, websites, or coaches do you suggest or follow in your lifting, and what would you suggest other lifters do to learn more about lifting?

Powerlifting USA magazine was great; I read that a lot, and I would use the workouts found in there. There wasn't a ton of stuff available when I was coming up. I read Hatfield's book, anything powerlifting related. Once I got to college I had access to a lot more information. I was also in contact with some of the best lifters in the world, lifters like Ed Coan, Kirk Karwoski, Jeff Douglas, my main training partner; I learned a lot from them.



Journals are important to powerlifters

Honestly I am not sure I get all of the stuff that is out there now. The Westside stuff — I tried that but it seemed it was more based toward training in the gear. Louie Simmons is knowledgeable and it seems like all he has ever tried to do is help people but that style didn't work for me. Mike Tuchscherer is really knowledgeable and he has a good system, I blended some of his stuff with Sheiko. I actually met Boris Sheiko at Worlds; we started communicating through the Internet so I got to learn more about his stuff.

What was it like competing in a World Championship?

When we would compete at the World championships, I — and I think most team members — was friendly with the other countries. We didn't hate each other. Sometimes the language barrier got in the way, sometimes you didn't really understand the other people, but we didn't hate them. When it was competition time you could cut the tension with a knife, but after the meet you could hang out and have a good time. I even became good friends with some of those lifters.

What do you attribute your personal success in powerlifting to?

My dedication — I never quit. I was lucky to have good people around me, people who were supportive — sometimes people didn't understand what powerlifting was but they were still supportive. My mom for example, watched me twice; both times she had her hands over her face, but she still supported me. I had a never quit attitude, even after I might have a brutal loss in a competition; you just come back and get better.

What advice would you give to someone who was just beginning to take up powerlifting?

Take the time to learn the technique, be consistent, never quit. Technique will overcome anything; find the right one that fits your body style.

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

Same advice as above — now it is the time to become a student of the sport. Learn for yourself — don’t just believe what everybody tells you. The best people have to figure things out for themselves.

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

Be prepared to put in the extra hours. When I was in-season I was training 20–30 hours per week, on top of having a regular job, family, etc. You need more work to continue to make gains.

Are there any changes that you would make to powerlifting if you had the power to do so?

If I was the czar of powerlifting I would do my best to unify it and make sure all lifters were lifting under 1 set of standardized rules. I think the IPF rules work well; they often get criticized for being dicks about the judging, etc., but in my opinion they offer the most consistent judging around.

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

Quit reading, put the damn book down, and go train.

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.

Wade Hooper
Photo credit: Wade Hooper

Pedro Mejias smashing over 600 lbs drug free
Photo credit: Doug Jantz Photography

Chapter 5

The Bench Press

The bench press is the second lift performed in a full powerlifting competition, and it is the main test of upper body strength. It is also arguably the single most popular exercise in the gym and the most common measurement of total body strength to a layperson. When you find out someone lifts weights, the next question is always “How much do you bench?” The bench press is also the most common lift that is performed as a single lift event, a competition where only one lift is offered. In a single lift bench press competition, the participants just compete in the bench press and do not perform the squat or the deadlift. It is quite common for a powerlifting competition to include both a full power meet (all three lifts) and a single lift bench press meet.

While the bench press is a very cool exercise, it should be noted that in a powerlifting competition it is the least important lift. This is because the amount of weight lifted in the bench press is almost always significantly less than in the other two exercises. If one competitor benches fifty pounds more than another, that is quite a big difference on the bench press. But that difference can be made up in the squat and the deadlift due to the large amount of weight being lifted in those two events. In other words, if one competitor benches 10% more than another one, but the other competitor deadlifts 10% more than the first person, and they both have the same squat,

the one with the better deadlift will win the competition.

> ***How much do you bench?***

Types of Bench Presses

There are many different types of bench press exercises. The most common one is a wide grip or regular bench press. For this exercise your hands are about elbow width (24–30″) apart and you just perform a regular bench press, bringing the bar down to your chest and pushing it back up. A closegrip bench press is where you take a more narrow grip, hands usually about 6–12″ apart and perform a bench press. This is more challenging because you have a greater range of motion and worse mechanical advantage. It also stresses some smaller muscles (the triceps) while taking some stress off the larger muscles (the pecs). You can perform a reverse grip bench press, where you use a supinated grip and lift the weight. This is often, but not always, illegal in competitions. It places more strain on the wrists and the triceps but some people are stronger with a reverse grip than a regular grip. There is a super wide grip bench or an illegal wide grip bench where you take a wider than normal grip (31″+). And there is a partial bench press where you do not perform the full range of motion; you use only a certain part of the range of motion. When describing the technique of a good bench press, I am referring to a regular bench press, as that form usually allows you to lift the most weight.

> ***Align yourself so that your eyes are under the bar.***

Equipment Set-up

There aren't too many adjustments to manipulate when setting up a bench press; however, hopefully your gym has a bench that does have an adjustable rack height for the barbell — if it isn't adjustable then it should have several options that allow the lifter to choose a height that works best for them. The rack should be high enough to allow the lifter to lift the bar out of the supports with minimal effort, but not so high as to mess up the lifter's form during the lift off.

First thing to do is to sit down on the bench. I like to pull my shoulder blades back (retract), puff up my chest, and then lie down on the bench. Align yourself so that your eyes are under the bar or in that general area. If you are too far back (hips toward the end of the bench press), it will be very hard to lift off the bar from the racks. If you are too far forward (throat under the bar) then you are likely to hit the racks as you press the bar, and that can be distracting and detrimental to the lift.

Proper Technique

I am going to give you two general sections on technique for the bench press. The first part is a guide to performing the majority of the reps you complete in training. The second part is a guide for performing your one rep max.

Hand Position

When your body is in position, take a grip on the bar. You want to use a closed grip (thumb opposite your fingers) as an open grip is illegal in most federations and is also significantly more dangerous. It is called a suicide grip for good reason. You want to be able to take the exact same grip each time you use the bar, so you need to find a way to make it consistent. Using the rings on the bar is quite useful for this. A good starting grip is to put your pinky on the ring. If you have longer arms or a thicker chest you might try going out a finger or two. If you have shorter arms or a thinner chest you might try moving in a finger or two. I find pinky on the ring, full thumb from the grip, and half thumb from the grip to be most grips used by most raw lifters.

(Left) Pinky on the ring

(Middle) Full thumb distance from the knurling

(Right) Half thumb distance from the knurling

Normally you want your forearm to be perpendicular to the bar or out just slightly when the bar is resting on your chest. Varying significantly from that angle will generally put you in a weaker position. If you are used to one specific grip and decide that you want to train with another grip, don't expect to change grips and lift the same weight immediately. Only make minor changes to your grip; one finger width at a time is optimal, and in this way

gradually move your grip in or out. You have to be more cautious when you start to move your grip out and the weight is heavy, because that will place more stress on the shoulder joint and the pec muscle. Remember that in a competition any grip that has all of the fingers outside the rings (more than 81 cm apart) is illegal, at least one finger must be covering up the rings for it to be legal.

Foot Position

Once you lie down, pay attention to your feet. If you are benching heavy, you want your feet to be relatively far apart with your toes pointed out; a 30–45 degree angle usually works well, similar to a squat stance. You also want to be as stable as possible, so keep your feet flat on the floor. Find a foot position that is comfortable; your feet should remain motionless throughout the entire set. In a competition, the feet are not allowed to move once you begin the lift. In most federations it is illegal to bench up on your toes, so be sure to keep your feet flat. Generally, you want your knees right over your toes. The more you bring your feet under you, the better arch you'll get, but this also makes it easy to thrust your butt up off the bench, which is incorrect and undesirable. Try to imagine that if somebody attempted to push you off the bench, you could hold yourself in place. The heavy weight will try to move you if you let it.

Keep your feet flat.

Head Position

Head position for the bench press is fairly straight forward. Rest your head on the bench pad itself (this is often mandatory in competitions); keep a neutral spine — your head should be in line with your body and straight. When you are going heavy you might find it useful to drive your head back into the bench and having a packed neck is likely a strong position (this will tend to happen naturally when you lift your chest and pull back and set your scapula).

General Form

Once you have good body position and a good grip, you are now ready to lift the weight. Make sure your shoulder blades are pulled tightly together before you lift. Squeeze your rhomboids, traps, and rear delts together, trying to make your back hard and firm. This will lift your chest, which decreases the range of motion and makes the bench press easier. It is important not to lose this position as you lift off the bar. To help with this, either get a lift off from a partner, or make sure you are far enough under the bar so you can easily press it up. Once the bar is lifted off, a lot of people like to line it up with something on the ceiling. They will then press the bar back up toward that same spot on the ceiling with each rep.

Right before you complete the lift off, you should take a deep breath in and hold it. Once the bar is set over your chest, exhale and then inhale again. Inhaling a deep breath also serves to raise the chest slightly. Hold your breath as you lower the weight to your chest, and then exhale as you are pushing it back up. Normally you want to exhale through the sticking point or the difficult part of the range of motion for you. For most people this is about halfway up. If you are performing multiple reps, inhale on the way back down and then repeat; you don't have to hold the bar straight to inhale each time, just on the first rep.

When you descend with the bar, the position of your elbows in relation to your body is extremely important, both for lifting performance and shoulder health. You do not want your elbows in line with your shoulders (upper arm at a 90 degree angle); this puts too much pressure on the shoulders (namely the rotator cuff) and also it does not put you in a strong position. You want to tuck your elbows in to your body, so

You want to tuck your elbows in to your body.

that your upper arm is at a 45–60 degree angle to your body. You should bring the bar down so that it touches the highest point on your chest. By highest point I mean the point closest to the ceiling when you are lying down. This point is usually right around nipple level for both male and female lifters. The bar should touch your sternum; it should not land on your stomach and it should not land near your clavicle (collarbone). Once the bar touches your chest lightly, press the bar back up, following the same path as the descent. Keep your elbows tucked at that angle for the beginning of the press.

You want to slightly arch your back during the lift. This will raise the chest slightly, and the higher the chest, the shorter the range of motion. There doesn't need to be excessive tension in your lower back, and your back should not cramp up during the exercise. Normally simply pulling your shoulder blades back and then lifting your chest is enough to produce this arch.

As you are pressing the bar upward, you may find that you have increased strength if you press the bar upward and backward slightly, so the bar travels in an arc going back up over your head. If the bar begins at the middle of your chest, as you press it up, if you also move it slightly back, the bar will end up over your upper chest or your clavicle. This arc should not be excessive but should follow a natural range of motion. The main problem with arcing the bar as you press is that when performing multiple reps, you can end up bringing the bar down in the wrong position. If you do choose to arc the bar, make sure each rep is the same and that the bar touches your chest in exactly the same position each time. Do not let your elbows flare out too much as you arc the bar up, no more than 90 degrees,

The bar travels in an arc going back up over your head.

and definitely keep your elbows tucked in at the bottom of the lift.

Wrist position during the bench press doesn't get talked about much, but I feel it is important. It can be useful and effective to roll your wrist slightly forward as you push the bar up, right at or before you reach your sticking point. As you bring the bar down to your chest, tuck your elbows in, and here you may let your wrist roll backwards just a little bit. Do not make it excessive or you can end up in a weak position. Then, as you push the bar up, roll your wrists forward a little. When in benching position, rolling the wrist forward means rolling them toward your hips (flexing the wrist). This can take some getting used to, because you are rolling the wrist forward as you are pressing the bar up and backward, but I find that little movement can help get the bar past your sticking point. You will probably find that with light or moderate weights the wrist roll is unnecessary, but it can be very useful on heavy sets or when you get tired. It takes a relatively strong wrist and forearm to be able to do this with heavy weight; you may have to add some specific forearm training to your regime to do it well.

A side note: it is also hard to do this move on an old or rusty bar, because you want the bar to roll just a little bit but the sleeve of the bar (the thicker part holding the weights) should not move. Old bars often have rusty sleeves, so when the bar moves, the entire sleeve moves, which moves the weight and that is not desirable. Luckily, in a competition, the quality of the bars is (hopefully) pretty good.

Roll your wrist slightly forward as you push the bar up.

Leg drive is important as well. Try to drive with your legs so that some force from your lower body goes into the bar. Think about pushing your upper body back to the racks that are supporting the barbell. Once you have some weight on the bar you likely won't be able to move your body, but that

momentum from your legs can help give you some extra pounds on your bench. Just make sure that leg drive doesn't cause your butt to come off the bench.

Leg drive is important as well.

When you are performing multiple reps and training to increase your bench press, you want your form to be good on every rep. Allowing your form to get sloppy to squeeze out an extra rep will actually teach the body that when the weight gets heavy and/or hard, you should adopt that form to lift the weight. Of course, when you go heavy, that is when you want your form to be the best. My rule of thumb when training is to try to make your last rep of each set perfect. It doesn't always work, but if your last rep when you are the most tired is perfect, then when you are fresh your max should be perfect. Of course, if you just perform one rep then it is your last and first rep so it should be perfect anyway. You want each rep to look the same, from the warm-up to the toughest work set. If you watch good lifters, both in the warm-up room and on the platform, they make each rep look the same. That is the goal. To help you achieve that goal, here are some simple guidelines to remember when performing the bench press, either in the gym or at a meet.

You want each rep to look the same.

1 Rep Max Form Specifics

Of course, powerlifters are ultimately interested in having the heaviest possible 1 rep max on the bench, and their training is geared toward improving that max. When you are performing a one rep max bench, the same general rules apply, but there are some things you can do to squeeze out a few extra pounds. Your body position should be the same whether you are doing a bench press for practice or for performance, but for performance it is okay to arch your back. There are some benefits and some costs to doing this. When you arch your back, you raise your chest up, which decreases the range of motion and therefore makes the lift easier. It is similar to performing a decline bench press. One of the risks of arching your back is that you are more likely to injure your lower back, which, of course, is not good. Arching your back is legal in powerlifting competitions; however don't confuse that with raising your butt off the bench, which is not legal. You must have your upper back and butt on the bench, but your lower back can be off the bench. Some people are more flexible in this area than others.

To get a good arch, people do many things and you can experiment with light weight or just the bar to see how they feel. There are two popular ways to get a good arch in your back. In the first, lie down on the bench and put your hands up on the supports to hold your body in place. From that position, drive with your legs. Since your upper body isn't moving and you are pushing with your legs, your back will arch.

A second way to get a good arch is to lie down on a bench, put your feet on the bench and then lift up into a bridge position and arch your back. From that position slowly lower one foot and then the other to the ground, holding the arch in your back and allowing your butt

You must have your upper back and butt on the bench.

to touch the bench. Make sure that you do not allow your butt to leave the bench during the lift, even for a moment, because then you could not count the lift.

Remember that even though arching your back is legal in a competition, that doesn't mean that you *should* do it. If you feel pain or major cramping, that can detract from your mental state and it may interfere with the lift. Of course, if you actually got hurt, that would mess up lifting for a longer period of time, and benching an extra 10 pounds would not be worth it. Personally, I would use a minor arch but nothing that would make my back cramp up. Long-term lifting has always been more important to me. If you do use the arch when you max out, you still should not use this on most of your training sets. Arch only if you are going to complete a heavy set of five or fewer reps; three or fewer is even better. Use the arch enough to practice it so it feels comfortable, but do not use it all the time, as the chance for injury is too significant. In addition, a large arch decreases the stress on the pecs, which are precisely the muscles you are trying to train.

The widest legal grip in a competition is 81 cm.

When you max out you want to slightly change your breathing as well. Once again, inhale before the lift off and then hold your breath as you get the bar in position over your middle chest. Exhale and then take a big inhale again. Hold your breath as you bring the bar down to your chest, and continue to hold your breath as you press the weight upward. Exhale either at the end of the lift or once you know mentally that you can do the lift. If you choose to hold your breath make sure you do this only on sets of very low reps — three or fewer reps. Breathe in between each rep; do not hold your breath for the entire set. Any set that is more than three reps requires you to breathe normally as described earlier. If you do not

breathe, you will run out of oxygen and then energy on your longer sets.

When choosing a grip on the bar, the same general rules apply to maxing out as to regular lifting. Most but not all people are stronger with a wider grip; you should experiment with different grips to find what works best for you. Wider is better, because a wide grip requires a smaller range of motion; however, if you go too wide, you will be weak at the bottom of the lift. Going too wide can also be rough on the shoulders. Keep in mind that sometimes a grip that feels good with a light weight will not feel good with a heavy weight.

As previously mentioned, the widest legal grip in a competition is 81 cm, which is putting your pointer finger on the ring on the bar. You can't go any wider than that. This can be disadvantageous for some very tall people, but those are the rules. The best starting point is a grip where your forearms are perpendicular to the bar at the bottom of the lift or at your sticking point, whichever one feels better. Once again make sure you are using a closed grip —your thumb on the opposite side of the bar from your fingers. This is good for safety, so the bar can't fall out of your hands and crush your chest. An open grip is illegal in most competitions, although some federations have different policies on this.

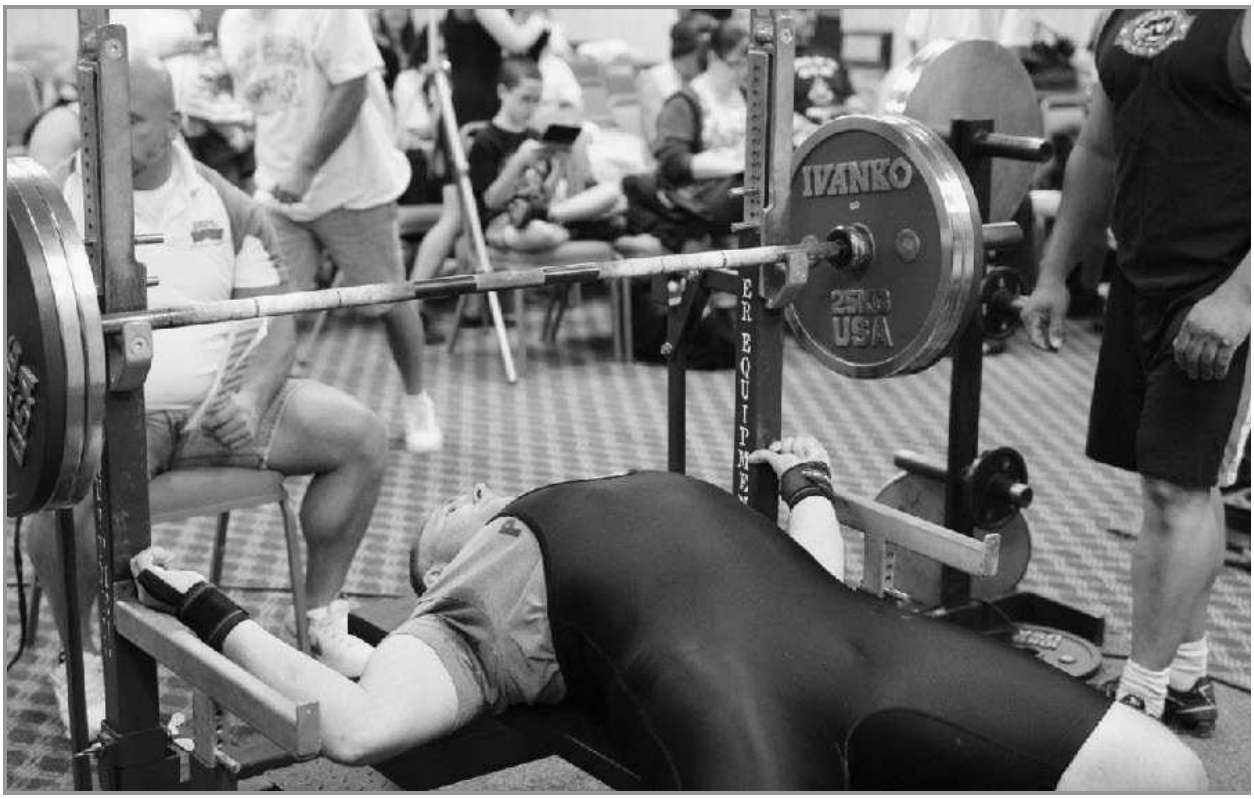
As you drive the bar upward, you want to keep your upper back muscles tight, which in turn keeps your body tight. Someone once told me to imagine that the bar was immovable, and you are trying to press your back into the bench as far as you can. Of course, the bar is what it is moveable so it will go up. When pressing the bar up, you can attempt to drive with your hips. There is nothing wrong with this as long as your butt does not come off the bench. You are trying to channel the energy from your hips into the bar to make it go up. This comes more naturally to some people than others.

For a one rep max, you will normally want to arc the bar back up as

described earlier. Often gripping the bar tightly will help the muscles in the arm contract, particularly the triceps. I have heard it suggested that you should imagine you are trying to separate the bar (pull it apart) as you press up to encourage your triceps to fully contract, but I have to admit, that has never worked for me. You want to keep your wrists generally in line with your forearm as you press up; however, it is a good idea to try to roll the wrists on your max. As previously described, when you bring the bar down to your chest, let the wrists roll back just a little bit, and then roll the bar forward as you press up. Keep your shoulder blades locked together; do not protract your shoulders as you press the bar up.

Things to do when performing the Bench Press:

- Do get a stable base
- Do keep your feet flat on the floor
- Do use a closed grip
- Do make sure your grip is even on the bar
- Do take a deep breath when you first bring the bar down to your chest
- Do tuck your elbows to about a 45–60 degree angle as you begin to press
- Do keep your body stable
- Do keep your shoulders retracted and your chest up high
- Do exhale during the sticking point or just after it
- Do roll the wrists slightly as you press up
- Do arc the bar up and slightly back as you press up
- Do drive with your legs if you can



▶ **1: Get set up**

▶ **2: Take proper grip**

▶ **3: Lift off, start with arms fully extended, big inhale**

▶ **4: Controlled descent, elbows tucked**

- **5: Know where the bar is going to land**

- **6: Pause the bar on chest, stay tight, wait for Press command**

▶ **7: Drive, keep elbows tucked initially, use lats and leg drive**

▶ **8: Flare elbows**

▶ **9: Push weight back toward face**

▶ **10: Squeeze bar hard, drive toward lockout**

▶ **11: Full extension, wait for Rack command**

Photo credit: Vadim Snitkovsky

Things NOT to do when performing the Bench Press:

- Don’t let your butt come off the bench
- Don’t dance with your feet, they should be stable
- Don’t slam the bar on your chest
- Don’t cut the range of motion short (DO touch your chest)
- Don’t flare your elbows out to the side at the bottom of the ROM
- Don’t let your head come off the bench
- Don’t look to either side or watch one arm
- Don’t let one side of the bar go up (or come down) faster than another side
- Don’t use an open grip (thumb on the same side as your fingers)
- Don’t excessively arch your back
- Don’t let your wrists bend backwards too far
- Don’t shift your weight to one side as you press

To Pause or Not to Pause

When you compete in a bench press competition, you must bring the bar down, rest it on your chest and hold it motionless. Then, once the bar is visibly still, you can press it back up. In some organizations the referee will yell, “Press!” as a signal that you can push it up; in others you must just pause it yourself and you decide when to press. Going off of the press command makes the bench press a little bit harder. The pause at the bottom dilutes the stretch reflex which in turn makes you a little bit weaker. Most people can expect to lose about 5% off of their max because of the pause. Some will lose more, some will lose less. If you lose a lot because of the pause, especially once you get used to it, then your form may not be as good as you think it is, and you may well be bouncing the bar off of your chest instead of lightly touching it.

Pausing the bench press eliminates any momentum. It is used in a competition to distinguish between just touching your chest and bouncing the weight off of your chest. If you are going to compete in a competition, or if you want to see what your true max is, at some point you will want to try a pause rep. However, that does not mean that all of your reps should be performed with a pause. If you do that regularly, it could ruin the stretch reflex and actually make you weaker. My personal recommendation is that powerlifters should pause at least half of their reps and all of them if they want to; however, athletes who want to train the stretch reflex should pause only a few of their reps.

If you do rest the bar on your chest, or even when the bar just touches your chest, make sure the bar does not sink into your chest. Some people relax as the bar hits their chest, and this makes it much more difficult. Keep your upper body very tight so that the bar is just lightly resting on your chest, and then when it is time to press, explode upward and drive the bar to the

lockout. Jennifer Thompson shared the insight that to stay tight the bar should touch your shirt without touching your chest.

> ***To stay tight the bar should touch your shirt without touching your chest.***

Flexibility/Mobility Issues in the Bench Press

The bench press doesn't require a high level of flexibility or mobility in the shoulder girdle to perform well. However, lifters can still have issues that might affect the bench press. Chronic strength training can tighten up the affected tissues. This has a benefit of increasing joint stability but it has a negative in decreasing flexibility of the same area. The most common flexibility issue is the inability of the lifter to touch the bar to chest particularly with lighter weights. This will first appear when performing inclines. If this is the case the lifter can try widening their grip and also definitely needs to focus on stretching the pecs (both), the lats and teres major, the front delts, subscapularis, and the biceps. I find stretching both arms at the same time to be more effective for increasing upper body flexibility. Performing overhead presses (with good posture and ROM) will help maintain proper shoulder girdle flexibility. It is also important to maintain proper muscle balance with the antagonistic muscles (rhomboids, middle traps, rear delts) to help even out upper body strength. A floor glide stretch with light weights is particularly effective at stretching all of the affected muscles at once to help increase bench press specific flexibility.

Common Problems During the Bench Press

Walk into any commercial gym and you'll see the bench press being butchered in any number of ways. Following the form guidelines presented previously will help fix almost all of those errors, but some of the more common technique mistakes are addressed here.

Poor set-up – Just lying back on the bench press and pressing it is not ideal; lifters should focus on getting their whole body tight and in particular setting the shoulder girdle so it is ready to receive the bar.

Dancing with the feet – As lifters struggle to get their last reps in a set, often they will start dancing with the feet. This means they are not getting any leg drive and they are actually wasting energy by sending some of it to their legs in an unproductive fashion. Making them aware of this problem is step one to fixing it, standing on their feet while they bench can also help teach them to keep their feet glued to the floor.

Watching the bar – Some lifters will watch the bar descend (thus raising their head up); cue the lifter to drive their head back into the bench to keep their head on the bench the entire time.

It is also important to maintain proper muscle balance.

Watching their weak side – Some lifters will look at their weak side as they press the bar up, thus turning their head. This can earn a red light and it is not productive; cue lifters to fix their gaze at a spot on the ceiling and to not deviate from that spot. Holding their head so it can't move can also fix this problem.

Using their chest as a springboard – The sternum has a good amount of

cartilage and it can be a bit springy; some lifters really master this and get a large amount of assistance by slamming the bar into their chest and catching it on the rebound. A strict paused bench press will really kill this lifter's bench poundage. Simply practicing the pause, with much lowered weight, will be necessary to develop the strength at the bottom that the lifter is missing. They will likely notice a nice increase in their hypertrophy as well after fixing this mistake.

Raising their butt up in the air – The good news is this lifter is using leg drive; the bad news is it is being misdirected. The lifter should focus on driving their body up to the bench supports, not just driving their hips up in the air. If necessary belting the lifter to the bench can help fix this problem.

Wrist falling backward – This can cause the lifter to lose strength on the way up. Cueing them with the light weight to keep their wrist straight (like they were punching the ceiling), strengthening the forearms (particularly the flexors) and wearing wrist wraps all can help fix this issue.

Cues for the Bench

Just like with the squat, there are some cues that you might utilize to direct your performance in the bench. Again, just pick 2–4 of them to focus on. They are listed in order of occurrence.

Strong Set-up
Big Arch
Get Tight
Good Lift Off
Big Air
Controlled Descent
Know where it is going to land
Stay Tight
Explode
Leg Drive
Lats
Flare the Elbows
Squeeze the Bar
Drive
Hold

Common Competition Mistakes

During the bench press in a competition, there are several common problems that seem to occur time and time again. I will go over these issues and how to quickly fix them, so that they don't happen to you. You don't want to waste valuable energy attempting or even completing a lift that will get you red lights.

A lifter gets mentally ready to bench
Photo credit: Vadim Snitkovsky

The first issue is a poor lift-off. A lift-off is where a spotter/partner hands the bar off to you so you can take it in the start position, with your arms extended. Getting a lift-off is generally a good idea both because it saves you energy, and also lifting the bar off the racks with heavy weight may put your shoulders in a poor position. The drawback to the lift-off is that

you don’t get to really feel the weight until it is right over you. In most competitions you can have the person of your choice give you a liftoff. It is ideal to practice with this person beforehand so they know what kind of lift-off to give you. If you don’t have a partner, you can either ask another competitor or you can ask one of the spotters to give you a lift-off.

While it is important to have help with the lift-off, this help can also cause problems, either when there is not enough assistance or too much. When the spotter doesn’t give enough help, he either makes the lifter work too hard to lift the bar off the racks or more commonly, he lets go of the bar too soon before the lifter is in the ready position.

A spotter should lift the bar off the rack and help you bring it over your chest with your arms straight. Once you are in the ready position and the bar is motionless and stable, then the spotter lets go of the bar. Normally the spotter will say, “Your weight” or something to that effect to signal they are letting go of the weight. They should not just help you lift the bar off the racks and then let go immediately — that leaves you basically doing a pullover with your max, which is not ideal.

The second problem is when the spotter lifts the bar off too much. In this scenario, they help the lifter get the bar out of the racks and bring it over the chest, which is good, but they take so much of the weight that they end up plopping the weight on to the lifter and it feels very jarring. It also creates the sensation of the bar feeling heavy and that can be detrimental to lifting the max weight. The spotter should not be doing all or even most of the work; they are just helping the lifter to lift the bar out of the racks and then bring it over their chest. Once the lifter has the bar in that position, the spotter signals that they will let go and then removes their hands. At this point the spotter needs to get out of the way so the head judge can see.

Another common problem in a competition is the failure to pause the bar

on the chest. This is more common in organizations where you are responsible for pausing the bar yourself. If you get the press command, then once the judge says “Press” you can lift it up, whether it has been a good pause or not. Of course, if you lift it before you get that command, even if it has been a good pause, it will not count. In a competition environment, you are fired up and ready to go, maybe a little nervous, and who wants to take a long pause with their max? But sometimes when you are fired up you think you had a good pause and in reality it was a touch and go. A good idea is to count “one thousand one” once the bar is on your chest and then press it up. It doesn’t need to be longer than that, in fact you might get away with going shorter, but if this is a problem for you, it is better to make sure the pause is nice and long. Occasionally training with the pause in the gym will definitely help you get used to this.

If your butt comes up off the bench, even for a brief moment, the lift will not count. If your butt is coming off, it means that your form was not very good during your training and your training partners (or yourself if that is all you have) should have noticed this. The best way to keep the butt down is to move your feet out away from the bench (straighten your knees). The legs don’t need to be completely straight, but straightening the legs decreases leg drive and makes it harder to lift the butt up. This does make getting an arch harder and it makes the lift harder in general, but I would rather fail because of lack of strength than fail because I couldn’t follow the rules. If you are benching on a thicker, fluffier bench in the gym (the pad may be 3–4″ thick), that can mask a slight lift of the butt so check carefully if you do train on a bench like that. If you do, and then you compete on a thin bench with a firm pad, you may get

Count “one thousand one” once the bar is on your chest and then press it up.

red lighted for moving your hips.

When you press the bar upwards, if it extends unevenly or if it locks out unevenly the lift will not count. Some organizations are stricter on this than others. If you are having a problem with this, check a few things. First, make sure your grip was even on the bar. Second, make sure the weight was even on the bar. Even in a competition sometimes the spotter/loaders make a mistake and put the wrong weight on the bar. It just happens, so do yourself a favor and be on the lookout for that. If both of those things are good, then the problem is you, and generally this means you have one side that is stronger than the other side. The strong side does not always go with your usual dominant side. For example, I am right handed and my right bicep is stronger than my left, but when I perform a bench press, my left side is stronger than my right.

Make sure your grip was even on the bar.

It can and will take a long time to fix a muscular imbalance. In a competition, you can try a short-term fix which might help you make your next attempt. The side that is slow pushing the weight up is your weak side. You want the weak side to lift less weight and the strong side to lift more weight. Get your usual grip and then move both hands one, not more than two, fingers over in the same direction. Move your hands in the direction of the weak arm (if my right arm is weak, I move my right and left hand over one finger to the right side of the bar). Moving over one finger is normally sufficient; go two only if you had a really big difference in strength. This means you do not have a symmetrical grip on the bar, which is usually undesirable, but for now that is okay, because you do not have symmetrical strength.

This technique can work to help you get that last attempt. However, I

want to be very clear that this is only a short-term fix. If you do this regularly, you will just make the problem worse and worse. If you notice this problem in the gym or in a meet, you need to address it and fix it permanently as soon as possible. If you just move your grip, and consider that a permanent solution, you will likely tear a pec or at least make the imbalance much worse in the future. See Chapter 6 for ideas on how to fix the issue.

I would strongly suggest you measure your bench grip.

Another common meet problem is that powerlifters sometimes take an uneven or improper grip on the bar without knowing it. The primary cause of this is that the bar they are using in a meet is different from the bar that they train with in the gym. Sometimes the rings on the bars are not in the same spot. On a good competition bar the rings should always be 81cm apart, but in smaller local competitions or just at your home gym, the rings may not be set at that distance. If you always lift with your pinky on the ring and then you go the meet and put your pinky on the ring, if the rings are different now your grip is different. To correct this issue I would strongly suggest you measure your bench grip, from pointer finger to pointer finger. Then bring a tape measure with you and set yourself up on the bar in the competition with the same grip. Now you will know with confidence that the grip is the same that you have been using in your training.

In some competitions, the bars have three rings on each side of the bar, instead of just one. Those bars are made more for Olympic lifting but sometimes they are used in a powerlifting competition. If you are used to just one ring, this can throw you off. Generally, the middle ring on each side of the bar is the appropriate ring to measure up with. Sometimes I have seen lifters put one hand on the proper ring on one side of the bar but then put the

other hand on the wrong ring on the other side of the bar. You would think this wouldn't happen, but when someone is fired up and nervous, they don't always think clearly and simple mistakes can happen. Personally, I don't think Olympic bars should be used in powerlifting competitions, but if they are, now you will know what to do with them. This is another instance where knowing the measurement of your grip would be helpful. Many lifters line their grip up with some aspect of the knurling, for example, they will set up so they are a thumb distance away from the knurling. The placement of the knurling is not uniform for all bars and this can seriously mess up where a lifter puts their hands on the barbell.

Some other issues with grip can arise. I have noticed that sometimes lifters put their hands on the bar in the proper position, and then as they psyche themselves up, they open and close their hands and sometimes actually move their hands in or out. Often it is just one hand that is moving; it may only be an inch, but that one inch can make a difference. Do whatever psyche-up routine you want, but make sure that your grip at the end of that routine is where you want it to be.

The placement of the knurling is not uniform for all bars.

During the bench press, as with all the lifts in a competition, you must follow the commands. Not all organizations use the same commands, but for the most part, you will be given two or three commands. Once the head judge says the "bar is loaded" you have one minute to approach the bar; get on the bench, and with a lift-off, lift the bar up off the racks. Normally you start with your arms straight and the head judge says, "Start!" Sometimes there won't be a start command, but if your organization requires a start command, then you must wait for it.

Once you bring the bar down to your chest, you must pause before you

press it back up. The rule for the pause requires a definite, visible pause where the bar is motionless, so even if the bar is sitting on your chest, if it is shaking or rocking back and forth, that is not motionless. Then you may either press it up on your own, or you will get the “Press!” command. If your organization uses a press command, you can’t press it up before you hear that command. Once you press the bar straight up, lock out your arms and hold the bar in position.

When the bar is held in position with your arms locked, you will get the rack command, which means rack the bar. Once you get the “Rack!” command, the lift is effectively over, and the spotters can help you into the rack if you need it. It is very important to wait for the rack command; I have seen many good benches that didn’t count because the lifter failed to wait for the rack command, often on their first attempt. I try to visualize that I want to push the weight up and hold it there to show everybody what I just did. Sometimes I think people are so happy they just benched it, they want to rack the weight and celebrate, and they forget to wait for the rack command. Push it up and hold it there. If this is a problem for you, you can practice this in the gym by having a partner give out the commands, and you can get in the habit of holding your last rep on your work sets for one second with your arms straight. This is not physically difficult to do; you just have to remember to do it.

The final problem that often gets red lights in competition is unstable feet during the exercise. If this happens to you, then you simply haven’t been practicing good form in the gym. Your feet should remain motionless as you lift, no matter what the weight is. If you have a lot of leg drive, it is more common for the feet to move, so a quick fix for this is to move your feet out in front of you, the same solution for solving your butt lifting up off the bench. Make sure the bottom of your shoes are clean and not dusty or

covered with chalk or baby powder as that can make them more likely to slip. If the platform itself is dirty, you can ask that the spotters/loaders clean it off with a damp cloth.

If you can catch these common problems before you lift, you can have successful bench press attempts. Three good bench press attempts and nine white lights will give you some good momentum as you head into the deadlifts.

> ***It is very important to wait for the rack command.***

Technical Rules for the Bench Press

Listed below are the technical rules for the bench press with some explanations following them. Again these rules come from 100% RAW Powerlifting; the most notable difference in this federation vs. some others is the lack of the "start" command in the bench press.

Bench Press:

1. **The bench shall be placed on the platform with the head facing the rear of the platform.**

2. **The lifter must lie on his back with head, shoulders and buttocks in contact with the bench surface. The feet must be flat on the floor (as flat as the shape of the shoe will allow). His hands and fingers must grip the bar positioned in the rack stands. Full and false grip is allowed (if a lifter chooses to use a false grip, they must announce prior to the start of the lift). This position shall be maintained throughout the lift. A reverse grip is not allowed.**

The lifter must keep the head and butt on the bench, the feet flat on the floor. Open grip is okay; reverse grip is not.

3. **To achieve firm footing the lifter may use flat surfaced plates or blocks to build up the surface of the platform. If a lifter is in need of a plate under his feet he/she may not use any which will be used during the competition. Plates used in the warm-up room may be used. In the event that one plate does not provide enough height, it is the lifter's responsibility to provide their own set of blocks. The blocks must be inspected by the event's meet director prior to the competition.**

The lifter may use plates or blocks to raise the feet up.

4. **After correctly positioning himself, the lifter may enlist the help of the spotter/loaders or their coach in removing the bar from the racks. The lift-off, if assisted by the spotter/loaders must be at arm's length.**

The lifter may get a lift-off but the lift must start with arms straight

5. **The spacing of the hands shall not exceed 81 cm measured between the forefingers (both forefingers must be within the 81 cm marks and the whole of the forefingers must be in contact with the 81 cm marks if maximum grip is used). If in the case of some old injury or anatomically the lifter is unable to grip the bar equally with both hands he must inform the referees prior to lift-off for each attempt and if necessary the bar will be marked accordingly.**

Hands can't be more than 81 cm apart

6. **After removing the bar from the racks, with or without the help of the spotter/loaders, the lifter may begin the lift by lowering the bar to their chest (the chest, for the purpose of the rule, finishes at the base of the sternum/breastbone) where, once it becomes motionless, the Head Referee will signal an audible "Press." If the lifter has a hearing defect, a prearranged signal must be agreed upon between the head referee and lifter (example: the head referee physically touches the lifter for the press and rack command).**

The bar will be lowered to the chest (not the belly), held motionless, and the lifter will receive a "Press" command.

7. **The lifter must then return the bar to arm's length with no excessive/immoderate uneven extension of the arms. When held motionless in this position the audible command "Rack" shall be given together with a backward motion of the arm.**

The lifter will press the bar relatively evenly and then hold it until the "rack" command is given.

8. **If anatomically, the arms cannot be fully extended, the lifter must inform the Head Referee prior to their first attempt.**

If the lifter can't lock their elbows they must inform the head judge prior

to lifting.

Causes for Disqualification of a Bench Press:

1. **Failure to observe the Head Referee's signals during or at completion of the lift.**
2. **Any change in the elected lifting position during the lift proper (e.g., any raising movement of the head, shoulders, or buttocks from the bench, or movement of the feet on the floor/blocks/ plates/or lateral movement of hands on the bar.)**
3. **Heaving, or sinking the bar into the chest after it is motionless in such a way as to make the lift easier.**
4. **Any pronounced/exaggerated uneven extension of the arms during the lift.**
5. **Any downward movement of the whole of the bar in the course of being pressed out.**
6. **Failure to press the bar to full extension of the arms at the completion of the lift.**
7. **Contact with the bar or the lifter by the spotter/loaders between the Head Referee's signals, in order to make the lift easier.**
8. **Any contact of the lifter's feet with the bench or its supports.**
9. **Deliberate contact between the bar and the bar rests support.**
10. **Failure to comply with any of the items outlined under the Rules of Performance.**

All of the contested exercises are important.
You can't neglect anything.

Benefits of the Bench Press

If the bench press is perhaps the most popular exercise performed in a commercial gym, it must have some significant benefits, right? From a powerlifting point of view, it is true that the bench press generally contributes the least to one's total, but we are only talking about 3 lifts here; all of the contested exercises are important. In a sport where victory or defeat might be a difference of 10 lbs or less, you can't neglect anything. In addition, if one isn't built for one lift, one may be built for another and you have to take advantage of your natural biomechanics to most improve your total. In general lifters with shorter arms and a barrel chest are more ideally suited for a big bench press, but I have seen enough long armed, elite benchers to acknowledge there are many exceptions to that guideline.

From a health point of view the bench press offers numerous benefits. Sometimes people knock the bench press for not being functional — meaning it does not mimic what we do in real life. While there is some truth to that (and remember the same can be said for squats) increasing one's bench press ability transfers over quite well to almost every other upper body pressing motion there is. Getting better at the bench press makes push-ups easy, makes using dumbbells easy, makes performing shoulder presses easy — if it improves all of those things in the gym it very likely will transfer to other abilities as well. Studies have consistently shown that starters in sports that have a strength component (like football) are stronger than those sitting on the bench. If you ever run into someone who poohpoohs the bench, ask them what their favorite exercise is for strengthening the upper body. I am willing to bet the bench press will help increase one's ability in that exercise better than their favorite exercise will help increase the bench press.

The bench press is a great mass builder for the upper body, particularly for the chest, shoulders, and triceps. These muscles don't only look good,

they can help in a large number of sports as well — from wrestling to basketball to soccer. A bigger, stronger athlete is less likely to get pushed around on the field and can better impose their will on an opponent.

The bench press strengthens the bones of the shoulder girdle, arms, wrist, and hands — the wrist is often injured during falls as someone tries to catch themselves. Because the bench press isn't usually limited by mobility issues (which the shoulder press might be) almost all populations can perform it. It is a very versatile exercise that is easy to judge, has a hugely positive transfer to any other upper body exercise, and it deserves its place as one of the big 3.

A bigger, stronger athlete can better impose their will on an opponent.

Apparel for the Bench Press

Dress for the bench is less important than for the squat and the deadlift; the guidelines are generally pretty simple.

T-shirt – Wearing something over the elbows can be distracting when performing upper body exercises, makes it harder to judge the proper range of motion, and is illegal in all federations. Lifters usually do have to wear a T-shirt under their singlet in a meet due to hygiene rules.

Shorts – This is the standard gym wear but one can bench in pretty much anything — shorts, sweats, even jeans or work pants. In a competition the lifter will be wearing a singlet which allows the judges to see if their butt leaves contact with the bench press pad.

Shoes – Regular tennis shoes, sneakers, or cross trainers generally work well for the bench press. The main thing is that the shoe is comfortable and has a good grip. When the lifter drives with their legs you don't want the shoe to slip off the ground, thus losing force and also earning a red light for foot movement.

Wrist Wraps – This is the most popular form of powerlifting apparel used on the bench press. Lifters will very often wrap their wrists when performing heavy working sets or maximal attempts; the wrap will help the wrist stay stable and also will support the forearm. Generally start by wrapping the wrist a little low and then with each wrap gradually move up the wrist. Most organizations don't allow the thumb loop to actually be on the thumb when lifting; get in the habit of taking that off once the wrap is secure. If you like to wrap your wrists it is worth investing in a pair of good solid wrist wraps from a powerlifting-specific store versus those found in a regular sporting

goods store. APT, Inzer, Titan, and EliteFTS all sell high-quality wrist wraps. Different organizations have different rules about how long wrist wraps can be; 50 cm seems to be the standard length that most accept.

> ***You just can't hide a good bencher in a T-shirt.***

Summary

I am not totally sure why the bench press has become the common exercise for the lay person to assess strength. Maybe it is because that is one lift they have performed themselves? Maybe it originated as a way of cheating when asked the question “how much do you press” decades ago when the press implied the standing military press? Personally I think it is most likely because when you are good at the bench press, that is the toughest thing to “hide” in normal clothes. When you are standing next to a 400+ lb bench presser, you are very likely going to know it. The thick, broad chest; the muscular arms; likely accompanied by a stand tall posture — you just can’t hide a good bencher in a T-shirt or office clothes. This isn’t automatically true for someone built to squat or deadlift; a discerning eye will catch the mandatory physical changes that those lifts bring about but it won’t “punch you in the face” so to speak. For better or worse, the bench press is a lift that defines how strong one is. As powerlifters, we might as well accept that, embrace the lift, and work hard to dominate it.

Beginning Bench program for a normal adult male

Week 1	45x12	75x10	75x10	75x10
Week 2	45x12	75x10	75x10	85x10
Week 3	45x12	75x10	85x10	95x10
Week 4	45x12	75x10	95x10	105x10
Week 5	45x12	75x10	95x10	115x10

Beginning Bench program for a normal adult female

Week 1	25x12	35x10	35x10	35x10
Week 2	25x12	35x10	35x10	45x10
Week 3	25x12	35x10	45x10	50x10
Week 4	25x12	35x10	45x10	55x8
Week 5	25x12	35x10	45x10	55x10

Note: The regular barbell in a gym is 45 lbs; often gyms will have

preloaded smaller barbells of 15, 25, and 35 lbs that can be used if 45 lbs is too heavy for that workout.

Many people reading this will be well beyond the beginner stage; if so, they should refer to program design chapters in this book for a more detailed and advanced bench press routine.

Interview with Jennifer Thompson

Jennifer Thompson is the strongest pound-for-pound female bench presser in the world. When she talks, you should listen. This interview was conducted via email. I sent her these questions and these are her responses to them.

Provide us with a brief history of yourself.

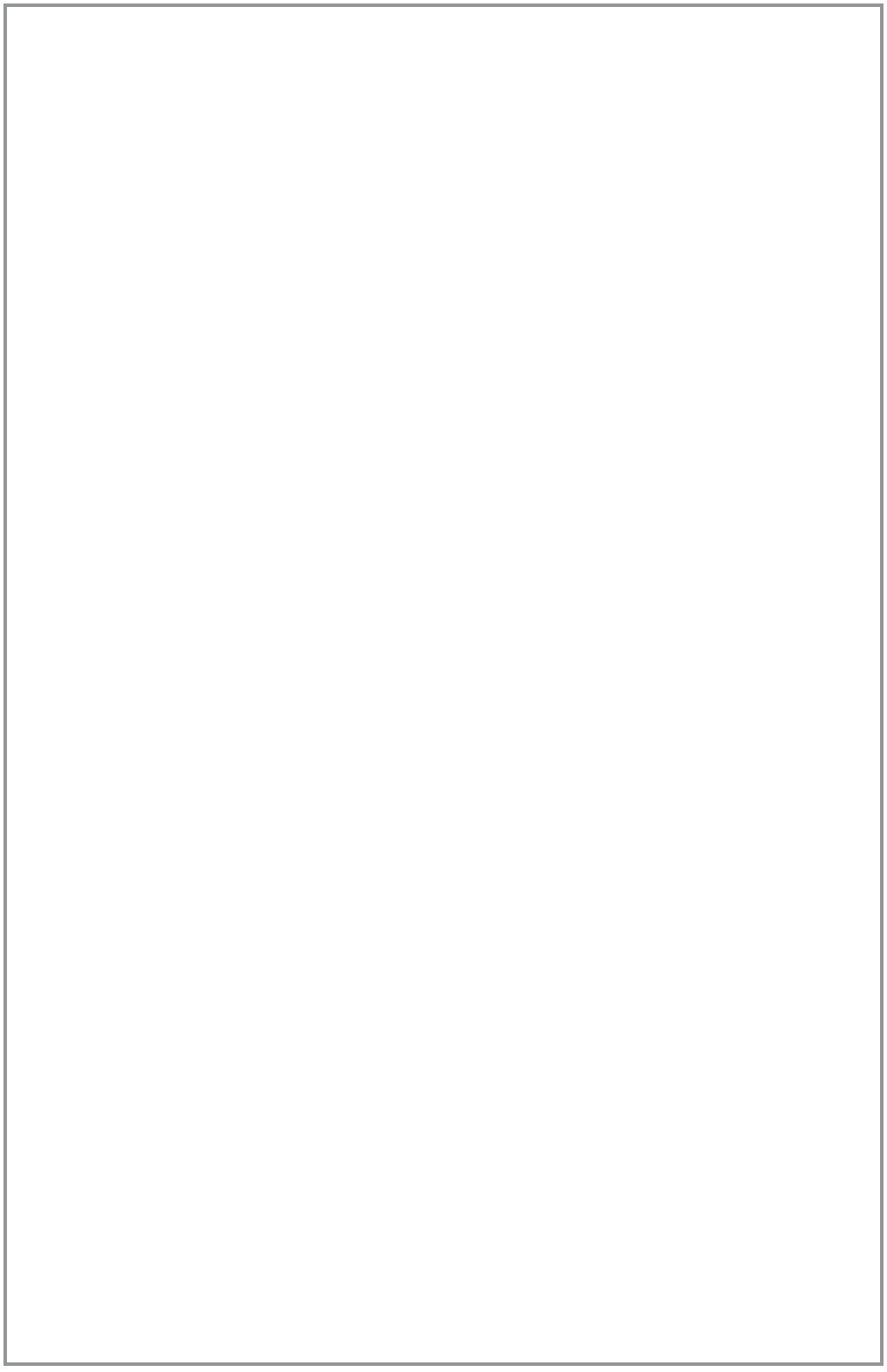
I am currently 39 years old, I was born in Southfield, MI and I spent most of my formative years in Rochester, MI. I am currently a high school algebra teacher and I live in Denver, NC. I train in the basement of my home with my husband, Donovan.

What are your best lifts?

My best raw bench press is 301 lbs in competition. My best equipped bench press is 331 lbs in competition. We do heavy holds with the bench press where we unrack the weight for 15 seconds. I have done this “static hold” with 515 lbs. My best bench workout was three sets of flat bench with 265 lbs.

List some of the titles and awards that you have won.

- Currently I have 16 USAPL National titles
- 4 IPF World Bench Press titles
- 6 IPF Silver medals
- 10 World Records
- 54 American Records
- Highest raw bench press coefficient 2.28
- Lightest female to bench press 300 lbs raw



Jennifer Thompson brings home Gold for the US

Photo credit: Jennifer Thompson

My proudest moment is the 300 lb competition lift. I have worked at it for years and it finally came to fruition. It was a world record and made me pound for pound the best female raw bencher.

When did you start training; when did you first compete; what were your first competition lifts?

I started seriously training in 1999. I sort of messed around in the gym to keep in shape, but 1999 was my first competition. It was a competition in my hometown of Detroit, Michigan. I almost bombed out in the squat with 185 lbs. I was too short on the first two and I think they were being gracious on the third. Then I benched the American record of 254 lbs and I did a deadlift of 295 lbs. I had actually never deadlifted before this meet.

How much weight did you lift the first few times you tried the bench press?

I first started benching in college. I started with the dumbbells because the bar was too heavy. I quickly moved onto reps with 65 lbs.

What was your first training program like and how far did that take you?

Looking back, my first training program was awful! We were lifting six days a week with one day off. We did every body part three times a week. We spent a lot of time being overtrained. My numbers never increased too much and I really started dreading the gym.

How much were you benching when you first hit a plateau in the exercise? How long did you plateau there? What did you do to get past that plateau?

For a couple of years I would work my bench up to 210, 215 every cycle — I just couldn’t break those numbers. I was stuck at these numbers for about a year and a half. Once I had completed college and wasn’t working two jobs, I started seeing a change in my lifting. I started teaching middle school and had more of a regular schedule where I could get good, consistent sleep. I think people discredit how important sleep and diet are.

Give a history of the progress you have made in the bench press since you started lifting.

In the beginning, like most athletes, I made big gains quickly. It was so long ago, it is hard to remember. But I do remember hitting 135 for reps of five and that was such a huge success to have the big plates on the bar. I worked out with my husband and a bunch of his buddies so they were really excited they didn't have to change the weights. A few years after that I made my way up to 225. We started incorporating our speed and heavy weeks. After we made this change I started moving up to 250. Now I was the one having to take weight off for my husband's buddies! Hitting these weights was beyond my wildest expectations. These last few years the gains have been smaller. About five pounds a year in my sets, but I guess when you are 135 lbs and repping out 270, that's not too bad.

What do you feel is key to being successful in the bench press?

The key is having good form. Getting a good arch to raise your chest up toward the bar. Having good hand placement. Squeezing your shoulder blades together to stay on top of your shoulders as high as you can. Digging your heels in for optimum leg drive. The other key is the speed of the bar off of your chest. A lot of athletes miss this. The study of physics says force equals mass times acceleration. In order to generate the optimum force, you have to have speed. We really work on this in training. Digging in hard with your heels, keeping your chest super tight, just waiting to spring the bar off. Then using as much force as you can generate to push the bar to the top as fast as you can. When I am going for a max lift, if I don't generate enough speed I'll miss.

What is your current training program like (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

We have a couple of different training programs that we do. One is purely to build our core muscle strength. I do this on a cycle that I am not training for a competition. It starts with reps of ten and over a 12 week cycle you work your way down to five reps. See the workout spreadsheet at the end of this interview.

When I am on a competition cycle, I work out on an eight day cycle alternating a heavy and a speed workout. We train hard for two hours, two days in a row and then take two days off. On my heavy week we have what we call static holds. On the bench we unrack the bar with a super heavy weight and hold it for 15 seconds. The idea is that you are working all of your stabilizer muscles and it gets your body used to holding heavy weight.

On our speed week we use bands to really work on the speed aspect of the lift. We wrap a thick blue band underneath some dumbbells on the bench. This way when you are pressing the weight off of your chest, you have to be fast because the weight gets progressively heavier to the top. We also practice speed presses. I take a light weight, usually close to my heaviest warm-up, and hold it for an exaggerated pause on my t-shirt (you should be so tight that the bar touches your t-shirt, not your chest). Once given the “Press” command you try to push the bar to the top as fast as you can. We do this for five individual reps.

Most of our exercises consist of five to eight reps. We start off with a weight five pounds higher than the last workout. If we start struggling the spotter will help you complete the lift until you have obtained the required reps. One thing that we do is we make sure the spotter maintains the same speed of the bar that the person lifting it is using.

Jennifer exploding up with a nice bench
Photo credit: Jennifer Thompson

It drives me nuts when I go out of town and ask someone at the gym to spot me. Most of the time they leave you hanging with this weight you can't you finish. They are yelling at you, "you can do it, finish it, finish it!" They finally grab it before it comes crashing down on your face. Now you have taxed yourself so badly on this one rep the rest of your workout is awful.

If different from above, what do you feel the best way to train for the bench is for a normal powerlifter (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I always train the same way whether I am training for a three lift meet or a bench only competition. Studies show that when you squat it releases endorphins into your system which helps spur on muscle growth and recovery. Your legs are such a huge muscle using them helps your entire body with growth and recovery.

What do you think of training with a high frequency (3+ times per week) in the bench press? Have you done this, what were your results, and do you believe in it or not?

For me three times a week is too much. I find that I get overtrained and tend to level out quicker. My workouts are pretty intense; it takes me two days of recovery before I am not super sore in my muscles.

Even though we bench only once every eight days, our tricep day is really another bench day since we incorporate close grip bench, board presses and lock outs. We try to use this day to work on the top end of the bench press. So I guess technically we work on our bench twice a week, but with different exercises. It is important to work on all of the different muscles used in the bench press through different exercises.

I find that if I go all out on every exercise this once every eight days system works great.

You have been mentally ready as well as physically ready to lift big weight
Photo credit: Jennifer Thompson

What are your favorite assistance exercises for the bench press? What areas do you feel they work? How often do you do them? How heavy do you go? Do you feel there is a direct correlation to any of those exercises and your lift (e.g. your bench is always 100 lbs heavier than X exercise)?

Our assistance exercises consist of tricep work: close grip bench — flat and decline — (I lift about 40 lbs less than my flat press), tricep extensions, lock outs (6 inches from the top with about 25 lbs over my max for 5 reps), board presses (three sets of five for what I can max with), dips (to failure), pull downs and ins for the back. Believe it or not, a good bencher uses their lats to push the weight off of their chest, so back exercises are important, too. My rule of thumb for workouts is anything I do the long straight bar with I do for 5 reps; anything else is 8. If it is not the long bar, you are working a small muscle group and you want to really work it out.

The bench uses mostly chest, but triceps and lats are important so I think if you are looking for maximum output you have to work every aspect of the lift: All the muscles involved, the top-end strength, and the bottom-end strength.

It is time to go to work
Photo credit: Jennifer Thompson

What are your thoughts on training until failure on the competition lifts? How often would you do that? What about training until failure on the assistance lifts? How often would you do that?

We max out on the bench press in our speed week. I practice three competition lifts on these days so that I am prepared for the meet. It also gives me an idea of where my strength is. When I am planning my competition cycle I pick a number I want to end my bench at. Then I work that number back 5 lbs each speed week. So on my very first speed week I know what my max lift is for that day (it is usually about 25 lbs off of where I want to end the cycle at). We hit three competition lifts at the beginning of the workout. The first lift is 85% of the max (third) lift. I practice like I am in a competition with a “Press” command. The second lift is 95% of the max lift and then I hit the max for the third. Each lift I am focusing on my set-up and speed. My last cycle I had myself ending at 300 lbs bench and I just missed a 302 lbs in the competition. Each week I get stronger and stronger and gradually build my strength toward my goal.

Each exercise I have a number of reps I want to achieve and if I get stuck I will get spotted on the remaining reps. I train to failure only on my last exercise of the day, whether it is dumbbell presses, dips or pull ups. I don’t go to failure on my other exercises because I find that after the first exercise, my workout is pretty much a lost cause. I gave everything I had to that exercise and I don’t have anything else left for the rest of the workout.

What injuries have you faced and how did you overcome them?

Most of my injuries have come from water sports which in turn then had a negative effect on my lifting. I had to have a cadaverous ligament replacement of my ACL after a bad crash on a wakeboard (I also broke my ankle wakeboarding). I lost a lot of leg drive in my bench for a while.

I also had a pinched nerve in my neck that caused me to lose feeling in my left arm and atrophied my muscles. It got to the point where I couldn't lift my arm over my head (I think this was from the whiplash of hard falls on my wakeboard). So I had to take time off for neck surgery where they shaved down some bone spurs on my vertebrae and made new holes for the nerves to grow through. But really I came back hungry and trained even harder. I haven't had any problems with my lifting because of these surgeries, but I do not wakeboard anymore.

Other than that, I have had a few pulled muscles here and there, but mostly have stayed injury free. I believe this is because I listen to my body and I use really good form on all my exercises. I get as many reps as I can with good form; otherwise, I get the reps spotted. I find that most of the time, athletes get injured in those final reps and they are up on their toes, butt off the bench, doing whatever they can to get the rep.

How important do you feel that nutrition is to powerlifting performance?

Nutrition is a huge part of powerlifting. The workouts can get you only so far; if you integrate good nutrition and supplements, it will get you even further. You have to get enough protein in your diet to build muscle. I try to get 1.5 grams of protein for every pound I weigh. That is about 200 grams for me. As I have gotten older I have had to decrease my carb intake. I try to keep carbs in the same amount as my protein intake or less. When I was younger I kept my meals in a 2:1 carbs-to-protein ratio. Now I am more like a 1:1 ratio. I want to make sure I have enough carbs for energy, but not so much that I gain fat.

My day consists of little meals all day long. I start with a 50g protein shake (Quest Ana Pro), protein bar (Promax); lunch is usually lunch meat, cheese, eggs or dinner leftovers, another protein bar or greek yogurt, dinner and another shake. I keep my diet pretty clean, but splurge here and there. I find if I keep myself on this tight schedule I do better. I like to grill chicken and I can cook it a hundred different ways. We also eat a lot of ground buffalo since it is lean and high in protein. I replace it with anything I would use ground beef for.

What do you usually do with your body weight and nutrition to prepare for a powerlifting competition?

If I have to lose weight I just start dropping the carbs about six weeks out. I keep a record of my intake of the day with the carbs and protein and evaluate it. I monitor my weight daily and make adjustments. If it is an international weight class I don't have to cut weight so I don't change a thing.

The day of the competition I have a very specific eating regime I like to keep to. After I weigh-in, I drink a SSP Post drink. It replaces my electrolytes quickly. I then have Maple Brown Sugar oatmeal with a banana. I drink a SSP Pre drink twenty minutes out from lifting and drink it in between lifts. After the squats I eat another banana and a granola bar with water. I drink another Pre before and during my bench. After that I have some cookies and water before I deadlift. I find bananas are the key to a good competition day.

How important do you feel that supplementation is to powerlifting performance? What kind of supplement program do you currently follow? Are you sponsored by any supplement companies and if so, what are they?

After I had our son, Brody, I had been in a real slump for several years. My numbers weren't going anywhere. Dennis Cieri is a friend of mine and he asked me to be a case study for his new product The System. This was my training cycle right before the 2010 IPF World Bench Press Championships in Texas. When I ran out of supplement I was begging Dennis for more to take me all the way to the competition. So he sent me some of his own supply — great guy. So I won my fourth IPF World titles and have been using it ever since.

I didn't realize how important supplements were until I started using The System by SSP Nutrition. This program has been a game changer for my lifting. I take the Pre drink before and during the first half of my workout. This is the creatine mix with some caffeine and amino acids. It is amazing how energized and focused it has made my workouts. I take the Post drink after; it has tons of amino acids with some protein for a quick recovery and the Maintain is a protein drink for later. I have tried all sorts of supplements for years and this is the first time I have noticed a significant change. I also think you get what you pay for. SSP Nutrition products use the best ingredients science has to offer; you won't find a bunch of fillers like in other products.

The queen of the bench press explaining how to do it
Photo credit: Jennifer Thompson

What are your thoughts on powerlifting equipment (gear) in powerlifting? How do you incorporate gear into your training? Are you sponsored by any equipment manufacturers and if so, who are they?

When I started competing, there was no other option than equipped lifting. No one lifted raw and if you did, you were at a significant disadvantage. I tried all sorts of gear and found that Inzer Advance Designs worked the best for me. It was more forgiving than the other options out there. It took years of working with the gear and conferring with other athletes to find the best way to use it.

When I am competing in a geared competition, I use the gear every speed week. You have to be comfortable with it and it takes some time to learn how to manipulate the equipment to work to your advantage. The difference in your raw to equipped numbers is so large that you really need to use it often to know where your numbers are.

What are your thoughts on steroids in powerlifting? Do you compete in drug-tested competitions? How do you feel about the effectiveness of drug tests for catching those whom use steroids?

I compete only in drug-tested competitions for several reasons. One, I don't use them so I want a fair (as fair can be) playing field. I don't compete in any competitions that are not drug tested, because I don't want there to be any question about my use of it. If you associate with people that use steroids you will be guilty by association. That is just how society works.

If an athlete wants to use steroids that is their choice and I really don't have a problem with it as long as they lift in competitions that allow it. It becomes cheating when you choose to compete in a drug free competition and you are not. I seriously question the character of athletes who do this. To compete against someone you know is clean and you are secretly not, it just makes you an asshole.

As far as the effectiveness of the drug tests, the urinalysis tests are good. It is the timing that we have to work with. I believe steroids stay in your system for one month. So we need to get good at testing people out of competition during their training to really identify athletes using.

However, it seems like science is always a little behind those who create these drugs. Steroids is a problem, but now, mostly on the international level, we are seeing growth hormone. As of right now, there is only an expensive blood test that is available. It is my understanding they are close to creating a more cost-effective test to identify growth hormone. I could think of a few people I would like to sign up for this test.

How would you feel about powerlifting being united? Would you compete in powerlifting if it was united and raw? Would you compete in powerlifting if it was united and single ply? Would you compete in powerlifting if it was united and multi-ply? What do you think the key to unification is?

Honestly, the only way I think you would ever see this happen is if powerlifting got into the Olympics. Only then would people be willing to come together and work under the same organization with a set of rules and regulations. Until then, it will never happen. There are too many variables in this scenario. There are too many people making money running different organizations and competitions. There are too many athletes that want to show up to a meet where they are the only lifter in their category to get their medal. It saddens me to see this. For me, I would be happy to take fifth if I went to a competition, had the day of my life and took fifth to the best powerlifters in the nation. Now people won't come to a National event if they don't think they will win. It irritates me to see organizations hold a "World Championships" and one guy from Canada shows up. Then they walk around spouting about being a World Champion and their flight was two deep — ridiculous. It belittles our sport and the meaning of achieving such a high standard.

What is your thought about the importance of having a workout partner(s) or teammates in helping you train for powerlifting? Whom do you currently (or did you) train with?

Having workout partners is priceless. They are good for motivation, timing between sets, checking form, bouncing ideas off of. Currently, I work out in my gym basement with my husband (he is the ringleader), Amanda Padgett and five high school football players who live in our area. I have to say, getting these kids to come workout with us has been the best. They are young, motivated and full of energy. I truly enjoy these times we hang out, work out, and razz each other in the gym.

What is your thought about keeping a training journal while powerlifting? Do you currently (or did you) keep a training journal?

We keep a spreadsheet of our workouts. On the spreadsheet we have our exercises listed at the top. Each workout we record the weight we used and how many reps we achieved. I don't know how you would be successful and not use some sort of documentation. I write down if I need to go up the next workout or I need to stay at that weight and work at it.

One of the rules we have in our gym is that if you don't improve your sets two consecutive workouts you have to drop 15–20 lbs. on that exercise. I think this has been one of our keys to success. If you are pushing the same weight over and over and not improving, you are getting slow. The idea is to drop the weight and really work on the speed of the lift. We have found when we do this we will quickly pass the number we were previously stuck at.

What books, websites, or coaches do you suggest or follow in your lifting and what would you suggest other lifters do to learn more about lifting?

There is a great site for women who want to powerlift, but men could benefit as well. It is called Powerlifting for Women http://goo.gl/O2wqu. I have a section in it as does the great Sioux-z Hartwig-Gary. It has great workout routines, how to prepare for a competition, videos on the three lifts, interviews. I also have read *Body, Mind, Master: Training for Sport and Life* by Dan Millman. I have taken a lot of pieces of this book and used it when I am training and on competition days.

As with all things on the Internet, you have to be careful what you read and believe. Also, with powerlifting, we are all different in our strengths, genetic build up and our living situations. So you have to pick and choose what is going to work for you.

What do you attribute your personal success in powerlifting to?

Mostly, I think my success is based on a few things. One, is that I am incredibly dedicated. I don’t ever miss a workout. If I am tired and I have a bad day, I take it out in the gym. Two, I go all out every workout. There is not a time that I am not sore from the workout the day before. I am constantly tearing apart the muscle and rebuilding. Three, I have Donovan Thompson, who believes that I can accomplish incredible things and constantly voices it. Four, I am constantly changing my routines, I talk with other athletes, listen to my body and try new things.

What do you feel is crucial to being successful in powerlifting, both in and out of the gym?

The most crucial thing is that you're willing to believe in yourself. There is not room for doubting or constantly questioning what you are doing. You have to believe you can do it; you have to believe you are the best. If you don't have confidence, it is pretty tough to walk out on the stage and have the best performance of your life. If you believe for a minute you can't make the lift, you won't. The mind is a very powerful thing.

What advice would you give to someone who was just beginning to take up powerlifting?

Find someone who is seasoned and learn really great form in the three exercises. To have great form it requires a lot of repetition and constant reminders of what you need to do. It all starts from the beginning. It is hard to go back and correct things. Also, don’t jump too big too fast. When you jump too big too fast, your form is the first thing to go and then your speed is the second. You get a much better workout if you can push three sets of five versus three sets of a slow three.

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

It is all about self-evaluation. What is working? What isn't working? What is my weakness? How can I make it a strength? Take advantage of competitions and talk with other athletes and find out what they are doing. Most powerlifters love talking about our sport and are happy to share.

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

When you are an advanced lifter, you have to constantly evaluate as well. Every competition my husband and I have a debriefing session. Did we pick the right numbers? How was my weight? Did that affect my lifting? Did I have enough speed when I pressed the bar off the bottom? You never stop learning no matter how many years you have into the sport. There is always new science, new techniques, new supplements.

The bar nears lockout

Photo credit: Jennifer Thompson

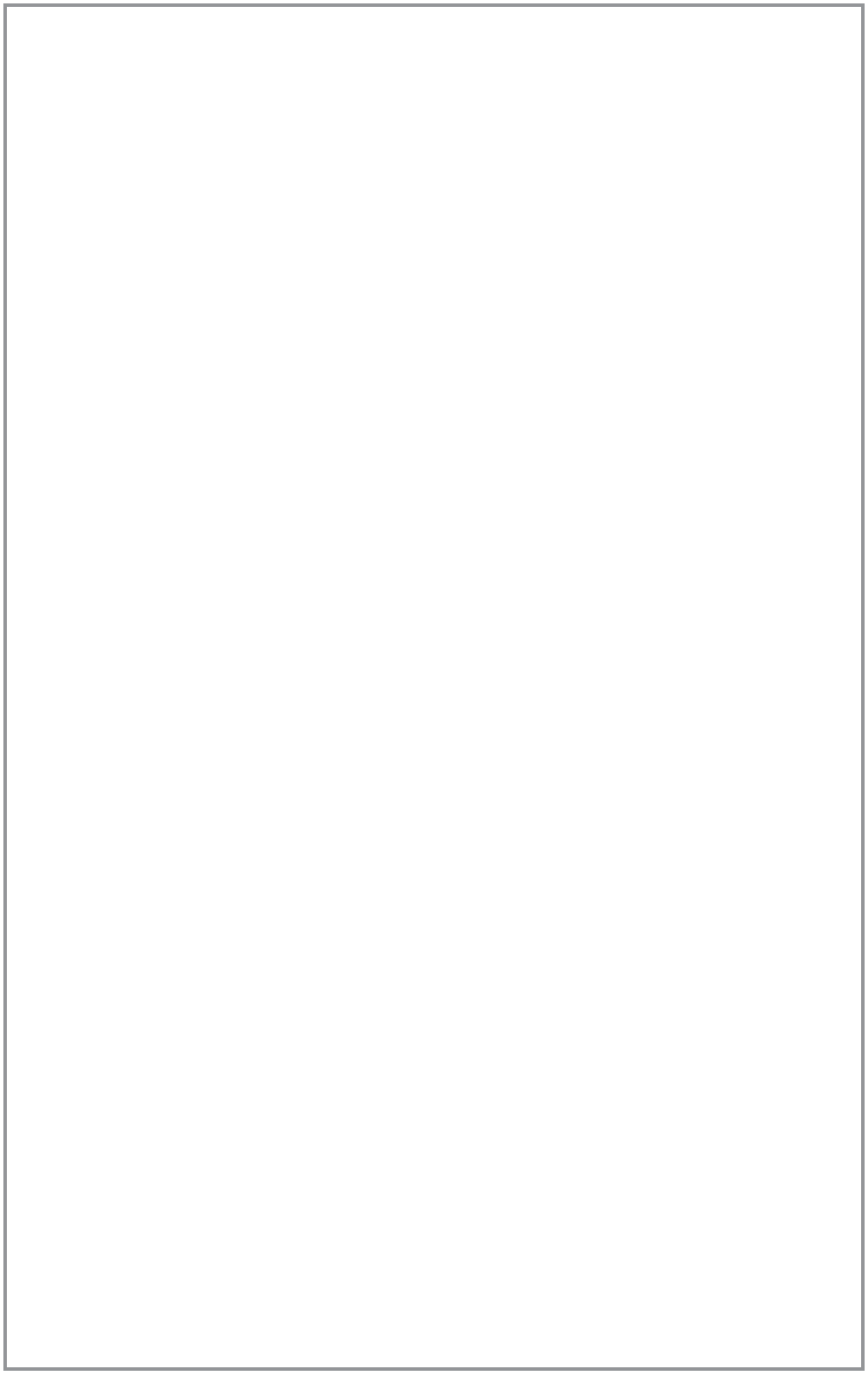
As a female, do you have any specific advice for females who either currently compete or are looking to compete in powerlifting? Do you think females need to follow a different program than a male?

As a female, you need to break the mold that girls shouldn't be in the gym; they shouldn't have muscles. This is an old and changing view, but we still face it from time to time.

If you are a female that does compete, I think it is important to keep some attention on our part of the sport through social media and other outlets. We need to celebrate our accomplishments and be good role models for girls. If we can do this, our end of the sport will see new interest and grow. Remember you won't look like a man by lifting heavy weight unless you are taking male hormones.

The training aspects are recognizing that we do have weaknesses in our physical make-up. We lack shoulder strength and all the little stabilizer muscles in the chest. We do have really good hip strength and a lower center of gravity, which is helpful in the squat and deadlift (this actually improves after having children). However, we naturally have knee instability because usually our quads are way stronger than our hamstrings. This is why you see a lot of young women's knees move in when they are pushing on the squat.

I follow the same workout that all the males do in our gym. I don't think being a female really makes much of a difference when it comes to programming. The heavy holds we do helps with the chest stabilizer muscles, but the guys need to do these, too. We need to build up our hamstrings, but again so do the men. Women really need to work core strength to maximize the hip strength.



Jennifer Thompson

Photo credit: Jennifer Thompson

Are there any changes that you would make to powerlifting if you had the power to do so?

I would make one federation that everyone lifts under. We could have raw and single ply with drug testing (taking drugs is actually illegal). I honestly don't understand the multi-ply lifting; it takes "ridiculous" to a whole new level. All the money would go to running the organization and supporting its workers and athletes in international competitions. Meet directors make a lot of money off these competitions, which they deserve; it is a lot of hard work. However, if we ran the organization more like a business and took the control away from individuals, the group would benefit as a whole.

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

Honestly, it is a wonderful sport. It is a sport you can compete in until you hit the dirt; you don’t time out in your 20s and you can start it at any time. It never gets boring; there is always something to achieve. It is full of awesome people. There is competition among athletes, but you’re still friends afterward. It contains the three main weight lifting exercises that the public understands. It is the truest measure of strength.

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.

Jennifer Thompson Workout Program

CHEST

HEAVY

HOLD	1x15
BENCH	3x5
INCLINE	2x5
DECLINE PRESS	2x5
UPLOAD PRESS (bands)	2x5
DUMBBELL PRESS	2x8

SPEED

SINGLES	3x1
BANDS	3x5
INCLINE	2x5
DECLINE PRESS	2x5
SPEED PRESS	2x5
BOARD PUSHUPS	2 sets

LEGS

WALKOUT	1x15
SQUAT	3,5 (95%),7 (85%)
REVERSE HYPER	3x8
HAM GLUTE RAISE	3x8
or	
HAMSTRING CURLS	3x8

SINGLES	3x1
BANDS SQUAT	3,5 (95%),7 (85%)
BOX SQUAT	3x5
or	
FRONT SQUAT	3x5

TRICEPS/SHOULDERS/CALVES

6″ LOCKOUTS	3x5
CLOSEGRIP BENCH	3x5
PUSHDOWNS	3x8
DIPS	2 sets
SHOULDER PRESS	3x5
DUMBBELL MILITARY	3x8

BOARD PRESS	3x5
DECLINE CLOSEGRIP	3x8
TRICEP EXTENSIONS	3x8
or	
STRAP PUSHDOWNS	3x8
SHOULDER PRESS	3x5
LATERAL RAISES	3x8

SEATED CALF	3x15	SEATED CALF	3x15
STANDING CALF	3x15	STANDING CALF	3x15

DEADLIFTS/BACK/BIS

DEADLIFTS	3,5 (95%),7 (85%)	SPEED DEADLIFT(PLAT)	5x3 (80%)
RACK PULLS	2x5	OPPOSITE DEADLIFTS	2x5
GOOD MORNINGS	2x8	ROMAN DEADLIFTS	2x8
PULLDOWNS	2x8	PULLUPS	2x8
PULLINS	2x8	T-BARS	2x8
BAR CURLS	2x5	DUMBELL CURLS	2x8
PREACHER CURLS	2x8	INCLINE CURLS	2x8

SQUAT	RACK	BENCH	
PINK 65–0(65)	PINK 70–45(25)	PINK 5–30(25) I	P&G 15—80(65)
GREEN 85–0(85)	GREEN 100—60(40)	GREEN 10—50(40)	P&B 35—115(80)
G&P 140–0(140)	G&P 160–60(100)	BLUE 30—85(55)	G&B 40—135(95)

Bench Specialist Dennis Cieri pressing 525 at 198
Photo credit: rawpowerlifting.com

Chapter 6

Increasing the Bench Press

Everybody loves the bench press. Well, unless you suck at it, and then you might hate the bench press. But for most non-lifters and causal lifters, the bench press is the single most common benchmark for strength. The question “How much ya bench?” gets asked countless times at parties, bars, sporting events, even at family gatherings. My goal for this chapter is the next time you are asked how much you bench, you can respond by saying “more than you.”

Just like with the squat, there are many reasons why a lifter fails in the bench. Understanding those reasons can then help us design programs to fix those problems.

The bench press is the most common benchmark for strength.

Technique

Technique is just as important with the bench press as it is on the squat. For a detailed description of the proper technique for a bench press, please refer back to Chapter 5. Here I outline the key points to focus on.

- Keep chest up and shoulders retracted throughout the lift
- Position yourself with your eyes under the bar
- Plant the upper back and head hard into the bench
- Arch lower back as able and desired
- Plant feet flat on the ground, heels dug in
- Grip the bar with a strong grip, evenly spaced
- Get set for the lift-off
- Inhale, lift the bar off the racks and bring it over your chest with arms straight
- Exhale if necessary, inhale again and lift the chest even higher
- Lower the bar under control to the chest, nipple level
- Keep the elbows under the bar and the upper arms tucked in
- Keep wrists relatively straight; they may bend back slightly
- Lift the chest to meet the bar
- Pause the bar on the chest, stay very tight and keep the bar motionless
- Drive the bar off the chest using your chest and lats
- Use your leg drive just before the sticking point; keep hips down
- Begin to flare your elbows slightly just before the sticking point
- If desired, roll your wrists slightly forward just before the sticking point
- Press the bar up and back so it finishes above the upper chest
- Press the bar evenly until the arms are locked straight
- Hold the bar in the finished position for a moment
- Rack the bars with arms straight upon completion of the lift

Neuromuscular Coordination Techniques

To help focus and improve your technique, first and foremost try to make every bench press a good rep. You don't have to pause all of your benches but for the most part they should look like competition presses. Don't let your feet move or your hips come up when training and then expect to perform flawlessly on the platform.

To help increase the force generated at the beginning of the exercise and through the sticking point, lifters use a variety of techniques usually with a focus on bar speed. One way to work on this is to perform a dynamic day, which is a key component of the Westside Barbell Club training philosophy. On this day you use lighter weight (~50–70% of the 1RM) and you press the weight as fast as you can to increase the amount of external force demonstrated by the bar. Lifters will sometimes include plyometric drills as a part of their training for the same effect. These can be performed on their own or occasionally in between bench press sets. Common plyometric drills are clap push-ups and their variants, medicine ball throws, medicine ball catch and throws, and bench press catch and throws (either free weight or on a machine). The idea here is to teach the muscles to fire all at once and to hopefully teach the body to recruit the more dormant, larger type IIB muscle fibers during the lift.

Just like in the squat, bands and chains can be used on the bench press to teach lifters to accelerate through the sticking point with the idea of the resistance increasing over the range of motion (this is called accommodating resistance).

Principle of Specificity

The principle of specificity also applies to the bench press. If you want the lift you are doing to improve the bench, it has to train some specific part of that lift. Exercises that most closely match the bench will include a barbell traveling in a similar bar path. They may match the ROM on the bench or they may just include part of it. Remember the problem of partials is that you only get stronger in the ROM you train in — if you spend all of your time doing rack presses, for example, then you will get stronger on them but it will not improve your actual bench very much.

Exercises and their benefits

Paused Bench Press – This is the regular competition powerlifting bench press, which includes pausing the bar on the chest for a second. This lift, combined with touch-and-go's, will make up the majority of your training for the exercise. I would suggest you primarily use the pause bench press if you are weak right off the chest and/or if your pause bench press is more than 5% lower than your touch-and-go bench press. Pausing your reps works great for increasing the size of your chest as well.

Touch and Go Bench Press – This the common gym bench press; the form is as described above but instead of pausing the bar on the chest as one does in a competition, the lifter just lightly touches their chest and then immediately presses the bar upward. This version is a little bit easier than the pause bench press. The benefit of this version is twofold, you are using slightly more weight than you would with the pauses, and it will train the stretch reflex more directly, which can contribute to your strength even during a pause. I would suggest you train primarily with touch-and-go reps if your pause bench is essentially the same as your touch-and-go bench press (which means you are getting nothing out of the stretch reflex).

Board Press (1–5) – The board press is the best way to perform partials on the bench, in my opinion. You lay a piece of wood, usually about 2″ thick and 6″ wide on your chest. The number in front of the board press is how many pieces of wood you have. You bring the bar down, pause it on the board briefly, and press it up. The big benefit of the board press is that it exactly mimics a bench and when the weight is on the board the weight goes through the board and is transferred to the chest, so the lifter must stay tight the entire time. Of course, the more boards used, the easier the exercise. Generally you want to use the boards so that the lifting is beginning just before your sticking point. Board presses are also easier on the shoulder joint and don't take as long to recover from compared to a regular bench press. And if a person is interested in primarily doing partials for whatever reason, I would suggest they use a board to keep it consistent. You can get creative with your boards; basic wood works fine but I have seen people use yoga blocks, foam rollers, rolled up carpet, just about anything that is reasonably sturdy and at the same time will not cause tremendous discomfort when on your chest.

(Left) A 3 board press

(Middle) A foam pad to mimic the board press

(Right) A foam roller to mimic the board press

Rack Press – The rack press is a bench press performed in a power rack.

Usually the bar starts on the rack (like a rack squat), and the pins can be adjusted to make the bar higher or lower. You might find it more beneficial to perform your rack presses with a negative, in which case set it up so you lift the bar off the hooks (the same things that hold the bar during a squat); bring the bar down to the safety bar, pause, and press it back up. This more closely mimics a real bench and it allows for preloading and pre-stretching to occur. The negative of a rack press compared to a board press is that it is easier to shift your form so that it doesn't really mimic a real bench. For example, on a high-rack press, a lifter might really flare their elbows and start with the bar over the throat. The bar will generally never be in that position on an actual bench, but with a rack press that might be the best position to lift it from. Even if the lifter gets better at that movement, it is unlikely that skill would transfer over to a real bench. It is important to try to mimic the bar path of the bench on the rack. The second negative of a rack press is when the bar is on the racks, the lifter can relax and this also does not mimic a real bench; the lifter must stay tight the whole time, this is why I prefer a board press over a rack press. Regular performance of rack presses also tend to cause elbow/bicep pain, be on the lookout for that if you include this in your routine. One benefit of a rack press is if you are lifting by yourself, the rack press is safer; if you fail nothing happens — the weight just rests on the rack.

Floor Press – This is a bench press performed on the floor. The weight lifted is usually about the same or slightly more than a bench press since it is a smaller ROM; it is a lot like a 1 or 2 board press. Lifters usually lie on a pad; be sure not to slam the elbows into the floor on the way down but use a controlled negative, pause on the floor briefly, and then explode back up.

Incline Press – The incline is a classic lift that focuses more on the upper chest and anterior delts, and less on the triceps. It is harder than a regular

A floor press

bench due to using less muscle mass and an increased ROM. Lifters usually lift about 20% less on the incline than the bench, but that can vary considerably. You should find your max on the incline at some point; see what relationship it has to the bench, and see what happens to your bench if the incline moves up. That way you can tell if this lift is a good indicator of your bench strength or not.

Decline Press – The decline version places more emphasis on the lower chest and the triceps, less on the upper chest and the anterior delts. It is generally easier than a bench press, on the order of about 5–10%. If a person mainly trains on the decline, they will find the regular bench tough to do due to the increased ROM; the decline is essentially just a partial ROM bench. Also many lifters raise up the decline bench to further change the angle, which is unnecessary and usually counterproductive. You might be able to lift more weight that way, but the greater the angle, the less transfer it will have to the real bench. Any decline more than 15–20 degrees is unnecessary. The decline is useful for teaching lifters how to tuck their elbows during a regular bench press.

DB Press – The dumbbell press has the benefit of forcing your arms to work together to match their pressing power; it is useful if one arm seems stronger than the other. It also has a slightly greater ROM than the bench because the hands can come together at the top. The dumbbell press can work more

stabilizers than the bench, although keep in mind that the weight lifted with the dumbbells is significantly less, which also means the prime movers are working less. Remember that the less stable the joint is, the less muscular power it allows to cross it. The dumbbells increase instability, which is one reason why your bench is much higher than just combining the weight of the dumbbells. You can perform dumbbells on the incline or decline as well, and you can use a neutral grip if you need practice tucking your elbows in or your shoulders are bothering you.

Band Bench Press – Performing a bench press with bands attached has the same training effect as a band squat. Generally use heavy dumbbells to hold the bands down, or place an extra bar under the bench press and then loop the bands around that bar.

Reverse Band Bench Press – Now the bands are pulling the bar up, making it easier. They give a lot of help at the bottom and less help at the top; this works more at the top of the ROM.

Chain Bench Press – A bench press using chains to create accommodating resistance.

Ideal grip for a closegrip bench press

Closegrip Bench Press – A bench press with a closer grip than normal, which focuses more on the triceps. This grip is usually easier on the shoulder joint as well. Most lifters lift about 5–10% less on the closegrip bench than a competition bench. The hands should be just outside the chest. For most lifters this means putting two fingers on the smooth part of the bar and two fingers on the grip, or about 8–12″ between the pointer fingers. The hands should not be super close together as that will really tax the wrists. Spotters should be aware that failure occurs more quickly with this bench than a regular bench press.

Reverse Grip Bench Press – A bench press performed with a reverse grip, or a supinated grip (the regular bench press grip is pronated). Sometimes this is legal in competition but usually not. For most lifters this grip is much harder than a regular bench. This places a lot of pressure on the wrists; it also emphasizes the triceps significantly and it teaches the lifter to tuck their elbows. Giving yourself a lift-off can be challenging with this grip. Generally your grip is about the same width as a regular bench — just flip your hands around.

Joint stability is very important in the bench press.

Joint Stability

Joint stability is very important in the bench press. The main joints under load are the wrist, the elbow, and the shoulder joint. As discussed earlier, an instable joint will shut off the muscles that cross that joint. The more stable a joint is, the better, from a strength perspective.

The wrist is under load as the weight of the bar is transferred from the hands down to the arms to the upper body. Weak wrists and forearms can give way under significant load, usually by bending backwards on a bench press. Having strong forearms will help (mainly flexors), as will having a strong grip. You can perform exercises to strengthen your wrist along the lines of what arm-wrestlers, martial artists, and racquet users would do. The simplest way to increase wrist stability is use a wrist wrap, which may or may not be legal in your federation.

The elbow is also under a load during the bench press. The elbow is generally a pretty solid joint and it doesn't usually limit the bench, but it can, especially if it is injured through other means. Most lifters interested in the bench will also be training the triceps, which is good, assuming the correct exercises and volume are used, but it is important to train the biceps as well, as they will help keep the elbow stable during a bench press. You can wrap the elbow or use an elbow sleeve to increase stability, although these items are currently not allowed in any powerlifting federation. Keeping the tendon of the tricep healthy is also paramount; you always want to warm-up well when pressing and training triceps, and use caution with throwing movements or movements that require a jarring lockout. Even something simple like bowling can bother a lifter who is not used to that motion.

The shoulder is the joint that is usually the most problematic during the bench. The shoulder is the most flexible joint in the body. That is great for throwing a baseball, but it is not good for benching 2.5x bodyweight.

Because the shoulder is so flexible, it relies primarily on muscles for stability. If these muscles are not strong or if they are injured, the shoulder joint will not function well. Often the rotator cuff is the weak link in shoulder stability, but it could be any number of muscles. In addition, if the strength of the muscles gets out of balance then that can also lead to shoulder problems. If your eccentric strength is about the same as your concentric strength, or if you feel like there are parts of the negative that you cannot control and the bar is essentially free falling, that is an indication of poor joint stability. In addition, if you require a lot of warm-ups and stretches to get ready to bench, if just lying down now and benching the bar or 135 would seem challenging, that is problematic.

Shoulder stability is not the easiest thing in the world to improve, but you can try a variety of things. Gymnasts are known for having good benches without even training them and they usually have very good joint stability. Training with exercises like push-ups, dips, and flys on the rings (or, if you are adventurous, the iron cross) can help build stability. DB presses might help, negatives can help, 1-arm push-ups can help, and direct rotator cuff training might help as well. You need a combination of a higher level of instability combined with the ability to still lift a reasonable amount of weight.

An increase in bodyweight, even if it is mainly fat, will usually yield a higher bench press.

Bodyweight

As with the squat, an increase in bodyweight, even if it is mainly fat, will usually yield a higher bench press. The additional tissue around the shoulder joint in particular increases its stability and allows one to lift more weight. The thicker you get, the less the bar needs to travel to hit your chest. This also means as you lose weight, even if it is mainly fat and not muscle, it is likely your bench will drop, especially if you are already advanced. It will be up to you find which weight class you are the most successful in and which one you enjoy the most. Use the classification charts in this text to see what kind of lifts you would need to perform to be competitive in a variety of weight classes.

Muscles Involved

It is very useful to know the muscles involved in the bench press, as with any exercise. While the bench press can benefit from a solid leg drive, the majority of the movement is produced by the muscles in the upper body. Because there is some confusion as to what role certain muscles have in the bench press, I have listed all of the major muscles in the upper body and listed their relative contribution to the bench press. A score of 5 means the muscle is the agonist or the prime mover. A 4 means the muscle is a strong synergist; it will produce actual movement and it will respond to the exercise. A 3 is a weak synergist or a strong stabilizer. The muscle will contribute a little bit to the exercise and it might show some development from the exercise. A 2 means the muscle is a stabilizer during the exercise. It is contracting during the exercise but it is not producing active movement and it will usually not respond to the exercise. However, if this muscle is injured it can significantly limit the weight lifted. A score of 1 means the muscle is relatively inactive during the exercise.

Muscle	**Contribution**	**Exercises to Strengthen the Muscle**
Pectoralis Major	5 – Horizontal Shoulder Adduction/Flexion	Bench Press, Pause Bench, Incline (any), Dips, Flys
Latissimus Dorsi	4 – Initial Horizontal Shoulder Adduction/Flexion	Pull-ups/Chin-ups, 45° Bent Over Row, 90° Bent Over Row, Dumbbell Row, Pulldowns, Cable Rows
Anterior Deltoid	4 – Horizontal Shoulder Adduction/Flexion	Military Press (any), Push Press, Front Raises
Middle Deltoid	3 – Horizontal Shoulder Adduction	Lateral Raise (any)

Posterior Deltoid	2 – Shoulder Stabilization	Rear Delt Raise (any), Rear Delt Machine
Teres Major	4 – Initial Horizontal Shoulder Adduction/Flexion	Same as Lats
Pectoralis Minor	2 – Scapula Stabilization	Protractions, Push Aways
Serratus Anterior	2 – Scapula Stabilization	Protractions, Push Aways
Traps	1 – Scapula Retraction	Shrugs (any), Retractions, Depressions, Deadlifts, Farmer's Walk
Rhomboids	1 – Scapula Retraction	Pull-ups, Lat Pulldowns, Cable Row, DB Row
Coracobrachialis	3 – Horizontal Shoulder Adduction	Bench, Flys, Cable Xover
Subscapularis	4 – Horizontal Shoulder Adduction	Bench Press, Cable Xover, Internal Rotations, Flys
Supraspinatus	2 – Shoulder Stabilization	Lateral Raises (pinky up)
Infraspinatus	2 – Shoulder Stabilization	Chin-ups, DB Row, External Rotations
Teres Minor	2 – Shoulder Stabilization	Chin-ups, DB Row, External Rotations
Triceps	4 – Elbow Extension	High Board Press, Closegrip Bench, Dips, Rack Press, Skull Crushers, Tricep Pushdowns
Biceps	2 – Elbow/Shoulder Stabilization	EZ Standing Curl, DB Curl, DB Hammer Curl, Reverse Curl
Forearm Flexors	2 – Wrist Stabilization/ Flexion	Wrist Curl, Wrist Roller, Grippers
Forearm Extensors	1 – Wrist Stabilization	Reverse Wrist Curl, Wrist Roller

Smaller muscles will recover faster than larger muscles.

Optimal Frequency of the Bench Press

The bench press does not require the same skill as the squat. This in general decreases the necessary frequency for which the bench press must be trained. However, the bench press involves smaller muscles than the squat, and smaller muscles will recover faster than larger muscles. This increases the possible frequency for the bench press. Anecdotally speaking, the bench press tends to respond better to higher vs lower frequencies. Presented below is a theoretical training guide for the optimal frequency for the bench press.

As with the squat, relative newcomers should start off benching at least once and up to three times a week to work on their technique and neuromuscular coordination. This will be combined with only moderate intensity. Please note that the intensity presented on this chart refers to total intensity (see Chapter 4 for a more detailed explanation) and not just a % of the 1RM. Then as the lifter becomes more skilled, the frequency will decrease. The lifter will want to be training intensely to learn how to recruit the high-threshold motor units, the lifter will be getting stronger and the greater weight will produce more damage to the muscle, and the lifter will be gaining muscle, which will require a longer recovery. Now the lifter should be benching once or twice a week. Once technique is near optimal and the strength levels are high, the lifter will have to reduce the frequency of intense training even more. However, to make up for this reduced volume, the lifter can increase the frequency of lower intensity training, so the lifter might

bench press 2 times a week but go heavy only once every 4 times.

The performance of other exercises can affect the recovery of the muscles that perform in the bench press. The most common exercise is the military press. Dips, inclines, dumbbells, and plyometrics on other days can all affect recovery. Personally I have found that I can bench heavy once a week and generally recover fine. I have found that I can bench heavy on one day of the week and often bench relatively heavy on a second training day if it is 3–4 days later and still recover okay, especially if it is a slightly different exercise like a closegrip bench. I have found I recover very well with one heavy bench day a week and one medium bench day a week. I can bench 3 or more times a week but the intensity has to be modulated smartly; for me that is something like the "Super Bench Program" outlined below. However I have found that performing the military press intensely interferes with my recovery on the bench. I usually cannot recover from benching heavy just once a week and then three or four days later performing a heavy military press; it is too rough on the shoulder girdle. Even though the weight is lighter in a military press, it is a much greater ROM and it can put a lot of load on the shoulder girdle, particularly the front delts.

Most people find that regularly benching once or twice a week is optimal. Once a week allows for plenty of recovery but may not provide optimal frequency. Twice a week gives you that extra practice on the bench without going overboard. Your programming needs to be very good to bench more than twice a week with significant intensity, but it can be done.

Your programming needs to be very good to bench more than twice a week with significant intensity.

Specific Training Routines

6 Week Program — This is a very solid 6 week program that I and others have gotten quite good results from; it should have you performing your 1RM for a double by the end of the plan. This plan is performed just once a week and it can be used with almost any barbell lift; it seems to work the best with squats, bench, and curls. This type of plan can also be followed in the general training phase if you want to up the intensity a notch.

http://www.t-nation.com/free_online_article/most_recent/a_6week_squat_bench

Plateau Breaker Program — This program has the lifter focusing on either maximal effort or repetitive effort to break through a bench plateau.

http://www.t-nation.com/free_online_article/sports_body_training_performance/

Super Bench Program — This is the most successful bench program I have ever followed. It was originally created by Christian Thibadeau and I have since modified it after testing it on myself and various lifters multiple times. This program tends to give an honest 20–40 lb increase in 8–10 weeks, which for an experienced lifter is an awesome increase. This is the program that finally took my bench to more than 400 lbs, which isn't spectacular for a powerlifter — I am well aware of that — but given that my bench had been stuck in the mid 3's for more than 10 years, it was a significant accomplishment for me. I suspect some of you reading this might be in a similar predicament.

This program is relatively radical; the lifter is benching heavy 3 times a week and one of those days the lifter is performing 3 separate bench workouts! That is a total of 5 bench workouts in one week. The negatives of this program are that it has a reasonable chance of overtraining in the form of

shoulder injury or biceps/elbow tendinitis (light hammer curls 2–3 times a week seems to help with this) and that it is kind of hard to follow without someone knowledgeable about the program guiding you through it, which is why I have written several articles about it and included tons of videos and resources. Here they are:

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http://www.t-nation.com/free_online_article/most_recent/8_weeks_to_super_be

http://www.t-nation.com/free_online_article/most_recent/how_i_benched_400

I have found that you essentially can't start too light with this program. It is set up in such a way that it will likely be new to you and thus you'll adapt to it. If you go up 10 lbs a week on everything that is a 70–90 lb gain over 8–10 weeks, even if you underestimate everything by 50 lbs, that is still very nice progress in the end.

Pedro Meijas displays impressive strength
Photo credit: Doug Jantz photography

Summary

The bench press is an interesting animal. One tends to experience pretty solid gains on it for a while, but a plateau on this lift is much more likely than with the other 2 competitive lifts even at relatively early stages of training. I don't believe one should try to specialize in increasing the bench press too early. The bench press responds very well to added hypertrophy in the upper body — bodybuilders are often the closest in strength to powerlifters in the bench press due to both their added size and the way they train. Programming for the bench has to be pretty specific once a plateau is reached, and one has to take care of the shoulder girdle to ensure that long-term training can take place to really reach one's potential.

Like all lifts the bench must be attacked; try to train with someone stronger than yourself — just seeing someone else suffer and move heavier weight than you can cause that switch to flip in your mind and can help motivate you to attack the weights just a little bit harder. If you happen not to be great at the bench press, don't shy away from it — embrace that fact, go after it, and share what helped your success with other lifters. We are powerlifters — we don't get to pick and choose what lifts we want to do; we need to dominate all of them to truly be successful. When gym goers and athletes are looking to increase their bench to improve their looks and performance, they should be looking to the powerlifters for the answers to those questions. It is one of our lifts after all; who knows how to train the bench press better than we do?

Paul Bossi throwing around some weight
Photo credit: Doug Jantz

Interview with Paul Bossi

Paul Bossi is an elite powerlifter who specializes in the bench press. He has benched 475 at 198; 515 at 220; and 530 at 242, all raw and drug free. He is also the president of the 100% RAW Powerlifting Federation, which promotes raw, drug free powerlifting. This interview was conducted via email; I sent him these questions and these are his responses to them.

Provide us with a brief history of yourself.

I am 45 years old and I was born in Somerville, MA. I lived there for 6 years before moving to Wilmington, MA where I grew up. I currently live in Camden, NC and I train at the Fitness Warehouse — a gym I own in Elizabeth City, NC.

What are your best lifts?

My best lifts are the following: 181 lb class-410 lbs (Shirt); 198 lb class475 (RAW) 505 (Shirt); 220 lb class 515 (RAW) 565 (Inzer blast Shirt); 242 lb class 530 (RAW) 605 (Denim Shirt). I have lifted RAW only since 2004 and all my shirt lifts were using old technology shirts.

List some of the titles and awards that you have won.

I have won the World Championships in the following organizations (100% RAW, AAU, WDFPF, WNPF, APA). My 1st World title was in 1992, and I have won a world title every year in at least one organization from 1992 to 2011. I have won National titles in 100% RAW, AAU, WNPF, APA. There have been so many titles between the World and National I cannot tell you without doing some major researching.

When did you start training; when did you first compete; what were your first competition lifts?

I started training when I was in 8th grade by lifting weights in my basement in 1979, and my 1st competition was in 1984 when I weighed 155 lbs and benched 300 touch-and-go in a non-sanctioned event in Lowell, MA. My next event was in 1989 when I learned there was an organization called the APA in Vermont that held meets. I went there and lifted RAW in the 181 class weighing in at 175 and benched 315 RAW. I had the bug and have never stopped since.

How much weight did you lift the first few times you tried the bench press?

Me and a bunch of my friends were playing around seeing who was stronger in my basement using my brother's weight set. I weighed 110 lbs and benched 162 lbs RAW. That was my very first day of lifting ever and because I wanted to be a professional football player I have never stopped. Unfortunately pro ball was not in the cards for me.

What was your first training program like and how far did that take you?

All we did were 3 sets of 12 reps. We did a basic push/pull routine, meaning chest shoulders and triceps 3 days a week and back and biceps 3 days a week. We did 1 leg exercise each day.

How much were you benching when you first hit a plateau in the exercise? How long did you plateau there? What did you do to get past that plateau?

My first plateau was 275 lbs for 5 reps; it took me about a year to break that. I just kept working hard and pushed to failure every time I worked out and it finally came. When I went to college I lifted with guys who were stronger for the first time so it made me work harder because I always wanted to be the strongest. It eventually paid off for me.

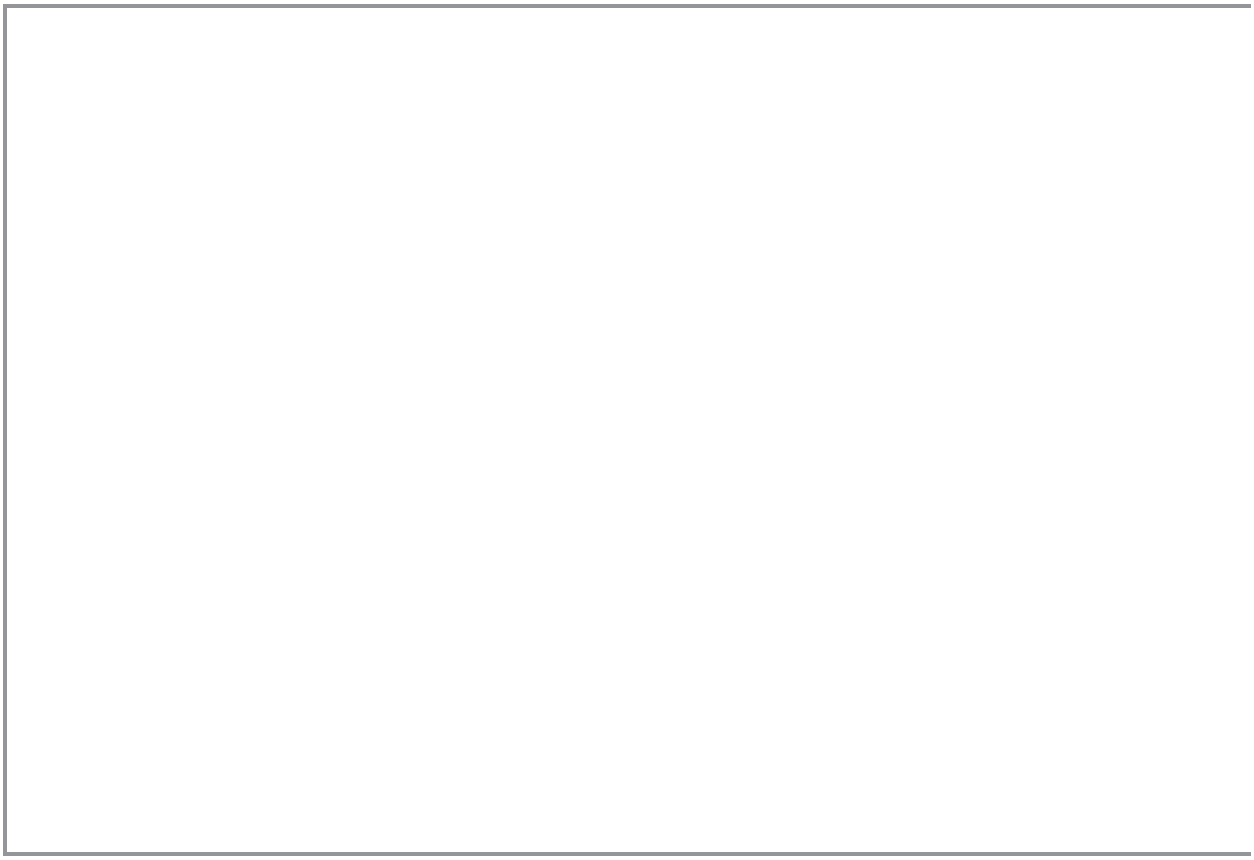
Give a history of the progress you have made in the bench press since you started lifting.

Well, I started off with a 300 touch-and-go in 1984; then in 1989 I got a 315 with a pause at 175 lbs. Then I used a shirt and it took me until 1992 before I hit 400 lbs in the bench. I was stuck there for 2 years as I even went backwards to 390 lbs. In 1994 I developed a routine I named the Reverse Pyramid and I shot up to 440 in 2 months. Then 465 in 5 more weeks and then 480 in another 6–8 weeks. I had realized I was not on a good routine and that I was overtraining, so I wracked my brain and used bits and pieces from articles I read and things I learned on these videos that used to come out every month called "Powerlifting Video." These videos were awesome as they had live footage of lifters across the country and had interviews of the top lifters in the country as well. I gathered all the information I learned and sat down and came up with a routine and hoped it worked. I went from benching 2 times a week and doing all my other lifts 2 times a week to just each body part once a week. In my workout on the 3rd week I saw progress and I made a 10 lb gain. I have been at a plateau for 2 years so when I saw these gains I thought I might have been onto something. As history would have it, I was, and the weight kept coming. In 1995 I hit my first 500 lb bench with a shirt in a competition in New Hampshire. I was in the 198 lb class. Now here I thought the sky was the limit but little did I know I would only make a 5 lb gain over the next 6 years. I was playing semi-pro football and between that and teaching and coaching, it did not leave me much time to train. In 2001 I took some time off from coaching as I was burnt out and I stopped playing ball as well. I went to school and taught and then to the gym after each day. That is when I finally got my bench to move again. Less stress and more time to relax paid big dividends for me as I made my climb back up to the top. In 2002 I hit a 515 with a blast shirt in the 220 lb class and in 2003

I got a 565 with a blast shirt in the 220 class. In 2004 I begin my RAW quest and got rid of the shirts. I hit a 500 RAW in 2004 in the 220 lb class; in 2005 I got 515 in the 220 lb class and then in 2006 I got 530 RAW in the 242 lb class. In 2007 and 2008 I was injured with a hernia and broken wrist and still won world titles hitting 475 and 485, respectively. In 2009 I had surgery to repair the hernia so I can be healthy again. In 2009 I got a 512 before I got a minor pec tear on 518. In 2010 I got a 512 but that was shortly after another injury. 2011 I sat out because of a pec tear again to the same spot, and I was coaching again and did not have the time to train. If I can stay healthy this year, I am looking at getting back to the 520–530 range in 2012. I realize I am not getting any younger and want to stay on top as long as I can before I have to submit to younger, stronger guys.

What do you feel is key to being successful in the bench press?

I feel the key is to have a good routine and follow it. You need to be consistent and not miss workouts and always push yourself to the next level. I call it “Push to Failure” which I instill into the kids I train.



Paul prefers the false grip when benching

Photo credit: Paul Bossi

What is your current training program like (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I train 3 days a week right now. I do bench on Monday and I do only flat bench, incline and decline on this day. I follow my Reverse Pyramid routine of 1 triple and sets of 8–11 and then do my pause sets. On Tuesday I do biceps and back, and on Wednesday I do shoulders and triceps. For legs I do a set here and there during the day in my weight training classes. I normally do leg curls, leg extensions and some squats on the hack machine. I hate the leg press as it is a useless exercise and does not make you a better athlete. I train athletes so I am very conscious to make sure the exercises I use make gains and make the kids a better overall athlete.

If different from above, what do you feel the best way to train for the bench is for a normal powerlifter (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I only promote my routine because I tried many routines and nothing has made me as strong as this routine. Everyone who has ever used my routine has gotten very strong on it, although it is very taxing on your body and it is not for everyone.

What do you think of training with a high frequency (3+ times per week) in the bench press? Have you done this, what were your results, and do you believe in it or not?

I did this when I was younger, in my teens, and, yes, I got stronger. But as a kid your body has an unlimited supply of testosterone, unlike lifters in their mid 20s and older. This is why I feel each routine needs to be tailored to the specific person for their age group as well.

What do you think of training with a medium frequency (2 times per week) in the bench press? Have you done this, what were your results, and do you believe in it or not?

I have done this as well, and it worked, but I hit a plateau and needed to change this. Everybody is different, and what works for me may not work for you. A lifter needs to try it and see if they see results and if they do stick with it; if not, move on, and shock your body with a new routine.

What do you think of training with a low frequency (1 time per week or less) in the bench press? Have you done this, what were your results, and do you believe in it or not?

This is where I am now in my life, and I love it.

Who's the Boss?
Photo credit: Paul Bossi

What are your favorite assistance exercises for the bench press? What areas do you feel they work? How often do you do them? How heavy do you go? Do you feel there is a direct correlation to any of those exercises and your lift (e.g., your bench is always 100 lbs heavier than X exercise).

When I am preparing for a competition I know I need to stick with the free weights and stay away from the machines. The machines will maintain my strength, but they don't get me stronger. I normally lose a little when I use the machines for assistance exercises, but they also give my central nervous system time to heal from the pounding it endures throughout the year from the heavy lifting. My favorite assistance exercises would be incline and decline bench, close grip bench, shoulder press and curls. These exercises are the ones responsible for building my auxiliary assistance muscles and allow me to hit big numbers in competition.

What are your thoughts on training until failure on the competition lifts? How often would you do that? What about training until failure on the assistance lifts? How often would you do that?

I train to failure in all my lifts whether core or assistance exercises. Now when I say “push to failure” I am speaking about going to the point where you cannot get another rep on your own. I am not a fan of having your spotter help you with 1–3 more reps. Hell no, that is dangerous and a good way to hurt yourself. When I do my set of 7–11 reps and if I get 8 reps, that means I cannot get number 9 and it was all I had. All of my sets are sets of 7–11 reps. I only do one set a week of 3 reps in the bench press, I use the triple (3 reps) to maintain my top-end strength. My Reverse Pyramid routine you can get online on my website www.RawPowerlifting.com. If you do less than 3 reps on the bench on a regular basis you will overtrain and deplete your ATP and Glycogen, and you will get weaker and go the wrong way. I try to explain this concept to my kids and tell them not to try to hit their maxes each week as this will have an adverse effect on your body.

What injuries have you faced and how did you overcome them?

I have had a hernia 2 times, broken wrist and some of the worst pain you could ever imagine in my shoulder. I had surgery for the hernias but just let time heal the other three.

How important do you feel that nutrition is to powerlifting performance?

I preach if it runs, swims or flies, eat it. That is protein and little carbs. I tell kids who want to get strong to eat foods with lots of protein in it. I do not follow a strict diet but I sure wish I did.

What do you usually do with your body weight and nutrition to prepare for a powerlifting competition?

I normally try not to lose more than 7 lbs for a competition because it makes me very weak. I have lost 12 lbs before in 3 days, and I was so weak I almost bombed out of the competition. I cut out water and drink very little, eat salad and dry tuna in it. I boil chicken and have egg whites. Then the day before, providing we have an early weigh-in I put on the suit and shed as much water as I can. Normally I can shed 6–7 lbs of water in a couple of hours. Fortunately the last 8 years I have not really had to worry about that as I moved up a weight class and got a little stronger. Now I am on a quest to get back down to 198 from 242 and stay there so I can look better and be healthier. I am aware I will lose strength, but I am willing to sacrifice that.

> *Author's Note: For additional info on dropping weight and maintaining strength, see* *Chapter 13**.*

How important do you feel that supplementation is to powerlifting performance? What kind of supplement program do you currently follow? Are you sponsored by any supplement companies and if so, what are they?

I have 2 sponsors whose supplements I use (Tribustol and SSP Nutrition). That is all I need to keep my body healthy and fit.

Getting set for 525
Photo credit: Paul Bossi

What are your thoughts on powerlifting equipment (gear) in powerlifting? How do you incorporate gear into your training? Are you sponsored by any equipment manufacturers and if so, who are they?

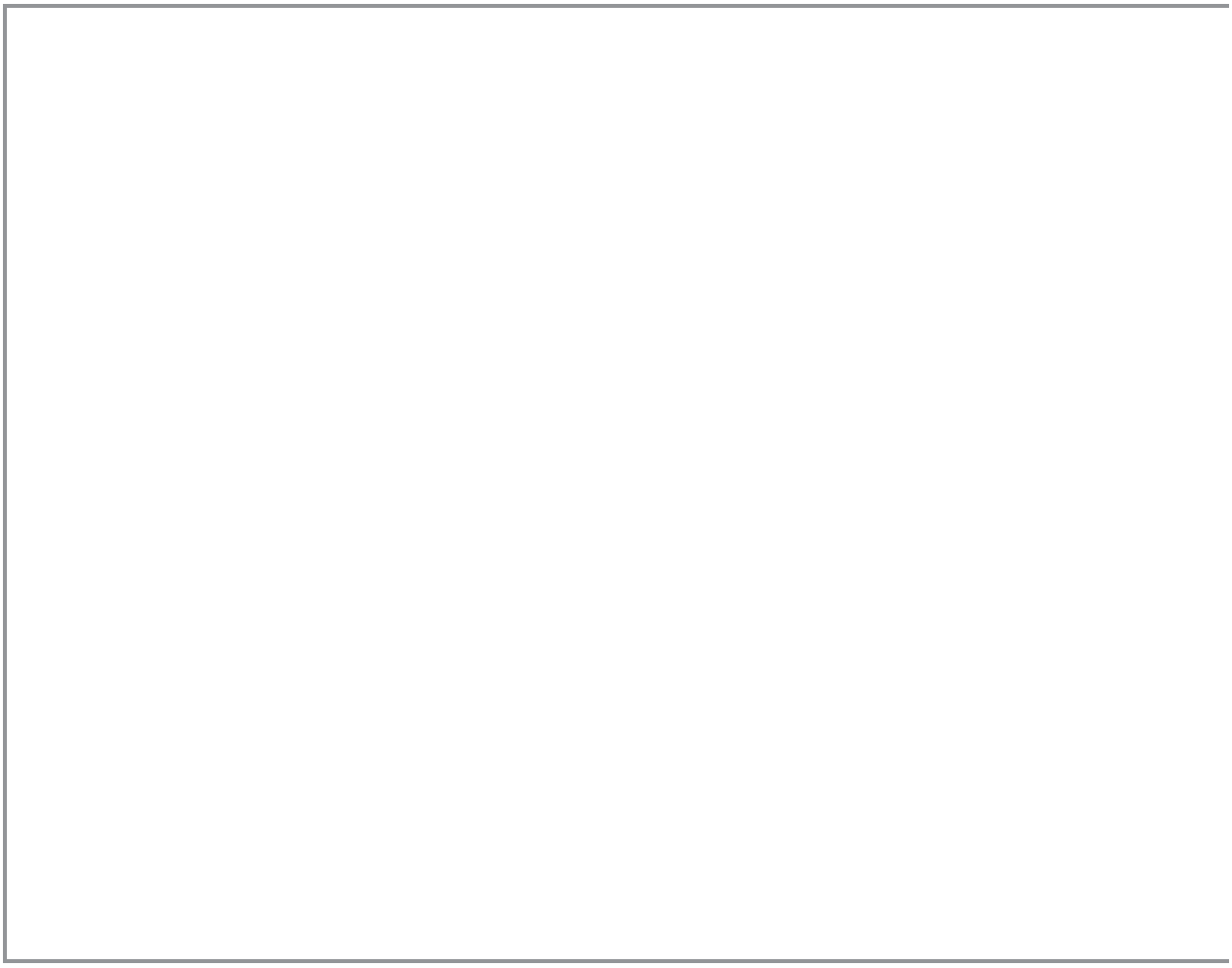
I used to wear gear when I was younger, but after becoming the president of 100% RAW Powerlifting Federation in 2003 I feel I would be a hypocrite if I used equipment now. I have no problems with guys who use it. I just choose to lift RAW.

What are your thoughts on steroids in powerlifting? Do you compete in drug-tested competitions? How do you feel about the effectiveness of drug tests for catching those who use steroids?

I personally do not use them and I promote drug free Powerlifting. I know guys who do use them, and I have no problems with them as long as they do not try to lift in a Drug Free RAW event. I am not into cheating and if a guy uses them and tries to compete against guys who are natural and clean for a trophy, I have an issue with that. There are plenty of federations that do not test, which is good for those who choose to go that route. I feel our tests do a really good job of catching lifters on banned substances.

How would you feel about powerlifting being united? Would you compete in powerlifting if it was united and raw? Would you compete in powerlifting if it was united and single ply? Would you compete in powerlifting if it was united and multi-ply? What do you think the key to unification is?

I would love to see Powerlifting be unified, and we have taken a step in that direction by combining 100% RAW Powerlifting and ADAU Powerlifting in 2012. Unfortunately I do not ever see full unification happening as there are too may greedy people out there. I would love to lift in it as there would be more competition. I do not have the answer to the unification process; if I did I would be promoting it. It is a tough and touchy subject with many people.



It is tough to accomplish anything worthwhile alone
Photo credit: Doug Jantz

What is your thought about the importance of having a workout partner(s) or teammates in helping you train for powerlifting?

You have to have a good partner in order to get stronger. If you have a team and you all train together that would be the best situation you could ever ask for. I would tell any lifter if you have a chance to be part of a team and train with them, then go ahead and do it. The experiences and fun times you will have are priceless.

What is your thought about keeping a training journal while powerlifting?

As a beginner I think all lifters should keep one to see what is working and what is not. Also you can see some progress and you know what the amount of weight each lift is that you did last week. I personally do not have one as I know exactly what I am doing for weights.

What books, websites, or coaches do you suggest or follow in your lifting and what would you suggest other lifters do to learn more about lifting?

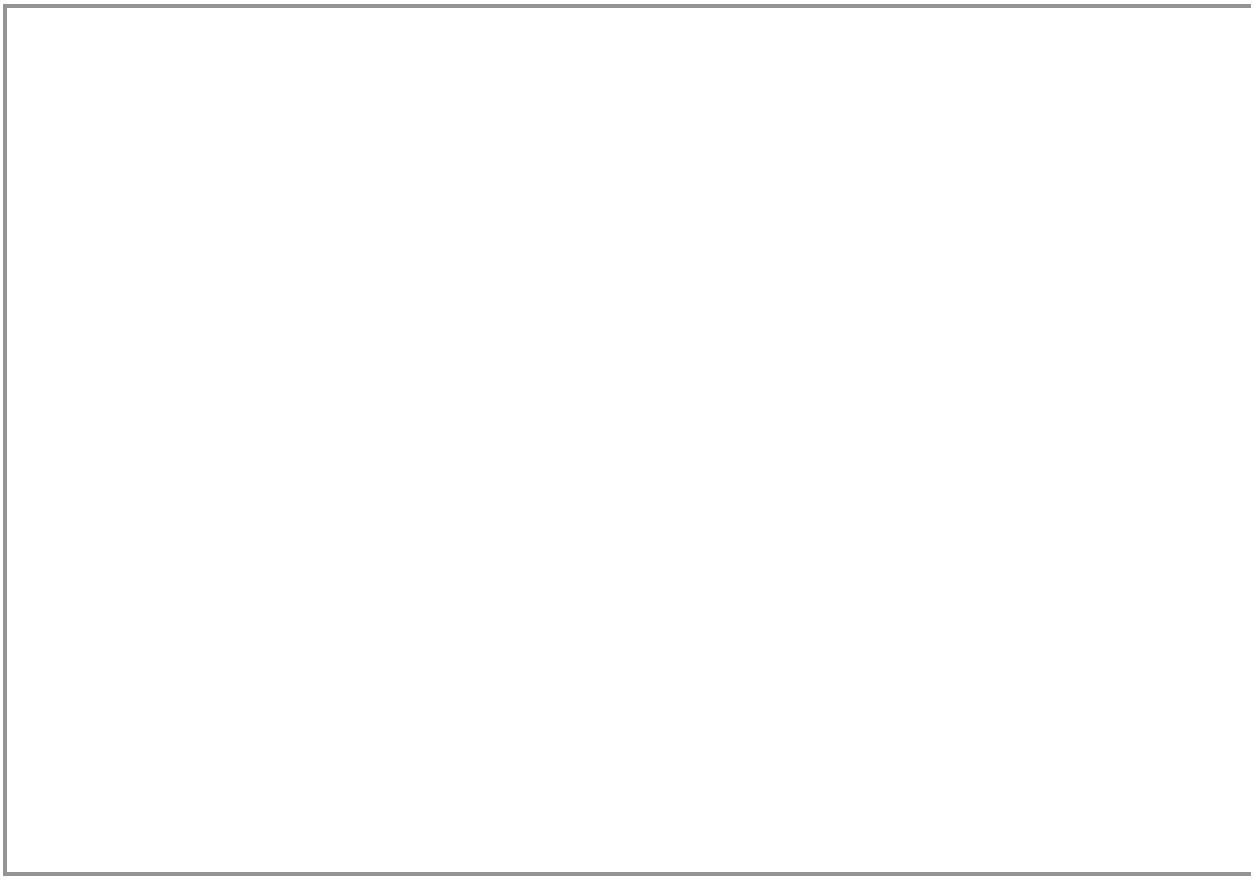
The best thing to do is find someone who is successful and ask them for tips and ideas. There is a lot of junk on the internet and also some lifters do not realize a routine for a guy on steroids is totally different for a guy who is clean and drug free. A drug-free guy cannot recover like a person on steroids. Make sure you get your advice from someone who is the same type of lifter you want to be (drug free or not).

What do you attribute your personal success in powerlifting to?

Hard work and dedication are key to my success. The ability to understand that plateaus are normal and the drive to be the best. Be consistent in the gym and do not take off long periods of time. I might take a week off every now and then to let my body rest, but that is about it. When you're in the gym it's about quality workouts, not quantity.

What do you feel is crucial to being successful in powerlifting, both in and out of the gym?

I tell my kids I train that I want them to be better citizens first and then athletes second. If you're a knucklehead people will not respect you, but if you're a good person people will respect you and support you.



Paul Bossi is built well for the bench press
Photo credit: Doug Jantz

What advice would you give to someone who was just beginning to take up powerlifting?

Work hard, be dedicated and don’t give up. It takes a long time to get results, and you only get out what you put in. Be consistent and have good quality workouts.

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

The same advice as a beginner.

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

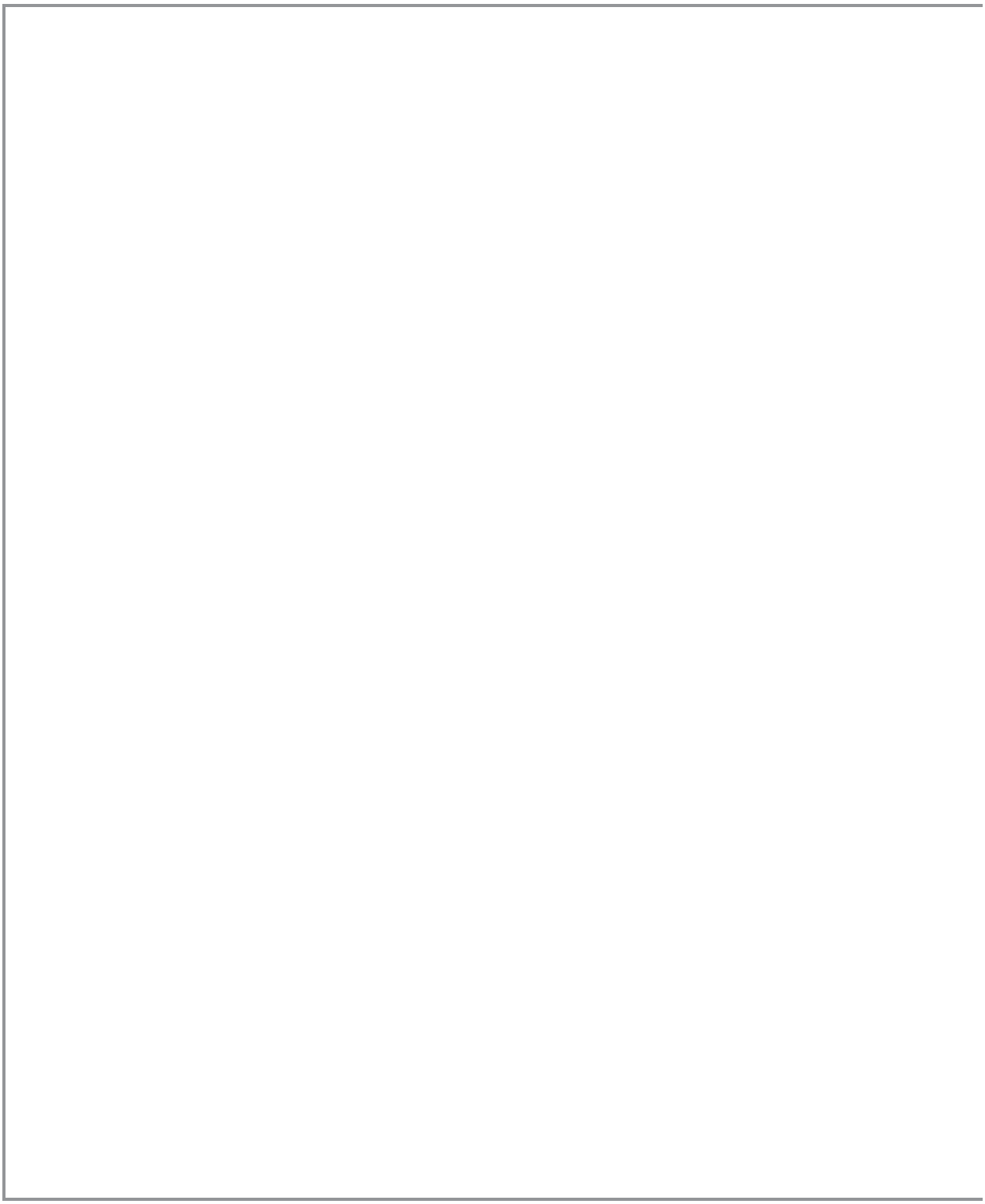
Still the same advice; it will come but you cannot give up. Just keep working, maybe change a few things around and seek someone who has been there, and see what advice they might give you on tweaking your routine.

Are there any changes that you would make to powerlifting if you had the power to do so?

I would love to unify it if I had the power, time and knowledge on how to.

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

Check out the 100% RAW Powerlifting federation. www.RawPowerlifting.com.



Everything you will ever need to know is there. We have a forum for you to ask other lifters, and you can email many of our state chairmen or

committee members for help and advice.

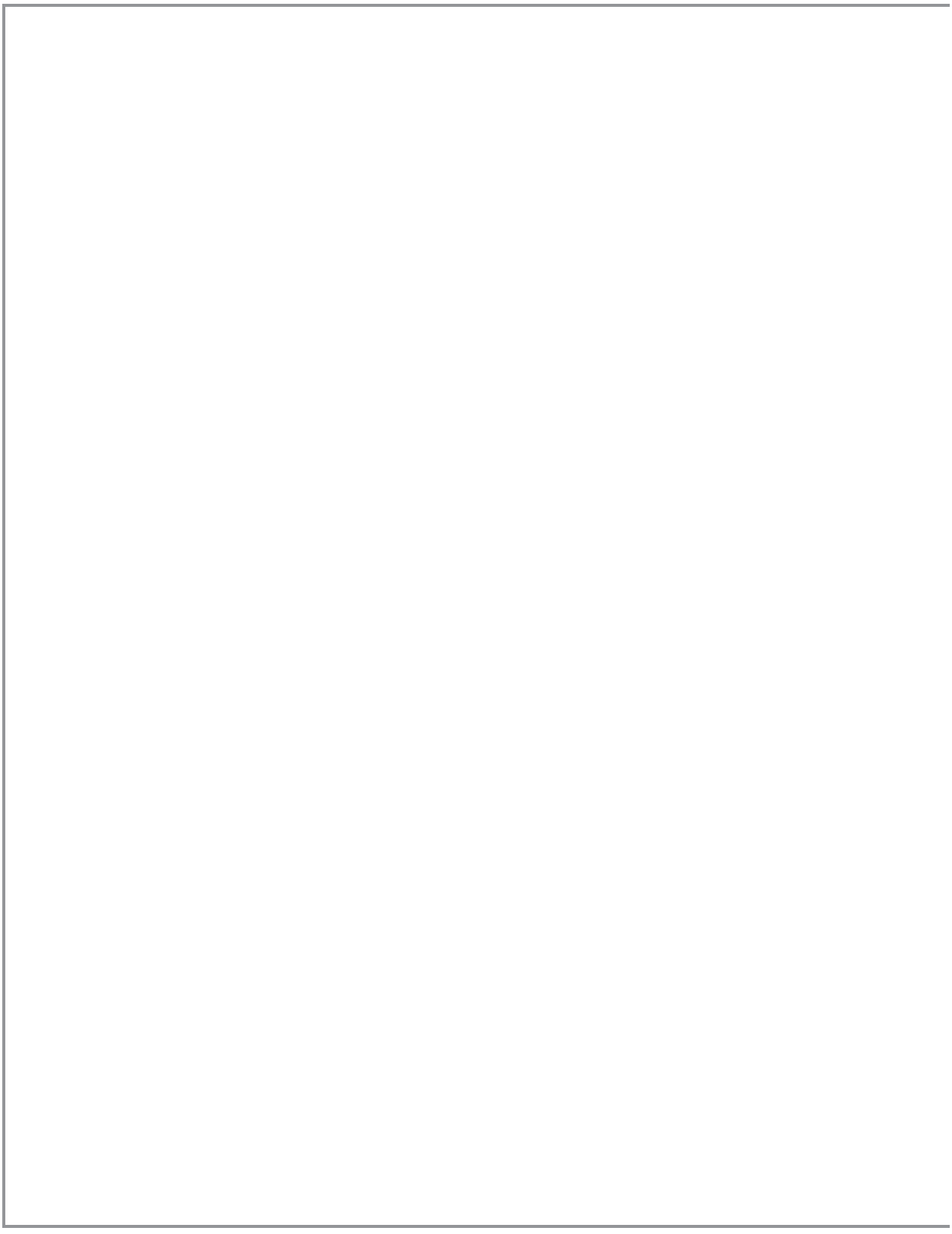
Paul Bossi, President of 100% RAW Powerlifting Federation
Photo credit: Paul Bossi

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.

Chapter 7

The Deadlift

Probably the most famous saying in powerlifting is “The meet doesn’t start until the weight hits the floor.” The deadlift is always the last exercise performed in a powerlifting competition. Generally competitors can lift more weight in the deadlift than they can in the squat or the bench press. A good deadlift can either make or break your total. The saying means that the deadlift is often the key lift that determines where the lifter places in a powerlifting competition.



Kirk Karwoski with perfect deadlift technique

Photo credit: Kirk Karwoski

The deadlift is one of the coolest exercises there is. It is so simple. There is a weight, bend down, pick it up, and stand up straight with it. Yet it is so grueling. Much like a tough set of squats, completing a brutal set of deadlifts will leave you gasping for breath and seeing stars. The deadlift is the simplest of the big three exercises, meaning it requires the least amount of skill. The deadlift is the most natural of the 3 lifts. Essentially all people, regardless of age, will perform a deadlift during their day basically every day. Every time you bend down to pick something up off the ground, from your shoes to your laundry basket to a child to your dog to your couch, if you pick something up you are deadlifting it. In addition your ability to deadlift remains for a relatively long time after you stop deadlifting. If you stop squatting, even after you have been squatting for a long time, your ability to squat very heavy or perform a max set decreases relatively quickly. You will notice a difference after just one or two weeks; take a couple of months off and when you start back it will almost feel like you have never done the exercise in the first place. Your ability to bench press remains at a moderate level; if you stop benching it begins to go away but assuming you stay strong in other exercises you will still be able to bench a decent amount of weight for a while. The deadlift ability remains the longest. If you stop deadlifting you can still come in a year later and deadlift a decent amount of weight. This is because the neuromuscular coordination required for the deadlift is the least specific of the three exercises and the basic motor pattern remains the same even when you stop deadlifting. If you stop squatting, for example, that motor pattern degrades relatively quickly.

Even though the deadlift has the lowest skill level of the three exercises, that does not mean it is easy to learn or

Many see the deadlift as the single best test

that it should be taken lightly. Many people see the deadlift as the single best test of brute strength there is. Just like the squat, extra time should be taken to learn the proper form with the deadlift and to spend time with the light weights, building your foundation, before you move to the heavy weights. This can be tough to do because people can often deadlift a good amount of weight right from the start, and it is fun to lift 2 or 3 plates on each side in a short period of time. But then your form breaks down as you try to go heavier and heavier, and you either hurt yourself or get frustrated by the lack of progress and quit the exercise. If you take the time in the beginning to learn how to perform the exercise properly, it will be much safer for you, and ultimately you will be able to lift much more weight in the long run.

of brute strength there is.

Equipment Set-up

The deadlift generally doesn’t require any sort of special equipment to do; all you need is a barbell and some plates. An exception to this is if one is first learning how to do the deadlift and isn’t using much weight; you don’t want to practice the deadlift with the empty bar sitting on the floor. This will make the bar too low and will make it very difficult to work on proper form. Instead the bar needs to be raised up. Many gyms will have training discs which are the size of a 45 lb plate but they don’t actually weigh that much; more commonly they are 5 lbs or less. You can also deadlift inside a power rack if necessary to raise the barbell up; the proper height of a deadlift is 8″ off the ground, which is the radius of a 45 lb plate.

As a side note sometimes loading/unloading the weight can be a pain. If you roll the deadlift up on a 2.5 or 5 lb weight on one side, that will make sliding the weights on and off much easier and will save your back for the real fun — which is picking all of that weight up at once when it counts — when you are deadlifting it.

Types of Deadlift

There are 4 main types of barbell deadlifts; 2 are commonly performed in competitions and 2 are generally viewed as accessory movements. The conventional deadlift is where the lifter will take a more narrow stance and they will grab the bar with the hands outside of their legs. A sumo deadlift is when the lifter takes a very wide stance and grabs the bar on the inside of their legs (at the bottom of the range of motion). Those are the 2 lifts performed in competitions; unless specified a deadlift implies a conventional deadlift. A Romanian deadlift and a stiff legged deadlift are assistance exercises that place extra emphasis on the hamstrings and erectors and are harder than conventional deadlifts.

Lifters can also change the range of motion performed on a deadlift. A deficit deadlift is where the lifter will stand on something to increase the range of motion, thus making the lift harder. This is commonly combined with stiff legged deadlifts for flexible lifters. A rack pull or partial deadlift is where the bar is elevated above the normal starting position. This allows the lifter to work on a specific part of the range of motion and it also allows the lifter to work on overload as it is easier than a traditional deadlift performed from the floor.

Proper Technique

The deadlift is a deceptively simple exercise, and once you learn how to perform it with good form it will come naturally to you, at least with the lighter and more moderate weights. But sometimes it takes a while to learn that proper form. It all starts with the lower back. During the entire deadlift, when you are lifting the bar up and setting the bar down, you want your lower back to be slightly arched. At the very least it should be flat, but slightly arched is better. You do not want your lower back to get rounded, as though you were bending forward to touch your toes. This will put you in a weak position and it is more likely to cause injury to your spine. The proper way to arch your back is to lift your chest up toward the ceiling and then pull your shoulder blades back together. This will cause a slight arch in your upper back. Next tilt your butt backwards and upwards, so that the top of your butt goes toward your lower back. Imagine that a pencil was rolling width-ways down your lower back and you had to try to catch it with the top of your butt. That motion of pushing your butt out will cause you to arch your lower back.

Foot Position

To start a deadlift walk up to the bar (the bar should be on the ground), and get your feet set. You want your feet to be slightly narrower than shoulder width apart. Most people have their toes pointing out slightly. Make sure that you are standing close to the bar; it should almost be touching your shins when you are standing up straight. Personally I normally line up the outsides of my shoes with the beginning of the knurling on the bar, but you want to find a foot position that works for you. Remember, with this type of deadlift (Conventional), the wider your feet are the higher you will have to lift the bar, so keep them relatively narrow but wide enough so that you can generate power at the bottom.

You want your feet to be slightly narrower than shoulder width apart.

Hand Position

Once your feet are set, lift your chest and push your butt back, thus arching your lower back. Maintain that position and bend forward at the hips first as you reach down and grab the bar. Once your hands get lower than your knees, it is okay to bend the knees significantly. The knees will also move forward, and in a deadlift it is okay for the knees to travel in front of the toes since your butt is not going that low. Reach down and grab the bar so that your hands are just outside your legs. You want your hands to be as narrow as possible but not so narrow that they rub against or lay on top of your legs during the exercise. In a conventional deadlift your hands will be on the outside of your legs. The wider your hands are, the higher you will have to lift the bar, and the harder the exercise will be.

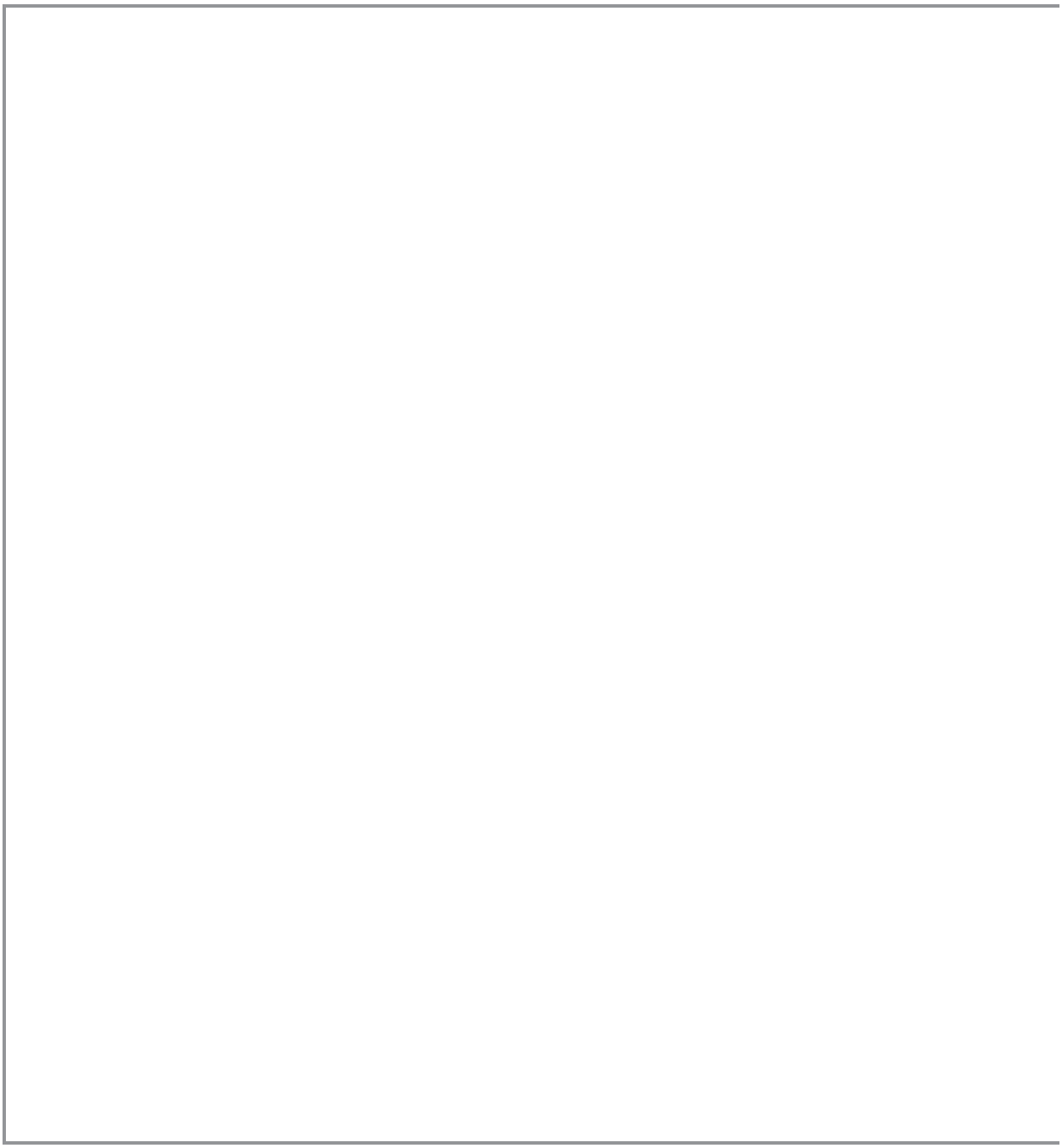
Head Position

Proper head position is a point of debate among lifters. Once you have grabbed the bar, pack your neck by pulling your chin in (making a double chin) and driving the base of your neck back into your traps. Almost everyone agrees with this. From here it gets more debatable. Some experts suggest just maintaining that position, keeping the head in line with the spine and thus eye gaze will be looking down slightly, likely at about a 45 degree angle or at a spot about 10 feet in front of you. I personally don't advocate that, although if that works for you, that is fine with me. After watching literally hundreds of elite deadlifters perform the lift and specifically noting their head position, I found that at least 75% of the lifters lifted with a chin up position. I define this as having the chin held high when the bar is at knee height or higher. Indeed some lifters look up as much as possible. Often with good lifters that there will be noticeable downward head movement as the neck flexes once the lifter is upright and is in locked out position. I believe that looking up, more specifically driving the chin up, will help you lock out the bar and will help prevent the upper back from rounding. However, much like the squat, it is possible that individual biomechanics might differ from lifter to lifter, and you will ultimately have to find what works best for you.

I believe that looking up, more specifically driving the chin up, will help you lock out the bar.

General Form for Performing the Deadlift

As you bend down and reach for the bar, try to maintain that arch in your back. Some lifters find that by raising their arms up over their head before they bend down, they are able to get a better arch than by keeping their arms down by their side. Either way, once you grab the bar, use the bar to stabilize yourself and lift your chest and push your butt back again to make sure you have a slight arch in your lower back. If you are very flexible, this does not have to be excessive. At the bottom position you should be able to see your chest fully in the mirror, and your hips should be higher than your knees but lower than your waist. Once you are set take a deep breath in, and then lift the bar up. You should be looking straight ahead or up slightly; most people prefer to look up. Right before you lift the bar, squeeze your core muscles to make your trunk very tight and stable. An easy way to visualize this is to imagine someone was going to punch you in your stomach and you would tighten up those muscles so it wouldn't hurt very much. You want that same tension when you begin the deadlift. It may take some practice to learn how to squeeze your core without losing the arch in your lower back. Some people find it useful to press their core out against their belt as though they were trying to pop the belt off of their body.



A lifter raises their arms before the deadlift to better arch their back
Photo credit: Vadim Snitkovsky

As you lift the bar, drive with your legs but maintain the stability in your upper body. Visualize that you are lifting the bar with your upper body first. What often happens as the weight gets heavy is that people first straighten

their legs out, thus raising their hips, and then they lift the bar up, thus mimicking a stiff legged deadlift. You can get away with this with the light weights, but you won't be able to lift heavy weight with this form. You want to always practice with good form to ensure that proper form is ingrained in your head. When you are lifting heavy weight, do not allow your hips to come up first. As you are lifting the bar you are pulling up with your upper body and driving with your legs. The goal is for your legs and your trunk to straighten out all at once. If you lock out one thing before the other it often kills your momentum, and you will not be able to complete the lift.

As you near completion of the rep, it is important to drive your hips forward by squeezing the glutes. Generally you begin doing this when the bar has cleared your knees. You want your butt to be under and in line with your body at the top of the lift, not sticking out behind you. This hip drive allows you to fully activate your glutes and hamstrings, which are very powerful muscles. If you have a weak lockout it is possible your glutes are not firing properly or you are not performing this movement correctly. Basically you are taking your upper thigh near your groin and driving that forward into the bar as the bar comes up.

Once you have completed one rep, the descent is very important because you want to end up in the proper starting position so you can perform another rep. Even if you are doing just one rep you still want to lower the bar properly because you can injure yourself if you round over to set the bar down. In addition, in a competition you must lower the bar under control to avoid the lift getting red lighted. That means don't drop it from the top or slam it down. Once you are at the top and you are locked out, you want to begin lowering the bar by first bending at your hips, as though you were taking a bow on stage. Keep your chest up and keep looking straight ahead. Once the bar clears your knees, bend the knees and allow them to move

forward. Maintain the arched position of your back the whole time. Go down until the bar lightly touches the floor and then reverse directions and lift it up again. Do not allow your back to round while you set the weight down.

Much like the bench press, you will have to decide if you want to pause each rep on the floor when deadlifting or if you would rather just perform touch-and-go reps. Each method has benefits which are outlined below:

Benefits of Pausing

- Mimics competition setting (the first rep is always paused)
- Eliminates bouncing off the floor to gain momentum
- Is harder
- Gives you a chance to reset your position on each rep
- Is probably beneficial if you are weak off the floor

Benefits of Touch and Go

- You can complete more reps this way
- It forces many muscles, namely the upper back, to make an extended powerful isometric contraction which is what those muscles have to do in a competition. Pausing allows some muscles to rest and relax every few seconds.
- It forces you to lower the weight with good form
- Is probably beneficial if you are weak at the lockout position
- Takes advantage of the stretch reflex, which causes greater muscle recruitment

Deadlift Form Specifics

As you get more advanced, the form for the deadlift remains basically unchanged. In the ideal world, a 135 lb deadlift should look similar to a 500 lb deadlift. If possible, try to bring your stance in a little bit; ultimately it should be narrower than shoulder width apart if you are performing a conventional deadlift. When people are first learning the lift they tend to feel more comfortable with a slightly wider stance, but once you have the movement down and the necessary flexibility, then try to bring your feet in a little closer. The wider your feet are, the farther up you have to lift the bar (note that this is the opposite of what happens in a sumo deadlift). A good rule of thumb to find the proper foot position in the deadlift is to have a lifter take one step and then perform a maximum vertical jump. One step means that one foot can move once, but you have to jump starting with both feet on the ground. Wherever people put their feet when they jump is often a good starting point to place your feet during the deadlift, because that is your natural power position. Another method I have heard of to determine foot position is to have someone hang from a bar a foot or so off the ground and then drop onto the ground, landing with both feet. The position they put their feet into to land (where the feet touch the ground) may be a good starting point for their foot position during a deadlift. Those two guidelines apply to the conventional deadlift.

In addition try to bring your hands in as much as possible. They should be just outside of the legs, a tiny bit wider than shoulder width apart. The wider your grip is, the higher you have to lift the bar to lock it out.

As you become more proficient at deadlifting and begin to lift heavier weight, you will want to use an alternating grip, where one hand is palms

Save the alternating grip for heavy, working sets.

up and the other is palms down. It doesn't matter which one you choose to be up; just the pick the most comfortable position and stick with it. Personally I have my right hand face down and left hand face up, and I am right handed. This is the most common scenario. Having an alternating grip will help your hold on to the bar. Amazingly just reversing one hand can increase grip strength by 100 pounds or more, so it can make a pretty big difference. The negative of using an alternating grip is that it can cause uneven development of your back muscles, particularly the erectors and traps. I would save the alternating grip for heavy, working sets and use a regular pronated grip for all other sets. It is possible to try to rotate what hand faces up, and you can do that to even out your muscular development. However, don't do anything that will mess up the motor pattern of a proper deadlift, and you will probably find that one hand position is significantly more comfortable and stronger than another. In addition, if you have just been using one alternating grip style (say right hand down, left up) for a long time, do not expect to be able to flip your grip (right up, left down) and lift the same amount of weight right away; that is asking for injury. If you do change your grip, start light and allow your body to adapt to that new position by gradually increasing the weight you lift with it.

Some powerlifters use a hook grip for the deadlift, but this is pretty rare. A hook grip is like a closed grip, but instead your thumb goes under your fingers, opposite the bar. The thumb is in direct contact with the bar, and then your first two fingers are on top of the thumb, holding it in place. Ultimately this can be a very strong grip, and it has the advantage of putting the body in a symmetrical position. However, the big negative is that this grip is very uncomfortable to learn especially with heavy weights. Because the alternating grip is strong enough to support the weight, it is generally adequate and feels more natural. The hook grip is commonly used in Olympic Lifting, where an

alternating grip is not feasible (heavy snatches, cleans, and jerks with an alternating grip would not work) because Olympic Lifters have no choice but to use a hook grip.

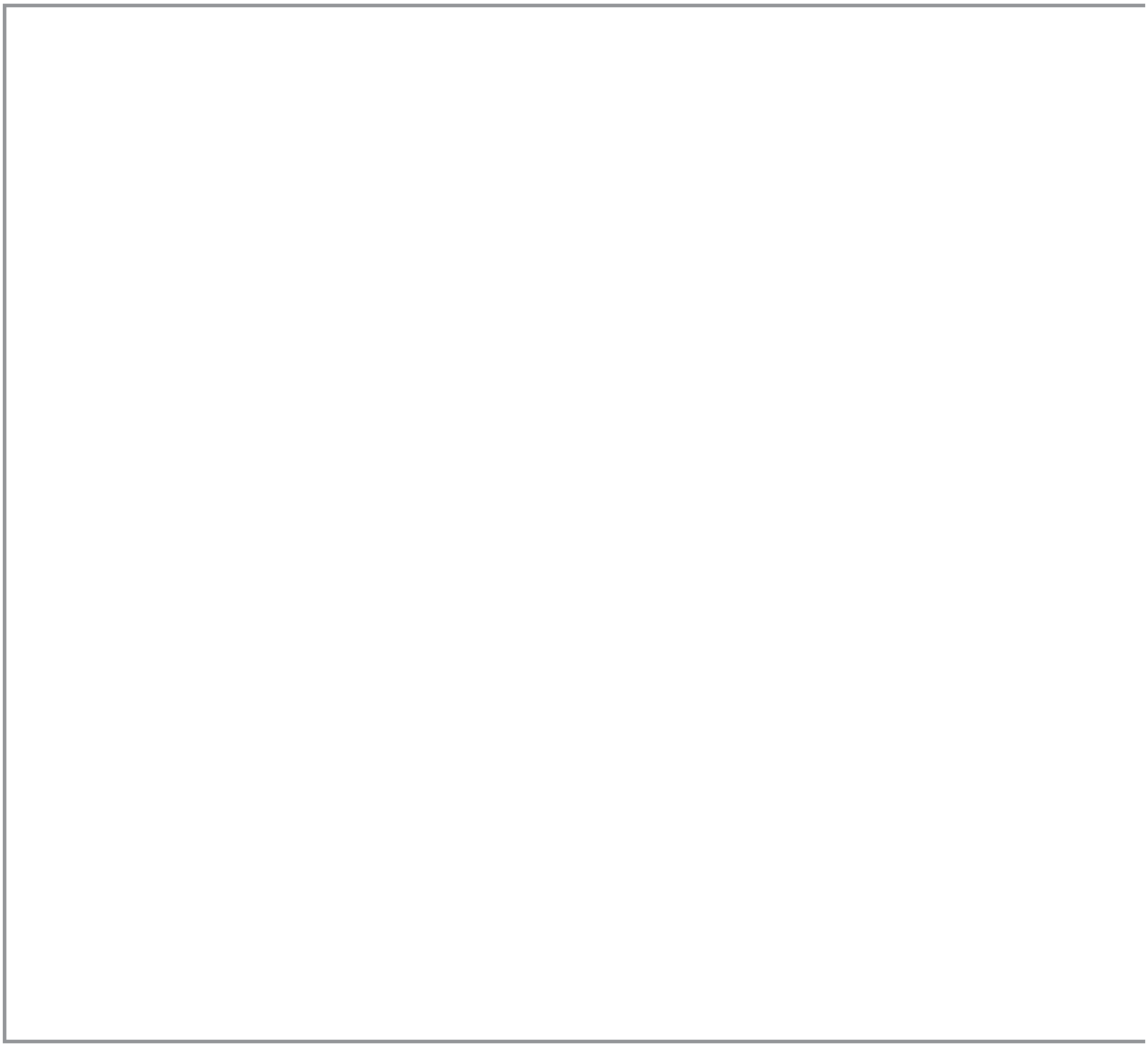
Things to do while Deadlifting

- Keep your chest up
- Keep a flat or slightly arched lower back
- Have your shins very close to/touching the bar
- Place your hands just on the outside of your legs
- Use an alternate or hook grip when the weight is heavy
- Use chalk when you are going heavy
- Pull the slack out of the bar before you begin the lift
- Keep the arms straight
- Lead with the upper body
- Look straight ahead or up
- Keep the chin high as the bar approaches and moves past your knees
- Drive the hips forward after the bar clears the knees

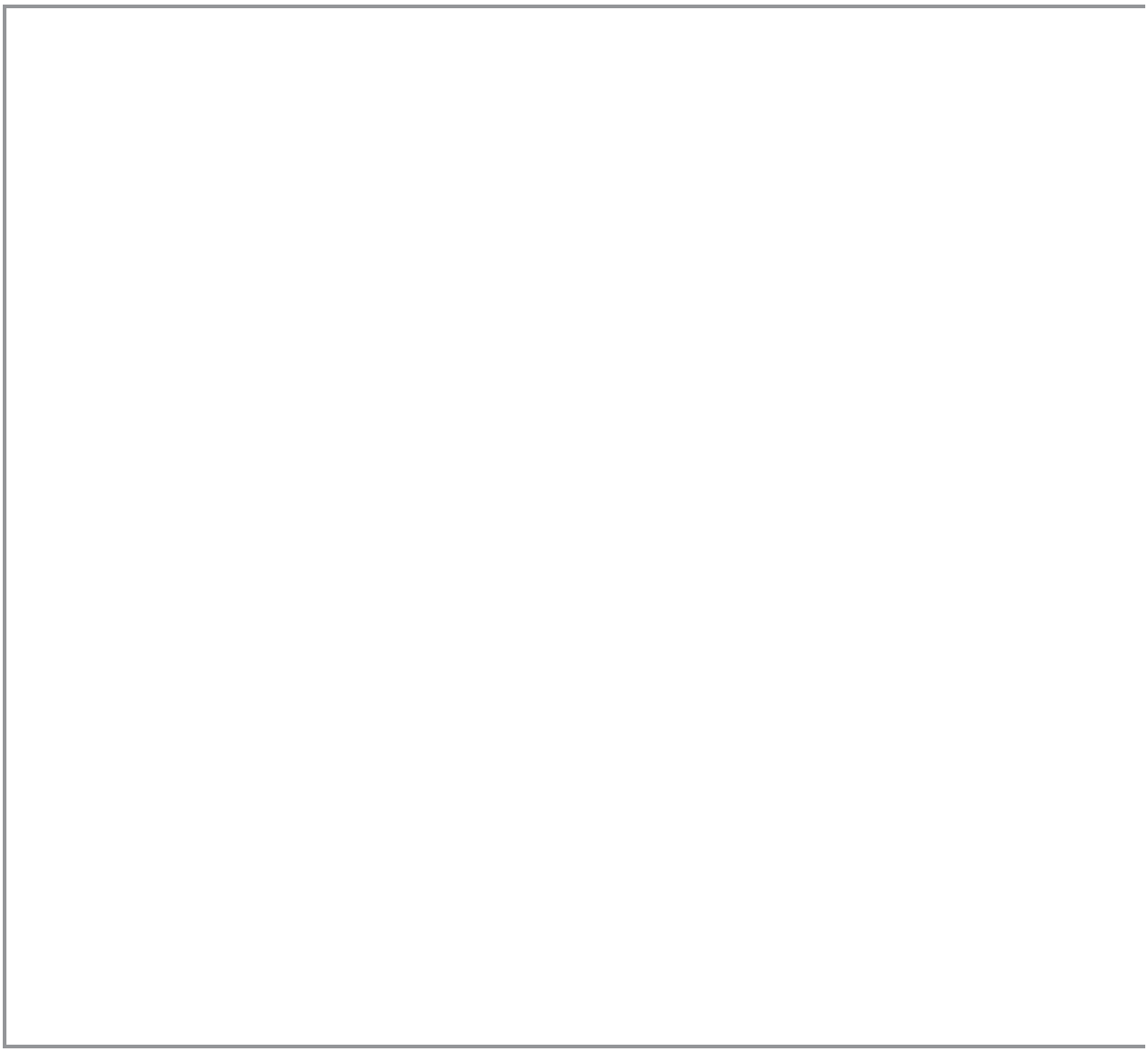
Things NOT to do while Deadlifting

- Look down excessively
- Round your lower back
- Jerk into the bar to get it moving
- Use a stance that is too wide
- Take too wide of a grip
- Put baby powder on your hands
- Put chalk on your legs
- Squat down very low to start the lift
- Let the bar go out over your toes or away from your shins
- Go up on your toes

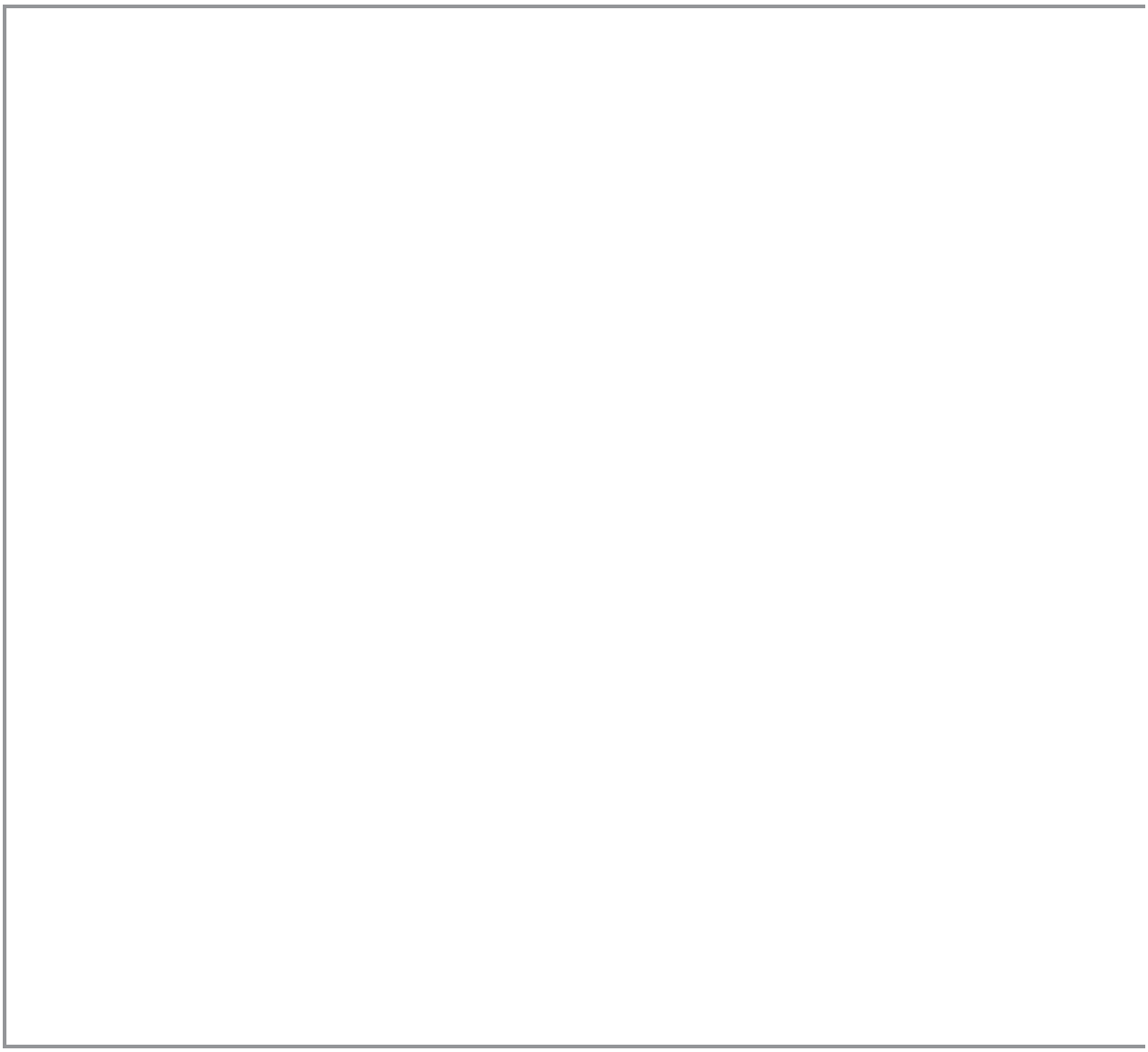
- Let your knees move toward each other
- Bend your elbows
- Hitch the bar by using your legs to support the weight
- Lean back excessively at the top
- Shrug the bar at the top
- Slam the bar down/drop it from the top (in a competition or commercial gym)



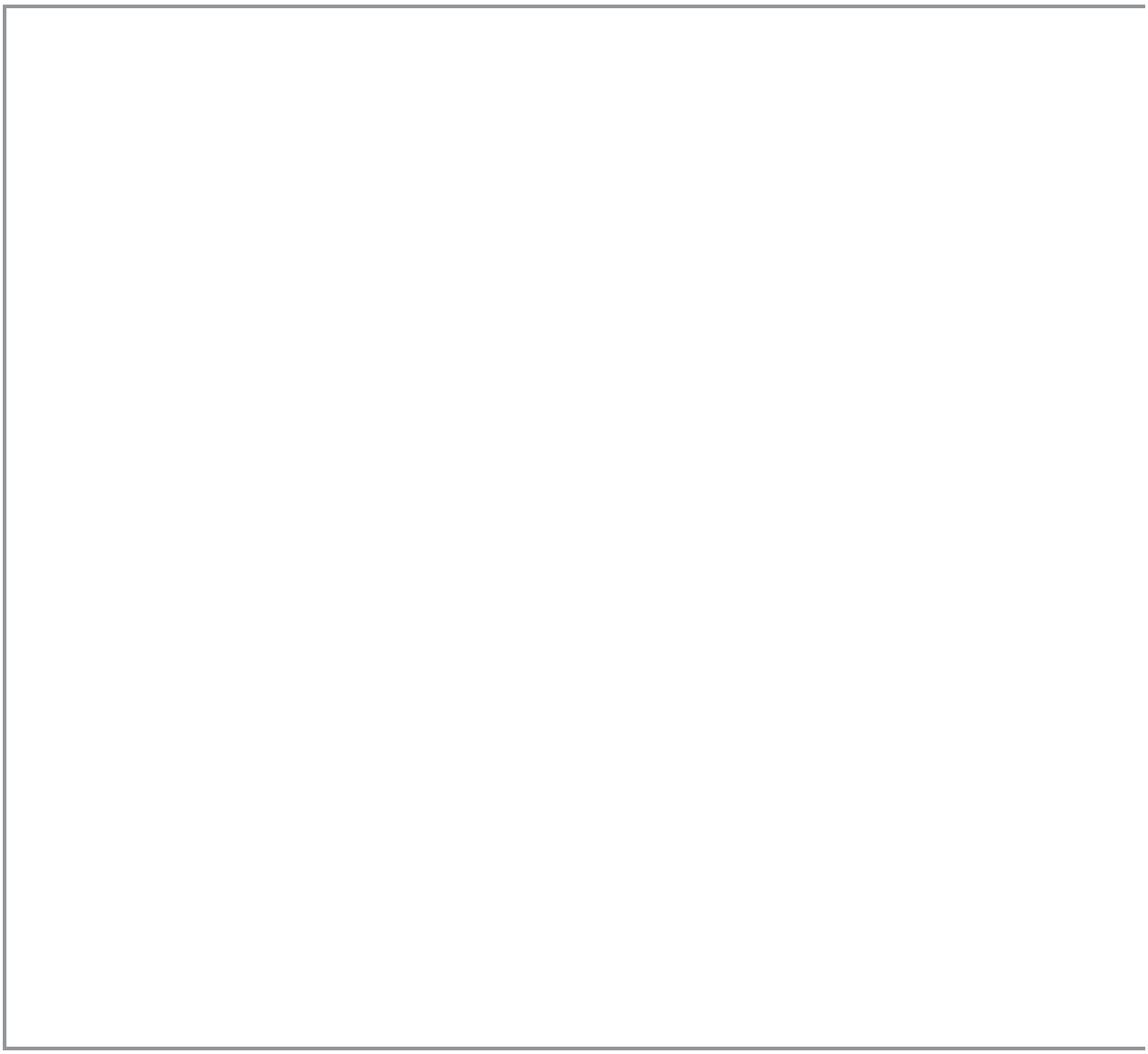
▶ **1: The lifter takes their grip and straightens their legs**



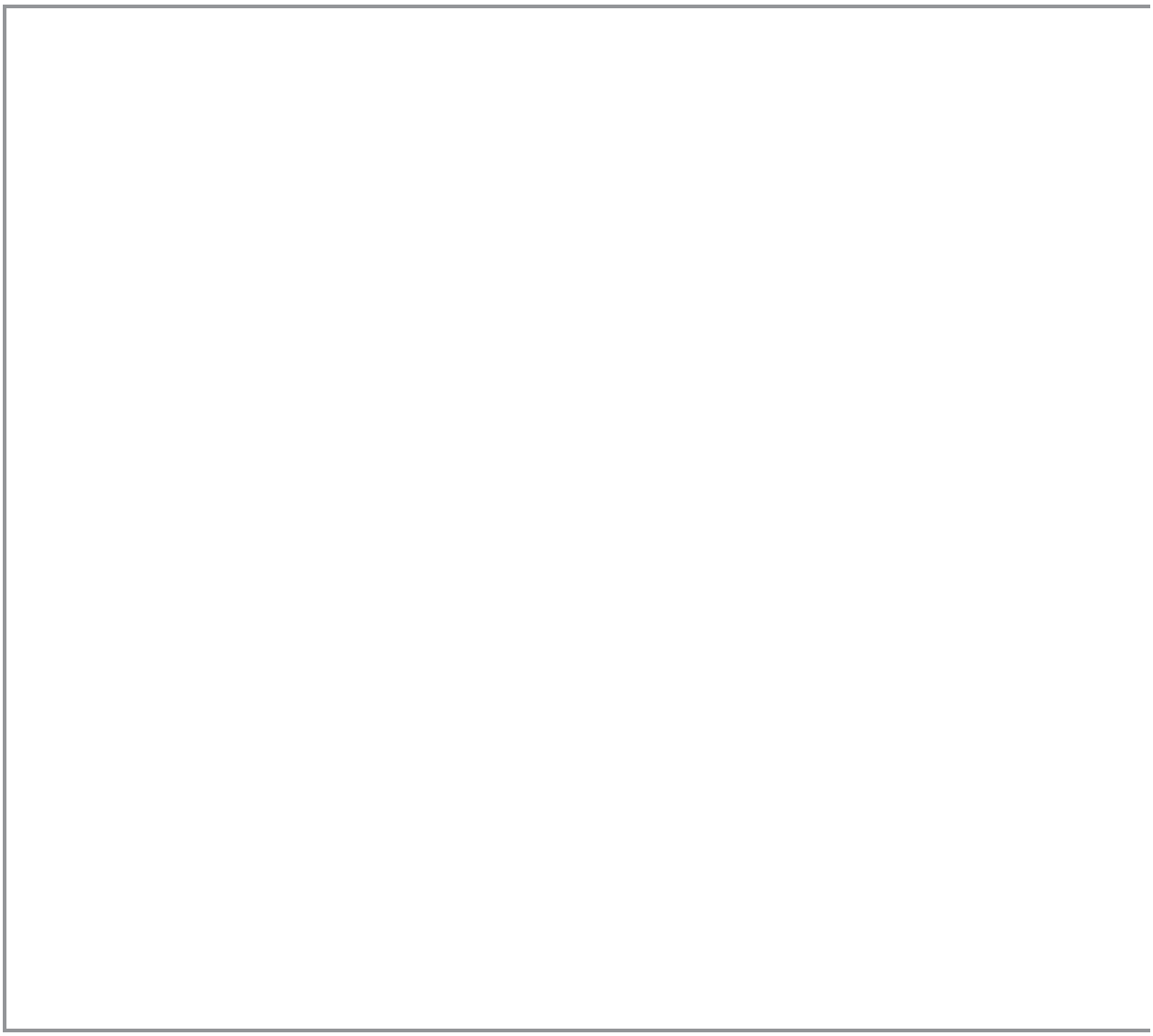
▶ **2: The lifter quickly bends down**



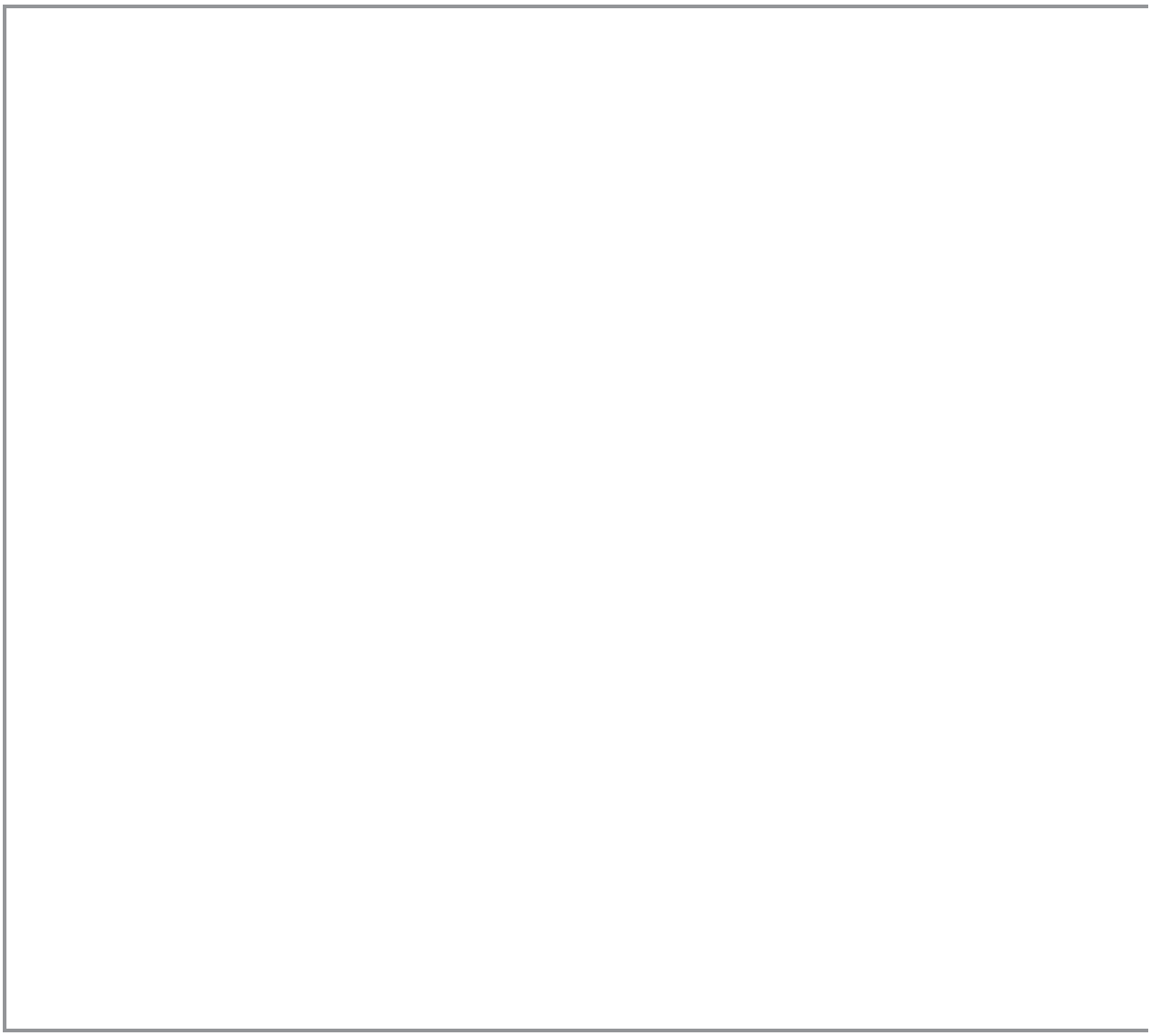
▶ **3: The lifter drops into starting position: chest up, back slightly arched, bar next to shins**



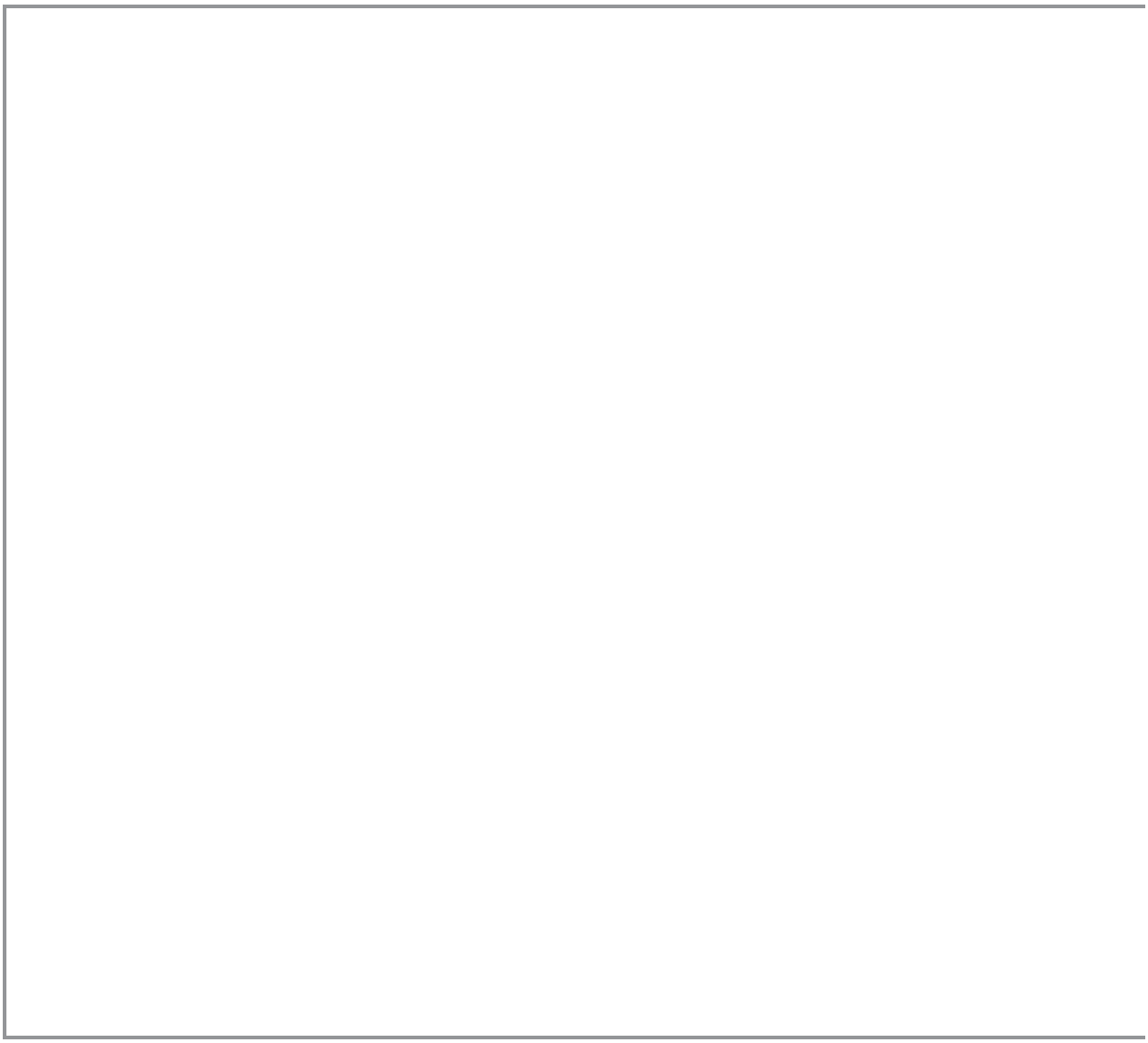
▶ **4: The lifter initiates the pull**



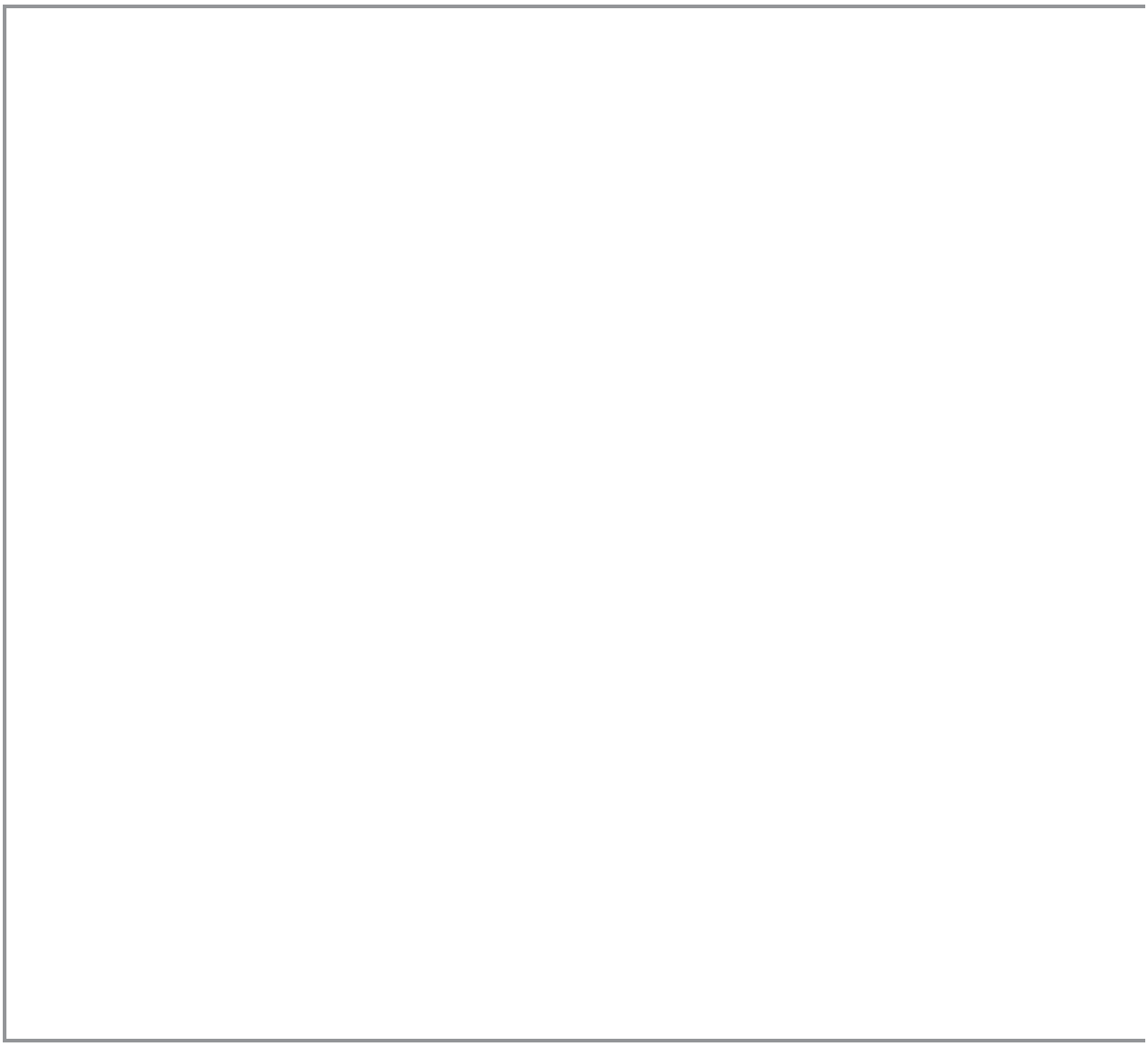
▶ **5: The lifter keeps the bar close their body**



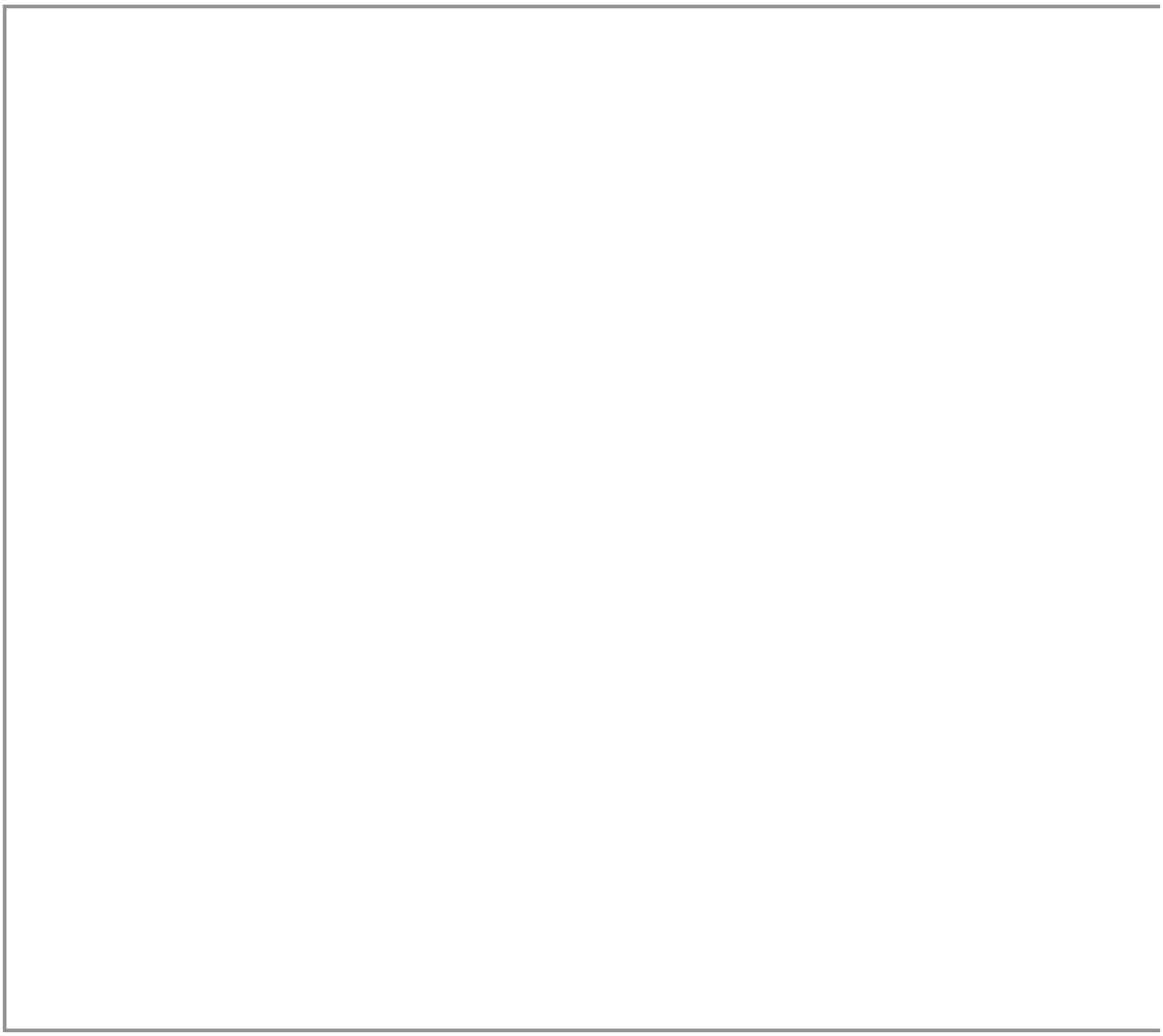
▶ **6: The lifter leads with their upper body**



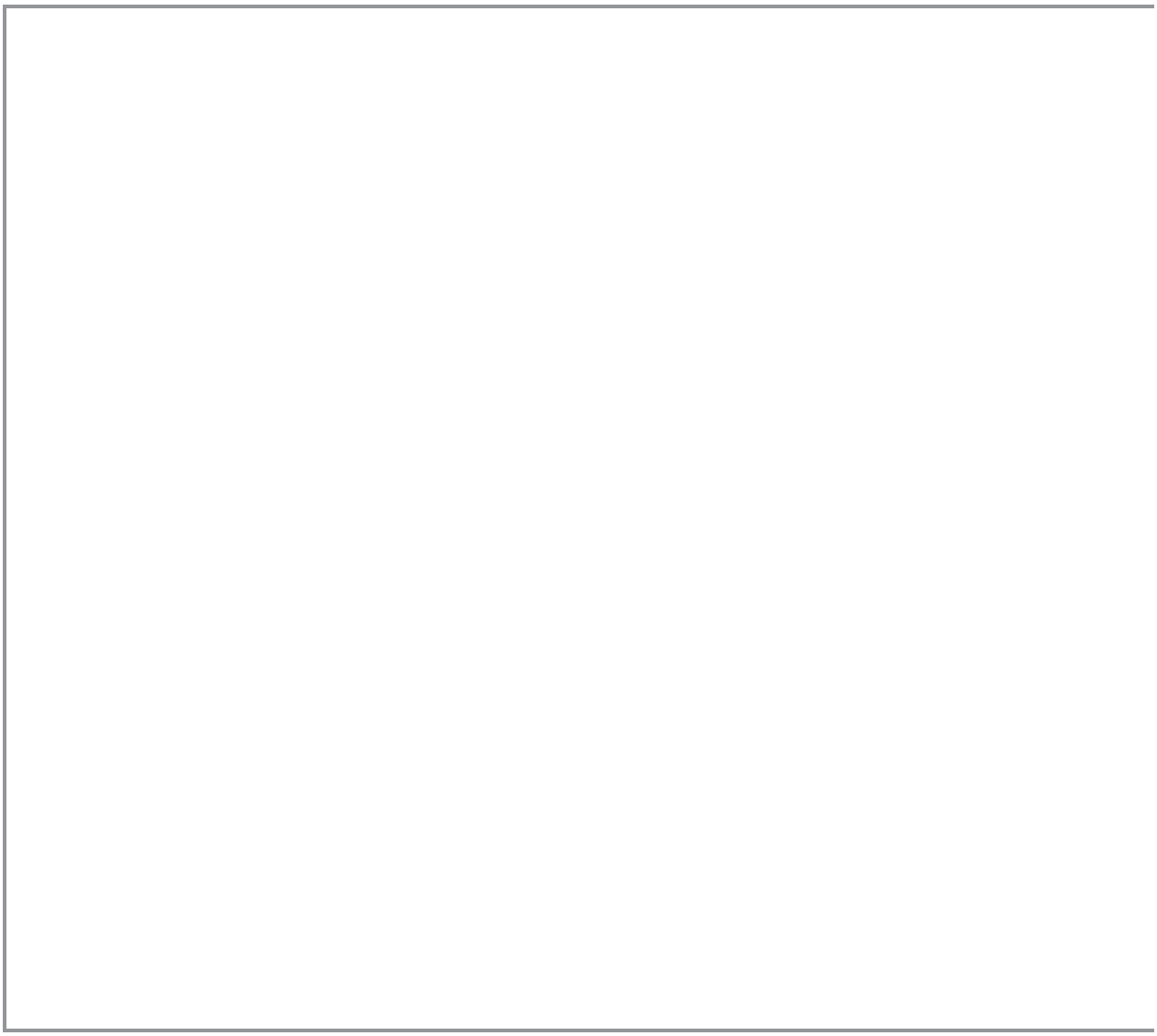
▶ **7: The chin is held high particularly when the bar is around knee position**



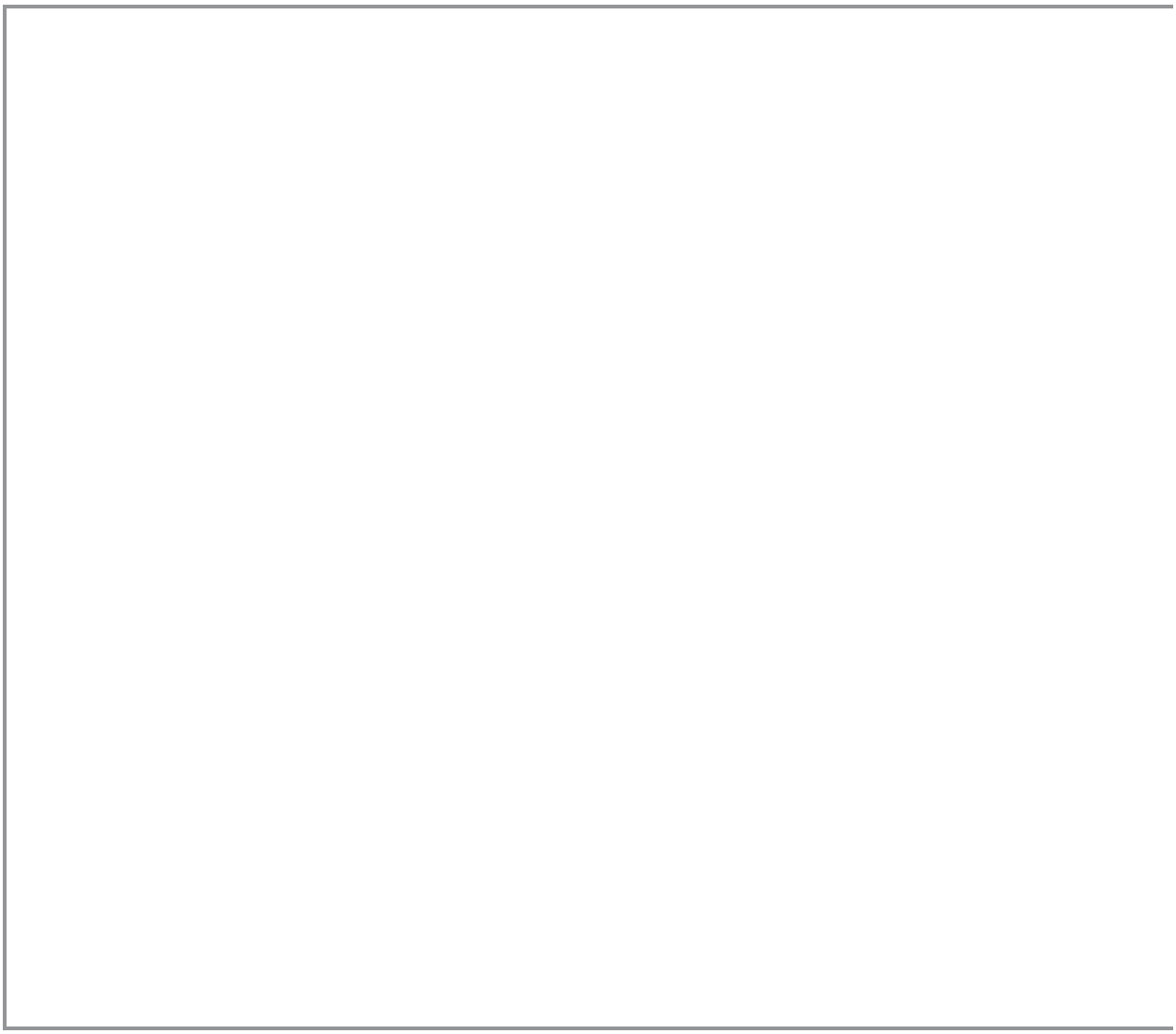
▶ **8: The lifter begins to push the hips through**



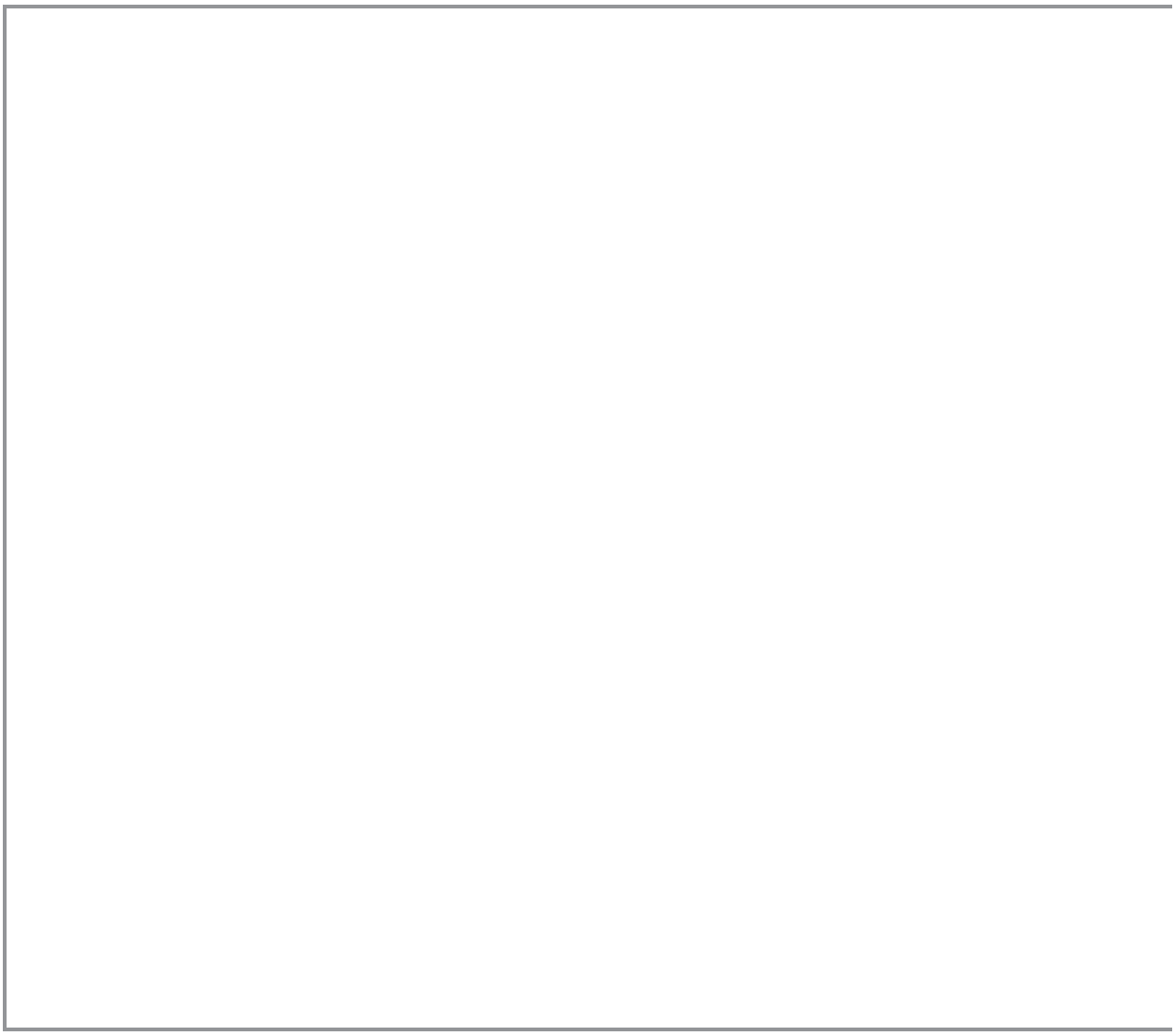
▶ **9: The lifter works to extend their trunk**



▶ **10: The lifter continues to straighten**



▶ **11: The lifter pushes the hip through to avoid a hitch**



▶ 12: The lifter squeezes the glutes and sets the shoulders back
Photo credit: Vadim Snitkovsky

Note: This lifter uses the leg pump in the beginning to achieve a stretch reflex; that is not necessary. If that is not your style simply start at image 3.

This lifter gets slightly out of position in image 9 as *the hips and knees are more extended than the trunk — that happens easily with heavy weight — but because he maintained his trunk position (slight arch, shoulders back, chest up, chin high) he* was *still able to lock out the weight when he reached his sticking point.*

Flexibility/Mobility Issues for the Deadlift

The conventional deadlift doesn't require the same mobility as the squat, and if you perform the lifts in the same workout often, the warm-up routine you use for the squat will suffice for the deadlift. The two main areas that limit a lifter's performance in the deadlift are tight calves, especially since the lifter is likely using a flat sole, and an inability to arch their back in the bottom of the lift. The same methods used to loosen up the calves in the squat will work in the deadlift. In addition, if you really want to attack the calves, you can sleep in a special sock called the Strassburg Sock. This is a sock with a strap attached to the toes that will pull the toes up to the shin. The Achilles tendon is big and thick; it needs a lot of work to effectively lengthen it. It isn't super comfortable sleeping in these socks, and you look pretty goofy when you have to get up to use the bathroom in the middle of the night, but they can help.

To improve one's ability to arch the back (both upper and lower) you can use mobility drills to increase the range of motion in the lumbar and thoracic area of the vertebrae. Lying on a foam roller with the foam roller perpendicular to the body can help. Practicing bridging can also help. Raising the arms above the body makes it easier to arch the back.

Common Problems

There are a lot of common mistakes that beginners make when performing the deadlift. Here are some of them explained in detail.

Rounding the lower back – As previously stressed, keep your lower back slightly arched. Focus on that, and watch yourself in the mirror from the side with very light weight while you perform the deadlift. Make sure your lower back is not rounding. Don't watch yourself from the side when you go heavy, or you might strain your neck. This arched-back position does not come naturally for some people. It is always a good idea to get someone who knows what they are doing to show you how to deadlift (just make sure they know what they are doing!). Have them focus on your lower back while you complete a set and they can give you feedback on how it looks. Memorize what it feels like to be in the proper position, and stick with the light weight until you can do that naturally.

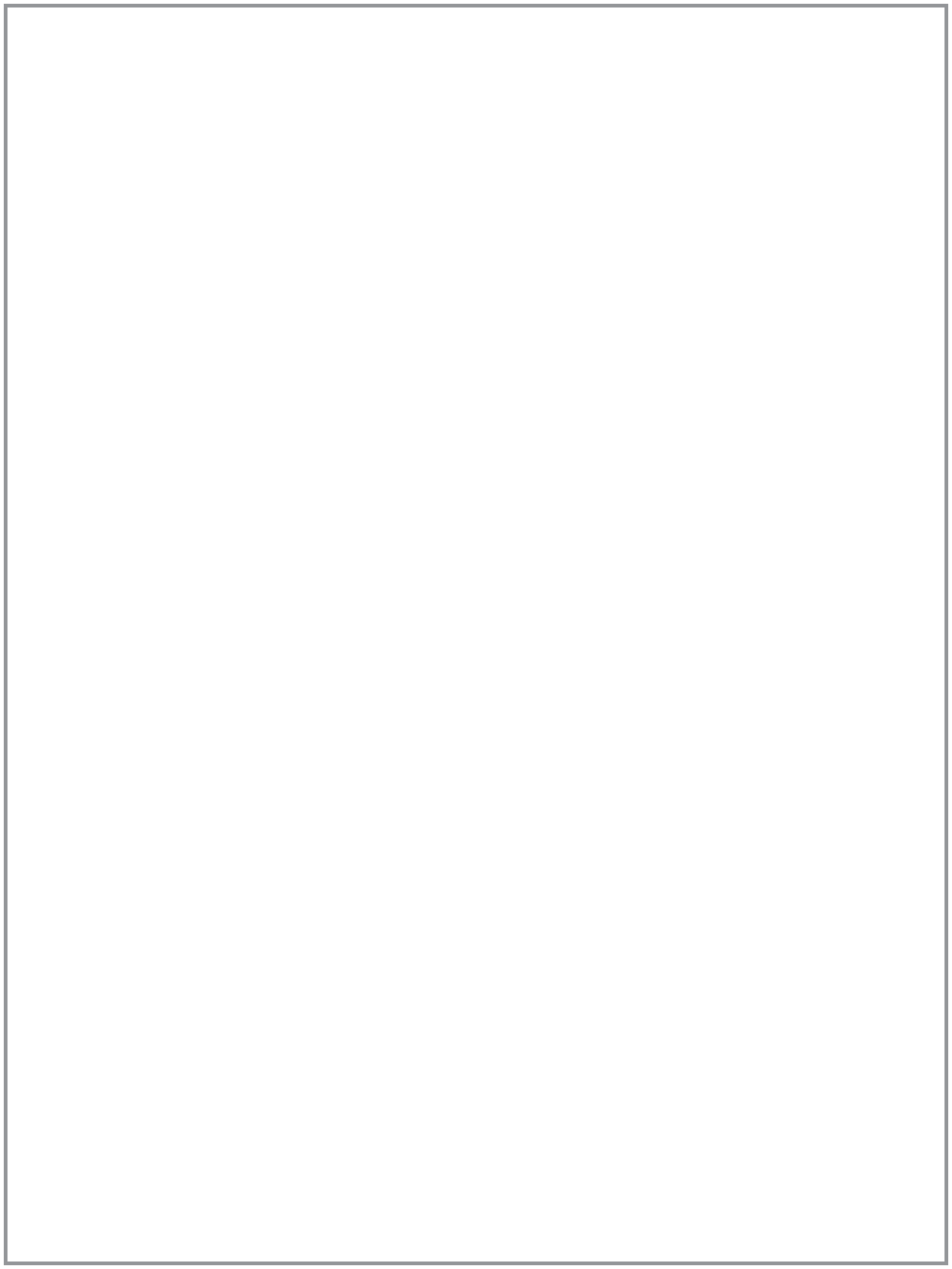
Bending the arms – Some lifters try to pull the bar up with their arms as though they were performing a row. This is not effective and once you get strong you will not be able to lift the same weight with your arms that you will be able to lift with your body. Keep your arms totally straight and relaxed. Think of them as hooks that are holding onto the weight and attaching them to your body. The only part of your arms that you need to focus on is your grip; keep that tight on the bar. Imagine you were going to carry some heavy suitcases through a long hallway in an airport. You would not want to carry the suitcases with your arms bent, or you would get very tired very quickly. Instead you would let your arms hang and simply hold onto the suitcases while you walked. You are doing the same thing with the deadlift. Just hold onto it while you pick it up.

A lifter bending his arms a bit too much in the deadlift
Photo credit: Vadim Snitkovsky

Starting too far away from the bar – When you begin your deadlift, make sure that you are standing very close to the bar. Your shins should be touching or they should be very close to the bar. If you start with the bar far away from you, then your first motion will be to pull or the roll the bar back to you, and you will be wasting a lot of energy by doing that. In addition, it places a lot of stress on your lower back to lift with the weight far in front of you; keep the bar in close to you at all times. It is not uncommon for the bar to rub against your shins, even drawing blood. Powerlifters often look at these wounds as badges of honor in their war against the iron.

Beginning with a dip to get momentum – Some people get down and grab the bar and get into position. Then when they are ready to go, they drop their whole body down an extra inch or two, often bending their arms in the

process, and then they pull up hard, as though they were trying to lift the bar off the floor as fast as possible. I do not like the dip before the deadlift for two main reasons. First, preloading is an important factor in force development. When you bend down, grab the bar and tense your body; that is preloading. If you dip right before that, you are removing the preloading and that has been shown to decrease force production, sometimes significantly. The second problem with this method is that you can only really do this on the first rep, because you do not need to pause in between each rep. Generally you want your reps to look the same, each one a mirror image of the other. Significantly altering your form on one rep is not desirable. I don't like this method because often that dip and the subsequent pull seems to get people out of proper position quickly, especially as the weight gets heavy. And finally I don't like this because in all my years of watching powerlifting competitions, I have never seen anybody with a good deadlift use this method. I understand that physiologically the reason to perform the dip is to prestretch the muscles, and I am huge believer in that, but in this particular exercise, the benefit you get from prestretching is outweighed by the loss of preloading and the likely breakdown of form. It should be noted that I am not talking about a slight dip in the hips which is common and can be valuable during the deadlift and is shown in the previous sequence in the first 2 images. The "dip" I am referring to is a more of a total body dip where a lifter drops their body (head, shoulders, everything) a few inches and then tries to spring up into the bar, as though they would use their momentum of moving their bodyweight up into the bar to get it started. The latter type of dip or movement I just described should be avoided but is common with people first learning how to deadlift.



There will be some casualties in the war with the iron

Not using your legs – When people first learn how to perform the deadlift, they are often pleasantly surprised at how much weight they can lift. Most people can lift more on the deadlift than they can on any other free weight

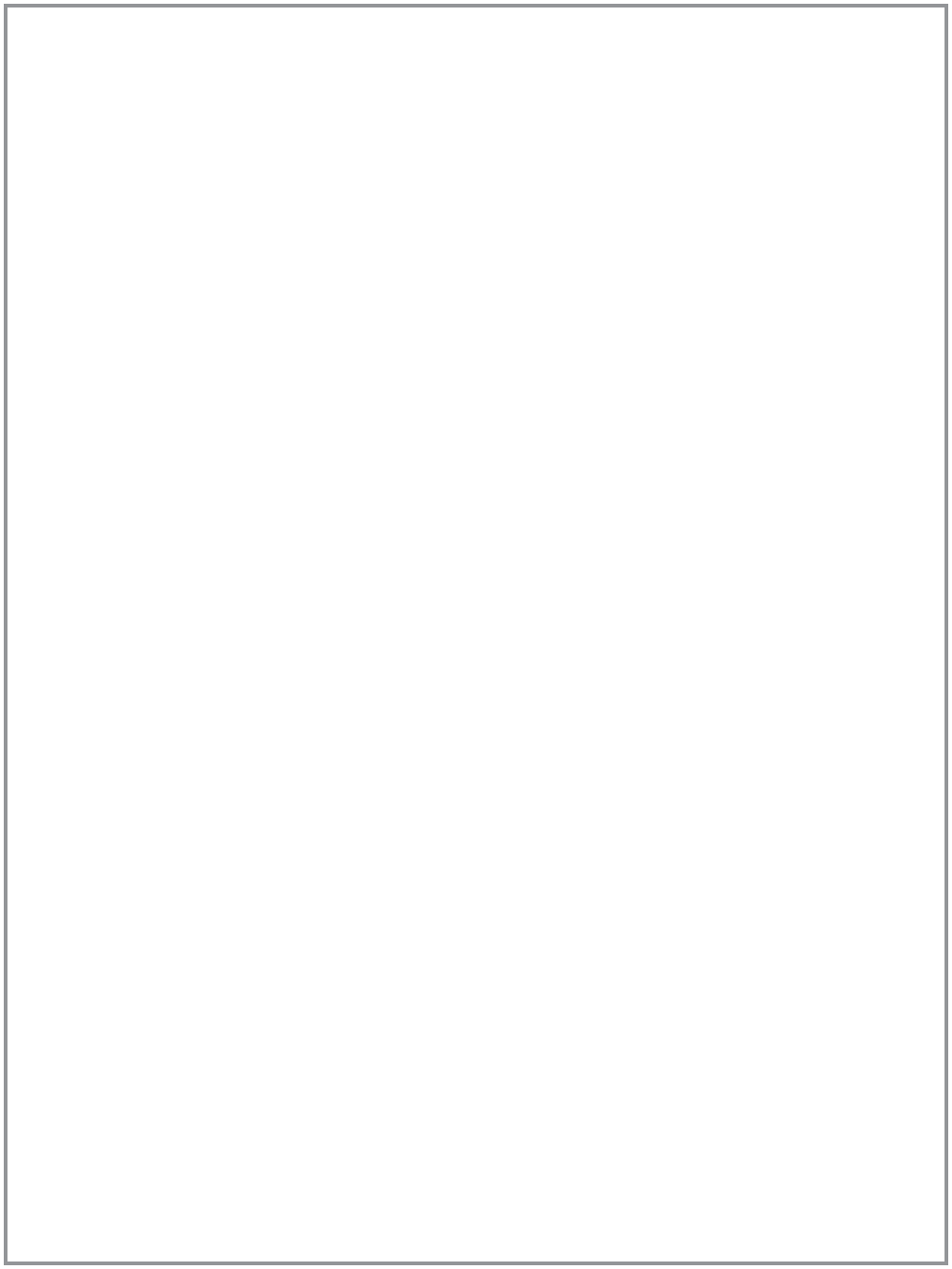
exercise. As they begin to go heavier and heavier, they rely on their back more and their legs less. This is a mistake because the legs are strong, and you want to use them as much as you can. A lot of people begin the exercise in the proper position but they straighten out their legs first to get the bar moving. Now they are bent over significantly, their legs are almost straight, and the weight is only 3 inches off the floor. To complete the rep they will have to use almost all back and hamstrings, similar to a stiff legged or Romanian Deadlift. This is harder then a regular deadlift. When you are learning how to lift, you have to be patient and not rush the heavy weight. In addition, be patient during the actual rep itself. If you let your form break during the rep just to get the weight going, then later during the range of motion when you need to lockout the weight, you will not be able to because you are not in the proper position.

> ***Be patient during the rep.***

Common Cues for the Deadlift

When deadlifting it is useful to have some key cues in your head or called out to you from your coaches and training partners, to help you keep your form as you hoist up big weight. Outlined below are some common cues you might hear; they are generally presented in order of how the deadlift is performed. It would not work to focus on all of them, but pick 2–4 that seem to work well for you and zero in on them during your lift.

Get Tall
Big Arch
Good Setup
Take your Time
Tight Grip
Big Arch
Hips Up
Chest Up
Big Air
Brace
Pull
Lead with the Upper Body
Look Up
Hips
Lock the Knees
Hold



Jennifer Thompson utilizes the sumo deadlift form
Photo credit: Jennifer Thompson

Sumo Deadlift Technique

Since powerlifting is all about lifting the most weight possible, you should experiment with sumo deadlifts and see how they work for you. At the very least, sumo deadlifts are a good supplemental exercise to help conventional deadlifts (and vice versa), and you may find that you can lift more weight with sumos than with conventional. Most people are relatively close in strength between the two deadlifts; usually there is about a 50 lb or 10% difference in strength. Something that I have found is that when completing lighter rep work, the sumo deadlift feels easier for me, but when going for a heavy one rep max, the conventional deadlift feels better. I point this out because I don't want you to feel good using one version of deadlifts only to find that when you try to go heavy you are not successful.

The set-up for the sumo deadlift is very important, in some ways more important even than the set-up for the conventional deadlift. This is because most sumo deadlifts are made or broken by how you get the bar off the floor. You have to find a body position that works for you, but in general follow these guidelines.

The set-up for the sumo deadlift is very important.

Foot Position

Approach the bar and place your feet quite wide, with your toes pointing out at about a 45 degree angle. Generally your feet will be under the rings on the bar, but that depends on a variety of factors. The wider you go, the shorter your range of motion will be, thus making it easier, but the wider you go the more power you tend to lose at the bottom of the lift (the start). You can go only so wide with your feet because ultimately you will hit the weights. I would not recommend practicing your sumo deadlifts with your toes very close to the weights because as you set the weight down, you can shift during your set (often the feet slide outward slightly) and you can literally set the weight down on your own toes, causing a potentially serious injury. Your stance will usually be wider than your squat, often 2–3 times as wide as the conventional deadlift.

Your stance will be wider than your squat.

Hand Position

Once you are in the bottom position (chest up, body basically upright, knees out, toes out, butt low, shins vertical), then allow your arms to hang straight down from your shoulders. Your hands will be inside your legs (that is the definition of the sumo deadlift), and your grip will be narrower than a conventional deadlift. For the average-size person it seems that the standard grip is to have two fingers on the knurling of the bar and two fingers on the smooth bar — not the knurling in the center of the bar but the part of the knurling that begins after the bar is smooth. This is about 12–18 inches wide. Remember that your grip is important in a sumo deadlift just like in the conventional. I would not take a grip that does not utilize the knurling at all, because then it will be very hard to hold onto a smooth bar when the weights get heavy.

Head Position

The suggestions for head position in the sumo deadlift are very similar to those given in the conventional deadlift. Look straight ahead or up, with a packed neck. If you wish to try holding your chin high as the bar approaches or clears your knees, give it a whirl and see how it feels.

> ***You are attempting to stay as vertical as possible.***

General Form for the Sumo Deadlift

Once your have your feet set, make sure you are very close to the bar. I would recommend that your shins touch the bar. You want to lift your chest and arch your lower back by sticking out your butt, just like you did with the conventional deadlift.

As you maintain the arch in your lower back, lower your butt down while staying as upright as possible. In a conventional deadlift your upper body is often at a 45 degree angle or more to the ground, so there is significant forward lean. In a sumo deadlift you want your upper body to be 80–90 degrees to the ground; you are attempting to stay as vertical as possible when you get into position. Keeping your upper body upright will force you to lower your butt significantly more during a sumo deadlift than a conventional deadlift. You want the butt to be basically even with the knee, essentially your lower body will look like you just performed a very wide parallel squat. As you descend down toward the bar, you want to push your knees out. This helps keep your hips very close to the bar, which is essential if you want to be successful. From the side view your hips should be essentially under your upper body; often they are pushed out too far back behind the body. This can be caused by a lack of flexibility, and it will reduce the available power you can generate.

Reach down and grab the bar. Initially use a pronated grip, but once the weight gets heavy, then switch to an alternated grip. Grab the bar and tense your body. Squeeze your abs just like in the conventional deadlift, but particularly tense your lower body because this is where a lot of the power is coming from. Look straight ahead or up slightly and then drive with your legs while doing everything you can to maintain an upright upper body. In the ideal world your upper body is essentially in the locked out position at the bottom of the lift. You reach down, grab the bar and then straighten your

legs. Your upper body remains tight as you lock your knees out, and then you are fully locked out. Essentially from the side it looks like you completed a wide stance squat but instead of the weight being on your shoulders, the weight is in your hands.

Flexibility/Mobility Issues with the Sumo Deadlift

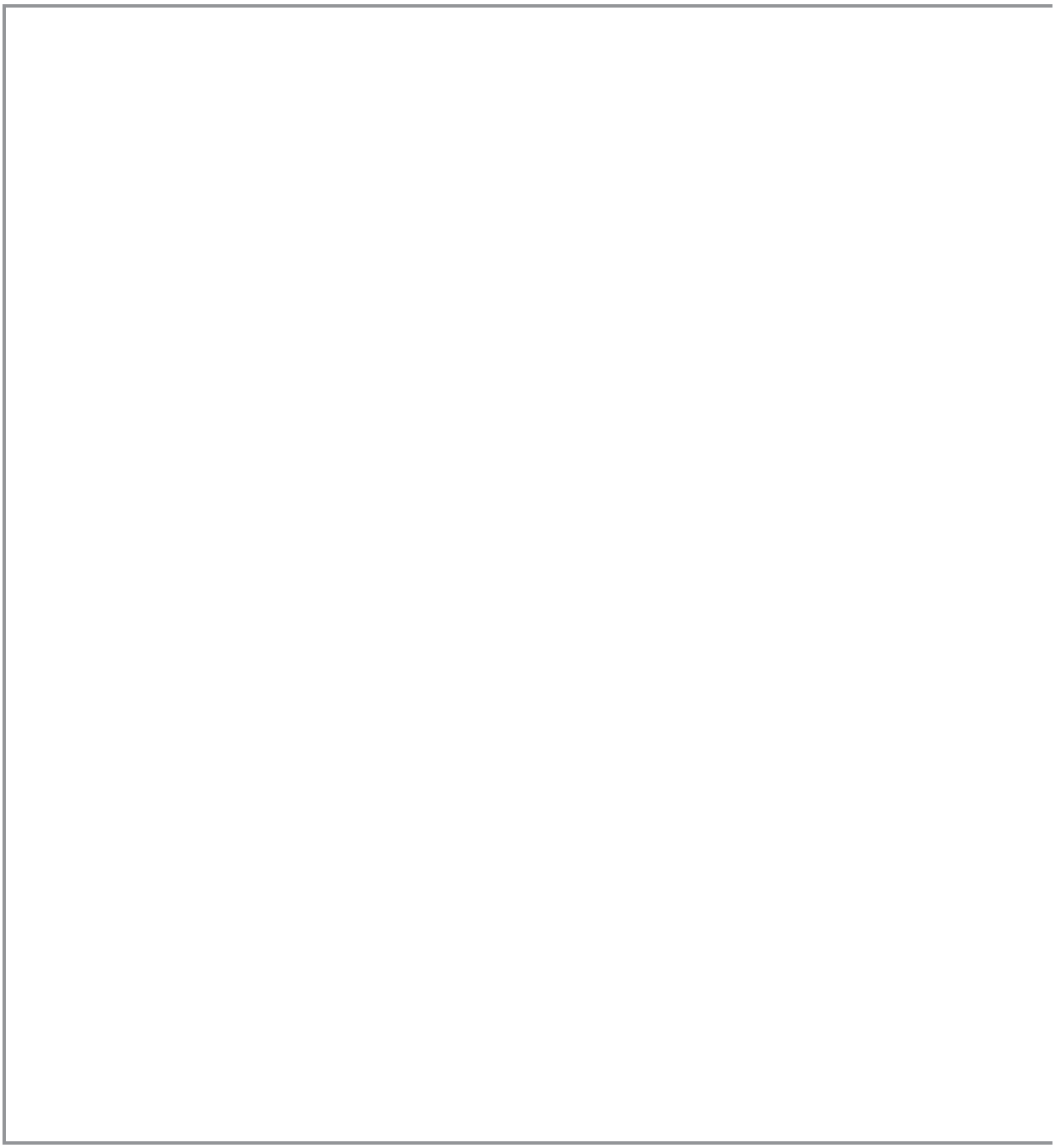
In my opinion a good sumo deadlift requires the most flexibility and mobility of any of the big 3. You can kind of “fake” your way with light weight, but once it gets heavy it is imperative that your form is spot on, or you are likely to hit a sticking point that you can’t get through. The sumo deadlift requires the same flexibility in the calves, and having a flexible back is an added benefit. The sumo deadlift places much greater demands on the adductors (inner thigh) and hips. To work on this both dynamic and static stretching of the adductors can be useful, mobility drills work well, and simple holds, where you get into the bottom position and hold it (start with your feet more narrow and then gradually work them out over time) work well, too. Try to look at videos of yourself and compare them to videos of elite level sumo deadlifters. Look at the angle of the knee, the thigh in relation to the body, how close the hips are to the bar, and see what areas you need to improve on to match their form.

> ***Your upper body is essentially in the locked out position at the bottom of the lift.***

Common Problems with the Sumo Deadlift

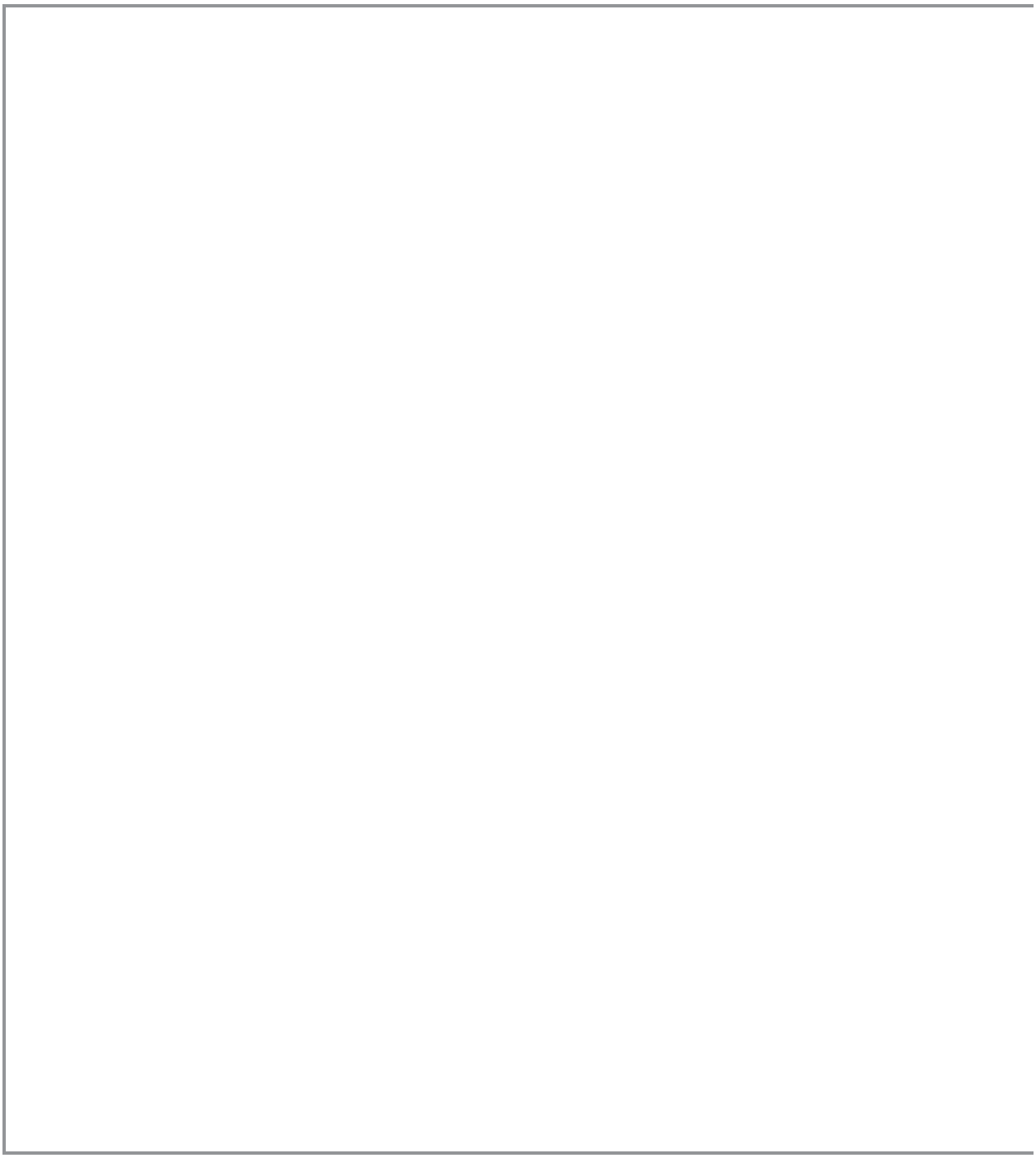
Hips too high – When you attempt a sumo deadlift it is very important to start the lift with your hips low so you can take advantage of your leg strength. If your hips start too high then you will have power from your legs for a little bit until they lock out, but you will have to bend over too far with your upper body to reach the bar. This generally results in the bar rising up several inches and then stalling with heavy weight. You probably can do this form with light weight, but it is not a good idea to practice your deadlifts with less than ideal form.

To fix this issue simply force your hips lower to start; again they should be parallel with the knee. The sumo deadlift requires a reasonable level of flexibility in the hips, adductors, hamstrings, and calves so it is possible you may be tight and having a hard time getting into position. Stretching out will help with this, and, temporarily, you can bring your feet in slightly narrower, and that will make it easier to get your butt lower.

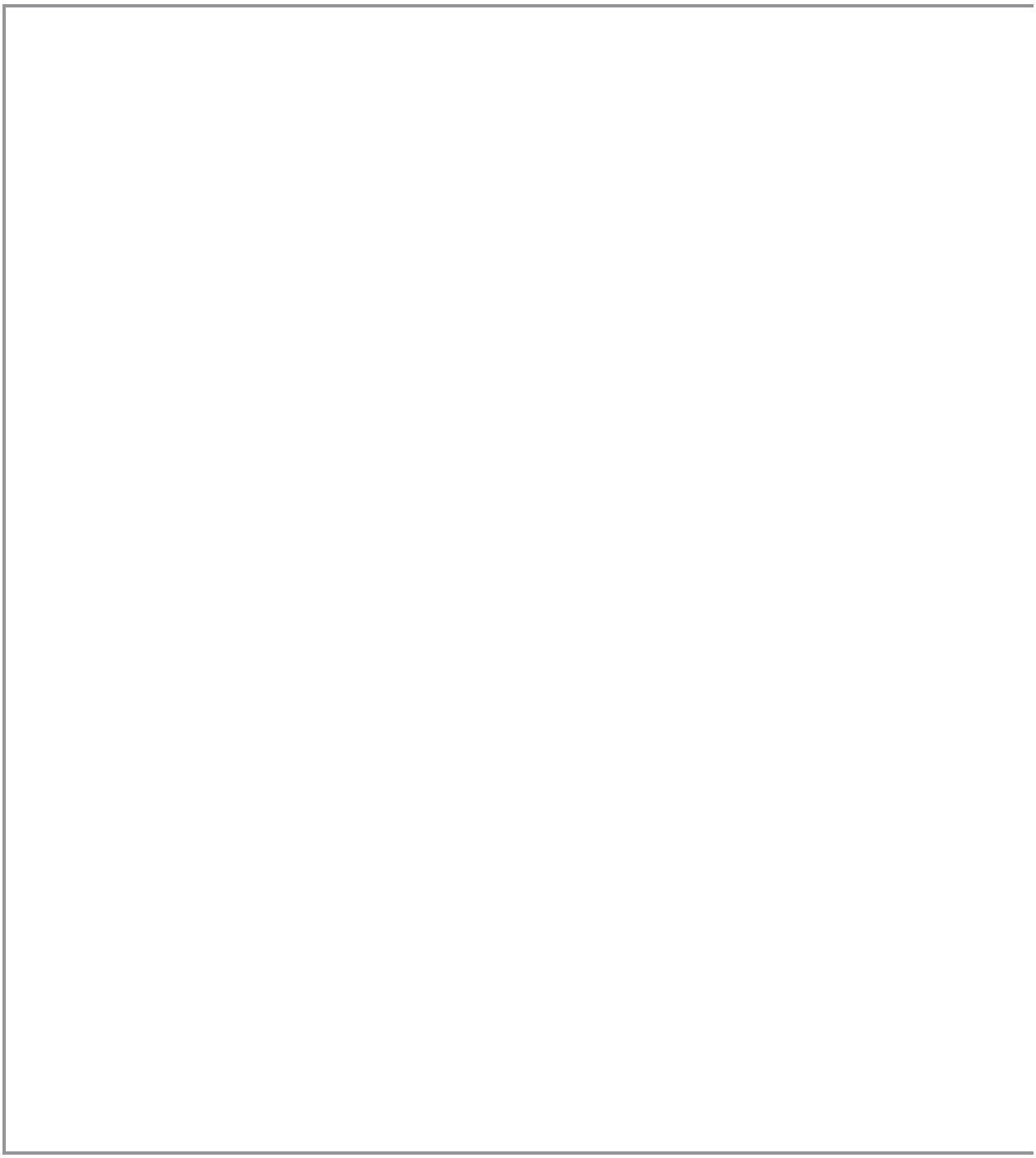


▶ **1: The lifter takes a grip on the bar, hands under shoulders and inside the legs**

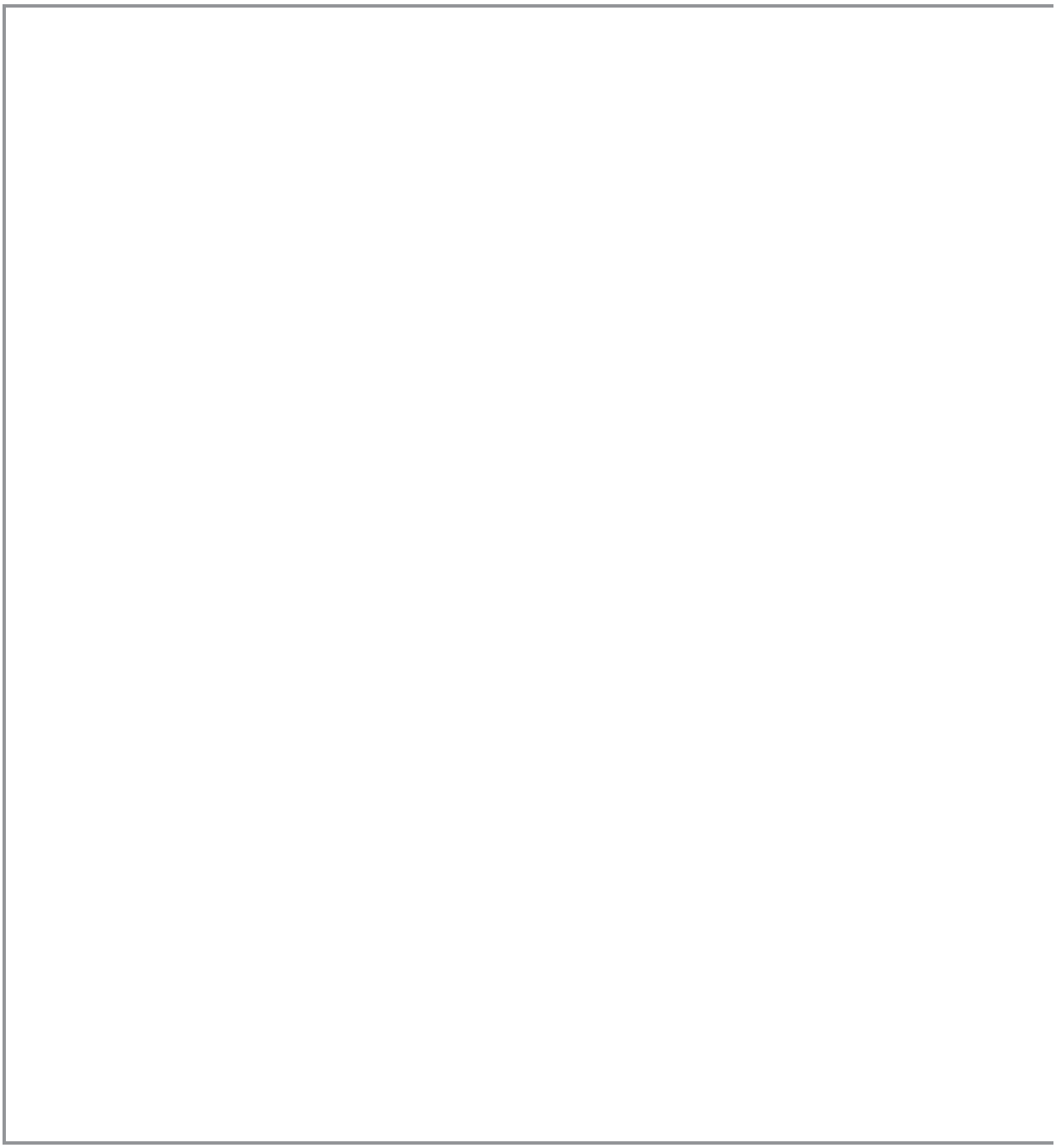
▶ **2: The lifter initiates the pull by leading with the chest**



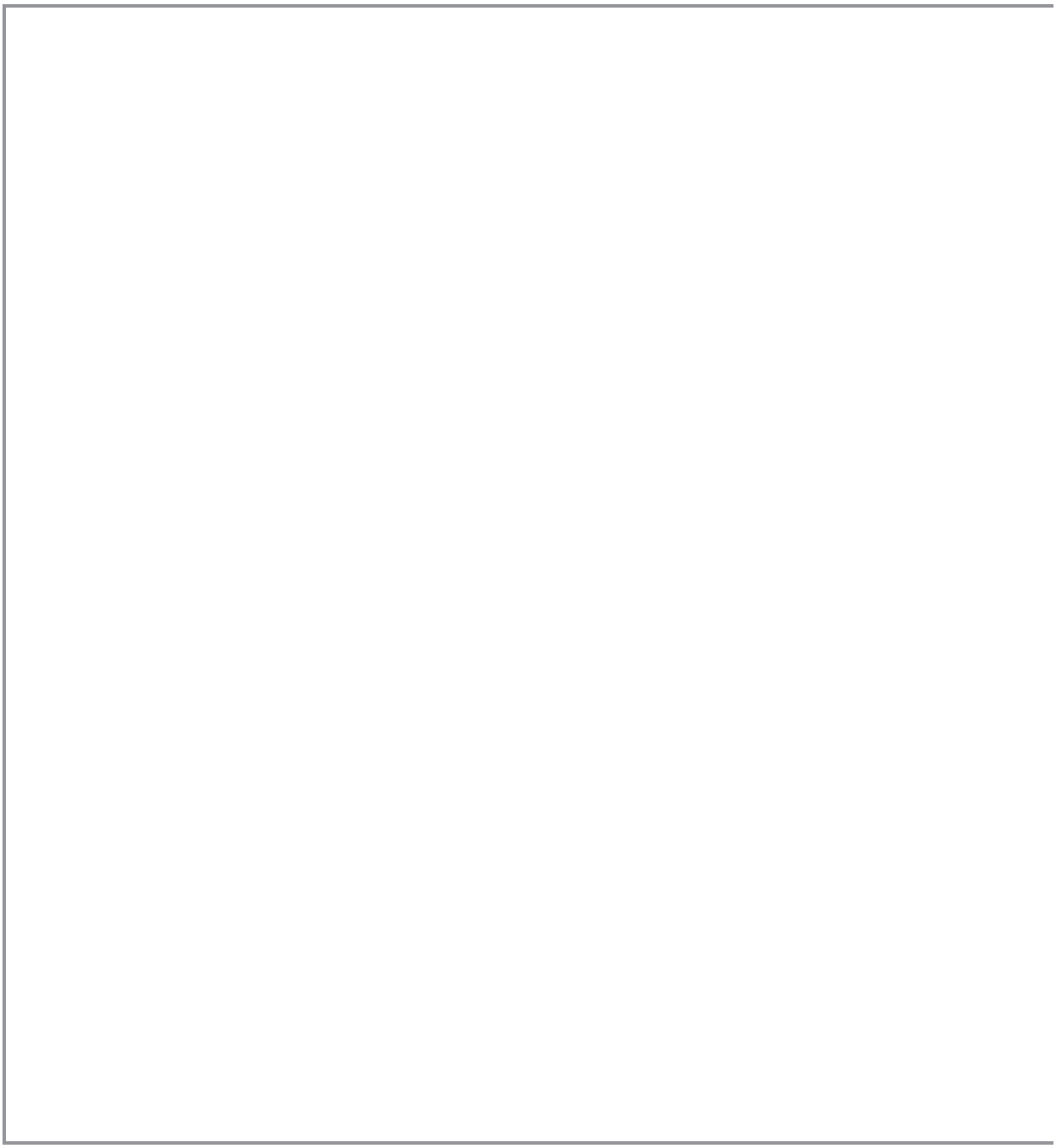
▶ **3: The lifter maintains trunk position while waiting for the bar to leave the floor**



▶ **4: The lifter has a slight form break as the back rounds very slightly**



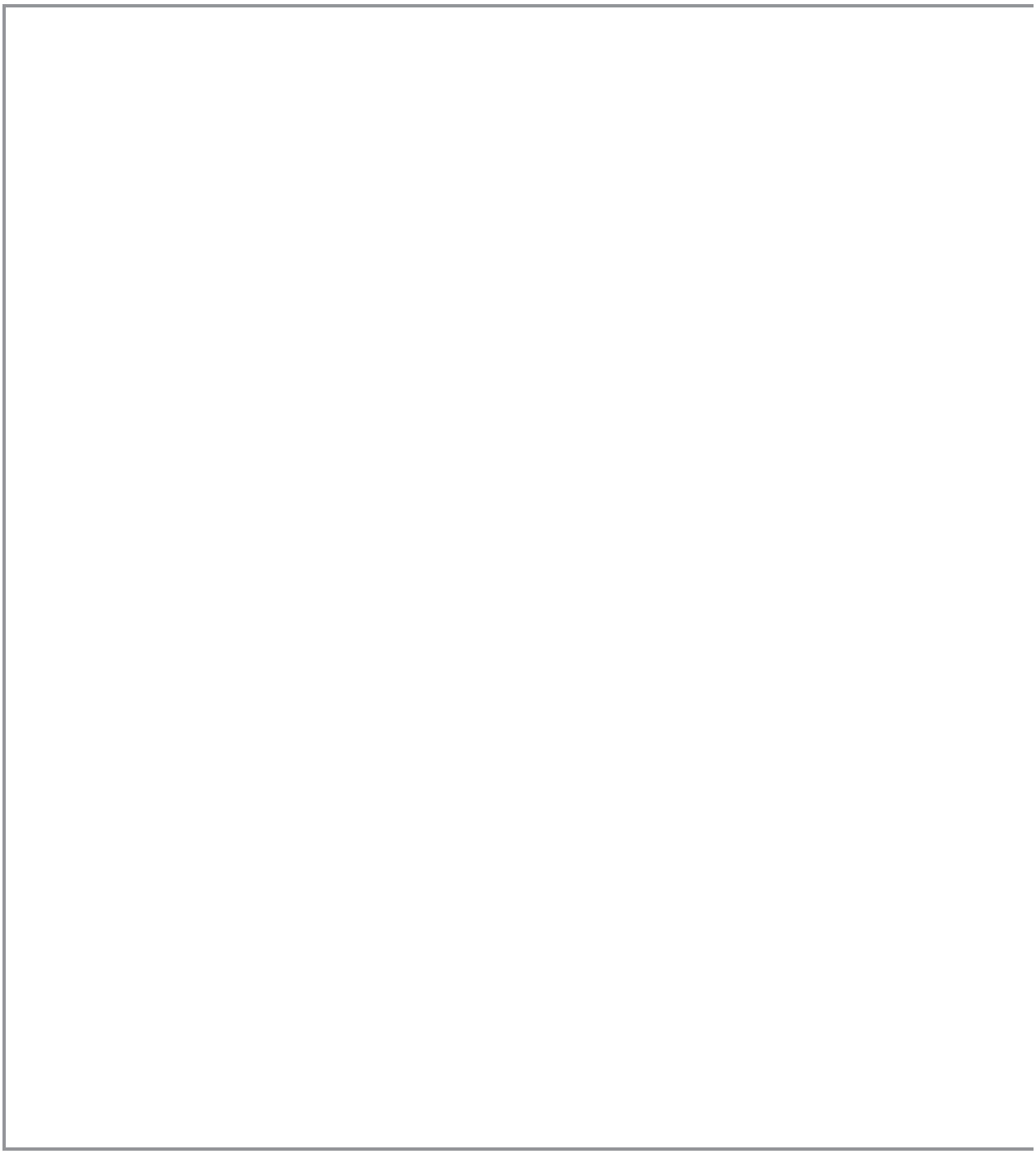
▶ **5: The bar clears the floor**



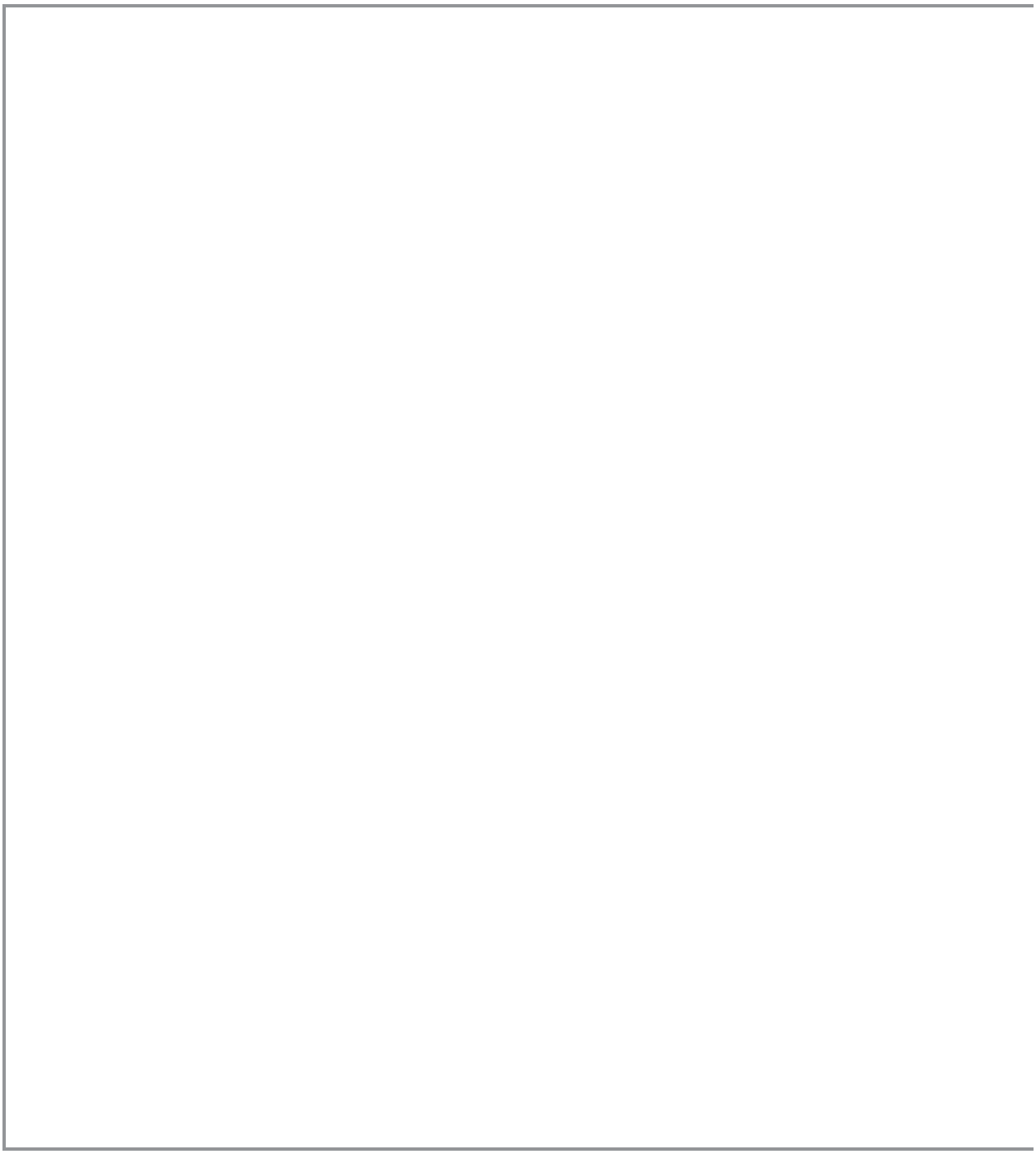
▶ **6: The lifter keeps the knees out and drives with the legs**

▶ **7: The lifter keeps the chin high as the bar nears the knees**

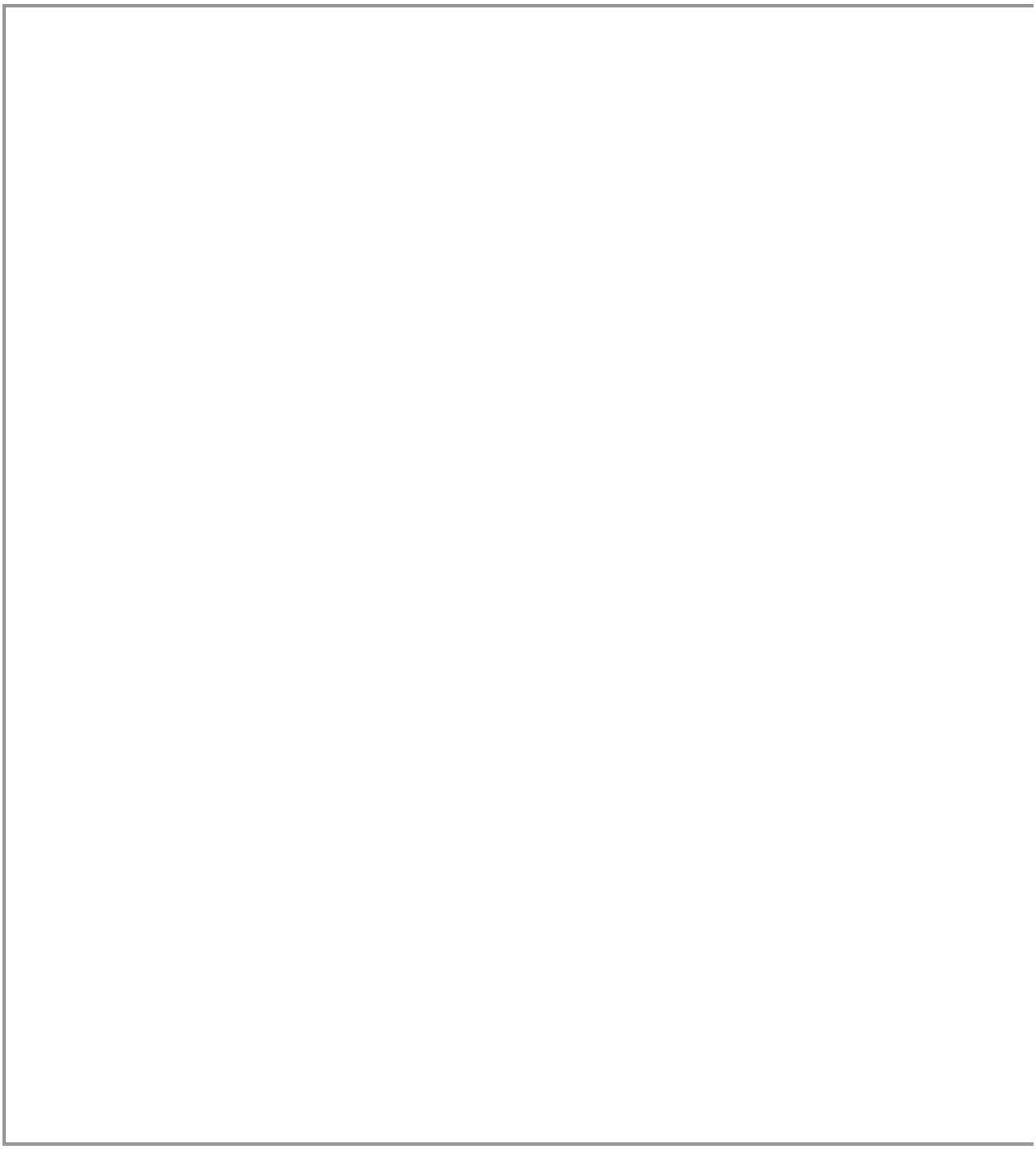
▶ **8: The lifter pushes the hips through as the bar clears the knees**



▶ **9: The lifter extends the trunk and continues to push through with the hips**



▶ **10: The lifter locks out the lift by locking the knees and leaning backwards slightly**



▶ **11: The lifter slightly tucks the chin and holds the finished position until the Down command is received**

Photo credit: Vadim Snitkovsky

Note: This lifter had 2 slight form breaks: he allowed his back to round slightly before the weight broke off the floor (image 4), and he locked out his knees before his trunk was in position (image 7) — both easy to do when lifting

maximal weight. But through strength and patience, he was able to overcome both of those slight mistakes and still successfully complete the lift.

Hips too far back – If your hips are too far behind the bar it will make you either lose power, or you will lean your upper body forward to compensate. To fix this issue first turn your toes out; they should be at a 45-degree angle at a minimum and perhaps more. Then turn your knees out to point in the direction of your toes. Push your hips forward; if necessary you can widen your stance. Your groin should actually be pretty close to the bar; if it is far away from the bar then your hips are too far back or too high.

Feet too wide – If your stance is super wide you can lose power at the bottom of the lift. If you simply can't budge the weight off the ground you may need to bring your stance in a little bit.

Feet too narrow – This will often feel good as you learn the exercise or lift with lighter weights, but, remember, the more narrow your feet are, the greater the range of motion (for a sumo deadlift). If your feet are really narrow then you are probably better off doing a conventional deadlift. A narrow stance will feel good when you are not very flexible, but over time you should work to widen your stance to take advantage of the benefits of performing a sumo deadlift.

Inability to lockout the weight out at the top – Nothing sucks more than pulling up a deadlift to near lockout and then failing to finish the job. All that work for nothing. If this happens it probably means you got out of position early in the lift. With a sumo deadlift it is very important to have patience and stay with your form. Keep your upper body upright, and don't let it round over as you pull. If you can't lock out your legs but your upper body is good then you probably need stronger legs. You can also try bringing in your

stance a little bit. If you can’t lock out the upper body but the legs are locked, then you need to start with and maintain better form throughout the lift, particularly when training with light weights. Work on your erectors, traps, the core, and your grip so your lower body can transfer its power into the upper body without the upper body changing its position. Just like with the conventional deadlift it is important to drive through with the hips by pushing them forward by squeezing the glutes at the top of the range of motion.

Common Competition Mistakes in the Deadlift

It is harder to make a quick fix to a problem in the deadlift than it is with the bench press or the squat. It is unusual to see someone miss a lift in the deadlift and then come back and make that same weight in their next attempt. Having said that, it is possible that you might make a correctable error in a deadlift attempt in a competition and still be able to fix the problem right away.

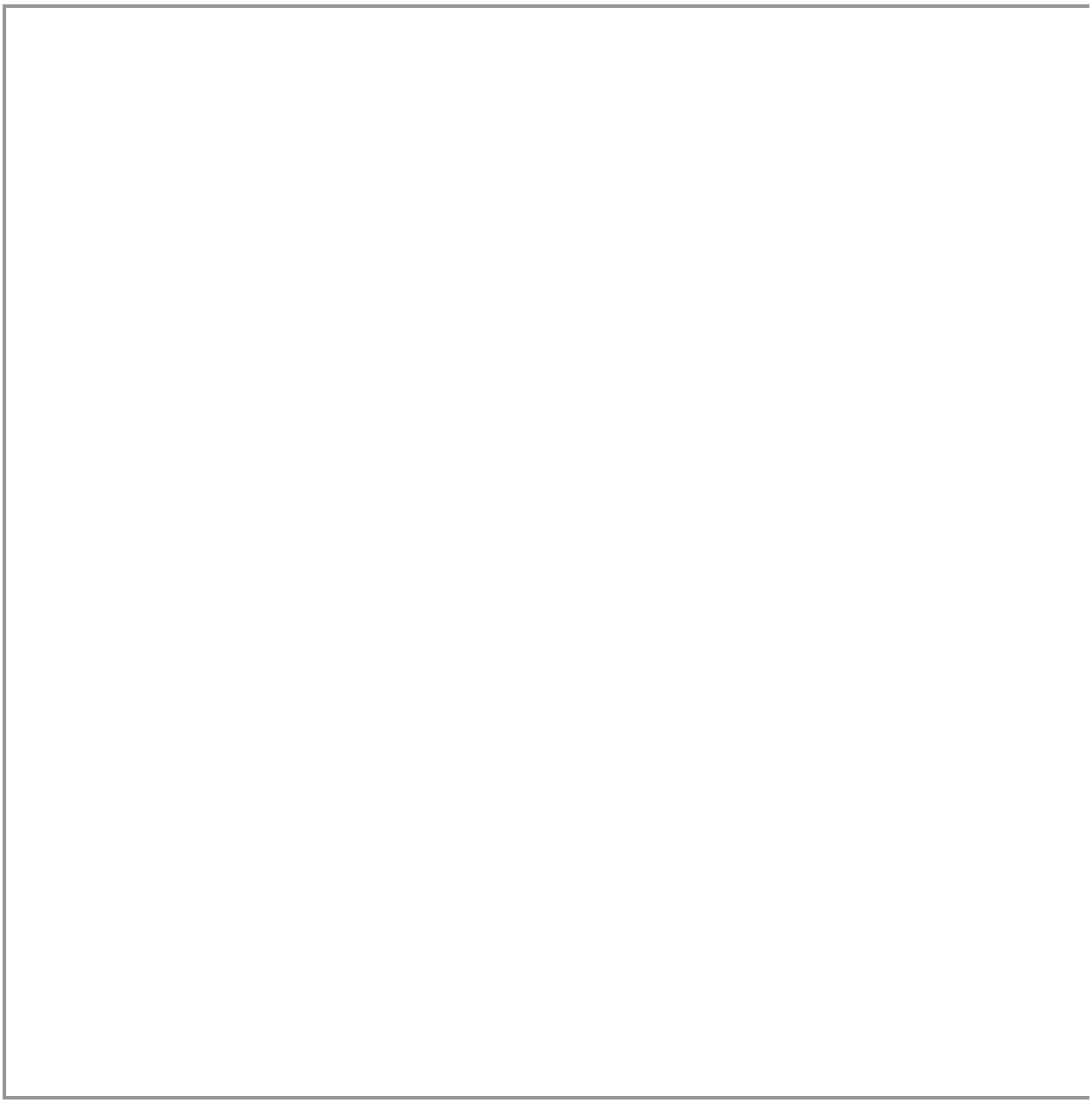
Starting too far away from the bar – When you set up to deadlift, get up very close to the bar. Your toes should be well in front of the bar, and your shins should be close to the bar if not touching it when you bend down. If you start off far away from the bar you will kill your initial momentum and you will be more likely to round your back (and injure yourself). If the bar is not set up the way you want it to be, e.g., it is crooked or not even on the platform, you are allowed to move the bar or ask the spotters to do so for you; don't be afraid to do that. Move the bar by rolling it; if you try to pick it up, then you may be disqualified, depending on how picky the judges are.

Bad set-up – It is very important that you get a good set-up before you begin the deadlift. This is even more crucial in a sumo deadlift. Sometimes getting a good set-up can be hard in a suit or if you rush it, so remember your cues and get a good set-up: back flat or arched, chest up, shins close to the bar, good grip, squeeze core, and drive with the legs as you pull with your upper body.

Lost grip – Sometimes you will have the strength to complete the deadlift but you will lose your grip as you pull it. Of course, the fix for this is to use chalk. Don't forget to chalk your thumb and the outside of your first finger. Make sure you don't get baby powder on your hands. If the bar seems dirty

with too much chalk or baby powder, you can ask to have the bar cleaned. This is particularly important if some of the people lifting before you have the bar rub up against their legs in the same place you put your hands, because then that part of the bar will often be covered in baby powder.

Uneven grip – It isn't uncommon to see lifters perform a deadlift with an uneven grip. I believe the use of the alternate grip adds to the confusion — lifters believe their grip is even and then just switch one hand, not realizing this can change how the bar is pulled. Lifters should use consistent landmarks on the bar and during their set up to make sure that the grip is even and in the right spot. The most obvious sign of an uneven grip is if the bar is tilted as the lifter pulls it up (this can also be a sign of a misloaded bar). This mistake is most common with the lifters who "grip it and rip it" and sometimes rush their setup, particularly when using a bar they are not used to.



An uneven grip or a misload can result in a tilted bar
Photo credit: Vadim Snitkovsky

Foot slips – Occasionally at the top of the deadlift a person's foot will slip, and the lift will be disqualified. Sometimes this happens because the person is really fired up and the weight is actually light (as in an opener), and the momentum catches the person off guard. This can also happen because the platform can get slippery. Both baby powder and chalk don't sit well on a

rubber platform, and that can make footing tricky, especially since the soles of most deadlift shoes don't have that much grip. Something that I would do that was pretty effective was to place a damp towel near the platform, and I would step on that before I got on the platform to lift. Of course you don't want the bottom of your feet to be really wet, but a little bit of water will absorb the baby powder and the chalk and generally helps to keep your footing.

Improper stance – Your stance can either be too wide or too narrow. If you are using a sumo deadlift, if you can't get the bar off the ground then try to bring your stance in slightly so you will have more power at the bottom. If you can't lock out the sumo, you might try to widen your stance just a little bit.

For the conventional deadlift generally most people have too wide of a stance. The wider the stance, the more you must lift the bar; try using a slightly narrower stance and bring your grip in at the same time. Admittedly, the middle of a competition is not the best time to be changing your stance and grip, but when it comes down to your third attempt and you missed the second one, you don't have many options left.

Setting the bar down – This part of the deadlift is often overlooked since you are performing just one rep but almost all powerlifting organizations have rules that state that you must return the bar to the ground under control. Sometimes lifters will either drop the bar from the top position or they will let it slam on the ground, and this can earn you red lights. They do this to prevent damage to the bar (so it doesn't warp) and to prevent damage to the floor. The weights should be returned gently to the floor in all situations, in the gym or during a meet, so this should be part of your training. But if you

get too excited and drop the weight, next time simply go down with the weight. When you are lifting a lot, you will make some noise; that is okay, but the bar should never be slammed down on the ground. The good news is that this problem has an easy fix.

Apparel for the Deadlift

There is not as much powerlifting equipment used in the deadlift as there is in the squat and bench press, but there are still some things that competitors like to use. The two most common tools, aside from chalk, are a belt and proper deadlifting shoes. A belt, as with the squat, can help support the lower back when lifting near maximal weights. It is possible that over reliance on a belt may weaken the abdominal muscles, just like with squats it is recommended to wear a belt only on your heavy work sets. Essentially all of your warm-up sets should be completed without a belt, and some of the work sets can also be performed beltless. Many powerlifters like to put on their belt for a tough set of 5 or less reps. This will keep your abs and lower back strong but will also allow you to spend enough time in the belt to be familiar with how it feels.

The second key piece of equipment is good deadlifting shoes. A good shoe for deadlifting has a very thin sole. This is because the thicker the sole of the shoe, the higher you have to lift the weight. A half an inch may not seem like much, but when you are lifting very heavy weights it can make a noticeable difference. The most common deadlifting shoe is a wrestling shoe, and they are easy to get at almost any place that sells sports shoes or online. Some people deadlift in slippers or just their socks, but that does not give your foot much support and is more likely to make you slip. Socks or bare feet are not allowed during a competition. If you have super tight calves you might find that deadlifting in Olympic Lifting shoes is more comfortable and gives you more power out of the hole.

Many federations are requiring their lifters to wear special deadlifting socks. These socks must be long enough to cover the shins; the idea here is to prevent a lifter from scraping their shins and getting blood on the bar which is a safety hazard for other lifters (and slows up the meet significantly). Any

shin high sock will work, and if the lifter has time to prep for this, they often buy cool powerlifting specific socks that match their singlet. If you are in a major pinch and have forgotten your socks, you can take a regular sock and cut a hole were the toes go and simply pull the sock up above your shins (you don't technically have to be wearing a sock on your feet when you compete).

It is worth noting that during the deadlift a T-shirt is optional; if you prefer to pull without a T-shirt on under your singlet, you can do that. This applies to both men and women. Females are required to wear a sports bra under the singlet.

Something else competitors do is they will put baby powder on the front of their thighs. This is to make their thighs smooth so there is little friction when the bar rubs up against their thighs as they pull the bar up. This is legal and there is nothing wrong with it; just make sure you don't get baby powder on your hands as it will ruin your grip. Baby powder has the opposite effect of chalk; it serves to reduce friction. Also don't make the mistake of putting chalk on your thighs as that will actually increase the friction and make it harder to pull the bar up your legs. Another thing you can do to reduce friction on your thighs is to shave your thighs. While this may not sound appealing to some, if you are particularly hairy it might be worthwhile.

Many federations are requiring their lifters to wear special deadlifting socks.

Some people wrap their wrists in the hopes of helping their grip. Personally I didn't like this because it would limit the blood flow to the hands, and if I wrapped them too early they would go numb and ruin my grip. Also you want your hands very close to your legs so that they just barely rub against your legs. A bulky wrap will rub even more on the legs and generally just get in the way. However, some very successful deadlifters have used this

method so you might want to try it out and see if it helps you. Remember that wrist straps (which go around the bar and hold onto the weight) and gloves are not allowed in a powerlifting competition.

The deadlift tends to have the greatest impact on the total.

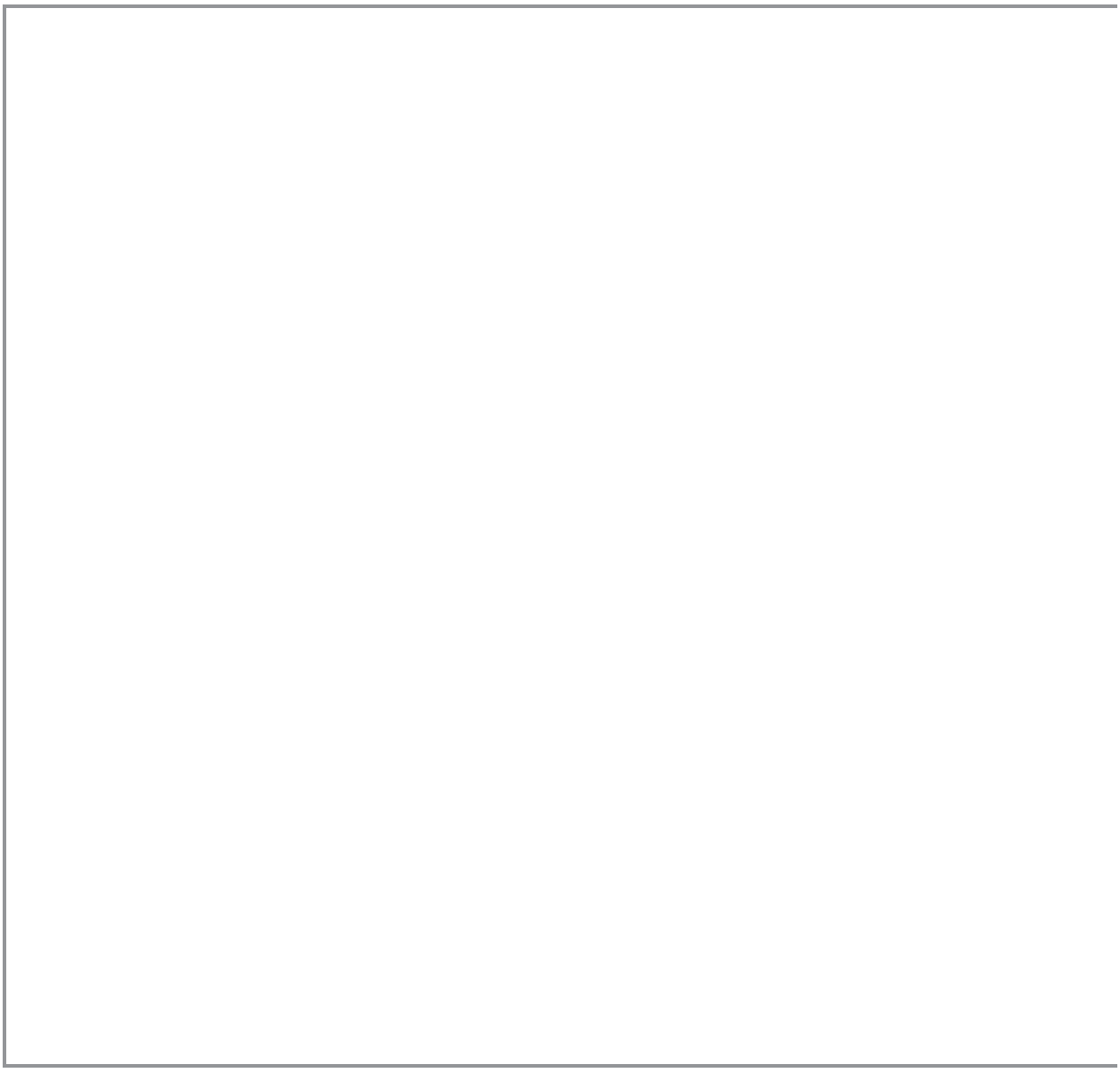
Benefits of Deadlifts

Deadlifts, like the other 2 competitive lifts, can be pretty brutal to train hard on, but common sense tells us what is the hardest in the gym will give us the best results out of the gym. From a powerlifting point of view, the deadlift is the lift a lifter will normally use the most weight on; thus, it tends to have the greatest impact on the total. One can often turn a 50 lb deficit after the bench into a 50 lb victory once the deadlift is added to the total. As such it is difficult to be a truly good powerlifter without a strong deadlift, and having a strong deadlift adds greatly to one's confidence during a competition — much the same way being a great finisher helps a 3 mile runner feel confident about their chances.

The deadlift offers almost countless health benefits as well. It does a great job of training many important muscles such as the erectors, multifidis, QL, the glutes, the hamstrings, the quads, the traps, and the forearm flexors, which are responsible for grip. It a structural exercise that loads up the bones of the spine and hip, thus increasing bone density and bone integrity. The deadlift hits the all-important posterior chain muscles — these muscles are crucial in developing sprinting speed and jumping ability, useful in almost all sports. Olympic Weight Lifters — who have to squat and deadlift regularly (they deadlift every time they pick up the weight off the floor) are known to have some of the best 5 M sprint times in the world, even compared to sprinters, and they are also known for having very impressive vertical jumps.

The deadlift stresses the neck muscles and the traps, useful for an everyday person alleviating tension in those areas and useful for any sort of contact athlete to protect their head and neck during a collision. It also develops grip strength, again useful in everyday life and in almost all sports from MMA to football to tennis. Unlike the other 2 lifts the deadlift is truly functional, meaning it does mimic what we do every day — we bend over

with no resistance, grab something, and pick it up. Practicing the deadlift teaches us how to lift something off the floor and builds up our core and our coordination in that activity, thus reducing the chance of a significant back injury during everyday life. The deadlift also has positive transfer to most other lower back exercises like good mornings and hyperextensions. The deadlift teaches the hip hinge, key in many sporting activities and everyday life, from grabbing the laundry basket to having sex. The deadlift is generally a bit easier to learn than the squat, and it is also more forgiving of previous injuries or postural distortions; people from most walks of life can begin deadlifting shortly after starting an exercise program assuming proper instruction and the ability to maintain proper form.



A lifter locks out a nice conventional deadlift
Photo credit: Vadim Snitkovsky

Technical Rules of the Deadlift

Listed below are the technical rules for the deadlift. These rules are specific to the 100% RAW Powerlifting Federation.

Deadlift:

1. **The lifter shall face the front of the platform with the bar laid horizontally in front of the lifter's feet, gripped with an optional grip in both hands and lifted until the lifter is standing erect.**

There is no "start" or "up" command; the lifter approaches the bar, grabs it however they prefer and picks it up until they are standing straight.

2. **On completion of the lift the knees shall be locked in a straight position and the shoulders back.**

The lifter's legs must be locked with shoulders back

3. **The Head Referee's signal shall consist of a downward movement of the arm and the audible command "Down." The signal will not be given until the bar is held motionless and the lifter is in the apparent finished position.**

The lifter must wait for the down command.

4. **Any rising of the bar or any deliberate attempt to do so will count as an attempt. Once the attempt has begun, no downward movement is allowed until the lifter reaches the erect position with the knees locked. If the bar settles as the shoulders come back (slightly downward on completion) this should not be reason to disqualify the lift.**

Once you try to lift it up once, that is your one try for that attempt.

5. **The Head Referee reserves the right to determine if a lifter has made a legitimate effort. Lifters are allowed to shake the weight, make jerks,**

or take their hands on and off the bar without penalty—providing the 60 second time limit has not elapsed.

It is okay to grab the bar and shake it to get fired up.

Causes for Disqualification of a Deadlift:

1. **Any downward movement of the bar before it reaches the final position.**
2. **Failure to stand erect with the shoulders back.**
3. **Failure to lock the knees straight at the completion of the lift.**
4. **Supporting the bar on the thighs during the performance of the lift. If the bar edges up the thigh but is not supported this is not reason for disqualification. The lifter should benefit in all decisions of doubt made by the referee.**
5. **Stepping backward or forward, although lateral movement of the feet or rocking of the feet between ball and heel is permitted.**
6. **Lowering the bar before receiving the Head Referee's signal.**
7. **Allowing the bar to return to the platform without maintaining control with both hands (e.g., releasing the bar from the palms of the hand).**
8. **Failure to comply with any of the items outlined under Rules of Performance.**

Summary

Take your time when learning how to deadlift. It is tempting to rush the weight because one is often naturally strong in the deadlift; you might be lifting 2 or 3 plates right away. But instead focus on your form, have patience, and watch your strength just continue to climb. One might be able to add 10 lbs a week to their deadlift for 30 weeks in the beginning of their programming! It isn't always easy to hit a big deadlift at the end of a long day after squatting and benching, but that is part of the challenge. Train hard, condition your body, eat right, and strengthen your mind and the possibility of that 9 for 9 day is well within your reach.

The deadlift, like the other two powerlifts, takes a while to master, but once you know how to do the exercise, you can often lift a lot of weight. Here is a sample beginning workout for a male and female of average strength who have never done a deadlift before.

It isn't always easy to hit a big deadlift at the end of a long day, but that is part of the challenge.

Beginning Deadlift program for a normal adult male

Week 1	45x12*	95x8	95x8	95x8
Week 2	45x12*	95x8	95x8	115x8
Week 3	45x12*	95x8	115x8	135x8
Week 4	95x8	115x8	135x8	155x8
Week 5	95x8	135x8	150x8	165x8

*Note: Use a normal range of motion with the 45 lbs; do NOT go all the way down to the floor.

Beginning Deadlift program for a normal adult female

Week 1	25x12	45x8	45x8	45x8
Week 2	25x12	45x8	45x8	55x8

Week 3	25x12	45x8	55x8	65x8
Week 4	25x12	45x8	60x8	75x8
Week 5	25x12	45x8	65x8	85x8

*Note: Use the normal range of motion on all sets; do NOT go all the way down to the floor.

Many people reading this will be well beyond the beginner stage, in which case, they should refer to program design chapters in this book for a more-detailed and advanced-deadlift routine.

Interview with Vince Anello

Vince Anello is a World Champion Powerlifter with an incredible 4x bodyweight deadlift; he is the first man in history to deadlift more than 800 lbs at under 200 lbs. He held the IPF world record for the deadlift in the 198 lb class for many years. This interview was conducted via email; I sent him these questions and these are his responses to them.

Vince Anello hoisting up a 4.3 x bodyweight deadlift

Photo credit: Vince Anello

Please give us a brief history of yourself:

I am 63 years old as of this interview. I was born in Cleveland, Ohio. I own and operate Anello Body Fitness, where I train private clients and work out myself. My web site is:

www.americanstrengthlegends.com

and this page will tell you more about me:

http://www.americanstrengthlegends.com/about.html

What are your best lifts?

In a competition I have squatted 750, benched 500, and deadlifted 821 officially; I pulled 880 in training, all at 198 lbs.

List some of the titles and awards that you have won.

- 20 World Records
- 10 National Titles
- 5 World Titles
- 1998 Induction Strength Hall of Fame York, PA

When did you start training; when did you first compete; what were your first competition lifts?

I started training in elementary school. I used to go in the woods and lift rocks and bricks. My first competition I benched 180, squatted 250, and pulled 310, I believe.

How much weight did you lift the first few times you tried the deadlift?

When I first started lifting, I thought 185 was a lot to Deadlift.

Did you experience any plateaus?

Through my career my deadlift plateaued at 400; then I reached 500. Then I was elated to get to 600. I set my goal at 700 and I made that, and then I wanted 800. It took me a couple of years and in 1975 I became the first man under 200 pounds to pull more than 800. Below are videos of my lifts:

http://www.youtube.com/watch?v=rQD9mHTLFCA

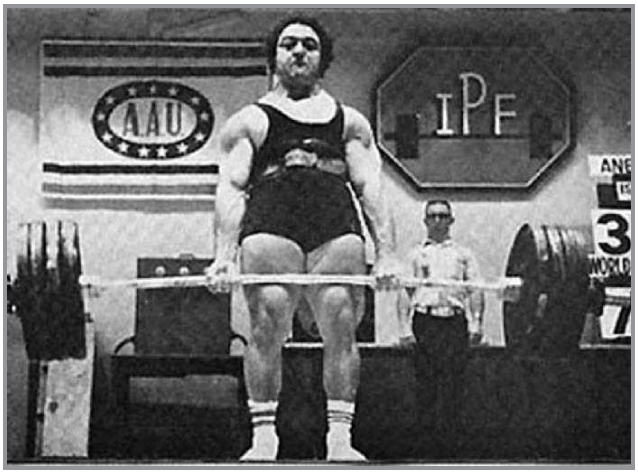
http://www.youtube.com/watch?v=hEU1Hyv4rX8

Provide a quick history of the progress you have made in the deadlift since you started.

I pulled 310 in 1966 to a meet best 821 in 1983. In 1976 I pulled 880 in training.

What do you feel is the key to succeeding in the deadlift?

There are hundreds of routines out there and champions have been produced by opposing theories; the common denominator is THE MIND!!! A Bulldog Mindset. Bite onto a goal and Don’t Let Go Until It Is Realized!!!



Vince locks out the deadlift
Photo credit: Vince Anello

What do you feel the best way to train for the deadlift is for a normal powerlifter?

In the deadlift I would train once a week. The stronger you get, the more rest is needed. Most overtrain. I would cycle my lifts.

What do you think of training with a high frequency (3+ times per week) on the deadlift?

I found one day per week on each powerlift. Sometimes I would deadlift once every 10 days. I am not a believer in high frequency deadlift training.

What do you think of training with a medium frequency (2 times per week) on the deadlift?

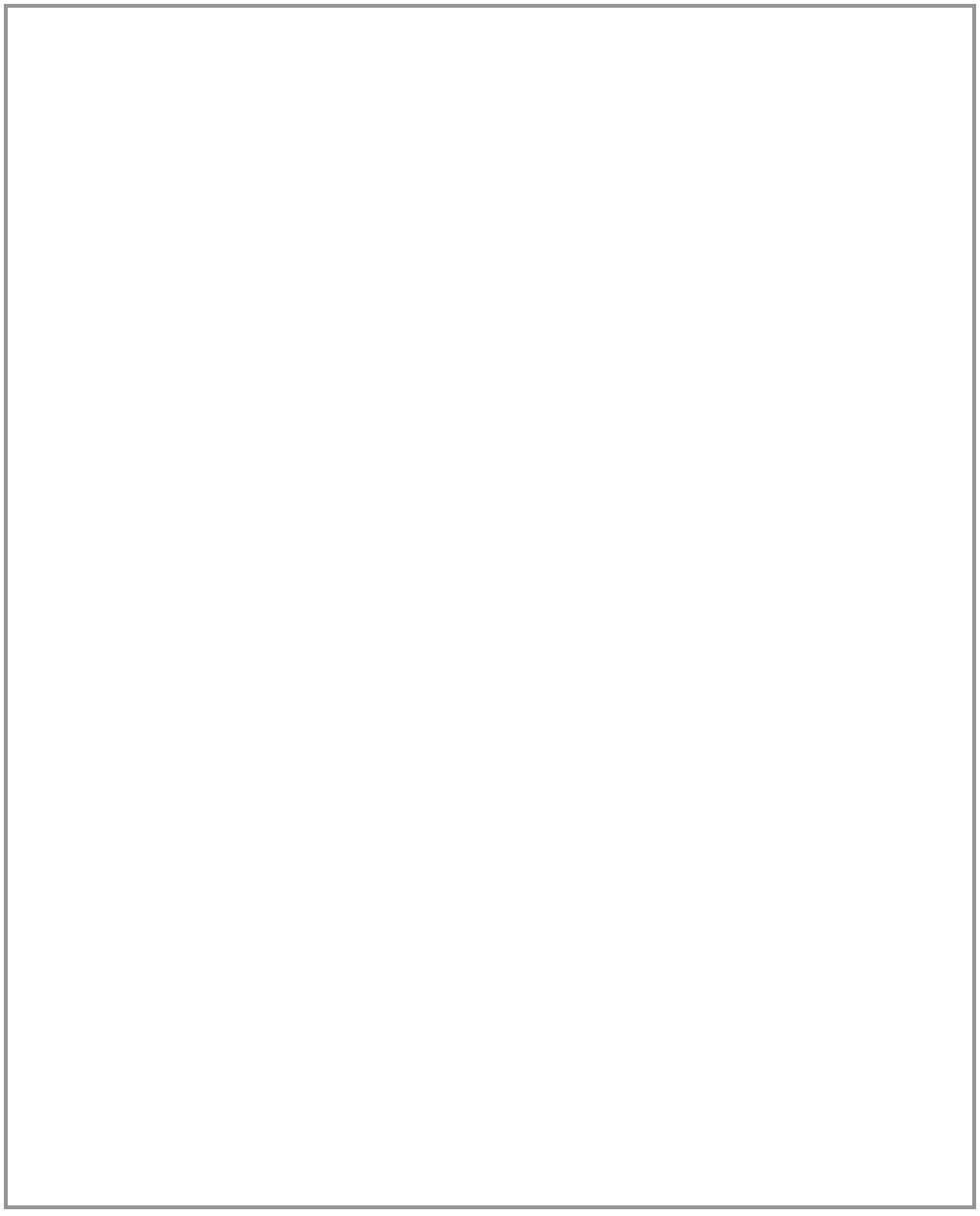
When I was really at a peak I would just go once per week on each lift; however, I do think you can do light assistance work on the muscle group a second time per week.

What do you think of training with a low frequency (1 time per week or less) on the deadlift?

This is what I believe to be ideal: as you get stronger you have to get more rest to allow recovery.

What are your favorite assistance exercises for the powerlifts?

For bench press I like flys and dips, also shoulders and triceps, even hammer curls for stabilizing muscles. For squats I like leg extension, leg curls, and hack squats, and for deadlift lots of grip work. I also like negative accentuated deadlifts — start with the bar up at the lockout in the rack, move backward and lower it first under control. You can combine this with a deficit deadlift and lower the bar to your ankles, I have pulled 750 in this fashion.



Vince bending the bar
Photo credit: Vince Anello

What are your thoughts on squatting and deadlifting on the same training day?

You have to squat and deadlift on the same day in the contest. If you squat one day and deadlift the next you won't have the endurance to do both in the contest.

What are your thoughts on training until failure on the competition lifts?

I have trained to failure in Bodybuilding but not in Powerlifting.

What injuries have you faced and how did you overcome them?

I have had knee injuries, shoulder, and back injuries. Light rehab work and rest is the best way to overcome the injuries.

How important do you feel that nutrition is to powerlifting performance?

I eat 7 times a day, small meals, and I try to get at least 30 grams of protein at each meal. Carbs I watched when trying to make weight for weight class as I believe a gram of carb holds 2 or 3 ounces of water (this may not be exact but I used this as an estimate).

What do you usually do with your weight and nutrition to prepare for a powerlifting competition?

I would drop carbs and sweat to make weight.

How important do you feel that supplementation is to powerlifting performance? What kind of supplement program do you currently follow? Are you sponsored by any supplement companies and if so, what are they?

I take protein supplements and vitamins. One supplement I believe in now is MonaVie. Two years ago I was demonstrating plyometrics to my athlete clients and I injured my knee. It bothered me for three months, I went to the doctor and they gave me drugs that I took for a couple of days, but I quit those because it made me feel terrible mentally. A client suggested MonaVie; after a week my knee felt better; a month later, no pain. I have not had any pain in 2 years. I know it has Glucosamine Chondroiton in it, but I have taken various forms before, and nothing seemed to help like MonaVie.

What are your thoughts on powerlifting equipment (gear) in powerlifting?

Gear should be used to prevent injury. When I competed it was just starting to become popular. I can’t give a fair evaluation of today’s equipment as I have not experienced it.

What are your thoughts on steroids in powerlifting? Do you compete in drug-tested competitions? How do you feel about the effectiveness of drug tests for catching those who use steroids?

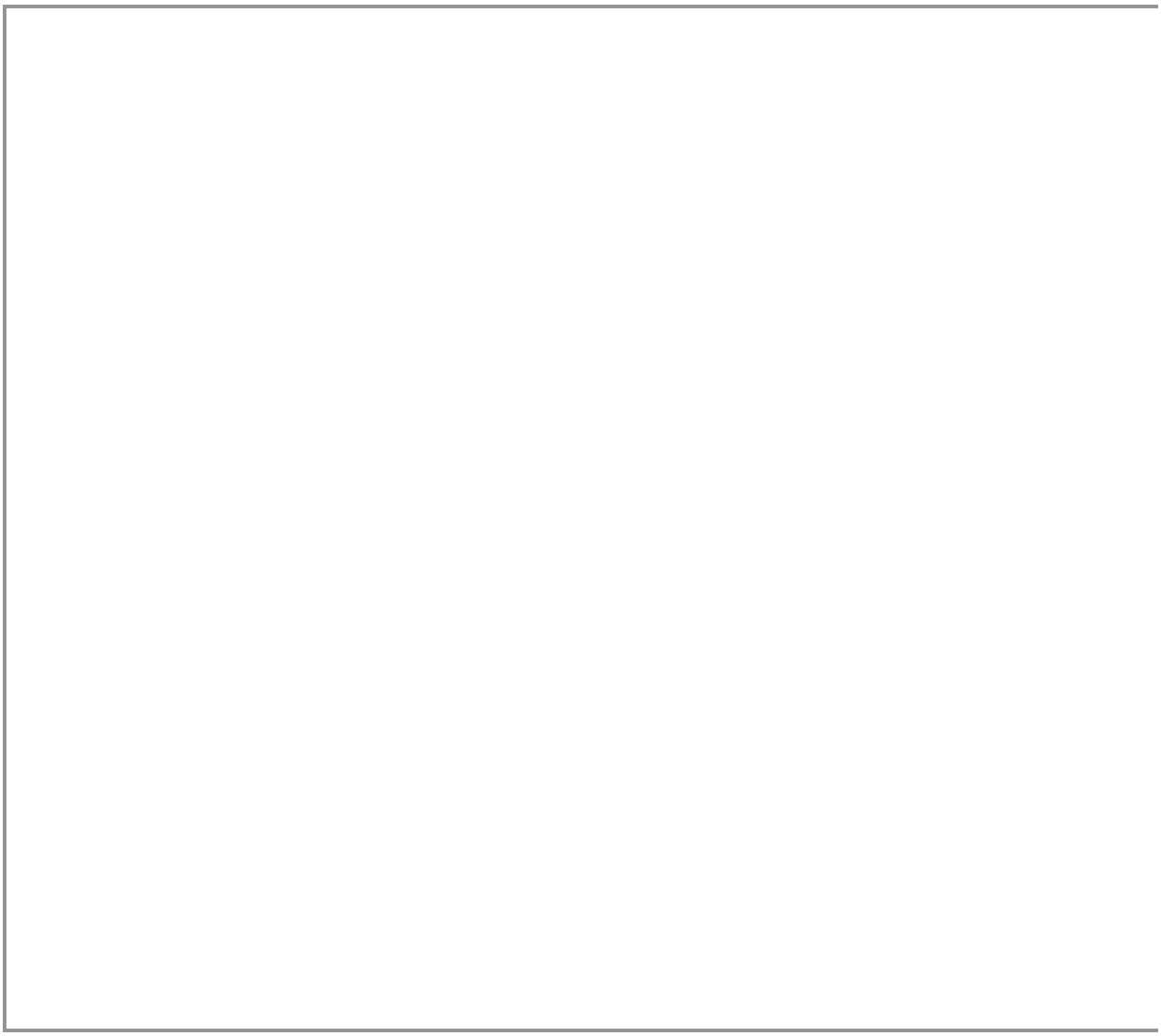
If you lift in a federation that bans drugs or is a drug-free federation, then you should not use drugs.

How would you feel about powerlifting being united? Would you compete in powerlifting if it was united and raw? Would you compete in powerlifting if it was united and single ply? Would you compete in powerlifting if it was united and multi-ply? What do you think the key to unification is?

I would like to see powerlifting united but there are so many different opinions and paths today that it would be extremely hard. I am retired now, and it is hard to speculate which path I would follow in my prime. I have lifted in an era that at its start did not allow even knee wraps to the full gear we have available now.

What is your thought about the importance of having a workout partner(s) or teammates in helping you train for powerlifting?

You need training partners in powerlifting for safety and motivation.



Bulldog Mindset — don’t let go
Photo credit: Vince Anello

What is your thought about keeping a training journal while powerlifting?

I kept a strict journal on all my training. Unfortunately I lost a journal in 1976 which documented my training that I used when I pulled 880 in training. I remember reading it in 1996, but since then I have moved and I must have lost or misplaced it.

What books, websites, or coaches do you suggest or follow in your lifting, and what would you suggest other lifters do to learn more about lifting?

Read and absorb as much as you can and find what works for you personally. You must follow your own path, not someone else's: "Be open to everything and attached to nothing." As we go through life we should be constantly experimenting!!!

What do you attribute your personal success in powerlifting to?

A BULLDOG MINDSET!!! GRASP ONTO A GOAL AND DON'T LET GO UNTIL IT IS ACCOMPLISHED!!!

What advice would you give to someone who was just beginning to take up powerlifting?

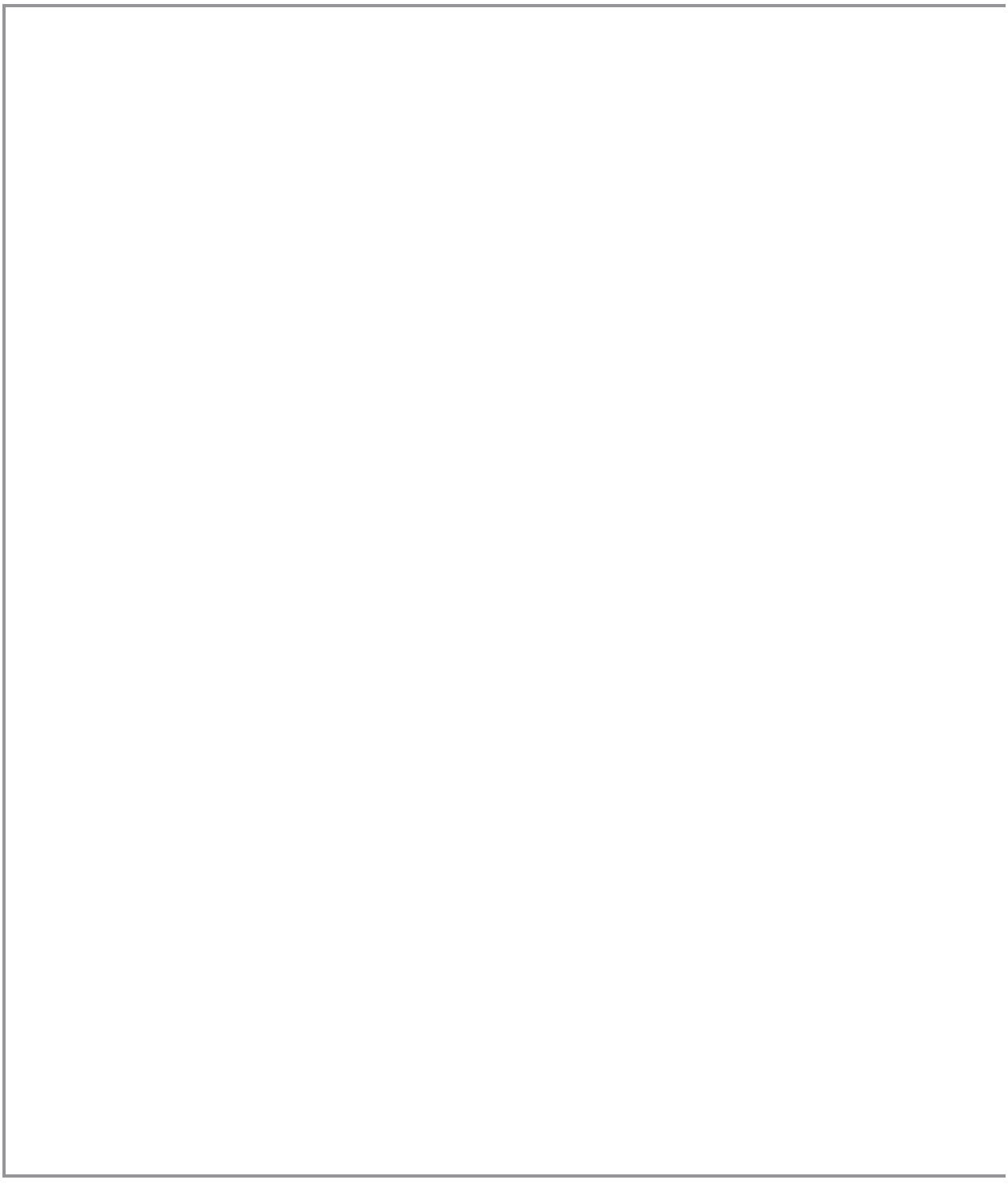
Go to contests and observe. Read as much as you can on training and the sport. Seek out an experienced coach.

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

Train the mind as much as the body!!!

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

The mind now is THE MAIN FACTOR IN SUCCESSS.



Vince Anello with his dogs
Photo credit: Vince Anello

Are there any changes that you would make to powerlifting if you had the power to do so?

I can’t and would not want to. I would like to see powerlifting be more popular with the public. I think a top champion powerlifter is just as much a great athlete as a champion basketball player!!! But the latter makes millions!!! LOL.

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

WE ARE ALL BROTHERS AND SISTERS IN THE STRONGEST SPORT IN THE WORLD REGARDLESS OF FEDERATION!!!! GOD BLESS YOU!!!!!!!

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers it is much appreciated.

Chapter 8

Increasing the Deadlift

The deadlift is a bad-ass exercise, no doubt about it. Its sheer level of brute strength makes it appealing to many a lifter. It is generally considered to require the lowest skill of the three lifts. The deadlift is the most natural lift, meaning untrained lifters will usually pick it up (no pun intended) the quickest, and they will have the ability to lift the most weight in the deadlift right off the bat. This is because the deadlift is the most functional of the three lifts, if you define functional as most mimicking movements we perform in everyday life. People are always picking stuff up off the ground, and thus people are always performing deadlifts. It is certainly not unheard of (although still rare) for a male who has never deadlifted before to lift double bodyweight on their first day of training for that exercise (usually this person will have been lifting weights for a while, just not deadlifting). Even though the deadlift is lower skill than the other two lifts, technique is still important. Refer back to Chapter 7 for a more detailed outline, but I am presenting here some bullet points to follow to ensure proper technique for the Conventional Deadlift.

A lifter locks out a nice deadlift
Photo credit: Vadim Snitkovsky

Technique

- Walk up close to the bar, shins essentially touching it
- Set feet in proper stance, often jumping position
- Raise arms above your head
- Create a significant arch in your upper and lower back
- Bend slightly at the hips and knees and incline forward; maintain the arch
- Lower the arms to the bar, maintaining the arch, hands close to legs
- Pull slightly into the bar to set chest and hips
- Pull the bar back into the body
- Set the body again for the pull
- Chest up, shoulders down and back
- Arch the lower back
- Tighten legs by pushing into the floor
- Tight grip
- Brace core, push against the belt if wearing one
- Head up, looking straight ahead
- Pull with the upper body to break the bar off the floor
- Drive hard with the legs, but don’t allow the hips to raise up first
- Keep the bar close to shins and thighs
- Hold chin high as the bar clears the knees
- Keep chest up
- Drive the hips hard into the bar once the bar clears the knees
- Neck returns to anatomical position as the pull nears completion
- Stand erect with knees straight, chest up, and shoulders back

Neuromuscular Coordination Techniques

To help improve neuromuscular coordination, there are certain techniques available that one can do. These options include:

Speed Deadlifts – This is a regular deadlift, but now you take 40–80% of your 1RM, with 60–70% being most common, and you try to accelerate it upward as fast as possible. Try to accelerate it hard just before you hit the sticking point, which isn't necessarily when the bar is on the ground. Of course, maintaining proper technique while doing this is crucial.

Cleans – Cleans can have both a positive and negative effect on the deadlift. The positive effect is that you must over pull the clean to get it to your shoulders. The beginning of the clean is very much like a deadlift, and because you have to explode into it, you will definitely activate the glutes and quads and hamstrings to get the bar moving that fast. The negative of the clean is that it is close enough to a deadlift to affect the motor program, but it could possibly negatively affect it. You tend to pull the hardest in a different spot in the clean vs the deadlift, and, of course, the weight is much lighter in a clean, especially if you are relatively untrained in it. The clean also requires a high level of wrist, elbow, and shoulder flexibility (along with smaller upper arms) and a lot of powerlifters have difficulty putting the bar in the rack position.

Bands – Just like with the other lifts, bands can affect the strength curve of a deadlift. Generally the bands are used to create tension as the bar is lifted off the ground. I find the bands help teach the lifter to accelerate through the sticking point, and they are useful if the lifter tends to fail near the lockout.

Principle of Specificity

The principle of specificity also applies to the deadlift. If you want the lift you are doing to improve the deadlift, it has to train some specific part of that lift. Exercises that most closely match the deadlift will include a barbell traveling in a similar bar path. They may match the ROM on the deadlift or they may just include part of it. Remember the problem of partials is that you only get stronger in the ROM you train in — if you spend all of your time performing rack pulls, for example, then you will get stronger on them, but it might not improve your actual deadlift very much.

A conventional deadlift
Photo credit: Vadim Snitkovsky

Exercises and their benefits

Conventional Deadlift – this is the standard deadlift, in which the hands are

outside the legs, the knees are bent and generally the most power is created. This deadlift can be done with a touch and go or a pause on each rep.

A sumo deadlift
Photo credit: Jennifer Thompson

Sumo Deadlift – The hands are inside the legs at the start of the lift. Generally the feet are spread wide, the toes out, and the lifter stays much more vertical than in a conventional deadlift.

Rack Pull – A deadlift where the bar is not on the ground but is elevated up on the pins. These pins are adjustable so the bar can be raised up an inch or two or many inches; the higher it is raised up, the easier it is. To have the best transfer of power to a real deadlift, make sure your pulling position is the same and don't let your knees travel forward.

Deficit Deads – A deadlift where the ROM is increased by standing on mats, an aerobics step, a block of wood, or using plates smaller than a 45. Generally increase the ROM by 1–4″; much more is likely to alter the start form too much.

Snatch Grip Deads – A deadlift with a snatch grip, which is much wider than a normal grip, thus making it harder and working the upper back more.

Trap Bar Deads – A deadlift with the trap bar, where you stand inside the bar and grab the handles, which are in a neutral grip position. This allows you to use more legs and stay more vertical. Too much focus on the trap bar will often have a poor carryover to a barbell deadlift. In addition note the handles are often higher than a regular deadlift, which also significantly affects the difficulty.

Fat Bar Deads – A deadlift on a fatter bar, thus increasing the role of grip strength.

Stiff Legged Deadlifts – A deadlift with the legs held straight, with the purpose of increasing the role of the hamstrings and isolating the lower back more. Good for increasing hamstring flexibility; can teach one to lift with improper form (legs straight).

Romanian Deadlift (RDL) – Like a stiff legged deadlift but the knees can bend; however, the shins can’t move forward, thus the hips are pushed back. Still isolates the erectors, glutes, and lower back but is a bit safer and usually easier than a Stiff Legged Deadlift.

Good Mornings – Like an RDL, but the bar is on your back in squatting position, ideally in low-bar position. Hits the same muscles as an RDL,

usually a bit harder.

Squats – Squats don't involve the same bar path as a deadlift, but they do make your legs and lower back stronger. Often lifters who have been squatting for a while, even without deadlifting, will have a solid deadlift. Once you get to elite levels the transfer of the squat will be less to the deadlift, but in the beginning it usually has a strong effect on it. The reverse is rarely true; a lot of deadlifting (by itself) does not usually create a strong squat.

Zercher Squat – This is where the bar is held in the crook of your elbows and you pick it up. Usually the bar is elevated on pins to make it easier. This causes a lot of pressure on the forearms and bicep insertion point; use this exercise with caution if you are also curling heavy. It hits the erectors, glutes, and hamstrings, some quads, and some arms.

Old School Hack Squat – This is where you stand in front of the bar, squat down, and grab the bar behind your legs. You then stand up and the bar slides up the back of your legs. The muscles hit are similar to a squat and a deadlift; of course, it is harder since it is behind you. I must confess I have never spent much time training on this exercise.

Joint Stability

This is usually not as much of an issue as it is with squats and bench press. Of course, a serious injury at any involved joint will limit your deadlift ability, but in general joint stability is not as much of a big concern. You want your ankles to be flexible enough to push the shin forward and yet keep the foot flat on the ground, without it rolling in or out. You need reasonable knee stability, but because the knee doesn’t bend much more than 45 degrees, most lifters’ knees can handle that well and still generate a lot of power, even if squatting is problematic. The hips and the lower back must be secure, but those are very stable joints that are held together mainly with bones, ligaments, and fascia.

Bodyweight

The increase or decrease in bodyweight usually has the least effect on the deadlift. This should not be surprising since we just discussed that joint stability is not as important in a deadlift, and one of the big benefits of increased bodyweight is joint stability. This is also shown by the fact that lighter weight lifters have pulled some numbers that are comparable with heavier lifters. Vince Anello pulled 810 lbs in competition and reported that he did 880 lbs in training at 198 lbs!

Mark Henry pulled 903 lbs at SHW, and he weighed well over 300 lbs. I don't share those numbers to say Mark Henry's lift was not up to par. A 900 lb deadlift is absolutely phenomenal, and according to some sources 903 is still the highest raw, drug-tested deadlift ever. The point is that some other super humans were able to come very close to that feat at significantly lighter bodyweights. The same is not true when comparing squats and bench press. Gaining bodyweight does little to increase the deadlift unless there is a large increase in strength along with it. Losing bodyweight doesn't have much of a negative. In some cases larger lifters find that their stomachs actually get in the way when they deadlift, and when they lose weight they can actually get into a more comfortable position and thus pull more.

Muscles Involved

Listed below is a table of the muscles that are involved in a deadlift. They are ranked on a scale of 1–5, with 5 being the most involved and a 1 being barely involved at all. It should be logical that to have a strong deadlift, you must make the muscles that are going to move that weight stronger. If a muscle has a 5 or a 4, it is working very hard in the exercise. It will get bigger and stronger just from that exercise, and it may also get sore after the exercise. A 3 is a medium level of involvement; it could be a weak point for a lifter, but usually it is not super significant to that exercise. A 1 or a 2 means the muscle is not that significantly involved, and as long as that muscle is not injured and assuming it is functioning properly, it should not limit the performance of the deadlift. Listed next to the muscle are some of the best exercises to improve the strength of that specific muscle. They may or may not directly transfer over to deadlift strength; that is more dependent on other factors (training status of the athlete, what the weak point in the deadlift is, skill level in the exercise, and correlation of improvement in that exercise and the deadlift).

Muscles involved in a Conventional Deadlift

Muscle	Contribution	Exercises to Strengthen the Muscle
Erectors and Multifidus	5 – Trunk Extension	Deads (any version), Good Mornings, Hypers, Reverse Hypers
Glute Maximus	4 – Hip Extension	Deep Squats (any), Deep Leg Press, Hip Thrusts, High Step-ups, Sumo Deads, Jumps, Sprints
Glute Med/Min	2 – Hip Stabilization	Fire Hydrants, Penguin Walks, Lying Leg Raises, Single Leg Work, Step Downs, Martial Arts Training (kicks)
Quads	3.5 – Knee Extension	Leg Press, Squats, Partial Squats, Smith Machine Squats, Hack Squats, Front Squats, Leg Extension
Hamstrings	4 – Hip/Trunk Extension	RDLs, Stiff DL, Glute/Ham Raise, Deadlifts, Good Mornings, Leg Curl
Adductors (magnus)	3 – Hip Extension/ Stabilization	Sumo Deads, Wide/Deep Squats, Wide/Deep Leg Press, Adductor Machine
Trapezius	4 – Shoulder Girdle Isometric Hold	Deads, Trap Bar Deads, Shrugs, Farmer's Walk, High Pulls, Cleans
Rhomboids	3 – Scapula Retraction Hold	Rows, Pull-ups, Lat Pulldowns, Retractions
Lats	2 – Shoulder Extension Hold	Bent Over Row, Chin-ups/Pullups, Rows, Pulldowns
Posterior Deltoids	2 – Shoulder Extension	Power DB Rear Delt Raise, Rear

	Hold	Delt Machine, DB Rear Delt Raise
Biceps	2 – Elbow Stabilization	EZ Curls, DB Curls, DB Hammer Curls, EZ Reverse Curls
Forearm Flexors	4 – Finger Flexion Isometric (Grip)	Wrist Roller, Wrist Curl, Holds, Grippers
Core (abs, obliques)	2 – Trunk Stabilization	Inverted Sit-ups, L-Holds, Cable Crunch, Ab Wheel, Hanging Leg Raise, Landmines, Rotations
Gastroc	1 – Knee Stabilization	Standing Calf Raise, Leg Press Calf Raise, Donkey Calf Raise
Hip Flexors	1 – Hip/Trunk Stabilization	Inverted Sit-ups, Decline Sit-ups, Hanging Knee Raises

Muscles involved in a Sumo Deadlift

Muscle	Contribution	Exercises to Strengthen the Muscle
Erectors and Multifidus	5 – Trunk Extension	Deads (any version), Good Mornings, Hypers, Reverse Hypers
Glute Maximus	5 – Hip Extension	Deep Squats (any), Deep Leg Press, Hip Thrusts, High Step-ups, Sumo Deads, Jumps, Sprints
Glute Med/Min	2 – Hip Stabilization	Fire Hydrants, Penguin Walks, Lying Leg Raises, Single Leg Work, Step Downs, Martial Arts Training (kicks)
Quads	5 – Knee Extension	Leg Press, Squats, Partial Squats, Smith Machine Squats, Hack Squats, Front Squats, Leg Extension
Hamstrings	4 – Hip/Trunk Extension	RDLs, Stiff DL, Glute/Ham Raise, Deadlifts, Good Mornings, Leg Curl
Adductors	4 – Hip Extension/ Adduction	Sumo Deads, Wide/Deep Squats, Wide/Deep Leg Press, Adductor Machine
Trapezius	3 – Shoulder Girdle Isometric Hold	Deads, Trap Bar Deads, Shrugs, Farmer’s Walk, High Pulls, Cleans
Rhomboids	2 – Scapula Retraction Hold	Rows, Pull-ups, Lat Pulldowns, Retractions
Lats	2 – Shoulder Extension Hold	Bent Over Row, Chin-ups/Pullups, Rows, Pulldowns
Posterior Deltoids	2 — Shoulder Extension	Power DB Rear Delt Raise, Rear

	Hold	Delt Machine, DB Rear Delt Raise
Biceps	2 — Elbow Stabilization	EZ Curls, DB Curls, DB Hammer Curls, EZ Reverse Curls
Forearm Flexors	4 — Finger Flexion Isometric (Grip)	Wrist Roller, Wrist Curl, Holds, Grippers
Core (abs, obliques)	2 — Trunk Stabilization	Inverted Sit-ups, L-Holds, Cable Crunch, Ab Wheel, Hanging Leg Raise, Landmines, Rotations
Gastroc	1 — Knee Stabilization	Standing Calf Raise, Leg Press Calf Raise, Donkey Calf Raise
Hip Flexors	1 — Hip/Trunk Stabilization	Decline Sit-ups, Hanging Knee Raises

To succeed in the deadlift, you must attack the bar.

Mental Attitude

To succeed in the deadlift, you must attack the bar. It is important to note that there is an inverse relationship between the skill level required for an activity and the arousal level necessary to be successful. The problem in golf, for example, is because the skill level is so high, arousal level must be relatively low. Golfers do well, start to get excited about the idea of winning (or afraid of the idea of losing), get over emotional and their performance goes down. The deadlift is a relatively low skill event. Because of this, arousal level must be high. Very high. It is not uncommon to see someone slapping a lifter's face to get them psyched, or a lifter will yell and roar to get fired up. Basically you want to imagine that you must run through a wall. Would you timidly approach a wall and kind of push up against it? Or would turn into a crazy beast and smash into the wall with all of your force? If the building was on fire and getting through that wall was your only way out, I sure hope you would go a bit "psycho" and smash through that wall. Picking up the weight is the same thing. You need to go a bit crazy to be able to lift that weight.

Get ready for a war with the iron

Get focused to pick up big weights
Photo credit: Vadim Snitkovsky

Optimal Frequency of Training for the Deadlift

Because the deadlift is a relatively low skill exercise, and because it is likely a powerlifter will be squatting with at least a reasonable frequency and the muscles involved in a squat and deadlift are very similar, the suggested frequency for the deadlift is generally lower than the other exercises. Lifters often find that they can train the deadlift every other week, every third week, even every fourth week and still be successful. I know personally in the program that I followed that led up to my lifetime best deadlift of 700 lbs, I was following a Westside cycle of squatting twice a week (once heavy and once light) and deadlifting only every third week.

Here is an interesting quote from Andy Bolton, taken from powerliftingwatch.com. In this meet Andy pulled 953 raw for a new (at that time) All-Time raw record. This is what he had to say about his training leading up to that competition:

> "I went to this competition not really knowing what I would be able to do, after only 6 deadlift workouts since my knee surgery in November 2009 [the competition was in May 2010 — 6 months after surgery]. I went there in my mind that I would have been happy with 900; that's why I did not use a suit. I wanted to hold it back and not get hurt again; so anyway warm-ups went well up to 660 lb, then my opener 770 then to mid 8's after that. It felt so so light so I wanted to pull more than anybody else out there with no suit and it felt easy 953."

Two things to note from this. First, he trained the deadlift only 6 times in 6 months and since this was after surgery, I am guessing a lot of those training sessions were not super intense. That is 6

The suggested frequency for the deadlift is lower than the other exercises.

sessions in 6 months most likely at <80%
total intensity to prepare for a max. Second thing is notice the percentage of the weight he used on his attempts. Let's assume his max was 953, since that is what he did — we know it was at least that.

Last warm-up =	660 lbs or 69% 1RM
First Attempt =	770 lbs or 80.7% 1RM
Second Attempt =	860 or 90.2% 1RM
Third Attempt =	953 or 100% 1RM

Making 10% jumps is relatively conservative (assuming you know your real max) but it turned out to be successful, and the benefit is that because the second attempt is only at 90% you a) will almost for sure be successful with it and b) will save a lot of energy to unleash on the third. Plus when you are in a meet and on the platform those submaximal weights really fly up. In looking at these numbers I can't help but see the similarity in weight selection to my own best deadlift day ever. I am certainly not trying to compare the impressiveness of my deadlift to that of a new All-Time world record; in fact it is the exact opposite. I am trying to show that a lesser lifter can often perform similarly if they match a more advanced (and wiser) lifter's percentage choices.

My deadlift numbers:

Last warm-up =	475 lbs or 67% 1RM
First Attempt =	540 lbs or 77% 1RM
Second Attempt =	630 lbs or 90% 1RM
Third Attempt =	700 lbs or 100% 1RM

Here is how the numbers would work out for 2 lifters using lighter

weight:

Deadlift	Goal 275 lbs	Deadlift	Goal 405 lbs
Last Warm-up	190 lbs	Last Warm-up	280 lbs
First Attempt	220 lbs	First Attempt	325 lbs
Second Attempt	250 lbs	Second Attempt	365 lbs
Third Attempt	275 lbs	Third Attempt	405 lbs

Presented below is a theoretical guideline as to the suggested frequency for deadlifting.

As with the chart for the squat and bench press, note the intensity listed here is the total intensity percentage, not just the percent of the 1RM. Unlike with the squat and the bench press, I don't believe the optimal frequency goes down and then goes back up, I think for the deadlift it just steadily decreases over time. Even for beginners I would not suggest deadlifting more than twice a week; once a week also usually works fine. Once a week works well for the next couple of years, and then as ability, skill, intensity, and muscle mass increase the lifter can experiment with lower frequencies. I would think that once every other week to once every third week would be optimal for lifters with a 2x bodyweight deadlift or better, especially if they are squatting with reasonable intensity as well. Because the skill is higher for the sumo deadlift, sumo lifters may find they need to practice that lift a bit more and thus have a

Even for beginners I would not suggest deadlifting more than twice a week.

higher frequency. If a lifter finds that they want more opportunities to practice and perfect their technique, then incorporating speed work more often can certainly accomplish that goal.

Specific Training Plan

You can't just blast away at deadlifts all the time and expect the weight to go up; you have to make your overall body stronger. Here is a 12 week plan to focus on the deadlift.

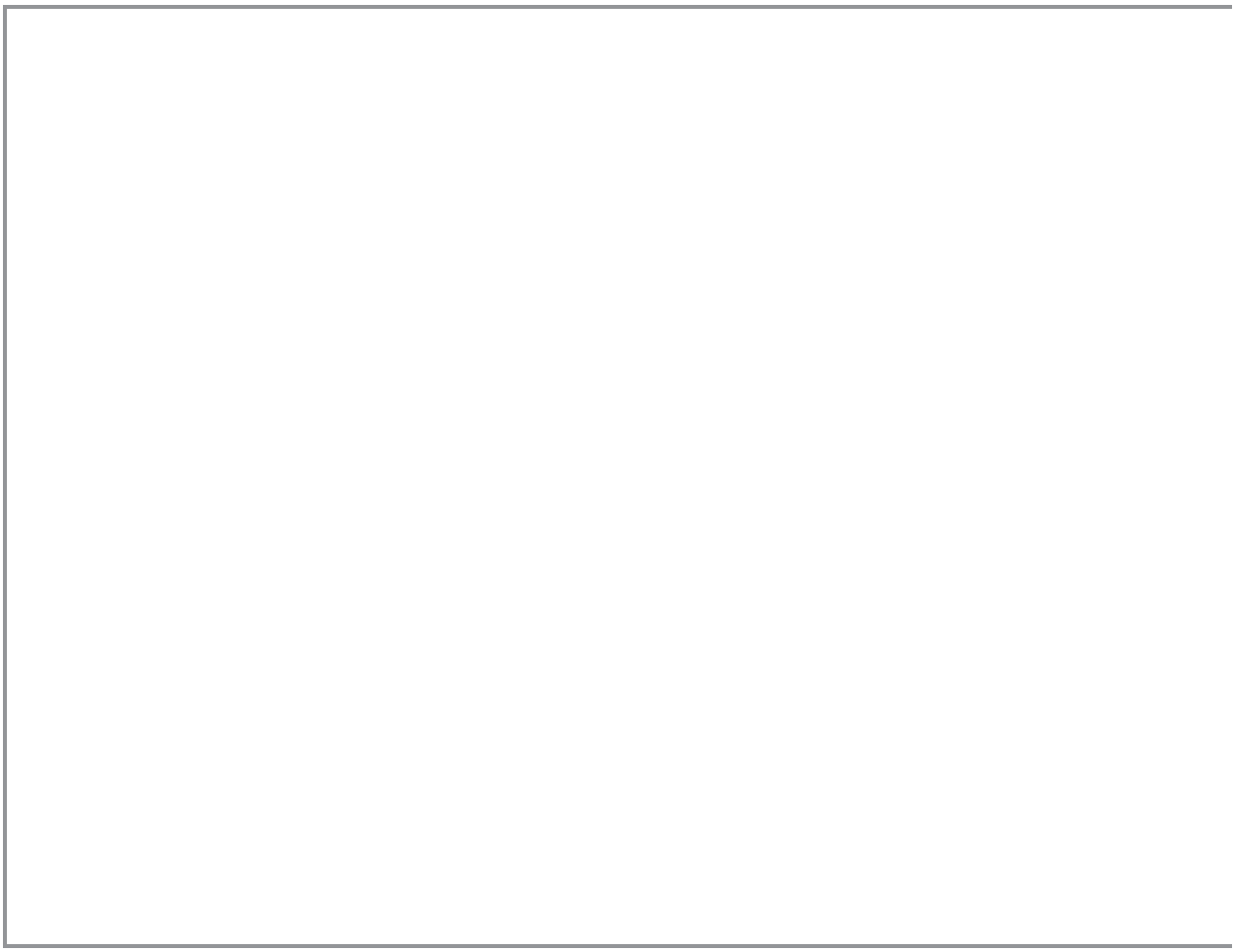
http://www.t-nation.com/free_online_article/most_recent/twelve_weeks_to_a_b

> ***Deadlifting heavy weight is serious business.***

Summary

You don’t have to hammer away with endless sets when training the deadlift, but instead channel that energy into intensity and perform the few work sets in each deadlift session with focus and aggression. Deadlifting heavy weight is serious business. I have seen many missed attempts in the gym due to lack of focus. Get the rest of your body strong and apply that strength to the deadlift. Believe in yourself and believe in your ability and get at that weight. All you have to do is pick it up. Can you?

All you have to do is pick it up
Photo credit: Doug Jantz photography



Sioux-z Hartwig-Gary with her famous jump before her pull
Photo credit: Sioux-z Hartwig-Gary

Interview with Sioux-z Hartwig-Gary

Sioux-z Hartwig-Gary is multi-time National Champion with the USAPL and an IPF Open World Champion. She has been lifting, and lifting very successfully, for more than 20 years. She competes both raw and in single ply gear, and she is a drug free athlete. This interview was conducted via email, I sent her these questions, and these are her responses to them.

Provide us with a brief history of yourself.

I am 42 years old. I was born in Watertown, South Dakota. My husband, Matt Gary, and I own a training facility, Supreme Sports Performance & Training, Inc. in Rockville, Maryland. I am also a massage therapist, personal trainer and office manager. I keep busy!

What are your best lifts?

My best raw lifts in the gym (done in the past few years) are 333 lbs squat, 181 lbs bench press and 319 lbs deadlift. In competition it is 308 lbs, 170 lbs and 319 lbs respectively. In competition I weighed 114 and in the gym somewhere between 114 and 119 lbs. My best geared lifts at 114 lbs in competition are 380 lbs squat, 226 lbs bench press and 369 lbs deadlift. In the gym I have benched 237 lbs. I also competed in a standing strict curl competition. I curled 75 lbs, I was hoping to get closer to my bodyweight but I injured my forearm and didn’t go heavier.

When did you start training; when did you first compete; what were your first competition lifts?

I started training in February 1991. I trained six weeks and then competed three times in four weeks. I weighed 104 lbs and in my first competition did 205 lbs squat, 105 lbs bench press and 205 lbs deadlift. Three weeks later I did 225 lbs squat, 120 lbs bench press and 250 lbs deadlift. You can see how learning better technique made a huge difference in my numbers in such a short time. I was not able to train much the rest of the year, did a little at the beginning of 1992 but truly started training without ceasing in August of 1992.

Sioux-z Hartwig-Gary about to explode on the squat
Photo credit: Sioux-z Hartwig-Gary

How much weight did you lift the first few times you tried the exercises?

I am an unusual person. I had actually learned how to bench press and squat when I was 14 years old. We were taught on a Monday, practiced on Wednesday and tested on Friday. (All the same week.) Weighing a mere 98 lbs I squatted 200 lbs and benched 100 lbs. Now, those lifts were not to competition standards but still quite impressive. Back then, “girls don’t lift” is what I was told and I did not do it again until I was almost finished with college. As far as when I lifted in 1991, I had squatted 185 the first week; most likely I could have done more but the guys I was training with started me with 65 lbs and progressed in tiny increments as they did not believe me when I said I had squatted 200 when I was 14 years old. I did a 165 lb ugly deadlift the one day they taught me before I competed and did a more proper 205 lbs. Again, very unusual, I was blessed with a natural God-given ability to lift weights. By the third time I tried I had done double bodyweight in the squat and deadlift and bodyweight in the bench press.

What was your first training program like and how far did that take you?

Honestly, I can't remember but most likely it was progressive overload. We had no idea where I was at strength wise, and I trained such a short time the first time I started to compete. After that I moved from South Dakota to Virginia and trained all by myself. I learned a lot from my competitors my first year, and my first real training program was progressive overload. I did that style for many years. As one of my first coaches, Captain Kirk Karwoski, used to say, if it ain't broke, don't fix it. I would simply add about five or ten pounds onto the end of my next cycle after a competition and do it all over again. Competition cycles were usually 8–12 weeks.

At what lifting weight did you first plateau in the exercise? How long did you plateau there? What did you do to get past that plateau?

I got very strong very fast in the bench press. I benched 172 lbs at a bodyweight of 104 lbs in less than six months of continual training. That was either late 1992 or early 1993. I never did that again in the 105 lbs class. I believe this was due to dieting. In training I did do slightly more but never in competition. When I started competing I naturally weighed 105 lbs. I stayed in the 105 lbs weight class until 2001, but my weight got up to 114 lbs. I was losing almost 10 percent of my bodyweight to stay in the class, and it zapped my upper body strength. I never really got past the plateau until I went up a weight class. Otherwise, I never really reached a plateau until 2003. I hit all time PRs (personal records) in 2003 and then had a bulging disc in my neck that stopped me from competing for a year. Until recently I could not get my deadlift back up. I just tied my PR at USAPL Nationals and feel I am ready to surpass it. I believe switching to singles and double in the deadlift helped me perfect my technique again and got me over the plateau.

Squat Time!
Photo credit: Sioux-z Hartwig-Gary

Give a history of the progress you have made in your lift since you started.

I started at 105 lbs in 1991 and within two months had done 205, 105 & 205 in competition.

Less than a month later I had done 225, 120 and 250 in competition.

I restarted in late 1992 and after getting a bench shirt and lifting regularly by March 1993 I had done about 265, 172, 300.

When I stopped competing in the 105 lbs class in 2000 my best competition lifts were 330, 172 and 330.

One of the best things to happen to my lifting was gaining so much muscle in 2000 that I did not make the 105 lb class for nationals in 2001. Weighing 109 lbs I did 330 lb squat, 187 lb bench and 347 lb deadlift. I had more in me but had thought I would be 105 lb and did not realize the strength I had. Within a year I had done 341 lb squat, 214 lb bench and 358 lb deadlift. My current bests in competition wearing gear are 380 lb squat, 226 lb bench and 369 lb deadlift.

What do you feel is key to succeed in the lift(s)?

1. Train with someone who is better than you whenever possible
2. Focus on learning proper form and technique before you add a lot of weight or gear.
3. Never say never. Our minds often stop us from doing things that our bodies are capable of doing.

What is your current training program like (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I train M, W, F and Sat. Currently I do heavy squats on Monday and vary my assistance exercises every three weeks on all days of lifting. Wednesday I have heavy bench, Friday is light squat and heavy deadlift, and Saturday is overhead press and light bench. I will not go into sets and reps because it varies. However I will state that our base training is 80% for typically five sets of two to five reps. As far as overload, the only overload is heavy handouts and board presses in the bench press and walkouts in the squat. The hand outs and walk outs are typically not more than 105% and are after the last heavy set of the day. I also typically do these for a few weeks toward the end of the cycle but not the week before competition.

If different from above, what do you feel the best way to train for the lift is for a normal powerlifter (routine, days per week, exercises, sets, reps, rest, etc.)? How often do you vary it? How do you introduce overload?

I recommend three to four days if possible and the most rest between your squat and deadlift days. Benching twice a week has made a big difference in my and other elite lifters' training so I recommend benching twice, once heavy, once light, even if it means adding it to the end of your squat or deadlift workout.

I will vary cycles every 12 weeks or so. However, if a training cycle goes really well and seems to be working still I will redo the cycle changing the numbers to my new meet PRs.

What do you think of training with a high frequency (3+ times per week) on a certain powerlift? Have you done this, what were your results, and do you believe in it or not?

Training with high frequency can be very rewarding but can also cause overuse injuries. I have done this and had amazing results. I went from 300 lbs squat to 333 lbs squat in one month doing a routine where I squatted four times a week. Great results but I did have overuse “injuries.” I did the same for the bench press and went from 165 lbs raw to 181 lbs raw in one month. In short, I believe it works in adding strength and perfecting form if done properly. I also caution people that it is not easy and may cause overuse injuries.

Sioux-z Hartwig-Gary dominating a deadlift
Photo credit: Sioux-z Hartwig-Gary

What do you think of training with a medium frequency (2 times per week) on a certain powerlift? Have you done this, what were your results, and do you believe in it or not?

I do train the bench and squat twice per week and believe this is the best frequency for me. Once does not seem to work as well, and more times does not seem to let the body recuperate as well.

What do you think of training with a low frequency (1 time per week or less) on a certain powerlift? Have you done this, what were your results, and do you believe in it or not?

Typically I only deadlift once a week and have learned that if I am squatting I can actually deadlift less than once a week and maintain my strength. However, a beginner should train once a week in the deadlift to perfect form and technique.

What are your favorite assistance exercises for the lift? What areas do you feel they work? How often do you do them? How heavy do you go? Do you feel there is a direct correlation to any of those exercises and your lift (e.g., your lift is always 100 lbs heavier than X exercise).

I love glute/ham raises (GHRs) to assist lower body. I vary doing high reps with bodyweight and sets of 5–8 with added weight. Other favorites for the lower body are: step-ups and lunges for single leg work, deadlifting with 35 lb plates, deadlift to knees and halting deadlifts. Typically the deadlift with 35s (or standing on a plate if you are conventional; I am sumo) will be up to 85–90% of my raw max and deadlift to knees and halting deadlifts are typically 80–85%. For bench press my favorites are the pull-up and board presses. I vary grips in the pull-up/chin-up and also add weight. Board presses I will do up to 110%. Typically it is either triples at a weight at or slightly above whatever I ended with for a single to the chest that day.

What injuries have you faced, and how did you overcome them?

I have been blessed with few lifting injuries but have dealt with a pulled groin, strained hamstring, and bulging disc in my neck. Anything else wasn't bad enough for me to consider it an injury. I pulled the groin skiing and had to narrow my squat stance and alter my technique. Time and perseverance is how I overcame.

The strained hamstring came at a very busy time so I had to work through it. I added my suit bottoms early in training to help protect the strain and when I was finally able, I trained lighter and gave it some rest. With the bulging disc I did only rehab work for two months — no weight training at all. I then started back benching only the bar and slowly added weights. Sometimes the hardest thing is admitting that you need time off. I believe that taking the time off and rehab was the only way to get back to my previous strength. My disc bulge happened in 2003, the year I won the IPF Open Worlds and USPAL Bench Press Nationals. I had never been stronger, and there was a point I wondered if I would ever get back to that strength. I was 35 when it happened. I can happily say that at 42, I have come back and am even stronger. My coach and husband Matt Gary has shown me that varying my training programs and intensity are key. Without his help I doubt I would have come back stronger than ever.

How important do you feel that nutrition is to powerlifting performance? What kind of nutrition program do you currently follow (calories, % of carbs, fat, protein, strictness)?

I believe nutrition is very important. When I am eating healthier I feel better and perform better. As far as calories — I am not sure. When I am cutting weight it is around 1500 calories but it's higher on heavy training days. I am quite strict the last month before a competition but otherwise since I typically eat healthy, low fat, medium carbohydrates and high protein (I am able to eat some chocolate, too). Sorry, I could not tell you percents. Samples of food that I eat regularly are oatmeal, sweet potatoes, apples, broccoli, chicken breast, turkey burger, lean beef, fish, and as a treat, english muffins.

What do you usually do with your weight and nutrition to prepare for a powerlifting competition (drop 10 lbs, stay the same, consume high carbs, etc.)?

I learned from my mistakes; dieting from 114 to 105 when I did not have much fat to lose was a mistake. I try to stay around 116–119 lbs in the off-season competing at 114 lbs. I never weigh more than one pound per week out from a meet. I will eat a few extra carbs the days I squat and bench press but otherwise I typically do not consume high carbs except the day of the meet. On meet day I worry less about protein and more about energy. I like to eat what I am used to so I will typically have a little eggs, oatmeal and fruit for breakfast after weigh-ins. During competition I nibble on a peanut butter/Nutella sandwich, chips or crackers, bananas and banana bread or something else that I can have small bites of that give me immediate energy and longer lasting energy. I also drink Emergen-Cs which are an electrolyte drink.

Sioux-z getting ready to deadlift
Photo credit: Sioux-z Hartwig-Gary

How important do you feel that supplementation is to powerlifting performance? What kind of supplement program do you currently follow? Are you sponsored by any supplement companies and if so, what are they?

I take a multi-vitamin and drink protein shakes for simplicity. I don't really consider the shakes a supplement but more a meal replacement. I try to eat 5–6 times a day and find it easier to have oatmeal and a shake for breakfast than to make eggs every morning. I have never been a supplement person.

What are your thoughts on powerlifting equipment (gear) in powerlifting? How do you incorporate gear into your training? Are you sponsored by any equipment manufacturers and if so, who are they?

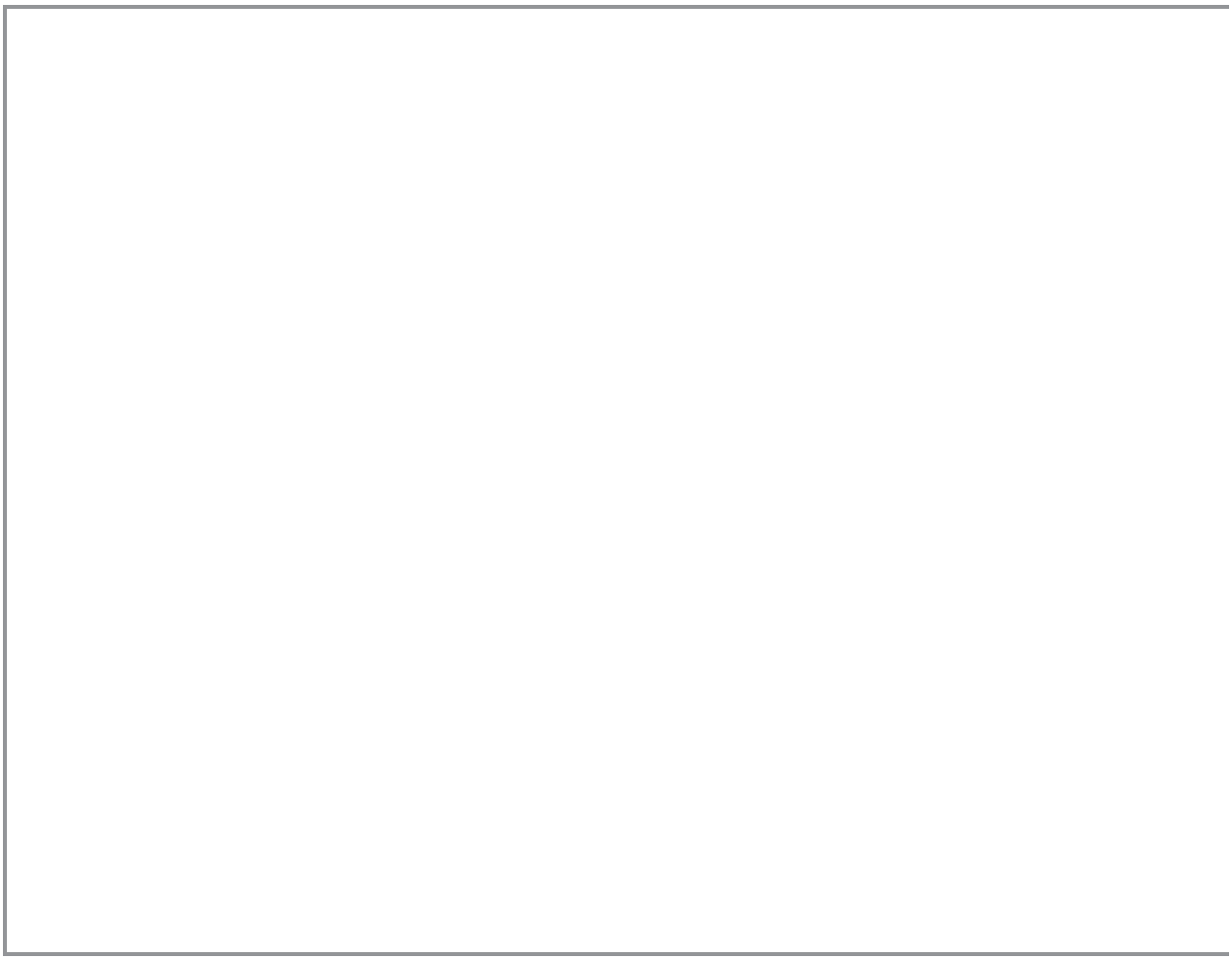
First, I think that anyone starting in powerlifting should lift raw for the first year or two so they can work on form and technique and get a good base strength. After that, I like to lift with and without gear. Gear certainly allows you to lift more weight and helps protect from certain injuries. I train raw in the off-season and when I am prepping for a raw competition. I now add gear about eight weeks out, sometimes more if the training dictates it. I find I need time to work in the shirt and "learn" it now that they have progressed so much. I struggle with getting the shirts to work correctly and sometimes wish they were not allowed. When squatting I add a belt around 60–65% of my max, knee wraps around 70–75%, suit bottoms at 80% and straps up 90% and above. I am sponsored by TITAN SUPPORT SYSTEMS. Pete Alaniz at Titan Support Systems has sponsored me since I won my first WDFPF World Championships (ADFPA affiliate) in 1993. THANKS, Pete!

What are your thoughts on steroids in powerlifting? Do you compete in drug-tested competitions? How do you feel about the effectiveness of drug tests for catching those who use steroids?

There is no question that steroids are in powerlifting as they are in many other sports. I wish they were not, but they are. I compete not only in drug-tested competitions but I am also out-of-meet tested every year. I probably average being tested 2–3 times per year. I think drug testing makes the playing field more even, but it is not perfect. There will always be people who find ways to beat the tests but at least when they test it's fairer. I believe the USAPL testing is fairly effective, at least with the more prominent lifters as they are out-of-meet tested which means they can't plan to pass the test. All that being said, I can respect a lift when it is performed to proper standards. I just don't believe that someone using steroids should compete in drug-tested organizations.

How would you feel about powerlifting being united? Would you compete in powerlifting if it was united and raw? Would you compete in powerlifting if it was united and single ply? Would you compete in powerlifting if it was united and multi-ply? What do you think the key to unification is?

First, I have competed in Raw Unity. It is a competition that has been held three years now and it had competitors from multiple federations. The rules closely followed USAPL, but there were early weigh-ins, no drug testing and rules a little lax. I would not compete in multi-ply but would do single ply.



Sioux-z keeps her chin high on all lower body lifts
Photo credit: Sioux-z Hartwig-Gary

What is your thought about the importance of having a workout partner(s) or teammates in helping you train for powerlifting? Whom do you currently train with?

I think it is essential to have good training partners if you want to excel. Currently I train with my husband, Matt Gary, and various lifters at our gym, Supreme Sports Performance & Training, Inc. A great training facility is key to success.

What is your thought about keeping a training journal while powerlifting? Do you currently keep a training journal?

WRITE IT DOWN. Yes, keep a journal; I do. It is important to be able to see what you did and didn't do when training was successful or when something went wrong.

What books, websites, or coaches do you suggest or follow in your lifting and what would you suggest other lifters do to learn more about lifting?

I do not go to a lot of sites or books as my husband, Matt Gary, does so much research I do not have to. We follow a lot of different Russian training programs.

As far as other lifters, find good coaches or training partners and utilize websites to ask other lifters programs they recommend.

What do you attribute your personal success in powerlifting to?

First and foremost God has given me the strength, ability and perseverance to succeed. After that, add in hard work, great training partners, a nevergive-up attitude, and the love and support of friends and family.

What do you feel is crucial to being successful in powerlifting, both in and out of the gym?

What I said above covers it. Work hard, never give up and believe in yourself.

What advice would you give to someone who was just beginning to take up powerlifting?

- Train raw for a year or two
- Find good training partners
- Don’t over train; have patience
- Have fun
- Go to seminars or get a knowledgeable coach if you can

What advice would you give to an intermediate-level powerlifter looking to improve his/her total?

Try a new training program. Mix things up a little. If possible, train with someone better than you or someone equal who can push you.

What advice would you give to an advanced-level powerlifter looking to improve his/her total?

Same — try a new training program. If you can train with lifters of your quality or above, you can improve by taking video and critiquing yourself or having an online coach.

Are there any changes that you would make to powerlifting if you had the power to do so?

I would unify the sport and get it into the Olympics. It is sad how splintered it has become and how many “World” champions there are. Many organizations have only 1–5 countries competing and yet have “World” championships. I would reduce it to one or two organizations. To appease everyone, one could be drug tested and the other not.

As a female, do you have any specific advice for females who either currently compete or are looking to compete in powerlifting? Do you think females need to follow a different program from a male or not?

Don’t be afraid that lifting in general or powerlifting will make you manlier. It is the drugs that make women look so manly, not lifting weights. I believe females can follow the same program as males. That being said, every person is different, and programs may need to be “tweaked” to each individual.

Do you have anything else that you would like to say to powerlifters and people interested in powerlifting?

Powerlifting is a great sport. You can compete against others, but truly, you are competing against yourself. Not everyone can be first place, but everyone can strive to hit new personal records. I have competed in various sports, and the camaraderie in powerlifting is wonderful. Train hard and have fun.

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.

(Left) Sioux-z Hartwig-Gary wins Gold

(Middle) Sioux-z Hartwig-Gary and Matt Gary

(Right) Sioux-z and Matt's Gym in Maryland
Photo credit: Sioux-z Hartwig-Gary

Creating that ideal program is part of the excitement — and the challenge — of powerlifting.

Chapter 9

Powerlifting Program Design

There are many, many methods and programs available to increase strength. It is beyond the scope of this text to go into detail about exactly how the body responds to exercise and how to create an effective program from scratch (if you want more information about that topic, please read the textbook that I wrote entitled *Fundamentals of Fitness and Personal Training* published by Human Kinetics). The plethora of programs available can also be quite discouraging — lifters tend to "hop" from one program to another in the hopes of finding that magic solution to increase their bench by 50 lbs in 8 weeks. They also constantly question themselves and wonder if they are on the "right" program at the right time. While the search for the best program must continue, one should not let that search affect the current training program one is following. Pick a program you believe in, follow it as best you can, put forth an effort that will make you proud in hindsight, and then see what happens on the platform. That is all one can do. Learn from that program, take notes and measurements, and over time your experience and knowledge will grow. Don't be afraid to try new things but don't jump on every fitness bandwagon that rolls by; remember, you can't argue with results. What kind of results did the person espousing the program actually get?

The point of this chapter is to help you — the lifter — narrow down

your choice of program selection. Think of this like a drop-down menu — when you select something, that will lead to certain choices and eliminate others. I can't promise that these programs are the 'best' for you at this moment in time, but I can say without hesitation that these programs have been battle tested by myself and my teammates, and they are all good and effective for their purpose.

You can't argue with results.

After a competition the body often needs to rest, the mind needs to recharge.

Where are you?

The first thing to know when selecting your next program is where are you now? To keep this simple there are 3 phases to choose from, each one should be pretty straightforward:

Phase 1: Post-competition — off season, or recovering from an injury. There is a saying that there is no off season in powerlifting, and there is some truth to that, but at the same time you can't just continually peak for a meet for 10 years straight. After a competition, especially as your level of advancement increases, you may want to back things off a bit. In addition if you are recovering from some sort of injury and you need to go a bit lighter, this is your phase.

Phase 2: General Strength Training

This is the phase that most lifters spend most of their training time in. They are training to get bigger and stronger but there isn't a meet right around the corner that they are prepping for.

Phase 3: Peaking Phase

This is when a lifter registers for a competition that is usually 6–12 weeks out and they design a program to get them ready to hit the best possible 1RMs on that day; the idea is to peak in strength and thus to maximize performance on the day of the competition.

Phase 1 — Post-Competition Phase

After a competition the body often needs to rest, the mind needs to recharge, and you may need to turn your attention to other things in life (school, business, family, etc.) that you might have been neglecting a bit in prepping for a meet. In general, post-competition workouts should have the following characteristics:

- Time Frame: 1–3 months (4–6 weeks most common)
- Lower intensity of your 1RM (<80%)
- Higher reps per set (8 or more reps, often 10–20)
- Shorter rest time in between sets (<2:00)
- Lower total time spent exercising (2–6 hours per week)
- Lower frequency of performing the barbell lifts (about 1 time a week give or take)
- More time spent on cardio and conditioning
- More attention to weak points such as mobility, muscle balance, stabilization, and injury prevention

If you are coming back from an injury but you are able to exercise the injured area, then I would suggest you follow the guidelines outlined in this article:

http://www.t-nation.com/free_online_article/most_recent/4week_routine_for_co

This is an effective 4 week routine laid out with a goal of preparing the body for more intense workouts after one has been cleared to exercise.

If you want to learn more about what happens during an injury, check out this article:

http://www.t-

nation.com/free_online_article/most_recent/stepbystep_approach_

If you are not injured and you are in the post-competition phase, then you should decide if you want to include the main barbell lifts in your program or not. If you don't want to include those lifts, then I would focus on bodyweight exercises, dumbbells, and machines.

If you do want to include the barbell lifts, then I would focus on using a lighter weight, perfecting technique and not really pushing yourself strength-wise during this phase. Remember the goal is rest and recovery. Linked below is a sample workout that would keep you familiar with the barbell lifts. This can be performed once or twice a week allowing time for other activities, and it can be completed in about 30 minutes or so.

http://www.t-nation.com/free_online_article/most_recent/5_by_5_by_5_in_25

> ***Think about where you want to be in 3–6 months after the competition and trust your gut.***

It should be noted that you don't *have* to go into the post-competition phase following a meet. Sometimes you perform well and you are stoked to keep training heavy and hard. Sometimes you perform poorly and you are pissed and you don't want to take time off. Both of those are common emotions 24 hours after a meet, but a week later, if your body is begging for a break, it is okay to listen to it. Generally younger, less experienced lifters will be more ready just to keep training hard after a meet, whereas the older, more experienced lifters will want to take some time off. The benefit of plowing through is that you don't have this detraining period which then takes time to "catch up" to where you were before. The negative

of plowing through (and thus the benefit of going easy) is that your body and your nervous system can recharge, your enthusiasm should hopefully return for lifting, and it might prevent injury in the long term. Think about where you want to be in 3–6 months after the competition and trust your gut.

Once this phase is over the lifter traditionally moves into the second phase, which focuses on increasing strength and size and preps one to handle a peaking phase.

> ***Hard work now will build up that total down the road.***

Phase 2: General Strength Training

In this phase the lifter is putting in their time under the bar, becoming stronger, becoming bigger, becoming more prepared for that next competition. They know that hard work now will build up that total down the road. Most lifters will spend most of their training lives in this phase (and those who never compete generally never progress beyond this phase).

In this phase the lifter will be training the competition lifts with some regularity; the only question is exactly how to set it up.

- Time Frame: 1–6 months (2–4 months most common)
- Moderate intensity of your 1RM (70–90%)
- Low to moderate reps per set (1–8 often)
- Moderate rest time in between sets (2–3 minutes)
- Reasonable total time spent exercising (4–8 hours per week)
- Reasonable frequency of performing the barbell lifts (1–2 times a week is most common)
- Some time spent on cardio and conditioning
- Some attention to weak points such as mobility, muscle balance, stabilization, and injury prevention

There are lots of good training programs out there to improve general strength levels in the big 3.

A more strength-oriented program is outlined below:

http://www.t-nation.com/free_online_article/most_recent/8week_basic_strength

A good strength/size combo plan is listed here:

http://www.t-nation.com/free_online_article/sports_body_training_performance

Other more popular programs that work well in this phase include the following:

Westside Barbell Routine (strength focus)

High Intensity Training (HIT) outlined and followed by Dorian Yates and shown in his video *Blood and Guts* (size focus)

Jim Wendler's 5/3/1 Routine

Brandon Lilly's Cube Method

Photo credit: Jim Wendler

Phase 3: Peaking Program

When an athlete has decided they are going to lift in a competition, they usually pick a date for their meet and then they count backwards to see how many weeks they have to prepare for the competition. More time is generally better, but it does increase the likelihood of burning out, peaking early, or injury if one trains too hard for too long. 6–12 weeks is the standard time for a peaking program, with 8–10 weeks being the most common. During this time the athlete is specializing in the competition lifts with a goal of performing optimally on the platform on meet day. Peaking programs generally follow these characteristics:

- Time Frame: 6–12 weeks (8–10 weeks most common)
- High intensity of your 1RM (80%+)
- Lower reps per set (5 or less most often)
- Long rest time in between sets (5+ minutes)
- Large amount of total time spent exercising (6+ hours per week)
- Reasonable frequency of performing the barbell lifts (2+ times a week is most common particularly for squats and bench)
- Little time spent on cardio and conditioning (unless cutting weight)
- Little attention to weak points such as mobility, muscle balance, stabilization, and injury prevention unless these will directly improve performance for the meet

When selecting a peaking program, the lifter will first choose a general routine that works for them. Once that is selected, then (and likely more importantly) the lifter will choose a specific set/rep/weight scheme designed to peak maximal strength for the day of the competition. Peaking programs are the hardest to plan out because one can peak too early and feel drained/slow/tired on the day of the meet; one can peak too late and not feel

ready for the 1RM attempts; or one can perform too much work and end up getting injured.

Listed below are several peaking programs with brief explanations about each one of them.

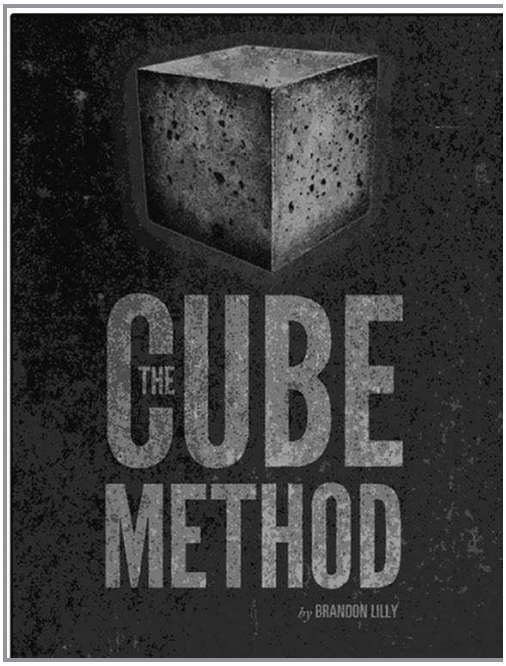


Photo credit: Brandon Lilly

Old School Peaking Program – This type of routine was popularized by such legends as Coan, Karwoski, and Gallagher so you know this works. These programs generally involve performing each lift just once a week, starting with higher reps and lighter weight and then each week building to lower reps and heavier weight. In my experience these work great the first couple of times you do them. In addition, it is crucial not to over-program them, and the lifter should not fail in the first several weeks; ideally the lifter doesn't fail at all during the program.

Here is a sample of this type of peaking program that is based off of something Ed Coan set up in *Powerlifting USA* magazine. It involves 2 work sets per exercise, each main lift is trained once a week, and generally 3–4 assistance exercises would be performed after the main exercise of the day. No warm-up sets are given; the lifter would need to program those in as desired.

week	sets/%age of 1RM/reps
1	2 @ 70.4 % x 10
2	2 @ 70.4 % x 10
3	2 @ 74.1 % x 8

4	2 @ 77.7 % x 8
5	2 @ 81.5 % x 5
6	2 @ 85.2 % x 5
7	2 @ 88.9 % x 5
8	2 @ 92.6 % x 3
9	2 @ 96.3 % x 3
10	2 @ 100 % x 2
11	2 @ 103.7 % x 2
12	1 @ 111.1 % x 1

Author's Note: I think an 11% jump in strength is pretty big for this time frame (that is taking a lift from 400 lbs to 444 lbs); one might be better scaling it back to a 5% jump which is still a very sizeable increase for any non-beginning powerlifter.

There are other programs available as well with the purpose of peaking a lifter for a meet. Some of the more effective plans include:

Smolov – A high volume, pretty high intensity plan that flirts with overtraining in an effort to really boost the max.

Sheiko – A percentage-based routine that generally has the lifter benching three times a week, squatting 2 times a week and deadlifting once a week. Sometimes you will perform the same lift twice in one workout. These workouts are kind of long (2+ hours) and can be a little boring; they also aren't conducive to training in a big group unless you break up into several smaller groups. But they can be effective, particularly in increasing the squat.

Here is a more detailed summary of Sheiko training

http://www.t-

nation.com/free_online_article/sports_body_training_performance/

And here is a link to a full 13 week program

http://www.t-nation.com/portal_includes/articles/2008/Sheiko13weekworkout.ht

Westside – Westside is listed in both general strength training and the peaking phase because it can do both. The workouts are focused around maximal effort which often involves performing just one rep; a lifter following the Westside system should be prepared for a meet with minimal notice, but it doesn't have to be reserved just for the peaking phase.

As a reminder, the information presented in this chapter relates to programming guidelines for the entire workout. In each of the chapters focused on increasing the competitive lifts (Chapters 4, 6, 8, and 19) there are specific programs set up for the individual lifts. Those programs would fall under either Phase 2: General Strength Training or Phase 3: Peaking.

Don't underestimate the effectiveness of performing a good program again but simply adding weight on the next go round.

Long Term Training

Your training will likely go in cycles if you continue to lift for any length of time (measured in years). You lift in a meet, then set up a post-meet plan, general strength training plan, then peak for another meet, and start the process all over again. Don't underestimate the effectiveness of performing a good program again but simply adding weight to the sets on the next go round. If it worked once for you, it is likely to work again with at least some level of effectiveness. I personally experienced a lot of success in a 2 year period when my total went from 1400 to more than 1600 by following this pattern (and my total had been stuck at 1400 for 2 years previous to this):

- Traditional Strength Training routine much like the one outlined in the Basic Strength Link, performed for about 2 months.
- Dorian Yates' HIT routine for 6 weeks.
- Westside Routine for 10 weeks, which peaked me for my meet.

That took about 6 months; once it was over I started it all again, just using a little bit heavier weight on the sets. It wasn't perfect but I got about 50 lbs on my total with each 6 month cycle using that set up; if you can repeat that a few times that is pretty nice progress.

If you are struggling just with the general template of setting up the workout routine (what to do on what day) then I would suggest you follow one of the 12 workouts outlined here:

http://www.t-nation.com/free_online_article/most_recent/12_programs_to_follo

If you are torn between trying high frequency or not with your lifts, then read this article outlining the pros and cons of each:

http://www.t-nation.com/free_online_article/most_recent/high_frequency_vs_lo

The very short synopsis of that article is that I think high frequency training will deliver results faster, BUT it is more likely to cause injury, it is hard to program for the masses, it leads to more burnout, performance drops faster once you stop it, and it doesn't seem to be correlated with long term lifting success, so I would use it with caution.

High frequency training will deliver results faster, BUT it is more likely to cause injury.

Finally you might be wondering what is with all of the T-nation articles inserted in this chapter? I wrote those articles for the same purpose as this book, to help aspiring lifters achieve new maxes and lift more weight than they are lifting now. I did get paid for all of those articles, and it is nice to have them get continued press, but I don't own T-nation or make money based on how many hits they get or anything like that. The bottom line is that those articles were already written so it saves me time — including them all would add 50 pages or more to this book (costing more to print and thus more to purchase), and that format allows for pictures and videos to be included to help clarify key points. I would urge you to read at least some of them and obviously carefully follow the more detailed plans. Workouts are much like recipes. I don't think new cooks should tinker with established recipes; sometimes little changes can really affect the outcome. A very experienced chef might feel more comfortable making changes and that is okay, but even for that person I think it is ideal to try the recipe as is before you modify it. The big difference in that analogy is that most things can be cooked in just one day; it will take 2–3 months on a workout plan for you to really be able to judge its effectiveness. Finally if you don't make any

progress after following a workout, you did learn something — you learned what doesn’t work for you. Don’t make the mistake of then repeating that with the hope that this time something will be different. Instead try something new — maybe radically new — and see what happens.

A lifter adds chains to the deadlift to build explosive strength and power.

Chapter 10

Powerlifting for Other Athletes

While this book was written primarily for powerlifters — people who will actually compete in a powerlifting competition — it is my hope that regular lifters, gym rats, weekend warriors, CrossFitters, and athletes who compete in other sports will see the value of including the powerlifts in their exercise program. It should be obvious from a simple observation of today's athletes that resistance training is contributing to their performance. Simply look at the bodies of today's athletes and compare them to athletes in those same sports 20, 40, 60 or even 100 years ago. Sprinters look like mini bodybuilders; the upper body of basketball players is cut and striated without being small; football players are larger and more powerful than ever; and athletes of all sports are carrying around more muscle than they have in the past. Not only is this muscle making them look better (in this author's humble opinion), it is helping their performance, as shown by the fact that ability in all of these sports continues to improve. That improvement is not due solely to weight training, but to think that lifting weights is not contributing to increased performance (and decreased chance of injury) is naïve.

Most athletes know that they should lift weights — I think that battle has been won. Now the question becomes, "What kind of program should I follow?" This is where there is still much confusion on the issue. Chances are

that the athletic ability you are demonstrating on the court or field requires a reasonable amount of skill (almost all sports are high skill activities, which is why it takes years to be good at them). It should be logical, then, that you want to perform activities in the gym that are relatively high skill. If getting better in the bench press makes you better at almost every other upper body exercise in the gym, it should make sense that the bench press will help out your upper body performance in your sport more than just about every other upper body exercise in the gym. The same goes for squats, deadlifts, and other free weight exercises. Becoming strong on the leg press does not mean that you will be strong on all lower body exercises, thus it is less likely that performing only a leg press will help out your leg strength on the field.

Three follow-up points to this concept: The first point is that beginners respond to everything. Performing a leg press for 3 months probably will help out your leg strength if you have never trained before, but devoting 3 years of your training to becoming good at the leg press will almost for sure not help you on the field as much as spending those same 3 years squatting.

Second point: Balance is an element of skill, and exercises that require higher balance involve more skill. However balance is a skill in and of itself, and the worst part is that balance is specific to a certain ability. By that I mean that balance developed in one area (for example, holding a certain yoga pose) is not necessarily transferable to balance in another area (for example, skating on ice). There is no one universal quality of “balance.” Someone might have good balance on a skateboard but that doesn’t mean they can walk a tight rope or tip toe full speed next to the out-of-bounds line in football. What this means in the gym is if you are performing an exercise that requires tremendous balance (such as squatting on a BOSU ball) you can certainly improve your balance on that exercise, but there is little evidence to indicate that the balance built by that exercise will then transfer over to

balance in another area. Combine that with the fact that the weight being used is so much less in balance exercise (and lifting more weights requires more skill as well), and it becomes likely that specific exercise will have very little positive transfer to the field. Those exercises should be primarily used to rehab an injured joint. This wave of so called “functional training” has been sweeping gyms for the last 10 or 15 years. It is starting to fade in popularity due to the fact that it simply doesn’t produce the same results as does a properly planned out resistance training program. It is appealing because the exercises look fancy; they are hard to an extent (hard to balance and they can get you out of breath), but they are not hard in the way a real resistance training program should be. That type of training also allows relatively weak and small people who would not usually seem to be experts in the strength and conditioning field (simply because they have not put in the necessary time, blood, sweat, and tears) to be experts because they can guide clients and athletes through those kind of workouts. You can’t argue with results. When a person can squat 500 lbs raw and deep, that is results. Benching 3 plates is results. I would so much rather line up and compete against a lineman who focuses on BOSU ball squats, standing cable chest press, and wall ball squats than one who focuses on squats, bench press, and prowler pushes.

Third point: It is often detrimental to sporting performance to try to exactly mimic a movement on the field in the gym and add to that movement considerable resistance. For example swinging a fake golf club attached to a pulley for resistance is a bad idea. Swinging a baseball bat with big rubber bands attached to it would not be good. It is okay to very slightly change the resistance (as in putting ½ lb weights on a bat when warming up) but changing the resistance considerably creates a completely new motor program. You will never see baseball players warming up with 100 lb bats on the sidelines. Getting good at swinging a 100 lb bat would not necessarily

transfer over to being able to swing a 2 lb bat, and indeed swinging the heavy bat would most likely alter the motor program and end up having a negative effect on performance. This can be seen very easily by spending 10 minutes shooting baskets with a medicine ball of about 10 lbs, and then try and shoot baskets with a normal basket ball. You will most likely find your shooting to be way off the first several minutes after returning to the normal ball, because the medicine ball has taught your muscles a new motor program to be successful with that ball, but that same motor program does not work with the old ball. I see this most often in the gym with the golf swing. Trainers are trying to help their clients overload the muscles in the golf swing by having them mimic that movement with resistance but it is usually either ineffective in general or actually counterproductive. Simply get the muscles involved strong and then teach them how to work by practicing that movement.

If you are an athlete and you want to train in the gym, you want to focus on the powerlifts, the Olympic lifts, some challenging bodyweight exercises, and a little bit of strongman and/or bodybuilding stuff thrown in. You also want to be in good enough shape to prevent fatigue from significantly affecting your performance especially near the end of the competition. What might surprise you coming from the author of a book entitled *All About Powerlifting* is that I am not advocating that you just devote your training to the development of maximal strength. My philosophy, which I believe is backed by clear science, is that you should become relatively strong *for your sport.* Once that level of strength is reached, you want to maintain that strength, improve your weak points, and use your training time to further improve your skill and refine your technique. The continued, linear development of maximal strength is not necessary and could actually be detrimental to sporting performance.

Presented below is a theoretical chart that depicts the relationship

between maximal strength (as measured by a free-weight exercise) and sporting ability.

The key themes from this chart are: the top line indicates maximal strength, which increases at a relatively rapid rate for the first few years of training, slows down but continues to improve as time goes by, ultimately following the point of diminishing returns. Sporting ability is depicted by the bottom line. It will improve once resistance training has begun, usually with a bit of a delay. As strength increases there is normally less and less transference to the sport (e.g., a football lineman will most likely notice a big increase in ability going from a 200 lb bench to a 400 lb bench; I don't think it will be the same improvement going from a 400 lb to a 600 lb bench). Once the theoretical maximum value of useful strength is met, then it is unlikely that additional strength gains will help the athlete. This is where the bottom line diverges into three parts. In some instances additional strength might be useful, especially if it is developed without compensations and without

sacrificing any other necessary abilities and thus sporting performance might go up. In many instances the additional strength developed will most likely become specific to that skill (as in a bench press), and, while the athlete might be able to bench press more, that doesn't automatically mean they can push away a standing, reactive opponent significantly better. If this is the case, the additional strength will not cause an increase in sporting ability. And in some instances the additional strength can actually cause a decrease in sporting ability. The body may learn to recruit the motor units in a very specific order during the lift, which is good for the exercise but might actually come at the expense of sporting ability. This might be seen when the squat continues to increase but the vertical jump actually decreases. The athlete might become slower and less powerful (on the field); they might develop more movement compensations; they might be devoting too much time to the exercise and not enough time to their specific sport; or they might find ways of increasing the gym lift without actually increasing the true strength (e.g., widening the grip on the bench might make you lift 10 lbs more due to leverage and decreased ROM, but your actual strength may not have changed, and those extra 10 lbs in the gym will most likely not help you on the field).

What is the theoretical value for the maximum usable strength for a sport? I don't think anybody knows for sure. Clearly certain sports and positions, such as being a lineman in football, require much more strength than other sports, for example, playing golf. I have broken the popular sports down into 4 categories. For Category 1 sports strength is not very important but that doesn't mean it should be completely neglected. As the category number increases, strength becomes more important. Category 4 sports are sports where maximal strength is a very big part of performance. For each of these categories I have set up what I believe to be a reasonable guideline as

the maximum value of usable strength in those sports. These guidelines are not based on years of scientific research (I don’t believe such research exists) but are based on many years of in-the-gym experience combined with tens of thousands of hours of observing all sorts of athletes and their strength abilities and noting their relationships. Ultimately the best thing you can do is improve your strength and see what happens to your sporting ability and then come up with optimal numbers for yourself. I would suggest you start with my guidelines and go from there.

Male Sports Categories

Category 4 Sports — Football (lineman, linebacker, tight end, running back); Shot put

Estimated average bodyweight of athlete — 200 lbs

	Decent	Good	Great
SQ	2.0/400	2.5/500	3.0/600
BP	1.5/300	1.75/350	2.0/400
DL	2.5/500	3/600	3.25/650

Category 3 Sports — Football (quarterback, wide receiver, defensive back, kicker); Wrestling; MMA; Sprints; Javelin; Jumps; Gymnastics

Estimated average bodyweight of athlete — 200 lbs

	Decent	Good	Great
SQ	1.5/300	2.0/400	2.5/500
BP	1.25/250	1.5/300	1.75/350
DL	2.0/400	2.5/500	3.0/600

Category 2 Sports — Baseball; Soccer; Basketball; Hockey; Skiing; Jockey; Tennis

Estimated average bodyweight of athlete — 180 lbs

	Decent	Good	Great
SQ	1.0/180	1.5/270	2.0/360
BP	.75/135	1.25/225	1.50/270
DL	1.5/270	2/360	2.5/450

Category 1 Sports — Golf; NASCAR; Billiards; Darts; Marathon; Triathlon; Swimming

Estimated average bodyweight of athlete — 180 lbs

	Decent	Good	Great
SQ	.75/135	1.0/180	1.5/270
BP	.5/90	.75/135	1.25/225
DL	1.25/225	1.75/315	2.0/360

Female Sports Categories

Category 4 Sports — Football; Shot Put

Estimated average bodyweight of athlete — 160 lbs

	Decent	Good	Great
SQ	1.0/160	1.5/240	2.0/320
BP	.75/120	.9/145	1.0/160
DL	1.5/240	2.0/320	2.25/360

Category 2 & 3 Sports — Wrestling; MMA; Sprints; Javelin; Jumps; Gymnastics; Baseball; Softball; Soccer; Basketball; Field Hockey; Ice Hockey; Skiing; Jockey

Estimated average bodyweight of athlete — 140 lbs

	Decent	Good	Great
SQ	.75/105	1.25/175	1.75/245
BP	.50/70	.75/105	1.0/140
DL	1.25/175	1.75/245	2.0/280

Category 1 Sports — Golf; NASCAR; Billiards; Darts; Marathon; Triathlon; Swimming

Estimated average bodyweight of athlete — 140 lbs

Decent	Good	Great

SQ	.50/70	1.0/140	1.5/210
BP	.50/70	.75/105	.90/125
DL	1.0/140	1.5/210	1.75/240

Both a bodyweight ratio and an absolute value have been given. When in doubt use the lower of the two numbers; thus a 140 lb male Category 1 athlete would need a bench of 1.25 x bodyweight or 175 lbs to be considered great.

A Note to CrossFitters

CrossFit is sweeping the nation — the CrossFit games are on TV (and it is tough to argue with the fitness level displayed by the competitors), the number of CrossFit gyms (boxes) are increasing by the hundreds every year, and CrossFit is helping intense fitness become more mainstream. I believe that anything that brings some aspects of powerlifting to the public eye is a good thing, and I think there are lots of positives one can take away from CrossFit. Powerlifters in particular would benefit taking a page or two from the CrossFitters, including a greater emphasis on conditioning, incorporating more bodyweight exercises, utilizing some lifts that emphasize mobility and flexibility, and their focus on nutrition (although I don't think one needs to go full Paleo to reap significant benefits). CrossFitters can take a page from powerlifters when it comes to how to properly perform the lifts — both for safety and performance — and how to best program the lifts. CrossFit is known for not intentionally specializing in anything, and I can understand that. However, if a specific weakness is holding an athlete back from maximal performance, there is nothing wrong with *prioritizing* that weakness until it no longer remains a limiting factor.

If my ankle mobility is what is limiting my athletic performance, I would be foolish not to prioritize addressing that issue, correct? Well, if you are a CrossFitter and you believe your maximal strength is what is holding you back, you would be foolish not to prioritize your strength. That is where this book comes in. You can utilize the techniques and the programming here; follow a strength building mesocycle (2–4 months long) to work on that weakness. Repeat that process until your strength is no longer your limiting factor. Once you are strong then it just takes practice and some conditioning (luckily conditioning improves much faster than strength) and you will find many of the movements performed in CrossFit to be much easier.

Male CrossFit Standards — bodyweight 190 lbs

	Decent	Good	Great
SQ	1.25/240	1.75/335	2.25/430
BP	1.00/190	1.25/240	1.50/285
DL	1.75/335	2.25/430	2.75/525
Press	.60/115	.85/160	1.10/210

Female CrossFit Standards — bodyweight 135 lbs

	Decent	Good	Great
SQ	1.2/160	1.65/225	2.0/270
BP	.75/100	1.0/135	1.25/165
DL	1.5/205	2.0/270	2.5/335
Press	.50/65	.75/100	1.0/135

Because CrossFit is focused around the lifts themselves instead of a "potential" transfer over to the field, athletes hoping to perform well in competitions should strive to achieve at least the good level, and ultimately the best athletes will be at the great level or higher. As long as the strength training doesn't take away or come at the expense of other capacities, CrossFitters should strive to become as strong as possible with good form on the lifts. You will note the Press has been added to the standards listed because it is such an integral part of that sport (the CrossFit Total is derived from taking one's best squat, deadlift and press). This is also why the standards for the bench are lower in comparison to the other lifts — being proficient in the bench press doesn't gain a CrossFitter quite as much as proficiency in the other lifts (and the Olympic Lifts) would.

What Do the Numbers Mean?

A score of below *decent* means that there is a high probability that potential physical performance is limited due to strength. In simple terms the athlete is weak and that weakness may be limiting them.

A score of *decent* or better is what the athlete should attain, at a minimum. At the decent level the athlete is probably about as strong as most competitors on the field. Their strength will not be an advantage but it should not be much of a disadvantage.

A score of *good* or better is what the athlete is striving for. A good level means the athlete will be stronger than most of the other athletes in the game. Their strength level will most likely give them an advantage in certain situations. The athlete should examine their abilities and try to determine if additional strength would be worth the considerable time and effort it will take to build it.

A score of *great* means the athlete is exceptionally strong *for their chosen sport.* This level of strength will be rare in that sport. The athlete should focus on maintaining strength, but most of their efforts should be on increasing other areas like endurance, speed, power, cardio, flexibility, specific sport skills, or whatever else is necessary to excel in their sport. It is unlikely that additional strength acquired as demonstrated by the barbell would serve a useful purpose on the field, particularly in Category 1, 2, and 3 sports.

A score of above *great* means the athlete is super strong for their sport. It also means that there is a high probability that they are spending too much time strength training and those efforts could be better spent on other activities related to the sport. The high level of strength may also be due to more specific adaptations to the lifts themselves, and their strength should be observed on the field and not just trusted because it is high in the gym. It is

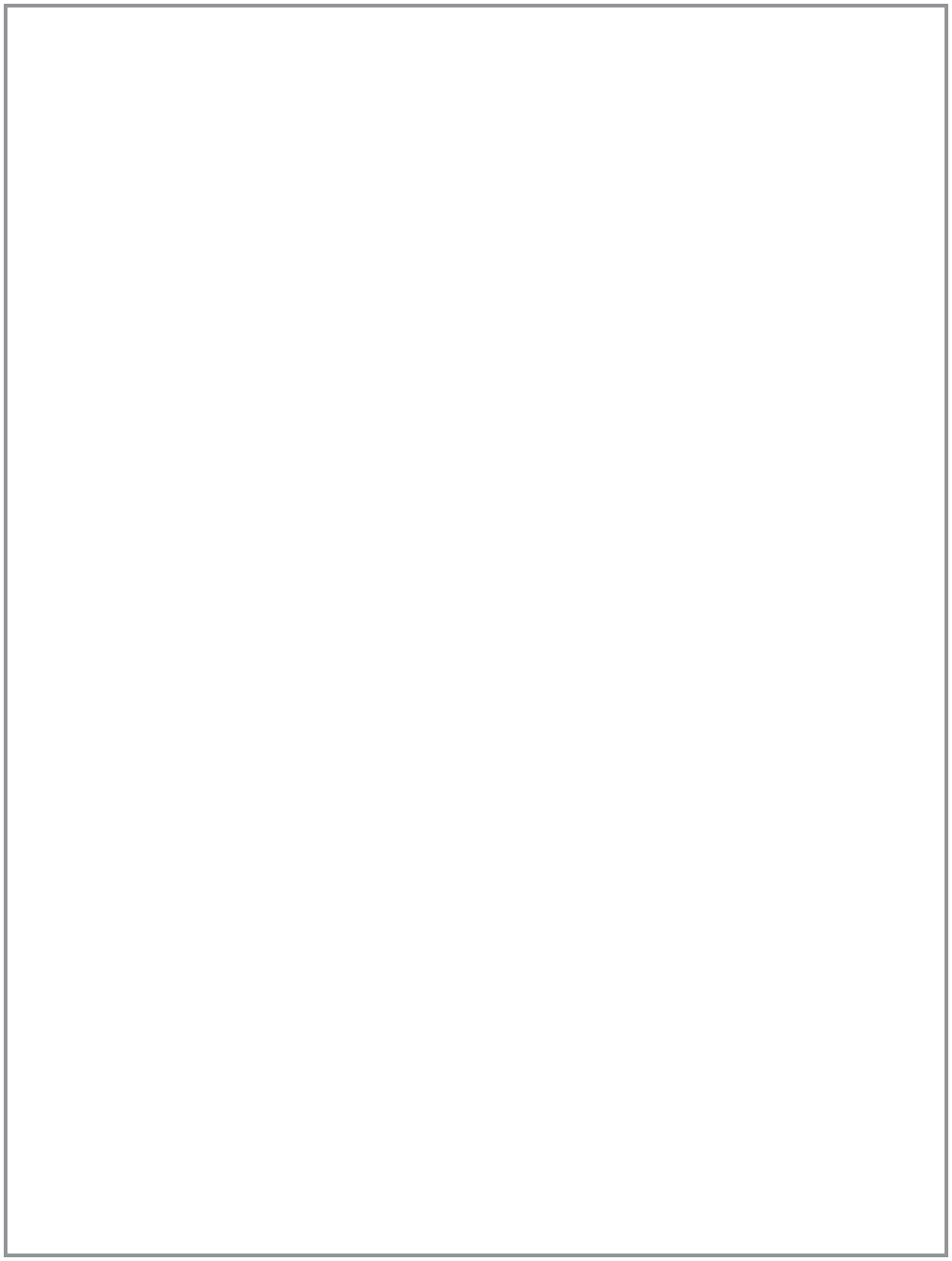
highly unlikely that any additional strength developed in the gym would transfer positively to their sport. Now, if they want to come over and give powerlifting a try, that is another story.

Clearly the terms “decent,” “good,” and “great” are somewhat subjective. I don’t claim that these numbers are backed by extensive scientific research because it simply isn’t there, but that does not mean they are not grounded in truth. Hopefully this is relatively obvious. It takes a certain level of strength to be able to perform certain activities. Picking up a glass of milk requires very little strength but if you don’t have enough you can’t do it. Swinging a golf club requires a little more. Jumping over a pole the height of your body requires even more, and blasting into an opponent who is your equal bodyweight and driving them backward requires even more.

Note that those terms are specific to their sport classification. No strength athlete or powerlifter would consider a 1.25 x bodyweight bench press for a male to be great or even very good, but that term and definition is not for a strength athlete. For a Category 1 athlete like a golfer or a NASCAR driver, that would be a very nice bench press and in my opinion, it is unlikely that a bench press higher than that would serve them any positive purpose.

I can’t tell you how many clients and students I have had who have put 20 yards on their golf swing without even specifically trying to improve it. Instead that improvement came from a few months of working out with weights, without performing one single golf-specific exercise. As their strength improved, so did their performance. It is a not a perfect linear relationship, and, as should be clear from the above categories, strength is more or less important in certain sports. My goal is here to provide an athlete other than a powerlifter some useful classifications as to where they fall in the strength realm for their sport and to show that most likely, unless they are walking around already very strong, spending 3–12 months improving their

strength will almost for sure improve their performance. Just to be clear, I am not saying that strength alone is the determining factor. For a Category 2 athlete I have a “great” level of strength, but that doesn’t mean I am a good skier just because I am strong enough. I am terrible at skiing because I never practice it and thus don’t have the skill necessary to do it. Strength is just one of many spokes in the wheel of performance, but when it is missing it is very noticeable.



Chapter 11

Conditioning and Supplemental Training for Powerlifters

Conditioning for Powerlifters

Cardiovascular endurance is the ability of the heart and lungs to provide oxygen to the body. You can measure your cardiovascular endurance with a score called your VO2 Max, which stands for the maximal volume of oxygen that you can deliver to your body. VO2 Max scores will be a two digit number, usually ranging from about 20 to 80. The number itself indicates how many milliliters of oxygen you can deliver to each kilogram of your bodyweight each minute. A higher score means you have better cardiovascular endurance.

All athletes need some form of cardiovascular endurance. If your VO2 Max was zero you would be dead — you would not be delivering any oxygen to your body. VO2 Max is super important to aerobic athletes as it is a great way to quantify how effective their heart and lungs are. VO2 Max is of some importance to a powerlifter, and it can be an indicator of general health. It is true that due to the way the individual lifts are performed in a competition (since we complete 1 rep and get a lot of rest in between attempts), your VO2 Max does not have much to do with how much weight you can lift. For that reason, and others discussed later, cardio has sometimes become the “six

letter word" to powerlifters, to be grouped in the same category as salad and shallow squats. But that is not necessarily wise.

Even though VO2 Max does not have much of a direct effect on powerlifting, performance up on the platform, it can affect our ability to train to become as strong as possible. In addition most meet days are long and tiring and those out of shape will find their energy waning when it comes time to deadlift. Powerlifters do need to pay attention to their VO2 Max; it is just not as crucial for us as it is for aerobic athletes.

Those out of shape will find their energy waning when it comes time to deadlift.

Basic Rankings of VO2 Max Scores

	Men under 40 years old	*Women under 40 years old*
Very Poor	<20	<20
Poor	21–30	21–27.5
Average	31–39	27.5–35
Fit	40–49	36–45
Very Fit	50–59	46–55
Excellent	60–69	56–65
Elite	70 +	65 +
World Record	~85	~80

As you can see, the higher the number, the better the score. Women on average have a VO2 Max of about 5 points lower than men. Elite truly means elite; most athletes (even professional athletes) will not have a VO2 Max in that level. Generally only endurance athletes like marathon runners, cyclists, cross-country skiers, and tri-athletes ever train enough to get into that range.

A powerlifter's goal is not to have as high of a VO2 Max as possible, but instead to be fit enough to handle any type of weight training workout that the lifter might perform. Since lifting weights is primarily anaerobic (there are a lot of starts and stops to it) a super high VO2 Max is not necessary (and indeed the training necessary to produce that might be counter productive). Listed below is my guideline for the proper aerobic capacity for powerlifters combined with a basic level of fitness. Notice the categories are broken into weight classes. This chart is mainly for males, I have not noticed too much of an issue with females being too out of shape for powerlifting, but in general a female should be able to match a male in terms of cardio performance given the weight class: a 181 lb female should be able to walk at 4.0 mph at a 5% incline for 30 minutes. I believe if a lifter can meet or

exceed the standards presented here, their cardio ability will not impede their ability to workout strenuously. However, if the lifter can't meet these standards, down the road they may find their strength, their workouts, and their health suffering as a result.

Weight Class	**<165 lbs**	**166–198 lbs**	**199–242 lbs**	**243 lbs +**
VO2 Max Score	50 + VO2 Max	45 + VO2 Max	42 + VO2 Max	40 + VO2 Max
Test: Treadmill Workout	4.0 @ 10% 30′	4.0 @ 5% 30′	4.0 @ 2% 30′	3.7 @ 2% 30′

Note: Treadmill information is written as speed at a certain incline for a certain time. 4.0 @ 10% 30' means walking at 4.0 mph at a 10% incline for 30 minutes (without holding on to the machine) without taking a break.

> ***Someone in good shape will cope with fatigue better.***

Other Benefits of Cardio for a Powerlifter

Having a good VO2 Max is useful for a powerlifter. This will be reflected by the lifter's ability to recover after a brutal set. Someone in good shape will be ready to go again in a couple of minutes; someone in poor shape will need many minutes and perhaps more to feel like they are recovered. If you get exhausted after one or two very hard sets, this, of course, affects your entire workout. You will be limited in the amount of volume you can complete, which will limit your results. In addition powerlifting competitions themselves are long and often draining. They basically take a full day, often starting at 6 or 7 am (to weigh-in, get ready, etc.) and the lifting itself is often spread out over 6–10 hours or even more. You have to get amped up, lift heavy, then relax, then do it again, relax, do it again and by the end of the day, you can feel pretty spent. Combine that with the fact that there is little chance to truly relax or eat good food and the lifters can be tired by the time deadlifts roll around. This is why they say it is harder to hit a PR bench and deadlift in a full meet than when you just perform those lifts by themselves. Someone in good shape will cope with that fatigue better — they will still have energy to burn when it comes to the last event as opposed to feeling like they are running on fumes.

Cardio, even light cardio, promotes blood flow in the body. Your body builds capillaries which are small blood vessels that deliver blood to the body. Increasing blood flow brings in nutrients and oxygen and helps remove waste products. This can promote recovery after a workout. Lifters often find that walking the day or two after squats can help the leg muscles recover from that workout faster than doing nothing at all.

Cardio can help build cartilage in the body. Weight training in general is good for the joints of the body, strengthening them and making them less susceptible to injury. However powerlifting is weight training taken to the

extreme. There is some anecdotal evidence that long-term, extreme lifting can damage the joints. The research in this area is unclear. Long-term intense exercise doesn't seem to cause degenerative joint disease in animals, but we don't know about humans. Once your joints hurt from arthritis, we do know that doing nothing often makes them feel worse and some activity is definitely beneficial. If the intense weight training is rough on the joints, easy cardio can help the joints recover. I know personally than when I lift heavy and skip the cardio, after a while I often feel stiff and slow just moving around. But if I perform cardio and lift heavy I feel more fresh and mobile.

Of course cardio can help a lifter control their bodyweight. While it is true that powerlifters don't need to be as lean as bodybuilders to be successful, it is also true that excess bodyfat after a certain point does little to help one's lifts. Certainly bodyfat in the unhealthy ranges (>22% males and >35%o females) should be avoided by powerlifters interested in their long term health.

Cardio can help build cartilage in the body.

In addition to potential performance benefits of cardio, it bears mentioning that cardiovascular fitness is a basic component of overall health. I personally believe that powerlifters should be representative of most athletes and in good general health. Being strong does not offset the negative health effects of being fat, out of shape, and consuming a terrible diet.

Negatives of Cardio

While staying in decent cardio shape is wise for a powerlifter, too much cardio can be detrimental to performance. Several studies have shown that intense aerobic exercise causes a decrease in maximal strength, and stopping intense aerobic exercise often increases strength.

Fiber Type Conversion – One potential negative of performing too much cardio is that intense cardio will target some of the type II muscle fibers. Powerlifters want a lot of type II muscle fibers and in particular they want the type IIB muscle fibers (sometimes called type IIx). It is thought that regular intense cardio might cause a conversion of the type IIB muscle fibers to the type IIA muscle fibers, which have a better aerobic capacity but are not as strong. It is also possible that an even greater transformation of the type II fibers to the type I fibers might occur although research is not completely conclusive in this area.

Loss of Fuel – When you perform cardio you are burning up fat and glucose as your fuel. The amount and type of each fuel depends on the time and intensity of the cardio. The more intense it is, the more glucose you will burn. You get your glucose from glycogen, which is the storage form of carbohydrates in the body.

If you exercise hard but don't have much glycogen in your body, then the body will turn amino acids into glucose for the required energy. These amino acids can come from the protein in your food or the protein in your body — this means you might break down your muscle to fuel your activity.

Intense weight training will rely primarily on glucose for the fuel; fat is not powerful enough to fuel most powerlifting workouts. If you are depleted in glycogen from intense cardio, then you may not have enough energy for your workouts, and either your performance will suffer or you will burn

muscle for energy (or both).

If you combine intense weight training (as one would when getting ready for a powerlifting competition) along with a lower calorie diet (such as dieting down to make a weight class) combined with performing hard cardio (to lose weight and lean up) you really run the risk of burning a lot of muscle for energy. The cardio programs in this book are designed to boost performance, cause weight loss, and to maintain the amount of muscle the lifter has as best as possible (some muscle loss is usually inevitable when significant bodyweight is lost).

You might break down your muscle to fuel your activity.

Cardio can be High Impact – I know I just said that cardio is good for the joints, and for the most part it is. But cardio can be relatively high impact, particularly jogging, running stairs, or anything with a lot of jumping. As mentioned previously, competitive powerlifting is rough on the joints. Add to that a high impact form of cardio and it can be hard for your joints — particularly the knees, lower back and hips — to recover. In my personal opinion, jogging and powerlifting don't mix, particularly with heavier lifters. I would be very hesitant to suggest that any powerlifter jog on a regular basis, and if you are over 200 lbs I would strongly suggest against it. I know there are some successful powerlifters that do run, particularly in the military, but they are the exceptions, not the norm. Another measure would be how hard it is for you to jog a mile or two with no breaks. If that is something that comes very easy to you and you could jog 2 miles no problem, then running may not be as bad for you. If you would find running a mile or two to be quite challenging, then jogging is probably not a good idea for you. The harder something is for you, the greater the adaptation response will be for you to become proficient at it. If jogging is hard it might be

because you have a lot of type II fibers in your legs and you might respond by altering those fibers through cardio training. If jogging is hard for you, you will burn more glucose to fuel that activity and that might end up burning a good amount of protein as fuel.

Instead of jogging, my cardio mode of choice would be walking. A brisk walk can be relatively hard, and performed at an incline, it can be very challenging. It is low impact and it should promote recovery without stressing your ability to recover; it is normally low enough intensity that muscle will not be burned as fuel under most circumstances. In addition walking, in my opinion, is the single most functional exercise there is, since it is something that humans are "required" to do a decent amount of every day, even in today's sedentary society. You can make walking even tougher by using a weight vest or a weighted back pack. I would not go much above 20% of your bodyweight when adding weights, or you may find it is too much compression on your shoulders and back.

The harder something is for you, the greater the adaptation response will be.

Cardio, GPP, and Work Capacity – Cardio is the ability of your heart and lungs to provide oxygen to your body. People who have great cardio have the ability to work at a high level of intensity at a continuous pace for a long period of time. GPP stands for General Physical Preparedness. GPP measures how prepared your body is for the physical activity that you are going to perform or might have to perform. It is sometimes referred to as your work capacity, which is the amount of work that you are able to perform, usually measured by how long it took you to complete that work.

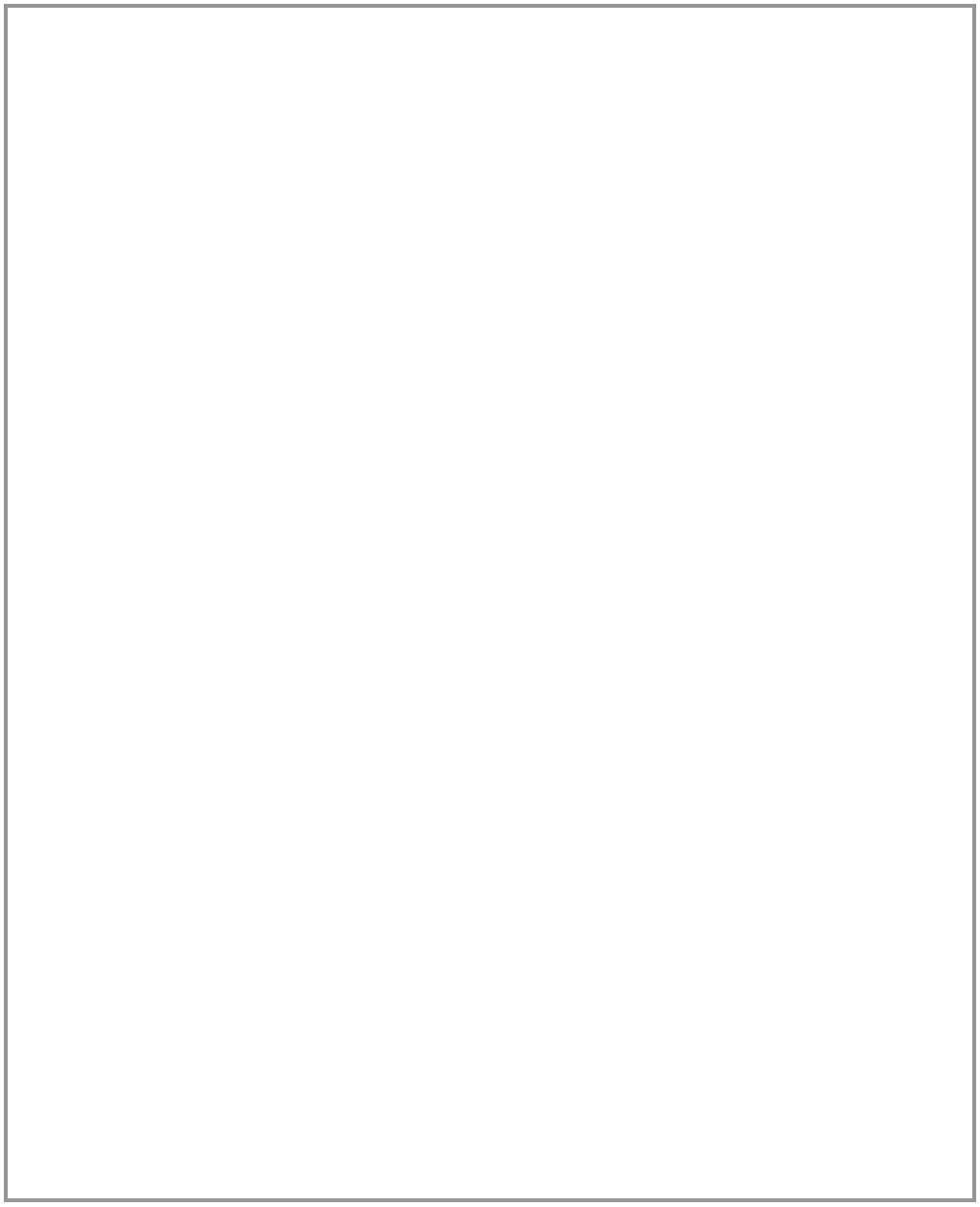
GPP and Work Capacity involve a greater degree of strength than pure cardio ability. An excellent marathon runner would have a great cardio

system, but at the same time their strength is probably pretty low. If you were going to hire a group of movers to move you out of your house, I don't think stereotypical elite marathon runners would be the type of person that you would want to hire. They would have a good enough cardio system so they won't get tired, but they are too weak to move the heavy stuff. Instead you would want somebody who is pretty strong but still has a good work capacity; that person could lift one heavy thing after another, not at a rapid pace but a nice, steady pace. At the same time you would not want a strong but out-of-shape person as that person would get exhausted after lifting one heavy piece of furniture, and you might find them sitting on the couch instead of moving it.

While your cardio and GPP can be built through more traditional methods (such as the treadmill, stairmaster, running, etc.) it is beneficial to make this type of training more specific to powerlifting. The most popular type of GPP is to drag a sled. Much like a work animal, continually performing this activity will significantly improve your fitness levels. Most cardio and GPP efforts also seem to have a restorative effect on the body. In simple terms this means you generally feel better afterwards than you did before. The common rationale given is that the increased blood flow helped heal any damage in your body. GPP, like cardio, can be of any level of intensity from very light to super intense. The more intense it is, the more you have to plan your workouts around it instead of the other way around. In addition, remember the principle of specificity. Getting good at a certain form of GPP, while impressive in one way, will not necessarily translate to lifting more weight in a certain lift so don't devote all of your efforts to it. Listed below are

Most cardio and GPP efforts seem to have a restorative effect on the body.

common forms of GPP that powerlifters often perform with a brief description about them.

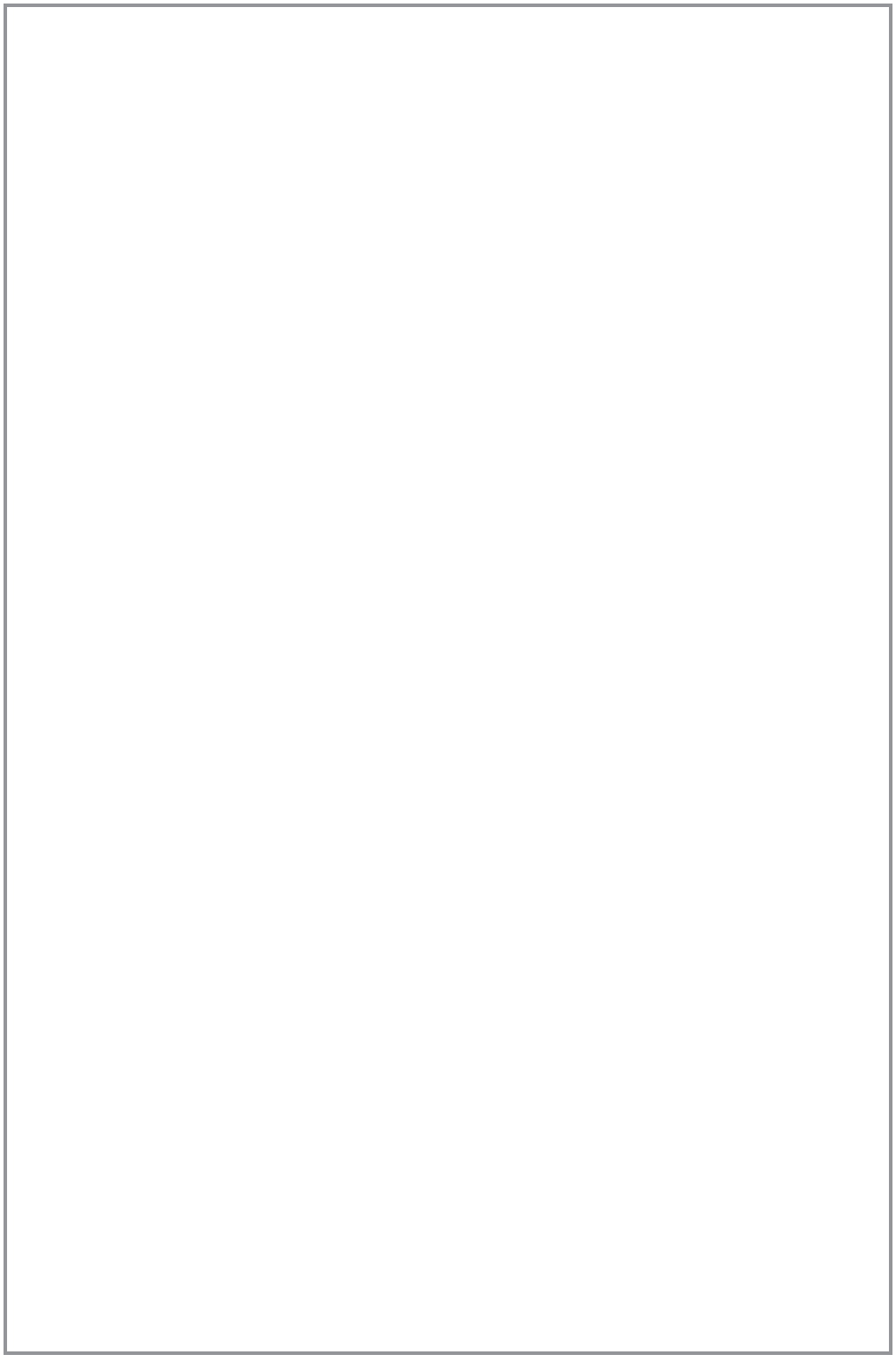


A homemade sled

Drag a Sled – The sled can be dragged on pavement, grass, carpet, and artificial turf. You can use store bought sleds or homemade sleds (see

Appendix A for how to make a cheap sled). Anything of weight can be added to the sled to increase the resistance. The sled can be dragged with ropes attached to the body, feet, or hands and you can drag it forwards or backwards or you can use your arms and legs to move the sled. The negatives are it is often a seasonal activity or can be affected by the weather; it can be inconvenient to find a sled with a bunch of weights or bring it along with you; and honestly you get a lot of weird looks if you do this in a populated spot. Still it can be very effective at building your work capacity and perhaps improving your strength at the same time.

Sprints – Sprints are a classic exercise that have a lot of benefits. They are training the type IIB muscle fibers which are the same ones we are trying to hit in powerlifting. They teach a person to be fast and explosive, which is useful in almost all situations. They can actually build muscle, and they are great for burning calories and fat. Most powerlifters sprint using a distance of 5–200 Meters, with <100 Meters being the most common since it is more sport specific. The shorter the sprint, the more reps you would perform. You might complete 20 5 Meter sprints (which is just working the start), 8 40s or 3 100s or something like that. The negatives with sprints are they are high impact, and they can be hard on the joints especially if you are heavy. They are hard on the nervous system so they can lead to overtraining. Extensive sprinting can burn some muscle when combined with other training. You have to have a good location to sprint (track or grass is best); it is often a seasonal activity, and it carries a relatively high risk of injury. The longer you have gone without sprinting full speed, the longer you should take to warm-up (in both minutes and days) to go full speed. Sprinting up a hill is a quick and easy way to make this significantly more challenging, more power related, and it might reduce the chance of injury.

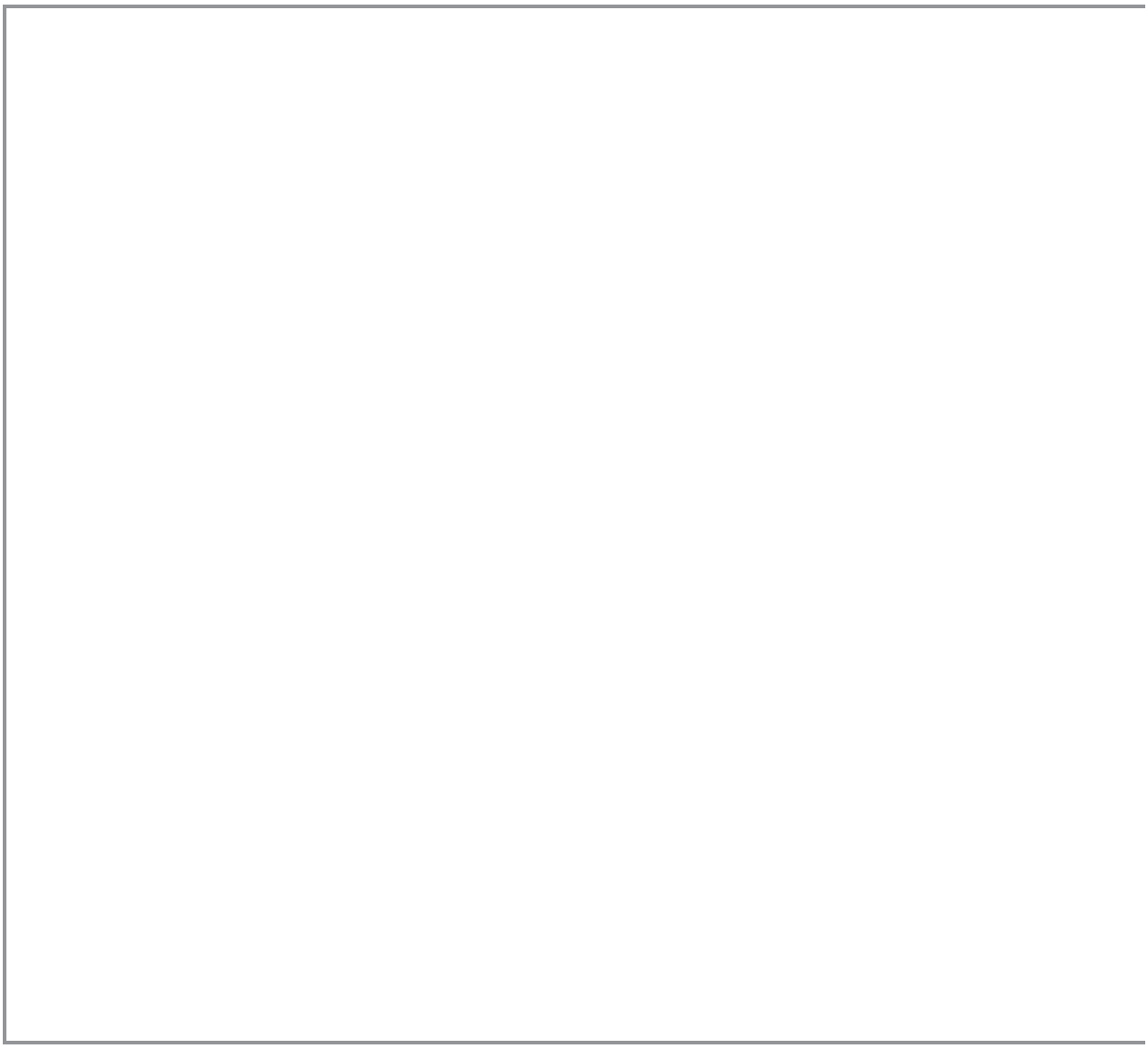


A strongman pulling a truck

Strongman Exercises – One of the big differences between powerlifting and strongman events is that strongman requires significantly more cardiovascular endurance than traditional powerlifting. But due to the high strength required for the strongman events, powerlifters are naturally drawn to those activities. Common activities like tire flips, farmer's walks, and stone carries all can build work capacity. Keep in mind though the greater the intensity of the exercise outside the gym, the greater the stress on the recovery system.

Jump Rope – This is an age old classic that always improves fitness levels. However, if you are heavy or if you are totally new to learning how to jump rope, it may be too much of an impact for your knees and lower back. Check your form if you have some joint problems; keep the knees soft, just rotate the wrists and don't jump too high. Using the rope in intervals of 30 seconds on, 30 seconds off is a good place to start.

Hiking – Getting outside can feel great — to take in fresh air, to be out of the gym, to actually go somewhere as opposed to walking on a treadmill, it can make you feel good to be alive. The varying ground and incline can add to overload, as well as using a weighted backpack. Hiking is extremely functional and practical; humans essentially always need to and are designed to walk everywhere. It can also be something you can do with others as a social activity. Carrying or pushing your kids can give you an added workout and help you introduce them to the physical culture.

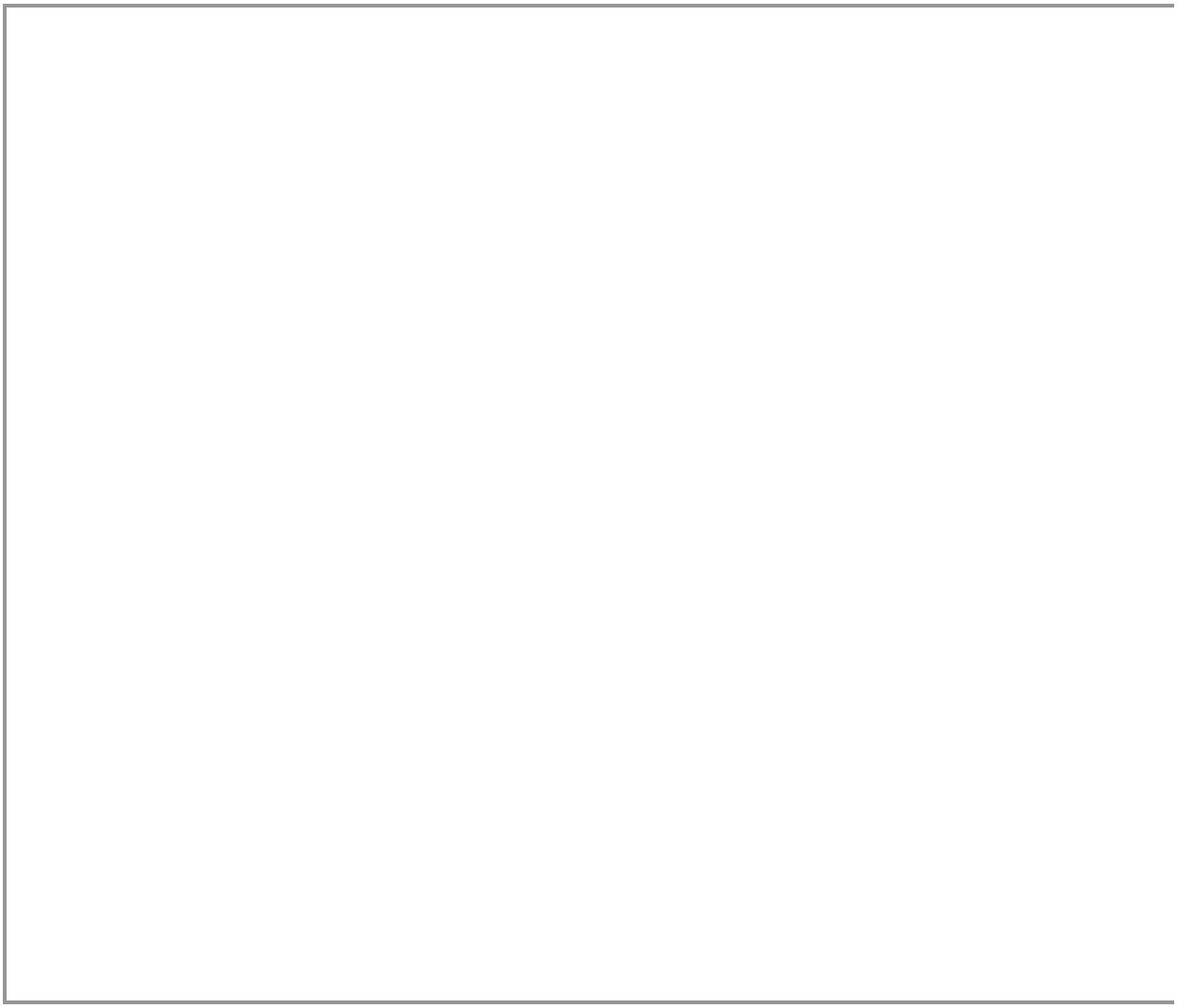


Hiking can be a family activity

Push Something – To get the upper body more involved you can push something. This can be a Prowler (special type of sled), it can be a sled with ropes that you hold in your hands with your arms straight, or it could be a car. The body position is very similar to pulling a sled, but, when pushing, the force is transmitted through your arms and shoulders. Generally pushing something is harder than pulling the same object, all other things being equal.

Sledge Hammer – Working a sledge hammer or similar implement can be

very taxing. Most lifters choose to hit the hammer into a large tire so the tire can absorb the force. You can use different swings (overhead, diagonal down, sideways, alternating the lead arm, etc.) to hit different areas of the body. Use the whole body to make it more explosive and taxing. You can also chop wood if you have access to this and you wish to do so. For both of these activities, watch performing too much too early and getting bad blisters on your hands. Even hands that are tough from years of holding a barbell can still be worn down by doing a very different activity (principle of specificity again); start with low volume and work your way up,



A lifter pushing a truck

Jumps – Put some jumps together or perform them fast enough and they start to tax the body quickly. Frog jumps (jumping forward for a number, backward for the same, one side and the next) are very effective. Short rest broad and vertical jumps or jumps repeated are also good. These can be hard on the knees; use caution if you are heavy or have knee issues.

Complexes – Perform a series of lifts with a barbell of the same weight all in a row with no rest. For example, complete the following:

Deadlift
Power Clean
Military Press
Squat
Good Morning
Bent Over Row

Perform 8 reps of each (or whatever number you want). Complete one series — that is a complex — and perform 1–5 rounds. My only negative about this is it might cause some fiber transition, and I don't like being so tired that super light weights feel hard in the exercise I will also go heavy in. The body sometimes remembers how the lift went. When you squat 500 usually the next squat workout the warm-ups feel light. But with complexes, most people are using 45–95 lbs. I don't want you to squat 95 lbs and have it feel heavy and then in 2 or 3 days squat 500; 225 might feel heavy. You can get around this by not performing any of the powerlifts in your complex.

Tabatas – This is a system of training where you perform 20 seconds of work, rest 10 seconds, and then go again. Usually four to eight rounds are

I find reverse tabatas to be effective and more specific to powerlifting.

performed, and since each round is 30 seconds long, it takes 2–4 minutes. It sounds pretty easy, but it can be very brutal. The first two rounds are easy, but then it gets tough. You can use whatever lift you want; the front squat seems to be the lift of choice; of course, smaller muscles are not going to cause the same level of exhaustion as larger muscles. I find reverse tabatas (10 sec on, 20 sec off or even 5 sec on 25 sec off) to be effective and more specific to powerlifting.

Combine the Methods – With conditioning you are limited only by your creativity, your mind, and your heart. There is no rule that says you can't combine lifting and sled dragging, or sled dragging and farmer's walks at the same time, or squats and jumping rope. Usually as powerlifters we are training for strength, and one of the fundamental rules is that you want to be relatively fresh for the start of each set so you can go as heavy as possible. That is true when trying to develop strength, but if you are training in order to develop your conditioning, then all bets are off and there are fewer official "rules" to follow. Make it hard, get out of breath, and the body will adapt to it.

What should I do?

You want to know what kind of cardio or conditioning *you* should do? The simple answer, in my opinion, is walk. Walking will burn calories yet burns very little muscle, improves your fitness, improves your recovery ability, improves your health, and it will not negatively impact your workout. It is also very difficult to overtrain from walking assuming you don't go crazy on the incline right away and develop shin splints. In my opinion, regular brisk walking is the single best form of physical activity a person can perform to improve their health.

Of course what kind of cardio you should do really depends on your specific goals. Here are some common goals and my suggestions for them (please note in the following section the terms cardio and conditioning are interchangeable).

The simple answer, in my opinion, is walk.

Goal: Improve VO2 Max to help get through powerlifting workouts

Perform cardio for 15–30 minutes 2–4 times a week, at the end of your regular workouts or on a separate day. Brisk walking is good, as is the elliptical machine (any version; they are low impact), biking (see how it affects your squat), and rowing. Interval training is okay. It is okay to combine two or more methods; for example, you might walk for 20 minutes and then row for 5 minutes. Your heart rate should be 70–80% of your max. It should be hard, you should be sweating, but you should feel like you will be able to complete it. Additional moderate intensity walking would only help.

Goal: Maintain my VO2 Max as it is already decent

Perform 15–30 minutes of cardio 1–2 times a week at the end of your regular workouts or on a separate day. Brisk walking is good, as is the elliptical

machine (any version; they are low impact), biking (see how it affects your squat), and rowing. Interval training is okay. It is okay to combine two or more methods; for example, you might walk for 20 minutes and then row for 5 minutes. Your heart rate should be 70–80% of your max heart rate. It should be hard, you should be sweating, but you should feel like you will be able to complete it. Additional moderate intensity walking would only help.

Goal: Use cardio to lose fat to lean up and make a weight class

Perform 30–60 minutes of cardio 3–9 times a week (4–5 is average). Ideally perform this cardio in the morning before you eat any food. If you can't do that, perform it after you workout or before you go to bed. Walking is ideal as other forms of cardio may cause too much muscular adaptation (in a bad way) or burn muscle. Occasionally performing another form of cardio or combining them is okay. Keep the intensity less than 70% max heart rate. Do not use a sports drink containing carbs before, during or after cardio. You should be able to talk comfortably while performing this type of cardio.

Goal: Maintain VO2 Max as I gain weight

Performing cardio and gaining weight can be tricky. On one hand, it is probably hard for you to gain weight, and performing cardio means you just have to eat even more calories. On the other hand, as you gain weight your fitness level will decrease anyway, plus sometimes cardio can serve to stimulate the appetite. If you are under 30 years old and in decent shape, you can hit cardio just once a week to try to maintain your fitness, I would go for 5–30 minutes; the shorter it is the more intense it should be. If you are over 30 years old or not in great shape to begin with, then I would complete cardio 1–2 times a week for 15–30 minutes each time as you try to gain weight. Trust the results you are seeing; if you are gaining too much fat, perform more cardio or ease back on the diet. If you aren't gaining the weight you want, decrease the number of cardio sessions or increase the diet accordingly.

What about HIIT?

HIIT is high intensity interval training, a form of cardio, not to be confused with HIT (only one I), which is a training philosophy related to lifting weights. HIIT is getting extolled in the media currently as being a great way to lose weight, and it does have many benefits. Essentially you perform cardio for a shorter amount of time (usually <30 minutes), with some fast and some slow intervals. Because it is pretty intense, there is a large aftereffect when the body is trying to recover from the workout. It may take 3, 4, 5 or more hours for the body's systems to return to normal after a HIIT workout, and during that time you are burning a lot of fat as your metabolism is elevated. HIIT also has positive effects on your VO2 Max.

HIIT is a fantastic way for the general population to perform their cardio to lose weight. However, I don't feel that HIIT is that beneficial for a powerlifter for the following reasons. First, any real powerlifter who is training hard (and any competitive bodybuilder as well) is already creating a huge after-effect with the weight training workouts (this is called EPOC). This is why weight training is generally more effective at changing body composition and producing long-term weight loss than cardio. You can have only so much of an after-effect; performing additional brutal things will not continue to elevate the metabolism even more — instead it will lead to overtraining.

HIIT relies on glucose for its fuel as you are performing the intervals. If you are low on glucose from dieting, poor eating, or a hard weight training workout then you will release cortisol to turn protein into glucose for fuel. This is a catabolic action and can lead to loss of

Weight training is more effective at changing body composition than cardio.

muscle. The leaner you are the easier it is to use muscle for fuel (since you have less fat). In general the body will use what fuel it has the most of. For most people that is fat but for some lean people it is muscle.

The HIIT intervals, while relatively short and intense compared to normal cardio, are too long to really hit the type IIB muscle fibers and to teach you to synchronize their firing. Instead they focus on the type IIA fibers and may cause a conversion from the type IIBs to the type IIAs since HIIT is definitely intense enough to hit those fibers.

And finally HIIT drains your recovery ability almost as much as intense weight training. Combining the two, especially while on a diet, which means you have limited fuel for recovery, can easily lead to overtraining. HIIT training is the type of thing where you do it for 2–3 weeks and feel great and then continue it for 3 months and feel run down and get an injury. For a powerlifter I would recommend only sporadic use of HIIT to lose weight and an increase in carbohydrate intake if you do use it on that day.

There are many types of programs out there and many methods available to improve conditioning. Listed below are two links that are particularly specific to powerlifting, but feel free to do your own thing; just watch for A) the results you get and B) how it affects your performance on the platform. If you aren't happy with either one of those, then give these methods a try.

T-nation article focusing on how to program in walking to build fitness and lose bodyfat:

http://www.t-nation.com/free_online_article/most_recent/get_ripped_get_walkir

T-nation article focusing on cardio for strength athletes:

http://www.t-nation.com/free_online_article/most_recent/cardio_for_strength_at

Supplemental Training — Plyometrics, Abs and Forearms

> *The purpose of plyos is to increase power.*

Plyometric Training

I primarily use plyometric exercises for lower body but you can perform plyos for upper body as well. When powerlifters are training plyos, they can be programmed in at many points in the workout. Plyos are power focused; thus, they often go in the beginning of the routine, but the principle of priority does state that you should do what is most important to you first, which is often squats. You can perform your plyos after your first main exercise or two, or you can do them near the end of the day. If you do the latter, your power output and even your form may be a bit compromised; however, for our purposes that can still work out.

The purpose of plyos is to increase power, learn to recruit the type IIBs, and also to maintain/increase speed, agility, and mobility. Sometimes powerlifters find if they lift only heavy, over time they start to feel a bit slow and stiff, since the speed of the powerlifts are relatively slow (.4—3.0 seconds) compared to many athletic events (.1—.3 seconds). If you are using plyos to increase power, then you do want to perform them early in the workout with frequent breaks and long recovery periods. However, if you are using them for mobility and agility, then you can do them near the end of the workout and you can still receive those benefits; they can even be a type of conditioning. Generally they would make up 5–20 minutes of the workout. Watch the total volume you complete, and look for soreness in the knees, hips, ankles and lower back the day or two after the workout. Reduce volume if you seem unusually sore or have more joint pain than normal.

Listed below are some possible lower body plyometric exercises that would benefit a powerlifter (and most likely just about any athlete):

Sprints (20–200M — usually perform a max of 400M total and break it up into chunks)

5M Sprints — to practice the start, complete 5–20 total sprints

Get-up-and-go sprints (sprints starting with your back/front on the ground)
Reaction Sprints
Box Jumps
Jumps over a box
Depth Jumps
Lateral Hops/Jumps
Frog Jumps
Broad Jumps
Vertical Jumps
Triple Jumps
Shuttle Runs
Agility Drills
Kettlebell Swings

Please note: for any sprints or high expressions of power, it is imperative to warm-up properly. Being quite strong and attempting to sprint full speed when it has been a long time since your last sprint is a recipe for disaster. You might even take 4–8 weeks where you sprint but only at 80–90% of full speed (do not run next to anybody during this time — you will be too competitive and you won't pace yourself).

You might take 4–8 weeks where you sprint only at 80–90% of full speed.

Training the Core

Most powerlifters perform some form of direct core training. If you are happy with your current program and getting results, then feel free to continue it. I will present some guidelines here that you can follow if you want to try them out or experiment with something new. The type of ab training that powerlifters, even very successful powerlifters, perform varies greatly. Some feel that heavy squats and deads give plenty of core stimulation and that direct ab work is better left to the bodybuilders. Others feel that increasing core strength might transfer over to a stronger squat and deadlift. As with all assistance work, I will tell you to ultimately put that hypothesis to the test. See where you are now and how you perform on the platform. Then chart the improvement in that exercise and see how it affects your performance on the platform. For example, let's say you can start off by performing 20 sit-ups before you fail and you can squat 400 lbs. Now you train abs for 6 months and if at the end of that time you can rep out 60 sit-ups and your squat is 500 lbs, there is a reasonable chance that the stronger core helped you squat more. Let's use another example. Taking the first person, if they could now complete 60 sit-ups but their squat was still 400 lbs, that means that their ability to perform sit-ups has little transfer to their ability to squat. Please note it doesn't automatically mean that their core is either strong or weak; it just means for that person sit-ups don't help their squat at that stage in their training. It could be that they have a weak core and sit-ups simply didn't target it, at least not how it is functioning in the squat.

Most people find training the abs one to three times a week to give the best results for strength and power. I think two times a week going reasonably hard is probably optimal. Generally the abs

I prefer attempting more challenging exercises and not endless reps.

are trained at the end of the workout;
however, I have often trained them in the beginning and found that worked for me. The logic for finishing with them is that they are stabilizing muscles; why make them tired and then go hit squats and deads, etc.? I would agree with that and I would usually not start with them on a leg day. For me personally however, I never really liked training abs. I also worked out pretty hard in general; by the end of the day when I was tired I was then supposed to go do something I didn't even enjoy? I would end up half-assing it or I would skip it all together. I started to train them in the beginning and found I put more effort into them, I didn't skip them, and I was still looking forward to the rest of the workout once I was finished with them. While training abs early can fatigue the core, in reality the abs recover very fast. By the time you warm-up for your next exercise most likely your abs will be feeling fine and they will adapt over time.

Abs can be very boring to train but that is usually because people just use the same old bodyweight exercises and they adapt to them quickly. Then the lifter must add lots of reps to be effective. I prefer attempting more challenging exercises and not endless reps. Usually I suggest a person perform 2–4 core exercises per workout, with an average of three sets for each exercise.

You can break the abs up into the following sections: upper abs, lower abs, obliques. I also like to perform some flexion, some rotation, and some stabilization to train the core's varied functions.

Flexion Upper Ab Emphasis

Sit-ups, Decline Sit-ups (weighted), Cable Crunch, Inverted Sit-ups (hits both upper and lower)

Flexion Lower Ab Emphasis

Leg Raises, Leg Thrusts, Hanging Knee Raise, Hanging Leg Raise, L-holds,

Dragon Flags, Weighted Pull-ups with weight held by feet

Rotational Movements

Woodchops, Rotations, Sword Swings, Sledgehammer movements, Landmine, Martial Arts, Unilateral Incline Knee Ups (flexion with a twist)

Stabilization Movements

Ab wheel, Planks (many versions — go for more challenging instead of longer times, weighted), Squats, Deads, Standing Military Press, Isometric Oblique Holds with bands or cables, core braces

How to Train the Forearms

Forearms are important for a powerlifter, primarily for the grip strength during the deadlift but also for stability on the bench press and the curls. I also think that an impressive set of forearms just looks studly and conveys power and strength — and who doesn't want that?

The muscles of the forearms are broken into 2 main categories: forearm flexors, which are on the palm side of your forearm and forearm extensors, which are on the back side (hairy side) of your forearm. The forearm flexors are responsible for wrist flexion (bending it forward when your palm is facing away from you) and finger flexion (grip strength). The forearm extensors are responsible for wrist extension (bending it backward when your palm is facing away from you). The forearm flexors are the most important section of the two for a powerlifter (and for most athletes).

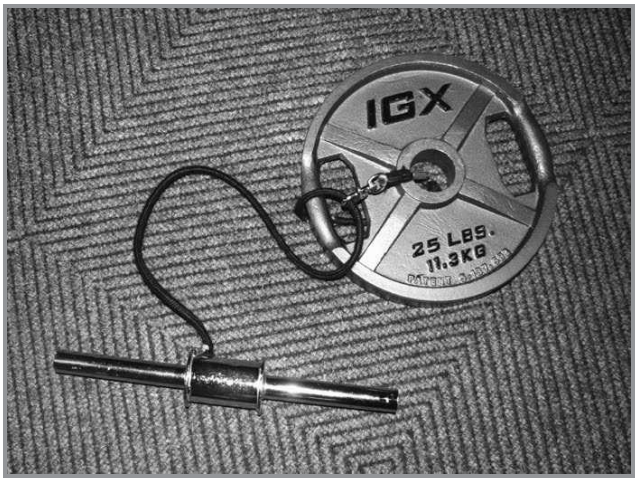
Forearms are important for a powerlifter for grip strength.

Wrist Flexion Exercises – Barbell Wrist Curls, Cable Wrist Curls, Wrist Roller Forward

Finger Flexion Exercise – Grippers, Wrist Roller, Fat Bar Training, Back Exercises with challenging grips — limited fingers, ropes, towels, not using straps, Deadlifts, Shrugs, Pull-ups, Rows, Hangs

Wrist Extension Exercises – Reverse Barbell Wrist Curl, Reverse EZ Bar Curl, Wrist Roller backward.

Forearms can be trained once or twice a week, usually at the end of the workout. Use caution if you program them in the day before a deadlift workout. If you are already lifting very intensely and then you add in intense



An example of a wrist roller

forearm work, you can overtrain your grip, forearms, or elbows — be on the lookout for tendinitis in these areas. Like the calves, the forearms will be stubborn; they will respond quicker to strength than to size, but you can make good progress in both of those areas over time. Martial Artists, Arm-wrestlers, and Climbers often have strong hands and a strong grip, and they usually have some very unusual but effective exercises for increasing grip strength.

Pick a few exercises that you believe in.

Summary

The purpose of the supplemental exercises and conditioning workouts is just that — to supplement the workout and help improve potential weak points that might hold one's performance back on the competitive lifts. Supplemental exercises can be included in a workout that includes the big lifts or they can be a separate workout themselves, it is really up to you. Pick a few exercises per area that you believe in, perform them at least weekly, and incorporate gradual progressive overload. Take note of your performance on these lifts and see if you believe they will help your performance on other activities. Once you plateau in a supplemental lift, switch it up and repeat the process with another exercise of a similar, hopefully even more challenging, nature. Don't major in the minors, however; being able to perform 10 good dragon flags or a 1 arm pull-up is much less impressive if you can't also squat double bodyweight for reps.

Chapter 12

General Nutrition Information

The food you eat can have a very powerful effect on how you perform in the gym and on the platform. Nutrition also has a significant effect on how you look, how you feel, and how healthy you are. Given the undeniable importance of all of those things, nutrition is a key part of the puzzle that shouldn't be overlooked. Following a good nutrition plan doesn't mean that you have to keep track of every single calorie you consume, but it does mean that you should have a heightened awareness of what is going into your body and how it will affect you.

A good sized breakfast to start the day

> ***Nutrition has a significant effect on how you look, how you feet, and how healthy you are.***

The nutrients we consume have three broad functions in the body. One is to provide us with energy; to give us fuel to accomplish our daily activities whatever they may be. Of course, the fuel required to complete a brutal 2 hour leg day is different from the fuel required to sit on the couch. The second function is to build and repair the tissue in our bodies. The cells of the body are constantly dying and they must be replaced. The building material to replace those cells comes from the food that you eat.

You literally are made up of what you eat. The third function of nutrients is to help regulate our metabolism. Your metabolism is all of the metabolic and chemical processes taking place in your body that are necessary to keep everything running smoothly. A person's metabolism is generally measured in how many calories they burn in a day (we will figure out your specific metabolism in the next chapter). Certain nutrients are able to help that metabolism run the way it is supposed to.

There are 6 essential nutrients to the body. In nutrition the term "essential" means you must consume them because the body does not make them, and it is necessary to life. The 6 essential nutrients are:

- Carbohydrates
- Protein
- Fat
- Vitamins
- Minerals
- Water

The first 3 nutrients are called macronutrients because they are relatively large in size. They provide us with energy because they all have calories. Vitamins and minerals are called micronutrients because they are small in size, and they do not have any calories in them. Water is also calorie free, and it is crucial in helping our metabolism run smoothly.

Carbohydrates – Carbs are made up of the elements carbon, hydrogen, and oxygen, which is where their name comes from (they are sometimes abbreviated CHOs for the same reason). The building blocks of carbs are

The fuel required to complete a brutal 2 hour leg day is different from the fuel required to sit on the couch.

saccharides, which are units of sugar. The main function of carbohydrates is to provide the body with energy, particularly high intensity energy.

Carbs come mainly from plants. Anytime you are eating a plant product, it will have some carbs in it, and often that is the primary nutrient. Plants are able to capture the energy from the sun using photosynthesis. We can't do that, but we can eat the plants and thus get their energy. Very common sources of carbs include potatoes, grains, bread, rice, pasta, fruits, and vegetables. Animal products usually don't contain many carbohydrates with the notable of exception of milk, which has a fair amount of milk sugar in it.

Carbs are broken down into 2 main classes, simple and complex. That term relates to how easily the body can digest certain carbs. Simple carbs are made up of only 1 or 2 sugar units, and they are easy for the body to digest. Complex carbs are made up of many sugar units stuck together; they are harder for the body to digest. The goal of the body is to take the food that was consumed and break it into its building blocks. When you eat carbs you are trying to break them down into individual saccharides. If a food is easy to digest (simple carbs) it is broken down and usually absorbed rapidly. If a food is harder to digest (complex carbs) it is broken down more slowly and then absorbed more slowly.

Carbs come mainly from plants.

The general recommendation is to eat more complex carbs and fewer simple carbs — and while that is good advice it does not show the entire picture. Not all complex or simple carbs are created equal. Simple carbs usually come in two main forms, regular sugars and fruit. Regular sugar, like that found in cookies and cakes and many other foods, should generally be avoided although an occasional treat is likely fine. Sugars are easy to digest, provide little long

term energy, and are easily stored as fat. While they do taste good, they usually don’t carry very many vitamins and minerals, and thus they are not considered nutritionally dense. There may be some times when it is preferred to eat mainly simple sugars (discussed later in the chapter), but certainly they should not make up the bulk of the diet.

Fruits should not be lumped together with Twinkies, as common sense will probably tell you. Fruits are usually digested slower than regular sugar, and thus they are much less likely to cause an energy spike and subsequent drop. They usually do contain a reasonable amount of vitamins and minerals, they are pretty low in calories, and they are tasty — a reasonable amount of fruits in the diet is to be encouraged. When you hear people knocking simple carbs, know that fruit should be excluded from that category.

Complex carbs are also not a case of black and white in terms of health. In general the recommendation is to eat a reasonable amount of complex carbs, and that is sound. However, complex carbs can vary greatly in quality. For example, the carbs found in a bowl of brown rice and the carbs found in a bowl of Doritos are both complex carbs, but again common sense should tell you that if you were to eat one and then go workout hard for 2 hours, you would have a different reaction eating the rice than if you ate the Doritos. When it comes to complex carbs you are looking for primarily unprocessed complex carbs. A simple way of looking at that is how close the carbohydrate is you are eating to its natural form, the way it is found in nature. Vegetables are a great example of an unprocessed carbohydrate. Generally the broccoli you are eating is very similar to how it was found when it was growing. Peas and corn and potatoes and rice are all pretty good examples of unprocessed carbs. If you have a choice, going with whole grain and/or brown is better than the white version (of rice or bread), although in my opinion the differences are not as great as they are often made out to be. Bread is

somewhat processed; it depends on what type of bread you are eating. And, of course, things like Doritos and potato chips and other general junk foods are very processed.

There are several problems with processed food. First, as they go through the procedure of being made, much of the original plant is stripped away. This causes the food to lose a good amount of its vitamins and minerals, turning the food into “empty calories” or a food that is calorically dense but not nutritionally dense. It is very easy to overeat processed food. They generally taste good, and they usually have a very little amount of fiber. They don’t fill us up until we feel bloated and have overeaten. They often have a high level of sodium to taste better and last longer, and a lot of excess sodium can cause a myriad of problems. While processed foods are usually technically complex carbs, they are pretty easy for the body to digest because so much stuff has been stripped away. It is almost like the machine that makes the food has pre-digested it for you. When a food is rapidly digested, it enters the blood stream quickly and raises the blood sugar levels rapidly. The body wants to maintain a pretty even level of blood sugar, so if the sugar is too high, the body will release insulin to use up the sugar and bring the levels back to normal. If the sugar is spiking, you have to release a lot of insulin, which in turn brings the level down rapidly as well. This causes that “crash” feeling you can get after eating a lot of food, and it also can cause a lot of the carbs to be stored as bodyfat, which, of course, is usually not desirable.

When you eat carbs, your body will use up a little bit of the carbs right away to boost up your blood sugar, but that requires only a very small amount of food (a few bites). The additional food that you have eaten, assuming it is

How close is the carbohydrate you are eating to its natural form.

digested and absorbed, will have to be stored for later. The body can store carbs in two ways. The body prefers to store carbs as glycogen. In the ideal scenario, any excess carbs eaten will be packaged up (glycogen is a complex carb) and then stored inside the muscles and the liver to be used later. Later on, when you need energy to maintain your blood sugar or to fuel more intense activity (such as a workout), the body will break down that glycogen for energy. In general, most people can store about 200–400 grams of glycogen in the muscles and about 100 grams of glycogen in the liver. A well trained athlete will store more than an untrained individual. This level of glycogen will usually last for about 1–2 hours of intense activity; the more intense the activity, the shorter it will last, although if you start off full, you almost always have enough for at least an hour of high intensity energy.

If the body is low on glycogen and you consume carbs, generally those carbs will be stored as glycogen. However, once your glycogen levels are full, continuing to consume carbs will do little good (it is like pumping gas into a full tank — it just goes somewhere else — fat). The carbs are more likely to be stored as glycogen if they are from unprocessed, complex sources of carbs.

If the glycogen levels are full, or if the carbs are digested and absorbed very rapidly, the body will generally send the carbs to be stored as fat. Storing a nutrient as fat is the default mode when the other needs have been met. You are more likely to store an ice cream sundae as fat than you are a bowl of rice, and you are more likely to store your third bowl of rice in that meal as fat than the first bowl (because the first bowl or so probably went to fill up your glycogen).

Fiber is another type of carbohydrate that is a complex carb, but we don’t generally digest fiber. Even though we don’t digest fiber, it has a lot of valuable functions in the body. It helps keep your system running well and

regular, it helps prevent constipation, it helps you feel full, it helps slow down the digestion rate of any other nutrients, it can bind with bile and cholesterol and eliminate excess amounts, and it has an antioxidant effect. Fiber is found in primarily unprocessed carbs like beans, peas, and broccoli. It is a good idea to try to eat some fiber (5–10 grams) at every meal. Since powerlifters often eat a lot protein, it is good to eat a lot of fiber as well to offset some of protein's negative effects.

The RDA (Recommended Dietary Allowance) for carbohydrates is 45–65% of your diet, meaning that about half of the calories you consume should be in the form of carbohydrates. As we have stressed, those carbs should be primarily complex, unprocessed sources of carbs. To limit the amount of simple sugar you consume, a good rule of thumb is to keep the amount of simple sugar to lower than 10–15% of the diet, not including fruit or milk. The RDA for fiber is 38 grams for men and 25 grams for women under the age of 50. The fuel factor of carbs is 4, meaning that for every gram of carbs you eat, you receive 4 kilocalories of energy (simple and complex are the same).

For powerlifters the RDA values for carbs work pretty well. You will need a reasonable amount of carbs to fuel intense weight training workouts. A diet of any less than 40% carbs would be considered a low carb diet and would generally not be recommended. In addition, you will need a reasonable amount of fat and protein in your diet, so going above 60% carb would also not be recommended. Humans are extremely adaptable and can perform well on many different types of diets. If you are unsure of what to do, start with the guidelines presented here and then modify them over time to suit your needs.

Protein – Protein is made up of the same elements as carbs (carbon, hydrogen,

Humans are extremely

oxygen) plus it also has nitrogen and sulfur. Those 5 elements are considered essential to life, and protein is the only macronutrient to provide nitrogen and sulfur; thus it is essential.

adaptable and can perform well on many different types of diets.

The building blocks of protein are amino acids. There are 20 common amino acids found in the foods that we eat. There are 9 indispensable amino acids, which means you must eat those amino acids because your body cannot make them. The other 11 amino acids are considered dispensable because your body can make them from the other 9, but it would still prefer to eat the normal form as opposed to have to alter something to make something else. A protein that contains all 9 indispensable amino acids is called a complete protein; one that doesn't have all 9 is called an incomplete protein.

Some protein is found in virtually all foods. The primary source of protein for most people is animal products. If you are eating an animal product, it will contain mainly protein (and fat), and you can be assured that it is a complete protein. Plants do contain protein but in smaller amounts; some have more protein than others. Most grains do not have much protein, veggies have some, considering their low calorie total (usually they have about 1:2 or 1:3 ratio of protein to carbs); some plants like nuts or soy have a pretty good amount of protein. The vast majority of the protein found in plants is incomplete because it is missing an indispensable amino acid. This does not mean it is useless, just that the quality is usually a little less than an animal product. In addition you can combine two or more incomplete proteins together to form a complementary protein, which is when the two or more incompletes put together provide all 9 indispensable amino acids. Beans and rice are an example of this, as is peanut butter and bread.

The primary purpose of protein is to build and repair the body's tissues. Tissue is not just muscle but basically all of the body's tissue, including the skin, hair, nails, organs, bones, and muscles. Remember, the cells in your body are all dying at a pretty rapid rate, and they must be replaced. Even if you are not gaining weight and building muscle, you still need some protein to continually rebuild and maintain your body.

The primary purpose of protein is to build and repair the body's tissues.

Protein can also provide energy if the body needs it. The body will prefer to burn carbs and/or fat for its energy source, but if those sources are low and energy is needed, it will use protein. Protein will mainly back up carbs as a source of high intensity energy. Protein provides the same amount of energy per gram as carbs: 4 kilocalories per gram. However, it takes more energy to turn protein into fuel so it is not as efficient of a fuel source as carbs (this can be a good thing if you are trying to lose weight).

Protein also helps regulate the metabolism of the body; it helps maintain the proper amount of water in your blood; and it helps maintain your immune system because you use protein to build your white blood cells.

Protein can be stored in the body, it is stored as muscle. Protein has the only functional storage form of energy, as you can use the extra muscle you have to help you do things (such as lift weights) and you can burn it for energy if necessary. If your body is low on energy, using up a lot of fuel, or not getting enough protein, you will use your current muscle to make up the difference. This is why if you get sick or go on a diet, you tend to lose a lot of muscle.

Excess protein will be stored as muscle up to a point. First, there must be a stimulus to create that new muscle tissue. Just sitting on the couch and

eating a bunch of steak is not going to cause your muscles to grow, but if you go exercise and create that demand, then some of the extra protein will go to repair and build the muscle. However, excess protein can also be stored as fat; unfortunately protein is not a nutrient that you can eat unlimited supplies of. Protein also tends to travel together with fat; if you are eating a lot of protein, you may be eating a lot of fat as well, which will mean your total number of calories might be high.

The amount of protein required is an often debated question. The RDA for protein is 10–35% of your diet or .8 grams/kilogram of bodyweight per day. The thought process behind giving two recommendations is that, since protein is used mainly to build and repair tissue, the amount of tissue you have (your weight) should have an effect on how much protein you need. The RDA used to be 10–15% protein, but in the past few years, it has been modified and updated; however, the .8 gram/kg recommendation has not changed. In general the .8 gr/kg is usually about 10% of the diet, and as such it is on the lower end of the recommendation.

Since this book is for powerlifters, it is assumed that you are working out pretty hard and that you are not sedentary. Assuming that is the case, it is has been well documented that low protein intake will hinder recovery and the ability to build new lean mass. For active people, 10% protein intake or .8 gr/kg is too low in my strong opinion. The NSCA (National Strength and Conditioning Association) recommends a percentage of 20% protein for active people or an amount of 1.4–1.7 gr/kg of bodyweight, essentially doubling that RDA. There are some who feel that level is still too low. It is not uncommon to read in fitness magazines of

My own personal suggestion would be to eat a diet of 20–35% protein or .75-1.50 gr/lb of bodyweight.

athletes eating 30%+ protein or 1–2 gr/lb of bodyweight.

My own personal suggestion would be to eat a diet of 20–35% protein or .75—1.50 gr/lb of bodyweight. The .75 gr/lb recommendation will line up essentially with the NSCA's recommendations. The very common guideline of 1 gram per pound of bodyweight works pretty well, but much more than that is probably unnecessary and may have negative effects on the body. Drug use (steroids) may increase the need for protein, but since I am assuming you are drug free, that is not entering into my recommendations. While a few nutritionists may cringe at the idea of 30% protein, since it is probably 3 times more than what they were taught in school, it is still within the RDA, and that should not raise any red flags.

	RDA	**NSCA**	**Media**
% Protein	10–35%	~20%	30%+
Amount of Protein	.8 gr/kg/day	1.4–1.7 gr/kg/day	1–2 gr/lb/day

Listed below are 3 sample people, and you can see once we run the numbers how many grams of protein they would take in each day according to the different recommendations found above. In this example, the following three people are all involved in intense weight training.

1. 120 lb (54.5 kg) female eating 1800 cals a day

   RDA – 10% = 180/4 = 45 grams of protein a day (much too low if active)
   RDA – .8gr/kg = 54.5 x .8 = 43.6 grams of protein a day (again too low)

   NSCA – 20% = 360/4 = 90 grams of protein a day (better)

NSCA – 1.6gr/kg = 54.5 x 1.6 = 87.2 grams of protein a day (okay)

Media – 30% = 540/4 = 135 grams of protein a day (seems good)
Media – 1gr/lb = 120 x 1 = 120 grams of protein a day (seems good)

My opinion — anywhere between the 90 and 135 grams of protein a day seems reasonable. The lifter could try both ends of that range and see how they felt and how they performed.

2. 181 lb (82 kg) male eating 3000 cals a day

RDA – 10% = 300/4 = 75 grams of protein a day (much too low)
RDA – .8gr/kg = 82 x .8 = 65 grams of protein a day (very, very low)

NSCA – 20% = 600/4 = 150 grams of protein a day (better)
NSCA – 1.6gr/kg = 82 x 1.6 = 130 grams of protein a day (still a little low)

Media – 30% = 900/4 = 225 grams of protein a day (top end)
Media – 1gr/lb = 181 x 1 = 181 grams of protein a day (good)

My opinion — eating somewhere between 150 and 200 grams of protein a day would seem to be ideal.

3. 275 lb (125 kg) male eating 4750 cals a day

RDA – 10% = 475/4 = 119 grams of protein a day (very low)
RDA – .8gr/kg = 125 x .8 = 100 grams of protein a day (very low)

NSCA – 20% = 950/4 = 238 grams of protein a day (better)
NSCA – 1.6gr/kg = 125 x 1.6 = 200 grams of protein a day (a little low)

Media – 30% = 1425/4 = 356 grams of protein a day (top end)
Media – 1gr/lb = 275 x 1 = 275 grams of protein a day (seems best)

My opinion — eating somewhere between 225 and 325 grams of protein would seem to be good, with 275 probably ideal.

There are some negatives to eating too much protein. First, protein is a costly nutrient — your food bill will add up. Supplements help with the cost, but my own personal opinion is that the effects of supplements pale in comparison to that of real food. Protein can be high in fat, so you need to pay attention to how much fat you are eating. Protein is low in fiber, and it is important to make sure you are eating plenty of fiber during the day. A higher amount of protein can cause constipation; again the fiber will help with that. Excess protein can also cause calcium to be leeched from the bones, although a lot of protein sources provide calcium as well, but it is something to be watched. Protein is harder to digest than some other nutrients, and it uses up more water in that process. In turn if you are eating a fair amount of protein, you want to drink a reasonable amount of water at the same time. There is debate if protein is hard and/ or bad on the kidneys. It is pretty clear that if you have kidney problems, you will probably have to watch your protein intake. Some experts feel that the excess protein is bad for the kidneys but many others feel that excess protein would have no negative effect on healthy kidneys. I do not know the answer to that myself, so all I can do is relay information back to you. In addition 30% is within the RDA, and the RDA is generally pretty conservative, so it is unlikely that a 30% protein diet would cause any problems. I would also say that almost all active people should eat at least 20% protein.

There is an inverse relationship between the number of calories that you are eating and the percent of your diet coming from protein. The more

calories you eat, the more energy you have coming from carbs and fat, you can get away with less protein. A person trying to gain weight can probably eat 20–25% protein. The fewer calories you eat, the greater the percentage of protein you need to build and repair your tissue. In addition it is easy to lose muscle, and extra protein will help prevent that. A person dieting will probably eat 25–35% protein. The more muscle you have on your body, the more protein you will need to maintain that level.

It is ideal to spread your protein intake throughout the day. You can have your largest intake of protein as your post-workout meal (see below). Listed below are several common foods and the amount of protein found in each one. A very general recommendation is to take in 20–50 grams of protein at each meal.

1 egg = 6–7 grams of pro (half in yolk, half in white)
1 ounce of milk = 1 gram of protein (same for skim, 1%, whole, etc.)
1 ounce of red meat = 6–8 grams of protein, the leaner it is, the more protein it has
1 ounce of chicken = 6–8 grams of protein
1 ounce of pork = 6–8 grams of protein
1 ounce of turkey = 6–8 grams of protein
1 ounce of fish = 6–8 grams of protein
1 ounce of nuts = 6–8 grams of protein
1 serving of veggies = 3–5 grams of protein
1 serving of starch = 2–4 grams of protein
1 serving of fruit = 0–2 grams of protein

The type of protein that you are consuming probably does have some effect on how your body uses it. There have been several tests to measure the quality of proteins available. In general, eggs score the highest, and milk scores the second best. The animal products do pretty well (all seem

somewhat similar), and then plants only do okay.

As a powerlifter you need more protein than a normal person, but you do not need to gorge on protein. Having some protein at each meal seems to have the most positive effects on the body. Eggs, milk, lean animal products, nuts, and vegetables should provide the bulk of your protein intake. Supplements can be used when necessary, but my suggestion would be that they provide no more than 20–25% of your daily protein intake with a definitive maximum of no more than half your total intake (e.g., if you consume 200 grams of protein try to limit the protein to 50 grams and definitely limit it to less than 100 grams in supplement form).

I would say that almost all active people should eat at least 20% protein.

Where you are in the lifecycle will also affect your protein requirement. It should make sense that an 18 year old male trying to gain weight and build muscle would need more protein than a seasoned 40 year old man trying to maintain weight and not gain fat, even if both are lifting hard. Personally I would say that, as I have gotten older, I think my need for protein has decreased a bit, but again 20–30% seems to be ideal in most cases. An interesting point of information: human breast milk is considered to be very anabolic and an infant could thrive on it alone for the first 6 months to 2 years of their life; however, it is only 6% protein (cow's milk is significantly more). Something to think about when toying with super-high-protein diets (breast milk varies but it is usually 52% fat, 42% carbs, and 6% protein).

Salmon and sweet potatoes will provide high quality carbs, protein and fat

Fat – Fat is made up of the same elements as carbs: carbon, hydrogen, and oxygen. The main type of fat that is found in the food you eat is called a triglyceride, which stands for 3 fatty acids combined with one glycerol. Fatty acids are the building blocks of fat, and there are three different types of fatty acids.

Saturated fatty acids are usually found in animal products. These fats are a solid at room temperature, like butter or lard. Saturated fats are considered the “bad” fats because they are the ones associated with heart disease and strokes, but that is a term to move away from. Saturated fats don’t have to be completely avoided but should be eaten in moderation, and the more natural the saturated fat (like grass-fed beef), the better.

There are two types of unsaturated fatty acids, both of which are generally found in plants. A monounsaturated fatty acid is most common in olive oil. Polyunsaturated fats are found in vegetable oil. Both types of fats

are a liquid at room temperature. These fats are considered the “good” fats because they do a lot of positive things for the body, but, again, in nutrition, it is tough to label things as good or bad without looking at the whole diet.

There are two essential fatty acids (EFA’s) which the body needs to function properly. The first is Linoleic Acid, or Omega 6 fatty acid, and the second is Linolenic Acid, or Omega 3 fatty acids. Both of these EFAs are polyunsaturated fatty acids.

Fats do a lot of things in the body, some negative and some positive. Fats provide you with the EFAs, which are necessary for life. Fats carry the fat soluble vitamins (A, D, E, K), which are also essential. Fats help add to the flavor in food, and they help you feel full for a long time. Fats cover the nerves in your body to allow them to conduct electricity faster and work more efficiently. Fat helps keep you warm through thermal regulation, and it helps protect the body by acting as a layer of cushion around the vital organs. Fats help lubricate and pad the joints so they continue to work well. Your body often uses fats and lipids to create hormones which are vital in directing the body and telling it what to do. Finally you use fat to build the cell membranes in the cells in your body, which are like building the walls to the cell. A strong cell wall means a healthier cell that can keep bad stuff out and good stuff in.

Powerlifters need fat in their diet.

Fat is very dense in energy. Each gram of fat provides 9 calories per gram, which is more than twice as much as protein and carbs. Fat is mainly a low intensity fuel; it is usually burned together with carbs to provide energy for the body. Fat is stored in the adipose tissue. Adipose tissue on a human looks pretty similar to adipose tissue on a chicken. It is yellow and spongy like the fat you might peel off of a whole cooked chicken.

One of the big problems with fat is its name. It just seems automatic that if you eat any fat, you will get fat, and this is not the case. It is also never the goal to have a diet of zero percent fat, nor do you have to eat the same amount of fat in your diet as the bodyfat you desire. Fat should not be severely restricted, nor it should not be eaten without limitations; both situations can be bad.

Powerlifters need fat in their diet. We need the energy to help get us through long workouts (with carbs being the main fuel), and many powerlifters need to eat a reasonable amount of calories throughout the day; the sheer volume of food would be enormous if it were all super low fat. In general powerlifters should eat a minimum of 20% fat in their diet. Unless you naturally eat very lean foods, a person would have to make a conscious effort to eat a diet 20% fat, as that is pretty low. The average American diet is usually 35–45% fat depending on what you read. It is unadvisable to go below this level (20%) of fat in the diet for several reasons. The first is that a very low fat diet will lead to a decrease in testosterone, which is not good at all for a person who lifts heavy weights. We want to maximize our testosterone levels, not minimize them. The second reason is that a low fat diet in females can negatively affect their menstrual cycle and their fertility. When you eat a low fat diet you are telling the body there are not enough resources around to support you properly, and the body is going to begin to shut down any systems it sees as unnecessary. To perform as well as possible, we want to maximize our body's systems, not minimize them. Finally a very low fat diet has been shown in some populations to actually increase blood lipid levels and cholesterol count, which can be markers for heart disease.

While fat is crucial and should be a minimum of 20% of the diet, that does not mean you can consume unlimited

A very low fat diet will lead to a decrease in

amounts of fat each day. The RDA for fat is 20–35% of your diet, and both of those ranges make sense for powerlifters. It is probably unnecessary to eat much more than 35% fat in your diet; that would begin to cut down on the amount of carbs and protein you could have. Any diet that is 30% or less fat would be considered a low fat diet; don't think that 30% fat is a huge amount. Eating too much fat generally means too many calories, which can lead to an increase in bodyweight, usually decreasing your relative strength and perhaps bumping you up to the next, more competitive weight class. Carrying around a large amount of excess weight can lead to certain types of cancer, stroke, hypertension, joint problems, and type II Diabetes. Eating too much fat often means eating too much of the wrong types of fat and that could be hard on your heart.

testosterone.

There is one type of fat that powerlifters and pretty much all people interested in their good health should avoid, and that is trans fats. Trans fats are unsaturated fats that have been chemically altered, and they tend to be a solid at room temperature. More and more studies are showing the negative effects of trans fats, and it would not surprise me if we learn they are even worse than we previously believed. The negative effects of trans fats include worsening the blood lipid levels (increasing the fat in the blood), decreasing the body's ability to metabolize the EFAs, decreasing a woman's fertility level, and altering the cell membranes. Remember that fatty acids are used to help build the cell membranes. Trans fats are chemically altered, and they do not make good building blocks for the cell membranes. It is like using weaker, abnormally shaped bricks to build a house; the walls of the house will be weaker and more likely to break.

With nutrition a good rule of thumb is that moderation is key. You can have some ice cream — just do it moderation. You can drink a coke — just

do it in moderation. That used to be my approach with trans fats; however, with studies showing significant negative effects with subjects consuming just 3–4 grams a day of trans fats, it seems that moderation is no longer a good policy for this particular nutrient. Instead the general goal is to avoid trans fats as much as possible. Trans fats are primarily found in processed foods and junk foods; look for hydrogenated or partially hydrogenated vegetable oil in the list of ingredients.

Powerlifters do need to eat some fat in their diet. Good types of fats generally are found in eggs (whole), whole milk, leaner cuts of animal products, butter, fish, and almost any natural plant product such as nuts, beans, and avocados. As with the other nutrients, it is best to spread your daily intake of fat relatively evenly over your meals as opposed to consuming a very high fat meal all at once. This is particularly true before you workout; a high fat meal will probably leave you feeling full and sluggish. As a very general guideline most people should eat about 7–25 grams of fat per meal. If at all possible try to eat organic, healthier sources of fat as studies show that the type of fat contained in those sources (for example milk from grass-fed cows) has a better fatty acid profile than those cows fed grain and kept on feed lots.

A good rule of thumb is that moderation is key.

A sample pre-workout meal of salmon, baked potato and broccoli

Pre-Workout Meal

As a powerlifter (or any athlete) what you eat before you workout can have a very strong effect on how you perform during the workout. Eat well and you should have long lasting and relatively constant levels of energy. Eat poorly and you can be weak, shaky, run out of energy, feel lethargic, and you might even throw up. When you are first learning how your body responds to food, try to pay attention to what you eat and then how you feel during the workout. That alone will teach you how your body responds to certain foods and what are the best types of foods for you personally to eat.

In general active people should eat about 4–6 times a day. That includes regular meals and snacks if you eat them. To me a meal is anytime food goes in your mouth. Try to space the meals about 2–4 hours apart so you will have pretty regular intervals when you are eating. This will help keep your blood sugar relatively constant, it will prevent insulin spikes, and it will help prevent you from getting extremely hungry. It is when we are very hungry and craving certain foods that we tend to make poor food choices.

Your pre-workout meal should be consumed 2–4 hours before you workout. The larger the meal or the more intense the workout (squats and deads), the more time you need to digest the meal. A small meal might be digested in 2 hours or so, and with a less intense workout, having a bit of food in your stomach doesn't matter that much. If you workout early in the morning you probably won't have time to eat 2–4 hours prior; the best solution is often to eat a good-sized dinner and have a very small meal (like some oatmeal or a bagel and peanut butter) 45–60 minutes before the workout.

The most important nutrient in your pre-workout meal is carbohydrates. You want the bulk of the carbs to be complex, unprocessed carbs. Some of the best examples include rice, potatoes, sweet potatoes, and oatmeal. Pasta

and bread can also work pretty well, assuming you don't overeat them. You are looking for a good source of starch that will fuel the high intensity exercise and prevent you from crashing during the workout. Consuming some vegetables along with the starch is a good idea as well; the veggies are good for you, of course, and also add in some good carbs that are slowly digested.

You can have a small amount of simple carbs in your pre-workout meal, but it should not be excessive; less than 25% of the total carbs should be simple (including fruit). There may be some sugar in the meal you eat or you might have a piece of fruit or some juice with your meal, and that is okay. However, if the simple carbs make up too much of the meal, you will probably have some energy for 15–30 minutes and then begin to feel weak and tired. While fruit is good for you, fruit alone is a very poor pre-workout meal. You can have one piece of fruit or a cup of fruit juice with your pre-workout meal, but I would not have any more than that.

Most people should consume 50–200 grams of carbs in their pre-workout meal. You would eat 50 grams if you are on the small side (125 lb female, for example), or if you didn't have as much time to digest your food. You would eat 200 if you were on the larger side (250 lb male, for example), or if you had a long time lapse between eating and working out. If you eat much more than 200 grams of carbs, that is a pretty big meal and will probably slow you down. If you eat less than 50 grams, you are likely to run out of energy during the workout, assuming it is pretty intense.

Active people should eat about 4–6 times a day.

It is important to eat some protein before you workout, but you do not want a lot of protein. Protein takes longer to

Only you can be the judge of what types of

digest, and too much protein may slow you down in the gym. In addition, if you eat a lot of protein you may not have as much room for the complex carbs you need. You also want the protein source to be pretty lean to prevent excessive fat. A chicken breast, turkey, a piece of fish are all good sources of protein before you workout. You want to eat some protein to help maintain your plasma pool of amino acids. This way if your body needs to use amino acids during the workout, hopefully it will use the amino acids in the food you have eaten, and it will not take them from your body by breaking down your muscle tissue. The longer your workouts are and the leaner you are, the more likely this is to happen. Most people should eat between 15 and 50 grams of protein before they workout.

foods work the best for you.

You should consume some fat before you workout to help maintain your energy levels, but too much fat may slow digestion and may leave you feeling too full. In general the pre-workout meal should be pretty lean, and you normally do not have to make an effort to add any fat to it; the fat will come along with the protein and in the starch source. If the food is very lean, adding a tablespoon of olive oil, butter, or salad dressing can add in healthy fat. Eating 5–20 grams of fat will work well for most people.

Ultimately only you can be the judge of what types of foods work the best for you. The food should taste good, it should be relatively healthy, and it should give you fuel for your upcoming workout. I like to say “you can’t argue with results”; if you are performing very well and feeling great, you must be doing something right. If you are not performing well and if you are feeling tired and sluggish, even if you are eating the supposed “right” foods, something needs to change. Below is a list of foods that tend to work well before a workout and foods to generally avoid.

Foods to Eat	**Foods to Avoid**

Chicken Breast	Cold cereal
Turkey Breast	Donuts
Fish (salmon, tuna, trout, etc)	Pastries
Potato (baked, mashed)	PopTarts or the like
Oatmeal	Large amounts of milk
Sweet Potato	Soda or equivalent (including diet)
Rice	Large amounts of fruit
Lean Red Meat	Fast Food
1 piece of fruit max or 1 glass of juice	Junk Food
Bagel	Fried Foods
Olive Oil	Any desserts
Any veggie (peas, carrots, broccoli, etc.)	Protein Bars
	Protein Shakes
Salad (will need some starch with it most likely)	Chinese Food*
	Mexican Food*
	Pizza

**Note: It is not that the foods from these cultures are necessarily bad, but the Americanized version of them found in restaurants is not good before a workout.*

Listed below are some examples of what a complete pre-workout meal might look like for given lifters.

	Female 130 lbs or on a 1500-2000 Cal Diet	**Male 180-220 lbs or on a 3500 Cal Diet**	**Male >220 lbs or a on 4250 Cal Diet**
Option 1	3–4 oz of Chicken Breast 4 oz of mashed potatoes 2–4 oz of broccoli Water	4–8 oz of Chicken Breast 6–8 oz of mashed potatoes 4–6 oz of broccoli	10–12 oz of Chicken Breast 8–12 oz of mashed potatoes

		8 oz of Orange Juice	6–8 oz of broccoli 16 oz of Orange Juice
Option 2	3–4 oz of Salmon 4 oz of Orzo Small Salad Water	4–8 oz of Salmon 6–8 oz of Orzo 4–6 oz of asparagus 8 oz of Gatorade	10–12 oz of Salmon 8–12 oz of Orzo 6–8 oz of asparagus Small Salad 16 oz of Gatorade
Option 3	1 small bagel (½ a large bagel) 4–5 slices of turkey 1 piece of cheese 2 slices of tomatoes 1 tbsp mayo 1 piece of fruit Water	1 large bagel 4–6 oz of turkey 2 pieces of cheese 4 slices of tomato 2 tbsp of mayo 1 piece of fruit Water	1 large bagel 6–8 oz of turkey 3–4 pieces of cheese 4 slices of tomato 2–3 tbsp of mayo 1 piece of fruit 1 yogurt Water

Fruit juice is not a good choice for a sports drink.

During the Workout – Of course it is unusual to eat anything during the workout, but if you are working out at an intense level for a pretty long time, you may want to consider a sports drink. This can be particularly useful if you are trying to gain weight (if you are trying to lose weight, do not use a sports drink). You should also consider a sports drink if you were not able to eat very well before the workout as your energy and glycogen levels will probably be low.

Generally getting a drink that is primarily simple carbs is what most people choose. Gatorade, Powerade, Carboforce, and Surge Workout Fuel all work pretty well (Surge is my personal favorite). Most people will want to consume 25 to 100 grams of carbs during the workout. Generally it is better to sip the drink during the workout than to chug it all at once. Often the

commercial sports drinks are over sweetened to make them taste good; an option is to drink half of it, fill it back with up with water, and then drink the rest. Fruit juice is not a good choice for a sports drink because of the way the body digests fructose, and sodas (diet and regular) are also a very poor choice. Of course, water is good, but it will not replenish lost carbs and/or electrolytes.

Some sports drinks now contain protein, and there is some research that consuming protein during the workout can have an anti-catabolic effect. Unfortunately these drinks tend to taste poorly. Another alternative is to consume some branched chain amino acids (BCAAs) before and after you workout (usually in pill form) to get a similar effect. The longer your workout, the more important this is. In general if you workout for less than an hour, you probably don’t need a sports drink. If you workout for more than an hour, you may want to consider one.

Why do I feel sick when I workout?

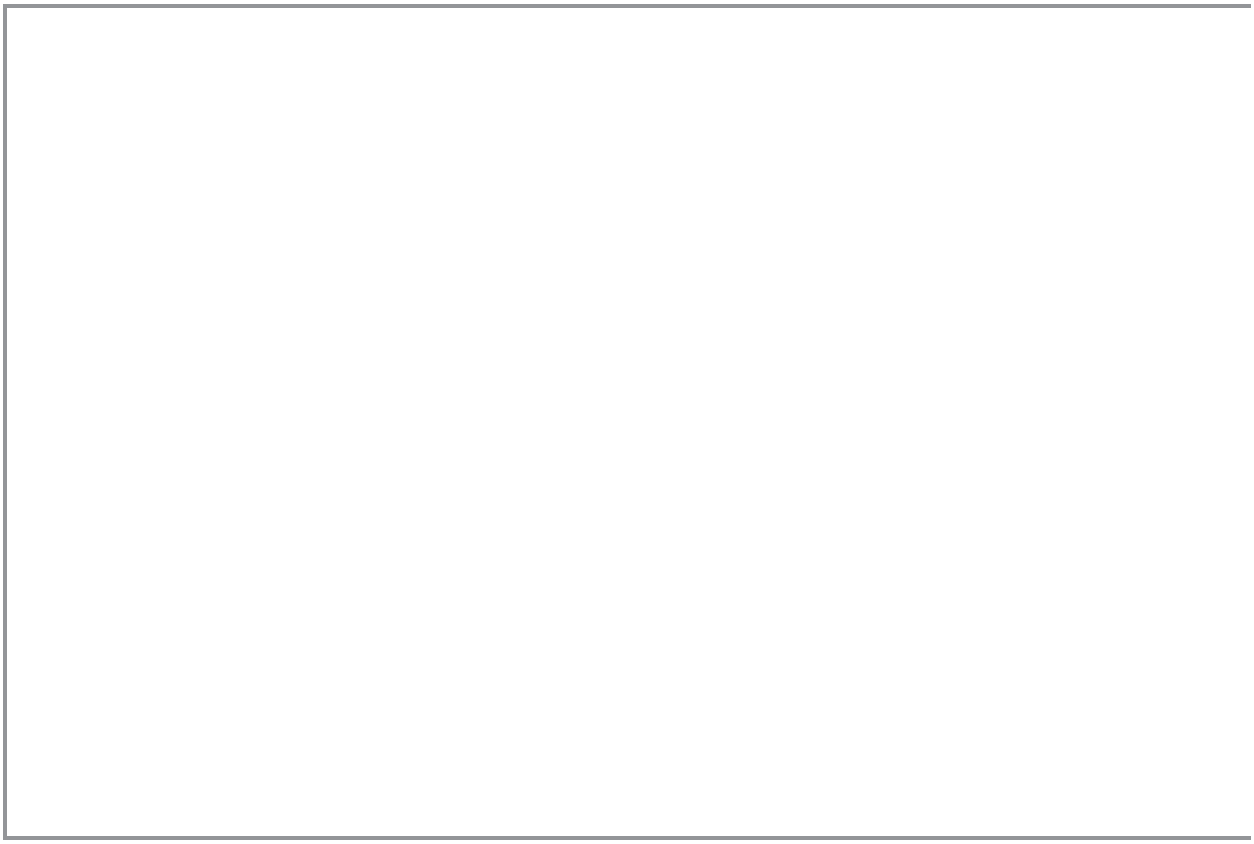
One reason you might feel sick when you are training is that you have eaten either too close to the workout time or even during the workout. The reason you don't want to consume food during a weight training workout is that when you are working out, the blood is sent out to the muscles. When you are digesting food, the blood is sent to the stomach and intestines. Activity takes priority over digestion (imagine if you had to run away from a lion after eating a big meal — you would want to be able to do so), so if you are active you are sending your blood away from the GI tract. If there is a lot of food in your GI tract but not much blood there to help it digest, the body will purge itself of the food, so you will throw it up, kind of like a fish that is recently caught will often puke up the other food in its stomach under stress.

Another reason you might feel sick, particularly on leg day, is the fluid ball phenomenon. Your stomach is basically where your center of gravity is. When you are squatting or training legs, the force from your legs acting on the ground is driving your legs up. The weight on your back is driving your body down. These forces meet at your center of gravity. In addition you have to squeeze your core (abs, obliques, TA, erectors and the diaphragm and the pelvic floor muscles) to maintain stability, and everything inside the abdominal area will be compressed. All of these pressures will act on the stomach and anything in it, and if there is a reasonable amount of food in the stomach, the body will eject it under these circumstances. Leg training is also much more draining in general because of the size of the muscles involved.

The third reason why you might feel sick during a workout is not because you have eaten too recently but because you did not eat enough or you did not eat the right food for the workout. If you don't

Leg training is much more draining because of the size of the muscles involved.

eat well, your blood sugar will drop, often quickly, when you exercise intensely. As your blood sugar drops below normal, you become hypoglycemic. When this is happening, you often feel weak and shaky, you look pale, you feel sick and tired and out of energy, and you may find it hard to concentrate. If this happens to you, consume some simple carbs as soon as possible. Gatorade or orange juice normally works very well. If you catch this very early and consume an adequate amount of food, you may be able to continue to workout. Normally you will have to either reduce the intensity of the workout or simply stop the workout all together. This is very rarely a serious condition (unless you are diabetic or try to drive when you have low blood sugar), but it may make you feel poorly for several hours.



Chicken thighs with skin, baked potato, and peas is a staple for me

Pre-Competition Meal

Your pre-competition meal (the meal you eat before a competition) should be similar to what you eat before your workout, particularly your hardest workout. The big challenge is that you may be in an unfamiliar place and you may not have access to the same restaurant or grocery store or even cooking possibilities as you would at home. You might be staying in a hotel that does not have cooking facilities. In addition, most powerlifting meets start between 8 am and 10 am, so you will probably not have 4 hours to digest your meal, and you might need to begin warming up even prior to that time. Generally you will weigh-in in the morning as early as possible, then eat, try to digest and then begin the day.

A common change made to the pre-workout meal to make it a good pre-competition meal is to reduce the amount of protein. Protein takes longer to digest, and you may not have the same amount of time as usual to digest your food. In addition it is natural and normal that you will be nervous the morning of a competition. When you are nervous, you often don't have much of an appetite, and your digestion will be slower than normal. Carbs are usually the easiest and the quickest food to digest, but make sure you don't eat a bunch of simple carbs, or you will run out of energy. Powerlifting meets can often start in the morning and run into the evening so you do need to eat something that can tide you over for as long as possible. The list of foods to avoid for a pre-workout meal still applies to a pre-competition meal.

The morning of a competition is not the time to try something new — do not try a new breakfast or a completely different routine. That is what training is for. You want to choose foods that taste good to you, are easy to digest, give you

The morning of a competition is not the time to try something new.

fuel throughout the day, and that you believe will make you strong.

Some pre-competition meals that I have used myself are Kraft Macaroni and Cheese; Spaghetti O's; a turkey bagel sandwich; eggs, potatoes and peas; spaghetti mixed in with some meat and veggies. The first two might look a little odd, but they are good if you have to travel. If you have access to a hot plate (which you can bring with you) and/ or a microwave, you can make Mac and Cheese or Spaghetti O's. They taste pretty good, have a reasonable number of calories, are pretty easy to digest, contain mainly carbs, and have enough pro and fat to give you some lasting energy. A bagel sandwich is good because there are often bagel places near meets that are open early for breakfast (be aware that they don't always open as early on Sat. or Sun. as they do on Mon.–Fri.). You could even buy it the day before and keep in the fridge if you wanted to. Eggs and potatoes are good if you have more time to cook. You probably want to eat 2–4 eggs (whole) with a ½ to 1 baked potato, which you can top as you desire. I would strongly suggest you avoid things like donuts and pastries (which I see with surprising regularity the morning of competitions), as they do not provide lasting energy at all; if you want a donut, eat as many as you want *after* the meet.

A good time to eat is shortly after you complete your squats or bench press.

Nutrition intake during a meet

Because powerlifting meets are often quite long (8–12 hours is not uncommon), you probably will need to eat something solid during that time to maintain your energy levels. A good time to eat something solid is usually shortly after you complete your squats or bench press or both (the meet is over after you do deads). You will normally have about a 90 to 270 minute break in between lifts. Make sure you have an idea of when you are lifting next if you leave the facility to go get food. Remember you have to warm-up before you lift, so plan your time accordingly. In general you want to eat a pretty small amount of food so it will quickly digest. Half a turkey sandwich or a banana or something like that generally works pretty well. I would often make some pasta with tomato sauce and throw in some chicken and a hot dog or two, and I would eat that as breakfast and throughout the day. Don't eat too much or you will feel bloated or uncomfortable when you are lifting. In addition you definitely want to drink a sports drink throughout the day of a powerlifting meet. For a full meet, a common recommendation would be to consume 4 small bottles during the day, 2 of which would be a sports drink and 2 of which would be water. Generally the order would be as follows: sports drink, water, sports drink, water or something like that. I would not suggest eating supplement-type foods like powerbars or protein bars or shakes or something like that because those types of food often don't digest that well. When in doubt whole food is usually the best choice. Avoid things like sweets and candy bars until after the meet. If you have to eat fast food, eat a small amount of it; if you would usually get 2 burgers, a large fries, and a large coke during a meet, get one smaller burger (like a double cheeseburger), a small fries, and a water.

Post-Workout Meal

Once you finish an intense workout, it is very important that you consume some good quality food to help replenish the nutrients that were used up during the workout. If you workout hard, generally the pre-workout meal and the post-workout meal are the two most important meals of the day.

Timing is important for the post-workout meal. Most experts agree that it is important to eat within 2 hours after completing the workout, generally the earlier, the better. Once you have finished working out, your body is depleted of nutrients. It will act like a sponge, and it will do a better job of absorbing the nutrients at this point.

It is important to eat all of the macronutrients in your post-workout meal, but the most important nutrient is protein. During an intense weight training workout, you have damaged your muscles and tissues. Protein is vital to build and repair this tissue so it is ready the next time you need to use it. You should have a good serving of protein after a workout. A minimum of 30 grams of protein is important to help with recovery, and you can go up to 80 or 100 grams. There is a persistent myth floating around that the body can absorb only 50 grams of protein at one time. There is no scientific or practical evidence from this. The body was designed to be very effective at absorbing nutrients once it gets them in the system. The idea that early man would make a kill — which might happen only once in a while — and that he would be able to absorb only 50 grams of protein (less than 8 ounces) makes no sense from an evolutionary point of view. It is true that you want to spread out your protein throughout the day for ideal absorption. It is also true that if you are eating 4–6 meals a day, if you consumed 50 grams of protein at each meal, that would be 200–300 grams of protein, which would meet almost all people’s needs. But that does not mean that once 50 grams of protein are absorbed, there is a shut off mechanism and no more can be taken in.

It is also crucial to eat carbs after your workout. During the workout you have depleted your glycogen storage, and you want to replenish that before your next workout. Post-workout you should eat 50 + grams of carbs, all the way up to 250 grams for the big boys out there. In general we want to focus on consuming more slowly digested carbs, but there is some research that shows that eating faster digesting carbs right after the workout will make for better absorption. This doesn't mean you should wolf down 2 or 3 Twinkies after you train, but it does mean you can be a little looser on your carbohydrate selection. Personally I believe some simple carbs are good after the workout, but the bulk of the carbs should still be mainly the good, unprocessed complex carbs.

> ***Most experts agree that it is important to eat within 2 hours after completing the workout.***

Fat is important to eat post-workout; it will tend to come along with the extra protein and carbs. If you are going to eat extra fat, it is better to do it post-workout than pre-workout. Most people should consume 10–50 grams of fat post-workout.

It can be very tempting to consume a huge meal after you workout as a reward for your hard effort. If you do this rarely, it is probably not a problem, but if you do this regularly, when you eat a large meal it will make you feel full and tired, and then you will probably skip out on your subsequent meals for the day. For example, I generally eat 4 or 5 times a day. Let's say I am going for 5 meals a day. I will usually eat 2 meals, 1 at 8:30 am and the other at 11 am, and then I workout at 2:30. I am finished with my workout around 4 and I try to eat by 4:30. If I eat a giant meal, I am not hungry, and I may stay full until I go to bed, so that is only 3 meals that day. On the ideal day, I would eat at 4:30 pm, then again at 7 pm, and then again something small at

11 pm. If you are trying to gain weight, it is very important not to skip meals; even though you had a large meal with a lot of calories, you still missed two other meals, and ultimately you probably didn't eat enough calories and enough of the right food you are trying to eat.

Personally I tend to view foods as "pre-workout foods" or "post-workout foods." Pre-workout foods are foods that are ideal to eat before a workout. Since I workout in the middle of the day, I consider both of my meals before the workout to be pre-workout meals, but I am the most strict with the meal right before the workout. I would never think about eating cookies or drinking a milkshake or having a bunch of fast food on that meal because I know if I have an intense workout, I will perform poorly and more than likely get sick. Pre-workout foods are also fine, and often ideal, as post-workout foods. However, post-workout I am less strict, and if I was going to have cookies or a milkshake or fast food, that is when I would do it. In my opinion the worse a food is as fuel, the more meals you need to have between it and your workout. Desserts or junk food are not good pre-workout meals; even if I was working out in the evening I would still not want to eat those foods at any time during the day. It would be after I worked out that I would even consider those foods, or I might save them for a non-training day. One benefit of this method is if you are strict on your pre-workout meals, that is just fewer meals that you can cheat on, and it makes the chances of you following a stricter diet a little better.

The more active you are, the more water you need to drink.

Water

Water is an essential nutrient to the body. One could argue it is the most essential nutrient since a person will die the quickest without water as compared to the nutrients (most people can go about a week without water and about a month without food). Water does not have any calories, and it does not provide energy. It is important in regulating the metabolism of the body and for providing form and structure in the body. Water is also used to help digest food, transport nutrients, and to help control the temperature of the body.

It is important to drink liquids throughout the day to stay properly hydrated. Water intake can come from the liquids that you can drink and the foods that you eat; some foods have more water than others. Things like fruit, vegetables, milk, and most carbohydrates carry a lot of water. Most fat and protein sources do not contain much water.

You have probably heard that you should have 8 glasses of water a day. While this is a widely used recommendation and there is probably nothing wrong with it, there is little scientific evidence to support this recommendation. The thirst mechanism can be slightly delayed, meaning you might be a little bit dehydrated once you are thirsty, but in general drinking when you are thirsty is a good and easy way to judge your hydration level.

The more active you are, the more water you need to drink. You should consume some water before you exercise, generally 8–16 oz 1–2 hours before you workout. During the workout you can and should drink water; a guideline is 4–6 oz every 15 minutes. And once you complete your workout, it is also a good idea to consume more water — another 16 oz once you are done. Of course, the longer the workout, the more intense the workout, and the more you personally sweat will have an effect on how much water you need during the workout. In general a level of 2% dehydration can lead to a

decrease in performance. The easy way to test this is to weigh yourself nude before you workout and then weigh yourself nude after you workout (sweaty, wet clothes can alter the reading). If you have lost more than 2% of your bodyweight during the workout, then you are a little dehydrated and you should consume more water during the workout. If you have lost only 0.5 or 1%, you are probably okay. While powerlifting workouts are very intense and can be long, because of the longer rest periods, lifters don't normally sweat profusely during the entire workout, so dehydration is not as big of a deal for us as it is for endurance athletes. On long meet days, however, it is important to stay hydrated.

> ***Weigh yourself before you workout and then weigh yourself after.***

Vitamins/Minerals

Vitamins and minerals are considered micronutrients. They are essential but they are required by the body in much smaller amounts than protein, carbs and fat. Vitamins and minerals do not have any calories and thus they do not provide energy for the body. They are useful for the building and repair of tissue, and for helping to regulate the metabolism.

There are many different types of vitamins and minerals. The vitamins are broken down into fat soluble and water soluble vitamins. The fat soluble vitamins are vitamin A, D, E, and K. Fat soluble means that these vitamins dissolve in and are transported by fat. Vitamins always have multiple functions in the body. This next section is simply meant to provide you with a brief overview of some important aspects of each vitamin and the key minerals.

Vitamin A is otherwise known as Retinol and it is most associated with the proper functioning of the retina in the eye. Deficiency in vitamin A can cause night blindness and other eye problems. Vitamin A is usually found in brightly colored, yellow, orange, and green vegetables.

Vitamin D is not a real vitamin but instead is actually a prohormone because the body can make it under certain situations, namely with exposure to sunlight. Vitamin D is important in absorbing calcium and phosphorous and it helps the bones form. It may also be involved in the immune system. A deficiency of vitamin D is called rickets. Good sources of vitamin D are sunlight, fish, milk, and eggs.

Vitamin E is one of the body's primary antioxidants. Antioxidants are substances that help fight free radicals in the body. Free radicals can destroy cells, so antioxidants help kill the things that might kill your cells. Vitamin E is also good for your skin. Good sources of vitamin E are nuts, vegetable oils, sweet potatoes, mangos, and avocados.

Vitamin K is most associated with the ability of the blood to form clots. It is also associated with bone development. Vitamin K is found in green leafy vegetables.

The water soluble vitamins are the group of B vitamins and vitamin C. There are 9 specific vitamins related to the B group of vitamins; sometimes these are referred to as a number like B6. They are: Thiamin, Niacin, Riboflavin, Pyridoxine, Folate, Cobalamin, Pantothenic acid, Biotin, and Choline. The B vitamins are associated with regulating the metabolism and energy production. It is for this reason that some people think that vitamins provide energy, but in fact vitamins do not. If you are lacking in certain vitamins, you can feel sluggish and lethargic, but vitamins do not give you energy; they allow you to use the energy that you already have. If you want to prove me wrong, you can just take a bunch of vitamins to a desert island with no food and see how long you last. Most B vitamins are found in meats, nuts, and milk.

Vitamin C is probably the most popular vitamin. It has many functions; its main one is helping to maintain the strength of the soft, wet tissues in your body such as your gums and your gastrointestinal tract. It is also an antioxidant along with vitamin E. Vitamin C is found in citrus fruits and vegetables. A deficiency in vitamin C causes scurvy.

Eating good quality meats, milk, eggs and nuts will be helpful.

There are 25 key minerals that the human body needs to function optimally. They are broken down into two categories. A major mineral is one that we need more than 100 milligrams of each day. Calcium and sodium are two of the seven total major minerals. A trace element is a mineral that we need less than 100 milligrams of each day. Iron and selenium are two trace elements (there are 18 of them). Minerals

have functions similar to vitamins; they help build and repair the various tissues in the body, and they help regulate the metabolism. It is not in the scope of this book to discuss in detail each of the various minerals.

In order to ensure that you are eating enough vitamins and minerals, the key is to make sure you are eating a variety of mainly whole food that is relatively unprocessed. Eating good quality meats, milk, eggs and nuts will be helpful. Eating vegetables or fruits at most every meal is a good guideline, and choosing complex unprocessed carbohydrates should cover the bases. Some lifters like to take vitamins to cover anything that their diet might have missed. There is not a ton of evidence as to the effectiveness of multi-vitamins for either performance or health, but I have found personally that they do make a difference primarily in health, with the main result being a decreased frequency and intensity of illness. There is a popular quote which is “The average American thinks 4 colds and 1 episode of flu each year is normal” the point being that you should not be getting sick anywhere near that often, and if you are, you are most likely deficient in certain vitamins. I know this was true for me. I had read most of the studies that show that vitamins don’t do much so I never really took them, but I was getting sick reasonably frequently. My wife actually suggested that I start taking vitamins, and so I did. Truth be told I took them to prove her wrong and myself right, being the egotistical SOB that I am. In this case I was happy to be wrong, as my rate of illness dropped significantly once I started the vitamins (I was keeping a multi-year illness log to keep track so the results were very clear). I decided to follow Bill Starr’s suggestions for vitamin and mineral intake that he outlines in his

Some lifters like to take vitamins to cover anything that their diet might have missed.

classic book, *The Strongest Shall Survive,* which I highly suggest you read. Here are his suggested dosages for the vitamins. They are well above the RDAs for each nutrient, but I don’t think that is negative. I have been consuming this amount for years with no apparent adverse effects and readily apparent positive effects. It is surprising and impressive to me that he made these recommendations 30 years ago and they still stand strong today.

Starr's Recommendation for Vitamins and Minerals

Vitamin A – 25,000 units
Vitamin D – 4000–5000 units
Vitamin C — Natural preferred 100—200 mgs/min; 4000 mg of ascorbic acid
Vitamin E — 1200 units

Thiamin, B1 — 100 mgs
Riboflavin, B2 — 60 mgs
Niacin, B3 — 100 mgs
Pyridoxine, B6 — 200 mgs
Pantothenic Acid, B5 — 100 mgs
Folic Acid – 5 mgs
Biotin – 100 mcg
B12 – 500 mcg
PABA – 100 mcg
Inositol – 500 mcg
Choline – 500 mcg

Calcium – 2 grams total (Calcium gluconate or calcium lactate better)
Phosphorus – 4 grams max (1:1.5—1:2 of calcium)
Magnesium — 1 gram (1:2 to calcium)
Potassium — 5000 mg (about equal with sodium)
Sodium — <10000 mg
Chlorine — 10000 mg
Copper — 5 mg
Zinc — 1 mg
Cobalt — 5 mg
Iron — 15 mg

Iodine — .15 mg
Manganese — 10 mg

One thing I would add to this is, as a powerlifter, you might consider adding in some Omega-3s to your diet, probably 2.5 to 5 grams a day additional would be ideal.

I have found that the multi-vitamin/mineral made by AST: Multipro 32x does a good job of mimicking Starr's recommendations; just consume 2—3 of those vitamins a day and supplement with the few things it is missing.

Summary

My final note on nutrition is that I know some of you may have rolled your eyes a bit when reading about the importance of nutrition. I know there are some powerlifters out there who seem to subsist on nothing but fast food and junk food, and some of them are able to perform very well on the platform. I agree that you can't ignore that evidence, and I would never say that someone can't be successful because of a poor diet. But even if someone is performing well and eating poorly, that does not mean that what they are eating is *optimal*. It might be okay, but it might not be the best way to go about things. I believe these recommendations provided here will maximize one's performance on the platform. I am not saying that every meal you consume must be perfectly strict. One of the joys in competing in powerlifting (as opposed to bodybuilding) is that we don't have to be unbelievably strict with our diet all of the time. A benefit of working out like a monster is that your body can handle less-than-ideal food and having some ice cream or fast food or occasional junk food is not the end of the world; it might even be that periodic ingestion of that type of food is actually good as it shocks the body and provides it with caloric overload.

I love powerlifting and I love the idea of performing on the platform and dominating the competition, but I also love life and health. I cannot condone diets that I think will contribute negatively to one's health in the long run. I know when a person is young, they are feeling indestructible, and they can get away with a lot of stuff; lifting weights may be the most important thing in their lives. It is my hope that lifting weights remains important throughout one's life, but, most likely, as one gets a bit older and starts a family, one's health will start to become

We are the strongest of them all; we should lead by example.

more important. There is a saying in bodybuilding which is “look big in the coffin” which essentially represents the “gain size at all costs” attitude. To be a successful athlete I think you need a bit of that attitude — huge squats are not for the faint of heart — but I don’t think it is worthwhile or a good idea to literally throw one’s health away in pursuit of a bigger total. That is why I can’t condone the use of steroids and I can’t condone the crappy diets that I sometimes see suggested for powerlifters. One’s health is easy to ignore or downplay when you are feeling great, but as soon as it starts to go away, its importance can’t be overstated. I also think of powerlifters as being role models in the fitness world. We are the strongest of them all; it is natural that people will come to us for advice and we should lead by example. I personally would never have been content if I performed well on the platform but I could not do a pull-up, or I could not walk around the mall without getting winded. I am not saying you have to agree with my philosophy, but I will tell you the power of powerlifting is greatly diminished when a normal person sees someone who is admittedly strong but that same person is thought of as being “fat and out of shape.” Personally I always wanted to be super strong, ripped, and in great shape. I am not saying I always achieved that ideal, but it was my goal, and nutrition is a big part of that.

Chapter 13

Changing Your Bodyweight

Powerlifting is broken into different weight classes, ensuring that lifters compete against other lifters of similar body size. It is not really fair if someone who is 198 lbs has to compete against someone who is 275 lbs, as the heavier person is almost always stronger (although don't tell that to Ed Coan). If one wishes to be as competitive as possible, then your goal will be to choose the weight class in which you fare the best against your competitors. Throughout your lifting career, you may decide it is time to increase or decrease your weight. You might do this to be more competitive; to try to lift more weight; or just because you decide you want to look a certain way. The goal of this chapter is to help explain how you should go about changing your diet and/or exercise routine if you want to change your weight.

The number one thing that affects how much you weigh is the number of calories that you are eating each day. A calorie is a unit of energy (heat). It is the way we measure how much energy a food contains. The technical definition of a kilocalorie is “the amount of heat necessary to raise 1 Liter of water 1 degree C.” When you are trying to change your weight, you have to affect your energy balance. Energy balance is the number of calories you have coming into your body from food and liquids compared to the number of calories that you burn each day. Altering how much you eat is the way to change energy intake; altering the amount of exercise you perform is the main way to change energy output.

(Left) A meal with significant calories

(Right) A meal with much lower calories

How many calories you do burn each day?

If we want to change our weight, we have to know how many calories we burn on an average day. There are many different formulas available for us to figure this out. None of them are perfect, but the goal is to get a close estimate of how many calories you are using up. Then we can make the correct nutritional choices to either increase or decrease your weight. The formula that follows is one that I have used extensively in the past on myself and others, and in my opinion, it is the most accurate one I have seen. The formula involves a few steps, but I will try to make it as simple as possible.

Metabolism Formula

To figure out your metabolism, you will need to know the following information:

- Gender
- Weight
- Activity Level
- Calories Eaten Per Day
- Age (optional)

The first thing we are going to do is estimate your BMR, which stands for your basal metabolic rate. This is an estimation of how many calories your body burns while it is at rest. It may surprise you to learn that the body burns the majority of its caloric expenditure just to keep the system running. Your brain, heart, lungs, and other organs all require large numbers of calories to function properly. The main factors that affect your metabolism are your weight, your lean body mass, your age, your gender, your hormonal level, and your current state of health.

Step 1: Find the **BMR** (how many calories are burned at rest)
Find weight in kilograms = weight in pounds divided by 2.2

Ex) 198 lbs/2.2 = 90 kg

Males use this formula: Weight in kg x 1.0 x 24 = BMR

Females use this formula: Weight in kg x .9 x 24 = BMR

Ex) 90 kg x 1.0 (male) x 24 = 2160 cals/day BMR

Once you have found your BMR, you now need to estimate how many

calories you burn through additional activity. Regular daily activity like walking, driving a car, doing the dishes, etc., all burn some extra calories during the day. Of course, intense weight training and cardiovascular workouts also burn a lot of calories. I will assume the majority of you reading this book are interested in powerlifting and workout pretty hard, so you will not choose a sedentary lifestyle. Most powerlifters are either in the light or moderate category. Do not choose the activity level based just on how hard your workouts are; remember there are 168 hours in a week, and even if you workout really hard for 6 hours a week, that is still 162 hours to sit around and not do much, so think about how hard your workouts are and what you like to do during your other time. If you just can't decide which activity category you are in, you can go halfway in between each category. For example, if you are not sure if you are light or moderate, you can go light-moderate which would be 35%.

Step 2: Find calories burned through **Activity Level**
Estimate Activity Level

- Sedentary 20% — Rarely works out, job involves little movement; activity is the equal to walking about 1 mile a day
- Light 30% — Sedentary job but works out a few times a week OR more active job but does not workout, activity is equal to walking about 3 miles each day
- Moderate 40% — Somewhat active job and works out regularly; activity is equal to walking about 6 miles each day
- Heavy 50% — Rare; very active job, constant physical labor OR long distance aerobic athlete; activity is equal to walking about 12 miles each day

BMR (from Step 1) x Activity Percentage (20–50%) = Activity Calories

Ex) 2160 x 40% (Moderate) = 864 activity calories

Step 3: Find the Thermic Effect of Food (**TEF**)
This is the energy used up simply to digest food

Calories Eaten Per Day x 10% = TEF

Ex) 3250 x 10% = 325 TEF calories

Step 4: Add them up to find **Total Metabolism**

BMR + Activity Calories + TEF = Total Metabolism

Ex) 2160 + 864 + 325 = 3349 Total Metabolism

Total Metabolism is how many calories you burn each day at your activity level

Step 5: Adjust for age (optional)

30–49 years old: –3% 50–70 years old: –7.5% 70+: –10%

Ex) 3349 – 3% (35 years old) = 3248.5 calories per day

Your total metabolism tells you about how many calories you burn each day at your current level of activity. It also tells you how many calories you would need to eat each day to maintain your current weight at your current activity level. Often people want to change their weight or body composition, so knowing a person's total metabolism is the key starting point to then make adjustments to help someone gain or lose weight.

Gain Weight

If you want to gain weight, it is assumed that you are talking about primarily gaining muscle (lean mass). Generally, as your weight goes up, so do your lifts. It may not be a proportional increase in strength, e.g., if you can bench 400 at 198 and then you can bench 420 at 220, the 400 at 198 is a better bench from a competitive standpoint. But 420 is still higher than 400, and now the lifter is lifting more weight at the new body weight. Occasionally the weight gain will increase strength out of proportion to one's current relative strength ability. At one point I added 20 lbs of bodyweight, and my bench increased 50 lbs, and I was not benching 2.5 x bodyweight when this happened. Bodyweight usually has the greatest effect on the bench press, a pretty big effect on the squat, and not as much effect on the deadlift. It also has a very minimal effect on the curl.

If you want to add muscle to your frame, two things must happen. First, you must create the need for more muscle. You do this by working out, primarily through resistance training. This will (hopefully) release hormones like testosterone which will tell the body to add lean muscle mass. The second thing that must happen is you need a caloric surplus. You need to eat extra calories in order for you to have enough energy and building blocks to create that new muscle. Both of these things need to happen to gain quality weight. If either piece of the puzzle is missing, then your results will be seriously compromised.

It takes about 2200 calories to build a pound of muscle.

Since you will be working out hard lifting weights, I will assume there is an adequate stimulus to build muscle. You can't just sit on the couch and watch Strongman competitions on ESPN, eat a lot of food, and then hope to be big and muscular. But the focus of this

section is on nutrition. If you want to gain weight, you must eat more calories than you burn.

It takes about 2200 calories to build a pound of muscle. This number is not as universally agreed upon as the amount of energy in a pound of fat. The fuel factor of protein is 4, not 9, thus the numbers of calories in a pound of lean tissue will be significantly less than fat. However, the energy required to build muscle is significantly higher than to store fat. Building muscle takes a lot of energy; storing fat is something the body was designed to do as efficiently as possible. The number "sounds" about right to me, and, from practical experience working with myself and hundreds of students and clients, the results it yields means it is pretty accurate.

A realistic amount of muscle that most people can gain is ½ to 1 pound of muscle a week. This is not sustainable forever, but for a common weight gain period (2–4 months) it works quite well. Men will generally gain muscle faster than women due to their greater levels of testosterone. The longer you have been training and the further away your weight is from your natural weight, the harder it will be for you to gain and then maintain that weight. For example, my natural weight when I am working out is right at 198 lbs. My goal for this year is to move up to the 220 lb weight class and see how being at that bodyweight affects my strength levels. To keep it simple, if we use those exact weights, that is a 22 pound weight increase, and something like that would probably take 6 months or so to achieve. I lose weight easily so it will take some work to stay at 220 lbs; that would also be the heaviest I have ever been. If I wanted to continue to gain weight and get to 242 lbs, even though that is still a 22 pound weight increase that will take much longer and require even more effort since that weight is so far away from my current natural weight. In addition I have been training for a long period of time so my gains, in both strength and size, will come slower than if this was

my first year or two of training.

Your body requires an extra 2200 hundred calories a week to build one pound of muscle. Since you are not going to try to build that muscle in one day but instead you are hoping for a relatively even period of muscle growth, if you divide 2200 calories by 7 days in a week, you get 315 calories. You need to consume an extra 315 calories each day, every day, to gain 1 lb of muscle a week. This is, of course, combined with intense weight training to help ensure that most of the weight gained is muscle. There is no guarantee, nor should you expect that all of the weight gained will be muscle. As you gain weight, you will gain some fat; it should not be a ridiculous amount, but you will gain some fat. A ratio of 70% muscle to 30% fat or better is generally desirable. The only way to really know exactly how much muscle and fat you are gaining is to take your bodyfat before you gain the weight (assuming it is accurate) and then take it again after you gain weight and then see what the difference is.

A common misconception when you are trying to gain weight is that the extra calories you eat must all come from protein. This is not the case. Remember that carbs have a protein-sparing effect, which means that as you eat carbs you are saving the protein so it all can be used to build and repair tissue. Generally the more calories you are eating (to gain weight), the higher the percentage of carbs. 50% is the minimum, up to 65% carbs. The more calories you are eating, the lower the percentage of protein, somewhere between 20–30% protein, and your fat intake will stay relatively constant at 25–35% of the diet. While it is absolutely true that you do need some good quality protein to build muscle, it does not need to be in excessive amounts. Excessive amounts to me would be anything greater than 1.5 grams per pound of bodyweight.

To gain weight use the following chart as a reference. Every 10 pounds

your weight changes, you should recalculate your total metabolism and then adjust it properly. Once you have reached your goal weight, you should also recalculate your metabolism with the new goal of maintaining weight as opposed to gaining weight. If you wish to skip the formula and just want a super simple guideline to figure out caloric intake when gaining weight, take your current bodyweight and multiply it by 20, and that is how many calories to eat to gain weight (a 200 lb male would need to eat 4000 cals/ day to gain weight).

You need to consume an extra 315 calories each day, every day, to gain 1 lb of muscle a week.

Weight Gain

Realistic Weight Gain is .5 to 1 pound of muscle per week

It takes approximately 2200 calories to build a pound of muscle

Desired Muscle Gain	Caloric Adjustment
+ .5 lbs of muscle per week	+ 157 calories per day
+ 1 lb of muscle per week	+ 315 calories per day

Ex) Goal is to gain 1 pound of muscle per week
3248.5 calories per day + 315 calories = 3563.5 calories per day

Some keys to follow when trying to gain weight:

- Don’t skip meals
- Don’t eat gigantic meals (this fills you up and tends to make you skip

Every 10 pounds your weight changes, you should recalculate your total metabolism.

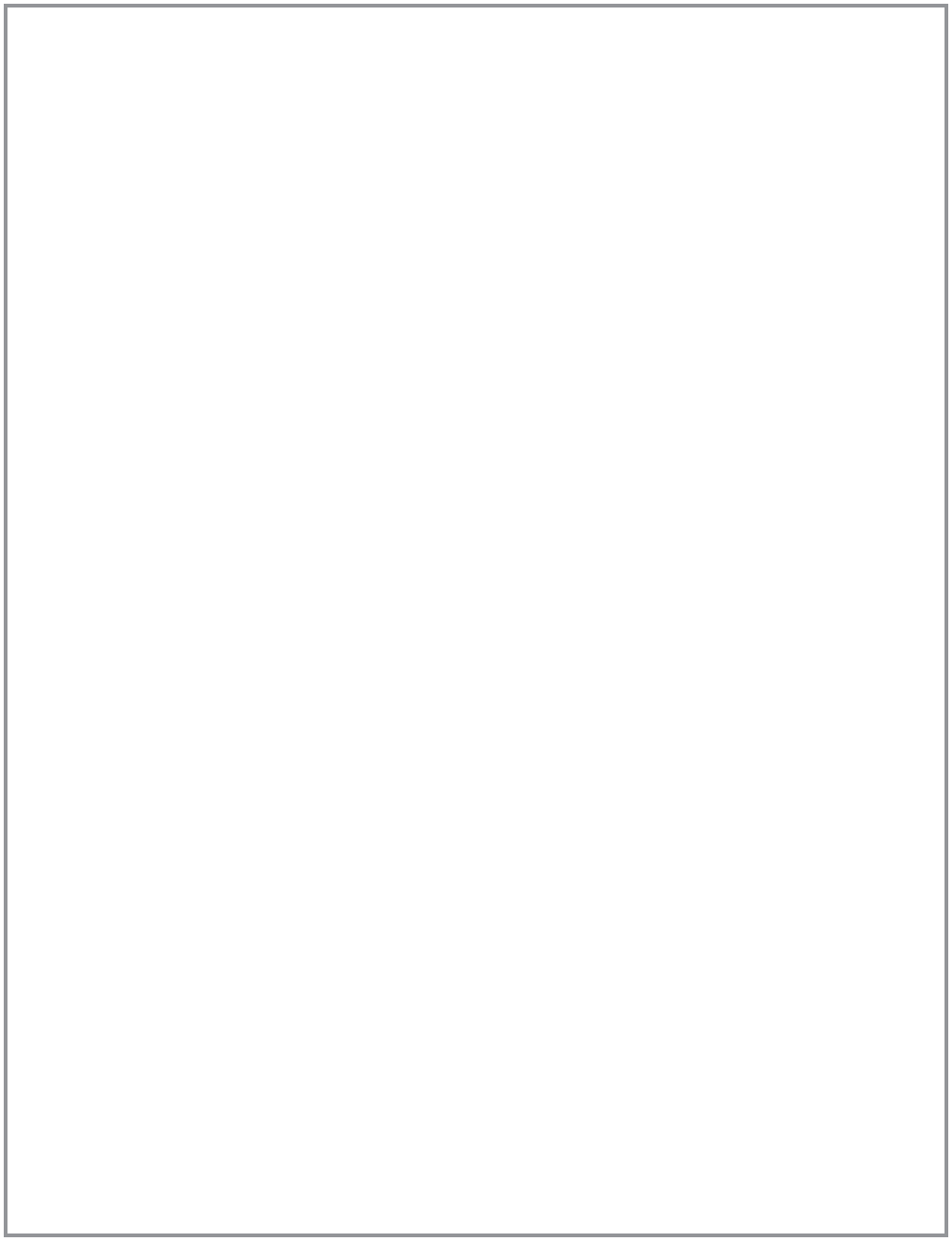
the next meal or two thus resulting in a smaller caloric intake and poor nutrient timing)

- Plan your meals and/or food intake ahead of time
- Have access to good quality snacks/small meals (e.g., nuts, milk, fruit)
- Weigh yourself every day
- Don’t worry about how your abs look (for the moment)
- Eat a good breakfast
- Try to consume a lot of lean meat, whole milk, whole eggs, nuts, and potatoes

Lose Weight

If your goal is to lose weight, then you must create a caloric deficit. That means that the total number of calories you burn each day (through your BMR, activity level, and TEF) must be above the total number of calories that you consume each day. This will cause your body to use its sources of stored energy to make up the deficit, and hopefully the majority of that stored energy used will be adipose tissue (fat).

One pound of bodyfat contains about 3500 calories. Fat has more calories per pound than muscle because of the relatively low amount of water in fat and also because the fuel factor for fat is 9 kcals/gram compared to protein's 4 kcals/gram. Since you don't try to lose 1 pound of fat all in one day, the general guideline is to make a reduction in your daily caloric intake to lose the desired amount of weight. A realistic weight loss goal is to lose 1–2 pounds of fat a week or 1% of your bodyweight per week. If you are pretty light, say, 125 lbs, it will probably be too aggressive to try to lose 2 pounds of fat in a week and maintain that level for any length of time. At the same time, if you are quite heavy, say 350 lbs, you could try to lose 3.5 pounds of fat each week or, honestly, even a bit more. The more fat you have, the quicker you can lose it, but remember the quicker the weight comes off, it tends to go back on at a rapid rate as well.



At some point most lifters experiment with trying to decrease their bodyfat significantly

Just to be clear, I am not saying that a person cannot lose more than 1–2 pounds of *bodyweight* each week. Your body contains a tremendous amount of water, and you can manipulate that level if you wish to lose a lot of weight

in a short period of time (see the section on water manipulation for more info). What I am saying is that if you want to lose *bodyfat*, then you can lose only so much fat at once, and a good goal for that is 1–2 pounds a week.

Weight Loss

Realistic Weight Loss after the first week:1–2 lbs of fat per week OR
1% of bodyweight per week

1 lb of fat contains approximately 3500 calories

Desired Fat Loss	Caloric Adjustment
—.5 lbs of fat per week	–250 calories per day
–1 lb of fat per week	–500 calories per day
—1.5 lbs of fat per week	–750 calories per day
–2 lbs of fat per week	–1000 calories per day

Note: Do not follow a diet below 1200 calories per day

Ex) Goal is to lose to 1.5 lbs of fat per week
3248.5 calories – 750 calories = 2498.5 calories per day

This person should eat about 2500 calories per day to lose 1.5 lbs of fat per week

It is possible to try to lose ¼ or ⅛ of a pound of fat per week, and over a very long time period, that could add up, but with such low levels it becomes almost impossible to measure. If you lose ¼ of a pound of fat per week, that equals just one pound a month of fat loss. Given that most people normally fluctuate 2–3 pounds on either side of their weight, it would be 4 months before you would really start to notice a measurable change in your weight.

A meal with lower carbs to promote fat loss

It is noted in the chart above that you should not go below 1200 calories a day. Anything below that level is a considered a Very Low Calorie Diet (VLCD), and you should seek nutritional counseling if you are eating that small of an amount of food. It becomes very difficult to get in all of your essential nutrients with such a low number of calories. In addition, if you are working out hard (which I am assuming you are), then it would be very rare to eat less than 1200 calories. Most women will lose a reasonable amount of weight on a 1250–1500 calorie diet; lots of females engaged in powerlifting might even need a few more calories. Most men will lose a good amount of weight on a 1500–2000 calorie diet. Using the formula above is the best way to figure out how many calories you should be eating each day (to lose or gain weight), but a super simple weight loss guideline is to put a zero at the end of your bodyweight (multiply weight x 10). Consuming that number of

calories should produce weight loss, especially if you are active. If I weigh 200 lbs, eating 2000 calories should produce some weight loss.

> ***A super simple weight loss guideline is to put a zero at the end of your bodyweight.***

When you are losing weight, you will be on a diet, and there are three big negatives to that from a powerlifting perspective. The first is that you might lose some muscle. If a person just diets and does not exercise at the same time, that person can expect to lose a fair amount of muscle as they lose weight. Sometimes as much as 50% of their weight loss will be muscle. The best method to prevent this muscle loss is to workout hard and provide the stimulus to build and/or retain the muscle. This will tell your body that it needs to keep the muscle as you are losing weight. This method is not foolproof, and the leaner you are trying to get, the more muscle you will lose on the way. In general the body will tend to shed what it has more of. For the majority of people, they will have more fat, thus, that is what tends to get used up. However, if you are muscular and lean, you will have more muscle than fat, and then the body begins to burn muscle more quickly.

The other thing you can do to prevent this is to eat an adequate amount of protein. When you are on a diet, the percentage of protein that you are eating should go up, not down. It isn't that you are necessarily eating more grams of protein each day, but since you are eating a fewer number of total calories and you need about the same amount of total protein, the percentage will go up, usually by 5–10%. When you are on a diet, you generally want to eat 25–35% protein.

The second negative of being on a diet when you are powerlifting is that you will be a little low on energy, and that can make long, grueling weight lifting workouts more challenging. It will be very important to eat mainly

complex carbs particularly before and after your workout so you will have enough energy (glycogen) to train hard. You can't waste any of the carbs you are eating when you are on a diet by eating crappy food. Generally you should eat 40–60% carbs while on a diet.

The third negative is that, often, as you lose weight, your strength goes down. This isn't just because you might be losing muscle, but sometimes as you lose weight your leverages can decrease and the stability of the joints can decrease. Generally the bench press is the most affected by loss of bodyweight, then the squat, then the deadlift; the bicep curl is the least affected by bodyweight. Of course, your goal as a powerlifter is to find the weight class in which you are the most competitive. If your bench goes down only 10 pounds when dropping from the 220 weight class to the 198 pound weight class, that would normally be considered a good trade off. Carrying around a lot of extra fat is that not that beneficial to powerlifting, and, of course, it is bad for your health. Most male powerlifters seem to do well with bodyfat levels between 5 and 15%; females are usually 15–25% bodyfat.

Listed below are some helpful keys to follow when on a weight loss program

- Plan what you are going to eat a day in advance (I like to decide in the evening what I will eat the next day for each meal)
- Prepare your meals ahead of time (example — cook everything on Sunday)
- Weigh your food
- Serve the food on smaller dinner plates
- Wait at least 15 minutes after eating to get seconds
- Avoid/eliminate foods you have problem eating in moderation

As you lose weight your leverages can decrease and the stability of the joints can decrease.

- Avoid/eliminate alcohol from the diet, particularly beer and hard liquor
- Avoid eating at restaurants
- Avoid/eliminate trans fats
- If you are eating out, decide what you are going to order before you enter the restaurant and write it down or tell someone
- If served a larger meal, immediately divide it in half and eat only half of it at that time; put the other half in a box or move it off your plate and/or out of your sight
- Try to have a serving of vegetables and/or fiber with every meal
- Avoid fast food
- Avoid heavily processed food
- Lose weight with a group of friends/family at the same time
- Eat a piece of fruit like an apple or banana before you eat your main meal
- Eat on a regular schedule; don't allow yourself to get extremely hungry
- Eat relatively small amounts; you should never get full or stuffed when trying to lose weight
- Weigh yourself every day to chart progress

If you are trying to lose or gain weight, I would suggest you plot out your proposed weight loss or gain on a calendar. First, pick a realistic goal for weight loss or weight gain that you think you can achieve. Then pick a target day of the week where you will have access to a scale, and hopefully your eating habits will be the best for a few days prior to that day. For most people the target day is either Monday (for those people who eat the best over the weekend) or Friday (for those people who eat the best throughout the week).

For example, imagine we have a male powerlifter who is 195 lbs currently, and he wants to drop some fat and compete in the 181 lb weight

class. He needs to lose 14 pounds; hopefully those pounds will be primarily fat. If he decides to lose 1.5 lbs a week, it should take about 10 weeks to achieve his goal. Let's say he decides to make his target weigh-in date on Friday. Remember, he will still be weighing in every day or at least most days a week, but if he hasn't hit his target weight, he has to hit it by Friday. Listed below is what might happen to his weight over the 10 week period.

Week	Monday	Tuesday	Wednesday	Thursday	Friday
1	195	**193.5**	196	194	194
2	195	194	194	**192**	193
3	193	193	194	193	**190**
4	192	190	**189**	188	190
5	189	188	189	191	189
6	187	**186**	188	187	186
7	187	185	186	**184.5**	185
8	185	185	**183**	184	184
9	185	184	182	182	**181.5**
10	183	182.5	182	183	**180**

Note: Bolded numbers indicate the day the target weight for that week was reached.

As you can see from the above chart, on the first week, he hit his goal on the second day of the diet. That is a good way to start off. Once you hit your target weight during the week at any time, you have achieved your goal for that week. Then you must focus on the next week. Of course, just because you hit your goal doesn't mean you can go crazy and pig out; otherwise, that will make reaching the goal weight for the following week even harder.

This person made their goal every week except for week 5. That does happen occasionally, and, if it does, then you must redouble your efforts over the next few days and be extra strict to make sure you lose the weight. Note

that even though the goal weight was achieved on most weeks, there was still a natural fluctuation and sometimes the person was a couple of pounds heavier than the target weight for that week. That is totally normal, and you want to be mentally prepared for the fact that your weight will fluctuate, and sometimes it goes up a little high. This is one of many reasons it is a good idea to weigh yourself every day. You will learn what kind of fluctuations your own body makes, and you will know if you are on the high end of the fluctuation or the low end. If you weigh-in only once a week, you will not know where you are in that natural weight cycle.

When we are trying to lose weight, we are generally trying to decrease bodyfat. I realize that we haven't talked as much about body composition, for a couple of reasons. One, it is much harder to measure whereas using a scale is very easy (make sure you always use the same scale for your weight). The second is that for powerlifters you have to make a certain weight class and you must weigh a certain amount on that specific date, so it is valid to talk about weight. And finally if you are following a decent diet like the one outlined in this book, and you are engaged in relatively intense weight training (as you should be), then you can be pretty assured that the majority of weight you will lose will be fat. Of course, taking your body composition at the beginning of the plan and at the end will help tell you for sure what happened. In addition, paying attention to your lifts will give you an idea of what kind of weight you are losing. It is natural for your lifts to go down slightly as you lose weight, especially if you are already somewhat lean. However your relative strength (pound for pound) should increase. In the above example, if our 195 lb powerlifter could bench 350 at 195, that is being able to lift 1.79 lbs per bodyweight. If he drops to 181 and can now bench 330, he can lift 1.82 lbs per bodyweight, so his relative strength went up. Generally if your relative strength is going up, you will be more competitive

in your certain weight class. If our powerlifter's bench had dropped to 300 at 181, that is only 1.65 lbs per bodyweight, and he would probably be more competitive at 198 lifting 350 than 181 lifting 300. You can use the Lifter Classification Charts found in the appendix to see what you would need to lift at various weight classes to maintain or increase your strength-to-weight ratio.

When you are trying to gain weight, the same approach can be followed. If you are eating a reasonably clean diet and working out hard, the majority of the weight gained (assuming it is at the recommended pace) should be muscle. When you gain weight, your lifts should go up; if you are able to maintain your relative strength (or perhaps even improve it) as you get heavier, that is great progress. If our 195 lb powerlifter decided to gain 20 lbs and compete weighing in at 215, if he could still lift 1.79 times bodyweight, he would bench 385 lbs, and that bench of 385 at 215 is more impressive than 350 at 195. The heavier you are, the harder it is to maintain your relative strength.

It is a good idea to weigh yourself every day.

A good measure when gaining weight is to watch your waist size. If it increases significantly, you may be gaining too much fat. It will increase a little, and your abs may not look quite as good while you are gaining weight. Use your pant size (or dress size) as an indicator. If you can no longer fit into your pants because your waist is too big, then you should probably monitor your bodyfat along with your weight to see what is happening as you gain the weight. You don't want to gain 20 lbs and have 15 of those pounds come from fat.

A good measure when

gaining weight is to watch your waist size.

Water Manipulation

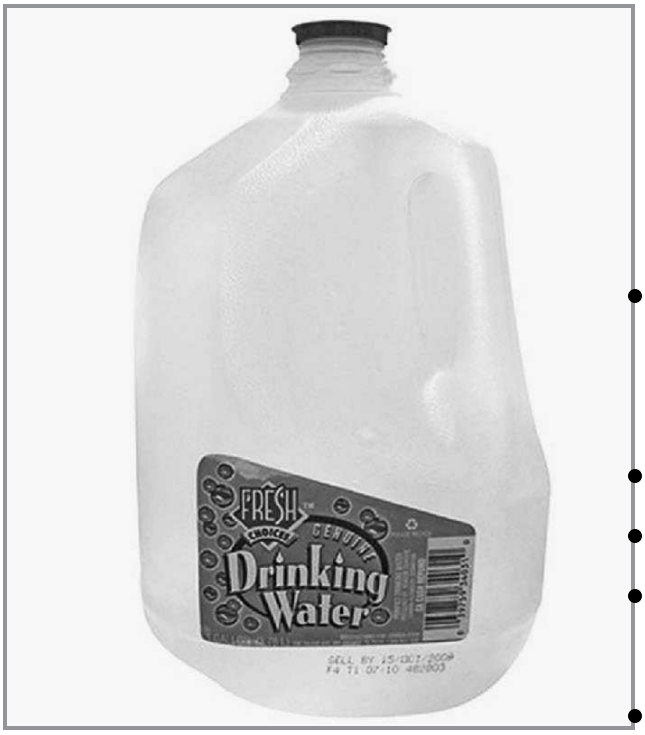
It is possible to use water manipulation to modify your weight to help you weigh a certain amount at an exact period of time. Different powerlifting federations have different rules about the time there can be between weigh-ins and when the lifting starts. The strictest federations require no more than 2 hours between weigh-ins and the start of the competition; other federations allow 18 or even 24 hours between weigh-ins and the actual competition. Common sense tells us it is easier to try to lose a significant amount of water weight if you have more time to rehydrate. For most lifters I do not recommend any sort of serious weight cut if they have a 2 hour rule as that very often makes them feel weak and tired, and if you do it wrong, it can mess up 3 months of serious training in 1 week. For those lifters I instead recommend that they simply follow a more traditional meal plan; if you want to lose 20 lbs and you have a 2 hour weigh-in, lose about 1.5 lbs a week for 10–12 weeks, and then you can lose the last bit by water manipulation, but trying to lose 20 lbs in that last week is too risky in my opinion.

If you have 18 or more hours to rehydrate, then you can use slightly more drastic measures to lose the water. I want to be clear from the outset that I am not advocating these methods as being healthy or "good for you"; they are simply a means to an end. I don't think there is enough information on either side of the equation to declare them as either good or bad, although clearly one can lose water the wrong way and end up with harmful and/or negative results.

Generally if the goal is to manipulate one's water level, it is a two stage process. The first stage is approximately a week long. The second stage is done right before the actual weigh-in.

First Stage

The goal here is to overhydrate. Consuming excess water is a strong diuretic,



Drink water to lose water

and you will lose water weight (temporarily) by drinking excess water. Assuming weigh-ins are on Friday for a Saturday competition, this plan is 5 days long.

- Sunday: Drink 1 gallon of water in the second half of the day (after 2 pm)
- Monday: Drink 2 gallons of water
- Tuesday: Drink 2 gallons of water
- Wednesday: Drink 2 gallons of water
- Thursday: Drink 2 gallons of water. Before bed consume 1 whole lemon
- Friday: Do not eat or drink anything until weigh-ins

Consuming excess water is a strong diuretic.

A note about the water consumption. Drinking 2 gallons a day is not impossible, but it is not convenient, either. It is very important that this amount be spread out evenly throughout the day. Here is how I do it. I calculate the hours I will be awake for the day; let's say it will be from 8 am to midnight, which is 16 hours. There are 128 ounces in one gallon; with 2 gallons we will be drinking 256 ounces. Consume only water during this phase: no soda, fruit juice, coffee, etc. Divide 256 ounces into 16 hours, and this comes out to 16 ounces per hour. If you drink 16 ounces (2 cups) per hour, every hour, for 16 hours straight, you will consume 256 ounces. It is imperative that the water consumed is spread

throughout the day. These are some techniques I use to ensure that this happens.

- Buy at least 1 gallon of water from the store.
- Begin drinking it when you wake up. That gallon must be finished by the halfway point in your day (at 4 pm in our example). Then start another gallon (or refill the first one) and continue.
- Because it is easy to get behind, I usually try to drink 3 8 oz cups each hour to give myself a bit of cushion.
- Bring your gallon of water with you wherever you are going. Look at the clock, 3 cups in 1 hour means every 20 minutes you drink 1 8 oz cup.

I find simply drinking room temperature water to be the easiest thing to consume large amounts of.

DO NOT TRY TO PLAY CATCH UP AND DRINK HUGE AMOUNTS OF WATER IN A SHORT PERIOD OF TIME — this can be very harmful and even fatal.

This amount of water should have you peeing like crazy — that means the procedure is working.

Weigh yourself in the morning, midway through your day, and before bed to get an idea of how you are responding the water intake. Personally I lose about 2 lbs a day (or 1% of my bodyweight) when I do this (the most noticeable loss is comparing the evening weight to the morning weight the next day).

Diet — During this first stage, you will want to modify your diet. The key element is to modify/reduce carbohydrate intake. Carbohydrates, when stored as

Sweating out the water through physical activity is the WRONG approach.

glycogen, hold water, by most accounts at a 1-to-4 ratio. Most people have about 1 pound of glycogen in their body (powerlifters likely have a bit more) which means that is also holding 4 pounds of water attached to it. By modifying carbs we can drop off 5 lbs of bodyweight with relative ease.

The plan is relatively simple. Consume normal amounts of protein and fat but on Sunday to Thursday consume only a maximum of 50 grams of carbohydrates total **per day**, mainly in the form of veggies and natural, unprocessed carbs like sweet potatoes. By the end of the week, one should be glycogen depleted from this process.

Stage 2

If you still need to lose some weight, the morning of the weigh-ins (I would suggest 10 lbs or less), you can further manipulate your water by forcing yourself to sweat it out. Let me be very clear on this, however. Sweating out the water through physical activity (e.g., putting on lots of clothes and performing cardio excessively) is the WRONG approach. You may make weight, but it will almost assuredly leave you tired, weak and burned out — remember, you are getting ready for a powerlifting competition here; you want to feel big, strong, powerful, and fresh. Instead you have to make your body sweat passively, simply as a response to heat. This is pretty draining in and of itself, but you can recover rather quickly from this (again, I do NOT recommend this method if you have only 2 hours to weigh-in).

There are 2 good methods to force yourself to sweat. Option 1 is to use a steam room or a sauna (I find the sauna to be slightly more effective but use whichever one makes you sweat more). Spend about 15 minutes in the sauna, come out for 5 minutes, and repeat. Weigh yourself regularly to ensure that you don’t lose too much weight, but be aware that because you are already dehydrated (from following the stage 1 approach), you will not sweat as

much as you might think. If you are getting very hot, you can put a cool, damp towel on your head or neck, but don't drink anything, and some people report showering at this stage can make them gain weight through water absorption via the skin. It is nice to have a partner with you during this period to keep an eye on you — it should go without saying that if you passed out in an empty sauna in this condition, bad things could happen.

The second option, if you don't have a sauna, is to use an immersion bath. Fill up a big bathtub with very hot water, essentially as hot as you can stand it but not so hot that it literally burns you. Most regular bath water will not get hot enough; it is likely that you will need to boil water and then pour that into the tub along with the regular water to increase the temperature. Immerse yourself in the tub for 10 minutes; everything should be submerged except for your face. Get out for 5 minutes to cool off, and repeat the procedure. This should make you sweat profusely. Weigh-in at regular intervals until the desired weight is lost.

Of course, try to go to the bathroom (both #1 and #2) before the official weigh-in. I have read about lifters and other athletes using enemas to help with weight loss, but I cannot provide any firsthand knowledge of that. I would not suggest or encourage taking any diuretics or laxatives or strong stimulants to promote weight loss — these may be illegal in your federation, and they may have negative side effects. Keep in mind that what bodybuilders do to lose weight and look shredded on the day of the competition isn't necessarily ideal for what an athlete who has to perform at their best physically should do.

Rehydrating

Here comes the fun part — getting that fluid and energy back into you. This part is crucial when it comes to performance.

You have to make your body sweat passively.

Your goal is to gain every single pound that you lost, starting that week prior, before you step on the platform. If you can, gaining even an extra pound or two will make you feel even better. If you are unable to gain all of the weight back, even if you are just 2 lbs away, expect your performance to drop.

In the rehydration phase, you are focused on 3 big things — water itself, electrolytes, and energy. Here is how I like to do it.

My favorite rehydration drink is a combination of 10 ounces of Gatorade, 2 servings of Surge Workout Fuel, and 10 ounces of water. The Surge Workout Fuel does a great job of providing fast absorbing energy (carbs) and electrolytes, the Gatorade has some of that, too, and the Gatorade makes it taste good (I don't use the artificial sweeteners that come with the Surge as I am allergic to them).

Your goal is to gain every single pound that you lost.

I alternate the above drink with standard water (about 20 ounces). I try to drink 3 of the Surge drinks from the time I weigh-in until I go to bed (I usually pound one immediately after weigh-ins, but if I drink too much of that I will get stomach cramps) and then I pound 1 bottle of plain water. I also try to get in at least 2 meals — 3 is better — after weigh-ins but before bed — it is better to eat 2 or 3 smaller meals than 1 giant meal that leaves you so full that you can't eat again until breakfast. Eat something salty to help with the water retention, I find French Onion Soup to be great. Keep in mind your stomach may be sensitive in this state and after following the diet for the past week.

The morning of the competition, eat your usual breakfast. I like to sip the Surge drink throughout the competition, again alternating that with water. I would likely consume 2 of those drinks during a full meet along with at least 2 bottles of water and at least one medium sized meal (after squats).

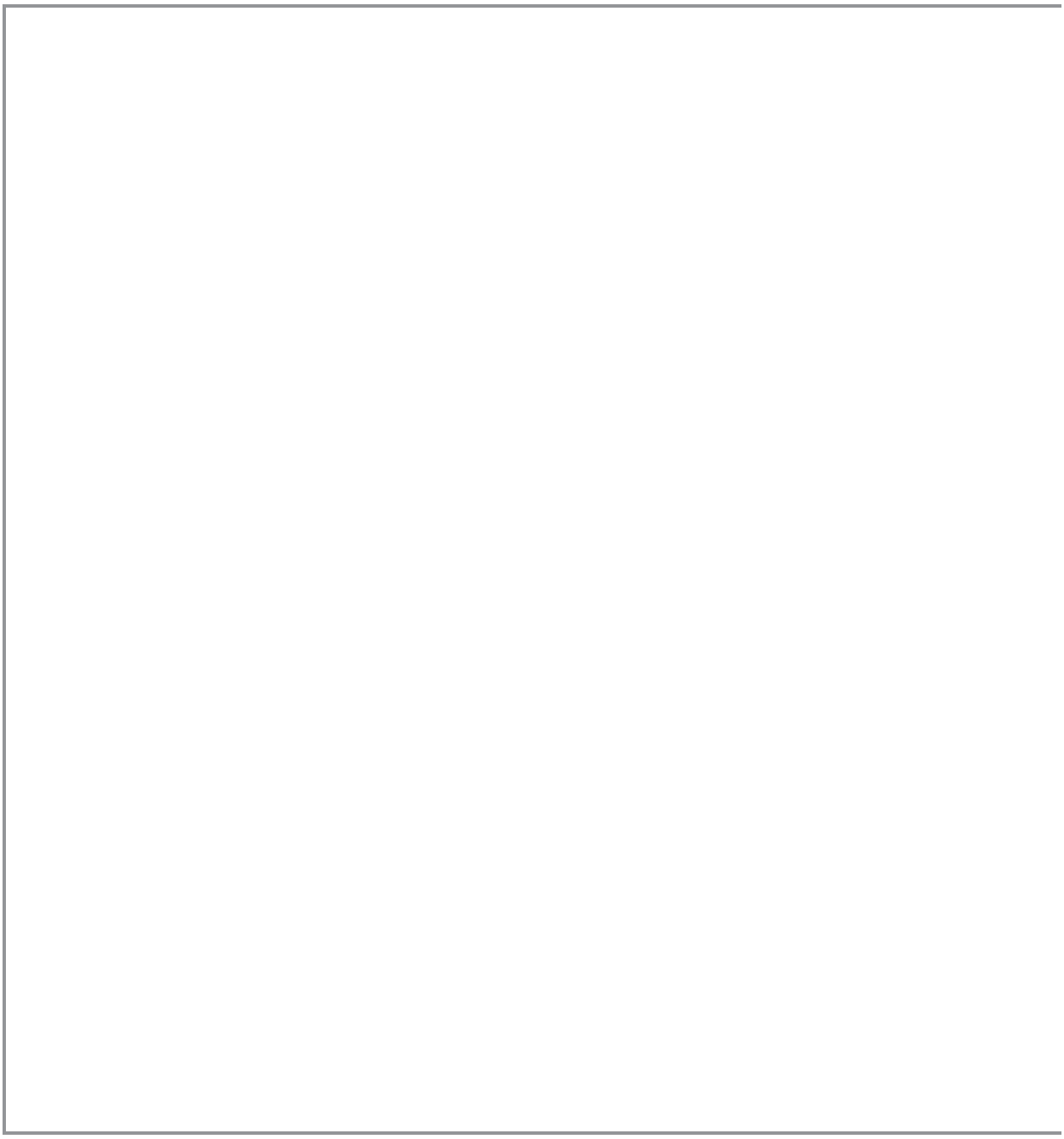
I would recommend water manipulation ONLY for those lifters who meet *all* of the following criteria:

- Seasoned lifter with at least 2 competitions under their belt at their natural weight
- A lifter who has realistic competitive aspirations, meaning a good shot at placing in the competition or a chance at setting a record (e.g., I see no reason for a Class IV lifter to do this)
- Lifters who are otherwise healthy with no known health problems (heart problems, diabetes, pregnant, etc)
- Adult Lifters only — I would not recommend this method for kids, teens, or older adults
- I would water manipulate a maximum of only 2 times per year
- The methods listed here are moderately extreme as is; combining these guidelines with other methods/modalities/tools at the same time may lead to negative health complications

Powerlifting is an intense sport. Lifters/readers use the methods presented here AT THEIR OWN RISK!

Summary

You will have to decide what bodyweight you wish to compete at. This is likely going to be affected by how much you want to weigh, the weight class you are most competitive in, and how easy it is for you to gain/lose weight (and how much time and effort you want to put into it). Over the course of your career, it is likely you may compete in multiple weight classes. At the highest level of competition it is important to be aware of the methods and modalities available to manipulate your weight. It is also important to note that improper use of these methods — or simply an unusual individual response to them — can significantly decrease performance. You will have to weigh the potential risk versus the reward and decide for yourself what course of action is best. Let your performance and your health serve as your guides through this process.



Earning a trophy is nice, but setting PR's is what powerlifting is all about.

Chapter 14

Powerlifting Competitions

The Day of the Meet

The purpose of this chapter is to help you, the lifter, be as prepared as possible on the day of the meet. Being prepared can help alleviate some (not all) of the nerves that are common pre-meet, and the more experience you acquire, the more routine the meets will become. It is very easy as a novice to forget to bring something important; to not know an important rule; or just to be confused about how the day is going to go. That is what this chapter is for. Every meet may be slightly different but in general, this chapter should cover the common occurrences found at most meets.

Lifters almost always taper their training.

Tapering

A huge part of being ready for a competition comes from all of the hard work you completed in the months and years leading up to that competition. Knowing you followed a solid training program, worked hard, and got results should give you confidence as you go into the meet. However in order to be able to perform at an optimal level, lifters almost always taper their training intensity as the meet approaches. Tapering means a decrease in training intensity and volume which allows the body to catch up to all of the hard training that you have been doing. It is generally unwise to train very hard the week of or the day before a meet.

Most lifters taper anywhere from 3 days to 2 weeks before the competition. Ultimately you must find a system that works well for you. If you are unsure about how you will respond, I suggest you follow the routine outlined below as it has worked very well for almost all of the lifters who followed it.

2 Weeks Out from the Meet

For me the week before the week of the competition (7–14 days out) is usually one of my toughest training weeks, at least in terms of percentages of the 1RM. I will usually work up to a weight that is equal to or a bit higher than what I am planning on lifting for second attempt, especially for squats and bench presses. If you are not using singles at this point then using a conversion chart of the weight lifted combined with the reps should equal that second attempt or a bit more (this is assuming a reasonably conservative attempt selection — see Chapter 16 for more information). Then in the meet I would hope to best that number on my third attempt and hopefully that will result in a new personal record as well. While I am going intense in the gym this week, I am not attempting an all-out one rep max so there should always be a little left in the tank. Remember it is easier to max in the gym than a

competition — not harder — save something for the meet.

In terms of specifics, I found that performing a very heavy squat somewhere between 11 and 14 days out from the meet worked well for me. It was close enough to ensure I was still used to the feel of heavy weight when it came time to compete, and it was far enough away to allow me plenty of time to recover. For the bench I found that a similar time frame, perhaps reduced by a day or two, also worked well. For deads I liked more recovery time; my heaviest deadlift/lower-back exercise I found should be about 17 days before the meet for me. That was my best lift, and I was using the most weight on that one. In general stronger and bigger lifters will need more time to recover, while lighter and smaller lifters will need less time to recover. More experienced lifters usually have increased confidence and the knowledge that their lifts will be there when the time comes. Newer lifters often like to lift a bit heavier closer up to the meet to give them confidence going into the meet.

1 Week Before the Competition

Stronger and bigger lifters need more time to recover.

I generally suggest lifters lift weights on 2 days during the pre-competition week, usually on Monday and Tuesday if the meet is on Saturday. You can keep that the same or push it back one day if the meet is on Sunday. Then I prefer a pretty easy cardio day on Wednesday, followed by two full rest days on Thursday and Friday (stretching and light cardio is okay if necessary to make weight).

In terms of specifics, I like to squat on Monday or Tuesday. I will complete my usual warm-up (bike, foam roll, dynamic stretch — whatever you normally do); then I perform all of the warm-up sets that I plan on doing at the meet, and I will work up to my opener (I am moderately conservative

when it comes to numbers; my opener is usually 85–90% of my true max, so hitting just one rep is not hard and I should be fully recovered in 5 days or so). Then I will complete 2 or 3 more leg exercises, the same ones I have been using all along — the week before a meet is not the time to experiment with a new exercise or training methodology. The intensity will be low; it might just be 2 or 3 warm-up sets and that is it, or maybe it is just an easy first work set with lower reps repeated a few times. The whole goal is to stay in the groove, stay loose, and feel fresh, but you do not want to get sore, tired or burned out from the exercise. You should not be anywhere close to failure in this point of your training. Personally I usually don't see the need to perform deads in the pre-comp week, but if you want to do them, you can complete all of your planned warm-ups and maybe go up to your opener. When you go through your planned warm-ups on any lift, it is a good idea to time them so you have a rough idea of how long you need to warm-up at the meet.

For bench, this would probably be done on the same day or the day after squats. I follow the same guideline: complete your warm-up routine, complete all of your planned warm-ups with the sets and weights and reps, and then work up to your opener or close to it. Follow that up with a few upper body exercises, maybe one more for chest, one or two for back, one or two for arms, and call it a day. Again on the assistance work the intensity and volume are low. Generally the attitude I take is if I think it will promote a feeling of readiness and help prepare me for the meet, I will do it. If I think it might tire me out, drain me, or, worse, hurt me, then I will not do it.

The week before a meet is not the time to experiment with a new methodology.

If everything is going well you should be feeling pretty strong during the

pre-competition week, and you may want to test yourself in the gym. Remember it is easier to max in the gym than in a meet, and, frankly, nobody cares what you did in the gym, so save that energy and intensity for the platform. Competition day is the time to test yourself for real, and it is coming very soon. Having said all of that, sometimes lifters are a little burned out at the beginning of pre-comp week, they know the weights are relatively light, and during training on those days, they feel kind of weak and they get scared about the meet. That is natural. Even moderately light weights, for you, will feel heavy if you don't attack them. Since you know you are squatting only 85 or 90% for 1 rep, you tend not to get fired up for that set, and suddenly that weight feels heavy. It is still 3 or 4 or 5 hundred pounds or whatever it is, so it can feel heavy. My point here is you should not abandon the plan at this point. There is nothing you can do in your training in the last few days to greatly improve your performance on the platform, but you can do things to mess it up. Each meet is an experiment; even if you start to lose faith in the program or how you will perform, chalk it up as that. Give the program an honest test by following the guidelines here, go for what you are supposed to do, and see what happens. If things work out well it was probably just nerves. If they don't, then at least now you know what not to do, and that is actually valuable knowledge in and of itself.

> ***You can't win if you are not on the platform.***

Be There

First and foremost, know where the meet is located and make sure you know the correct address, the date and the time. No matter how good of a powerlifter you are, you can't win if you are not on the platform when you are supposed to be there. This often brings up the question of should I attempt to drive to a meet the morning of or should I go down the night before? There are pros and cons of each situation.

Driving to a meet the morning of has two big pros. First you get to spend a night in your own bed, where you are comfortable and familiar with your surroundings. You get to eat food you are comfortable with, and the second big positive is you don't have to pay for a hotel room so it will save money.

However getting to the meet a day in advance has several positives as well. You are not stressed about making it to the meet on time — you never know on a longer drive exactly what traffic will be like. Driving itself can take a toll on you; think of how tired you can get from a very long drive, even a short drive (1–3 hours) can be a bit taxing. If you are going with friends/family, it can be a bit of an adventure, and, so, while the cost of the hotel might be annoying, I would suggest that you strongly consider that option as your best one, as you want to be as strong and as fresh as possible on the day of the meet. If the drive is 4 or more hours long I would very strongly advise getting there a day in advance; if it 2–4 hours I would still suggest that, and if it is less than 2 hours then I would leave it up to you. One of the problems with driving there is even if things go well on the drive, it is common to use up some nervous energy wondering how the traffic will be or if the car will breakdown or whatever. No one likes it when something out of a person's control affects how they perform; removing the drive is removing one more thing that might mess you up. If you are there the night before I

also suggest driving by the actual meet location if you are unfamiliar with the area. Then you can time how long it will take you to get there, and if you get lost it is much better to get lost going there at night (when it doesn’t matter) then when you are going there in the morning and stressing about missing weigh-ins or the rules briefing, or, worse, your warm-up time for squats.

Be Prepared

When you are competing in a powerlifting competition you want to be prepared for anything that could happen. Things come up that are hard to predict; listed below is a list of things that you will need or might need. It is hard to remember all of these things at once when you are packing up your stuff; the goal of this list is to ensure that you don't forget anything important. I have broken the lists down into prepping for each lift, and I have also included a list if you compete in gear. As you can see if you do choose to compete in gear, that greatly adds to the amount of stuff you must bring with you to the meet.

List of things you must bring to all competitions

- ✓ Singlet — this must be worn by all competitors for all lifts
- ✓ Belt — this is optional; no Velcro or 6″ wide belts allowed
- ✓ T-shirt — usually mandatory in the squat and bench, optional in the deadlift
- ✓ Shoes — whatever you prefer; some like to wear different shoes for different lifts
- ✓ Chalk — there will be some at the meet, but it is nice to have your own for warm-ups or in case they run out

List of things to bring for squats

- ✓ Squat shoes
- ✓ Socks you like to squat in
- ✓ T-shirt you like to squat in
- ✓ Comfortable underwear — usually briefs
- ✓ Knee sleeves if desired and allowed

If using Gear

- ✓ Wrist wraps

- ✓ Knee wraps (already rolled up)
- ✓ Squat suit
- ✓ Groove briefs
- ✓ Suit slippers

List of things to bring for bench press

- ✓ Bench shoes
- ✓ T-shirt you like to bench in

If using Gear

- ✓ Bench shirt
- ✓ Wrist wraps

List of things to bring for deadlift

- ✓ Deadlift shoes
- ✓ Deadlift socks (some feds require socks that cover the shin)
- ✓ Baby powder
- ✓ T-shirt you like to deadlift in (T-shirt is usually optional on deads)

If using gear

- ✓ Deadlift suit
- ✓ Erector shirt
- ✓ Suit slippers

Good idea to bring some or all of the following items

- ✓ Water bottles with water
- ✓ Sports drinks
- ✓ Supplements you are taking
- ✓ Cooler for drinks/food
- ✓ Money for T-shirts, food, etc
- ✓ Camera and/or camcorder with batteries/charger

- ✓ Warm-ups and attempts written out beforehand
- ✓ Workout Journal
- ✓ Sweats to wear before/after lifts (you may want layers as some meets are cold and some are hot)
- ✓ Food to eat (fruit, peanuts, sandwich, pasta, etc)
- ✓ A small towel
- ✓ Clothes to wear after the meet
- ✓ Your breakfast for the day of the meet if necessary
- ✓ iPOD/Music if you like that
- ✓ Screwdriver to adjust lever belt if necessary
- ✓ Tape measure for setting up grip on bench press bar if necessary (rings are not always in the same spot)
- ✓ Tylenol/Alleve just in case
- ✓ Liniment if you apply ointment to your joints
- ✓ Sports tape/band aids
- ✓ Kick ass attitude
- ✓ Workout Partners
- ✓ Cheerleading squad

The Night Before

The night before a powerlifting meet can be a bit nerve wracking, especially if it is a big competition and you are really hoping to perform well. The night before my first real powerlifting competition, I heard some of the guys talk about weights they had seen people lift, and it made me want to pack my stuff up and head home. I thought I was going to be a laughing stock, but they had been talking about some of the best lifters in the country; most lifters were similar to me, some a fair amount better, some a fair amount worse, and in the end it worked out.

What you do the night before a competition is up to you. If you have already weighed in, you want to eat well — at least one and preferably 2 or 3 meals between the weigh-in and bedtime. They don't have to be huge; it is better to eat 2 or 3 smaller meals than one gigantic one. You can pretty much eat what you want, but eat things you are somewhat familiar with; tomorrow is not the time to wake up with a stomach ache from eating some strange food. In addition if you are in a place you are not used to, or, worse, a foreign country, be careful about the water (and ice) and the food as you don't want to get sick with food poisoning the night before a meet. This makes seafood a little bit less appealing to me at this time, since the chance of illness from that is just a tiny bit higher than other foods. This is also not the time to get drunk; one or two drinks might help you relax but watch it: if you want to party, do that the night after a competition, not the night before. Do you really want to mess up all of that hard training you did to get ready for the meet with a single night of drinking?

I think it is a good idea to make sure you are hydrated. If you have already weighed in I would consume at least one drink with carbs and electrolytes (I like Surge Workout Fuel from Biotest) to super-hydrate; this becomes more important if you lost weight to make weigh-ins.

Most lifters eat a fair amount of food and relax in their rooms, watch a movie, read a book, hang out with others, and then head off to get some sleep. You might take yourself through a stretching routine or some meditation/relaxation stuff to help you relax. Try to get a reasonable amount of sleep if possible — what you are used to or a bit more, although it often ends up being a bit less.

> ***I would consume at least one drink with carbs and electrolytes to super-hydrate.***

Whether or not you wish to have sex the night before a competition or engage in other similar activity is up to you (and your partner, of course). The benefit is it tends to have a relaxing and sleep inducing effect so it might help you get some sleep. It does burn off some energy and aggression however, and I imagine most lifters have at least heard stories about athletes abstaining from sex or other activities for a long period of time to increase their performance. To be honest I don't know what the answer to that question is. I have never abstained long enough myself to give it a real test. I would usually abstain anywhere from a couple of days to a week or two before a competition, although there have been times when I did not and I never noticed any significant difference. I would never do anything like that right before a competition; I always had a 4 hour rule for myself, which meant 4 hours before a serious workout I had to be "good." The evidence in this area is conflicting. There are a lot of "old school" coaches and athletes who believed that abstaining was effective. However evidence does show that married males who have sex at a more regular interval generally have higher levels of testosterone than unmarried males (whom, it can be assumed, are not having sex at the same frequency, although I suppose that is not a guarantee). It could also vary from person to person and in theory it might correlate to their

refractory rate, which is the length of time necessary for a person to be able to orgasm again after already having an orgasm. Some people have a very fast refractory rate and others much slower. If I had to speculate I would say those with a fast refractory rate would not need to abstain much; those who had a slower rate would do better abstaining for a while in theory. Generally refractory rates increase with aging.

As I stated before I simply don't know what the "correct" answer is, but I do think it is an interesting question. I am hesitant to dismiss "old school" knowledge because all they had to rely on was performance, but then again there are a lot of real myths when it comes to how old timers train. More recent evidence doesn't seem to suggest to a boost of testosterone or other hormones from abstaining but I would take that with a grain a salt. If you really want to know the answer to this question, then I would experiment and find out for yourself.

Bottom line is to try to get some rest, try to relax, try not to waste nervous energy, do not perform any exercise other than a very light walk, do not go and hit your warm-ups or test yourself or anything stupid like that; just relax and know that most of the hard work is over (for now) — you are simply going to show off that hard work that you did. Also if you are truly incredibly nervous about the next day, that could be a sign your openers are too high. You should be able to do them even if you had the flu (literally).

Practice some time management on this day.

The Morning Of

The morning of the meet is most likely a busy time, with a lot of getting ready and hurrying to end up waiting. Most likely you need to weigh-in, although some competitions will offer this the day or evening beforehand, and generally that is more advantageous to the lifter so you should take advantage of it if possible. If you have to weigh-in, generally the idea is to weigh in as early as possible, usually before breakfast; then eat once you are done so you will have time to digest before you lift. You want to practice some time management on this day to make sure things go smoothly. If you like to shower or shave or whatever your morning routine is, try to do this as you want to be awake and alert when it comes time to lift. Time how long that will take; know how long it takes to get to the competition. Keep in mind you may be checking out of your hotel room that morning; packing up always takes some extra time. Don't leave your singlet hanging up in the hotel closet — go over your checklist. You might put everything crucial in your gym bag or you might just get dressed in it now and then put your warm-ups on over it. Know what you are going to, or where you are going to eat, and approximate how long that will take.

Generally you will want to attend the rules briefing unless you are a very experienced lifter. The rules briefing usually starts anywhere from 8:15 am to 9:30 am that morning; it will be on your entry form. It is often delayed, but don't count on that. That usually takes 15 to 30 minutes, and then the lifters are usually given 15–30 minutes to begin warming up, with formal lifting to begin after that. Personally I always thought that warm-up time was a little short, but usually it is the lighter and weaker lifters in the first flight, and they didn't seem to complain about it, so maybe it was just me. I know before squats I always like 45 minutes to an hour to stretch and then complete my warm-up sets.

Weigh-in

The time of the weigh-ins will vary, with a maximum of 24 hours before the event and a minimum of 2 hours before. Usually the time will be indicated on the entry form, and it will be a range of 1–2 hours in most instances. For example, if a meet starts at 9 they might have weigh-ins from 7-8:30 or something like that. Generally you want to weigh-in as early as possible; of course, other people know this as well, and there is often a line to weigh in and sometimes it moves slowly. You can wear what you want, but when weighing in it is advisable to wear as little clothing as possible, usually just underwear. Even if you can make your weight class, fine; you still want to weigh-in as light as possible because in the event of a tie, the tie goes to the lightest lifter. Also if you have a chance for a best lifter award they will use your actual bodyweight to calculate your coefficient, and sometimes a few pounds can make a big difference.

The general guideline is to get to the weigh-in relatively early, weigh-in as soon as possible, and then eat/drink something as soon as possible afterward so you have time to digest your food. Normally you do not eat or dink anything prior to the weigh-in; however, if you know for sure you will make your weight class and you are worried that you will not have enough time to digest your food, then you can eat or drink something and then weigh-in. When eating something right before a weigh-in, don't calculate the weight of the food from the calories (3500 calories equals one pound of fat); instead just use its mass (how many grams it is). If you drink 16oz of water, eat 2 eggs and an apple and a baked potato, you will have easily added 2–2.5 lbs to your body mass even though that meal is far from 7000 cals.

Listed below are the top weights for each weight class. Keep in mind if you weigh-in on a digital scale, the highest amount for the digital scale is rounded up in the lifter's favor. For example, the max weight for a 90 kg

male is 198.25 lbs, but on a digital scale the max weight is 198.4 lbs (this is because most digital scales give out weights in increments of .2 lbs, so 198.3 is not an option).

Women			Men		
Weight in Kilos	**Weight in Pounds**	**Digital Scale**	**Weight in Kilos**	**Weight in Pounds**	**Digital Scale**
44	97.0	97.0	*60*	132.25	132.4
48	105.75	105.8	*67.5*	148.75	148.8
52	114.50	114.6	*75*	165.25	165.4
56	123.50	123.6	*82.5*	181.75	181.8
60	132.25	132.4	*90*	198.25	198.4
67.5	148.75	148.8	*100*	220.25	220.4
75	165.25	165.4	*110*	242.50	242.6
82.5	181.75	181.8	*125*	275.50	275.6
90	198.25	198.4	*140*	308.50	308.6
90+	198.25+	198.4+	*140+*	308.50+	308.6+

The digital scale is rounded up in the lifter's favor.

Of Note: The USAPL and IPF decided to redo their weight classes, and their weight classes do not reflect what is listed above. Most other federations have not followed suit, nor have the rankings on powerliftingwatch. com. Historically altering the weight classes makes keeping records, etc. near impossible and often wipes away records that represented lifetime achievement goals for certain lifters.

Rules Briefing

After the weigh-in is over, the rules briefing usually starts. This is the time where the meet director or similar agent will go over the rules expected on that day. This is particularly important if you are new to powerlifting or new to lifting in that federation. Of course, it is my hope that this book adequately prepares you for what to expect during a powerlifting competition, but it is still a good idea to attend the rules briefing to get a sense for what the meet and what the judging will be like. Remember that every federation is different and there can be slightly different rules that can make a big difference on the platform (like whether or not you need to wait for a "start" command in the bench press or not, or whether you get an actual "Press" command, etc.). Also things change over time; what may be a rule now may change later on. Attending the rules briefing will ensure that you are aware of the most up-to-date rules for federations, and if you have any questions, it gives you a chance to ask them.

The other benefit of attending the rules briefing is that you know when the meet is really going to start, because they will announce that at the end of the rules briefing. Stereotypically powerlifting competitions run late, sometimes they are on time, and rarely they are early, but things can change. You don't want to skip the rules briefing, estimate on your own when the meet will start and then show up and find that something was wrong with your estimations and your flight has already started or something. I am a bit of a control freak; personally if I don't know what is going on timing wise it starts to freak me out. Even if my estimations are right, if I don't have the actual information to back them up I spend so much nervous energy wondering what if things changed or moved faster or what have you, that any little benefit I might gain by relaxing in my room or going out to eat is lost by worrying about how things are progressing back at the meet. I would suggest

that you attend the rules briefing, but ultimately it is up to you. You can still lift in the competition if you skip the rules briefing, so it is not mandatory.

Order of Events

In traditional events the order of the lifts is always the same. You start off with the squat, then the bench press, and then the deadlift. If it is just a push/pull competition, it goes bench and then deadlift. If you are just benching or deadlifting then you will lift with the other lifters competing in that lift even if they are doing a full meet. When it is time to squat, all lifters at the meet will squat before moving on to the next event. The order of events is correct from a physiological standpoint. You want the highest skill stuff to come first, then you bench to give your lower body time to recover, and then you deadlift. You squat before you deadlift so a tired lower back doesn't ruin the squat.

Lifters at a meet will be broken into flights. A flight is a group of lifters, usually ranging from about 7 to 20 lifters. Depending on the size of the meet, it may be all the lifters in one weight class; it may be several weight classes put together; it might be just half of a weight class in a large competition; it might be all the teens and all of the females; or it might be people just performing a single lift like a bench press. It is up to the meet director's discretion. Most normal sized meets have about 30 to 80 lifters, and they are broken down into 2–4 flights. Very large meets will often run 2 or 3 days, but each lifter will lift on just one day.

Once a flight is established, the lifters are ranked by their opener in that lift from lightest to heaviest. When it is time to begin, the lifter lifting the lightest opener will start, and he will perform his first attempt; then the next lifter is up, and so on and so forth. Once all lifters in that flight have completed their first attempts, the second attempts begin. The lifters are again re-arranged in order

Lifters are ranked by their opener in that lift from lightest to heaviest.

from lightest to heaviest (the order tends to stay about the same but not always; don't get in the habit of thinking you just follow "that guy"; instead listen for when your name will be called) and second attempts are taken. The same thing happens for third attempts. Once that flight is finished, the next flight will begin and usually there is no intentional delay; one flight will move right to the next. There is often a delay when going from one exercise to another, usually about 15–30 minutes or so. This is necessary because the equipment for one lift must be removed, and the other lift must be set up. In addition this gives the referees and such time to go to the bathroom, grab a bite to eat, and just relax. If you are in the first flight that will perform the next lift, ask the meet director what kind of break they are taking as that will affect your warm-ups. Also it is not uncommon for that break to extend another 5–10 minutes past what the meet director said it would; that is just something to be prepared for.

The one confounding factor in the order of a powerlifting competition is the strict curl. Not all powerlifting competitions include an option for a bicep curl; it is not included in one's powerlifting total, but it can sometimes be added to other lifts like the bench press or one's push/pull total. Different organizations put the bicep curl in at different times. Some organizations start with the curl; at least it will come before the bench and the deadlift. Some organizations finish with the curl following the general order of contesting larger muscles first. Other organizations put it between the bench press and the deadlift with the idea that it is more exciting to finish a meet with a big deadlift than a big curl. The order should not affect your performance too much; just make sure you know what the expected order is. Many lifters choose not to curl, and the lift is easy to do, so that flight will usually move pretty quickly.

Make sure your own stats are right.

Awards

The awards at a powerlifting competition are almost always given after all of the lifting is completed. Powerlifting competitions tend to be long, starting around 9 am (with weigh-ins often much earlier) and then finishing anywhere between 4 pm for a short meet to after midnight for a long one. Sometimes lifters who have to leave will ask for their awards early, and an accommodating meet director might do this, but it can be inconvenient to do. The awards presentations themselves often take a while, about an hour or so, because the way powerlifting is currently set up most lifters receive a trophy because they are lifting in a specific division, with few actual competitors. It also takes a while for the meet directors to compile the results and put the lifters in the proper order and make sure everybody gets the right awards. The best thing you can do for yourself in this situation is to just make sure your own stats are right, that you are credited for all of the lifts you made, that your total is right, and that you are in the correct division. You won't automatically know how you placed unless you were keeping track of your competitors, but you should know your general standing and you can compare your stats against others when they are called out. If everybody makes sure their own info is correct, then everybody's info will be correct.

It is not uncommon to win multiple awards in one powerlifting competition. If you enter in a specific division, for example, an age group, then you would be eligible for a trophy or medal in that division. If you are also in the open division, you could win a trophy for that. If you compete in a single lift competition, then you could win something for each lift if you entered multiple lifts (like bench press and deadlift, for example).

Lifters often appreciate receiving something other than a traditional trophy to represent significant achievements

My First Meet

I remember my very first real powerlifting meet. It was the Commonwealth Games of VA, which was kind of like a state-level Olympics that included powerlifting, although, for this competition, you didn't have to qualify. I had competed in a regional meet in high school, but this was my first real sanctioned meet. I was 19 at the time; somebody at Gold's had told me about the competition 3 weeks prior. I had been training hard, loved to lift, but hadn't peaked or anything like that. I just decided to sign up. I called the meet director, who told me to get a "wrasslin' singlet" and head on down. I did the 4 hour drive and got there the night beforehand. They had a parade for all of the athletes involved in the Commonwealth Games in some stadium in Roanoke so I decided to participate in that. I found my way there and met up with about 10 other lifters partaking in the parade. We talked for a while and, of course, talked about lifting. The amount of weight that these guys were talking about seeing people lift made me want to pack up and drive home right then and there, but I didn't. I competed the next day and everything went well, although it started a little shaky. For some reason, even though I was a 198 lb lifter lifting reasonable weight, I was in the first flight, and I was the first person to lift. I opened up the whole event (in hindsight I think this was because I was a teen). I walked up on to the platform nervous but ready to lift and here is what I heard the meet director say: "Stop the meet!" He got up from the head judge's chair and pulled me aside. First, I was wearing a 6″ Velcro belt (at the time I thought my Shiek belt was top of the line) which was illegal (there was no rules briefing or equipment check at this meet to the best of my knowledge), and secondly my T-shirt was on top of my singlet, not tucked in the way it is supposed to be. The meet director let me borrow his own lifting belt (many thanks to Bill Lindsay for doing this), he got my outfit squared away, and I went out and promptly missed my first squat

because I moved my foot after I got the squat command. However the lift was easy, and I was a cocky teenager (even after all that I still thought I knew it all) so I went up in weight, got it, and the meet was smooth running after that. I ended up going 7 for 9, missing my first squat and third bench. I was very glad I didn't go home; there were some strong people but it was not overwhelming or intimidating, and I got hooked for life, so something must have happened right. And meet directors take note — they could have chosen to not help me, ridiculed me, left me to flounder on my own, constantly red-lighted me for the wrong equipment — any of those things, and my own personal history might have been very different. But they delayed things by 5 minutes (probably less in reality) to let a kid have a chance at a small, local meet (this was not Nationals), and he went on to compete in countless powerlifting meets and hopefully give back a bit to the sport as well. Powerlifting is a grassroots sport, and it is tempting to just cater to the old veterans who have been around for a while, but to ensure growth and development you have to cater to the up-and-comers as well. That definitely does not mean giving them white lights when they should get reds — that just sets them up for failure down the road — but it does mean pulling them aside and giving them a helpful hand when necessary.

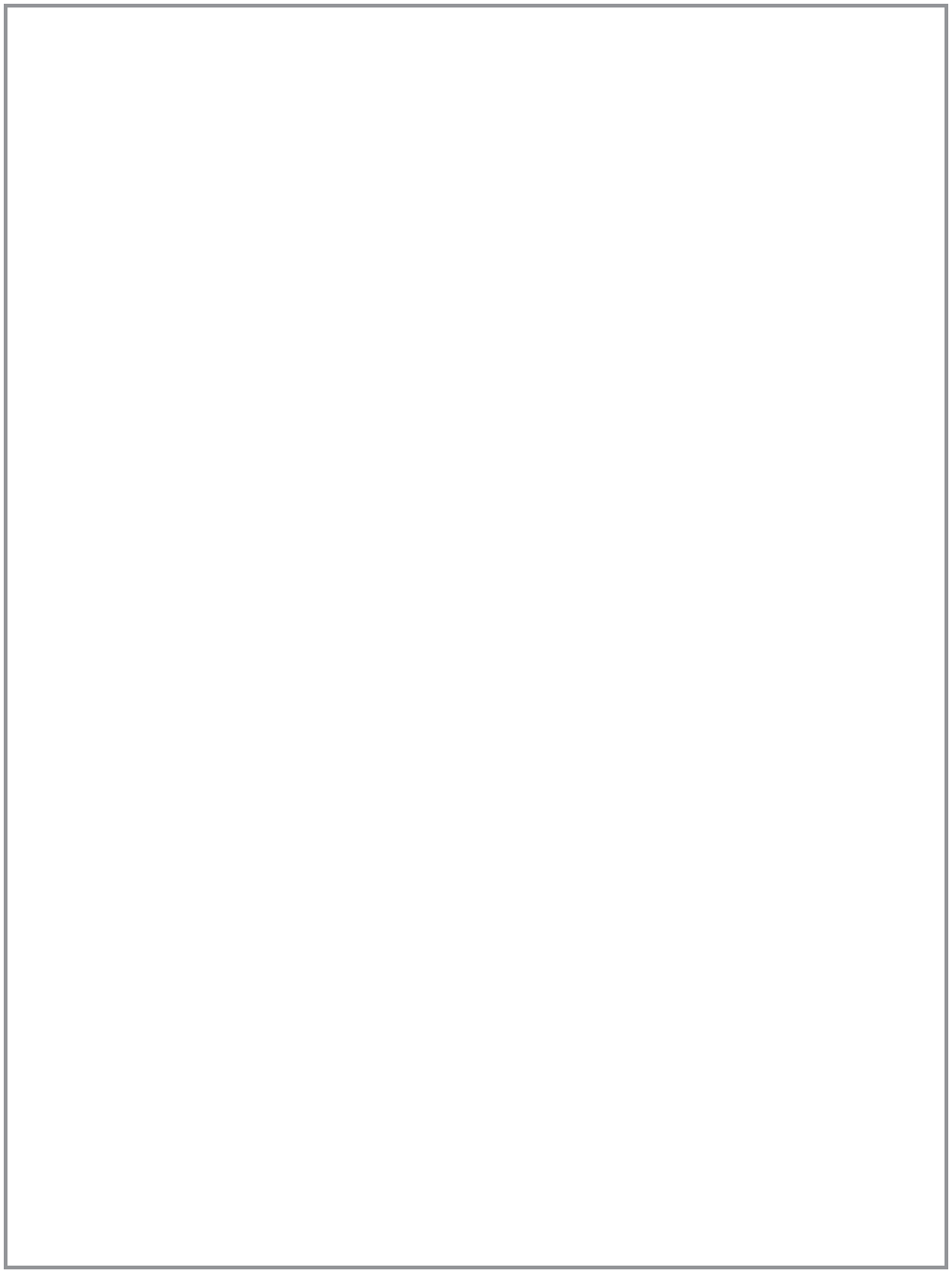
That does not mean giving them white lights when they should get reds.

Injuries

For some reason I often seem to get a minor injury right before a meet, and, to be honest, I think a lot of them are fairly mental. I believe the body is looking for a way out of something it knows will be strenuous; it is basically asking to not have to lift. Of course, for a truly serious injury, we will not work through that, but there are a lot of nagging injuries that feel kind of iffy that we do have to work through. That may not be the safe advice, but it is the advice that will get you standing on the top of the podium. Work around the injury in training as best you can the last few days or weeks; skipping an assistance exercise or modifying it will rarely mess up your total too much. And then finally, on the week of the competition, if you are going to do the meet, you just have to decide that you are competing, no matter what. I have woken up on meet mornings and felt like I pulled my back, or my shoulders hurt, and you just have to say it doesn't matter, I signed up for this thing; if I bomb out, so be it, and then go compete. And essentially you always end up performing just fine on the platform. I know at one meet when I was warming up for deadlifts, I pulled a muscle in my back (most likely my Multifidus or my QL) while I was simply changing the weight on the bar; I imagine the heavy squats prior had set that muscle up for an injury. It was not a minor injury — it affected my ability to bend down even without weight — but it was not debilitating and actually during the deadlift it didn't really hurt that much. I went on to deadlift, and on that day I had my best deadlift ever of 700 lbs at 198.

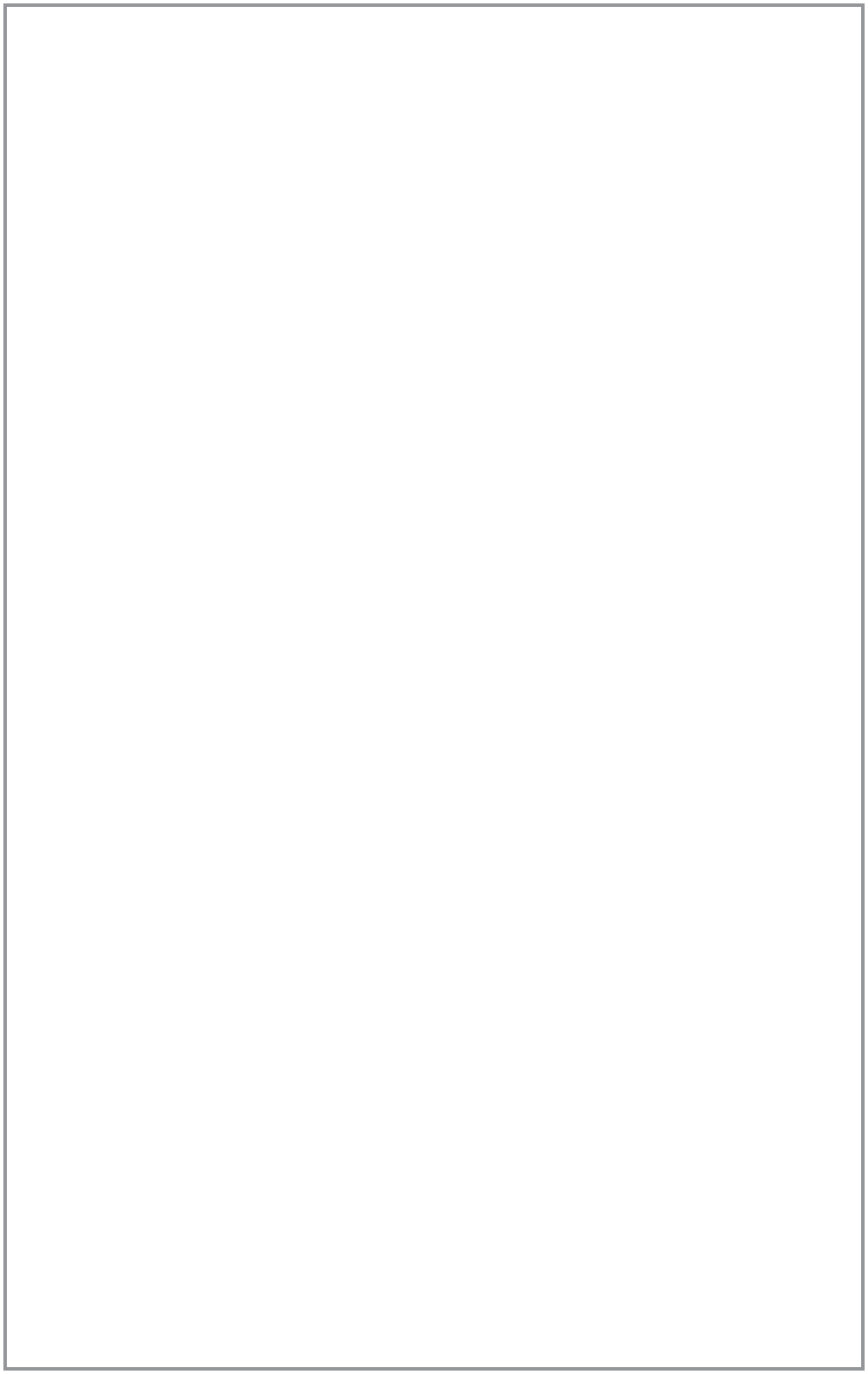
There is a classic footage of a great lifter, Ken Patera, on youtube where he breaks some ribs squatting and then goes on to bench press and deadlift significant weight, even though he can barely get on and off the bench because of his ribs. Athletes all over the world rarely get to compete only when they feel absolutely fresh and perfect, and powerlifting is no different.

Give your injury a very honest assessment, consider your lifting career carefully — I don’t believe in sacrificing one’s lifting career just for one big lift — and then once you decide, stick with that plan. If you decide to lift, you might as well attack it. There is probably an increased chance of injury if you lift tentatively or timidly than if you go all out.



Summary

Meets can be a lot of fun if you are reasonably well prepared, have trained hard and properly, and go for realistic attempts. I find meets to be a lot more fun when some friends/supporters come with you and help you and cheer you on. The meet is the proving ground — you are officially testing your training methodologies, programs, and strategy on the platform and you can’t BS your way out of a bad cycle. The weights don’t accept excuses. Give it the best that you can on that day, and if you aren’t happy with the results, then try harder and be smarter next time.



A lifter preps for the deadlift.

Chapter 15

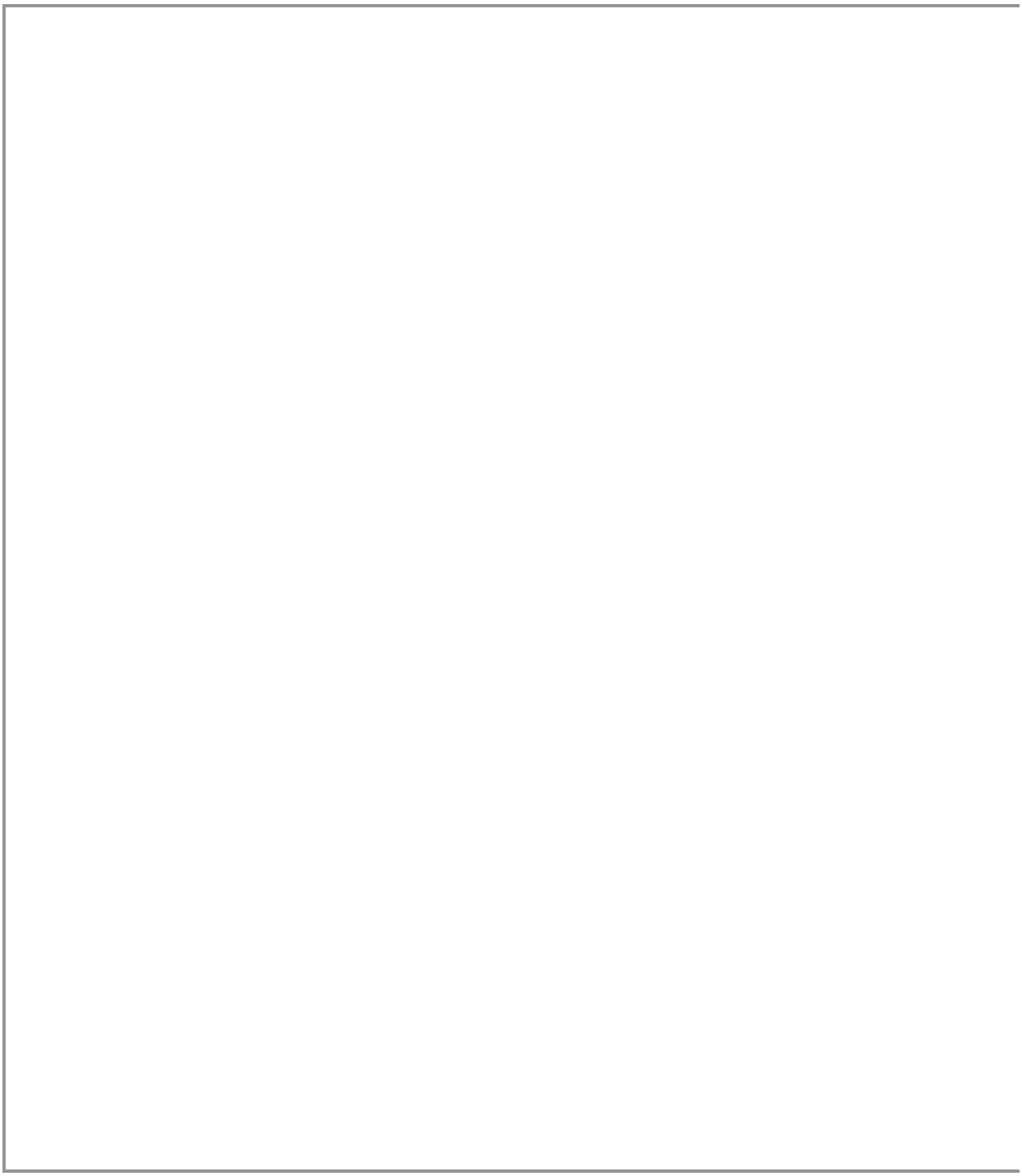
How to Warm-up for a Maximal Attempt

Even if you train hard for 5 or 6 months, and even if you eat correctly and make weight, you must still perform well on the day of the meet in order to fully succeed as a powerlifter. A big part of performing well on meet day is knowing how and when to warm the body up so it is ready to smash some serious weight.

There are two phases to warming up. The first is called a general warm-up, and this is where you increase blood flow throughout the body to increase body temperature and elasticity and to generally prepare the body for more intense activity. Examples of a general warm-up for powerlifting would be walking, riding the bike, jumping rope, performing jumping jacks, dynamic stretches, using the rowing machine, or marching in place. One thing to keep in mind is you want to be able to duplicate the kind of warm-up you complete in the gym on the day of the meet — if you love to ride the bike before squats, you may have a hard time doing that if the meet is in a hotel or a high school gymnasium that doesn't have a stationary bike. It is a good idea to perform a general warm-up before you lift. In all honesty, a general warm-up is a little less important on the day of a competition simply because you will most likely be pumped up and anxious to compete. That will serve as a mini

warm-up since your resting heart rate and blood pressure will be elevated. That aside, it is still a good idea to do something to prep you to lift, particularly if the first lift is going to be the squat, which requires the most warm-up. In addition, if you are older, if the meet area is very cold, or if you are simply having a hard time getting loose, a good warm-up can help with that.

Most general warm-ups are 5–20 minutes in length, particularly for the squat. It is easier to warm-up for the bench press and the deadlift, because fewer muscles are used in the bench, and less of an extreme range of motion is used in the deadlift. To warm-up for the squat, I would suggest riding a stationary bike for 3–10 minutes (if available) or taking a brisk (3.5–4.5 mph) walk for 5–10 minutes. You can often go outside and just walk around the parking lot to warm-up. Wear your sweats to help you get and stay warm. Once that is completed, a lot of lifters like to perform some dynamic stretching to prepare for the squat. The older you are and the more inflexible you are, the more important this is. Try to choose stretches that hit your target areas. Common problem areas are the hamstrings, quads, glutes, lower back, hip flexors, adductors, and calves. Foam rolling can be a good idea before lifting to help loosen you up. Static stretching is okay before you squat if you really feel it is necessary to help you get prepared to squat, but remember too much static stretching can actually weaken the tissues by reducing their elasticity. Once you have performed 5–10 of minutes of some type of cardiovascular activity and 5–10 minutes of flexibility exercises, you should be ready to start squatting.



Walking/running on the treadmill is an example of a general warm-up

A general warm-up for the bench press is less important. If you have squatted already that day, even if it was a few hours prior to bench pressing,

you will still have some lingering effects of that keeping you warm. Even if the bench is your first lift of the day, it is rare to see lifters spending much effort performing a general warm-up to prepare for the bench press.

Some lifters like to perform a general warm-up for the deadlift, but it is not as important as the squat. The deadlift occurs at the end of the day, and it is important to manage your fatigue; you don't want to waste energy by tiring yourself for the deadlift. Conventional deadlifts don't require much of a stretch to get into position, and a bit of tightness can actually be beneficial. Sumo deadlifts require a greater stretch and thus may require a warm-up similar to that of squats. The goal is to perform as little of a warm-up as possible and still feel completely ready to lift. Some dynamic stretches seem to work well to prepare the body to deadlift. It is generally not necessary to complete a general warm-up for the curl.

Once the general warm-up is completed, it is time to complete the specific warm-up. The specific warm-up is the most important warm-up, and its purpose is to prepare the body for the specific activity that it is about to perform. Most lifters like to perform the same specific warm-up each time they train the lift. This seems to help prepare the body mentally and physically for what is about to come. Remember that not only does your body need to be warmed up, but your nervous system needs to warm-up as well. This is not just specific to powerlifting but is very common in many sports. Field-goal kickers regularly practice kicking the ball on the sidelines before they try to kick during the game. Golfers almost always complete a practice swing before their real swing to get the feel of the activity. And powerlifters like to lift lighter weights on the same movement before they try to lift heavy weights. The heavier weight you are lifting, the more important a warm-up is.

Some lifters will perform light leg presses to warm-up for the squat

The specific warm-up is often broken into two sections. The first part is always performed before the exercise, no matter how heavy you are going. The second part is often based on what you will be trying to do during the exercise. For example, the first part of a specific warm-up for a squat might look like this:

Bodyweight squats — 12 reps
Front squat — bar x 8 reps
Back squat — 135 x 5 reps
Back squat — 135 x 5 reps
Back squat — 225 x 5 reps

A lifter warming up on the squat

Once that first section is completed, the lifter then completes the second part of the specific warm-up, if necessary. If the lifter was going to squat and their first work set was 275 lbs, no additional warm-ups would be necessary. If the lifter was going to lift 365 lbs for reps, then probably one more warm-up set would be ideal (295 would work well), although the lifter could complete 2 or 3 more warm-up sets if they felt it was necessary. If the lifter was going to max out, then several more warm-ups sets would be necessary.

Listed below are several common warm-up scenarios for men and women.

Specific Warm-ups Part 1

Male Squat	*Female Squat*
Bar x 8 reps	Bodyweight squats — 12 reps
135 x 5 reps	Back Squat — bar x 8 reps
135 x 5 reps	Back Squat — bar x 8 reps
185–225 x 3–5 reps	Back Squat — 95 x 5 reps
225–315 x 1–3 reps	

Male Bench Press	*Female Bench Press*
Bar x 8 reps	Bar x 8 reps
135 x 5 reps	75 x 5 reps
135 x 5 reps	75 x 5 reps
185–225 x 3–5 reps	

Male Deadlift	*Female Deadlift*
135 x 5–8	95 x 8
225 x 3–5	135 x 5
315 x 1–5	

The above are just guidelines that work for most people; you can feel free to modify them as you see fit. Once you have completed that beginning part of the specific warm-up, you then ask yourself what weight you are going to use on your work sets. Think about the difference in the weight you just lifted and what you are planning to lift. Ask yourself how many more warm-up sets you will need to feel ready to lift that weight; generally the answer is 0–3 more warm-up sets. Decide how many warm-up sets you want to use, and break up the weight difference relatively evenly among those sets. Listed below are some examples. These warm-ups are assuming the lifter has already completed the first part of the specific warm-up listed above.

135 is a common starting point for males on the bench press

Specific Warm-up Part 2

Male Squat 405 x 5 with 1, 2, and 3 warm-up sets (ending at 225 x 5)

Opt 1) 1 w/u set — 315 x 5

Opt 2) 2 w/u sets — 295 x 5; 365 x 1–5

Opt 3) 3 w/u sets — 275 x 5; 325 x 4; 365–375 x 1–3

Female Squat 155 x 6 with 1 and 2 warm-up sets (ending at 95 x 5)

Opt 1) 1 w/u set — 125 x 5

Opt 2) 2 w/u sets — 115 x 5; 135 x 3

Male Bench 275 x 8 with 1, 2, and 3 warm-up sets (ending at 135 x 8)

Opt 1) 1 w/u set — 205 x 8

Opt 2) 2 w/u sets — 195 x 8; 235 x 4–6

Opt 3) 3 w/u sets — 185 x 8; 215 x 6; 245 x 2–4

Female Bench 115 x 5 with 1 and 2 warm-up sets (ending at 75 x 5)

Opt 1) 1 w/u set — 95 x 5

Opt 2) 2 w/u sets — 90 x 5; 105 x 1–3

Male Deadlift 455 x 3 with 2 and 3 warm-up sets (ending at 225 x 5)

Opt 1) 2 w/u sets — 315 x 5; 385 x 1–3

Opt 2) 3 w/u sets — 295 x 5; 365 x 1–4; 415 x 1–2

Female Deadlift 185 x 8 with 1 and 2 warm-up sets (ending at 135 x 5)

Opt 1) 1 w/u set — 160 x 5

Opt 2) 2 w/u sets — 155 x 5; 170 x 2–4

The above scenarios are to help you prepare and warm-up for a certain **work set;** they are not necessarily ideal for a 1 rep max. If you are maxing out, as you would in a powerlifting competition, then you should perform your usual first specific warm-up and complete that with the following guidelines.

After initial specific warm-up (listed above)
Set 1 = ~30–50% of Opener x 5–8 reps
Set 2 = ~60% of the Opener x 3–5 reps
Set 3 = ~70% of the Opener x 2–3 reps
Set 4 = ~80% of the Opener x 1 rep
Set 5 = ~90% of the Opener x 1 rep
Set 6 = Do the Opener (ON THE PLATFORM!)

In terms of rest in between sets, you want to rest as long as necessary so you don't get tired or out of breath when warming up. However, you don't want to rest so long that you never get warmed up. In general rest 2–5 minutes on sets 1–3 and rest 5+ minutes on sets 4 and 5 (the set number refers to the chart listed above; you can rest 1–2 min between your initial warm-up sets). On the platform you will probably have about 10–15 minutes of rest in between attempts.

There is nothing wrong with starting with the bar and moving up from there

The above guideline works quite well and most lifters seem to perform well using it. However, some lifters prefer a less intense warm-up, and others prefer a more intense warm-up, almost like a mini workout. Keep in mind it is the number of *reps*, not the number of *sets*, that causes fatigue. Performing several sets with low reps and a lower percentage is relatively easy, but it still does a good job of preparing the body for a heavy set. Listed below is the standard guideline (as given above) along with a low volume version and a high volume version. In my experience newer lifters tend to like the standard and higher volume warm-ups; more advanced lifters tend to like the standard and lower volume warm-ups. Listed below is the entire specific warm-up; parts 1 and 2 are included here.

Keep in mind the numbers listed below are **Openers,** not third attempts.

Male Squat 450 lbs			**Male Squat 600 lbs**		
Standard	**Low Volume**	**High Volume**	**Standard**	**Low Volume**	**High Volume**
45x12		45x20	45x12		45x20
135x8	135x5	135x12	135x8	135x5	135x12
135x8	135x5	135x20	135x8	135x5	135x20
225x8	225x3	225x10	225x8	225x3	225x10
275x5	295x1	275x6	315x5	325x1	315x6
325x3	355x1	325x4	405x3	425x1	405x4
365x1	405x1	365x2	475x1	525x1	495x2
405x1		405x1	545x1		550x1

Female Squat 150 lbs			**Female Squat 250 lbs**		
Standard	**Low Volume**	**High Volume**	**Standard**	**Low Volume**	**High Volume**
BWx12		BWx20			BWx20
45x12	45x8	45x20	45x12	95x5	45x20
75x8	75x5	75x12	95x8	95x5	95x12
75x8	75x5	75x12	95x8	135x3	95x12

90x5	95x3	90x8	135x5	175x1	135x8
105x3	115x1	105x3	165x3	200x1	165x5
1 20x1	1 35x1	1 20x1	195x1	225x1	195x2
135x1		135x1	225x1		225x1

Male Bench 315 lbs			Male Bench 420 lbs		
Standard	Low Volume	High Volume	Standard	Low Volume	High Volume
45x12		45x20	45x12		45x20
135x8	135x5	135x12	135x8	135x5	135x12
135x8	135x5	135x20	135x8	135x5	135x20
185x8	185x3	185x8	205x8	225x3	205x10
225x5	225x3	225x6	265x5	275x3	265x8
255x3	275x1	255x3	315x3	325x1	315x5
275x1		275x1	355x1	375x1	345x1
295x1		295x1	395x1		370x1
					395x1

Female Bench 115 lbs			Female Bench 175 lbs		
Standard	Low Volume	High Volume	Standard	Low Volume	High Volume
45x12	45x8	45x20	45x12	45x8	45x20
65x8	75x5	65x12	75x8	95x5	75x12
65x8	75x5	65x12	75x8	95x5	75x20
80x5	95x1	75x8	90x8	125x3	95x10
95x2		85x3	115x5	150x1	115x6
105x1		95x1	130x3		130x3
		105x1	145x1		145x1
			160x1		160x1

Male Deadlift 500 lbs			Male Deadlift 650 lbs		
Standard	Low Volume	High Volume	Standard	Low Volume	High Volume

135x8		135x12	135x8		135x12
135x8	135x5	135x12	135x8	135x5	135x12
225x8	225x3	225x8	225x8	225x3	225x10
315x5	315x2	315x5	315x5	315x3	315x8
385x2	385x1	365x3	405x2	405x1	405x5
455x1	455x1	415x2	495x1	495x1	475x3
		455x1	585x1	585x1	545x1
					600x1

Female Deadlift 225 lbs			**Female Deadlift 315 lbs**		
Standard	**Low Volume**	**High Volume**	**Standard**	**Low Volume**	**High Volume**
95x8		95x12	95x8		95x12
95x8	95x5	95x12	95x8	95x5	95x12
135x5	135x3	135x8	135x8	135x3	135x10
160x3	165x1	160x5	185x5	185x3	185x8
185x1	195x1	185x2	225x3	235x1	225x3
205x1		205x1	255x1	275x1	255x1
			285x1		285x1

A lifter prepping for a heavy bench press

There a couple of guidelines to keep in mind when warming up for a maximum attempt. The first is that, when you make jumps in weight, the jumps should either be the same size or they should decrease in size as the weights get heavier. For example it is fine to warm-up with the following weight: 95, 135, 165, 185. It would not be good to go 95, 105, 115, 165, 180, 185. Having regular increments as you jump up in weight helps the body prepare to lift heavy, and it helps prepare the neuromuscular system for what the weight will feel like. If you get used to making 20 lb jumps and then suddenly you make a 50 lb jump, your first thought will be “Wow, this is heavy” and that is the last thing you want to go through your head when you are lifting. You want the weight to feel light, as that will build your

confidence.

Your final warm-up set should be between 85 and 95% of your first attempt. It should not be any closer than 5% of your attempt; otherwise, that is not a warm-up set, and it will do nothing but tire you out and make you weaker when you compete. If you are so nervous about what your opener is that you feel you must complete a lift within 5% of it, then your opener is too high.

A big mistake that is often made by newbies or inexperienced lifters is to complete their opener or perhaps even a higher lift in the warm-up room. The cardinal rule is this: You should never lift your opener or anything higher than it in the warm-up room. To do so simply wastes energy that should be saved for the platform. Sometimes it can be a little intimidating in the warm-up room, and people often perform too many reps, or they lift too heavy — sometimes even failing — to show off or impress their fellow lifters (or to make themselves feel better). I am all about throwing around some heavy weight with fellow lifters, but the time to do that is on the platform, not in the warm-up room. What you do in the warm-up room means nothing compared to what happens on the platform. Remember, it is not what you start off with that counts, but what you finish with that matters. That goes for competition lifts as well as warm-up sets.

Once you have your warm-ups planned out (I would suggest writing them down beforehand so it is one less thing you have to worry about on meet day), then you have to time them so you are properly warmed up and ready to hit your max at the time you have to lift. As a general rule of thumb, it takes lifters about 30 seconds — 2 minutes for each attempt in a competition. 1 minute per lift is a good average. Squats take the longest; the bench press and deadlifts tend to move more quickly. Competitions that allow gear take significantly longer than those that are raw.

If you are in the first flight of lifters, normally, after the rules briefing, you will be given 15–30 minutes to get warmed up. For most people this means as soon as the meeting is over, you go warm-up; you should already be in your singlet and attire at that time. If you require a warm-up longer than 30 minutes then you can skip the rules meeting. It is not mandatory to attend it, although it is often useful, and if you are a new lifter or new to that organization, it is definitely a good idea to attend the rules briefing. If you are in the second or third flight, then you can try to figure out when to begin warming up by looking at how fast the lifters are moving through their attempts. The very general rule of thumb is to begin your warm-ups when the flight in front of you starts their second attempts, but you may need to go earlier or a little later (it is rare to go much later than that).

For example, if there is one flight ahead of you and there are 13 people in that flight, than means there should be 39 attempts taken (not including passes or fourth attempts). Assuming it takes 1 minute per attempt, that means that, when the first flight starts, you have 39 minutes until your flight begins. Most organizations try to start the next flight within 5 minutes after the previous flight has finished, so don't expect a big break. There is often a 15 minute break or more in switching over from the squat to the bench press and the bench press to the deadlift.

Don't forget to find out where in your own flight you are lifting. If you are early in your flight, then essentially once your flight begins, you need to be ready to lift. If you are late in your flight and there are a fair number of people in your flight (15 or so), you might actually be finishing your last warm-up while those in your flight are starting with their

If you are new to that organization, it is definitely a good idea to attend the rules briefing.

openers, because you will have an extra 15 minutes or so before you are on the platform. It is always a nice feeling to see your name down at the bottom of the posted attempts board.

You should be able to estimate with reasonable precision how long it takes you to complete your warm-ups. This should include everything from putting on the right shoes to completing your general warm-up and then your specific warm-up. As I prepared for one meet, I got so anal as to record exactly when I did each warm-up set. Keep in mind in the warm-up room you will most likely have to share a squat rack or a bench press, so be a little flexible and don't freak out if you get a minute or two out of your rhythm; knowing that it takes you 45 minutes to get fully prepared and you like to hit your last warm-up set about 8 minutes before your opener is valuable information.

Most beginners rush their warm-ups in the warm-up room.

The tendency that most beginners have is to rush their warm-ups in the warm-up room. They get nervous about running late, they have nervous energy they want to burn off, and so they rush through their warm-ups. Often they see other lifters adding weight to the bar and they feel bad about having to take weight off the bar so they jump in before they are ready. You want to be calm and relaxed (as much as possible) in the warm-up room and have confidence in your plan. Don't rush the warm-ups, simply watch the clock and keep an ear out to listen to the progress that the lifters in the earlier flight are making, and you will be fine. You can do yourself a favor and try to generally choose convenient weights to use when you are warming up (by this I mean choosing to lift 185 lbs instead of 180). Feel free to be a little more anal with your last warm-up, so if you are supposed to lift 300 lbs and there is 315 on the bar, definitely take that third

plate off and put on 300. Even if it was 295 and you wanted to do 300, I would hunt around and find those 2½'s and put them on there.

It is possible that, if some unfortunate event happens on meet day, such as you fall asleep in your hotel room; you get confused on what flight you are lifting in; you get lost coming back from lunch; or something else happens, it might cause you to have a very limited time to warm-up. In that case, you still want to perform some type of warm-up, but you can use an abbreviated version of the warm-ups listed above. I want to stress that this is not the ideal way to warm-up, but we are no longer dealing with an ideal situation. In this case you want to hit 2–5 warm-up sets, with 10 seconds to a 1 minute rest in between each set. Complete 1–5 reps each set. Listed below are 3 sample versions of the abbreviated warm-ups:

Squat 405	Bench 315	Deadlift 495
135x5	135x3	135x5
225x2	225x2	315x2
315x1	275x1	405x1 (if time)
365x1 (if time)		

All of the above warm-ups should take 5 minutes or less for the total time, and some would take only 1–2 minutes.

Warm-ups are individual, and ultimately you must find what works best for you.

Summary

Warm-ups are individual, and ultimately you must find what works best for you. I have seen individuals hit a new max in the middle of a tough workout after many hard sets, and I have seen people walk into the gym, put a lot of weight on the bar, and lift it. I don't suggest either method, as I don't think either method is optimal. I like to say you can't argue with results. If you are constantly going 3 for 3 in each lift and setting PRs in each meet, you are doing something right. But if you are struggling with some aspect — getting only your openers or feeling strong in the gym but weak on meet day, it may well be due to your warm-up routine. I strongly believe in these methods, I use them myself, and almost all of the fellow lifters, teammates, and students I have tried these out with ultimately find them to be effective (even if they are initially skeptical). Try them out as they are written, and then start to modify them as you see fit, but always use the end result to judge their merit. That guideline of being centered on results will serve you well during your lifting career.

Proper weight selection can make or break your competition experience.
Photo credit: Doug Jantz Photography

Chapter 16

Weight Selection

Making intelligent weight selections in a powerlifting competition has a huge impact on what your total is. It is not always the strongest person who wins a powerlifting competition but often the one who chooses the best weights on that day. I have been on both the winning and losing end of that fact. I have beaten people who were stronger than me, meaning if they had made proper weight selection (such as trying to lift just 5 lbs more than my attempts), they would have won, but due to poor weight selection, I out totaled them. I have also been hung up on lifting a certain weight, and the problem with that is if you go just for that weight and miss it, often a weaker person will end up lifting more than you. I am not crying foul — I was beaten fair and square on that day, but it is eye opening to have a weaker person beat you in a powerlifting competition which is supposed to separate out the strong from the strongest.

When you enter into a powerlifting competition, it is important to set your mind for the competition. What is your primary goal in competing? For most people it is to compete against themselves and to continually improve oneself. For others it is to test themselves against other lifters to see where they stand. And for some it is all about winning and beating the other lifters. These separate goals are not mutually exclusive; you can compete to set some PRs and hope to do well at the same time. But it is important for you to know

what your real goal is. Don't just pay that lip service. People know they are "supposed" to say I am competing against myself, but if that is not how you feel, own up to it.

Here is an example of how your mindset might affect how you feel about a powerlifting competition. Let's say you have a previous best bench of 400 lbs. If your main goal is to win the competition, and the competition that shows up that day is pretty light, with a best bench of 320 lbs, you can lift 325 lbs, and you should be very happy since you achieved your goal, which was winning the event. Even though lifting 325 when you have lifted 400 is not very impressive, you still achieved your goal of winning so you should be content.

A powerlifting competition is supposed to separate out the strong from the strongest.

If your goal is to improve your own performance, then you should just focus on that and not worry about what others are lifting. If you lift 410, even if 5 other people beat you and you don't even place in the competition, then you should be happy because you had a good day and you achieved your goals. If you lift 340, even if that wins the competition, you should be very disappointed because 340 is nowhere near what you have lifted in the past.

Goals are generally grouped as either process goals or outcome goals. Process goals focus on the process (such as: I will workout 4 times this week, I will eat right, I will not miss a lift due to a technicality) and they are under the athlete's control. Outcome goals focus on the outcome of the event (I will win, I will place in the top 3, I will win the best lifter award) and they are not in the athlete's control. I could have the

Process goals are under the athlete's control.

best day I have ever had, but if the world record holder in my weight class shows up at the competition, and if he does well, I will lose, but that doesn't mean I should be disappointed. In general it is best to focus on process goals since you have more power over them. Hopefully continually achieving your process goals will lead to a good outcome.

In all powerlifting events you have 3 tries at each lift. Once you decide you want to lift a certain weight, you can never decrease that weight — you can only repeat it or go up. This means that basically everyone starts with a weight that they hope and believe they can do, and assuming they are successful with it, the lifter will increase the weight and try again. As you enter a meet, you want to decide if you want to be aggressive with your weight selection or conservative. In my experience being conservative usually yields better results. Occasionally when a lifter is aggressive, he has a great day when everything falls into place, but more often than not the lifter misses a few key lifts, gets disappointed, and winds up with a poor total.

The Opener

The opener is the first lift of the event, and it is usually a weight you feel quite confident you can do, even on a bad day. This is true whether you are going to be conservative or aggressive with your weight selection. Sometimes people take the attitude that if they can't lift a certain weight they don't care where they place or even if they bomb out, but if you feel like this I still think the opener should be conservative and not your goal weight for the meet. Making your opener builds confidence and calms your nerves. In addition you may not be as thoroughly warmed up as you thought so the opener really serves as a final warm-up set to the goal weight. You can get a feel for that particular bar, squat rack, bench press, and making your opener will keep you in the meet. If you miss all three lifts for an event, you "Bomb Out," which means you are disqualified from the event. It also means that any previous lifting you did that day does not count. If you set a record in the squat but then bomb out on the bench press the record is gone in the squat. Some people argue that you should start heavy with your opener, because that way, if you miss it, you still have two more chances to successfully lift it. There is some logic to that, but if it takes you three tries to lift the weight, either your form sucks or you are making some simple mistakes that are disqualifying the lift. Instead start light with the opener, get in the meet and relax, and then go heavier on the second attempt. If you miss that one, you still have another shot at it on your third attempt.

Making your opener builds confidence and calms your nerves.

The general rule of thumb is that your opener should be a weight that you can perform a triple with if necessary. If you like to make big jumps with your attempts, it might even be lighter than that. In order to select an opener, people often think about their overall goal

weight that they would like to lift in the meet. Assuming that this is the third attempt, the opener is normally 10–20% less than that. If it is more than 20% less, the opener is pretty low, and that means you will be making very big jumps. However, if the opener is more than 90% of your goal lift, then it may be too heavy. If you have overestimated your goal lift you might not be able to successfully complete the opener if you are not having a good day. The opener should be something you have done in the gym several times in your training cycle leading up to the meet, or, if you are a veteran, it should be a weight you are very confident you can lift. Kirk Karwoski's rule of 20 lbs less than your best triple is a good guideline in most instances (this works better with class II and above lifters).

	Female		Male	
Exercise	**Goal Weight**	**Opener**	**Goal Weight**	**Opener**
Squat	150	120–130	365	315–325
Squat	200	160–170	455	375–395
Squat	250	200–215	545	465–495
Bench	115	95–100	250	210–220
Bench	160	130–140	340	295–315
Bench	200	160–170	425	365–385
Deadlift	185	145–160	405	335–355
Deadlift	275	225–240	500	420–450
Deadlift	350	295–315	625	535–565

and third attempts
that wins meets.

Second Attempt

Once you make your opener, it is time for the second attempt. Remember that in most organizations you have about a minute to turn in your second attempt after the first one. Assuming you made the opener, you are going to go up in weight, but the question is, “How much?” The goal is to increase the weight enough so it has a significant impact on your total but not so much that you probably won’t be successful with the weight.

Deciding if you want to be aggressive or conservative with your weight selection will have the biggest impact on your second attempt numbers. If you want to be aggressive, you should go for a pretty big lift on your second attempt, perhaps a 5–10 lb PR (personal record) for you or the weight you were hoping to lift. This way, if you barely miss that weight or make a technical mistake, you can repeat it on your third try, and you will have another shot at that weight. If you do make your second, then the third attempt will be like a bonus round, and you can go for whatever you want.

Personally I recommend lifters take a more conservative approach. It is making the second and third attempts that wins meets, and you gain a lot of momentum and confidence by going 2 for 2 in your lifts heading into your third attempt. I don’t believe you need to plan for the possibility of missing the lift and giving yourself another try at it. If you miss it for strength reasons, it is rare (not unheard of) to come back and make the lift a second time around when you are more tired because you already failed at it once. Making a lift after you failed with it is harder with raw lifting. I also don’t believe you should plan to make a technical mistake. They do happen, but powerlifting is not complicated, and you should plan on always following the rules on all lifts. I believe second attempts should be heavy weights but weights that you are quite confident you can lift. I don’t believe they should be long shot weights that you are hopeful you can lift but you are not really

sure. Save that for later. If you go the conservative route and you still end up missing your second attempts (due to lack of strength), that means you do not have an accurate assessment of how strong you really are. It probably means that your opener should have been your second attempt and you should have opened up lighter.

If you were not successful on the opener, then the obvious decision is to simply repeat that weight. Hopefully it was just a technical or nerves related mistake and not a strength mistake; you might still go up in weight. If it was just a simple, fixable mistake, it is possible to increase the weight for your second even though you didn't get your first lift, but I would be very cautious doing so. If you miss your opener for a simple reason but increase the weight anyway, then if you miss your second attempt due to strength, you are stuck with that heavier weight for your third attempt, and you have a very good shot at bombing out, which really should never happen, barring injury. If you do decide to increase your opener after missing it, I would increase it only a small amount (5–10 lbs on bench, 5–20 lbs on squats and deadlifts) for your second. You have made one mistake already; it is time to be more cautious and not make a second mistake by being too aggressive with the weight.

If you missed your opener due to strength reasons, then you are in rough shape. The good news is you have 2 more tries to be successful with it; the bad news is the weight is not getting any lighter. For the future it should be a lesson that you have overestimated your strength and/or you did not have a good idea of how the lifts would actually be performed. This is most common by not realizing how deep the squat needs to be to count, by not realizing that a pause bench is much harder than a bounce bench, by not realizing that you

If you missed your opener due to strength reasons, then you are in rough shape.

shouldn’t start with your all-time high on deadlifts as your opener, and by not realizing that a curl up against the wall is vastly harder than doing it standing upright even if you are strict. However, my saying “I told you so” doesn’t do you any good at the moment. All you can do now is to try to maximize your performance by trying some of the simple fixes listed in each chapter on the lifts for how to immediately fix common mistakes. If they don’t work, then you are out of luck, but you can come back next time more properly prepared and ready to lift. If it means anything to you personally, I bombed out of my very-first-ever powerlifting meet in high school. I grossly overestimated my bench because I was used to a friend spotting me and saying “It was all you” as they did an upright row to help me bench (I am not blaming them; it was my fault but this is a common problem especially with young lifters). Maxing out the day before to see what I could do probably didn’t help the situation. But 40+ meets later, I haven’t bombed out since, so one bad day doesn’t have to lead into another.

Back to the original question. If you make your opener, how much do you go up? It depends on the lift in question, the amount of weight you are actually lifting, and, of course, how easy the opener was. If you are following the guidelines given here, the opener should hopefully be pretty easy. A simple test is you should feel like you could have doubled it if you had to. If it felt hard then you may not have been warmed up enough, or it may just have been a little heavy. The good news is that even if you weren’t warmed up enough for the opener, the opener can serve as a final warm-up, which should get you ready to go for the second attempt.

On the squat, males should up their opener by 10–50 lbs for their second attempt or increase it by 5–15%. Do not increase the squat by 5 lbs unless it was

One bad day doesn’t have to lead into another.

ungodly hard as that will just tire you out. A 7–10% jump, or 20–40 lbs, seems to work very well for most people. If you are already planning out the third attempt, generally the weight increase from the first to the second attempt should be equal to or often higher than the weight increase from the second to the third attempt. Don't do it the other way around and go up 15 lbs for the second and then 45 lbs for the third. Females should increase their opener by 5–35 lbs, with 15–30 lbs seeming to be the best choice for most women lifters. Even females should also not increase the squat by only 5 lbs unless it was super hard or they are only squatting like 60 lbs. Percentage-wise, females should make the same or just slightly less of an increase compared to males.

On the bench press you will increase the weights less than the squat simply due to the poundage involved. A 40 lb jump on the bench press would be considered very large; a 40 lb jump on the squat would not. Males should increase their opener on the bench by 5–35 lbs for their second attempts, with 10–25 lbs working well in most situations. Again limit a 5 lb jump unless you really feel it is all you have left, which it should not be with a properly planned opener. Percentage-wise, most lifters go with a 5–10% increase on the bench; 7% seems just about right in a lot of situations. Females should increase their bench by 5–20 lbs; it will be more common (although still not ideal) for females to increase the bench by only 5 lbs as that may make the difference for them. 10–15 lbs seems to work best for females.

On the deadlift you will generally make the biggest jumps in weight. Most lifters jump 20–60 lbs from their openers up to their second attempts when pulling, which is usually at or just below 10% of their lift. Females should increase the deadlift by 10–40 lbs; again 10% works well.

On the curl you will make the smallest jumps in weight. A 10 lb jump can make a big difference on the curl. Most lifters will jump 5–20 lbs on the

curl, with 10–15 lbs being the most common. Females should increase their curl 5–15 lbs. There should be 1.25 lb plates available at a curl competition (I think they should be available in all the events); don't be afraid to use them. For example, a 10 lb jump might be too much, but a 7.5 lb jump might be just right. A 5–10% percent jump seems to work the best for curls.

> ***Successful third attempts are what powerlifting is all about.***

Third Attempts

Now it is time to have some fun. Successful third attempts are what powerlifting is all about. Whether you decided to be aggressive or conservative, you should still be throwing around some serious weight on your third attempt. The only possible exception is if you are just going for the win and you are being very conservative since you are significantly stronger than your competition.

The third attempt should be a goal weight for you that you think you are capable of at that time, and/or it should be a new personal record for you. However, don't expect that just because it is the third attempt and people will be yelling at you that you will magically lift 100 lbs more than you have ever done before. You still want to be realistic, when selecting third attempts. You still want to have a pretty good shot (at least 50/50 if not better) that you will be successful on your third attempt. Failures at trying pie-in-the sky numbers are not what boosts up your total; making 2 or 3 third attempts does. In addition, if you can make your first third attempt, that begins to build very good momentum as you get ready for the next event. Making that next third attempt really builds momentum as the chance for a 9 for 9 day (every lifter's goal) becomes more of a reality. There is something I call "riding the wave" of successful attempts, which makes you feel good and strong and unstoppable. However, once you have failed at something, it can be defeating. Some lifters get more fired up once they fail, and they can come back on the next events and do well, but that is easier said than done.

On the squat, for the third attempt most lifters should increase their squat by 5–40 lbs, with 15–30 lbs being the norm. Females should jump 5–20 lbs, with 10–15 lbs being the norm. Percentage-wise a jump of 2.5%–7.5% works well

On the bench, a weight increase of 5–30 lbs works well, with a jump of

10–20 lbs being common. An increase of 2.5–5% is good. Females should increase the bench 5–20 lbs, with 5–15 lbs being common; even 2.5 lbs would be acceptable for a new max if that is available.

On the deadlift, increasing the weight 10–50 lbs seems good, with jumps of 20–40 lbs or 5–7.5% working well. Females should increase the deadlift by 5–30 lbs for their third attempts; 10–25 lbs is common.

On the curl, small increases in weight for the third attempt are in order. For males increase the weight 2.5–15 lbs, with 5–10 lbs being common. Females should increase the weight 2.5–10 lbs; 5 lbs works well.

Those lifts can serve as a starting point for choosing your attempts for your next meet.

If you have already competed in a powerlifting competition, you should have some successful lifts to your credit. Those lifts can serve as a starting point for choosing your attempts for your next meet. The first goal should be to set a PR if at all possible. This PR should come either on the second attempt or the third attempt. It might be on the second attempt if it has been a relatively long time (6 months plus) since your last competition, if you feel like you have made a ton of progress in your training, or your PR is from a second attempt in your last competition (meaning you missed your third). This PR should be 5–10 lbs heavier than your old record, no more (the curl should be 2.5–5 lbs). You want to walk away setting a new PR and thus increasing your total.

For example, let's say in your last meet, on the squat, you went 500 for the opener, 540 second attempt, and 560 third attempt, which was not successful. The 540 squat was your highest ever competition squat; that means that 540 is your PR. Now you are getting ready for another for competition. You want to evaluate why you missed 560 and how realistic it is

for you to squat 550, 560, or 570 at this meet. Be honest with yourself; don't be scared, but don't be foolishly optimistic, either.

You essentially have two choices when it comes to weight selection from the above example. It depends on how aggressive you want to be, how your training has been going, and again why you missed 560: Was it strength or a technicality? If you want to be conservative, if you missed 560 due to strength and your training has only been going okay, I would choose the following lifts:

490 opener, 530 second, 545–550 third

You have a very realistic shot at 550 or so, which increases your PR by 10 lbs, which is real progress. Remember 10 lbs every 4 months is a 30 lb increase each year; if you could do that on each lift, that is a 90 lb per year increase on your total, which is very significant. Fight and claw for the extra 5 or 10 lbs; they make a big difference in the long run.

If you decide you want to be aggressive, if your training has been going well, if you just barely missed 560 last time, and you really feel very confident you can squat more than 540, then I would choose the following lifts:

505 opener, 545 second, 555–570 third

This way you still set a PR on your second attempt; if you miss your third, at least you walk away with a PR and you have been somewhat successful. When you go for the PR on your second attempt, I would just increase it by 5 lbs unless your progress in training has been through the roof. Then you can see how the 545 feels, and you will have a shot at trying something really big on your third attempt. Of course this method is a bit more aggressive; if you can't do 545, you will only be left with 505, which

will really hurt your total, but if you can't do 545 since that is just a 5 lb PR, that will give you valuable feedback that something is wrong with your training and/or meet preparation. Also if you make a big jump after 545, say up to 570 and you just miss that, you are credited with 545. If someone in a similar situation chose the more conservative route and went with 490, 530, and 550 and was successful with all of their attempts, they have a 5 lb lead going into the benches. This is a case where the strongest person (a very near miss with 570 is probably stronger than success with 550) didn't lift the most weight on that day.

If your PR was made on a third attempt at your last meet, then usually you will want to go for a PR on your third attempt at your next meet. Assuming your lifting went well and you liked your weight selection, just increase all of your lifts by 5–15 lbs, and 15 lbs is stretching it. If your benches looked like this last time

345 opener, 365 second, 385 third

then this time around your attempts could look like this:

350 opener, 370 second, 390 third

With a weight like that, of course, it is tempting to want to go for 400 lbs; who doesn't want to bench 400 lbs if you never have before? But take your time getting there and leading yourself to that weight. If that same person decides to go 340, 370, 400 and misses 400, not only did they not set a PR but their bench is 370 and they have taken 15 lbs off of their last total. Again, lift what you can do on that day.

A 400 lb bench is a goal of many lifters
Photo credit: Doug Jantz Photography

It is possible to get aggressive and set a new PR on your second attempt if your training has been going very well and you think you can do so, even though your last PR was on your third attempt. This is more common with novices to powerlifting as initially their lifts go up by leaps and bounds and they might add 40 lbs to their squat in 6 months or something like that. In this case you would follow the aggressive outline for setting a small PR on your second attempt and then getting aggressive on your third attempt. Using the above bench press scenario, if someone's best was 385 but in training they had tripled that and they really thought they could go significantly heavier, I would select the following weights:

360 opener, 390 second (PR), 405–420 third

The principle is the same. One of your lifts should be a 5–10 lb PR from what you have done in the past, assuming your training justifies that. The question then becomes, “Is that going to be a second attempt, or is that going to be a third attempt?”

You might choose not to go for a PR for several reasons. If you are injured, you have lost weight, your training has been inconsistent, or if the competition is so tight that you can not afford to give away even a few lbs, then you may end up not setting a PR or repeating a weight you have already lifted. But for the majority of people who are competing in this sport to see what they can do, there is very little sense in repeating a weight you have already made successfully in a competition. Lifters rarely cheer and get excited about still being able to lift a certain weight; it is increasing your max that gets one fired up.

If you have never competed before or never attempted heavy singles in the gym, it is harder to select the appropriate weights but not impossible. There is something called a conversion chart, which is a way to estimate how much you can lift for a 1 rep max (or a certain number of reps) if you know you can lift a certain weight for a certain number of reps. In other words it should be pretty obvious that a female who can bench press 100 lbs for a 10 rep max will have a higher 1RM than a female who can bench press 100 lbs for a 3 rep max and fails on the fourth rep. We also know that both of these ladies should be able to do 105 without much doubt, but the chance of them doing 200 lbs for 1 rep is very unlikely. I use those examples to make the point that we all have a version of this conversion chart floating around in our head; we just need to make it a little bit more clear.

The idea behind a conversion chart is that every additional rep you lift (beyond your first rep) has a certain value in terms of weight. That value is

labeled the pounds-per-rep value, and thus every extra rep you perform means you could have lifted that many extra pounds. Here is an example.

If our female in the previous example can bench 100 lbs x 10 reps, and I tell you that the value of every extra rep is worth 4 lbs, that predicts her one rep max to be 136 lbs. You figure that out by calculating how many extra reps she did:

10 reps completed – 1 rep (remember you must do the 1RM 1 time) = 9 extra reps

Then you take the 9 extra reps and multiply it by the pounds per rep value

9 extra reps x 4 lbs/rep = 36 extra lbs

Then you take that number and add it to the original weight lifted

100 lbs lifted + 36 extra lbs = 136 lb 1RM.

– <50 lbs Reps 2 and 3 3 lbs each
Reps 4–12 2 lbs each

1	2	3	4	5	6	7	8	9	10	11	12
40	37	34	32	30	28	26	24	22	20	18	16
42.5	39.5	36.5	34.5	32.5	30.5	28.5	26.5	24.5	22.5	20.5	18.5
45	42	39	37	35	33	31	29	27	25	23	21
47.5	44.5	41.5	39.5	37.5	35.5	33.5	31.5	29.5	27.5	25.5	23.5

50–75 lbs Reps 2 and 3 3.5 lbs each
Reps 4–12 2.25 lbs each

1	2	3	4	5	6	7	8	9	10	11	12
50	46.5	43	40.75	38.5	36.25	34	31.75	29.5	27.25	25	22.75
52.5	49	45.5	43.25	41	38.75	36.5	34.25	32	29.75	27.5	25.25
55	51.5	48	45.75	43.5	41.25	39	36.75	34.5	32.25	30	27.75
57.5	54	50.5	48.25	46	43.75	41.5	39.25	37	34.75	32.5	30.25

60	56.5	53	50.75	48.5	46.25	44	41.75	39.5	37.25	35	32.75
62.5	59	55.5	53.25	51	48.75	46.5	44.25	42	39.75	37.5	35.25
65	61.5	58	55.75	53.5	51.25	49	46.75	44.5	42.25	40	37.75
67.5	64	60.5	58.25	56	53.75	51.5	49.25	47	44.75	42.5	40.25
70	66.5	63	60.75	58.5	56.25	54	51.75	49.5	47.25	45	42.75
72.5	69	65.5	63.25	61	58.75	56.5	54.25	52	49.75	47.5	45.25
75	71.5	68	65.75	63.5	61.25	59	56.75	54.5	52.25	50	47.75

76–100 lbs Reps 2 and 3 4 lbs each
Reps 4–12 3 lbs each

1	2	3	4	5	6	7	8	9	10	11	12
77.5	73.5	69.5	66.5	63.5	60.5	57.5	54.5	51.5	48.5	45.5	42.5
80	76	72	69	66	63	60	57	54	51	48	45
82.5	78.5	74.5	71.5	68.5	65.5	62.5	59.5	56.5	53.5	50.5	47.5
85	81	77	74	71	68	65	62	59	56	53	50
87.5	83.5	79.5	76.5	73.5	70.5	67.5	64.5	61.5	58.5	55.5	52.5
90	86	82	79	76	73	70	67	64	61	58	55
92.5	88.5	84.5	81.5	78.5	75.5	72.5	69.5	66.5	63.5	60.5	57.5
95	91	87	84	81	78	75	72	69	66	63	60
97.5	93.5	89.5	86.5	83.5	80.5	77.5	74.5	71.5	68.5	65.5	62.5
100	96	92	89	86	83	80	77	74	71	68	65

102.5–125 lbs Reps 2 and 3 5 lbs each
Reps 4–12 3.5 lbs each

1	2	3	4	5	6	7	8	9	10	11	12
102.5	97.5	92.5	89	85.5	82	78.5	75	71.5	68	64.5	61
105	100	95	91.5	88	84.5	81	77.5	74	70.5	67	63.5
107.5	102.5	97.5	94	90.5	87	83.5	80	76.5	73	69.5	66
110	105	100	96.5	93	89.5	86	82.5	79	75.5	72	68.5
112.5	107.5	102.5	99	95.5	92	88.5	85	81.5	78	74.5	71
115	110	105	101.5	98	94.5	91	87.5	84	80.5	77	73.5
117.5	112.5	107.5	104	100.5	97	93.5	90	86.5	83	79.5	76
120	115	110	106.5	103	99.5	96	92.5	89	85.5	82	78.5
122.5	117.5	112.5	109	105.5	102	98.5	95	91.5	88	84.5	81
125	120	115	111.5	108	104.5	101	97.5	94	90.5	87	83.5

126–150 lbs Reps 2 and 3 5 lbs each
Reps 4–12 4 lbs each

1	2	3	4	5	6	7	8	9	10	11	12
127.5	122.5	117.5	113.5	109.5	105.5	101.5	97.5	93.5	89.5	85.5	81.5
130	125	120	116	112	108	104	100	96	92	88	84
132.5	127.5	122.5	118.5	114.5	110.5	106.5	102.5	98.5	94.5	90.5	86.5
135	130	125	121	117	113	109	105	101	97	93	89
137.5	132.5	127.5	123.5	119.5	115.5	111.5	107.5	103.5	99.5	95.5	91.5
140	135	130	126	122	118	114	110	106	102	98	94
142.5	137.5	132.5	128.5	124.5	120.5	116.5	112.5	108.5	104.5	100.5	96.5
145	140	135	131	127	123	119	115	111	107	103	99
147.5	142.5	137.5	133.5	129.5	125.5	121.5	117.5	113.5	109.5	105.5	101.5
150	145	140	136	132	128	124	120	116	112	108	104

151–200 lbs Reps 2 and 3 — 6 each
Reps 4–12 — 4.5 each

1	2	3	4	5	6	7	8	9	10	11	12
155	149	143	138.5	134	129.5	125	120.5	116	111.5	107	102.5
160	154	148	143.5	139	134.5	130	125.5	121	116.5	112	107.5
165	159	153	148.5	144	139.5	135	130.5	126	121.5	117	112.5
170	164	158	153.5	149	144.5	140	135.5	131	126.5	122	117.5
175	169	163	158.5	154	149.5	145	140.5	136	131.5	127	122.5
180	174	168	163.5	159	154.5	150	145.5	141	136.5	132	127.5
185	179	173	168.5	164	159.5	155	150.5	146	141.5	137	132.5
190	184	178	173.5	169	164.5	160	155.5	151	146.5	142	137.5
195	189	183	178.5	174	169.5	165	160.5	156	151.5	147	142.5
200	194	188	183.5	179	174.5	170	165.5	161	156.5	152	147.5

201–250 lbs Reps 2 and 3 — 7.5 each
Reps 4–10 — 5.5 each
Reps 11–12 — 4.5 each

1	2	3	4	5	6	7	8	9	10	11	12
205	197.5	190	184.5	179	173.5	168	162.5	157	151.5	147	142.5
210	202.5	195	189.5	184	178.5	173	167.5	162	156.5	152	147.5
215	207.5	200	194.5	189	183.5	178	172.5	167	161.5	157	152.5
220	212.5	205	199.5	194	188.5	183	177.5	172	166.5	162	157.5
225	217.5	210	204.5	199	193.5	188	182.5	177	171.5	167	162.5

230	222.5	215	209.5	204	198.5	193	187.5	182	176.5	172	167.5
235	227.5	220	214.5	209	203.5	198	192.5	187	181.5	177	172.5
240	232.5	225	219.5	214	208.5	203	197.5	192	186.5	182	177.5
245	237.5	230	224.5	219	213.5	208	202.5	197	191.5	187	182.5
250	242.5	235	229.5	224	218.5	213	207.5	202	196.5	192	187.5

251–300 lbs Reps 2 and 3 — 8.5 each
Reps 4–10 — 6 each
Reps 11–12 — 5 each

1	2	3	4	5	6	7	8	9	10	11	12
255	246.5	238	232	226	220	214	208	202	196	191	186
260	251.5	243	237	231	225	219	213	207	201	196	191
265	256.5	248	242	236	230	224	218	212	206	201	196
270	261.5	253	247	241	235	229	223	217	211	206	201
275	266.5	258	252	246	240	234	228	222	216	211	206
280	271.5	263	257	251	245	239	233	227	221	216	211
285	276.5	268	262	256	250	244	238	232	226	221	216
290	281.5	273	267	261	255	249	243	237	231	226	221
295	286.5	278	272	266	260	254	248	242	236	231	226
300	291.5	283	277	271	265	259	253	247	241	236	231

301–350 lbs Reps 2 and 3 — 10 each
Reps 4 and 5 — 7.5 each
Reps 6–10 — 6.5 each
Reps 11–12 — 5.5 each

1	2	3	4	5	6	7	8	9	10	11	12
305	295	285	277.5	270	263.5	257	250.5	244	237.5	232	226.5
310	300	290	282.5	275	268.5	262	255.5	249	242.5	237	231.5
315	305	295	287.5	280	273.5	267	260.5	254	247.5	242	236.5
320	310	300	292.5	285	278.5	272	265.5	259	252.5	247	241.5
325	315	305	297.5	290	283.5	277	270.5	264	257.5	252	246.5
330	320	310	302.5	295	288.5	282	277.5	269	262.5	257	251.5
335	325	315	307.5	300	293.5	287	282.5	274	267.5	262	256.5
340	330	320	312.5	305	298.5	292	287.5	279	272.5	267	261.5
345	335	325	317.5	310	303.5	297	292.5	284	277.5	272	267.5
350	340	330	322.5	315	308.5	302	297.5	289	282.5	277	272.5

351–400 lbs

Reps 2 and 3 — 11.5 each
Reps 4 and 5 — 9 each
Reps 6–10 — 7 each
Reps 11–12 — 6 each

1	2	3	4	5	6	7	8	9	10	11	12
355	343.5	332	323	314	307	300	293	286	279	273	267
360	348.5	337	328	319	312	305	298	291	284	278	272
365	353.5	342	333	324	317	310	303	296	289	283	277
370	358.5	347	338	329	322	315	308	301	294	288	282
375	363.5	352	343	334	327	320	313	306	301	293	287
380	368.5	357	348	339	332	325	318	311	306	298	292
385	373.5	362	353	344	337	330	323	316	311	303	297
390	378.5	367	358	349	342	335	328	321	316	308	302
395	383.5	372	363	354	347	340	333	326	321	313	307
400	388.5	377	368	359	352	345	338	331	326	318	312

401–450 lbs

Reps 2 and 3 — 13 each
Reps 4 and 5 — 10.5 each
Reps 6–10 — 8 each
Reps 11–12 — 7 each

1	2	3	4	5	6	7	8	9	10	11	12
405	392	379	368.5	358	350	342	334	326	318	311	304
410	397	384	373.5	363	355	347	339	331	323	316	309
415	402	389	378.5	368	360	352	344	336	328	321	314
420	407	394	383.5	373	365	357	349	341	333	326	319
425	412	399	388.5	378	370	362	354	346	338	331	324
430	417	404	393.5	383	375	367	359	351	343	336	329
435	422	411	398.5	388	380	372	364	356	348	341	334
440	427	416	403.5	393	385	377	369	361	353	346	339
445	432	421	408.5	398	390	382	374	366	358	351	344
450	437	426	413.5	403	395	387	379	371	363	356	349

451–500 lbs

Reps 2 and 3 — 15 each
Reps 4 and 5 — 12 each
Reps 6–10 — 9
Reps 11–12 — 7 each

1	2	3	4	5	6	7	8	9	10	11	12
455	440	425	413	401	392	383	374	365	356	349	342

460	445	430	418	406	397	388	379	370	361	354	347
465	450	435	423	411	402	393	384	375	366	359	352
470	455	440	428	416	407	398	389	380	371	364	357
475	460	445	433	421	412	403	394	385	376	369	362
480	465	450	438	426	417	408	399	390	381	374	367
485	470	455	443	431	422	413	404	395	386	379	372
490	475	460	448	436	427	418	409	400	391	384	377
495	480	465	453	441	431	423	414	405	396	389	382
500	485	470	458	446	436	428	419	410	401	394	387

501–600 lbs

Reps 2 and 3 — 17.5 each
Reps 4 and 5 — 14 each
Reps 6–10 — 11 each
Reps 11–12 — 8 each

1	2	3	4	5	6	7	8	9	10	11	12
510	492.5	475	461	447	436	425	414	403	392	384	376
520	502.5	485	471	457	446	435	424	413	402	394	386
530	512.5	495	481	467	456	445	434	423	412	404	396
540	522.5	505	491	477	466	455	444	433	422	414	406
550	532.5	515	501	487	477	465	454	443	432	424	416
560	542.5	525	511	497	487	475	464	453	442	434	426
570	552.5	535	521	507	497	485	474	463	452	444	436
580	562.5	545	531	517	507	495	484	473	462	454	446
590	572.5	555	541	527	517	505	494	483	472	464	456
600	582.5	565	551	537	527	515	504	493	482	474	466

601–700 lbs

Rep 2 — 22.5 lbs
Rep 3 — 20 lbs
Reps 4–5 — 17.5 lbs
Reps 6–10 — 12.5 lbs
Reps 11–12 — 10 lbs

1	2	3	4	5	6	7	8	9	10	11	12
610	587.5	567.5	550	532.5	520	507.5	495	482.5	470	460	450
620	597.5	577.5	560	542.5	530	517.5	505	492.5	480	470	460
630	607.5	587.5	570	552.5	540	527.5	515	502.5	490	480	470
640	617.5	597.5	580	562.5	550	537.5	525	512.5	500	490	480
650	627.5	607.5	590	572.5	560	547.5	535	522.5	510	500	490
660	637.5	617.5	600	582.5	570	557.5	545	532.5	520	510	500

670	647.5	627.5	610	592.5	580	567.5	555	542.5	530	520	510
680	657.5	637.5	620	602.5	590	577.5	565	552.5	540	530	520
690	667.5	647.5	630	612.5	600	587.5	575	562.5	550	540	530
700	677.5	657.5	640	622.5	610	597.5	585	572.5	560	550	540

701–800 lbs

Rep 2 — 25 lbs
Rep 3 — 22 lbs
Rep 4–5 — 20 lbs
Rep 6–10 — 13.5 lbs
Rep 11–12 — 11 lbs

1	2	3	4	5	6	7	8	9	10	11	12
710	685	663	643	623	609.5	596	582.5	569	555.5	544.5	533.5
720	695	673	653	633	619.5	606	592.5	579	565.5	554.5	543.5
730	705	683	663	643	629.5	616	602.5	589	575.5	564.5	553.5
740	715	693	673	653	639.5	626	612.5	599	585.5	574.5	563.5
750	725	703	683	663	649.5	636	622.5	609	595.5	584.5	573.5
760	735	713	693	673	659.5	646	632.5	619	605.5	594.5	583.5
770	745	723	703	683	669.5	656	642.5	629	615.5	604.5	593.5
780	755	733	713	693	679.5	666	652.5	639	625.5	614.5	603.5
790	765	743	723	703	689.5	676	662.5	649	635.5	624.5	613.5
800	775	753	733	713	699.5	686	672.5	659	645.5	634.5	623.5

801 lbs+

Rep 2 — 30 lbs
Rep 3 — 25 lbs
Reps 4–5 — 22.5 lbs
Reps 6–10 — 15 lbs
Reps 11–12 — 12 lbs

1	2	3	4	5	6	7	8	9	10	11	12
810	780	755	732.5	710	695	680	665	650	635	623	611
820	790	765	742.5	720	705	690	675	660	645	633	621
830	800	775	752.5	730	715	700	685	670	655	643	631
840	810	785	762.5	740	725	710	695	680	665	653	641
850	820	795	772.5	750	735	720	705	690	675	663	651
860	830	805	782.5	760	745	730	715	700	685	673	661
870	840	815	792.5	770	755	740	725	710	695	683	671
880	850	825	802.5	780	765	750	735	720	705	693	681
890	860	835	812.5	790	775	760	745	730	715	703	691
900	870	845	822.5	800	785	770	755	740	725	713	701

Value of the Conversion Chart

Listed previously are various conversion charts. Their purpose is to estimate how much weight you can lift for a certain number of reps if you already know how much weight you can lift for another certain number of reps. For example, if you knew for sure a male could bench 290 for 1 rep, this chart estimates he could bench 255x6 reps. It is not an exact science, but it tends to workout pretty well in the majority of circumstances.

This chart highlights why it is so important to fight and struggle for each and every rep in a set. If this same male can move his bench up to 255x8, the chart estimates that now he might be able to bench press 305–310 lbs. That means the ability to do those extra two reps should result in an improvement in his one rep max of 15–20 lbs or so.

In order to compare two sets, the form and technique must be the same on both sets. If you are very strict on one set and then sloppy on another, the two sets are no longer comparable. Of course, if you receive help from a spotter you cannot claim that you are doing those reps yourself, so only use reps you can do on your own.

Problems with Conversion Charts

Conversion charts are not perfect, and they estimate some people's ability better than others. Generally the more skilled a lifter you are, the higher your conversion is, meaning that each rep you complete is worth more additional pounds. However, the more skilled you are, the more weight you are lifting and the additional weight has a higher conversion so it balances out a bit.

Large muscles and compound exercises have a bigger conversion than small muscles. Performing 3 reps on the leg press is more valuable than 3 reps on a tricep kickback. A leg press of 200 lbs x 5 will generally have a higher conversion than a bench press of 200 lbs x 5 reps. This means that if you figure out your pounds per rep for one exercise, that does not mean that is your pounds per rep on every exercise. For example, if you calculate that each rep you complete on the bench is worth about 7 pounds, that does not mean that every rep you hit on the squat or the deadlift is also worth about 7 pounds. It is different for each exercise.

Men generally have a higher conversion chart than women. Women tend to be better at reps than men; men are better at one rep maxes. A lot of this is due to practice; men tend to practice lifting heavy more regularly. My guess is that female powerlifters have a more equal conversion with male lifters.

There is some variability in the conversion chart itself. As you can see, each rep is not weighted equally. It takes more strength to go from lifting a weight once to being able to do a double with it, than to go from lifting a weight 10 times and now getting 11 reps. The closer the weight is to your one rep max, the more effort it takes to do another rep with it. As noted above, the more weight you are lifting, the more valuable each of the reps is. I have divided the weights into 50 or 100 lb blocks; however, at the ends of the blocks sometimes there is not a perfect overlap. If you look at the chart for 405–450 lbs, if you lift 363 pounds 10 reps the chart estimates that you can

lift 450 pounds for 1 time. However, on the following chart (455–500 lbs), if you can do 363 lbs for 10 reps it estimates that you can do just more than 460 lbs, a 10 lb difference. That is a flaw in the chart; I realize that. Even so, it is only approximately a 10 pound difference, and if we can be within 10 or even 20 lbs of the one rep max based off of a set of 10, I am very comfortable with that level of error. The fewer the number of reps you perform, the more accurate the chart is.

The fewer the number of reps you perform, the more accurate the chart is.

Let's say you look at this chart and your first reaction is it is all wrong and the sets you do don't seem to line up with the maxes predicted; either you can lift a lot more weight than estimated or a lot less. First, I would ask if your form is equal on both sets. Is your intensity equal on both sets? Was your level of fatigue equal (hopefully zero) on both sets? This will help ensure you are comparing apples to apples. Even if that is the case, it is still possible to throw away the charts and to come up with your own conversion chart that works for you on a specific exercise. All you need to know is the weight you lifted and reps you completed on two challenging sets that you did on a certain exercise. Each set needs to be a different amount of weight and a different amount of reps.

For example, imagine a person could bench 240 for 10 reps and they could also bench 290 for 2 reps. From that information, we can figure out what their own conversion chart is. First, you take the difference in weight on the two sets (290 – 240 = 50 lbs). Then you find the difference in reps between the two sets (10 – 2 = 8 reps). Then you divide the number of reps into the difference in weight to find the pounds per rep value (50 lbs/8 reps = 6.25 lbs/rep). What that means is that each rep that lifter does on the bench

press is valued at approximately 6.25 lbs. This means they should be able to be successful with a 1 rep max of 296 but probably not 300; they should be able to do a set of 8 for 252.5, and they should be able to finish a set of 12 with 227.5 lbs. Not only does finding one's conversion chart help you predict your one rep max, but it also helps you predict the appropriate weight for almost all heavy sets.

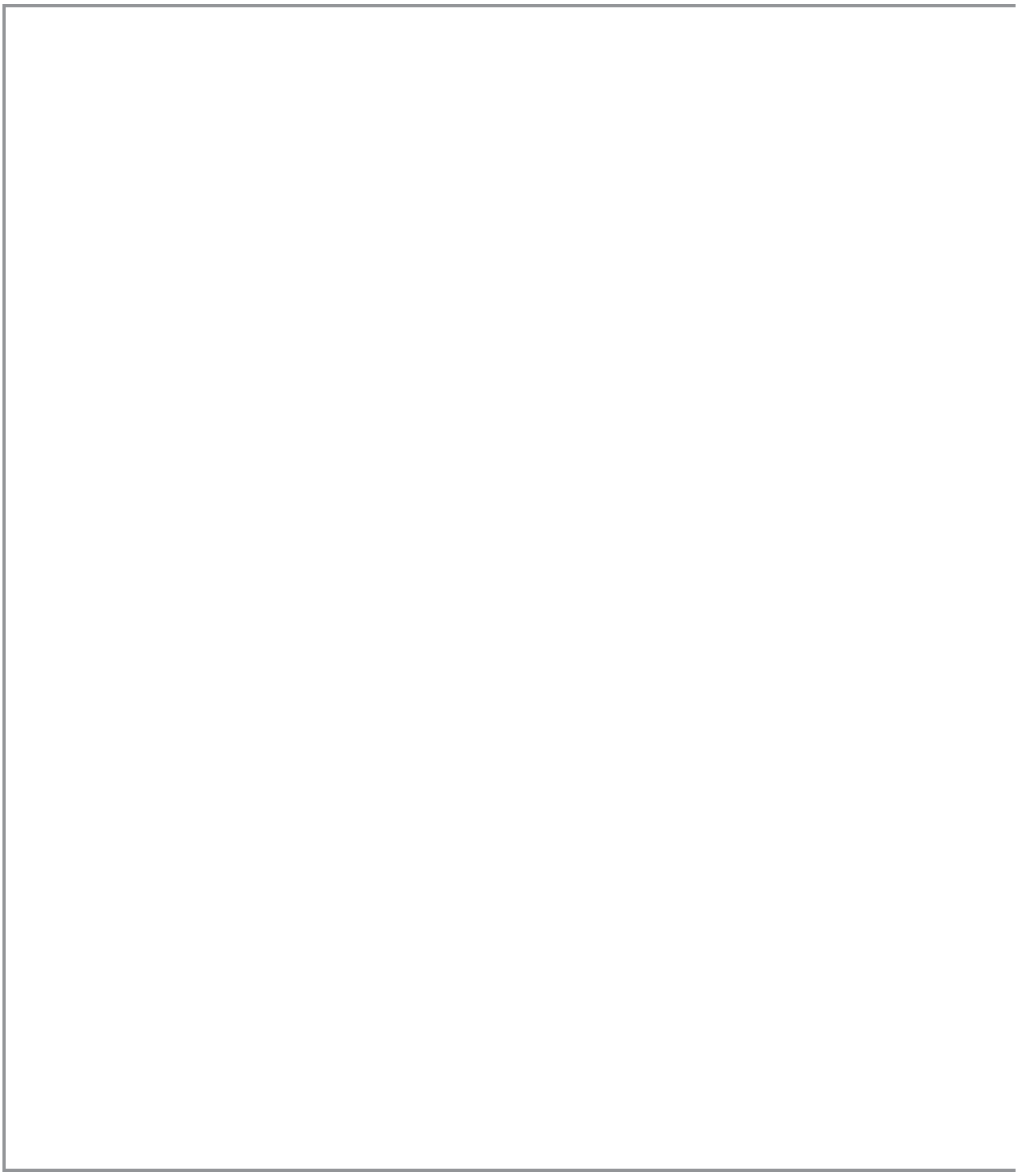
Keep in mind that figuring out your conversion chart is good only for a certain exercise. The same lifter in the above example would need to figure out his conversion for the deadlift; it would not necessarily be 6.25 lbs a rep and would probably be higher. As a side note, do not round up when looking at the conversion chart. It is tempting to see 363 lbs, which may be hard to put on the bar, and go for 365, but sometimes those 2 lbs can be enough to make you miss the lift.

The whole purpose of the conversion chart is to help you, the lifter, predict how much weight you may be able to lift in a competition. Keep in mind that it is predicting your one rep max, and, therefore, that number should be close to your third attempt, not your opener. The opener is normally something you can do for 3–5 reps if you had to, as explained previously. Also don't forget that sometimes the form in the gym is not always the same as the form in a competition, even for powerlifters. For example, on a set of 8 reps for squats you might not hit proper depth on all of them, which would make the set easier and thus over predict your max. On the bench you may not pause your reps, which would make it easier, and on deads if you bounce the weight off the floor, that will

The other benefit of knowing your pounds per rep value is that it helps motivate you to achieve additional reps with a certain weight.

make the set less accurate in predicting how much weight you can lift. Of course, in a competition you are (hopefully) a little more fired up so sometimes that compensates for the slight changes in form.

The other benefit of knowing your pounds per rep value is that it helps motivate you to achieve additional reps with a certain weight, which, in turn, improves your one rep max. Using the example above, our lifter could bench 290x2. If after training that lifter can now bench 290x4 and each rep is worth 6.25 lbs, that lifter's one rep max just went up approximately 12.5 lbs. This is why you fight and claw for those extra reps; when you can perform more reps at a lighter weight, your 1RM is going to go up.



Companies such as Elite Fitness Systems produce high-quality powerlifting equipment

Photo credit: Elite Fitness Systems

Chapter 17

Powerlifting Gear

Note: Sometimes the word "gear" is slang for someone using steroids, e.g., he is on "gear"; however, in this chapter "gear" is referring to different garments and equipment that a powerlifter might use in training and/or competitions.

There are different kinds of powerlifting gear, or equipment, that lifters will use in training and perhaps in competition. Sometimes the purpose of the gear is to help increase the safety of the lifter. Sometimes the purpose is to help the lifter lift more weight, and sometimes it is a combination effect. Different federations have different rules about what is allowed in a competition. In this chapter I will provide some basic information about what kind of gear you can wear, but be sure to know the specific rules for your own organization.

Lifters can choose from a variety of colors and styles to suit their personal taste
Photo credit: Adidas

Singlet – If you choose to compete in a competition, you will be required to wear a singlet, which is a one-piece outfit similar to what wrestlers wear, which is like a tank top and shorts combined. You will wear a T-shirt underneath the singlet. Generally it is mandatory to wear a T-shirt on the squat (and a good idea in general), and a T-shirt may be necessary in the bench press. The T-shirt is usually optional in the deadlift. The purpose of the singlet is to make it easier for judges to look at certain areas and see infractions, for example, how deep you are squatting or if your butt comes up during a bench press. A squat suit and deadlift suit counts as a singlet, but usually those things are so tight you will not want to wear them when you bench press and you will still need to have a regular singlet. If you lift raw the singlet will be the primary part of your outfit. You can normally have singlets in different colors and designs. Some organizations have rules about how long the legs can be, but most realize that the length of the legs makes

no difference if the singlet is not supportive. You can find a wide variety of singlets by searching online for wrestling singlets, and you can often pick out the colors and style that suit you best. Singlets usually cost around $30–50; you can also get them customized with words or designs if you wish. Most lifters choose to wear regular gym clothes when training and then don the singlet for the competition. If you are a bit anal (like I am), you may want to try wearing your singlet out a few times as it can slightly change the feel of the lift (mostly the squat) when you have it on. Granted you will get some odd looks in the gym if you start lifting in a singlet; if that bothers you, just put your singlet on under your regular gym clothes to still get the feel of it without standing out from the crowd.

(Left) A lever powerlifting belt

(Right) A single prong powerlifting belt

Belt – A lifting belt is the second most important piece of equipment to a powerlifter, behind the singlet, and most powerlifters will actually use their belts more than the singlets because they will use the belt in training and wear a singlet only during the competition. Lifting belts are legal to wear in basically all powerlifting federations, even raw organizations. Generally the belt cannot be more than 4 inches wide and 13mm thick (some belts sold at sporting goods stores are 6 inches wide), and it cannot have padding or anything stuffed in the belt. Also depending on the organization Velcro belts

are often illegal, although people don’t usually wear them because they provide less support than leather belts. The belt can be either a buckle belt or a lever belt. A buckle belt provides you versatility with regard to how tight you want the belt to be on that particular day or set, but it can be a pain to get the belt very tight or even to undo it. A lever belt will snap in position very easily, but it is not that easy to adjust once you set it (it takes about 5 minutes); thus it is not ideal if you like to constantly change how tight your belt feels from set to set, or if you share a belt with someone.

A belt is most valuable on the squat and the deadlift, as those two lifts place reasonable stress on the core and lower back. Some people also like to wear their belt when they bench press, and others do not; that is more a matter of personal preference.

Belts are not mandatory and there are successful lifters who have competed without them. Many Olympic lifters choose to lift without a belt, and they are capable of lifting tremendous loads. However, the majority of lifters do choose to wear a belt when competing. Some experts feel that over-reliance on a belt will actually make the core muscles weaker because the belt is there, taking up some of the load. This does make some sense. Others feel it is important to wear the belt because you have to learn how to use the belt. Belts do help maintain trunk stability through increased intra-abdominal pressure. I think you can appeal to both camps by following these guidelines, which are pretty standard when it comes to using a belt.

- Use a belt on heavy sets only. Don’t use a belt for warm-up sets or light sets (unless you are dealing with a current or reoccurring injury). Heavy is defined by you; it might be anything that is 5 reps or less, or it might be a certain weight, say a 400 lb squat, whatever seems to work for you.
- Use a belt on exercises that actually affect the lower back. That includes things like squats, deadlifts, good mornings, etc., but not things like

tricep pushdowns and leg curls.

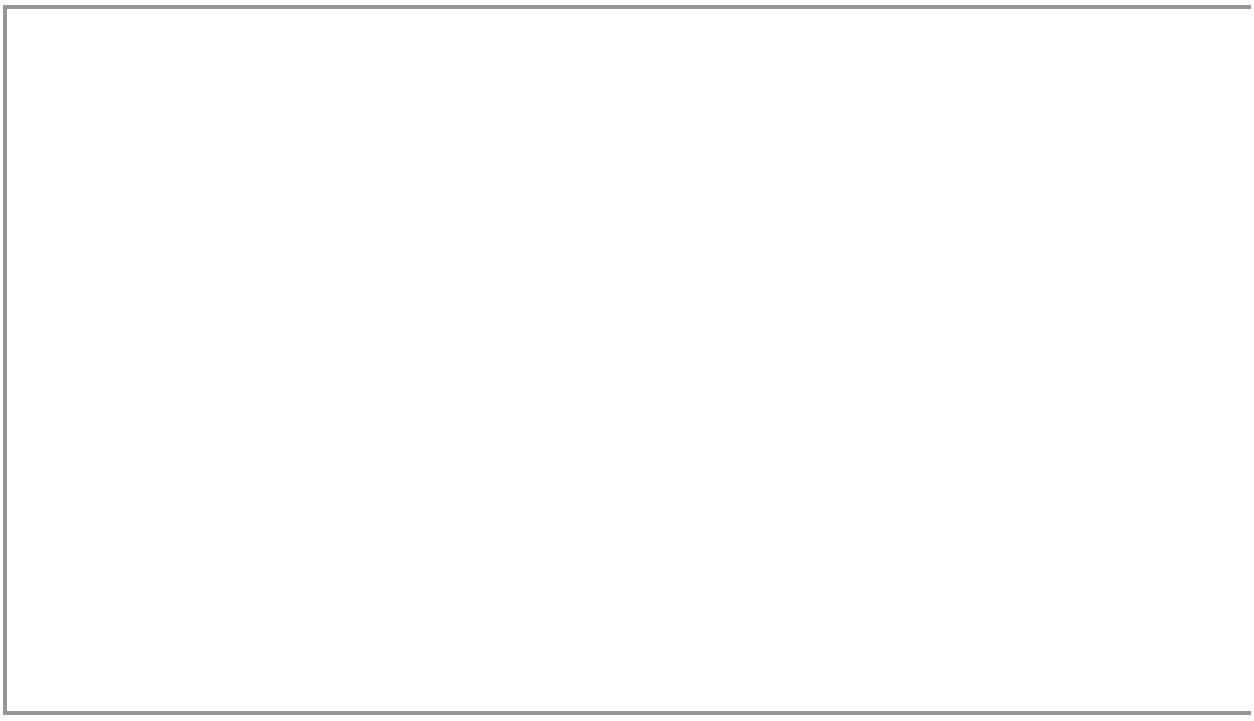
- Put your belt on for a heavy set relatively tight, and take the belt off in between sets. Don't just put your belt on in the beginning of the workout and wear it all the time until you finish the workout.
- If you do decide to lift without a belt and you have been using a belt, don't just go cold turkey, as that may increase your chance of injury. Gradually increase the weight you lift without a belt up to your max, but take your time doing this.

Belts can run anywhere from $20 to 100, but expect to pay near the top of that range for a good quality leather powerlifting belt. Those same belts can actually be somewhat uncomfortable when you first wear them because they must conform to your body, but, once they do, they are often quite comfortable. Good leather belts also last a long time; the cheaper ones tend to wear out. I have been using the same belt for almost 20 years, and it is still in perfect working condition (luckily my waist hasn't outgrown my belt). To maximize use of the belt you generally want to push your abs against the belt, as though you were trying to break the belt off of you by squeezing your core. This takes some practice to perform at the same time when you are lifting, but this will make your core very strong and rigid, which is what you are looking for.

Companies such as Inzer, Titan, and Elite Fitness Systems all make good quality powerlifting belts.

Chalk – Chalk is a powerlifter's best friend. It greatly enhances grip strength by increasing friction and drying up any water or oil that may be on your hand. It is useful in all exercises but of particular importance to the deadlift and the bench press. When you chalk up make sure you chalk up your palm, your fingers and thumb, and also chalk up the outside part of your pointer

finger on the second knuckle, wherever your thumb meets the pointer finger when you wrap your hands around the bar. This will help make sure that you have a good grip on the bar and you don't lose it during the lift. Be "gym friendly" and try to chalk up over a box or a bag so the residue falls in there instead of on the gym floor. Keep your hands right over the box so it doesn't blow around (a 12x12″ Tupperware container works great for this; if the container is only big enough to hold the chalk block it doesn't do a good job of catching the extra that falls off your hands). Keep the chalk as a block; don't break it up, and apply the chalk like you were drawing in on your hand. Finally there is no need to clap your hands or throw the chalk around unless it is a once-a-year occurrence or you are in your own gym. It is hard enough finding gyms that allow chalk; going crazy with it is what makes gym owners want to ban it.



Chalk: worth its weight in gold to a powerlifter

Baby Powder – Baby powder has the opposite function of chalk. It reduces

friction by making everything smooth and soft. Powerlifters use baby powder during very heavy deadlifts. The baby powder goes on the front of your thighs and perhaps the shins, wherever the bar rubs up against your legs on the way up. The baby powder can make the bar slide up your legs easier. Do not put chalk on your legs as this will increase friction and make the lift harder.

Something that is important to remember if you use baby powder is you do not want any baby powder to get on your hands, as that will negatively affect your grip. At a meet have a friend (who is not lifting in that group) put the baby powder on your legs or simply squeeze some of the powder on your legs and use the other end of the bottle to rub it in to your skin, but do not get any on your hands. You may want to wipe your feet off once you do that (on a damp towel); otherwise, the baby powder may stick to your soles, and you may find that makes the platform slippery.

If you lift raw (meaning without supportive equipment), the above pieces of equipment may still all be worn. However some or all of the items below may not be allowed; be sure to check with your specific federation so you know the rules before the competition begins.

Wrist Wraps – Wrist wraps are pieces of cloth that wrap around your wrist. They give your wrist and forearm added support and help you keep your wrist in a certain position. They are most popular in the bench press, since the wrist will be under a lot of strain with heavy weight. Lifters also like to wear them during the squat, primarily because the bar in low bar position can put a lot of strain on the wrists. Some lifters also wear them during the deadlift, with the belief that a wrist wrap helps maintain grip strength. The slight negative of wearing them during the deadlift is that they may rub up against your legs on the way up, and you may have to move your hands out wider, which increases the distance you must pull the bar. They can also be useful

on other exercises such as bicep curls, skull crushers, or anything that puts strain on your wrists.

If you lift in a federation that allows wrist wraps, check and see how long your wraps can be. The longer the wrap, the more support you can get from it. Generally either 50 cm wraps or 100 cm wraps are allowed. Wrist wraps are often made out of the same material that knee wraps are, which is either pretty hard and firm or somewhat soft and spongy. You can experiment and find what works for you. If you decide you want to wear wrist wraps, purchase a good powerlifting wrist wrap. Basic wrist wraps that you find in regular sporting goods stores are usually not very supportive. Companies such as Inzer, APT, and EFS make good wrist wraps.

For what it is worth, it is my opinion that wrist wraps should be allowed in raw competitions, although I don't see the issue as being a deal breaker one way or another. To me the key point is that the wrap is covering a non-moving joint, which means it will do very little to actually aid in lifting weight but it can add a reasonable amount of stability (much like a belt). In addition I have never seen someone in the audience or in a gym see someone lift a lot of weight and then go "Yeah, but could you have benched 450 lbs without those wrist wraps?" Lifters are allowed to tape their wrists in the Olympics, which produces a very similar effect.

Wrist Straps – Wrist straps are different from wrist wraps. Wrist wraps wrap around your wrist and give your wrist support. Wrist straps have a long thin strap that goes around the wrist and then goes around the bar and helps strap you to the bar. The purpose is to enhance your grip; you can hold much more weight with wrist straps than you can by yourself. The benefit of wrist straps is that if your grip is a limiting factor in how much weight you can lift, the wrist straps remove that weak link and allow you to more fully target a specific muscle. The negative is that they do not improve grip strength over

A pair of 50 cm wrist wraps

the long run, and in fact regular use probably decreases grip strength — which can be problematic for a powerlifter.

Wrist straps are illegal in all powerlifting federations. The only exercise they would be useful in is the deadlift, and part of the challenge is to hold onto the weight. However, just because you can't use them in a competition doesn't mean straps have no place in a powerlifting program. In my opinion there are a few exercises where your grip will almost always limit the amount of weight you can lift, in a negative way. Those exercises are shrugs, bent over rows, and probably Romanian deadlifts. I see no problem using straps on those exercises as you lift heavier weight. Do not use straps on regular (sumo or conventional) deadlifts and most other back exercises like pull-ups or pull-downs, as you should be able to hold onto that weight. If you limit your strap use to those exercises above, you should still develop a very strong grip that will enable you to hold onto big deadlifts. An exception to the deadlift rule is if you want to perform a regular deadlift with a pronated grip on both hands (perhaps to balance out uneven musculature). Then you can use straps for that, but be aware that you

Wrist straps: the strap should follow your thumb

may be weakening your grip by doing so. As a point of information, straps are often allowed in strongman competitions.

There is a "best" way to use wrist straps. First, make sure it is set up correctly on the proper hand. You do this by looking at your palm, and the strap should follow your thumb. Then when you strap yourself to the bar, you want the strap to go on the opposite side of the bar, and it should wrap toward your thumb. You should be able to make 2 or 3 loops around the bar with your strap. Once the strap is in position, put your fingers on top of it, and as long as you can keep your grip closed, the strap will lock you on to the bar. To release your grip, simply open your hands.

Shoes – As discussed in the chapters on each lift, you need to wear some form of shoes during each lift. When squatting, you want a shoe with a firm sole and good ankle support. Regular tennis shoes or running shoes usually have too much give, and lateral movement and should be avoided. The heavier you squat, the more important the shoes become. Specific squat shoes or Olympic lifting shoes are available if you choose to use them.

During the bench press (and the curl), a basic tennis shoe or sneaker will usually be fine. You want a shoe with good grip that feels comfortable so your feet do not slip when you drive your legs.

When deadlifting you want a shoe with a flat, thin, solid sole. Wrestling shoes or ballet slippers are the most common choice. A shoe with a big sole or a higher heel is generally not good because the higher the sole, the higher you must lift the weight.

(Left) Chucks are commonly preferred by powerlifters

(Right) Some lifters prefer to squat and deadlift in Olympic Lifting Shoes

The author's belt and deadlift shoes when they were first purchased. 2 decades later the belt is still going strong

Socks – Usually you can wear any type of sock that you want; however, most federations have a rule that the sock must cover the shin during the deadlifts. Normal socks don't go up this high; in order to be prepared you will need a pair of longer-than-normal socks to meet this requirement.

The following section discusses powerlifting gear that is definitely not going to be allowed in raw competitions but may be allowed in single or multi-ply divisions.

Assistive Equipment

Squat Suit – Some powerlifters like to wear a squat suit. A squat suit is a very tight singlet that is hard to put on but gives you support at the hips and on the upper body. It is similar to a bench shirt in that it is hard to bend down, but when you squat the suit helps you stand up. There are two general kinds of squat suits, single ply and multi-ply. Single ply simply means it is one layer of equipment; multi-ply means it is several layers of equipment sewn together. Different powerlifting federations have different rules as to what is legal. As time progresses the technology that goes into making powerlifting gear gets better and better, which simply means that the gear enables you to lift more and more weight. As of the time of this writing, a single ply suit normally adds 20–100 lbs to the lift, with an average of about 50 lbs or about 10% over the raw max. A multi-ply squat suit adds about 50–200 lbs to the lift, with an average of about 125 lbs or about 20% over the raw max. Over time, these numbers are only going to increase. If you choose to wear a squat suit, using a pair of suit slippers will help greatly in getting the suit on, especially if it is tight in the legs.

Groove Briefs – Groove briefs are a pair of thick, tight undergarments kind of like boxer briefs that serve to cross the hip joint and give the hip more stability and support. They are often worn under a squat suit and thus operate in conjunction with that piece of equipment. Groove briefs do not add as much help as a squat suit, and they are usually allowed only in multi-ply organizations. Groove briefs usually add about 20–50 lbs to the lift or about 5% increase over a raw max.

Knee Wraps – People often wrap their knees when they squat or when they train legs heavy. Knee wraps serve two purposes. One is they help you lift more weight. When you wrap your knees with your leg straight, this locks the

joint in that angle. When you bend your leg, the knee wants to spring back in to the straight position. This helps you lift more weight. The amount of assistance varies greatly, depending on the type of wrap used and how tightly it is wrapped, but tight powerlifting knee wraps add about 20–100 lbs to the lift, with an average of 30–50 lbs or so, or they add about 7.5% over the raw max.

The second function of a knee wrap is that it is supposed to help protect the knee by keeping everything tight and in place. There is little research to support this assertion although some people swear by them. If you are just using a knee wrap for support then it does not need to be very tight; probably a knee brace or knee sleeves are adequate. In fact there is some evidence that extremely tight wraps are detrimental to your knees because they restrict the natural range of motion and they do not allow good blood flow to the knee and calf. If you do use tight knee wraps, you put them on and take them off after each lift. In addition, try to time it so that once the wraps are on you are ready to lift. In a competition begin putting the wraps on when you are in the hole or in the wing, depending on how fast the meet is moving and how strict they are with the 1 minute time limit. If you have knee wraps on for too long your feet will go numb, and you may find the walkout and the squat itself to be more difficult.

Currently knee sleeves are an option and are allowable in most (not all) raw organizations. They are just tight neoprene sleeves that are pulled over the knee; you could wear them for the entire workout (unlike a wrap, which is taken off after each set). At the moment they provide warmth to the area, give a bit of stability and perhaps add a small amount of spring to the lift, on the order of 0–10 lbs or so.

Gloves – Some lifters like to wear gloves, but that is not common among powerlifters. Gloves are not allowed in any organization that I am familiar

with. In addition usually chalk on a bare hand will give equal if not superior grip to a glove. The main benefit of gloves is that they help prevent calluses from forming on the hand, but if you like to lift weights that is probably just something you will need to get used to. Ladies take note — real men like women with calluses.

Bench Shirts – A bench press shirt is a specialized shirt that some people wear to help increase their bench press. It is an extremely tight shirt that holds your arms out in front of you. Imagine you wrinkled up the front of your shirt so that you were trying to pull the seams of the shoulders together. This would pull your arms forward. That is what a bench press shirt feels like. Because your arms are pulled forward, the shirt provides resistance to the motion of going backward, which is, of course, what happens in a bench press as you bring the bar down to your chest. Since the shirt is tight and the material is somewhat springy, the shirt wants to bounce back to its original position. This helps to give you drive off of the bottom of the range of motion in a bench press.

Bench shirts are popular among powerlifters and anyone who wants to improve their 1 rep max on the bench press. They are popular among powerlifters because they work and because they are legal. They originated as an adaptation of the rules. You normally have to wear a T-shirt while benching, so people started to wear a very tight T-shirt to give them a little extra support. When you are competing sometimes even 5 pounds make a difference. Then people began creating extra tight shirts, and suddenly the bench press shirt was born. Now it has become a science of creating better and better bench shirts.

For the majority of the time bench shirts have been around they created a modest benefit to the lift. By “modest,” I mean they normally added about 10–50 pounds to a person’s 1 rep max on the bench press. That difference

was largely dependent on where you were weak. Someone weak at the bottom of the range of motion would get a larger benefit; someone weak in the top of the range of motion would get a smaller benefit. People liked the shirt because it enabled them to lift more weight and because it helped some people with old injuries keep their shoulders stable and bench heavy when they might not have been able to otherwise. They stayed legal in powerlifting competitions because powerlifting is a small sport. The sport relies on money from donors and sponsors, and most of that money comes from the people who make powerlifting equipment such as belts and wraps and now bench shirts. For powerlifting organizations to stop allowing bench press shirts would mean less money and perhaps death to that organization.

Currently bench shirts come in 2 forms — single ply and multi-ply. Single ply bench shirts as of this writing usually add 50 to 300 lbs to the lift or about a 25% increase over the raw max, with about 100–150 lbs being an average increase. It does take some skill and training to use a bench shirt and to adapt to the heavier weight. I am not saying that if you take a person who can bench 400 lbs and put them in a bench shirt they will hit 500 that first day. They might, but usually it takes some training to learn how to use the shirt most effectively. As noted, older bench shirts (before 1995 or so) added only about 10–50 lbs the lift in most circumstances.

Multi-ply shirts are like single ply shirts but are just a little thicker and stronger. They add about 100 to 400 lbs to the lift or give about a 35% increase over the raw max. In some cases the increase can be as much as 50%.

Deadlift Suits – Just like a squat suit is available to wear during a squat, a deadlift suit is available to wear during the deadlift. It is quite similar to a squat suit, but the tightness is in a different area, more focused on the upper body than on the hips. As of this writing deadlift suits have not progressed as

much as bench shirts or squat suits, but they still assist the lift. Generally they add about 25–100 lbs to the lift, with an average being 40–50 lbs or about a 10% increase.

Suit Slippers – Some of the supportive equipment can be very tight, particularly some of the squat suits. It will also take 10–20 minutes to get into some of the gear, and usually the assistance of a partner is required. If you are using a suit with tight thighs and you are having trouble pulling the suit up, you can wear suit slippers. These are just spandex leggings that you wear that allow the suit to be pulled up much easier. There is significantly less friction pulling the suit against spandex compared to pulling it against your skin (particularly if you are hairy). Once the suit is on you pull the suit slippers off; you do not wear them to compete. It is easier to get the suits off than to put on; it is not necessary to put the suit slipper back on to remove the suit, but you could if you wanted to.

Gear Use in Powerlifting

I had a tough time deciding how to write this next section. On one hand, I support all powerlifters and all lovers of the Iron Game. We are all working hard to improve our lifts, set PRs, and be as competitive as we can be. On the other hand, I have strong opinions about the use of powerlifting gear that I feel need to be expressed. They are opinions which I feel have a lot of evidence to support them. Gear has a significant effect on the sport and the direction it is headed in.

So my solution? I need to share those opinions, but at the same time my attitude is directed at the gear itself and not at the powerlifters who decide to use that gear. I think bench shirts are a bad idea, but that does not mean I think a powerlifter who decides to wear a bench shirt is a bad person or a bad powerlifter. The only situation that would make me feel that way about a person is if they cheated to win. My definition of cheating is to break the rules. Right now supportive gear is allowed by most federations and using it is not cheating or even wrong. But the real question is *should* the gear be allowed? Why it is allowed? And where will we go from here?

I feel that there are many negatives associated with the use of powerlifting gear in competitions and only a few positives. I am going to outline those issues as best I can in the hopes that you, the powerlifter — and the person who ultimately determines the future of the sport — will see these issues in a similar light to me.

Objectivity is Decreased

One of the beautiful things about powerlifting is that it is one of the most objective sports there is. A bench press is a bench press whether it is performed in China, England, or the USA. Follow a few simple rules that most people find easy to comprehend and judge, and lifts can be compared across the board. Some people suggest that comparison of yourself to others

is a “bad” thing and should be avoided. I disagree. That is what athletic competition is — a comparison of your abilities to that of others to see how you rank on a (hopefully) even playing field. While knowing where you stand compared to others is not the end all-be all of things, it is a valuable tool to give you a more accurate assessment of yourself. Gear ruins that objectivity. You benched 400 lbs — was that raw, in a blast shirt, in a HPHD Blast Shirt, in a denim shirt, in a Fury or a Rage or a Katana, was it single ply, double ply, Velcro, etc? But if it is a raw lift, then it is what it is. You benched 400 lbs, and assuming the basic rules were followed, you bench 400 lbs no questions asked.

Significance of the Records is Diminished

With the advent of gear, the historical importance of the lifts are now diminished. Records for a lift are being compared with another lift that is now performed under different circumstances. It is not fair to the guys of yesterday, who competed raw, that someone in supportive gear is able to set a new record in the same class wearing that gear. It will not be fair to the lifters of today when a lifter of tomorrow, wearing an even more advanced type of gear, sets new records.

I understand that the records issue can be a little cloudy. Steroids weren’t illegal until 1981 so some of the guys in the ’70s were using. Do we erase all records and start from scratch? Do we go back and try to piece together the raw records? I don’t know the answers, but I do know that we can’t keep comparing old records to the new ones using different gear. It is not fair, and it is not accurate.

Powerlifting is Not an Extreme Sport

Some people claim that powerlifting is a small, extreme sport that no one really cares about. They say let the lifters wear gear if that is what they want to do. First of all, I am not saying that bench shirts should be illegal to own

and if a lifter wants to go to a gym and max out in a bench shirt that is fine with me, but it should not be part of regular powerlifting competitions. I disagree that powerlifting is a small, extreme sport because millions and millions of people practice raw powerlifting by training the basic lifts in the gym. Joe Weider used to say that if you lifted weights to change the way your body looks, you are a bodybuilder. I say if you lift weights to make yourself stronger, you are practicing powerlifting. Certainly more than half of the people in America will at some time lift weights to increase their physical abilities if only to a small degree. It is true that it is a small number of people, in the thousands probably, who regularly use gear. I want powerlifting to appeal to a broad number of people and not just cater to a small group.

Gear Decreases Transfer of Skill to Other Sports

I say if you lift weights to make yourself stronger, you are practicing powerlifting.

The use of gear in powerlifting diminishes powerlifting's importance as a sport. Powerlifting is important because almost all athletes realize the importance of being strong, and they know that being strong will improve their sporting performance. As such, those same athletes then turn to the strongest athletes — us, the powerlifters — for advice on how to improve their strength. They do not need to become competitive powerlifters themselves, but they know that by training the squat, bench press, and deadlift with good form and solid routines, it should improve their sporting ability. However if we continue down the road of using gear in competitions, the training for that has become more and more specific to maximizing the gear itself. It used to be that everybody trained raw; then 2 or 3 weeks before a meet, you wore the gear to get to used to it and break it in, and you competed with the gear. The strongest raw lifter was the strongest

geared lifter, and everybody got basically the same benefit; everybody's bench was up 20 lbs, for example. Now the strongest raw lifter is by no means the strongest geared lifter. A lot of lifters will perform the majority of their workouts in the gear so they can learn how to best apply it. These programs will be more and more specific to training in gear, and they will have little to no impact on how to specifically improve raw strength. There is no evidence that being able to bench 400 lbs in a double ply bench shirt will help a football player play football. There is evidence that increasing a football player's raw bench press will help that person play football.

Gear Makes Powerlifting Less Likely to Be in the Olympics

I believe that powerlifting should be in the Olympics. To me there is no valid reason for powerlifting to be excluded from the Olympics, and, for Olympic Weight Lifting to be included. I think they both should be included. While I have not been specifically involved in the process of gaining Olympic status for powerlifting, it is my belief that the use of gear, along with the use of steroids, general politics, lack of unity, and lack of public support is why powerlifting is not in the Olympics. I believe that removing the gear will help build public support as more and more people watch and participate in powerlifting. I think that one of the major dividing lines in powerlifting and a reason for so many federations is the difference in gear that is allowed. Other than that, the federations are all basically the same with a few minor differences. If we remove gear, gain more public support, unite and follow the same drug-testing guidelines as the other Olympic sports, how could powerlifting not be included?

Powerlifting Gear Produces More Dramatic Improvements in Performance than Any Other Sport that Allows Equipment

Where and when will the improvement to the gear end? A bench shirt made in 2008 is vastly stronger than a bench shirt made in 1998. The curve of

improvement is increasing, not decreasing. It stands to reason that a bench shirt made in 2018 will be incredibly powerful if the current trend continues. What happens when someone can bench double or triple their raw bench because of a super powerful bench shirt? What is the point in that? Keep in mind that bench shirts are already adding up to a 50% difference in what a lifter can do. The highest raw bench press is just over 720 lbs. The best shirted bench press, as of this writing, is over 1100 lbs. This not only distorts the single lift that the gear is used on but also the total records that are kept are now inflated because each lift is higher than it should be. It is ironic because some of the single ply organizations take a superior attitude since they allow material in only one layer, yet the multi-ply gear of today is the single ply gear of tomorrow. The single ply bench shirts today add more than the multi-ply shirts from 10 years ago.

Gear is Changing the Nature of the Lifts

The gear is changing the individual lifts. Since the range of motion is shortest and there can be a lot of stored energy in the bench press, a bench shirt makes the most difference. A squat suit and tight knee wraps (and groove briefs) can make a very large difference in the squat. A deadlift suit and/or erector shirt can make a difference in the deadlift, but it is not as noticeable. In looking at the results of multi-ply combinations, it is now reasonably common for a lifter to lift the most weight in a squat, then the bench press, and then their weakest lift is the deadlift. Of course, the majority of normal raw lifters will lift significantly more in the deadlift than anything else, then the squat, and well behind both of them is the bench press. With the use of strong supportive equipment now the deadlift has become the worst lift and the squat and bench press are carrying more weight. This is “fair” in that it is not illegal, but it is distorting the nature of the competition because the weight lifted by the upper body exercise was supposed to have the least impact on

your total, and now that is not the case.

Gear Decreases the Advantage that Larger Lifters Have

Gear also gives a greater benefit to the lighter lifters. The benefit of being heavier is that the added weight creates stability at the joint, and, therefore, the heavier lifter can often lift significantly more weight. There should be a pretty large difference between the best 198 lb lifter in the world and the best SHW lifter. While there is still a difference in geared lifting, it is much less than the difference seen in raw lifting. In addition, if we continue to use gear, the lifting standards need to be changed. For many serious lifters, hitting a total of Elite is a lifetime goal. Yet hitting an elite total today, with the gear available, while still very impressive, is not the same level of accomplishment as was hitting elite 10 or 20 years ago. In addition, the formulas used to figure the Best Lifter award, which try to even out the bodyweights, were not created with today's gear in mind, and they are no longer as accurate as when they were applied to raw (or near raw) lifting.

Gear May Discourage Newbies from Starting Powerlifting

Gear inflates the numbers that people can lift. This usually has one of two effects, neither of which are desirable. Let's say a 165 lb teenage boy has a raw bench of 300 lbs, and he has never competed before but is thinking about getting into powerlifting. That would not be a world record, but it is a pretty good raw bench especially for that age and bodyweight. If he looks at the records and doesn't know how to distinguish a raw lift from a geared lift, he will see lots of 400 and some 500 lb bench presses in that weight class. But if he looks at the raw results, he will see a lot of 250 and 300+ lb benches and realize that while he may not win anything on that day he at least could be competitive. Often the inflated numbers scare otherwise qualified people away. The opposite could happen; he might do more research and find out about geared lifting, and then he might just dismiss those lifts altogether,

saying “Oh, well that was in a shirt; I don’t care about that.” That is not desirable as it discounts all of the hard work of the geared lifter. Either way it is a lose-lose situation.

Gear Adds to the Monetary Cost of Competing

Gear adds considerable cost to the price of competing in powerlifting. Powerlifting is a pretty cheap sport; even with gear it is not that bad, but one cool thing about powerlifting is once you buy a singlet and a belt and join a federation, you are good to go, and the singlet and the belt basically last forever or until you choose to change them. The gear can be pretty pricey, it loses its elasticity over time, and it doesn’t always fit right depending on your weight that day (even a few pounds — like cutting 10 lbs to make a weight class — will make a big difference in how it fits). Newer, better gear is coming out all the time, and if you want to be super competitive, you have to have the latest gear or you will be at a disadvantage. The use of gear can easily add several hundred dollars a year to the price of powerlifting and perhaps even more if you buy a bunch of stuff, get it tailored, and/or hunt around for the exact best match to your lifting style.

Gear Is Not a Necessary Part of Powerlifting

People who are in favor of using powerlifting gear in competitions often argue that other sports that use equipment have seen technological developments in their sport which have improved performance. For example a better pole vault will help increase the height jumped, a better golf club with increase how far the ball can be hit. The logic is “If other sports are doing it, why can’t we?”

The faulty part of that logic is that powerlifting gear is not necessary for a competition. The “gear” necessary to compete in powerlifting is a barbell, weights, a squat rack, and a bench press. Barbells have improved over time with the sleeves rolling better, bars having less whip, better knurling, more

evenly distributed weight, etc. Bench presses are more stable, squat racks stronger. I am sure there will be continued improvement in this area and I have no problem with it. If someone uses a slightly better bar in the future and can lift 5 more pounds because of it, so be it. I can never envision a situation where a better bar or set of weights would allow a lifter to increase their performance by 10% over what they can do today, let alone the 30–50% increase that some guys are getting out of their equipment.

Powerlifting gear like a bench shirt, squat suit, knee wraps, etc., is not necessary to perform powerlifting; it is something extraneous that some people like to use. A golf club is necessary to playing golf. You can't perform a pole vault without a pole, so comparing those types of sports with powerlifting is not a good analogy. And even in sports where equipment is an integral part of the activity, there are often lots of rules in place to make sure the equipment does not change or improve the performance too much. There are rules about how a golf ball can be made, what it can be made of, etc. There are rules about how a baseball bat can be shaped, what it is made up of, etc. I think any outside observer who saw the kind of performance enhancement that powerlifting gear gives to lifters, many of whom are already incredibly strong anyway, would suggest that that type of improvement was not appropriate and/or should not be allowed.

Gear May Increase Chance of Bombing Out

Powerlifting gear like a bench shirt, squat suit, knee wraps, etc., is not necessary to perform powerlifting.

I do not have absolute hard data to prove this, but I believe (and feel there is a lot of anecdotal evidence to support this) that the use of gear greatly increases the number of bomb outs during a powerlifting competition. Bombing out is when a lifter fails to get in one

successful attempt on a lift. When that happens not only do you get a zero for that specific lift, but you get a zero for your overall total, and any other lifts you might have done successfully don't count (even if you set a record). Excluding injury or something else that might affect a person that specific day, bombing out is generally the lifter's fault and is the result of poor planning when it comes to your attempts. This is true whether or not you are lifting in gear or out of gear. However there is more guesswork when it comes to what numbers to choose when you wear gear. This is because in the gear of today, what is relatively light and would make a good opener in gear is usually heavier than the lifter's raw maximum weight they can lift. In addition, with gear, sometimes the lighter weights don't always feel better. With raw lifting, if you feel confident you can lift 300 lbs, 250 lbs is going to be pretty light and you should be able to do that even on an off day. But with gear, if you think you can be successful with a 500 lb bench press as a max, you may find it hard to keep your form with a 400 lb bench press. As I mentioned before, even a small change in body weight (which can result from stress or from trying to make a certain weight class) can significantly affect how the gear feels and how much help it gives a lifter.

Bombing out is not good for anybody, except for the fact that it should serve to be a learning experience. Bombing out isn't good for the lifters — who wants to show up and get a zero in something they have been working hard to do (I know: it happened to me on the first competition I ever did)? It isn't good for meet promoters — generally when lifters bomb, they leave the meet feeling disgruntled and unhappy and some occasionally quit the sport altogether because of the embarrassment. And bomb outs aren't good for the sport itself if a large number of lifters regularly bomb out of the competition.

Gear Use is Not Consistent with the "You vs the Weight" Mentality of Powerlifting

I find the use of gear to be a little bit ironic. Powerlifters are known for being picky about how the lifts are performed. The judges are picky, and it is natural for lifters to apply those same standards to others lifting. After all, if someone says they can squat 500 lbs but you know in reality that is a half squat and you can really hit 500 lbs to full depth, there is a pretty big difference between the two lifts. Look at any youtube video of a wannabe lifter posting what they have done, and you will see lots of picky comments such as “You didn’t pause long enough on your chest,” “your butt came up one inch off the bench,” “the spotter had his fingers on it,” “you were wearing gloves and that is not allowed,” “your foot shifted slightly and that is illegal,” “your head moved,” “you didn’t quite touch your chest,” “you didn’t quite get to parallel,” “you hitched it,” “you used straps.” Whatever it is that is slightly wrong with the lift, chances are it will be pointed out. To me it is ironic that as a group we would be that picky about certain aspects of the lift, and then other aspects because it is within the rules we don’t care about what is happening. Yes, on youtube, when a person who benches 365 and their butt comes up a bit, they are showing an illegal bench and maybe that infraction aided the person by 10 or 20 lbs, but how is that different than a person who keeps their butt down but wears a bench shirt that adds 50–100 lbs to their lift? Why is a deadlift suit legal but wrist straps are not? Why are knee wraps legal but elbow wraps are not? Why is using a pad on the bar or wearing gloves considered cheating or wimpy but wearing a squat suit (which would have a more powerful effect on those lifts than either of those measures) is fine? How is it bad that someone lifts a weight raw but has a finger spot the whole time and that is illegal but another person lifts in a shirt that is helping more than the spotter would be able to? It is not consistent. To me you either lift raw and lift the weight yourself or you allow gear, and then any kind of gear that gives an advantage is fine, like gloves, elbow wraps, wrist straps,

etc.

Gear Provides an Artificial Increase in Strength

In my opinion using powerlifting gear is not consistent with the general message of powerlifting. Ask yourself why you compete. Many will answer that it is a way to see where you stand, to showcase your abilities, and to see what you are capable of. Powerlifting is the quintessential individual sport. It is nice to have a great team to root you on and to train with, but when you are on the platform it is just you and the weight, and that is it. And that is what builds a person's confidence. You do something hard and you do it alone and you are successful with it; you know that success came because of you. Hopefully in the future you will believe you can be successful again because you have done it in the past. But completing these lifts that have become so exaggerated from what we are actually capable of doing — it isn't the same message. Why would I or anybody want to say I benched 1000 lbs when in reality I (or anybody) could not come close to doing that weight without some sort of assistance? If you can bench 1000 lbs (or 9, 8, or 7 hundred) in a bench shirt, you have a phenomenal bench press. It is not the number itself that makes it great, it is what you can actually do. To me lifting 500 or 600 lbs raw is more impressive than lifting 800 or 900 lbs with a shirt.

Gear Changes the Weak Link and May Lead to More Severe Injuries

Excessive gear changes the weak point in the body. Normally the muscles are the weak link; if they can't generate enough force, you fail. When you fail because the muscles give way normally, nothing too bad happens (assuming you have a spot). Worst case scenario is you strain a muscle and even that heals relatively fast. But super strong gear improves the muscle's ability and can shift the weak point to the bones, joints, and ligaments. When you fail because of bone, joint, or ligament failure bigger problems ensue. Bones break, joints rip apart, and ligaments snap. These issues come without the

warning of muscular failure. You are good to go one moment, and the next you are in the hospital.

Some advocates of gear say that the lifts are safer when performed in gear. This is an “it depends” answer. I agree that most likely, and in the short term, the same lift performed raw or in gear is probably safer in gear. In other words I am probably more likely to hurt my shoulder benching 300 raw than 300 in a shirt. However, the lifts are not the same. Using gear allows the lifter to lift extra weight, weight the lifter is often not used to because in no other exercise does the weight approximate what they can do with gear on. Is lifting 300 lbs raw or 400 lbs in a shirt safer? I don’t know but it is certainly no guarantee that the gear is safer with significantly greater weights, and it may well be the case the gear is more dangerous. In addition, after long term training it is possible that over-reliance on the gear might weaken some supporting structures in the body. When some areas of the body are strong and some are weak, you have a greater potential for injury, so it is possible (although I agree it may not be the case) that prolonged use of gear is actually more dangerous than training raw.

Gear is Seen as Cheating

As I indicated before, to me the definition of cheating is breaking the rules to gain an advantage. In that way, gear is not cheating because it is allowed within the rules. But in another way, a more basic, fundamental way, gear is cheating. It is an artificial aid that allows one to lift more weight than a lifter could lift by himself. The first question asked by virtually every lay person who hears about what a bench shirt is and what it does is, “Isn’t that cheating?” The answer is “no” because it is within the rules, but a better question to ask is *why* is it within the rules? It might be interesting for me to find out how I much I can bench press with the aid of a spotter. Me and one spotter, both trying as hard as I could, how much can I lift? The spotter would

help me a lot right away, and if we practiced together we could perfect the art of “bench spotting” or whatever you want to call it. And if I got a super strong person to spot me then that would help even more. It would be interesting to see if together my spotter and I could lift 800 lbs, or 1000 lbs. It would be interesting to see if a group of any two people could lift 1500 lbs, but while that question is interesting, the basic, better question is, “What does that measure?” If I want to know how strong my chest is, we already have a great exercise called the Barbell Bench Press to determine how much I can lift. It doesn’t need to be improved. If my spotter wants to know how much weight he can lift off the ground, we have a great exercise for that as well. It is called the Barbell Deadlift. And if somebody wants to know how strong their lower body is, we have this great exercise called a Barbell Back Squat. The exercises don’t need to get any better, and there is no logical reason (other than financial support from sponsors and satisfying one’s ego as the numbers go up) for artificially increasing them through wearing gear.

It Is Hard to Go Back to Raw

Another negative of wearing gear is that once you get used to it, it is hard to go back to raw. I don’t think it is that hard on the body, but it is hard on the psyche. Once you have lifted a certain weight, there is usually less thrill in lifting less weight than that. If I squat 500 lbs, and then I squat 450 lbs, it is tough to get as fired up and excited about doing that 450 because I already did the 500. Once you take off the gear, your weights can’t help but go down, sometimes very significantly. Imagine you have already competed in powerlifting and you hit a shirted bench press of 500 lbs; that is pretty impressive. Now you decide to go do a raw bench press competition and you get 410 lbs. That is still a pretty good bench press, but when you are telling your friends and family that you benched 410 they might ask you “Isn’t that a lot lower than you did before?” or “Did you have a bad day?” and questions

similar to that. Then you have to explain (if you haven't before), "Oh, well, before I was wearing this shirt, and now I am doing it by myself, etc.," and it is just difficult to get as fired up for something that you have already done in at least some fashion.

Gear Often Requires Help to Get Into

Another negative of using powerlifting gear is that it usually requires help from a friend or training partner to help you get into the equipment and make sure it fits right. This can take anywhere from just a couple of minutes up to 15 minutes or so to make sure that some suits and shirts are on right, and often the assistance of more than one person is desirable. In addition that person usually needs to be pretty strong as essentially jamming someone in a tight shirt can be tiring and draining. This makes training in gear impractical to do yourself.

Gear Can Slow Down the Speed of a Competition

Using gear slows down the pace of powerlifting competitions, which are already pretty long. It is not a gigantic difference; I am not implying an 8 hour competition would be finished in 4 hours, but you might be able to shave an hour off of the usual time it takes to finish a competition. In raw competitions generally when the lifter's name is called and they get the "Bar is loaded" command, they are often pretty much ready to go. But when you are using gear, lifters are often trying to time when to put certain things on, such as knee wraps and the straps up on the squat suit, and if you make an error with that, it is better to be a little late so they will hear the "Bar is loaded" and then make final adjustments to their equipment before they go up to the platform. Those 15 extra seconds per lifter, done on each attempt, make a difference and add up over time.

Benefits of Using Gear

Almost no situations are completely black and white, and powerlifting gear is

no different. There are some benefits to using gear.

Gear May Allow Some to Lift When they Otherwise Might Not Be Able

Some lifters, particularly older lifters or those with injuries, may find that they can still lift heavy if their injured joint is covered with heavy, protective wrapping that gear would provide. This can allow some more people to be involved and stay involved with the sport.

Gear Companies Support Powerlifting Competitions

While making powerlifting gear is not a huge business, there are a couple of companies that make their living by selling gear to the lifters. These companies often give something back to the powerlifting community by sponsoring lifters, supporting competitions, and sometimes offering prize money. Some argue if that nobody was using gear, these powerlifting companies would go out of business, and powerlifting would suffer. I don't know for sure what would happen if everybody stopped using gear, and clearly, financially, for these companies, there would be some negative. But there is always a silver lining if one chooses to look for it.

Personally I believe that if gear was not used, the numbers of people lifting competitively would increase, possibly quite significantly. Those people would all need to purchase belts, etc., and so while some items would be purchased much less, others would be purchased much more. The companies might start focusing their efforts elsewhere, for example, on making cool singlets that are customizable or custom lifting shoes. They could start to run the powerlifting competitions themselves, as money can be made from a well run powerlifting meet. They could act as marketing/sports agents and try to get powerlifting more recognition, perhaps Olympic recognition or TV spots, which would increase equipment purchase.

Gear Lets You Go Heavier

There is little doubt about this one. Gear does let one go heavier, which some

say is good for the ego, and others say helps draw interest in powerlifting with the excitement of people lifting huge numbers. To the first point I say heavier is relative; right now, with gear, 800 lb benches are amazing and 500 lb benches are more commonplace. Take away the gear and 500 lb benches become amazing, and 300 lb benches are commonplace. To the second point I say that there is not that much public interest in powerlifting now; I don't think it would get much worse if we took away the gear, and it might get a lot better as people feel they can relate to the lifters better. Look at bodybuilding. There was greater crowd attendance and interest in bodybuilding in the late '70s and early '80s than there is now, even though the bodybuilders are bigger now. Getting more extreme is not always better, especially if it is through artificial methods.

Gear Helps Break Plateaus

This may be true. It is definitely easier to keep making progress with the gear so you feel like your numbers are going up but again to me it is an artificial inflation. Part of the challenge of the sport is figuring how to get those extra 10 lbs, and at top levels the gains will come slowly; that happens in any sport, powerlifting will not and should not be any different. And I have seen little evidence that using gear a lot helps one break through plateaus with raw lifts. Indeed with today's gear, athletes are finding the need to specialize and either focus mainly on raw lifting or mainly on geared lifting, and they are often unable to excel in both.

Gear Builds Toughness

This is probably true but it is also generally unnecessary. It is true that to put on super tight gear and then lift extremely heavy weights takes a big toll on the body, and you feel like you went through a war when you are done. I can respect the pain tolerance and the constitution necessary to be able to do that. But again the question must be asked, "Why? Why do that?" Powerlifters are

usually not lacking in toughness and fortitude. Brutal squat days and conditioning workouts and weight loss techniques and year round lifting all build a pretty good level of mental and physical toughness. Indeed some might say that getting up on the platform with just you and the weight, with no supportive gear as a crutch, is even more of a challenge. Not all may agree with that statement, but I do believe that powerlifting quickly weeds out the wimps and the pansies. The sport doesn't need to be any harder than it already is without a good reason for being so.

How Will the Use of Gear Be Judged in the Future?

How will history judge the use of powerlifting gear? I can't help but think it will not be looked upon favorably. It started as a little tweak here and there to maximize performance, and now it has snowballed greatly, some might say out of control. The technology will only get better, which means that gear will only get stronger and make more of a difference in the lifts. If we are ever going to put an end to the use of gear, we might as well rip the band-aid off so we can get a handle on the situation and start anew in the right direction.

My Position

I know I have talked a lot about this issue but I don't want there to be any confusion on how I personally feel about it. I believe that supportive gear should be eliminated from regular powerlifting competitions. My definition of supportive gear would be bench shirts, elbow wraps, squat suits, erector shirts, groove briefs, knee wraps, and deadlift suits. The general idea is that if the equipment crosses a moving joint (shoulder, elbow, hip, or knee) it should not be allowed because that is where you will see massive improvements in the amount of weight that can be lifted.

I feel that non-supportive singlets should still be worn to facilitate the judging. I believe that lifting belts and wrist wraps should be allowed. The

essential difference in this type of gear is that although it is supportive it does not cover a moving joint and thus will not give a large performance benefit. Instead the benefit is increased stability (which arguably can increase performance a little bit) and possibly increased longevity in terms of one's lifting career. Almost all lifters agree that a belt should be allowed even in a raw competition. There is more dissent concerning wrist wraps. I could go either way, but if I had my choice, I would allow them. Wrist straps should not be allowed as that removes grip strength from the lift, which is a part of the overall lift. Knee sleeves are another tricky one. They are basic wraps that pull-up around your knee to give it some warmth and support. They are not that tight, and it is easy to bend your knee with a sleeve on. However I could see them evolving into a more supportive form of gear or even people hiding knee wraps underneath them. I could go either way with knee sleeves, but, to send a consistent message, it would probably be better if they were not allowed.

Any kind of shoes for the squat, deadlift, or bench press are fine. I do feel that gloves should not be allowed. Lifters might try to spray stickum or other substances on the gloves, and a bare hand with chalk will give you a better grip than a glove anyway. Baby powder should be allowed to reduce friction on the deadlift. I would not have a problem with sports tape being used on a finger or wrist or shin (for the deadlift). In general I think that powerlifting equipment guidelines should be similar to the guidelines for Olympic Lifting which does not have any of the problems that powerlifting does when it comes to the overuse of gear.

Should be Allowed	Should Not be Allowed
Singlet	Squat Suit
T-shirt	Bench Shirt
Any shoes	Deadlift suit

Lifting Belt	Knee wraps
Chalk	Stickum spray
Wrist Wraps	Wrist Straps
Baby Powder	Vaseline
Sports Tape	Erector Shirt
Regular Underwear	Groove Briefs
Socks	Gloves

I can see powerlifting heading one of two ways in the future. The first, and my preference, is that the geared lifting simply fades out as more and more people lift raw. When winning the raw championships means being the real National and World champion and that is where most of the elite lifters are, I think the rest of the lifters will pretty quickly follow. Right now that group of elite level lifters is spread out pretty evenly among raw, single ply and multi-ply organizations. But hopefully 15–20 years from now raw lifting will be the thing to do, the records will be straightened out, and drug testing will be even better. I also think that following a conversion to raw lifting the number of total competitive lifters in the country will increase significantly. I think the price of the gear and how important being good in gear has become is holding a lot of people back from competing. Once the gear is gone, I think powerlifting will become even more popular as athletes around the world in a variety of sports realize the potential that getting stronger has to improve their performance. To find those golden keys of knowledge about strength, they will come to the powerlifters and trainers who know how to train raw. Powerlifting reached its heyday in the ’80s (when it was basically raw), with impressive lifters and growing public support. The big championships were on TV, and I remember reading that one championship, which spanned seven or eight days, got a total viewing audience greater than the Super Bowl that year (adding all of the days together compared to the one event of the Super

Bowl). That reflects the potential for high interest. That could draw money into the sport and that would draw even better athletes, higher records, greater competition, and more interest, and the snowball could grow and grow. I don't think powerlifting could ever eclipse football in terms of national interest because quite frankly it is not as exciting to watch, but that doesn't mean it shouldn't be on TV. The World's Strongest Man, CrossFit Games, and the Olympic Games do very well on TV. I think powerlifting could do the same with some human drama thrown in.

The second possible future I see is for powerlifting to continue on the way it is going now, with gear and no gear, until ultimately it splits into 2 separate sports. If lifters want to compete in gear against other lifters competing in gear, I have no problem with that. Remember I don't approve of the gear itself; I have nothing against the lifters who choose to use the gear. However, if that is to be the case, the names of the exercises need to be changed. They need to be changed so that a simple abbreviation of them does not cause confusion between the raw lifts, which exists now. Ask a powerlifter how much he benches, and he might say, "450." Then the follow up question is "Was that in a shirt?" so you can better understand the impact of that number. That should not continue to happen.

I believe the exercises should be renamed as follows. The squat should be the Suited Squat, indicating that a suit and knee wraps were used during the lift. The bench press should be called the Shirted Press (not Shirted Bench Press) to indicate a bench shirt was worn during the press. Efforts should be made not to confuse a Shirted Press with a real bench press, just like efforts should be made to separate a 3 board press from a real bench press. And a deadlift should be called a Suited Deadlift if a deadlift suit is worn. The final result of the competition should be called the Equipped Total. The best lifter award would go to the Best Equipped Lifter.

My own personal history:

I have worn a bench press shirt in competitions. I wore a single ply (1 layer) shirt and it added between 25 and 30 pounds to my bench press. I justified it because powerlifters are the only ones who pause their max bench press. Nobody in the gym pauses their max when they say how much they could do, so I felt like the two canceled each other out. Plus everybody else was doing it, and, let's be honest: when somebody asked me how much I could bench, I liked being able to add those 25 extra pounds; that could be a year's worth of hard work just because I put on that shirt.

But honestly I felt like I was cheating by wearing a bench shirt, not because it was against the rules (because it wasn't) but the people I was talking to in the gym weren't wearing one of those shirts. Plus there are so many people in the gym who exaggerate how much they can lift, and, frankly it is annoying. I have always prided myself on never taking credit for something I didn't do in the gym. If someone asked how much I could curl, I would say, "195" because that was what I did — not 200 or 225 because maybe I could or I was close enough. In fact I even took it to the point where, if I didn't perform the lift in a competition, it didn't count, so if I squatted a new max in the gym, I couldn't say I did that until I went to a competition and did it.

Now here I was telling people about an inflated number; it was legitimate but still inflated. I felt like if someone said, "I don't believe you," I wanted to just be able to go and do that lift or at least come very close to it, granted you can't set or match your max every day. But now if somebody said they didn't believe me, I would have to say, "Well, let me go home and get this special shirt that two people have to put me in and then I will be able to do it." To me it was getting out of hand, and that was just at the beginning of the shirts becoming very powerful.

To me lifting weights and competing is about expressing strength and power — your own personal strength and power. Individual competition is great because it is just you and the weight, and you know your success or failure depends totally upon you and your training. You can't blame anybody or make any excuses, but now wearing a special bench shirt is changing all of that. When wearing a shirt you couldn't move your arms; if you didn't have your max weight on the bar you can't even get the weight to touch your chest. When they show powerlifting on TV they make the lifters wear a T-shirt over their bench shirts and they even tape the T-shirt up because they know the audience will wonder why the big strong guy can't move his arms. You can't get into or out of the shirt by yourself, it normally takes one or two people who know what they are doing 10–15 minutes to get you into a shirt. To me it was ruining the pure joy that is lifting weights and the spirit of competition. I know that equipment used in sports gets better and better as technology advances, and I can accept that. But when equipment takes over the sport and changes it, then that is not progress. In addition, a bench shirt is not necessary equipment in powerlifting. It is totally optional. Necessary equipment for powerlifting is a barbell and weights, a bench press, and squat stands.

Chapter 18

Powerlifting Federations

When you compete in a powerlifting competition, you will either lift in an unsanctioned meet or you will lift with a powerlifting federation. An unsanctioned meet means the competition is not approved or follows any one specific organization's rules. Usually unsanctioned meets are relatively small, local affairs often held at a gym and put on by someone familiar with powerlifting to give others the opportunity to experience competing. The benefit of an unsanctioned meet is that a new lifter does not have to join an organization to lift on that day (thus saving the lifter anywhere from $15 to 40 dollars) and the meet director has the freedom to make minor adjustments to the general rules as he/she sees fit. For example if a new lifter had an unapproved piece of equipment, perhaps a wrist wrap that was too long or an improper belt, the meet director might decide to waive the rules so the lifter could lift on that day. If the meet was sanctioned and the meet director was being strict with the rules, that lifter would have to either not use that piece of equipment or any lift made with an illegal piece of equipment would not count. Sometimes there are larger unsanctioned meets, and the purpose is to allow elite lifters in certain organizations to cross over and compete against others of a different organization. Some organizations have rules that if you want to compete in their organization, that is the only formal organization you can lift in. To give lifters a chance to go against each other, there are

larger unsanctioned competitions.

The negative of performing in an unsanctioned competition is you may not know all of the rules beforehand (since they may not be published on the internet or available to the lifter) and it is really dependent on the meet director to set the tone for the meet. Some unsanctioned competitions allow sloppy lifts like high squats and touch-and-go bench presses; others are very strict and the lifts in those meets would pass in any organization. Another negative is you can't set any kind of official state or national record in an unsanctioned competition.

If you are going to lift in a sanctioned competition, then you will have to join that federation in order to lift in it. Federations do have a fair amount of overhead, so it is not just a quick way to make money but a necessary item if the federation is going to remain in business. Generally federation memberships last for 1 year — sometimes it is up to the end of the current year (e.g., if you join in August it might be good until December 31st of that year) and sometimes it is good for a full year; the expiration date will be on your membership card. Once you join a federation, you can compete in as many meets as you like with that federation in that time period and you will have to pay the membership fee only once. Each competition will also have an entry fee which you will have to pay each time.

Right now the current state of powerlifting federations in America is fractured. There are many different organizations, each with their own set of rules and records. Some are quite large and have a big base of lifters; others are smaller and only found in certain areas. It should be noted that when I say the state of powerlifting is fractured I mean

Face to face, I have always found powerlifters to be very supportive of each other.

that there is not one main unifying powerlifting federation which all lifters lift under. On a personal level I don't believe powerlifters themselves are very fractured, and you often see a powerlifter from one organization helping and cheering on a powerlifter from another organization. Of course, there are the usual Internet fights over generally petty things, but, face to face, I have always found powerlifters to be very supportive of each other.

There are literally hundreds of powerlifting organizations in the USA. I applaud all those who choose to compete in a powerlifting competition and those who are willing to give back to the sport by putting on a competition. I do not wish to offend any particular organization, and I do not have the space to list every single one; at the same time I think it is worth noting for the lifters some of the differences between the federations.

In general, powerlifting federations can be broken down into 3 broad groups based on equipment allowed. The first is a multi-ply federation. This means that this federation allows lifters to wear multi-ply powerlifting equipment such as squat suits and bench press shirts. Multi-ply simply means multiple layers of the same equipment to make it stronger and more supportive. Generally any kind of supportive equipment is allowed, with the exception of wrist straps and elbow wraps.

The benefit of a multi-ply federation is that lifters in this organization lift more weight than any other federation. There are some extremely strong, world class lifters lifting in multi-ply federations. Examples include 1200+ lb suited squats, 1000+ lb shirted presses, and even one 1008 lb suited deadlift by Andy Bolton. High level multi-ply lifters are regularly totaling more than 2500 lbs.

There are some negatives associated with multi-ply federations. The first, which some people see as good and

"Pro" division means there is no drug

others as bad, is that it allows multi-ply gear itself (see Chapter 17 for a discussion on powerlifting gear). The second is that lifters on steroids tend to gravitate more toward these types of events. Often there are "tested" divisions and "untested" divisions in a multi-ply federation. Tested means the lifter may undergo drug testing; "untested" (sometimes referred to as a "pro" division) means there is no drug testing performed. In some ways I respect the openness and the honesty of a lifter who is using drugs to lift in the non-tested division because at least they aren't trying to cheat and compete against lifters who are drug free. On the other hand, having a non-tested division basically condones the use of steroids, which are an illegal drug and might have negative effects on the body that we don't fully understand yet. In addition, my belief is that the more people see other people on steroids, the more people will take steroids. People in general and particularly athletes tend to be competitive, and it is natural to compare yourself to others. If someone looks really good or is squatting 200 lbs more than you, you can't help but think that if you started taking drugs, you might be able to do that or look like that as well (whether it is true or not). Bodybuilding is the only other high level sport that essentially condones steroid use by not testing for it, and I think most would agree that steroid use has gotten out of hand when it comes to professional and high level amateur bodybuilding. I do not want the same thing to happen to powerlifting.

testing.

The final negative with the multi-ply federations is in relationship to the judging. There are two issues with it. First the rules are not exactly the same when it comes to performing the lifts, particularly the squat. Most multi-ply federations are looking for a parallel squat as opposed to a below parallel squat. Squatting itself is the most subjective of the lifts, and the definition of parallel (as with below parallel) has its own problems. Does that mean top-of-

the-thigh parallel, femur parallel, or bottom-of-the-thigh parallel to the ground? Because the rules are different, this makes comparing a squat from one type of an organization to another quite difficult. In addition, if you do not have to go below parallel, many lifters find they can take a very wide stance and still be very powerful all throughout the range of motion in that lift, especially when wearing a suit. The wide stance reduces the ROM and allows even heavier weights to be used. The negative is that training for that type of a squat doesn't carry over as well as to any kind of athletic activity as would a traditional, full squat.

The second part of the judging issue is that the judging itself is loose, meaning it is not consistent from lifter to lifter. All organizations experience this to some degree, but, anecdotally, this seems to be more prevalent in the multi-ply federations.

Some of the more popular multi-ply federations are listed below. It should be noted that you do not have to use multi-ply gear to lift in a multi-ply federation, you could lift in single ply or raw if you choose to do so. I am not aware of any multi-ply federations that perform the strict curl as a competitive lift. Most multi-ply federations use a monolift when squatting.

- AFPF — American Frantz Powerlifting Federation
- APA — American Powerlifting Association
- APC — American Powerlifting Committee
- APF — American Powerlifting Federation
- IPA — International Powerlifting Association
- SPF — Southern Powerlifting Federation
- SSA — Syndicated Strength Alliance
- USPF — United States Powerlifting Federation
- UPA — United Powerlifting Association
- WCPF — World Class Powerlifting Federation

- WPC — World Powerlifting Congress

The second type of powerlifting organization is a single ply federation. This organization allows lifters to wear single ply gear. Lifters may wear a squat suit, a bench press shirt, and a deadlift suit but it must be single ply and not multiple layers.

The benefit of the single ply federations is that the largest single federation in the USA, which is the USA Powerlifting Federation (USAPL), is single ply. This federation has the most lifters, is found in the vast majority of states, and is associated with the International Powerlifting Federation (IPF). The IPF is the largest international powerlifting body. While there are many great lifters lifting throughout the available organizations, it is my opinion that, as a whole, the level of talent is the greatest in the USAPL and the IPF. This is reflected by the general dominance of the USAPL in the rankings in powerliftingwatch.com in single ply gear.

> ***Training for that type of a squat doesn't carry over as well as to any kind of athletic activity.***

The AAU (Amateur Athletic Union) is another large, popular single ply federation. I can say from experience that on the East Coast they run well organized and fun powerlifting competitions, I don't have experience with them on the West Coast, but I would imagine it is similar quality. Both the AAU and the USAPL usually offer a raw division for lifters not wearing equipment but they do not allow multi-ply equipment.

Negatives of the single ply federations, particularly the USAPL (and this line of thought applies only to some people) is that it is associated with the IPF. As discussed before the benefit is this allows them to represent the USA around the world. Some people feel the IPF rules are too picky, they change

too frequently, or there is too much politics involved in how the organization is run. I don't have any personal experience with those issues. The other problem is that since the USAPL is considered (by most) to be the most competitive organization, there is the most desire to win. If you win the USAPL Nationals, in my opinion, you can consider yourself the National Champion for that year. But the problem with this is since lots of people want to win, some people resort to steroids to improve their chances. The USAPL and the AAU do drug test, but, even at a tested meet, usually no more than 20% of the lifters are tested, and often it may not even be that much. Some people feel the tests are easy to beat, and others feel that you could use steroids for a year, get off of them for a brief period of time, and pass the test. There is out-of-meet testing, but it is not utilized on a regular basis. Don't get me wrong, I am in favor of drug testing, but one should realize that drug testing does not always mean drug free. In addition the IPF adopted new weight classes, which, in my opinion, was a step in the wrong direction for powerlifting to take, from both a historical perspective and from a future perspective.

Some people feel the rules in the single ply federations are too strict, particularly with the depth necessary on the squat at high levels of the squat. I do not feel this to be the case, although I do think clarification on the exact depth necessary, with more anatomical reference points, would help out both the judges and the lifters.

Drug testing does not always mean drug free.

To me the biggest problem with the single ply federations is the simple fact they are single ply federations. As I outlined in Chapter 17, I feel that ultimately the sport needs to move away from the use of gear and back to its roots. In addition the single ply gear of today is the multi-ply gear of

yesterday, and I would imagine as the technology gets better the single ply gear of tomorrow will be the multi-ply gear of today (the same gear that is often decried by single ply lifters currently as far too excessive). Since I feel there is the most cohesion in the single ply federations, in my opinion, I wish they would take stronger steps toward making a movement to becoming raw.

Listed below are several prominent single ply federations:

- USAPL –United States of America Powerlifting; has a large raw division
- AAU — Amateur Athletic Union; has a large raw division
- NASA — Natural Athletes Strength Association; has a large raw division; also contests the standing strict curl
- WNPF — World Natural Powerlifting Federation; also has a multi-ply division; contests the standing strict curl
- WABDL — World Association of Bench Pressers and Deadlifters — they only contest the bench, the deadlift, and the ironman total
- WDFPF — World Drug Free Powerlifting Federation

All of the above federations implement drug testing.

The third type of powerlifting federation is a raw federation. “Raw” means that the lifters do not wear any supportive equipment, with an exception made for a belt and sometimes wrist wraps and knee sleeves. Some people feel that raw lifters should not wear any equipment at all. I can understand that philosophy and would go that way if necessary. It is my opinion that covering a non-moving joint during a lift will not aid the lifter that much in lifting additional pounds, but it can add an element of stability and safety to the lift. Covering a moving joint, while perhaps increasing safety, has the ability to significantly increase the amount of weight being lifted as shown with a bench shirt, squat suit, and knee wraps. Following that guideline, it is my opinion that raw lifters should be allowed to wear a lifting

belt and wrist wraps (not straps). I do not think raw lifters should be allowed to wear knee wraps. Knee sleeves are a more tricky issue because they do cover a moving joint, but, at the current state of development, they do not add any significant pounds to the lift. Ultimately I think to send a consistent message, knee sleeves should not be allowed, but I admit I do use them myself in my training at certain points. At the end of the day, if the powerlifting federations unite and we end up fighting over whether or not knee sleeves should be allowed, that would be a good problem to have.

- 100% RAW Powerlifting — The primary unequipped federation. As already mentioned, some of the larger federations have a raw division but still allow other lifters to use equipment. 100% RAW does not allow any supportive equipment at all with the exception of a belt and wrist wraps. The word “raw” has a double meaning — “raw” meaning not using gear and “raw” also was an acronym for Redeemed Among the World, which is why it is capitalized. This federation initially started out with religious overtones, but that does not seem to be the case anymore, meaning the federation is no more or less religion based than any other federation I have lifted in. This federation also contests the strict curl up against the wall, and they recently joined with the ADAU (anti-drug athletes united). The federation conducts drug tests.
- Hardcore Powerlifting — This federation focuses on highlighting elite level raw powerlifters, often offering prize money to the winners. It is not tested.

One issue I do have with almost all powerlifting federations, particularly the smaller ones, is that each federation feels the need to keep its own list of National and World records, and they each must have their own National and World Championships every year. I understand the desire to do this both from the lifter’s point of view and the meet director’s point of view. Lifters

like it because they have a chance to win a cool trophy, a fancy title, and they might be able to break a record, which is motivating for their training. Meet directors like it because that is their biggest meet of the year (usually), and they will get the most business at their World Championships.

The big problem is it is simply not an accurate use of the term. Powerlifting is a sport that prides itself on accuracy. How much exactly do the weights weigh; do you have an accurate assessment of yourself and your ability? It is, in all honesty, a farce to say you set a World Record when a quick Internet search will show many lifters of equal bodyweight have already surpassed that mark. I myself have set 2 "World" Records in 100% RAW and the AAU at the Open level (1 in each fed). They were both good lifts, but I don't kid myself into thinking they were real World Records; to do so demeans the actual term and the people whom have really accomplished that.

My suggestion to help solve the problem is to drop the terms "National" and "World" championships from the smaller organizations and not to keep National and World Records for those federations. Instead just simply call it the _____ Federation Championship. That way, it is still an accomplishment. If you can win your weight class in the entire federation that is something to be genuinely proud of, but it does not mean you actually set a World Record or you are a World Champion in the process. Just keep records for your organization without attaching the title to it, so what used to be a World Record in a smaller federation now simply becomes a "federation record," which is more accurate and, I believe, still very satisfying.

Another possibility to help make the records more streamlined is for one organization to accept another's records. If the organizations basically have

As powerlifters we should look forward to challenging ourselves

similar rules and allow similar equipment, this is doable. It also returns the meaning to have a “State Record” in something. There should not be five 165 lb Open state record holders in the bench press in Virginia or any other state (because there are 5 main federations, for example). I know this would be a difficult first step for some to make, but it would be a step toward unification, and it would be a step toward greater competition, not lesser competition. As powerlifters we should look forward to challenging ourselves against other good lifters instead of going somewhere else to avoid them. I would much rather come in sixth place in the state championship and know that I was in actuality the sixth best person in the state than come in first place but hear or read about other lifters in the state lifting much more than me.

against other good lifters.

One other suggestion I would make to increase the level of competition would be to have all lifters automatically entered into the Open division in their weight class unless they specifically request not to be. Currently it is quite common for lifters to enter in just their age division, which is understandable, and it is fun to get a trophy, but at the same time sometimes this means the person who wins the Open was not necessarily the best in that division. I have been to a meet where I won the Open bench press for my weight class, but a junior in my weight class benched more than me. I did not deserve to get first place; I should have received a second place trophy (which is what I asked for). The junior was entered only in the junior division, and, thus, he wasn’t ranked in the Open division.

Having all lifters entered in the Open division would help create a little more competition at the local level, where it is needed more. First place loses its luster when you listen to an awards ceremony, and over 75% of the lifters are winning first place because they are the only ones in the specific weight

and age class. In addition the Open records should always be the highest record set. A scan of the powerlifting record books will sometimes show that the Open record is 500, and then the Master record is 525. If somebody lifted 525 in that weight class, whatever their age, that is the best lift for anybody in that weight class and it is the Open record. You can't have an Open record lower than a sub-category record; it does not make sense.

One negative of having everyone ranked in the Open division is that the meet director may end up giving out more trophies. This could be offset by limiting the trophies to first, second, and third place only (most go to fifth place currently), except for the National meets, which, generally, go first through fifth place. In addition many lifters end up winning 2 or more trophies in a meet. For example, I could compete in a push/ pull competition and I could compete with my total in the push/pull, just my deadlift, and just my bench press. Theoretically I could walk out of there with 3 trophies. That is cool the first one or two times you do it, but on your twentieth meet, it isn't necessary. I would suggest giving out one main trophy and then medals or something similar for secondary awards unless the lifter specifically requests a trophy (and has paid accordingly).

Lifters need to remember that powerlifting in general is a small sport. It costs money to run a powerlifting organization and to host a competition. Renting the space, having all of the equipment ready and having it be good quality, getting spotters and loaders, and purchasing the trophies all cost a lot of money; plus it is pretty time intensive. The meet directors need to get something back, or else they will stop holding the meets. Giving out 3 nice trophies to each lifter would severely cut into any possible profits for the meet director. One nice trophy and a medal or two is a better option in my opinion.

All of the powerlifting competitions that I have been to and watched

from various organizations have all been run by good people. They are powerlifters who love powerlifting and want to give something back to the powerlifting community. A powerlifting competition is generally a fun and friendly atmosphere. There is very little “talking smack,” and generally everyone is rooting each other on to set some PRs. Newbies will be on the platform with veterans, men lifting the same bar as women, and Masters going up against boys, all powerlifters there with one goal — to lift more weight than they have before.

> ***All of the powerlifting competitions that I have been to have all been run by good people.***

A Note to the Fans

Powerlifting is a great sport, but, admittedly, it does not have the same drama of a football game. There is a bit of monotony to the lifts and the meets can stretch out for a full day. Here are a few suggestions for fans in the stands.

Wear layers. Some powerlifting competitions are really hot with all of the lifters releasing some energy; others are cold in gyms that have the heat turned off during the weekends. You never know what it will be like.

Bring some snacks and drinks; it makes the time go by and it keeps you alert.

Cheer for the lifters while they are actually lifting. What commonly happens is a lifter's name is announced over the loudspeaker and people clap and cheer but then that dies down as the lifter is getting ready to lift. People then watch in anticipation to see if the lifter will be successful. This is the exact opposite of what most lifters want. We don't care if you clap when our name is announced, but right before the lift and as we lift that is when we need you to cheer. My rule of thumb was once the lifter grabs the bar with their hands (in any lift), start cheering and don't stop until they complete the lift. You might not be able to do that for every lifter but the point is there is nothing worse than lifting heavy in a room where you can hear a pin drop.

Try to predict how much weight a person can lift based on how their previous attempt went. If you see a stranger open up on the bench with 300, try to predict what their one rep max is from that lift or what they should go for on their next attempt. If the person smoked it, you might say they should go for 325, or if they struggled you might say they should go for 305 or whatever. Then see what happens on their next attempt. The beauty of this is that you will get to test and prove your hypothesis many times over. If you are at all into the fitness game, and I am assuming you are since you are reading this book, being able to predict how much weight a person can lift

based off of watching one or two lifts is a valuable skill to have, and personally I find it makes the time go by much quicker.

Get involved with the lifter. I am assuming you are going to a powerlifting competition to watch a spouse, friend, or family member compete. Ask if you can help them out in any way. Most commonly this consists of filming or taking pictures during the lift, but it could be anything. Most lifters will probably not want to talk much before they lift, but sometimes your presence can provide a calming effect. If you are there to support a lifter, don't miss their lifts. By that I mean don't walk out of the room a few minutes before they are supposed to lift because you might miss it. Even if you don't think you will miss it, the lifter might be watching you, and then they in turn will get panicky or upset if you leave because they don't know when you will be back. An exception to this is if you have small children and they are crying or being disruptive; then take them out of the room to avoid distracting other lifters who are trying to concentrate on the task at hand.

Every lifter is different, so at some point just ask the lifter what they want from you or what they expect. You might start off with suggestions about what you were planning on doing. Some lifters may be hesitant to ask you to come and watch them because they may not want to inconvenience you or make you do something you don't want to, but chances are, if you are important to that person, then that lifter will most likely want you there if at all possible.

Ask the lifter what they want from you and what they expect.

Summary

Every minute of every meet isn’t necessarily exciting, but it is what that meet represents that is. It represents training and working hard for a goal that is perhaps 10 years in the making, and this is the day when the lifter gets to go for it. That is exciting. Once a lift is completed in a meet and made official by the judges, it can’t be taken away from you. And sometimes it does get really exciting, coming down to that last lift of the day or one lifter just squeaking out another by 5 lbs. As with any sport, you can’t pick and choose to just watch the exciting moments; you have to be there to be engaged in it.

Interview with a Meet Director

I am including an interview with a meet director. I thought it might be beneficial for powerlifters and for those thinking about competing to get a glimpse on the other side of the sport and see what goes on behind the scenes. After all, without any meet directors and those putting on the competitions, there would be no sport at all.

John Shifflett
Photo credit: John Shifflett

Please give us a brief introduction of yourself

My name is John Shifflett; I have been putting on Powerlifting meets since December, 1985. I have put on more than 200 meets.

I have lifted in and promoted many different federations over the years. The bulk of my meets have been with ADFPA/USAPL, and now I do a lot with the 100% RAW Powerlifting Federation.

With the USAPL — I promoted the first 25 Virginia State Powerlifting Championships. I have put on a number of National Meets: Bench Nationals, Push/Pull Nationals, Masters Nationals (two times), Squat Nationals, Deadlift Nationals, and Raw Nationals. I have put on more than 50 Open meets, including the Virginia Open, Mid Atlantic Open and Eastern USA Open.

With 100% RAW — I have been putting on meets with them since 2004. The bigger meets have been the Ironman Nationals, Masters Nationals and The American Challenge. I usually host 6–8 meets per year.

For the Lifter:

What advice would you give to lifters in a competition, particularly newer competitors?

Open light; it is not where you start but where you finish that counts. Enjoy your time on the platform and try not to get too nervous. You have already put in the hard work in the gym. Get to know the other lifters. Lifters for the most part are willing to help and encourage other lifters. Find a good coach or a team to work with.

What are some common mistakes/errors you see when you host a competition?

Lifters open too heavy. Lifting in the meet is much different from lifting in the gym. Lifters need to take a look at the rules and the updates on the rules so they can be better prepared. Their training partners do not call the lifts in the gym like the judges call the lifts in the meet.

If people have questions before the meet, what is the best way to ask you?

Speaking personally here, email is much easier for me, I imagine different meet directors might prefer different methods. I can check email from anywhere most of the time and give them a clear and quick answer. I answer phone calls as well, but I am not home that often, and I am never home in the day time.

If people have questions during the meet, what is the best way/time to ask them?

The best time to ask questions is during the rules briefing. I try to be as free as possible during the meet, and I can generally answer them. There are times I will be judging, and they could check at the scoring table and see who else might be available.

If a lifter doesn't agree with the judges or misses a lift and isn't sure why, what would you suggest they do?

Check with the referees on the platform. They will tell you what they called

you for. Do it right after your lift so it is fresh in their mind, but don't do it when they are in the middle of judging another lift.

What can lifters do to help you out and make your job easier as a meet director (either before or on the day of the meet)?

Before the meet, it is very helpful to get your entry forms in as early as possible. This helps me to get the roster posted and the score cards and score sheets ready. I like to have the flights set up the week before the meet. It also allows me to make sure I have enough help and see if I need to add a second platform.

At the meet, just be ready to lift when it is your turn on the platform; turn in your attempts on time to keep the meet flowing.

Understanding Your Job:

How much time goes into setting up for a competition, both the first time around and once you have been doing this for a while?

It takes months to prepare for a meet. I like to have my schedule for the following year out by September 1. This allows the lifters to look over the schedule and see when and where the meets will be. I have to have all the locations rented and contracts signed and sanctions with the federation completed.

Once that is done, then it is on to the entry forms. I update and get the entry forms ready so that the lifters can look them over. I send them out to the federation website and to Powerlifting Watch. From that point on, it gets really busy sending and receiving the entry forms, posting the entries, ordering awards, T-shirts, etc. Once the deadline is in and I have a good idea who is lifting, then I can figure out how much help I will need setting up the meet, and I can work on getting the proper staff to run the contest. After the meet is over, it takes additional time typing up the results and sending them to the federation and to Powerlifting Watch. Once that is done it is time to update the records on the website.

How many people/helpers do you usually have at a meet?

I will go with a single platform meet set up; of course, 2 platforms means a lot more help is needed. We have around 15 people working at the meet. We use three for the scoring table. A team of 3–5 judges, a team of 3–5 spotter/loaders. Two people working at the door and the T-shirt table.

What are some big expenses you have to cover when you put on a meet?

The venue is a really big deal. Finding a location that is Powerlifting friendly is often hard to find. I have been blessed to work with a hotel staff that wants our business. Then you need to get awards for the contest. The lifters put a lot

of time and hard work in on their end, and they want nice awards for their effort.

The drug testing is also quite expensive.

Meet staff: the spotters/loaders are hard to come by. If you are willing to pay, they are more willing to help. It is a very hard job to constantly load and unload the weight. I generally have a group that works with us, and that is a huge plus!

The meet equipment used on the lifting platform and warm-up rooms: The calibrated plates, scales and bars are very costly. I try to update the most needed equipment each year.

Do you have a list of things you need to bring to be prepared?

Yes; it is a long list. The key things are having enough equipment for the lifting platforms and warm-up areas plus scales, score sheets, scorecards, chalk, awards and drug-testing equipment.

Do you usually make money when you put on a competition?

When I started in 1986, I made a promise to my wife Doris not to lose money on any contest. I have kept my promise for the most part. You can make some money on the contest if everything goes well. It is not a lot when you consider the time and effort put in. You have to have a love of the sport to promote it.

Do you make enough money from putting on competitions to make that be your full time job?

No, not even close. I am amazed how many people think I do. I manage a Heating Oil and Propane company, which is about 60 hours per week of work for me. There are a good number of meet directors who do it full time. It is hard work. I would love to put on meets full time if I could swing it. It is a true love of mine, and I really enjoy it.

What advice would you give to someone who was thinking about hosting a powerlifting competition?

Be ready to get a great team working with you. Start off small and put on some single lift or push/pull meets. Talk with your lifters and see what they want. Never cancel a meet no matter how small it is. The lifters train hard and need to count on you to have the meet once it is announced.

Do you have any pet peeves about running a meet or about lifters or anything you want to share?

If you have agreed to help out at a meet, keep your commitment. The meet director is counting on you and may have a really hard time getting help if you have to drop out.

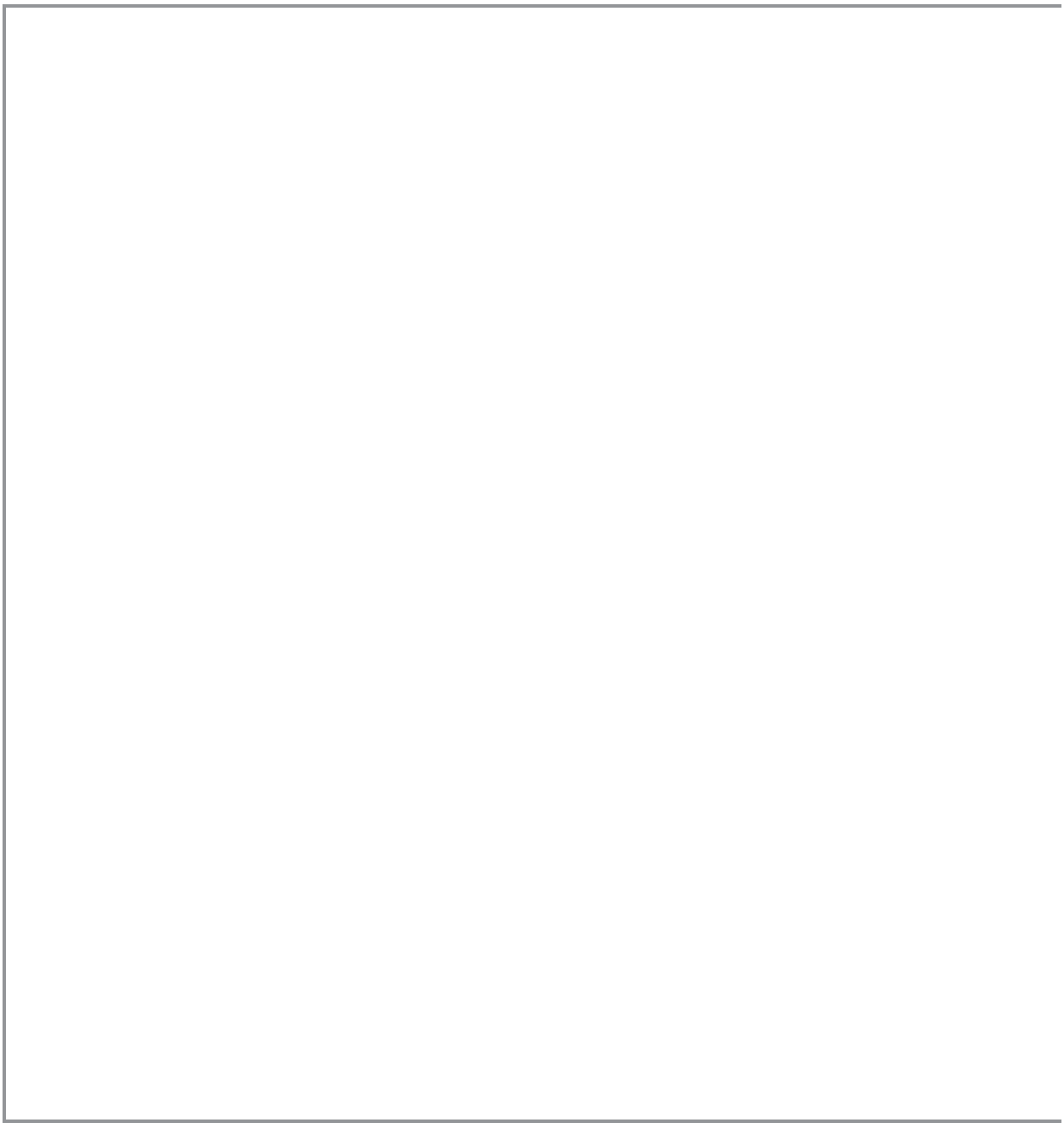
Anything else you want to share or stuff that you want those involved in the lifting community to know?

The meet director wants the meet to go smooth as much as you do. I really try hard to set the meet up so it will run smooth and be as quick as possible. I know my time is very valuable, and I know yours is, too.

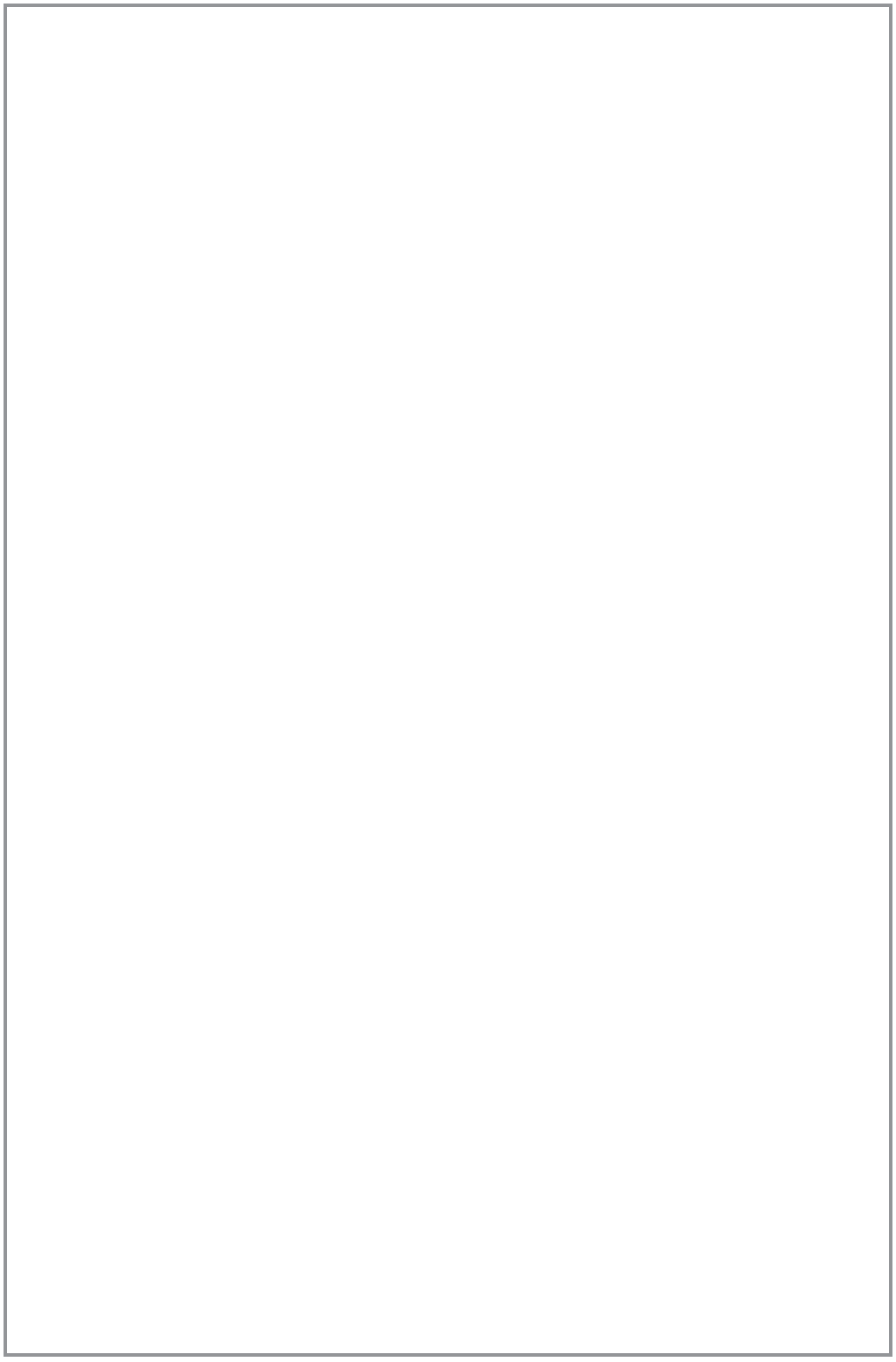
The Powerlifting world is full of good, decent, hardworking people. They are giving and will encourage every competitor to do their very best. I tell new lifters all the time no one will ever make fun of you for lifting a certain amount of weight or being a beginner; we all started at the same place.

If you are not lifting at a contest, see what you can do to help out!

Thank you very much for taking the time to share your extensive knowledge of powerlifting with me and our readers; it is much appreciated.



John Shifflett with powerlifter Kerry Self
Photo credit: John Shifflett



A lifter attempts a PR curl
Photo credit: Doug Jantz Photography

Chapter 19

The Strict Curl

A traditional powerlifting competition consists of 3 exercises: the squat, the bench press, and the deadlift. When all three lifts are performed that is called a full meet. Sometimes powerlifters will compete in only some of the exercises. Usually the squat is the one that is dropped. This is because the squat takes the longest to perform at a competition and requires the most equipment (and qualified spotters); the squat is the hardest exercise to do; previous injuries normally limit the squat more than any other exercise. If a lifter competes in both the bench press and the deadlift, that is called a push/pull competition or an Ironman competition, and the sum of those two lifts is added together to get the lifter's score. Some lifters compete in just one exercise, called a single lift competition. The most popular version of this is to just perform the bench press, but single lift deadlift competitions are becoming more popular, and occasionally you can find a single lift squat competition.

While most powerlifting organizations stick with the Big Three (as the squat, bench, and deadlift are known), some organizations have added a fourth lift: the strict curl. It is possible the curl can be added to the bench and the deadlift, where all three lifts are added together; this is usually called a Powersports competition or occasionally a Strongarm competition. The very first powerlifting competitions consisted of the Curl, the Squat, and the

Bench Press. The strict curl can also be contested as a single lift, either at the end of a full meet or along with other single lift events.

Types of Curls

There are two types of strict curls that can be performed in a meet. Only one will be used in a particular meet — if you are interested in competing in a curl competition, it would behoove you to find out which type of curl the lifters will be performing. The traditional strict curl is a standing bicep curl, using an EZ bar performed up against the wall. The butt and upper back must be placed against the wall, and they must remain against the wall throughout the entire lift. It is significantly harder to curl up against the wall as opposed to performing a freestanding curl. Generally lifters will lift 10–20% less performing a curl up against the wall as opposed to standing straight up, even when you are using strict form while standing up. The benefit of curling up against the wall is primarily for judging purposes. It is hard to judge a standing curl because the lifter may have just a little swing forward or backward; saying the body must remain vertical and erect is useful, but there is some gray area in that definition. Having the body remain against the wall makes the judging very easy — either you remained against the wall and you successfully curled the weight, or you didn't remain against the wall (curling the weight is irrelevant), and you didn't curl the weight.

A second type of curl is a standing strict curl, which is a bicep curl using an EZ bar, performed in a standing position. The legs must be held straight and locked. The weight is curled up near the chin, and the upper body must remain basically upright during the lift. A significant swing either forward or backward is considering cheating, and the lift will not count, although without the benefit of the wall this "swing" becomes reasonably subjective. It is okay on both types of curls for the elbows to move forward. It is this author's opinion that if curls are contested in competition, they should performed up against the wall to facilitate fairness of judging for all competitors across all federations.

If one is not competing, other types of curls can be performed. These include dumbbell curls, preacher curls performed on a preacher bench, hammer curls performed with the hands held in a neutral grip, and reverse curls with the hands in a pronated (palms down) position. Lifters can train curls on machines, cables, dumbbells, and barbells; various other types of resistance can be used as well.

It is significantly harder to curl up against the wall.

Equipment Set-up

Minimal equipment is necessary to complete a strict curl. Most curl competitions will involve the use of an EZ curl bar, which is the cambered bar most often used for curls. It is crucial that an EZ bar be used and not a straight barbell, as using a straight bar for bicep curls is extremely hard on the wrist joint. Regular use of a straight bar will often lead to tendinitis in the wrist and elbow joint. The heavier weight you use and the less flexible your wrist and arm are (bigger arms are generally less flexible) the harder the straight bar is on your wrist. Since you must train heavy on a regular basis to have a high 1 rep max, this is a recipe for disaster. I would even suggest that if someone were to compete with a straight bar on the bicep curl (which in my opinion should not occur), they should still train with an EZ bar on the majority of their sets to save their wrists and elbows. Most EZ bars weigh 15–20 pounds and, just like with a regular bar, you should count the weight of the bar. If you don't know how much your particular EZ bar weighs, just weigh it on a scale at the gym (EZ bars are actually often 16.5 or 22 lbs because they are made to be 7.5 or 10 kg).

The second thing you will need is a sturdy wall to lean up against (avoid leaning up against a gym mirror and/or just plan drywall as you might damage both of those structures). If you want to save yourself some effort, you can bring a flat bench over and set the bar on the bench, reducing the need to deadlift the bar up each time you lift it. Lift up against a sturdy and reasonably broad wall at least 12″ wide; curling up against a power rack, for example, will not give one the necessary support and is likely to result in lower weights being lifted. In a competition there should either be an official, sturdy curl platform or you can go up against the wall — once we had to curl up against the squat stands, and that was not at all appropriate.

Walls can be used if a curl platform is not available

Proper Technique

The curl requires the lowest skill level of any of the contested lifts, but that does not mean that technique is unimportant. Indeed it is reasonably common to see lifters make significant technique mistakes when performing a competition strict curl, which can be avoided by following these guidelines.

Proper hand position for a widegrip curl
Photo credit: Vadim Snitkovsky

Hand Position

Hand position in a strict curl is crucial for maximal performance. When using an EZ bar, lifters will have two main choices: a narrow grip or a wide grip. Lifters can choose either grip; most lifters choose the wide grip. However, I have seen exceptional performances with both grips — select the one that allows you to lift the most weight.

You can use a narrow grip by gripping the first bend in the bar. A narrow grip is generally (but not always) better for people with more narrow shoulders and/or with more flexible wrists. The narrow grip places more emphasis on the long head of the biceps brachii (the outer head) which may be useful for bodybuilding purposes but doesn't do that much for strength. A wider grip is used when gripping the second bend in the bar. This grip is normally (but not always) better for people with wider shoulders and/or less flexible wrists. I would say the wide grip is the grip most commonly used in bicep curl competitions, but there are enough exceptions that I would experiment with both to see which one you are stronger in. The wide grip places more emphasis on the short head of the biceps brachii (inner head). There is not a huge difference in strength between the two heads of the biceps; pick the grip that you feel the strongest on and use that one — don't worry too much about specific muscle recruitment of a part of the muscle.

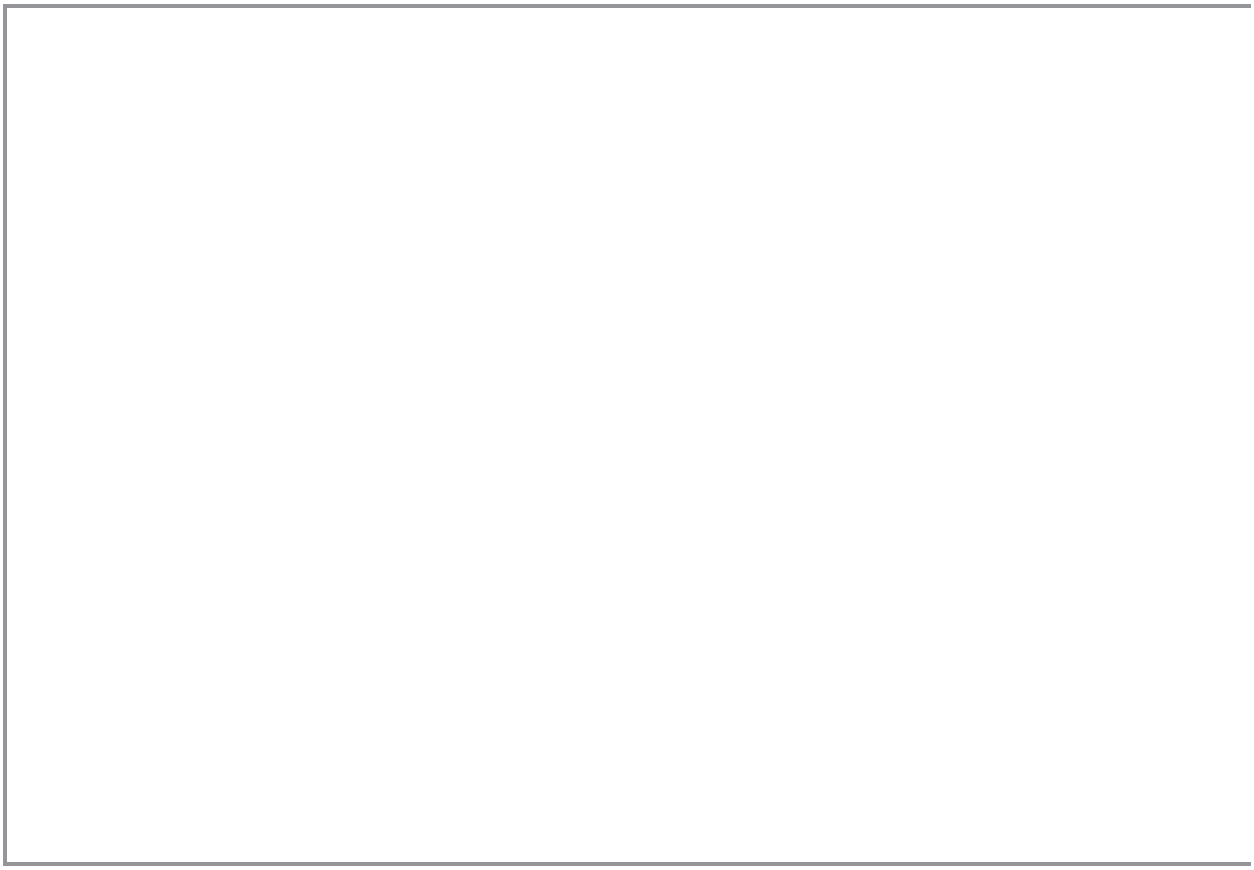
It is imperative that one grab the EZ bar in the correct position. When you grip the EZ bar, the angle of the bar must match the angle of your hand. The angle of the hand when supinated is such that the hand (going from the pinky to the thumb) points out slightly. The part of the bar being held must match this angle. Lifters will occasionally grab the EZ bar with the bar angled toward their body; this will feel awkward and will result in a poor performance. If it feels weird, it probably is.

Lifters should take a closed grip (thumb opposite fingers) and should

grip the bar reasonably tightly when curling.

Foot Position

Foot position is also important in a curl. The rules generally state that when a lifter is up against the wall, the heels must be no more than 12″ away from the wall. I would suggest you mark the floor where you train in order to know how far away your heels can be. The heels should be as far away as possible while staying within the rules. In addition, taking a wide stance, squat width or wider, will likely add support and will also decrease the chance of the hips coming off the wall. Pointing the toes out moderately will likely be the most natural comfortable foot, angle combined with the wide stance.



Proper foot position for a strict curl
Photo credit: Vadim Snitkovsky

With a standing curl, there is a distinct advantage to staggering your feet

by placing one foot in front of the other, thus I would suggest you use that method. Put one foot in front of the other and lock your legs. Generally you lock the legs because it is required by the rules, the legs must start and stay locked during the exercise. Staggering your feet helps with your balance. The weight will want to tip you forward as you curl it, so you in turn will try to pull it backward with your back. Imagine if someone were going to push you either from the front or from the back and you had to resist the movement. Would you stand with your feet symmetrical, about shoulder width apart, or you would stagger them? Of course you would stagger them; it is quite easy to push someone over if their feet are in line with each other, even if they are spread out. Staggering the feet makes you less likely to lean backward as you curl the weight.

Staggering the feet also gives another advantage. When you stagger the feet it breaks the line of the torso running to the legs. This makes it harder to judge if the body is staying completely upright or not. By staggering your feet, you can often start the curl by leaning forward 5–10 degrees. This lean forward can give you some crucial momentum which you can use to lift more weight. This is the main reason it is easier to curl standing up — even if you stay strict — than it is to curl against the wall. You can’t lean backward at all, but if you start leaning forward and then move backward just a bit as you curl, you end in the correct position, but you used some momentum to lift the weight. A staggered foot position makes this harder to notice and judge. Having your feet spread but in line with each other makes any body lean to the front or to the back very noticeable.

The negative of staggering your feet is that it causes you to twist your hips and spine. Twisting the spine at the bottom will cause it to twist back in the opposite direction to compensate for not being straight. Regular use of this method can promote an imbalance in the body, and it might lead to

injury. You can try switching the lead leg so one side does not become more dominant than the other. Personally I recommend that on your regular sets when training for the curl you perform the curl with the feet spread but in line with each other — the same position you would take for the majority of your standing exercises. Only stagger the feet when the weight is very heavy for you. The staggered foot position is not normally a hard position to get used to; most people feel very comfortable with it right away or after a bit of practice. Remember normally we want to be symmetrical when lifting weights to ensure the muscles develop evenly, but sometimes it is okay to be asymmetrical in the effort to lift more weights. The pros and cons of a staggered foot position apply to basically all lifts, not just the bicep curl. It should be noted that many athletic positions involved a staggered foot position due to the increased balance and stability in that position.

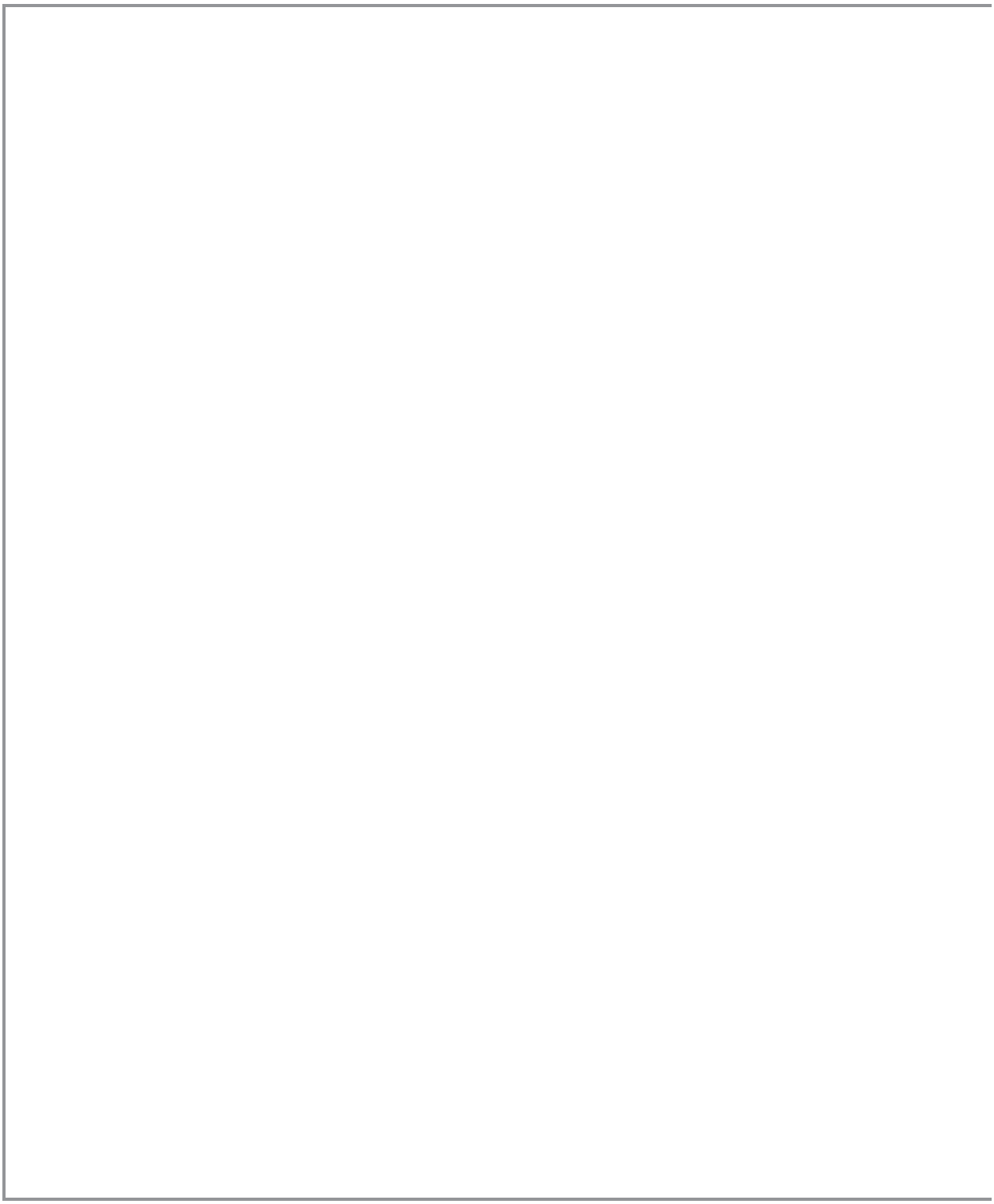
Only stagger the feet when the weight is very heavy for you.

Head Position

Head position in the curl generally follows one of two trains of thought. The first is to keep the head in line with the body and reasonably stable throughout the lift. I would suggest you perform most of your training reps like that. The second school of thought is to use a bit of momentum from the head and neck to assist with the curl. It is minor but it might make a small difference. In this example, start the curl with your head looking down (most commonly lifters look straight ahead, which will signal the judge to give the "Curl" command, and then you can look down after receiving the command). As you curl the weight up, look up at the same time. The logic is similar to deadlifting; by looking up, the chest and thorax will follow, putting one in a stronger position to finish the curl. Even if one is curling against the wall, there is generally not a rule that mandates that the head stay in contact with the wall — the head can move if you wish to do so. Keep in mind that significant head motion — especially looking to one side or the other — is associated with decreased neural drive during most lifting motions (e.g., looking to the side while squatting or deadlifting is particularly unwise and can result in injury).

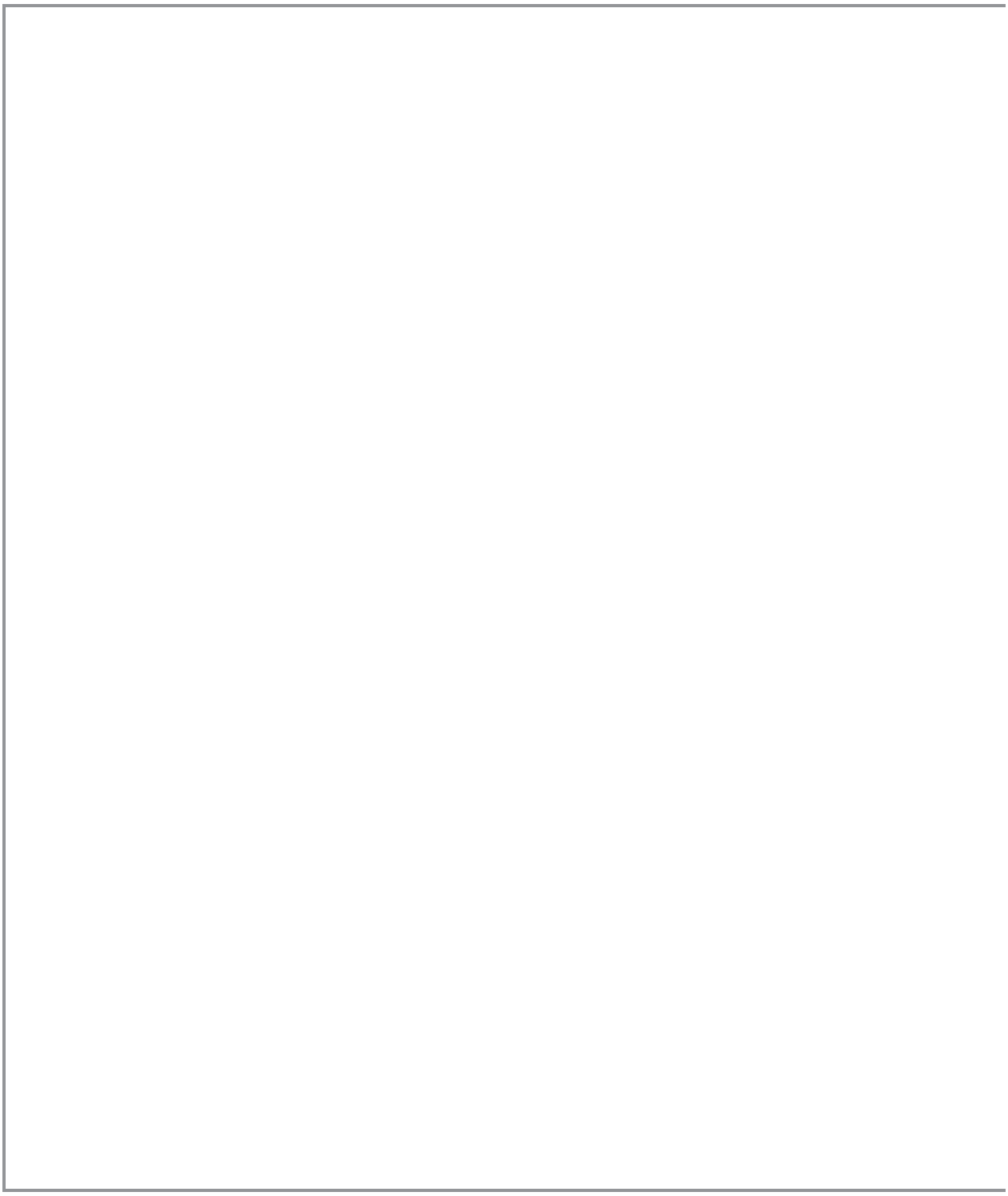
General Form for Performing the Strict Curl

To perform a wall strict curl in a competition, you will approach the EZ bar and take a grip on it. The same guidelines apply in terms of a wide or narrow grip to a wall curl as they do a standing curl, and generally lifters will use the same grip for maximum performance whether up against the wall or standing upright. Depending on the organization you lift in you may or may not get a command to pick up the weight. Once you pick up the weight, you will take a step backward and lean against the wall. The rules generally state that your butt and upper back/shoulders must remain against the wall throughout the lift. As you lean up against the wall, place your feet out in front of you. You will need to keep your legs straight the entire time — do not bend the knee. There is no benefit to using a staggered step when lifting against the wall, so have a symmetrical stance. Do not keep your heels very close to the wall; they should be pretty far out in front of you. If the federation limits how far your heels can be from the wall (often it is 12″), generally you want to be as far out as possible. You should feel like if the wall was suddenly removed, you would fall backward. In addition have a fairly wide stance. Having a wider stance makes it a little bit less likely to pop your hips off the wall, which is the number one problem in a curl. When you are leaning against the wall make sure you straighten your arms because you have to start the curl with your arms straight.

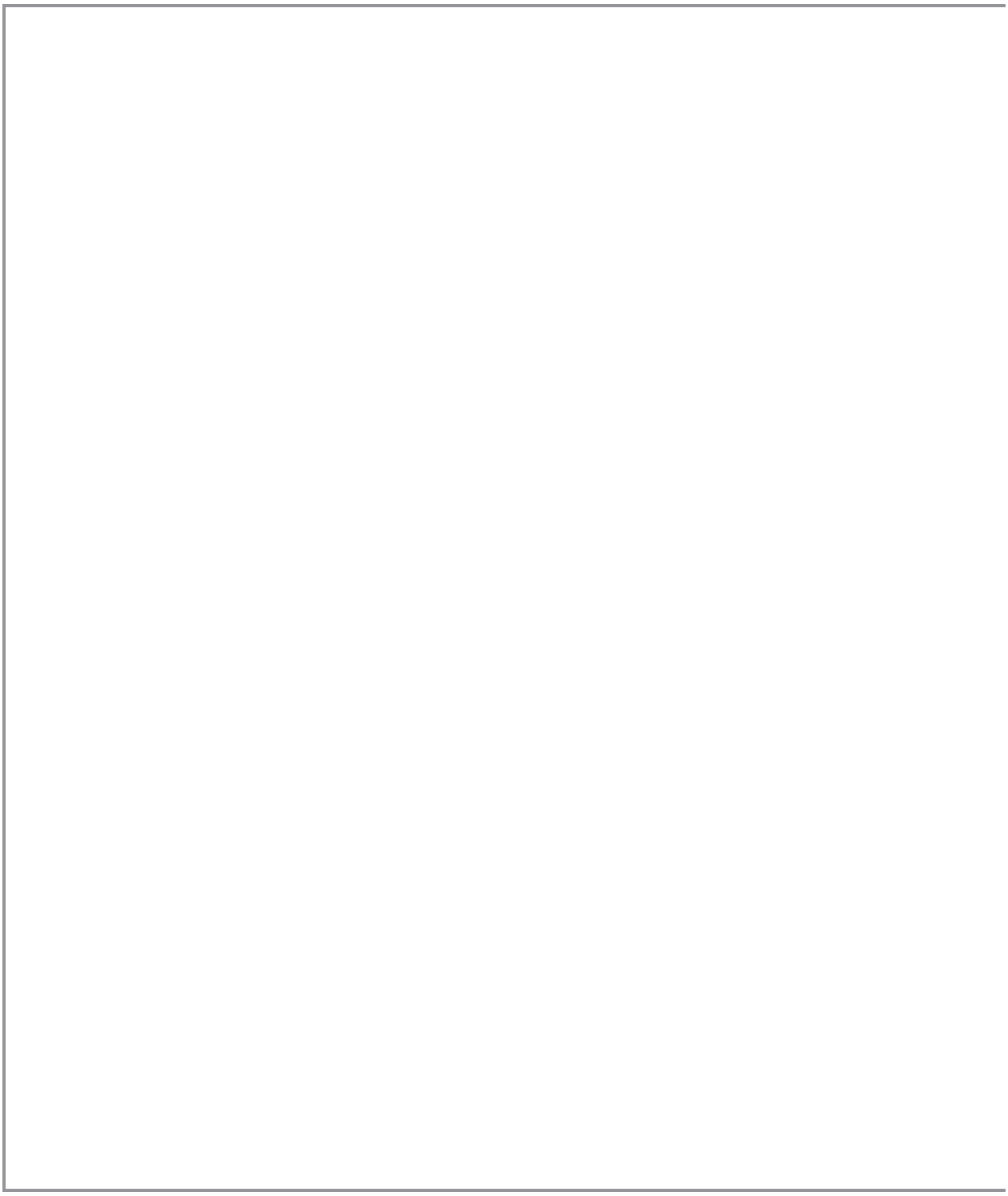


Proper start position for a strict curl
Photo credit: Vadim Snitkovsky

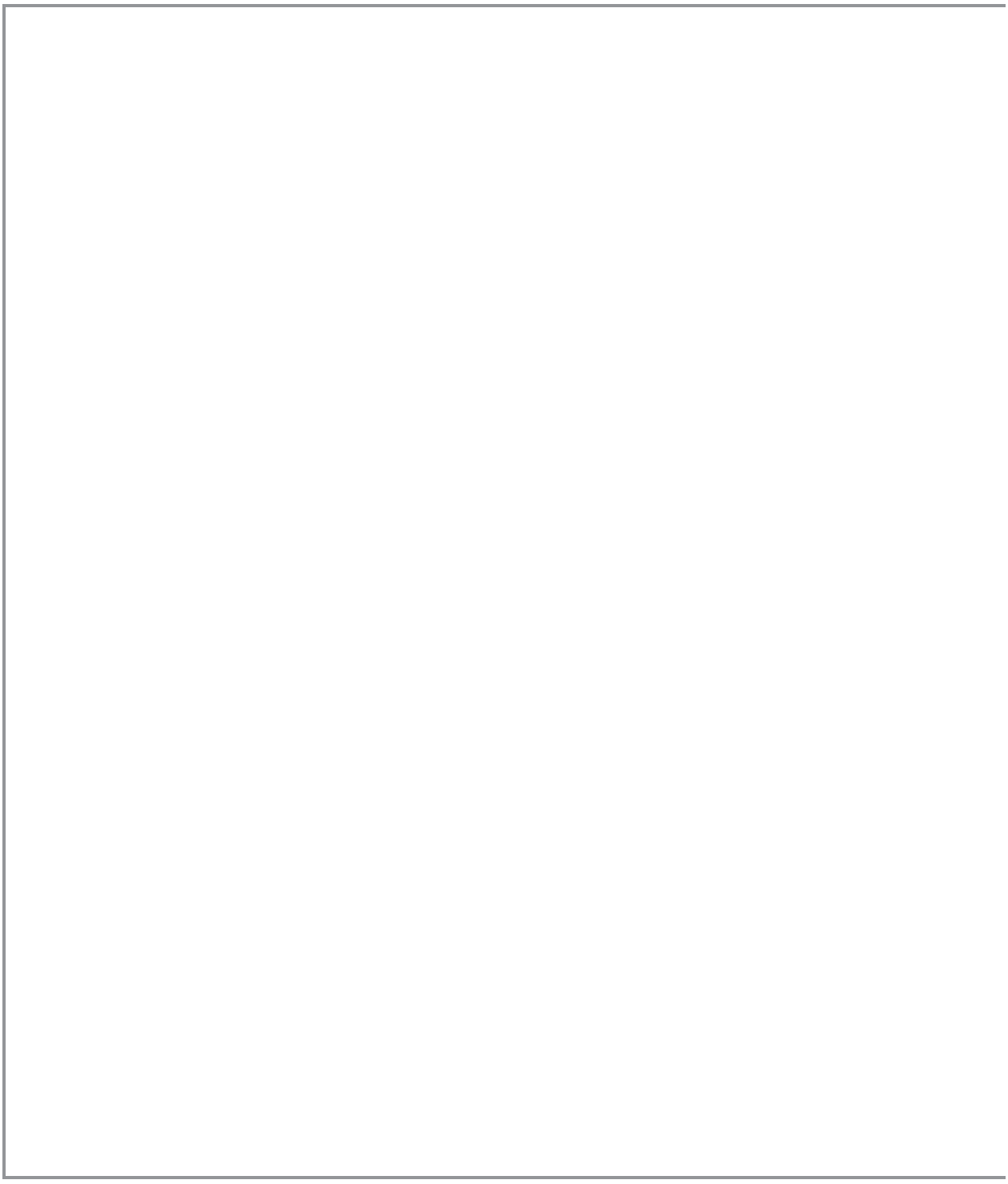
Once you are in the proper starting position, the judge will tell you to “Lift” or “Curl,” and you will curl the weight up. Take a deep breath in and hold it, brace yourself, and then curl upward. Your upper arms will likely move forward some, and this is fine. Push yourself back into the wall and be sure your butt remains against the wall; it will have a tendency to come off the wall when you are lifting heavy. You may find that curling your wrist back to you (flexing the wrist) is desirable once the weight is a little higher than halfway up. You will be lifting the weight up to your throat, nose, or chin. Once you have successfully completed the lift, hold the weight in that position. The judge will say “Down,” and then you can return the weight to your waist. You must keep your back against the wall on the descent; letting the weight drop rapidly can easily pull you off the wall. It is better to lower with control — if you have the strength to lift it up you have the strength to lower it with control. At that point you are finished, or the judge will say “Rack,” and you can set the weight down on the ground.



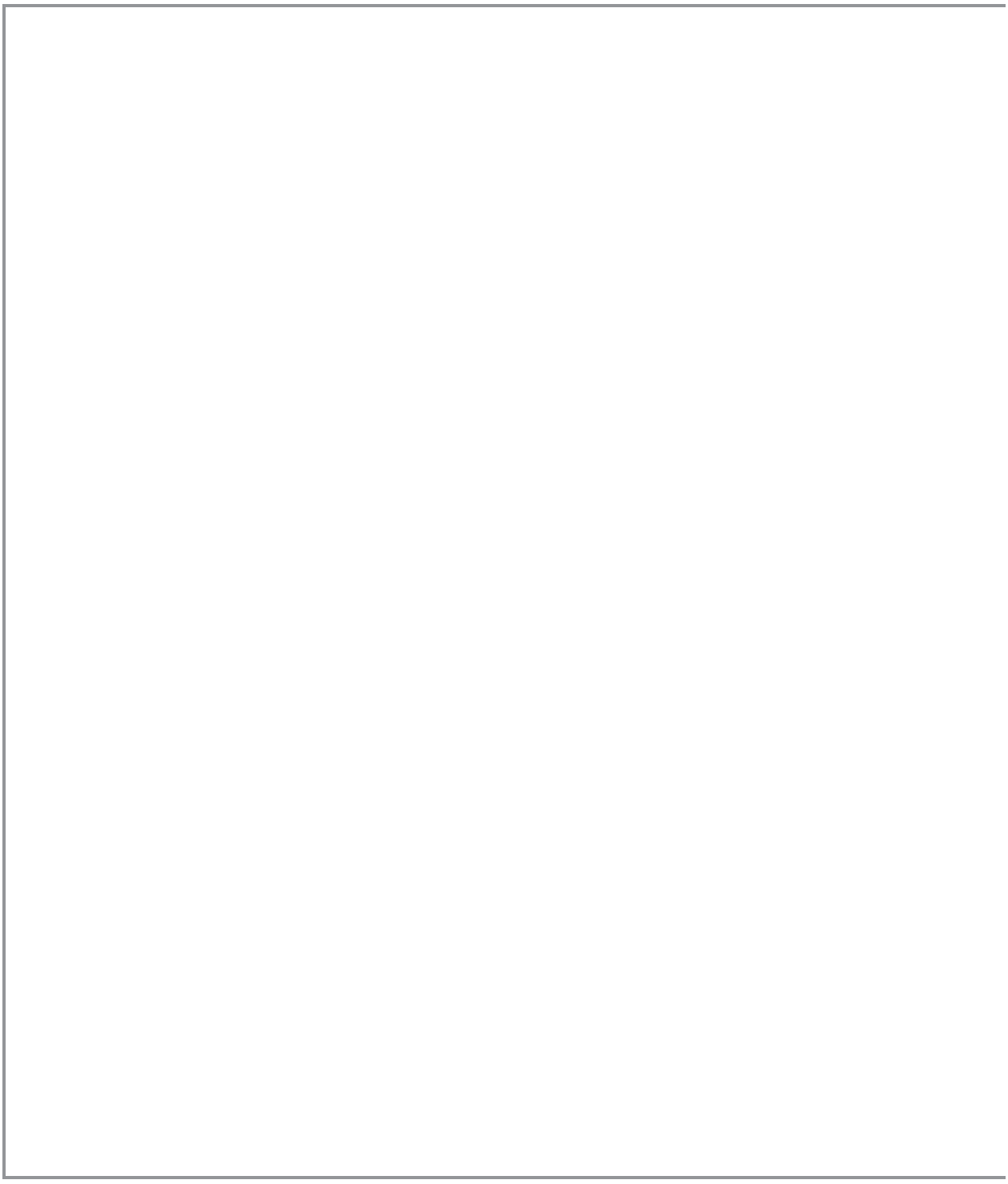
▶ **1: The lifter takes their grip on the bar**



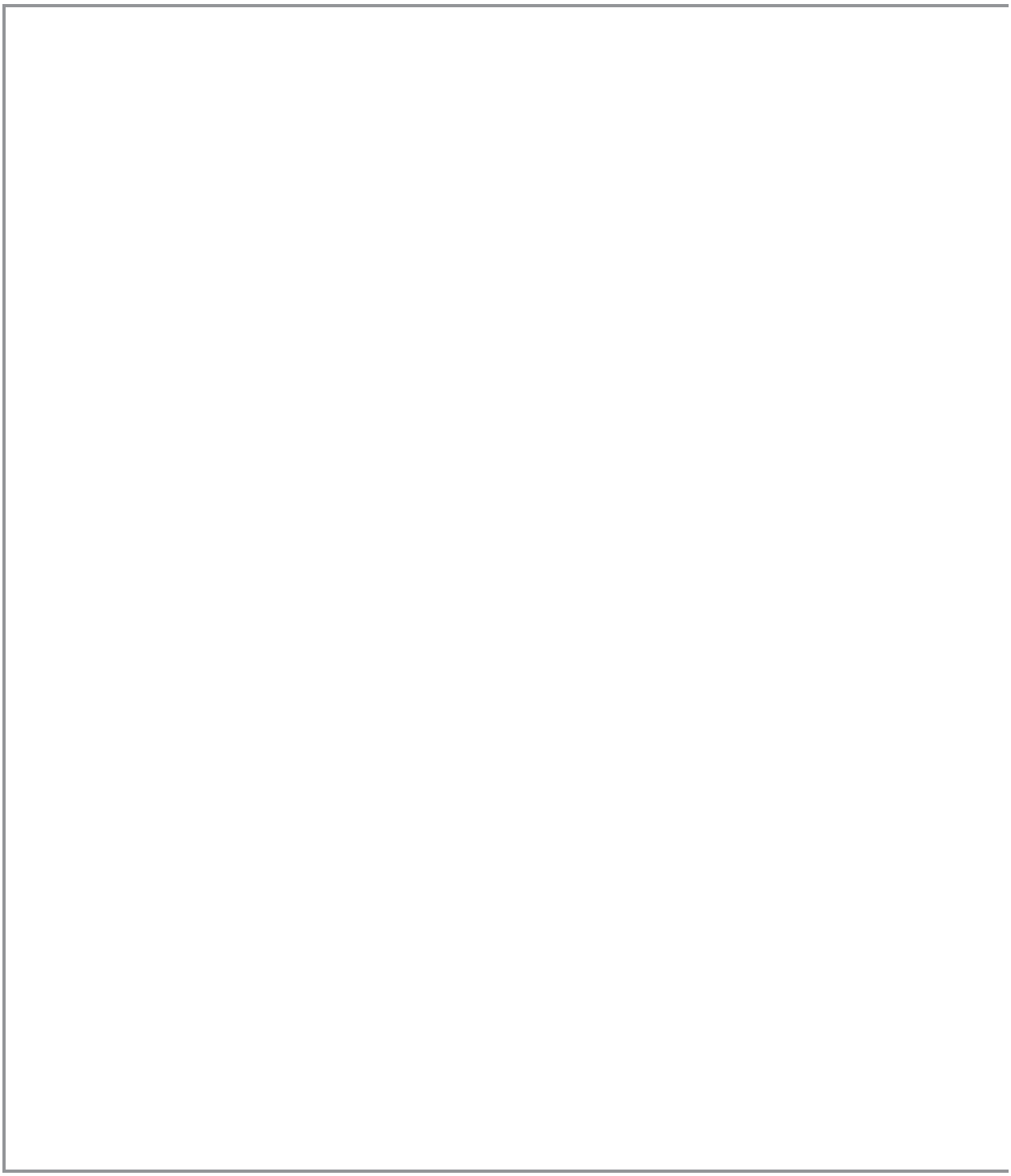
▶ **2: The lifter walks back against the platform**



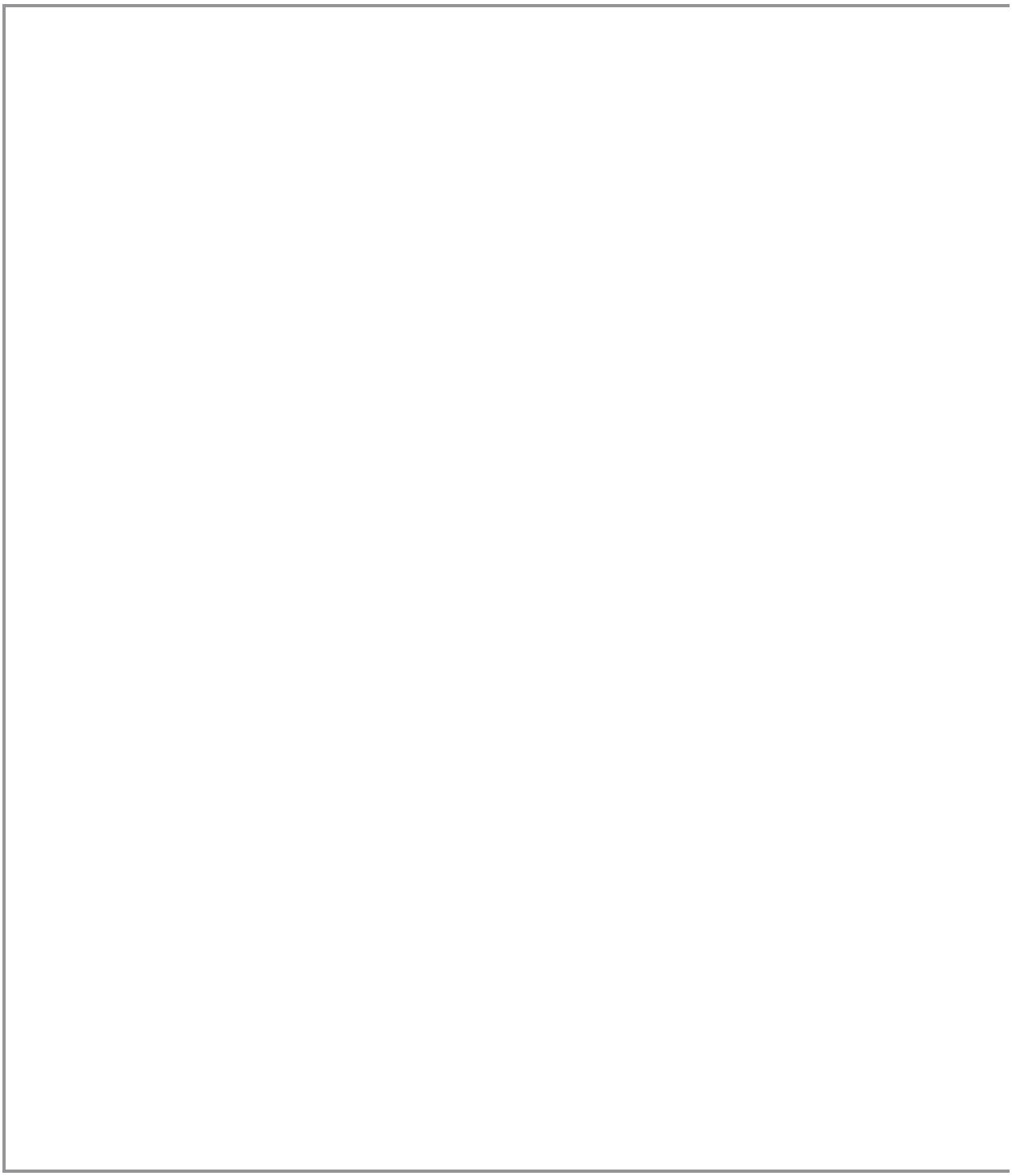
▶ **3: The lifter gets into the start position and waits for the start command**



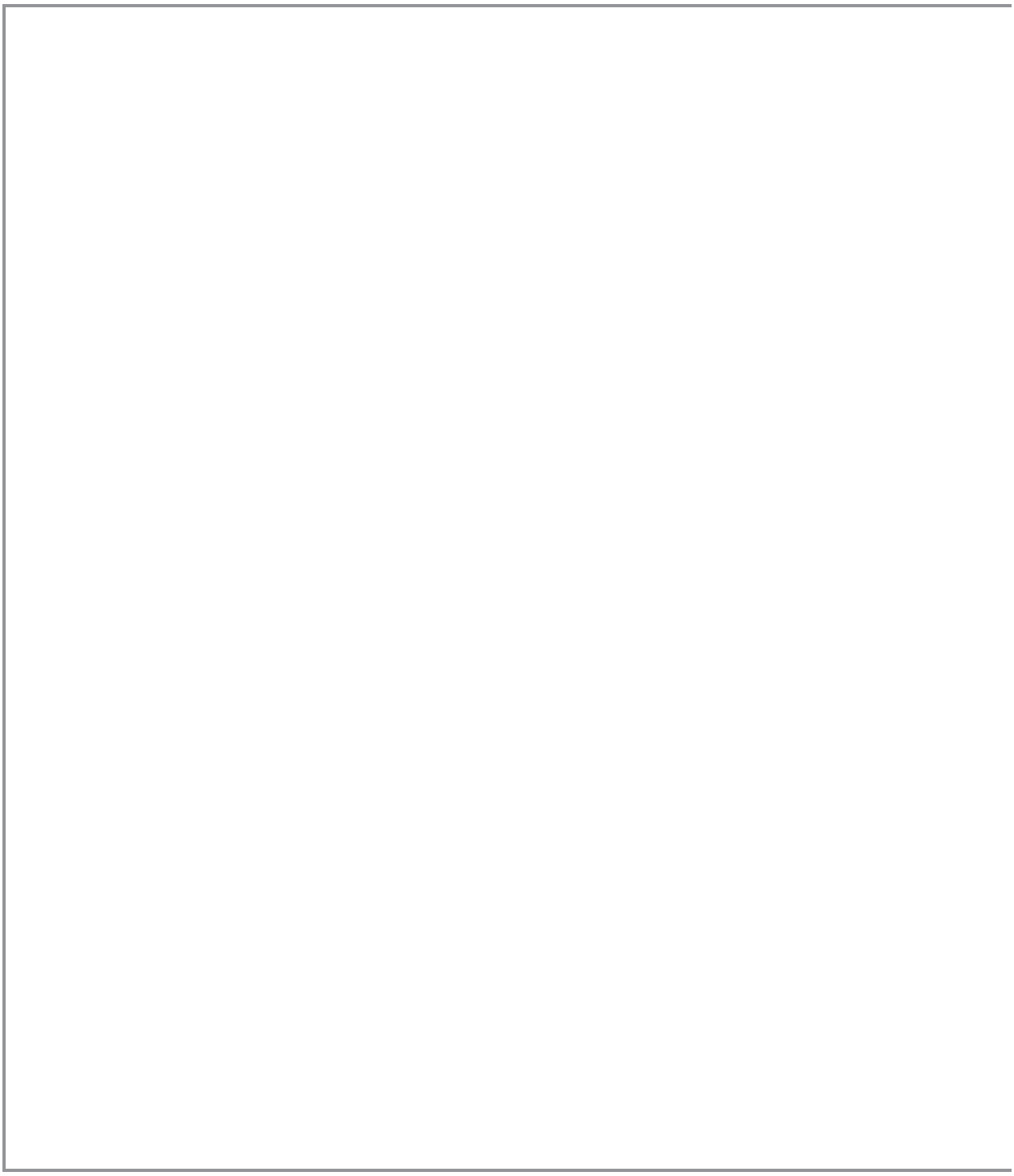
▶ 4: The lifter begins to curl, keeping the elbows back and the bar close to the body initially



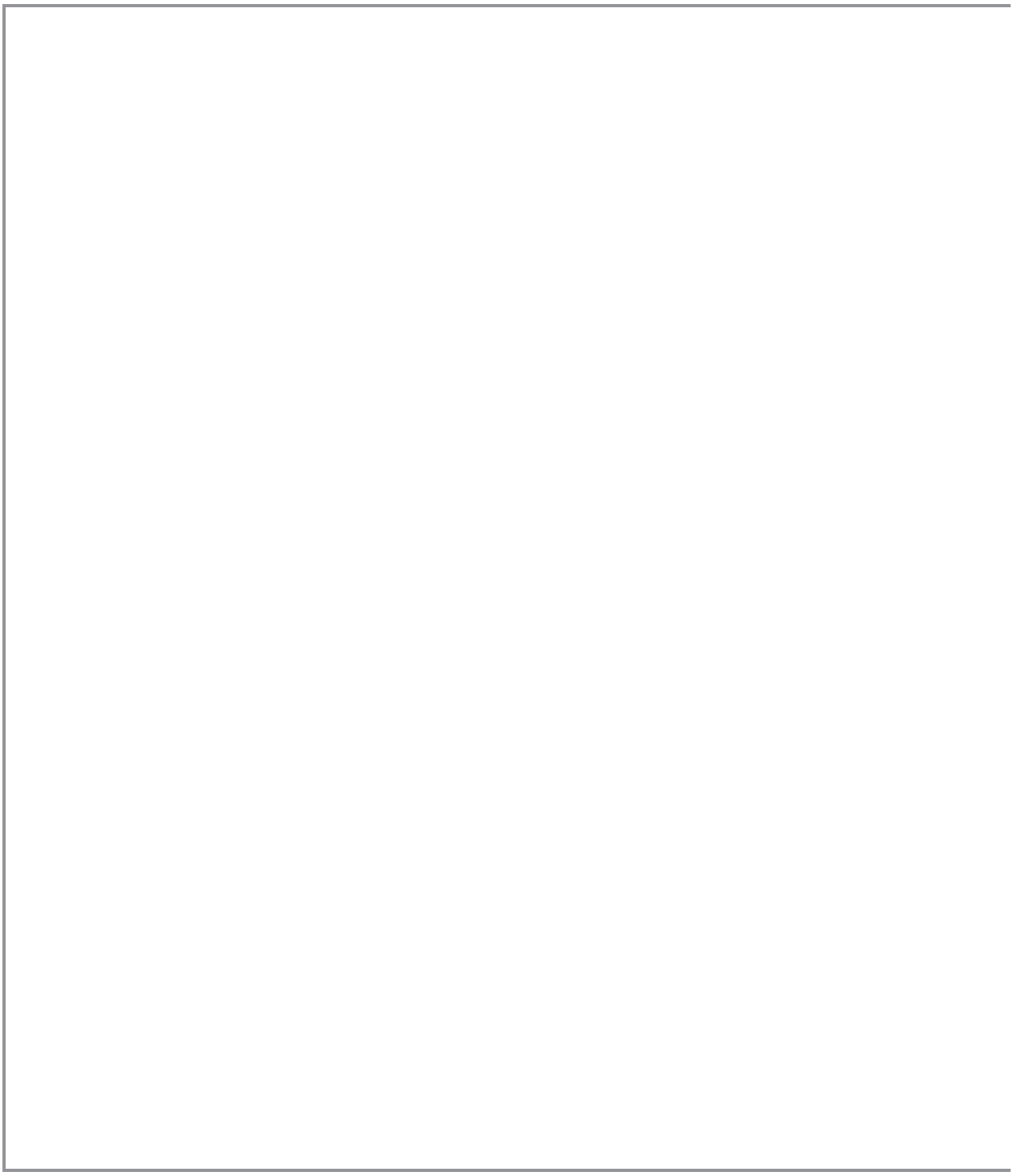
▶ **5: The lifter flexes the elbows and begins to bring the elbows forward**



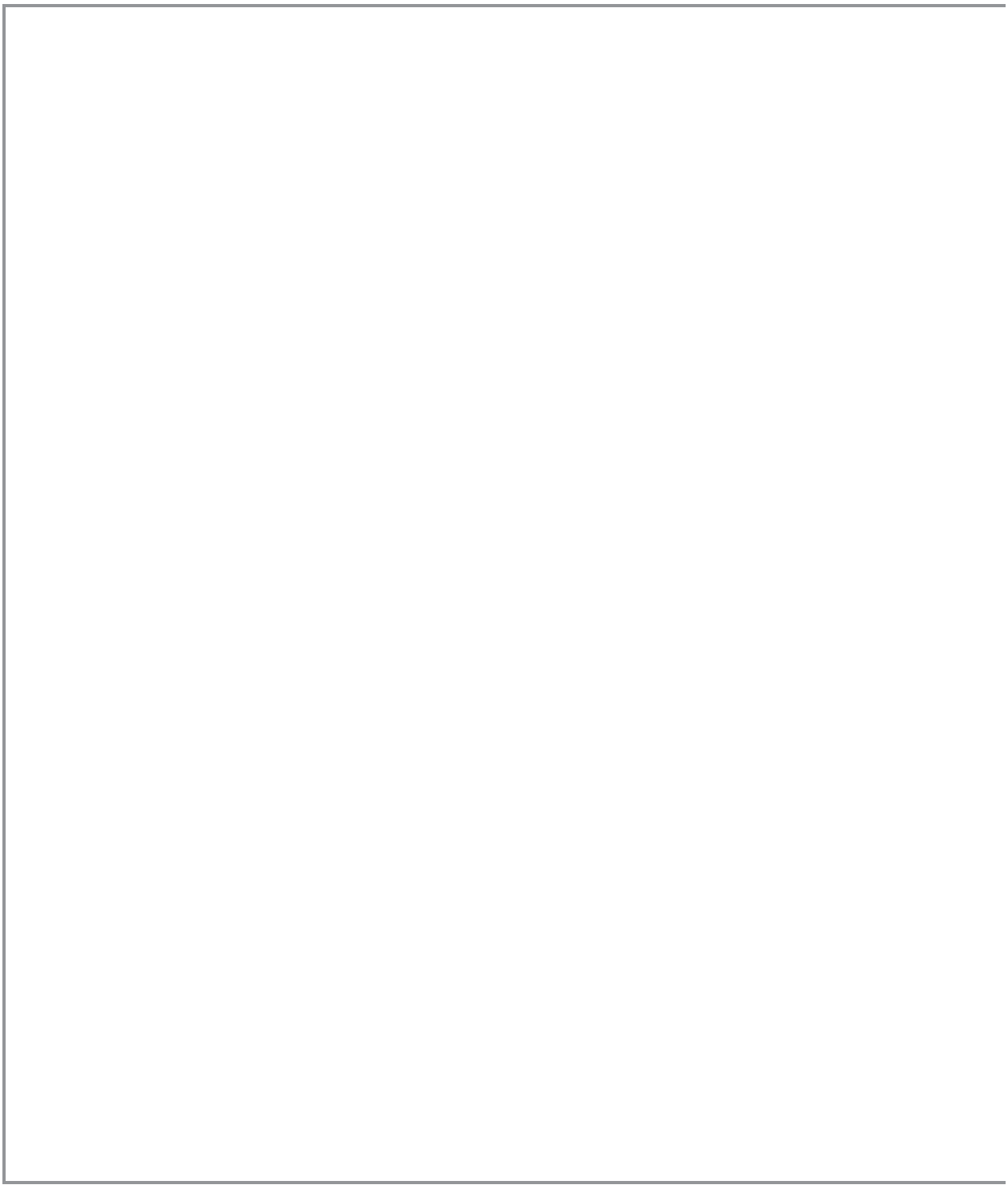
▶ 6: The lifter grips tightly and presses their upper back into the support at the sticking point



▶ **7: The lifter curls the wrist to reduce the moment arm of the resistance**



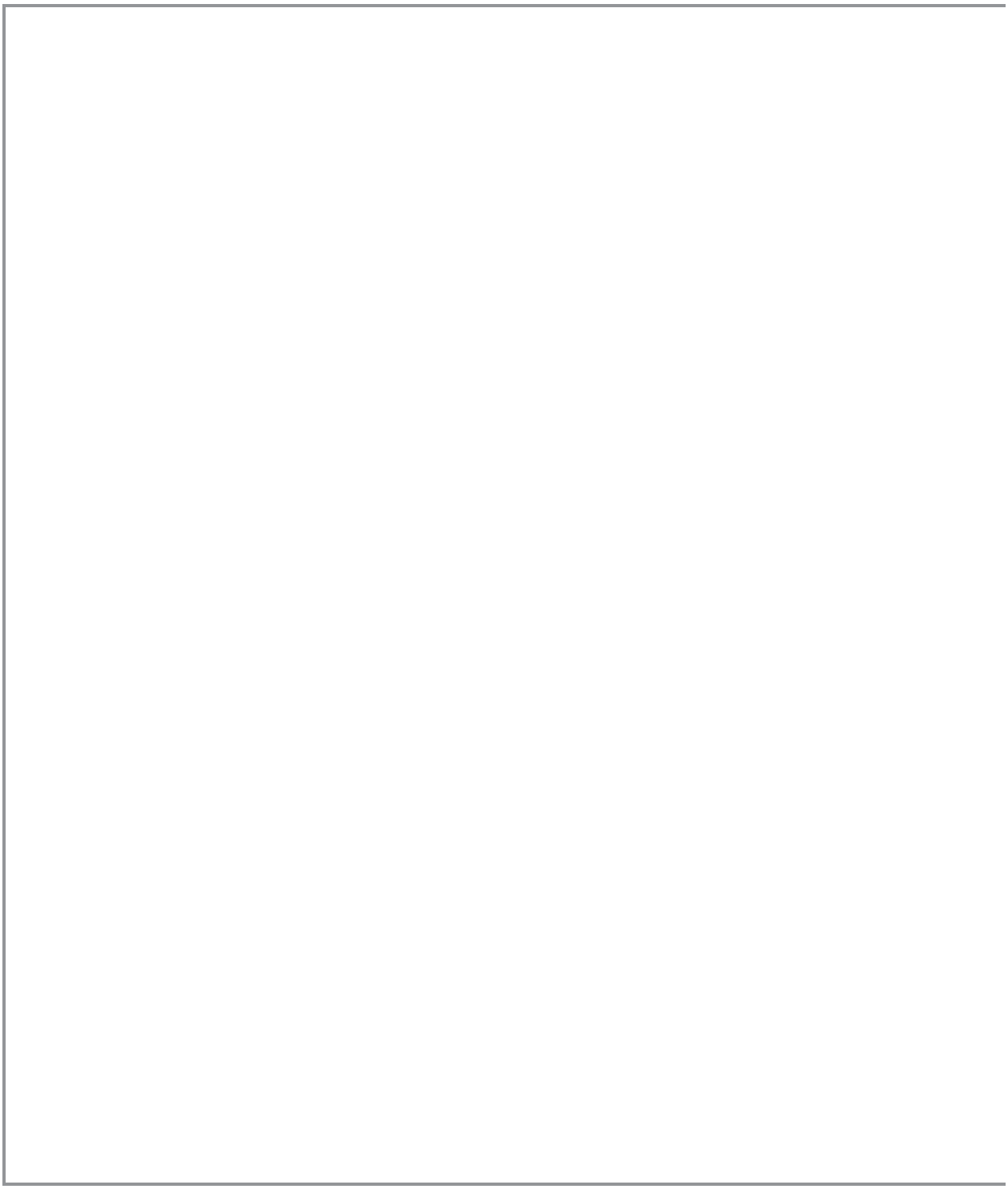
▶ **8: The lifter pushes the elbows slightly forward**



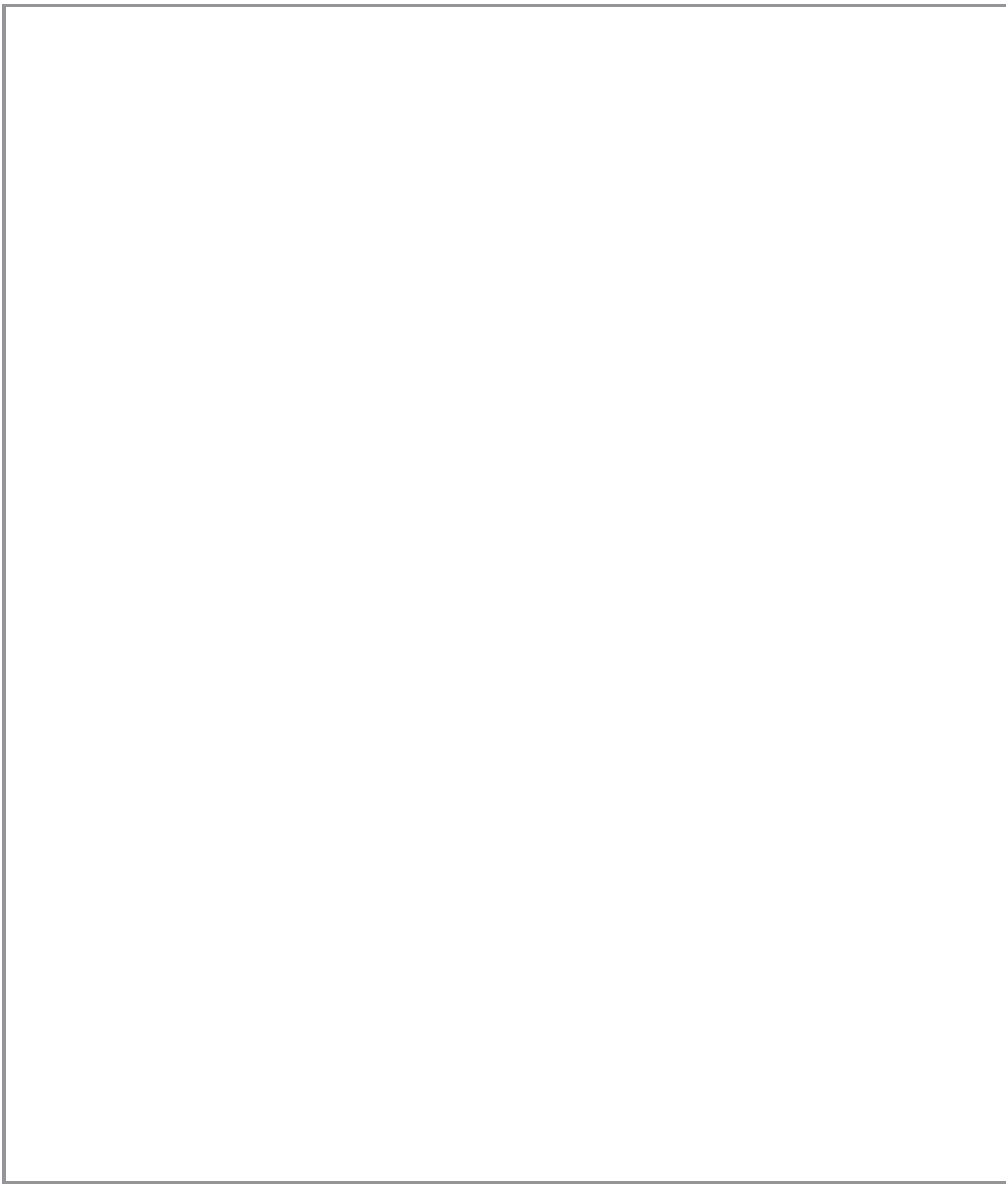
▶ **9: The lifter drives the bar toward the face**

▶ **10: The lifter holds the end position (the nose) and waits for the Down command**

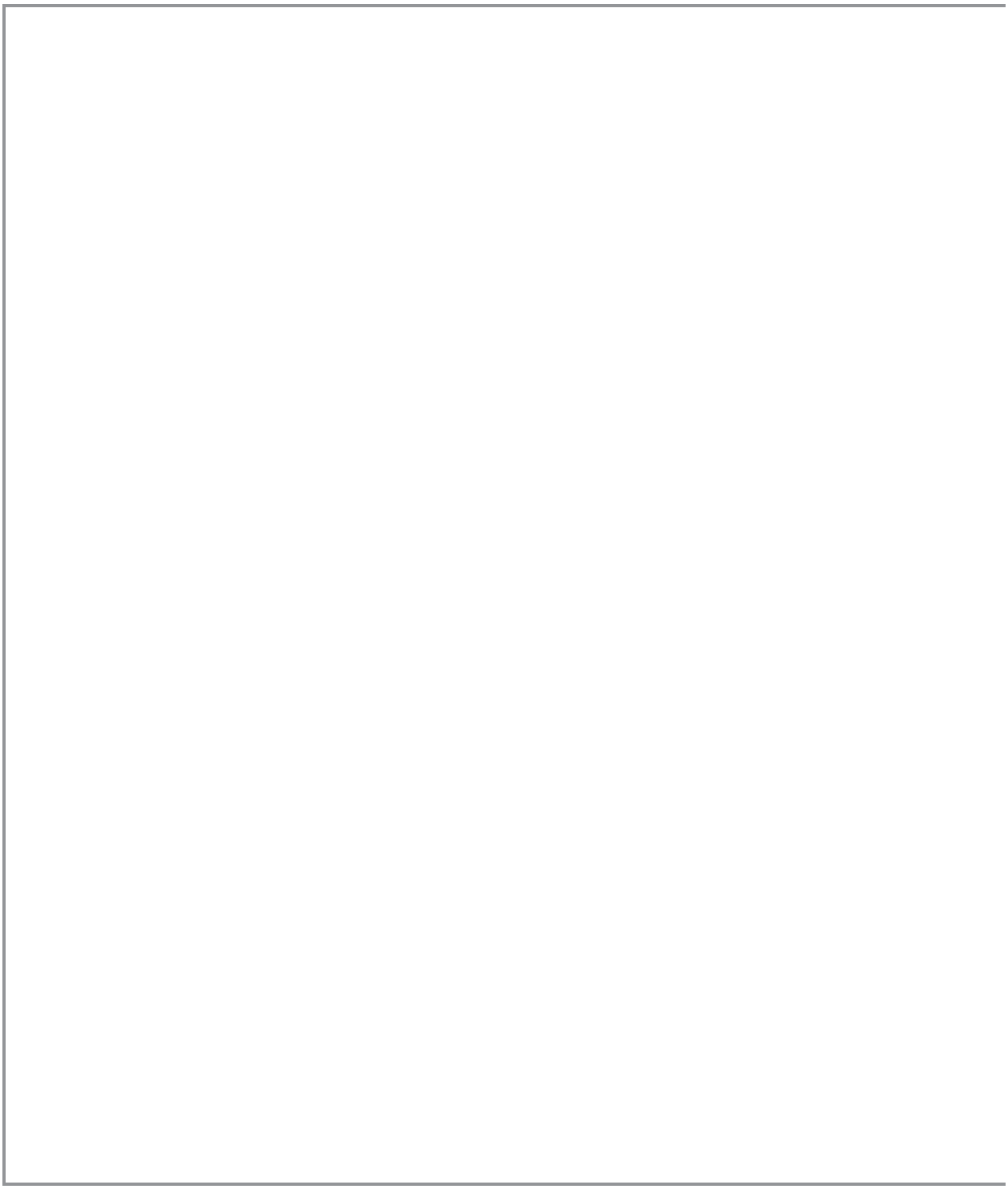
▶ **11: The lifter initiates a slow negative**



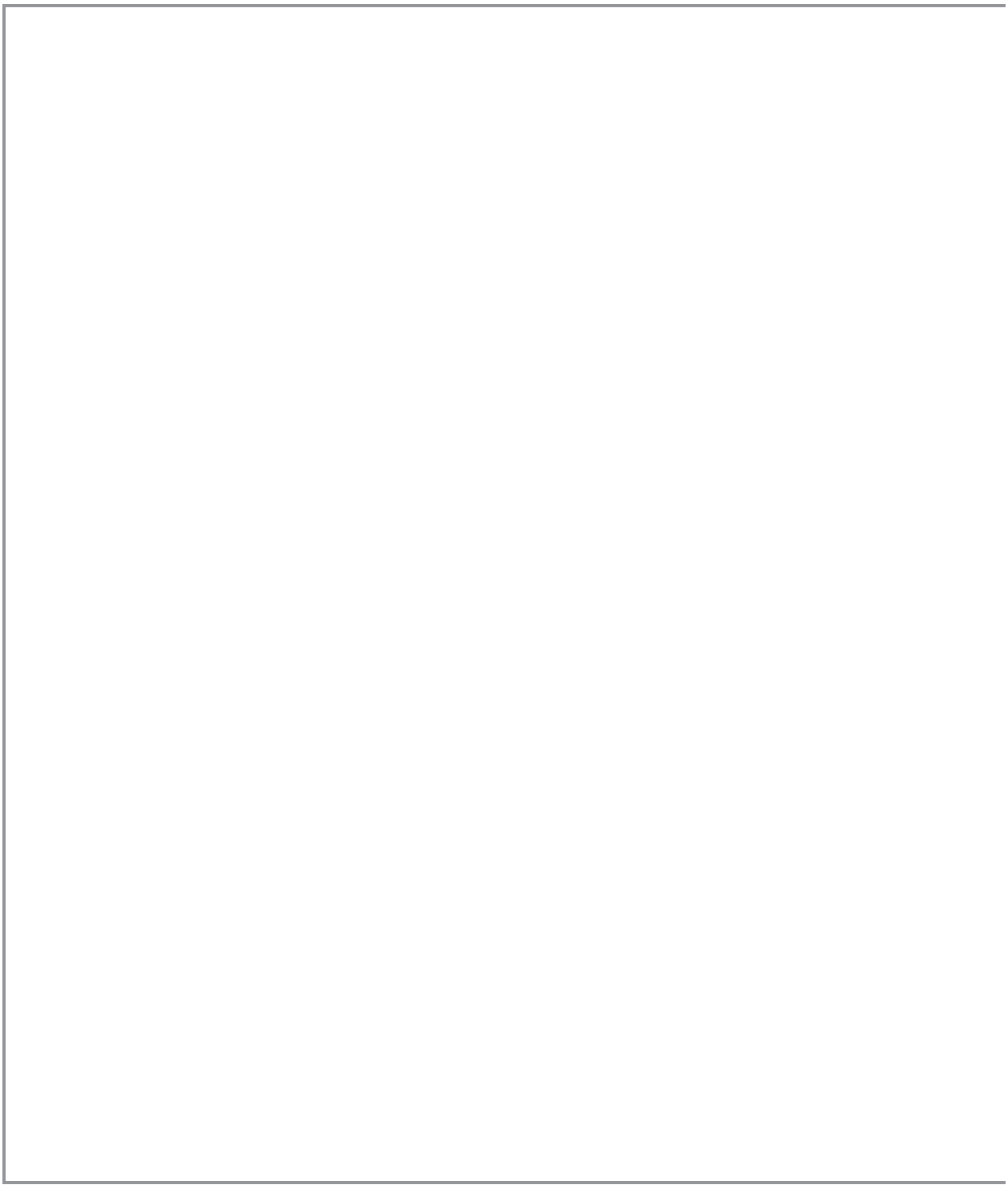
▶ **12: The lifter maintains the angle of the elbow and brings the elbows back to the body**



▶ **13: The lifter keeps their upper back against the wall during the negative**



▶ **14: The lifter holds the finished position and waits for the Rack command**



▶ **15: The lifter re-racks the bar**

Photo credit: Vadim Snitkovsky

Note: This lifter's set-up is slightly off; the feet should be significantly wider. In addition this series is missing the proper start position for receiving the curl

command which is arms and legs locked out, as seen in image 14 and also as was demonstrated previously.

Standing Strict Curl – As the name implies, you are standing straight during a standing strict curl. You will approach the bar just like a deadlift. Depending on the organization you lift with, they will either give you a command to pick the weight up, or you can just pick it up. Once you have your grip (which is palms up, of course; a reverse grip would be much weaker) stand up with the bar and hold it at arm's length, as though you had just finished deadlifting the bar. Now you want to get your feet set.

Allow the upper arms to move forward as you curl.

Once you have gripped the bar, lifted it up, and set your feet, you will get the command to "Lift" or "Curl" the weight up. Take a deep breath in, hold it, lean forward just a little bit if you can, and then curl the weight up to your chin area. The end position of the curl will vary depending on your own biomechanics, but most people finish the curl near the throat, nose, or chin. The bar should not be any higher than the forehead, and it should be higher than the collarbone. It is okay to allow the upper arms to move forward as you curl; in fact that is encouraged. If you try to pin your elbows by your side so the upper arm (humerus) doesn't move forward you will be very limited in the total weight you can curl. Most people finish with their upper arm having moved between 30 and 45 degrees. If the upper arm moves almost 90 degrees forward, then you probably moved it too much, and you probably lost power as you did so. As you curl the bar up to your face, continue to hold your breath to keep your trunk very tight. Once your elbows are bent about 90 degrees, you may want to curl your wrist toward you (flex the wrist) as you lift the weight. This will help bring the weight closer to your elbow, thus decreasing the moment arm of the

resistance force. I would do this only on very heavy sets with very low reps, as too much wrist movement will often aggravate the wrist. At no time should you let your wrist bend backwards during the curl.

Hold the bar near your throat, nose, or chin and the judge will say "Down." Return the bar to its starting position at arm's length. This is not hard and should not be a problem for you. Once the bar is in the starting position either the judge will say "Rack" indicating you should place the bar on the ground, or you can just set it down, as the lift is over.

If you do move your body during the curl, you lean forward right as you begin the curl, and, then, in the first half of the movement, you lean backward. This should be a relatively quick and explosive movement, but remember, this is a strict curl, so it must be subtle. The purpose is to get some momentum to drive the weight past its sticking point, which is about halfway up during the curl or just beyond that point. Even if you do get some body movement involved, you can't lean backward; you must start forward a bit. If you start totally straight up, any lean backward will result in your whole body leaning backward, and you probably won't get the lift. If you wait to lean backward until the end of the lift, it will tend to be a slower, more noticeable extension of the trunk. I should point out that in suggesting that you lean forward, the lean is initiated from the hips so your trunk is straight (not rounded); it is just inclined forward a bit (as though you were getting ready to perform a tricep pushdown). Then when you lean back the lean comes from the hips again — the trunk remains completely straight; it just reclines backward a bit. Throughout the entire lift the trunk will remain locked in position with the chest up; it will just tilt forward and backward slightly to get some momentum.

Things To Do while Strict Curling

- Lock the legs straight to start the lift

- Take a wide stance
- Have heels far away from the wall but within the limits stated by the rules
- Take a proper grip on the bar
- Drive your upper back and butt into the wall
- Start with your arms straight to begin the lift
- Take in a big breath before you curl
- Squeeze the bar hard
- Keep the forearms locked or slightly flex/curl the wrist forward
- Get your elbows under the bar as soon as possible
- Hold the bar at the top position
- Lower the bar under control

Things NOT To Do while Strict Curling

- Start with knees bent or bend them during the lift
- Take a narrow stance
- Have your heels close to the wall
- Grip the bar at an awkward or improper angle
- Exhale before you have completed the curl
- Allow your wrists to bend backward (extend) while curling
- Pin your upper arms back against the wall
- Use too much momentum to drive your butt or shoulders off the wall
- Expect to lift the same weight against the wall as you do when standing upright
- Let the weight free fall on the negative

Flexibility/Mobility Problems

The strict curl doesn't require the same flexibility or mobility as the Big 3. The joints stressed by the strict curl include the elbow, shoulder and the wrist. The wrist is the joint likely lacking the flexibility necessary for a good curl; many lifters lose the ability to fully supinate their wrist as their training career progresses. Forearm stretches can help with this, as can ART/ massage performed at the elbow and wrist. Lifters can use a barbell for curls but treat this as a stretch and go light. In addition performing back exercises with a supinated grip (chin-ups, 45 degree bent over row, reverse grip lat pulldown) all will force the wrist into that fully supinated position.

> ***It is hard not to equate muscular arms with strength.***

Common Problems

The most common problems that occur when people train the curl include: using too much weight, using a limited range of motion, and using an improper grip on the bar.

Regular gym goers like the idea of big arms and strong biceps; if you grew up watching Arnold movies, it is hard not to equate muscular arms with strength. In an effort to attain those “big guns” people often go too heavy and use a cheat curl involving significant body momentum. A little cheat can be fine but when the lift looks like a clean or a mini-seizure, that is not ideal. Of course, lifting up against the wall (and keeping your butt there) immediately fixes this issue.

The second big issue is using a limited ROM; in this instance lifters will not extend their elbow on each rep. One doesn’t necessarily need to fully extend the elbow on every training rep, but if one is curling and it really looks like the elbow is locked at a 90 degree angle and the movement is actually at the shoulder (and the trunk), again, that is not ideal. Chronic training like that can result in shortened muscles and tendons crossing the elbow, resulting in weakness when one does have to straighten the elbow (such as at the beginning of a competition strict curl).

Finally lifters often take an incorrect grip on the bar, and they will hold the bar in the opposite direction from what is ideal. This fact can be compounded because not all curl bars are the same — some have a very significant level of camber and others are near straight. Curlers should be aware that there is not a “standard” competition curl bar and that the angle of camber might vary from one competition to another; that is just the way it is currently.

Common Cues for the Strict Curl

When performing a heavy curl in the gym or in competition, it can be useful to have some cues in place to prompt optimal technique. Outlined below are some common cues you will hear; they are generally presented in order of how the strict curl is performed. It would not work to focus on all of them, but pick 2–4 that seem to work well for you and zero in on them during your lift.

Good Grip
Good Set-up
Wide Stance
Legs Straight
Arms Straight
Big Air
Brace
Curl
Squeeze
Curl Wrists
Head Up
Elbows
Hold
Lower Slowly
Wait

Not all curl bars are the same.

Common Competition Mistakes in the Strict Curl

Lifters will make a few common mistakes in a competition. Luckily, with some preparation, these can be relatively easily fixed. These mistakes include:

Not understanding the difference between a standing strict curl and a wall strict curl — many lifters will curl in the gym standing up, see the weights lifted in meet results, assume they will do well and then attempt those same weights up against wall and be quite disappointed with the results. Even a super strict standing curl is noticeably easier than a curl up against the wall.

Not being aware of the weight of their curl bar — this isn't a mistake made in competition per se, but it affects what happens in a competition. A lifter will curl in the gym and assume their bar has a certain weight, for example, assuming it is 25 or 35 lbs. Most curl bars are closer to 15 or 20 lbs; thus the lifter is overestimating what they can actually lift. In the curl, that amount of weight makes a big difference, and again the result is disappointing for that lifter. The solution is to actually weigh any curl bar you are going to train with on a reputable scale (usually found in the gym locker room); then you will know for sure what the actual weight is.

Various types of curl bars, all with different weights

Gripping the bar in the wrong spot — this has been covered at length, but don't let the excitement and nerves of a competition throw you off your game. Know exactly where and how you like to grip the bar, and make sure you take that grip on each rep. Proper grip position is just as important on the curl as it is on the bench, perhaps even more so.

Making too big of a jump in weight — A small increase in weight in the strict curl can make a big difference. You have to remember part of it is the percentage of what you are lifting — if you curled 150 lbs and then jumped to 160 for the next attempt, that is the same as someone benching 300 and going to 320 or squatting 450 and then going to 480 — not ridiculous but big enough to feel. The second part of that is, because the curl is an isolation exercise, the pounds per rep value of each rep isn't as significant. If you can squat 450 for 3, you know you have a decent amount of strength still left in

you. You might be able to curl 150 for 3, but that isn't a promise that you will be able to get 160. My standard advice for those performing the strict curl is to go up 10 lbs (5 kg) after their first attempt and just 5 lbs (2.5 kg) on the subsequent attempt. Federations should (and usually do) allow lifters to increase the curl by just 2.5 lbs (1.25 kg) if they wish, and even that weight can make a big difference at maximal poundages.

Allowing the butt to come off the wall — this can be fixed by two things; one (and the only thing you can do in the moment of competition) is to take a wide stance and drive your hips into the wall. The second thing is to lower the weight, but once the attempt has been turned in this is not an option. Don't let your butt come off the wall in training or that motor pattern is likely to show up again come competition time.

Finally lifters will often receive red lights for lowering the weights too fast and letting their upper back or butt come off the wall on the way down. To correct this, simply perform a slow negative with the weight and don't celebrate (or relax) until that final command is given and the lights are up on the scoreboard.

Technical Rules of Performance

Presented below are the official rules for the Strict Curl. Rules can vary from federation to federation, these particular rules apply to the 100% RAW Powerlifting Federation. These particular rules are focused on performing the strict curl up against the wall. A simple explanation follows if necessary.

Strict Curl:

1. **The lifter shall face the front of the platform. The bar shall be held horizontally across the thighs with the palms of the hands facing outward, and fingers gripping the bar. The feet shall be flat on the platform with the knees locked and arms fully extended. The lifter shall have their shoulders and buttocks firmly against a wall during the lift.**

The lifter will face the platform, using a supinated grip. The feet are flat, the legs and arms locked to begin the lift. The upper back and butt must be against the wall.

2. **After removing the bar from the racks, the lifter must move backwards to the wall to establish his starting position. The lifter shall wait in the starting position for the Head Referee's signal. The signal will be given once the lifter is motionless and the bar is properly positioned with your Head Up & Chin Up and arms extended fully down. The Head Referee's signal shall consist of an upward movement of the arm and the verbal command "Curl."**

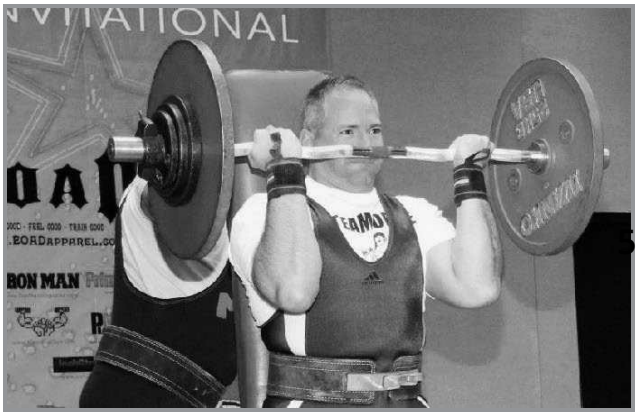
The lifter will grab the bar, move backward against the rack, and get into the proper position with head and chin up (looking straight ahead).

3. **Once the curl command is given the lifter must bring the bar up to the fully curled position (bar near chin or throat with palms facing backward). The knees must remain locked and the shoulders and buttocks against the wall throughout the entire lift.**

The lifter will curl the bar up; knees stay straight and body stays against the wall.

4. **When the lifter has reached the finished position, the Head Referee's signal shall consist of a downward movement of the hand and the verbal command "Down." The signal will not be given until the bar is**

held motionless and the lifter is in the apparent finished position.



A lifter in the finish position of the curl
Photo credit: Sabrina Tarbell

Once the lifter is in the finished position, they will receive the Down command.

5. At the completion of the lift, the knees shall be locked and the shoulders and butt firm against the wall, and the lifters will need to wait for the signal to replace the bar. This will consist of a backward motion of the hand and the verbal command "Rack."

Lifter must keep legs locked and shoulders/butt against the wall on the way down. They will then receive the Rack command to set the bar down.

6. The legs and hips may not be used in any way for momentum to complete the lift. Lifter may not lean back to assist in bringing the weight up. Any thrusting of the legs or hips for momentum is not allowed. The feet must remain flat and motionless throughout the lift.

You can't use your legs or trunk to help; feet must stay flat and can't slide during the lift.

7. Any raising of the bar or any deliberate attempt to do so will count as an attempt.

Once the lifter tries to lift the bar up, that counts as the attempt.

8. The lifter may, at the Head Referee's discretion, be given an additional attempt at the same weight if failure in an attempt was due to an error by one or more of the loaders.

If the head judge agrees, the lifter can retry the weight if the bar was misloaded.

9. **This lift will be judged by 3 referees.**
10. **The back part of the heel cannot be more than 12" from the wall. There should be a line (tape) designated on the floor where the heels cannot cross over.**

Heels must be 12″ or closer to the wall; there should be a mark designating this spot on the curl platform.

Causes for Disqualification of the Strict Curl:

1. **Any downward movement of the bar before it reaches the final position.**
2. **Leaning back to assist the lifter in raising the weight.**
3. **Shoulders or buttocks coming off the wall during the lift both while going up and down.**
4. **Failure to stand erect with the shoulders square and buttocks flat against the wall at the completion of the lift.**
5. **Failure to keep the knees locked and straight during the lift.**
6. **Failure to keep feet flat during the lift.**
7. **Stepping backward or any foot movement such as rocking the feet.**
8. **Lowering or racking the bar before receiving the Head Referee's signal to do so.**
9. **Bouncing the bar off the thighs or bending the back to assist the lifter in starting the upward motion.**

Does a Strict Curl Have any Value?

The question of the value of any exercise is relatively subjective. Does the bench press or the squat have any value? Of course, to a powerlifter, the answer is "Yes." I think we can make the argument that to a lot of people those exercises have value because when you get good at the powerlifts you are also getting good at a very large number of exercises. If you are interested in health, physical fitness and aesthetics, the classic powerlifts do have a lot of value. But what about a bicep curl?

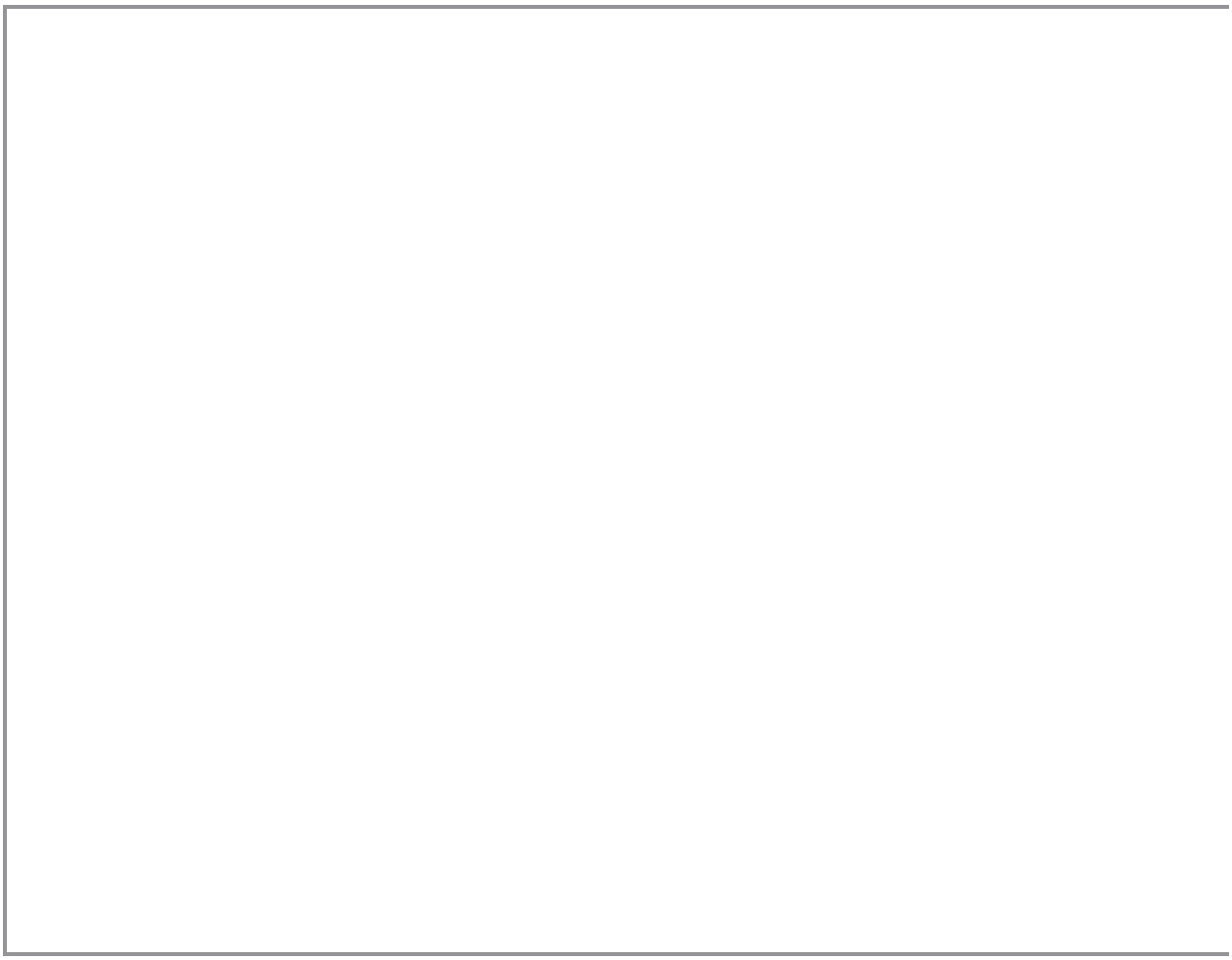
There are two common knocks against bicep curls. The first one is that the curl is an isolation exercise, which means it involves just one main joint. It is true that a bicep curl is generally considered an isolation exercise (although technically there is often movement at the shoulder and/or wrist as well as at the elbow), but to me that doesn't necessarily mean its value should be reduced. The only way to test the strength of a smaller muscle group without also involving a lot of other muscle groups is to perform an isolation exercise. You can't test the strength of the biceps without doing bicep curls. Things like rows and pull-ups would involve the biceps but would place more emphasis on the lats and other muscles.

The second knock against the bicep curl is that it is not a functional exercise. This comment is often applied to all isolation exercises with the observation that muscles don't work in isolation — they work together. The word "functional" has several definitions but to me the most applicable is, "Does the exercise mimic activities that we do in everyday life?" Using that definition, I would say that bicep curls are perhaps one of the most functional exercises one can do. Bicep curls would clearly be more functional than a

The idea that a bicep curl is not functional carries no weight whatsoever.

bench press and, as pointed out earlier, than a squat. Again, the definition of functional we are using is that the exercise as performed mimics activities of daily living. People perform bicep curls in real life all of the time — using concentric, isometric, and eccentric actions. Lifting your grocery bags out of the cart or out of the trunk of your car is a bicep curl. Curling your book bag up to your shoulder is a bicep curl. Holding a tray of food or a baby in the crook of your arm is a bicep curl. Carrying your laundry basket upstairs or carrying your spouse around the house (for fun) is a bicep curl. We use our biceps in their basic function — elbow flexion — all the time, and we often use them while the rest of our body remains relatively stable. To me the idea that a bicep curl is not functional carries no weight whatsoever.

It is true that you generally get more bang for the buck performing compound exercises (2 joints or more) versus isolation exercises. You hit more muscles and can usually lift more weight with compound exercises. But isolation exercises do have value, particularly for the muscles that are always going to be synergistic in the compound exercises. These muscles include the biceps, triceps, and hamstrings. Isolation exercises for the larger muscles like the pecs, lats, and glutes carry less value because the compound exercises for the same muscle groups generally hit that group better, but all exercises have their place.



Professional Bodybuilder Robby Robinson had some of the most impressive biceps of all time

Photo credit: Robby Robinson

Why a Bicep Curl?

It is a fair question to ask why a bicep curl should be included as the 4th contested lift and not another exercise such as a skull crusher or a calf raise? I am not sure there is just one concrete answer to that question. The bicep curl measures arguably the most important muscle not really measured in a powerlifting competition, which is the biceps. Remember, the big three hit almost every main muscle in the body with the two significant exceptions being the biceps and the calves. In an effort to measure true total body strength, the bicep curl does have some use by filling in a few gaps left from the big three. The bicep muscle is probably the most famous muscle there is; if you ask someone to flex a muscle without specifying which one they will almost always flex the bicep. Big arms (for better or worse) are symbolic of a person's strength. The bicep curl is a free weight exercise that can produce good aesthetic changes to one's physique, and it is a lift that is often practiced in gyms. You do not need much equipment to perform the lift in a competition, you don't really need any spotters for safety, and, as mentioned earlier, the lift is easy to judge especially if done up against the wall. Competing in something like a skull crusher would be harder to judge, would require good spotters, and there would be a strong correlation between the good bench pressers and those lifters good at the skull crusher; thus it would be somewhat redundant.

Apparel for the Strict Curl

You don’t need much special apparel for a strict curl. Comfortable shoes with good grip are important — you don’t want your shoes to slide on the platform as you are pressing into the wall. All lifters in a competition need a singlet. You can wear a belt if you wish while strict curling, but because one is up against the wall, a belt doesn’t have much effect. If you are performing a standing curl, the belt is much more important in bracing the core.

Wrist wraps are likely the most important piece of useful equipment for the curl. The curl places a lot of stress on the forearms and often lifters will let their wrists bend backward slightly, thus decreasing potential force production. The wrist wrap can help the lifter keep their wrist stable and thus help the lifter curl more weight. Chalk, of course, can help one keep their grip on the bar.

Wrist wraps are likely the most important piece of useful equipment for the curl.

Starting Out

Many lifters have performed curls in their training, but not many of them are used to completing a strict curl against the wall. Many are curious as to what they can actually strict curl; to find a very quick estimation of the strict curl maximum, I suggest the following:

Strict Curl for the average trained male

40 x 3–5 reps; then add 10–20 lbs and perform 1–3 reps. Repeat this until a challenging rep is completed or failure occurs. An example might look like this:

40x5, 60x3, 80x3, 100x1, 120x1, 130x1 (hard)

Strict Curl for the average trained female

20 x 3–5 reps (empty bar): then add 5–10 lbs and perform 1–3 reps. Repeat this until a challenging rep is completed or failure occurs. An example might look like this:

20x5, 30x3, 40x1, 50x1 (hard)

Muscles Involved

The curl stresses primarily the anterior muscles in your arms, along with muscles that stabilize the shoulder girdle. The standing curl increases the involvement of additional muscles. Listed below are the muscles working in a strict curl, their relative contribution on a scale from 1–5, with 5 being the most, and then some effective exercises for strengthening those muscles.

Wall Strict Curl

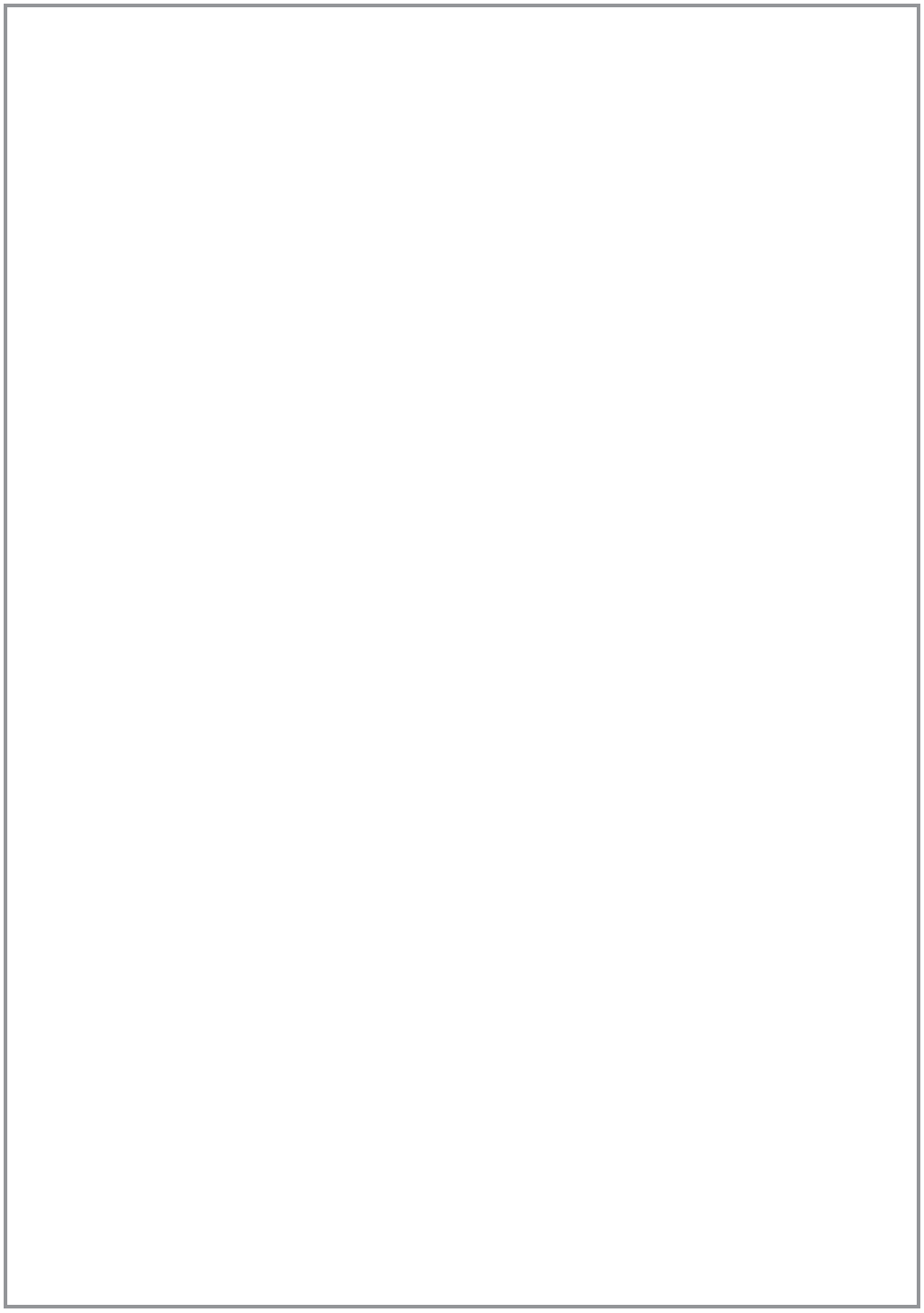
Muscle	Contribution	Exercises to Strengthen the Muscle
Biceps Brachii	5 – Elbow Flexion, Shoulder Flexion	Strict Curl, Standing Curl, Power Curl, Supinated Dumbbell Curl, Cable Curl, Machine Curl, Preacher Curl
Brachialis	5 – Elbow Flexion	Dumbbell Hammer Curl, Rope Cable Curl, Parallel Bar Curl, Machine Hammer Curl
Brachioradialis	3 – Elbow Flexion	EZ Reverse Curl, Dumbbell Reverse Curl, Zottman Curl
Front Deltoid	3 – Shoulder Flexion	Military Press (any), Push Press, Front Raises
Forearm Flexors	3 – Wrist Stabilization/Flexion	Wrist Roller, Wrist Curl, Grippers, Grip Work, Fat Bar Work
Pectoralis Major	2 – Shoulder Stabilization	Bench Press, Pause Bench, Incline (any), Dips, Flys
Latissimus Dorsi	2 – Shoulder Stabilization	Pull-ups/Chin-ups, 45° Bent Over Row, 90° Bent Over Row, Dumbbell Row, Pulldowns
Middle Deltoid	2 – Shoulder Stabilization	Lateral Raise (any)
Posterior Deltoid	2 – Shoulder Stabilization	Rear Delt Raise (any), Rear Delt

		Machine
Trapezius*	2 – Shoulder Stabilization	Deads, Trap Bar Deads, Shrugs, Farmer's Walk, High Pulls, Cleans
Rotator Cuff	2 – Shoulder Stabilization	Internal Rotations, External Rotations, Lateral Raises (pinky up)
Triceps Brachii	1 – Elbow/Shoulder Stabilization	High Board Press, Closegrip Bench, Dips, Rack Press, Skull Crushers, Tricep Pushdowns
Core	1 – Trunk Stabilization	Inverted Sit-ups, L-Holds, Cable Crunch, Ab Wheel, Hanging Leg Raise, Landmines, Rotations
Erectors*	1 – Trunk Stabilization	Deads (any version), Good Mornings, Hypers, Reverse Hypers
Forearm Extensors	1 – Wrist Stabilization	Reverse Wrist Curl, Wrist Roller

**If one is performing a standing strict curl, the traps and erectors receive much greater stimulus and become a key stabilizer/synergist in the exercise and will receive a training stimulus from that exercise.*

Principle of Specificity

Hopefully it is clear at this point that the principle of specificity reigns supreme when it comes to program design and exercise selection. This is particularly true for the Big 3 because of their relatively high skill demands compared to most exercises in the gym. Simply training the muscles involved will not automatically increase the competitive lift. A broader range of exercises are available for the curl due to the lower skill involved; however, an equally common misconception is that the bigger muscle always wins. It is certainly true that muscle size contributes to strength, but it is not the only factor, and having a big set of arms is not a promise of impressive curl strength. Of course, one does need some muscle there to get the job done, and when the sirens sing of big guns that is a tough song to ignore. Luckily as long as you incorporate sensible programming and progressive overload on the key exercises, you can have both size and strength in your arms. Listed below are some important exercises for increasing the curl.



A lifter hitting a federation record curl
Photo credit: Doug Jantz

Exercises and Their Benefits

Wall Strict Curl – This is the classic competitive lift. Performing a lift as it is performed in competition is very helpful to acclimate the lifter to what that lift feels like; it helps with visualization, and, most importantly, it will help the lifter select an appropriate weight for the competition attempts. The negative of primarily performing strict curls is that it can be easy to hit a plateau, and it is tough to work around that plateau while at the same time keeping the form very strict. In addition it is not hard to learn how to perform a curl against the wall; thus, it is not imperative to perform all of your training reps in this fashion.

Standing Curl – The standing curl allows you to lift noticeably heavier than the strict curl, and it does involve more upper and lower back as important stabilizers. You can use a bit of body English to complete the last few reps in a set which can help one get through a plateau and make continued gains.

Power Curl – This is a version of the standing curl in which some swing is intentionally used. The lifter should start leaning forward about 10–15 degrees, much like they were getting ready to perform a tricep pushdown. As they curl the weight up, they should stand up straight, thus using the momentum from extending their trunk to help lift the weight. The benefit of this form is that while momentum is used (and this is a version of “cheating”) it is quite safe on the body. Normally lifters start the curl with their trunk straight; then the weight gets hard about halfway up, and they lean back significantly, thus placing significant additional stress on the lower back. The excessive lean back position is to be discouraged in all of the lifts. The power curl allows one to get used to heavy weight, which is partly mental but also helps train the wrist, forearms, and elbow flexors; it does make even the heaviest strict curl “feel light” which can be motivating for the lifter.

Neutral Grip Curl – Using a neutral grip allows one to emphasize slightly different muscles in the arm. A neutral grip places more emphasis on the brachialis (muscle under the biceps) and the brachioradialis (muscle in the forearm) while slightly deemphasizing the biceps brachii. You can use either dumbbells or specialized barbells for this grip. With dumbbells it is generally called a hammer curl.

Dumbbell Curl – Dumbbell curls are effective for ensuring both arms are working equally and for highlighting and fixing any strength asymmetries one might have, which is more prone to express itself in a lift like the curl. If you lift both dumbbells at once, this is generally harder than using a barbell, and the total weight will be slightly less (there is not as much of a difference between dumbbell and barbell curls as there is between dumbbell and barbell bench press, however). If you curl just one dumbbell at a time, this is a bit easier because all of your energy can go to that one side and because you can slightly tilt your body to the working side, thus putting it in a more favorable position. Dumbbell curls with palms up (supinated) throughout the ROM will emphasize the biceps. Many people use a semi-supinated position and supinate only at the very top of the range of motion, but this doesn't hit the biceps very effectively; instead it works the brachialis more.

Preacher Curl – A preacher bench is a bench where you sit down and place your triceps on a pad that is slanted down about 60 degrees. You keep your upper arms on the pad and just perform curls, using EZ bars or dumbbells in most instances. The preacher bench is harder even than a strict curl against the wall because the apparatus doesn't allow any movement at the shoulder. However, lifters often significantly compromise the ROM and that, of course, makes the lift much easier. It is not necessary (and might be harmful) to complete an absolutely full range of motion on the preacher bench — when

you are at the bottom, it is easy to hyperextend the elbows with heavy weight in the hands. However, removing the last 10 degrees of range of motion is not an excuse to perform a little partial, in which the elbow opens up only 10–20 degrees from full flexion. Lifters regularly training like this will be sorely disappointed when it comes time to perform in a competition under the watchful eye of 3 judges. A guideline I use to ensure the ROM is at least adequate and will transfer over is that I want to see the entire front of my forearm (the smooth, non-hairy side) in the mirror when I am at the bottom of the range of motion.

EZ Reverse Curl – A reverse curl involves taking a pronated grip (palms down) and performing a curl, usually in a standing position. This is much harder than a regular curl and the weight will need to be reduced, usually by 25% or more. A reverse curl places significant emphasis on the brachioradialis and the brachialis; the biceps are not working that much in this exercise. It also stresses the forearm extensors as they work to keep the wrist from flexing during the exercise.

Zottman Curl – This is a specialized type of dumbbell curl in which the lifter uses a palms up position on the way up and, then, at the top of the ROM, they turn the palm down and use a pronated position on the way down. This type of curl will place a much greater emphasis on the brachio-radialis and the brachialis. It can be a little hard on the wrist with heavy weights due to the twisting motion required at the top.

Tim's Curls – Yes, this is named after me, I am as much of a glory hound as the next person. This involves performing supinated curls with dumbbells with a certain weight and reps, taking a short rest of 10–30 seconds, and then performing hammer curls with the same weight and reps (if possible). Rest

normally (1–2 minutes) and repeat. By supersetting the regular curls with the hammer curls, it gives the arms a great pump, and it also forces the deeper and smaller muscles (brachialis and brachioradialis) to get stronger while at the same time continuing to stimulate the biceps.

Partial Curls – Just as partials can be used in all other lifts, partials can be performed with curls as well. Set a bar up in the rack (although don't take up the rack for curls if someone wants to squat or perform other lifts in there); if it doesn't fit, then prop the bar up on something sturdy to raise the bar up, thus eliminating the first part of the ROM. In general start with the bar positioned so that your forearm is just below the parallel position (to the floor) or just above it. Partials can help improve strength in a specific range of motion; however, I would not make them my only method of training, or weak points in the ROM may develop.

Cable Curls – Cables can be used to train the biceps with a variety of grips: EZ bars, cable handles, 1 arm at a time, ropes, straight bars, etc. Cables tend to cause more constant tension, which is good for size. However, often the weight stack is not that heavy for the cables, and lifters need to bear in mind that, due to various pulley arrangements, the weights used are normally not comparable to free weights. In other words, curling 100 lbs for 10 reps using a cable is likely not equivalent to curling 100 lbs for 10 reps using an EZ bar.

Narrow Grip Curls – Taking a narrow grip with the palms up places more stimulus on the long head of the biceps. This head is also responsible for more supination, and it gives the peak of the biceps which is important for bodybuilders. The more the hand is supinated, the more the long head works. It is worth noting that the origin of the long head of the biceps (supraglenoid tubercle of the scapula) is reasonably vulnerable and can detach in certain

situations — lifters should use some caution and care when placing emphasis on this particular aspect of the muscle.

Wide Grip Curls – Holding the implement with a wider grip with the palms up places more stimulus on the short head of the biceps. This head tends to be more of a workhorse, and one's level of supination isn't crucial to activate this muscle.

21s – 21s is a name given for performing 3 sets of 7 in a row with no rest. 1 set of 7 involves using the full ROM, 1 set of 7 involves just the top half of the ROM, and 1 set of 7 involves just the bottom half. You can perform that in whatever order you wish: bottom, top, full ROM, and full ROM, top, bottom are the two most common arrangements. There is no law that says you can't perform 18s (3 sets of 6) or 24s (3 sets of 8) but 7 reps seems to work quite well — few enough reps to keep using heavy weight but long enough to make those arms burn by the end of the set.

Barbell Curl Throws – In this example the lifter will use a light weight (~30–60% 1RM) and they will curl the weight, but at about the halfway point, they will throw the weight up in the air and then catch it as it falls back to the ground. Much like a jump squat or an explosive bench press throw, the idea here is to turn on all of the motor units and muscle fibers to ensure that everything fires well when it comes time to max out. Start light with this exercise as it can be relatively jarring on the elbows

Compound Back Exercises – the biceps are an important helper (synergist) in all of the big upper body pulling exercises such as barbell rows, pull-ups, chin-ups, dumbbell rows, pulldowns, etc. Using a supinated or neutral grip places added emphasis on the elbow flexors.

Bodyweight

An increase or decrease in bodyweight is likely to affect the strict curl the least of any of the contested lifts. This is likely because the joint stability of the elbow, being primarily a hinge joint, is reasonably stable and because, as an individual's bodyweight changes, it will not have much effect on the involved joints in a strict curl. Practical evidence also backs this idea up. The 100% RAW Powerlifting Federation has the most complete strict curl records, with the curls performed up against the wall. For consistency's sake we will use their bench press records as well for a quick comparison:

100% RAW Federation Records

Weight Class	Bench Press Record	Weight lifted per lb	% Change	Strict Curl Record	Weight lifted per lb	% Change
165	418	2.533	—	148	.897	—
198	529	2.67	+26.6%	178.5	.902	+20.6%
242	600	2.479	+43.5%	187	.773	+26.4%
SHW	634	2.012	+51.7%	181.8	.577	+22.8%

It should be clear from the above chart that, as bodyweight increases, bench press strength increases as well, rather dramatically. When looking at the curl, it does increase but at a much less significant pace. It is also worth noting as of 2013 the record curl at 242 was higher than the 275, 308, and SHW weight classes.

Optimal Frequency

In my opinion the skill level of the bicep curl is most like the deadlift — it isn't that hard of a lift to learn, and, even without regularly curling one's ability to curl remains reasonably static particularly if one is training with weights in general. Indeed a competitor and friend of mine, John Franks, decided at the last minute to compete in the curl in a competition — he didn't think the curl was to be offered at this particular competition. Without specifically training for it, he ended up lifting his all-time best and the federation's All-Time best lift (it is his 187 lb curl at 242 that is listed above). I don't think too many lifters would have a chance at squatting a personal best on a whim if they hadn't been specifically training the squat.

In short that means training the curl intensely once or twice a week seems to work, and one can even go less frequent than that. The curl hits smaller muscles, and it will recover faster than some other lifts. Unlike the deadlift the curl responds to higher volumes of work (more reps per set, more total work sets per exercise, and more exercises per session). Think of combining a typical bodybuilding type arm program but using sets, reps, and the load of a powerlifter.

Unlike the deadlift the curl responds to higher volumes of work.

Power Curl

The technique for the power curl, sometimes called the cheat curl, is similar to the standing curl, but we are going to use more momentum. I would suggest you generally take a symmetrical stance and then stagger your feet once the weight becomes very heavy. Grip the bar as you would a regular curl; start with your arms straight and chest up. The key to the cheat curl is to then incline your trunk forward, as though you were taking a bow but you go forward only about 25 degrees or so. You will also have soft knees (not locked straight), and this will help you absorb the swing of your body. If you keep your knees locked, then you will likely swing too much, arch your back too much, or you will start going up on your toes. Start with a forward lean, and then use your hips and lower back to begin the curl by driving upright, much like the last bit of a power clean or a Romanian Deadlift. Continue to drive the bar with the biceps, hopefully getting past the sticking point (which is when the elbow is bent at approximately 90 degrees), and then finish the lift. The trunk will move through about a 45 degree ROM. If you start inclined forward about 25 degrees, you will go straight, and then you will continue to go slightly back to about 20 degrees or so.

There is a big difference between what I am suggesting and what most gym goers do. Most people start the curl with their body in a straight line. They perform a heavy curl, and, when they get stuck, they start to lean backward, often up to 45 degrees, which is too much and puts too much pressure on your lower back. What I am suggesting is that you do not start standing straight up; you start leaning forward so your chest is out slightly in front of the bar. Then, when you drive back, you will finish in a much safer position, which is leaning back slightly instead of having to significantly arch your back to complete the rep.

Some might ask, “Why perform a cheat curl at all?” Why cheat and why

not just go strict all of the time? I would respond by saying there are two types of cheating in the gym. One type of cheating is when you take emphasis off the muscle by using other muscles, or, worse, by changing your body mechanics. This is generally discouraged because it is both ineffective and often dangerous. An example of this is thrusting your hips up when you bench; that reduces the stimulus on your pecs and greatly increases the pressure on your back, both of which are negative. A second type of cheating is when you use some bigger muscles (in the case of the power curl, your erectors and glutes) to help spot your smaller muscles and to make them work harder. That is what is happening in a cheat curl. You are using your erectors to help make the biceps work even more. Of course this can be taken to an extreme, so one must still control a cheat curl; it does not mean that all rules of form are abandoned.

The other reason I advocate this exercise is that, quite simply, it works. Personally I feel that power curls are one of the best ways to improve the standing and wall strict curls. To me if you are always staying super strict it is too easy to hit a plateau on this lift, but by allowing yourself to use a bit of momentum (yet staying under control), you can keep the weight moving up, and you can keep the curl increasing. Then, as you get used to that new weight, you simply start to tighten up the form a bit.

Effective Routines for the Curl

The curl responds well to the classic bodybuilding style of training — hitting a muscle group once a week, sometimes twice, 2–4 exercises for that area, 3–5 work sets with a variety of reps included — some low to focus on strength, a good amount of medium rep work to focus on size and even some high rep work to target all of the muscle fibers. As one preps for a curl competition, it is likely wise to develop some sort of a peaking scheme. I have used the following programs or training tactics with good success:

My 6 week Routine is found here:

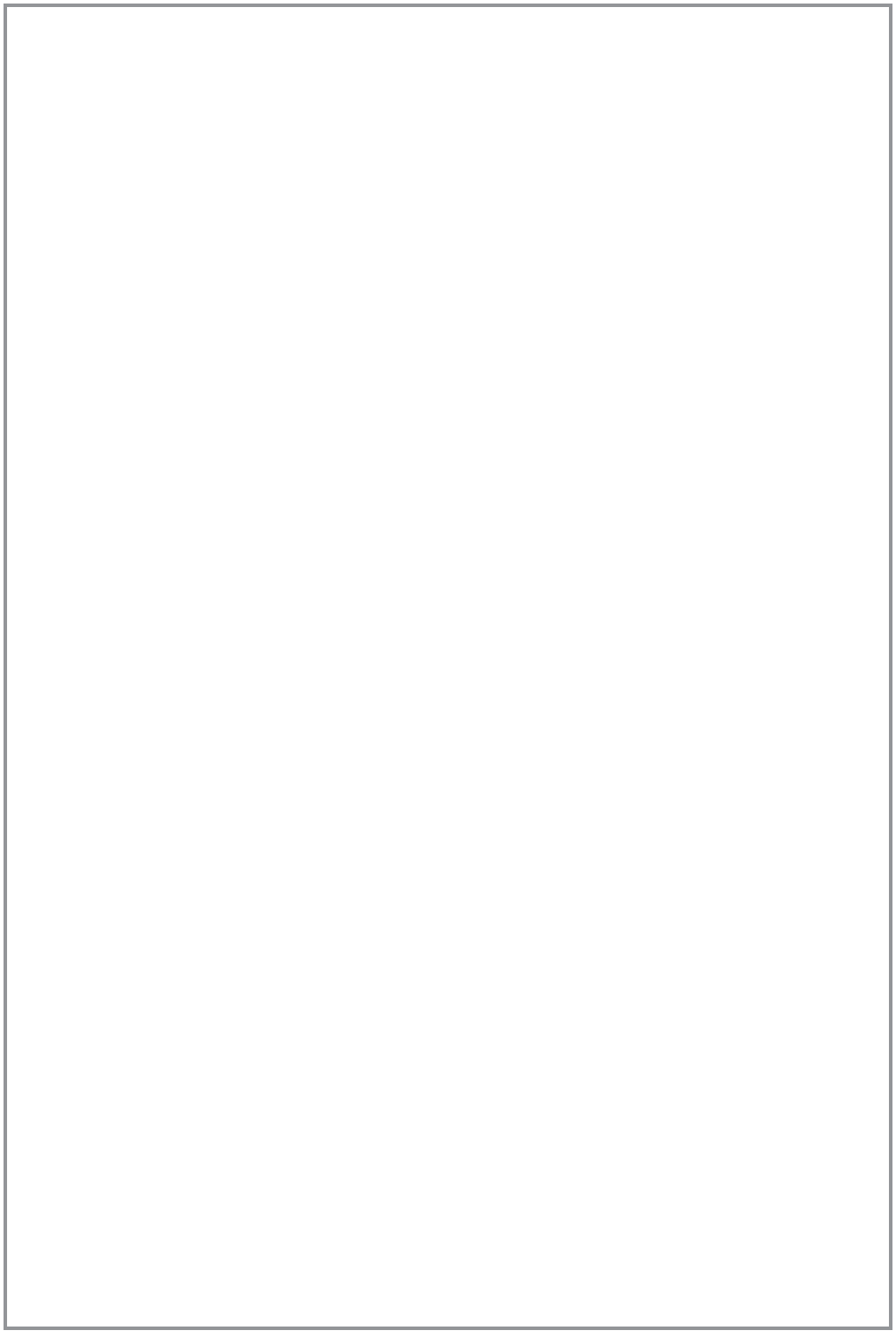
http://www.t-nation.com/free_online_article/most_recent/a_6week_squat_bench

A curl specific routine is found here:

http://www.t-nation.com/free_online_article/most_recent/strict_curl_for_max_w

Summary

Having strong and powerful arms helps everybody, from the powerlifter to the athlete to the dude (or dudette) walking down the street. Muscular arms are forever linked with fitness and strength. Focusing on powerlifting does not mean one has to forgo the idea that they can develop an appealing physique — far from it — many powerlifters over the years who have watched their body composition sport extremely impressive builds that not only look good but perform exceptionally well. There is nothing wrong (and likely something good) about spending some quality time training curls — just don’t do it in the squat rack.



Who doesn’t want impressive looking arms?

Appendixes

A. Frequently Asked Questions about Powerlifting

B. Newbie Mistake Checklist

C. Recommended Readings for Powerlifting

D. Powerlifting Related Websites

E. Raw Powerlifting – Female Classification Standards

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H. Tim Henriques Powerlifting Career Summary

I. You know you are a powerlifter when...

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Appendix A

Frequently Asked Questions about Powerlifting

I have to skip some workouts this week; what should I do?

If you have skip some workouts and you are in the middle of the training cycle, you don't want to just forget about the workouts that you had planned for yourself. What I would suggest you do is take the biggest lifts from each day and perform those exercises when you workout.

For example, imagine this week you had planned to train squats and other lower body stuff on Monday, bench press and upper body assistance work on Wednesday, and good mornings and more hamstring work on Friday. If you can get only one workout day in that week, you would take the most important exercise from each day and do that. Your workout would be squats (same weight and reps you were going to do), bench press (same program), and good mornings. If you had two days you would do the most important lower body stuff on one day and then the upper body stuff on the next day.

In another example, let's say I am going on a quick vacation this week. Right now I am training four days a week and am 13 weeks out from a competition. By going on vacation I have to miss 2 training days, but I can still workout two times this week. My current routine is a modified Westside Barbell Club routine. On Monday I am doing dynamic squats and other assistance leg work plus abs. On Tuesday I am maxing out on a 5 board press

and doing other bench assistance work plus abs. On Thursday I am doing a max set of deadlifts in the rack from 11″ height with average bands with other lower body work, and on Friday I am doing a Peters Bench program; this week is 2 sets of 3, with some other upper body work plus abs. I don't want to miss any key lifts. Remember most good programs build upon themselves, so what will I do?

Since I have two days to train, I would do the dynamic squats, the band rack pulls, and abs on Monday. On Tuesday I will do the 5 board press, the Peters Bench routine, one or two assistance exercises and abs, and that will be probably be it for the week (I'll do my cardio as usual on Wednesday). I will just take Thursday and Friday off and then come back on Monday hopefully fresh and ready to go and still feeling like I haven't really missed anything.

The bottom line is to pick out the key exercises and do them and skip the rest (this theme is actually useful when designing regular programs as well).

I can workout only 1 or 2 days a week regularly; what should I do?

If you can workout only 1 day a week, the best program I have seen for that in my opinion is on day 1 to do squats, bench press, one or two assistance exercises and abs if necessary. On day 2 (performed the next week) you do squats, bench press, and deadlifts and maybe one assistance exercise if you have time. Then you just go back and repeat day 1 and then day 2 again and so forth. So you squat every time, bench press every time, and deadlift every other time. Generally you will do a lot of sets for the main exercises to get the appropriate amount of work in. I do not feel that this is ideal by any means, and I don't think it is good for beginners or early intermediates at all because they need more work to build up their work capacity and neuromuscular coordination, but I have seen people make progress on this system.

If you can workout 2 days a week, you have more choices. The most

common routine would be to do an upper body workout on one day and a lower body workout on the other and then just rotate back and forth. Another option would be to do the routine listed above; that way you squat and bench twice a week and deadlift once a week. Or you could do a push day (chest, shoulders, triceps and core) and a pull/leg day (legs, back, lower back, and biceps). You will have to decide what frequency of training each lift works best for you.

Most powerlifters train with weights 4 times a week (occasionally more), a significant minority train 3 times a week, and a few get by with 1 or 2 regular workouts a week. Usually the extra sessions above 4 times a week are for weak point or work on other components of fitness aside from strength.

I am going on vacation; what should I do?

If you are going on vacation you have to think about how long the vacation is going to be, what you have access to train with, how much you want your training to affect your vacation, where you are in your training cycle, and when your last period of time off was.

If the vacation is 7 days or less and it is has been a while (2 months +) since you last had a similar period of extended passive rest, feel free just to take the time off if you want. If you want to be active, you can follow a modified program described below. You should not lose hardly anything (if at all) with a week off. In addition trying to train hard while on vacation poses a host of problems. You may not have the necessary equipment to do so, for example, squatting on a Smith machine is not the same as real squats. People often find they are weaker on vacation. This is because they are stressed from travel, they are in a new environment, they are eating different (and often less healthy) food than normal, and they are doing different activities which might tire them out. Going swimming, sitting in the sun,

walking around all day, going fishing or exploring, all can drain your energy.

The important people in your life (spouses, children, other family members and friends) have probably made some sacrifices for you to be able to devote the time necessary to powerlift. Hopefully they are not too resentful of those sacrifices. When you are on vacation that can be a time for you to be devoted to them. If you have a big workout scheduled, you will be distracted, you will be worried about what you are eating, and simply you will be gone for a while to complete the workout. This can try anyone's patience. My suggestion is this: do not create that problem by trying to get in a tough powerlifting workout on vacation.

If you do want to workout my suggestion is that you have some guidelines for yourself. I try to follow these guidelines if I am going to workout on vacation. First, keep it short, 30 minutes maximum. This is not the time for a 3 hour workout. Second, reduce the frequency and just go 2 or 3 times a week; you don't need to train every day. I would suggest you go right when you wake up or soon after breakfast to get it over with so it doesn't interfere with the rest of your day. Make it fun and intense. Try something you don't normally do. Do supersets or very short rest, bodyweight exercises, train arms, use machines, go slow, get a nice burn and a pump and get out of there. Only do a few exercises (this helps with the time thing); 4 is a good max. Just get the body moving and the metabolism going. For those of you who are "addicted" to exercise (when you don't feel like yourself if you don't workout) this will take care of that addiction, and you will have the rest of your day to do as you please. The boost in metabolism will help you burn off some extra vacation calories, and it will maintain your fitness level.

If you are going on vacation for longer than a week (like 2 or more weeks) then you probably do want to try to train at least a little bit if you can.

If you can have access to a gym, even if you have to drive a ways to do it, getting in one or two good days can make a big difference. Follow the previous guidelines for when you have a few days to train, pick the big exercises, and do them. If you are just trying to maintain your fitness level plyometrics like jumping squats, push-ups and their many variations, and sprints can help a lot. If you have access to a pull-up bar, dip station, or gymnastics rings, that is even better. Fast walks and staying active with recreational activities can maintain your cardiovascular fitness. Your squatting ability will be the first to go with time off, then your bench press (regular push-ups can help delay this a bit, preferably with weight — have someone sit on your back, kids seem to enjoy this and are the right weight usually), and then your deadlift will tend to maintain itself pretty well.

I am having knee pain when I squat; what should I do?

If your knees are hurting when you squat, first stop squatting. I know that sucks to hear, but training through joint pain almost always makes it worse in the long run. Get your knees, hips, and feet checked by the doctor if it persists to see if something is out of whack. Change your stance (either wider or narrower; one of them normally feels better). Temporarily you may squat higher (a box squat is good to control depth and keep it constant), but don't get in the habit of squatting high. Briefly change the exercise. Front squats or high bar squats normally feel better on the knees. Safety squat bars, especially if you hold on to the rack when you squat, are quite easy on the knees. Increase your warm-up procedure; try riding a bike for 5 minutes following by dynamic stretches — a foam roller may also be of assistance. Sometimes very light leg extensions or leg curls or leg presses for 50–100 reps prior to squats make the area feel better. Do more warm-up sets. Re-examine your form to see if something has changed. Change your shoes to a more sturdy shoe. Wear knee sleeves (Rheband makes the best in my

opinion) when you warm-up and when you lift, and then stretch the quads, hip flexor, hams, adductors, abductors, and gastroc after you lift and on off days. One or more of those things should help. But if that pain persists for weeks or months, get it looked at. Often if you catch a problem early enough, you can fix it, but if you go in and say my knees have been hurting for 3 years and I am here to have you look at them, there may be very little you can do at that point.

I am having shoulder pain when I bench; what should I do?

First, stop benching until the pain goes away. Try a variation of the bench press like the incline or decline; one of those normally feels better. Try dumbbell presses, particularly with a neutral grip. Temporarily you can limit your range of motion if going down to your chest hurts; gradually build it back up to normal. Board presses work well for this. Work on muscle balance. Your lats and upper back may be weak, and your external rotators (rear delts, infraspinatus, and teres minor) may be weak. Supraspinatus may be giving you problems by getting impinged. Spend a month doing rotator cuff exercises 3 times a week. Start stretching more. Get up against a wall, bend your elbows at a 90 degree angle and put your arms up against the wall like you were in the “Hands Up” position. Your butt, upper back, head, feet, and elbows and forearms (not fingers bent back) should be flush up against the wall. If you can’t do this, you are too inflexible and benching heavy will add to that problem. Stretch out the pecs, lats, front delts, biceps, teres major, and subscapularis. Do that almost every day. Strengthen the rhomboids, mid traps, rear delts, and IT muscles. Bring your grip in on the bench; normally closegrip feels better than wide grip. Use lighter weight and work on your technique, especially tucking the elbows to a 45 degree angle and arching the bar back as your press. Make sure you are not doing any upright rows or behind the neck military press as they tend to make the problem worse, and

deep dips should avoided as well.

What should my openers and attempts be? — see Chapter 16

The bar keeps falling out of my hands when I deadlift. — see Chapter 8

What should I eat before and during the meet? — see Chapter 12 & 14

Do running and powerlifting go together?

In my opinion running (longer distances) and powerlifting do not go together. The primary lift negatively affected by running is the squat with a secondary effect on the deadlift. The bench press doesn't seem to be affected much by jogging. There are several reasons for this.

The primary reason is muscle fiber type. In general you have two types of muscle fibers, type I and type II. Type I is smaller and weak but good at endurance. Type II is larger and more powerful but not good at endurance. We used to think that you could not change your fiber type due to training. We know for sure there can be a shift of the subtypes in the fibers and newer evidence is suggesting that long term training can cause a shift in the primary fiber type. Running will train your leg muscles to be more type I. That is good for running but bad for squatting. Squatting will train your leg muscles to be more type II. The bigger you are and the harder you find running in general, probably the more negative effect this will have. Some light weightlifters seem to be able to be pretty good at lifting and running, but that is the exception, and there is reason to believe that if they specialized at one they would get better at it. Running trains a different energy system from powerlifting. Running is all about the body's ability to use oxygen effectively; it is not really a muscularly intense activity. Powerlifting is all about the body's ability to develop force and strength rapidly. Oxygen plays almost no role in a 1 rep max (this is why you can, and often should, hold

your breath when performing a 1 rep max). You do need a decent cardiovascular system to be able to handle a tough powerlifting workout, but walking or using the elliptical can achieve that with less or no fiber type change. The reason is because walking is easy enough to achieve with the amount of type I's that almost everyone has now; you don't need more to get the activity done. But running is hard enough on the muscles that it will affect their make up.

Running is a fair amount of impact on the feet, knees, hips, and lower back. Powerlifting is also pretty intense on those joints, particularly the latter 3. The combination of heavy squats, deadlifts, and pounding the ground running can be too much for most bodies to handle.

My suggestion is this: ditch the running if you want to excel at powerlifting. Walk as your primary method of CV training, and you can throw in sprinting, sled dragging, and other methods to build your work capacity. Avoid running anything longer than 800 Meters straight. If you just compete in the bench press, this guideline doesn't apply as much to you.

What are the All-Time best lifts in powerlifting?

The use of powerlifting equipment, steroids (whether an athlete is tested or not), and different federations make having a quick list of records harder to compile than it should be, and that is a shame. Olympic lifting, for example, is more cut and dried having solved 2 of the three issues (there is no equipment and only one big federation); the best snatch is 472 lbs and the best clean and jerk is 581 lbs as of this writing (even OL has some issues because of restructuring the weight classes; some records were changed or erased — athletes have snatched 476 and clean and jerked 586 lbs before under competition rules). And these athletes are tested to IOC regulations, which is not perfect but it is at the same level as any other Olympic sport.

Men’s Best All-Time Squats

Weight Lifted	Lifter	Bodyweight	Year	Notes
934	Don Reinhoudt	SHW	1975	Raw, AAU, not tested
881	Blaine Summer	SHW	2012	Raw, IPF, tested
953	Mark Henry	SHW	1995	In wraps, WDFPF

Men’s Best All-Time Bench Press

Weight Lifted	Lifter	Bodyweight	Year	Notes
722	Eric Spoto	SHW	2013	Raw, SPF, not tested
710	James Henderson	SHW	1997	Raw, USPF, tested
705	Siamand Rahman	SHW	2013	Raw, Para-Olympian, tested

Men’s Best All-Time Deadlift

Weight Lifted	Lifter	Bodyweight	Year	Notes
1015	Benedikt Magnusson	SHW	2011	Raw, Hardcore, Not tested
939	K Konstantinovs	308	2009	Raw, AWPC, tested

Men’s Best All-Time Strict Curl

Weight Lifted	Lifter	Bodyweight	Year	Notes
225	CT Fletcher	280	n/a	Unsanctioned, Not tested
187.3	John Franks	242	2011	100% RAW, Tested

Men's Best All-Time Total

Weight Lifted	Lifter	Bodyweight	Year	Notes
2298	Don Reinhoudt	SHW	1974	Raw, AAU, Not Tested
2171	K Konstantinovs	275	2009	Raw, AWPC, Tested
2336	Mark Henry	SHW	1995	Raw, with wraps, WDFPF, Tested

Is powerlifting in the Olympics?

No, as of this writing, powerlifting is not in the Olympics. The weight lifting you see in the Olympics is called Olympic Weight Lifting and it consists of only 2 exercises, the Snatch and the Clean and Jerk, both of which are quite different from the 3 powerlifts.

What sport generates more power, Olympic Lifting or Powerlifting?

In general Olympic Lifting generates more power than powerlifting. This is because the bar, while having less weight, is moved a greater distance at a very high rate of speed in Olympic Lifting. Some very explosive powerlifters like the great Brian Siders may give some of the Olympic Lifters a run for their money in power generation, but in general powerlifting uses more weight but generates less power than Olympic Lifting.

How can I make a cheap homemade sled for GPP?

Here is an article that details how to build and use a homemade sled.

http://www.t-nation.com/free_online_article/most_recent/build_your_own_sled

Questions about filling out an entry form:

What does a crossover mean?

A crossover is when you compete in 2 or more categories. For example, you could compete in the Open division of your weight class and then crossover into the Master division (assuming, of course, you meet the requirements to be in that other category). This generally increases your cost to do the

competition, and it means you might have a chance to take home multiple trophies. Some federations have rules that you can't set a record unless you are entered into that specific division.

You may often also crossover into a single lift part of the competition. For example, if you're competing in a full 3 lift powerlifting competition but your bench press is pretty studly, you can crossover and enter the bench press competition itself as well. This way you are competing against those people who are performing the bench only part of the competition.

What if I miss my weight class?

If you miss your weight class you can still compete; you will simply be in the actual weight class that you weighed in for. For example, if you registered as a 181 lb lifter, but then at weigh-ins you weighed 183 lbs you will have missed that weight class, and you are now in the 198 lb weight class. You can still lift; you'll just be in a different weight class.

Can I weigh-in only once?

No, you can weigh-in as many times as you want during the set weigh-in period. For example, if you weigh-in initially and you are 8 oz too heavy, you can go to the bathroom, run around, whatever, and then weigh-in an hour later; if you make it, that second weigh-in weight counts. Be courteous to other lifters; if you miss your weigh-in, you need to get back in at the end of the line to weigh-in again. Keep in mind that weigh-ins do have an official start and end time; if you are still too heavy by the end of weigh-ins, you will not be able to make the lower weight class.

I missed the entry deadline for the competition; can I still compete?

That is up to the meet director, but it is worth contacting them to see if they can get you in. I would be very polite and ask if you can lift; you might make it clear that if they don't have a trophy for you at that time, you will be fine with that as they may have ordered an exact number and late entries will

throw that off. If they say no, so be it; you didn’t lose anything, but they might say yes. You will never know unless you ask.

Appendix B

Newbie Mistake Checklist

Squat

- **Depth Depth Depth**
- Moving feet after command
- Too many steps on walkout
- Wrapping knees too early if using wraps
- Hands too wide
- **Opening way too heavy**
- Not knowing the commands
- Thinking a squat pad is allowed

Bench

- Getting a spot without knowing it
- Not touching the chest
- **Not realizing a pause makes a big difference**
- Not waiting for commands
- Getting nervous and not setting up hands evenly
- Moving feet and/or butt
- Thinking gloves are allowed

Deadlifts

- Uneven grip on bar
- Setting up too far away from the bar
- Not arching back prior to pulling
- **Squatting down too low and jerking into the bar**
- Slamming the weight down
- Getting baby powder on your hands
- Putting chalk on your legs
- Having baby powder on shoes for sumo deads
- Thinking straps/gloves are allowed
- **Not realizing you might be tired at the end of the day and thus a bit weaker**

Curls

- **Not holding bar properly to start**
- **Not realizing the much greater difficulty of doing it against the wall**
- Not allowing arms to move forward
- Not waiting for both down signals
- Unlocked legs to begin with

In General

- **Not knowing you need a singlet or how to wear it**
- Having an illegal belt
- Thinking gloves, straps, and a bar pad are allowed
- Not knowing basic rules of the lifts and the commands
- **Opening up too heavy**
- **Warming up too heavy**
- Not realizing the amount of help a person gives you just by putting their hands on the bar

Appendix C

Recommended Readings for Powerlifting

General Powerlifting Related books

Book Title	Author
All About Powerlifting A book that tries to cover all aspects of powerlifting	**Henriques**
Starting Strength, 3rd Edition A good book on how to coach the bench, squat, dead, clean, and press and coach young athletes	**Rippetoe**
Power: A Scientific Approach One of the first modern "textbooks" to help people get strong	**Hatfield**
Maximum Strength A book with a 4-month detailed plan to build strength	**Cressey**
Westside Book of Methods A book about the Westside training methodology; kind of tough to follow	**Simmons**
Strong Enough A book of essays from someone with 30 years, experience in the gym	**Rippetoe**

Book Title	Author
The Purposeful Primitive	**Gallagher**
A book of essays about how elite powerlifters, bodybuilders, and nutritionists train	

Programming Based Books

Book Title	Author
Practical Programming for Strength Training	**Rippetoe**
A good book on how to design programs with solid practical info; relatively easy to follow	
The Cube Method	**Lilly**
A book that lays out how to follow the cube method	
5/3/1 2nd Edition	**Wendler**
A book that lays out how to follow the 5/3/1 program	
5/3/1 for Powerlifting	**Wendler**
A book that modifies 5/3/1 specifically for powerlifting	

Textbooks

Book Title	Author
Fundamentals of Fitness and Personal Training	**Henriques**
A book that explains how the body adapts to exercise and what kind of exercise one should to perform to elicit certain results – you will like this if you want to have a deeper understanding of how the body works – due to be published at the end of 2014.	
Supertraining	**Siff and Verkohansky**
The most comprehensive textbook on strength	

	training; complicated but great
Science and Practice of Strength Training	**Zatsiorsky**
	Second only to Supertraining, best text on strength training
Essentials of Strength Training and Conditioning	**Baechle & Earle**
	Generally the most accepted textbook in the fitness industry

Size Focused Books

Book Title	Author
Strength Training Anatomy, 3rd Edition	**Delavier**
	The best anatomy picture book; drawings of a person lifting weights
Bodybuilding: A Scientific Approach	**Hatfield**
	Similar to above; designed to help people get lean and defined
The Poliquin Principles	**Poliquin**
	Several guidelines from a popular strength coach
Muscle Revolution	**Waterbury**
	A general book on fitness with sample programs; favors total-body training
High Intensity Training the Mike Mentzer Way	**Mentzer and Little**
	A book on the HIT philosophy
Natural Bodybuilding	**Hansen**
	A book on bodybuilding, training, and nutrition for people not on steroids

Mobility/Flexibility Books

Book Title	Author
Stretching Scientifically	**Kurz**

Best textbook on flexibility from the Soviet bloc

Becoming a Supple Leopard **Starrett**

Great book on how to improve mobility from Kelly Starrett, founder of mobilitywod.com

Fitness Videos

Blood and Guts — Dorian Yates' workout
Unbelievable — Ronnie Coleman's workout
Pumping Iron — Documentary of 1976 Mr. Olympia competition
Pumping Iron II — Documentary of 1980's Ms. Olympia competition
Power Unlimited — Documentary on powerlifting, focused a lot on geared lifting
Real Muscle — John Hansen's (Natural Mr. Olympia) workout
Magnificent Mobility — Cressey and Robertson — about dynamic stretching
Bigger, Stronger, Faster — Bell Brothers — talks about the use of steroids in sports
Food, Inc — a movie that details food prep in America, a combo of Fast Food Nation and the Omnivore's Dilemma

Novels	**Author**
Gorilla Suit	**Paris**
A basically true story of bodybuilder Bob Paris	
Muscle	**Fussell**
A story of a person who becomes a bodybuilder for the wrong reasons	
A Portrait of Dorian Yates	**Yates**
Dorian's thoughts on fitness, nutrition, etc.; pretty good	
Mr. America	**Adams**
A biography of perhaps the founder of the fitness movement	
Brothers of Iron	**Weider**
A biography of Joe and Ben Weider, the founders of modern bodybuilding	
Steroid Nation	**Assael**
A book about the rise of steroids in sports and American culture	

A book about the life of Robby Robinson

Inspirational Videos

Rocky I-IV, VI
Miracle
Without Limits
Conan the Barbarian (and most Arnold movies)
First Blood
Predator
Terminator

Did I leave out your favorite book or resource? If so, contact me and tell me what it was, and I will consider adding that in to future editions.

Appendix D

Powerlifting Related Websites

Be sure to check out the following websites when searching for powerlifting related information:

Powerliftingwatch.com	A comprehensive website with up-to-date news, meet results, schedule, records, and rankings of all powerlifting competitions
T-nation.com	A website that provides free articles and forums dedicated to strength training; also sells some supplements
Elite Fitness Systems	A site that has articles focused on hardcore training, and it sells good-quality gym and powerlifting equipment; run by Dave Tate
Bodybuilding.com	A site that sells supplements and also has articles and forums
Netrition.com	A site that sells supplements at reasonable prices
Criticalbench.com	A site that promotes things related to strength and fitness
Inzernet.com	The site for Inzer Advanced Designs; sells powerlifting equipment and belts, etc.

Prowriststraps.com	APT's site for selling powerlifting-related gear
Eastbay.com	A site focused on wrestling; has a large variety of singlets
Reactivetrainingsystems.com	Mike T's site about powerlifting and his training methods
Ironaddicts.com	A site with a forum for Q&A
Ironmind.com	Sells high-quality strength-training-related equipment
Supertraining.tv	Mark Bell's site that has lots of videos and analysis of the lifts
Startingstrength.com	Mark Rippetoe's site focused on his work and discussions of things related to strength
Mobilitywod.com	Kelly Starrett's site; great for info for working on mobility

A note to the reader: This is not meant to be an exhaustive listing; the Internet is vast and new sites are being created daily. Google will likely help you find a specific site more readily than I can list them here; however, if you feel I have missed an important site, contact me, and I will consider adding the website to this list in future editions.

Appendix E

Raw Powerlifting – Female Classification Standards

Women's Squat

Wt. Class	*97*	*105*	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*198+*
ELITE	**179**	**189**	**204**	**240**	**250**	**260**	**279**	**295**	**305**	**330**
MASTER	163	172	186	218	228	237	254	268	278	300
CLASS I	147	155	167	197	205	213	229	242	250	271
CLASS II	131	138	149	175	183	190	204	215	223	241
CLASS III	113	119	129	151	158	164	176	186	192	208
CLASS IV	97	102	110	130	135	140	151	159	165	178

Women’s Bench Press

Wt. Class	*97*	*105*	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*198+*
ELITE	**120**	**135**	**143**	**160**	**170**	**181**	**192**	**206**	**210**	**225**
MASTER	109	123	130	146	155	165	175	187	191	205
CLASS I	98	111	117	131	139	148	157	169	172	185
CLASS II	88	99	104	117	124	132	140	150	153	164
CLASS III	76	85	90	101	107	114	121	130	132	142
CLASS IV	65	73	77	86	92	98	104	111	113	122

Women’s Deadlift

Wt. Class	*97*	*105*	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*198+*
ELITE	**248**	**259**	**285**	**304**	**321**	**353**	**363**	**380**	**385**	**405**
MASTER	226	236	259	277	292	321	330	346	350	369
CLASS I	203	212	234	249	263	289	298	312	316	332
CLASS II	181	189	208	222	234	258	265	277	281	296
CLASS III	156	163	180	192	202	222	229	239	243	255
CLASS IV	134	140	154	164	173	191	196	205	208	219

Women's Strict Curl

Wt. Class	*97*	*105*	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*198+*
ELITE	**50**	**58**	**65**	**72**	**76**	**79**	**83**	**88**	**94**	**99**
MASTER	45	53	59	65	69	72	74	80	85	90
CLASS I	41	48	53	59	62	65	67	72	77	81
CLASS II	36	42	47	52	55	57	59	64	68	72
CLASS III	32	37	42	46	49	50	51	56	60	63
CLASS IV	27	32	36	39	42	43	44	48	51	54

Women's Powerlifting Total

Wt. Class	*97*	*105*	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*198+*
ELITE	**535**	**573**	**611**	**665**	**703**	**756**	**813**	**839**	**859**	**918**
MASTER	487	521	556	605	640	688	740	763	782	835
CLASS I	439	470	501	545	576	620	667	688	704	753
CLASS II	385	418	446	485	513	552	593	612	627	670
CLASS III	342	361	385	419	443	476	512	529	541	578
CLASS IV	289	309	330	359	380	408	439	453	464	496

These lifting classifications apply to a drug-free lifter who is using only a belt and wrist wraps. The squat is below parallel, the bench press is paused, the deadlift is not hitched, and the curl is up against the wall.

Appendix F

Raw Powerlifting – Male Classification Standards

Men's Squat

Wt. Class	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*220*	*242*	*275*	*275+*
ELITE	**292**	**320**	**348**	**389**	**452**	**495**	**535**	**565**	**590**	**620**	**662**
MASTER	269	294	320	358	416	455	492	520	543	570	609
CLASS I	239	262	285	319	371	406	439	463	484	508	543
CLASS II	210	230	251	280	325	356	385	407	425	446	477
CLASS III	184	202	219	245	285	312	337	356	372	391	417
CLASS IV	161	176	191	214	249	272	294	311	325	341	364

Men’s Bench Press

Wt. Class	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*220*	*242*	*275*	*275+*
ELITE	**235**	**258**	**275**	**314**	**358**	**387**	**415**	**440**	**462**	**483**	**500**
MASTER	214	235	253	289	329	356	382	405	425	444	462
CLASS I	193	212	226	257	294	317	340	361	379	396	412
CLASS II	169	186	198	226	258	279	299	317	333	348	363
CLASS III	148	163	173	198	226	244	261	277	291	304	315
CLASS IV	129	142	151	173	197	213	228	242	254	266	275

Men’s Deadlift

Wt. Class	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*220*	*242*	*275*	*275+*
ELITE	**346**	**374**	**402**	**468**	**537**	**597**	**625**	**642**	**660**	**689**	**700**
MASTER	318	344	370	431	494	549	575	591	607	634	644
CLASS I	284	307	330	384	440	490	513	526	541	565	574
CLASS II	249	269	289	337	387	430	450	462	475	496	504
CLASS III	218	236	253	295	338	376	394	404	416	434	441
CLASS IV	190	206	221	257	295	328	344	353	363	379	385

Men’s Strict Curl

Wt. Class	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*220*	*242*	*275*	*275+*
ELITE	**85**	**100**	**115**	**130**	**142**	**151**	**160**	**168**	**175**	**175**	**175**
MASTER	77	91	105	118	130	138	148	156	163	163	163
CLASS I	70	82	94	107	117	124	132	139	144	144	144
CLASS II	61	72	83	94	103	109	116	122	127	127	127
CLASS III	54	63	72	82	90	95	102	107	111	111	111
CLASS IV	46	54	62	70	78	83	88	92	97	97	97

Men's Powerlifting Total

Wt. Class	*114*	*123*	*132*	*148*	*165*	*181*	*198*	*220*	*242*	*275*	*275+*
ELITE	**805**	**881**	**947**	**1124**	**1261**	**1396**	**1495**	**1587**	**1630**	**1710**	**1740**
MASTER	741	811	871	1034	1160	1279	1375	1460	1500	1573	1601
CLASS I	660	722	777	922	1034	1148	1226	1301	1337	1402	1427
CLASS II	580	634	682	809	908	1012	1076	1143	1174	1231	1253
CLASS III	507	555	597	708	794	879	942	1000	1027	1077	1096
CLASS IV	443	485	521	618	694	768	822	873	897	941	957

These lifting classifications apply to a drug-free lifter who is using only a belt and wrist wraps. The squat is below parallel, the bench press is paused, the deadlift is not hitched, and the curl is up against the wall.

Appendix G

Lifter Classification Information

This is applicable to all lifters who compete in ***raw, drug free*** powerlifting competitions in federations that enforce below parallel squats and paused bench presses.

Operational definitions of the following terms:
Elite: An exceptional lifter. For males this means the lifter is very likely in the top 10 in the nation for their respective weight class and the lifter may be close to a top 5 ranking among comparable federations for that competition year. For females this means the lifter is very likely in the top 5 in the nation for their respective weight class and the lifter may be close to a top 3 ranking among comparable federations for that competition year. Elite lifters generally place very well at local level competitions and will usually hold their own at National level competitions. It is not uncommon for Elite level lifters to have 10+ years of experience with serious powerlifting training. Approximately 1‰ of competitive powerlifters will reach the Elite level of classification.

Master: A very skilled lifter. For males this means the lifter is likely in the top 50 in the nation for their respective weight class among comparable

federations for that year. For females this means the lifter is likely in the top 20 in the nation for their respective weight class among comparable federations for that competition year. Master lifters usually perform quite well at local level competitions and may want to think about competing on a National scale. Master lifters are likely to have 6+ years of experience with serious powerlifting training. Approximately 10% of competitive powerlifters will reach the Master level of classification.

Class I: A skilled lifter. A Class I lifter is significantly stronger than the average person who engages in regular intense weight training. Class I lifters are likely to have 4+ years of experience with serious powerlifting training. A high percentage (~30%) of competitive powerlifters are at the Class I level classification.

Class II: A relatively skilled lifter. A Class II lifter is stronger than the average person who engages in regular intense weight training. Class II lifters are likely to have 3+ years of experience with serious powerlifting training. A high percentage (~30%) of competitive powerlifters are at the Class II level classification.

Class III: A Class III lifter is stronger than the average person. Class III lifters are likely to have 2+ years of experience with hard resistance training. A reasonable number (~20%) of competitive powerlifters are at the Class III level classification; this classification is common among teenage and upper level master lifters (50+ yrs old).

Class IV: A Class IV lifter is at the beginning stage for a powerlifter. Class IV lifters are likely to have 1+ year of experience with hard resistance training. A smaller number (~10%) of competitive powerlifters compete at the Class IV level classification.

FAQ about the New Lifter Classification System

Why were the numbers revised?

Initially the goal was simply to create a classification system for the individual lifts based off of the system that was currently being used to classify a lifter's powerlifting total. After further examination it became apparent that the current system was not quite describing what we were seeing on the platform. Just a few people each year were hitting elite at the lightweights, even with a large number of competitors, and a large number of people were hitting elite at the heavyweights; the terms were not balanced. An attempt has been made to more properly align the standards so that elite at 148 means approximately the same thing as does elite at 242. Standards have been established for each individual contested lift, and the numbers for the powerlifting total standards have been slightly revised.

How did you come up with the numbers?

The primary resource that was used to calculate actual numbers instead of expected numbers was the lifter rankings system on PLwatch.com. Information from the years 2008, 2009, and 2010 were primarily used. PLwatch.com does not separate by out federation standards and/or if lifters are drug tested, so we had to further examine the rankings based on those qualities. Those numbers provided the backbone for the Elite standards; however, several lifting formulas were also used, input was given from federation officials, and patterns were analyzed. The other rankings (Master through Class IV) were set up to maintain a similar standard in relation to the Elite rankings as occurred in previous years.

Why do we need 4 classes of lifters plus Elite and Master? Why not fewer classes?

Powerlifting is a sport that accepts all comers — some great lifters, some not so great. Some lifters are at the tail end of their sporting career, and others are just getting started. In addition general gym goers who lift weights but don't compete like to "look in" to get a sense of how they might fare in a competition. The 6 rankings help separate out those lifters who are at the top of their game from those lifters just getting started and everywhere in between. While it is true that the significant majority of full meet lifters will be Class II or above, the standards for the single lifts are higher, and many lifters will find themselves working hard to simply place on those standards, let alone shoot for a Class I or Master level or beyond. Multiple standards can inspire a lifter to get that extra 5 or 10 lbs on a lift; if one is so close to that next level it might be the motivation they need to hit that weight. Finally powerlifting as a whole is inclusive, not exclusive. It is not just a sport for the "super strong" among us; it is a sport for all those lifters who want to test themselves in a competitive setting and to see what their own personal limits

may be. If lifters disapprove of the lower classes, they can simply ignore them.

What's new about this?

First, the powerlifting total standards have been revised, although not greatly. In general the light weight standards have been lowered slightly, the middle weights are relatively unchanged, and the heavy weight standards have been raised slightly. Secondly there are standards presented for each individual lift, which is a new idea. The individual lift standards can be applied to both a full meet and a single lift competition.

Why don't the standards on single lifts add up the standards for the total?

The single lifts standards represent the ability to specialize; if one trains only the bench press, it will be easier to improve just that than if one divides up their resources among additional lifts. Thus a good all around lifter who is Class I in most lifts might actually be a Master lifter when examining their total because they have no weak points.

Why are the standards higher for females than males, relatively speaking?

Fewer women compete than men; it can be assumed that if more women were to compete, there would be more good lifters.

Why aren't the jumps between weight classes more even?

The jumps in weight between the weight lifted in weight classes was based on actual performance, not expected performance. For some weight classes and lifts, there was a minimal advantage in being just one weight class heavier; in others, there was a very significant difference in the performance of the lifters. The standards were set up in general to classify what lifters were actually doing on the platform instead of what was simply expected of

them.

I know a lot of people who can hit the Elite numbers. Are your numbers right?

It is natural with any sport that the more accomplished lifters receive more press and recognition. “Elite” in this sense does not mean one is guaranteed to be a National Champion or a World Class lifter, although both of those lifters would be elite. “Elite” simply means very good. When looked at nationally, there will still be a reasonable number of Elite lifters per year. If you think you have seen a lot of people train in the gym who could be Elite, it is likely you might not be familiar with the standards set forth to complete the raw powerlifts in a competitive setting, and you may not be aware if those other lifters are drug free or not. The rankings on PLwatch.com confirm that an Elite level performance to the standards set is relatively rare. If you think you really know some people who could hit these numbers but they don’t currently compete, suggest that they go and lift in a competition; they should be very competitive.

Aren’t these standards too high?

The standards need to be moderately high in order for them to mean anything. If 50 people in each weight class are hitting Elite, then it lessens the title. You personally might have your own definition of Elite and Master and so on; these standards are based on the operational definitions put forth at the beginning of this article.

Aren’t these standards too low?

They were based on the rankings on PLwatch.com. It is likely that if all competitive powerlifters competed in the same federation under the same standards, there would be more “Elite” level lifters than the numbers here, and the numbers might need to be revised. However, use of powerlifting

equipment, strictness of judging, and drugs all play a significant role in the numbers lifters can put up. If more people want to come and lift under these standards and prove them too low, so be it. Until that time, all we can use is the information we have.

Will you revise these standards?

Yes, the plan is to reexamine them in 3–5 years and see how they are holding up. If they are still adequately describing the type of lifting seen, then it will not be necessary to revise them. If many more lifters are achieving the designations put forth, then the standards can be revised.

What's with the curl?

The curl was actually included in the first powerlifting competitions in the 1960s but it was dropped after a few years, likely due to the time a full powerlifting meet takes as it is. The curl has been reintroduced lately in several federations to test the strength of the arms. These standards apply to a strict curl (up against the wall) using an EZ bar. For those who like this lift, it is meant as a guide; for those who don't like this lift, ignore it. The curl is not added to a lifter's total; it is a stand alone event.

Why don't the Elite numbers on Curls increase after the 242 weight class?

So far there has not been a noticeable increase in curl strength in the heavier weight classes, so the standards reflect that.

How did you come up with the curl numbers?

Because fewer people compete in the curl, the standards are higher. The Elite number would generally give a lifter the first or second ranking for the curl in that weight class for that year for that federation. As more people compete in the curl, those numbers might rise. The curl numbers can be revised in 3–5 years if necessary as well.

How come the Master lifter is closer to Elite than the other categories?

The Master lifter ranking is closer to the Elite lifter ranking from a percentage point of view than the other rankings because of the fact of diminishing returns. In the beginning lifters start off using low weight but make great progress. The more experienced the lifter gets, the slower the gains come. Once a lifter has reached a high level of proficiency adding another 5 or 10% to their strength can be quite a challenge.

If I am not at a Class IV level, am I too weak to compete in powerlifting?

Powerlifting is more about competing with yourself and bettering yourself than lifting a certain amount of weight. If you can lift the bar and you enjoy testing yourself in a competitive environment, powerlifting is for you. You will very likely find the atmosphere at a powerlifting competition to be quite supportive. Almost all good lifters started out pretty weak at some point so they can relate to where you might be now. In addition if you are older (50 + years old) or younger (<18 years old), of course, it is less common for those lifters to lift extremely high amounts of weight.

What can I do with this information?

You can use this information to assess your strengths and weaknesses as a powerlifter. If you are a Class I bencher but a Class III deadlifter, it means your deadlift is a weak point for you and likely needs more work. You can use these standards as a motivational tool to lift more weight. You can use these standards to see if moving up or down a weight class is likely a good idea for you. If you can move up a level by gaining or losing body weight, it is likely a good idea; if your relative classification goes down, then it is likely not a good idea. Finally you can acknowledge your accomplishments and hard work by purchasing certificates that designate your ranking with a particular lift in a certain competition for that year.

If you have any questions/comments/concerns or feedback about these standards, please contact **Tim Henriques at NPTITim@aol.com**

Appendix H

Tim Henriques' Powerlifting Career Summary

Year	Age	Weight Class	SQ	BP	DL	Curl	Day	Total
1994	17	165	295	0	345	—	5 for 9	
1996	19	198	405	285	515	—	7 for 9	1205
1996	20	198	445	305	535	—	8 for 9	1285
1997	20	198	485*	290	550	—	7 for 9	1325
1997	21	198	525*	315*	565*	—	8 for 9	1405*
1998	21	198	525*	345*	545*	—	6 for 9	1415*
1998	22	181	530*	330*	545*	—	6 for 9	1405*
1999	22	181	505*	325*	575*	—	6 for 9	1405*
1999a	22	220	—	320*	585*	—	4 for 6	—
1999a	23	220	535*	325*	600*	—	8 for 9	1460*
2000	23	220	540*	350*	610*	—	9 for 9	1500*
2000	24	220	—	335*	590*	—	4 for 6	—
2001a	24	198	495*	360*	625*	—	6 for 9	1480*
2002	26	198	550*	385*	650*	—	9 for 9	1585*
2003	26	220	535*	390*	675*	—	8 for 9	1600*
2003	27	198	545*	370*	700*	—	7 for 9	1615*
2004	28	220	—	363	677*	—	5 for 6	—
2006	30	198	—	330	618	—	4 for 6	—

2007	30	198	—	325	590	177.5b	8 for 10	—
2007	30	198	—	335	—	180b	5 for 6	—
2007	31	198	—	340	600	160	7 for 9	—
2007	31	198	—	—	—	150	2 for 3	—
2008	31	220	—	325	605	165	7 for 9	—
2008	32	220	—	336	584	154	7 for 9	—
2009	32	220	—	355	585	—	4 for 6	—
2009	33	198	—	342	551	159	7 for 9	—
2010	34	181	—	308	—	148	5 for 6	—
2011	34	198	—	347.5	—	—	2 for 3	—
2011	34	220	—	380	606	160	6 for 9	—
2011	35	220	—	—	—	162	2 for 3	—
2012	35	220	—	386	634	168	7 for 9	—
2012	36	220	—	402	—	170.8	5 for 7	—
2013a	36	198	—	—	—	171.3	2 for 3	—

* This lift/total was made wearing single ply gear

a – I was drug tested at this event

b – This curl was performed standing up, not against the wall. It was an open federation record

Notable Life Incidents

- Took 2 years off after 2004 to focus on arm wrestling and strongman
- In 2007 I had my gall bladder removed
- In 2010 I fell down the steps and suffered a 100% rupture of my right quad tendon
- In 2013 I had my labrum in my right shoulder repaired

Why did I include this?

I included this information so that you, the reader, would have full disclosure

about me and perhaps more importantly for you to see the ups, downs, and rate of progression that one might have as they progress through 2 decades of competitive lifting. I have experienced some downs — some in the form of injuries and others in the lack of progress; the 2 are likely tied together. For example, I hit my best squat 8 years after I started competing, and I hit my best deadlift 9 years after I stepped on the platform. The bad news is that I have competed in 9 additional deadlift competitions since that time, and I never really came close to surpassing my old max.

But there is some good news to be found. After 19 years of training hard, I hit my best bench press and achieved a lifetime goal of benching more than 400 lbs. After 20 years of training hard, I set the Open National Record for the strict curl. I can promise that none of those things would have happened had I given up or quit because progress didn’t always come at the rate that I hoped or expected. This also gives you an idea as to the success, or lack thereof, with my attempts and it should highlight how important it is to successfully complete as many attempts as possible on the platform.

Appendix I

You know you are a powerlifter when...

What follows are some funny one-liners and jokes that go with the line, you know you are a powerlifter when..." Hopefully you can relate to some of these, and they will put a smile on your face, or maybe make you LMAO. *Note: The vast majority of these are not original to me; I found them in various places on the Internet, and I am just paying them forward.*

You know you are a powerlifter when…

You plan business trips according to what big gyms are in the area

You get excited by popped blood vessels

220s and below are the little guys

Your wife asks if you liked the dinner she cooked, and you respond by giving her 3 white lights

You consider high reps to be anything above 4

You get a bicep cramp from talking on the cell phone for more than two minutes

You give short, concise verbal cues that are too loud during sex

You have a separate drawer/closet space just for meet T-shirts

You can redlight squats just by hearing the federation

You love the blood running down your shins, which had just healed from your last deadlift workout

You can give people advice from your own experience and not an article you just read on the Internet

You lift enough weight that people stop giving you advice at the gym

You show people your hands; they tell you to get gloves, and you just laugh at them in response

You can’t count over 3, but you know exactly what 585 looks like on a bar

You check your depth while taking a dump

You’ve dropped something and gone to pick it up... in a sumo stance

You've used a Home Depot card to buy training equipment

You're having sex and think... "hmm this will help teach me to pull my hips through better."

Wasting chalk is a sin, and when Icy Hot and sweat smell sweet

You own ballet slippers and aren't accused of being gay

You consider sex as GPP

You have quit a job because it interfered with your training time

You consider "hardgainer" a Latin word for "undereater"

You help someone you don't know, or your chief rival, at a meet

Before the trophy is in your hand, you are planning your next meet

You would rather go to the Arnold than the Super Bowl

You find squat bruises on the opposite sex hot

You have to visualize yourself getting out of a chair before you do it

You are the only one in your gym who knows how to use the GHR

The weight on the bar matters more than grades on tests

You think Ed Coan is a super hero

You set a bucket down next to the platform on squat day

You perform a box squat when you get out of every chair

Your good shirts are either from powerlifting meets, or they have equipment companies' names on them

You use a false grip when you push anything with a handle

You worry about developing "the white lung" from chalk and baby powder

Your idea of a rite of passage for a man is their first meet

You would pay *$$$$$* to see Dave Tate bitch-slap Tony Horton

You still move more weight on a deload week than the strongest guy in your gym

You view a “sissy squat” as any squat that does not hit depth

Your wife asks you, “What are you thinking about?” and you lie because you don’t want to tell her you’re thinking about your next bench workout instead of her or the kids

Your diet advice to others is simple — Eat More

You think it is normal to allow another man to smack you in the face

You get pissed when someone is using YOUR power rack during your normal workout time

You put more effort into a set of 3 than the average person puts into an entire workout

You go berserk when someone grabs or touches the bar just to “help you” when you are about to set a PR

You were up all night puking on your deathbed thus you decide to just make today a speed day

Your HD camera is covered in chalk

You curl what most people can bench

You row what most people can deadlift

You squat what most people can leg press

You actually know what a good squat looks like

People are afraid to spot you

You wish your doctor or PT would talk to you in terms of your percent of your 1RM

You get pissed that more random people don't ask you what your max deadlift is just by looking at you

You wish that all jobs required you to list your max squat, bench, and deadlift on your resume

You think it is normal to admire another dude's development of his traps, triceps, and glutes

You squat with a mouth guard

Your dinner-time conversations over the day's events revolve around the postings on T-Nation.com forums

You can say "Eh, it's only 600 pounds"

The manager at the gym comes out and tells you that a picture frame just fell off the wall in his office because you were doing deadlifts

You die a little inside when someone says, "I don't know; squatting deep seems bad for your knees"

Your GPP is carrying in all your groceries in just one trip

You try to eat like a bodybuilder, which lasts until lunchtime, at which point you realize that is a dumb idea because bodybuilders are weak, and you suddenly find yourself staring at 2 double burgers, large fries, and a shake

You can't stop thinking about your traps and core when carrying something

You're over 40 and have Five Finger Death Punch on your I-pod

You weigh yourself before and after you take a dump and keep track of the

largest difference and try to beat it

You take off your shirt after a tough deadlift workout and you are disappointed you didn’t blow out more blood vessels on your chest and back

You don’t have to add up the weights on the bar; you know how much it is just by looking at it

You are a little disappointed when you look down after a good set of deadlifts and you are not bleeding

Appendix J

The Future

What's next for powerlifting? I wish I could say with certainty. I do believe that fitness in general and powerlifting in specific are growing and gaining more public recognition after a decade or two of stagnation. I do believe the raw movement is in part contributing to this, along with the swell of CrossFit and the serious dichotomy that the American population displays when it comes to fitness levels.

I hope to sell a large number of copies of this book. One of the reasons is quite simple — I would like to make some money off of this venture and the more money, the better; I certainly am not going to deny that fact. I would also like to gain additional recognition in the field. Another one of my reasons is very simple as well. I love powerlifting. That is the real reason I wrote this book. Even if I get nothing back, I believe there is a need for this information, and, honestly, I *needed* to get this information out. Writing this book was a big bucket-list item for me. I believe this book will benefit current and future powerlifters. That is enough for me.

Because I love powerlifting, I want to continue to support the sport, and I want to see it flourish. Possible ideas I have to help the sport include:

- Conducting more interviews with elite level powerlifters, making them more accessible to the reader and also focusing on the details of things — for example, analyzing lifts and getting their commentary on what

the lifter should do to improve.

- Helping to promote powerlifting and increased competition by securing cash prizes worthy of some attention for top level lifters.
- Putting on seminars to help lifters fix their technique and learn about other training methodologies.
- Building a network of like-minded individuals drawn to the same purpose.

> ***I love powerlifting. That is the real reason I wrote this book.***

If this book is a success, it will facilitate those goals becoming reality. The success of this book is in your hands. If you liked this book, if you learned from it, if you think it benefits the sport of powerlifting — tell other people about it. I am not coyly alluding to that idea, I am directly asking for your help. Use social media to talk about it, review it where it is offered for sale, give it to your lifting partners and friends as a holiday present or ask for it as a present if you can't afford to purchase it, and mention it on various forums and such where other lifters congregate. In this age of the Internet, a grassroots movement can quickly build momentum, and I believe powerlifting is ready for its Tipping Point. What role will you play in powerlifting's future?

Matt Kroc epitomizes what powerlifting is all about.
Photo credit: Matt Kroczaleski

About the Author

Tim Henriques has been a competitive powerlifter for more than 20 years. For the past decade he has focused on coaching his powerlifting team, Team Force, which has won 2 National Titles. His full time job is teaching people to become personal trainers. He is the Director/Owner of the National Personal Training Institute of Virginia, which was founded in 2001. NPTI is a 6 to 12 month, 600 hour long accredited program in which students learn about the human body, how it works, and how the body responds to exercise. Tim authored the comprehensive textbook for NPTI: *Fundamentals of Fitness and Personal Training.*

Tim was born in Fairfax, Virginia. His mother was a kindergarten teacher and his father was a history professor, Tim was the youngest of 4 boys. Initially a rather poor athlete, he found weights and applied himself to the iron discipline. He graduated high school as Health and PE Student of the Year. He went on to James Madison University where he received his degree in Kinesiology with minors in Psychology and Coaching. He became a

Collegiate All-American Powerlifter while at JMU. He has set State, National, and Federations records with several prominent powerlifting organizations. He is lifetime drug free.

Tim lives with his wife, Christina, and their 3 boys: Nathan, Ryan, and Collin. His passion is fitness, and apart from his full time job as the Director of NPTI, he continues to personal train and coach athletes and non-athletes alike. He has authored numerous articles and he regularly lectures and presents on the topics of health and fitness. He continues to coach, train, and compete with his powerlifting team in the never ending and always fulfilling search for that next personal best.